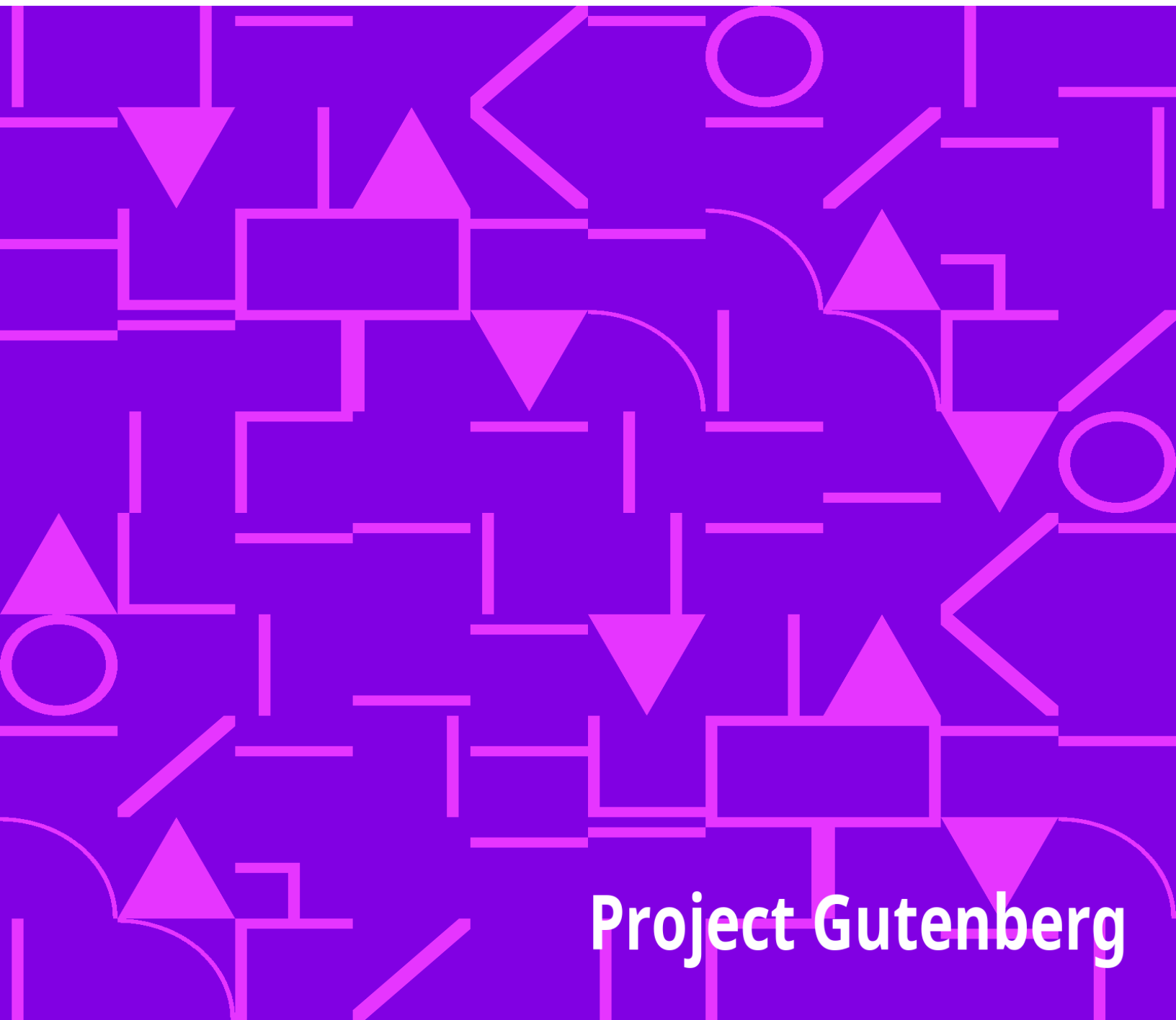


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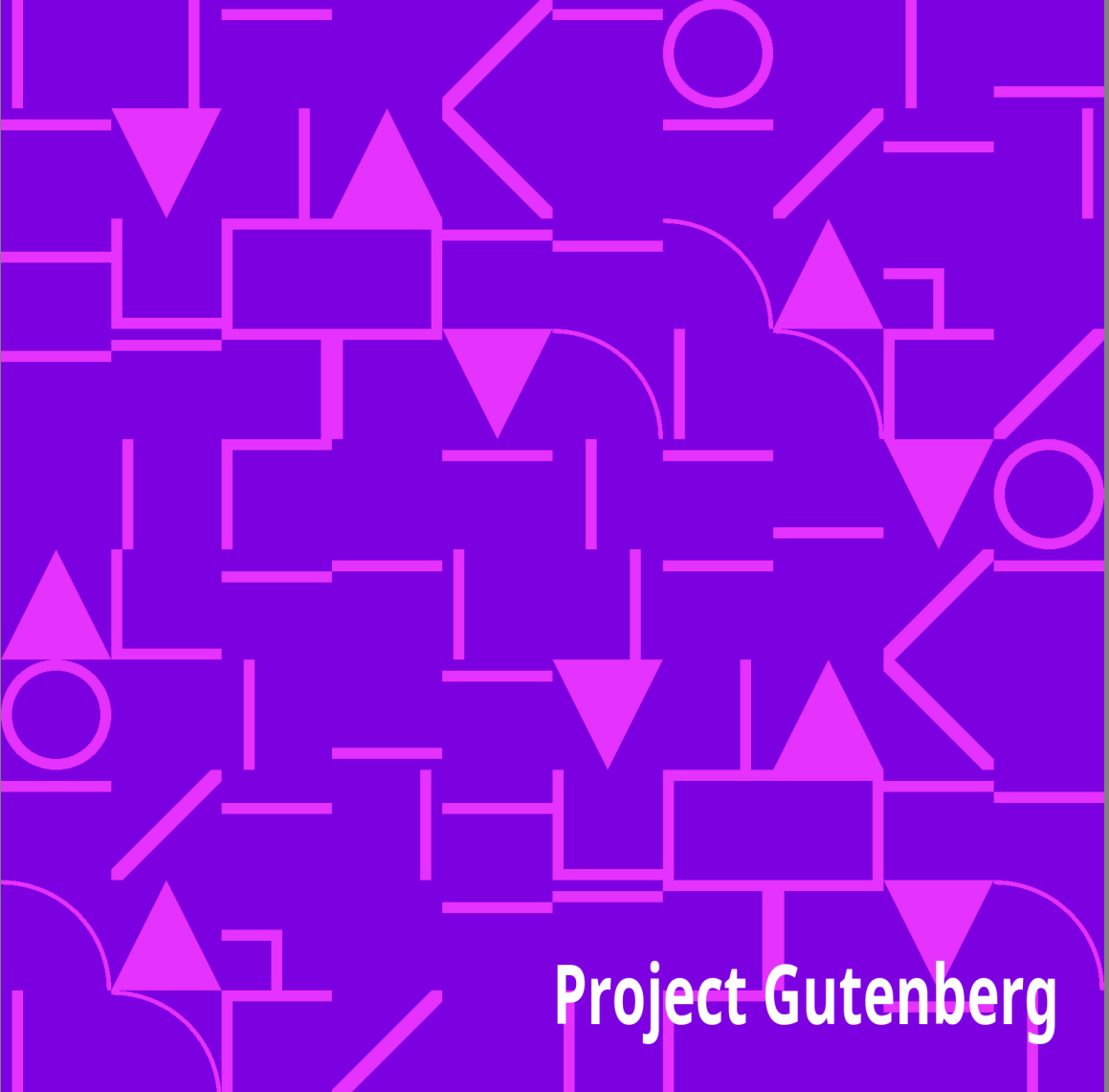
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note:

A few typographical errors have been corrected. They appear in the text like this, and the explanation will appear when the mouse pointer is moved over the

marked passage.

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VOLUME · IV · PART · 1



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KEY TO PRONUNCIATION

The method of marking pronunciations here employed is either (1) by marking the syllable on which the accent falls, or (2) by a simple system of transliteration, to which the following is the Key:—

VOWELS

ā, as in *fate*, or in *bare*.

ä, as in *alms*, Fr. *âme*, Ger. *Bahn* = á of Indian names.

a', the same sound short or medium, as in Fr. *bal*, Ger. *Mann*.

a, as in *fat*.

a", as in *fall*.

a, obscure, as in *rural*, similar to *u* in *but*, è in *her*: common in Indian names.

ē, as in *me* = *i* in *machine*.

e, as in *met*.

è, as in *her*.

ī, as in *pine*, or as *ei* in Ger. *mein*.

i, as in *pin*, also used for the short sound corresponding to ē, as in French and Italian words.

eu, a long sound as in Fr. *jeûne* = Ger. long ö, as in *Söhne*, *Göthe* (Goethe).

eu, corresponding sound short or medium, as in Fr. *peu* = Ger. ö short.

ō, as in *note*, *moan*.

o, as in *not*, soft—that is, short or medium.

ö, as in *move*, *two*.

ū as in *tube*.

u, as in *tub*: similar to *è* and also to *a*.

u", as in *bull*.

ü, as in Sc. *abune* = Fr. *û* as in *dû*, Ger. *ü* long as in *grün*, *Bühne*.

u', the corresponding short or medium sound, as in Fr. *but*, Ger. *Müller*.

oi, as in *oil*.

ou, as in *pound*; or as *au* in Ger. *Haus*.

CONSONANTS

Of the *consonants*, **b, d, f, h, j, k, l, m, n, ng, p, sh, t, v, z**, always have their common English sounds, when used to transliterate foreign words. The letter **c** is not used by itself in re-writing for pronunciation, **s** or **k** being used instead. The only consonantal symbols, therefore, that require explanation are the following:—

ch is always as in *rich*.

d, nearly as *th* in *this* = Sp. *d* in *Madrid*, &c.

g is always hard, as in *go*.

h represents the guttural in Scotch *loch*, Ger. *nach*, also other similar gutturals.

n', Fr. nasal *n* as in *bon*.

r represents both English *r*, and *r* in foreign words, which is generally much more strongly trilled.

s, always as in *so*.

th, as *th* in *thin*.

th, as *th* in *this*.

w always consonantal, as in *we*.

x = ks, which are used instead.

y always consonantal, as in *yea* (Fr. *ligne* would be re-written *lēny*).

zh, as s in *pleasure* = Fr. *j*.

THE NEW GRESHAM ENCYCLOPEDIA

VOLUME IV

Deposition of a Clergyman, the degradation of a clergyman from office, divesting him (in churches which do not, like the Church of Rome, hold the indelible nature of orders) of all clerical character.

Dépôt (dā'po or dep'ō), a French word in general use as a term for a place where goods are received and stored; hence, in military matters, a magazine where arms and ammunition are kept. The term is now usually applied to a military station situated in the centre of the recruiting district of a regiment, where recruits for this regiment are received and where they undergo preliminary training before joining their unit. In America it is the common term for a railway station.

Deprivation, the removing of a clergyman from his benefice on account of heresy or misconduct. It entails, of course, loss of all emoluments, but not the loss of clerical character.

De Profundis, in the liturgy of the Roman Catholic Church, one of the seven penitential psalms, the 130th of the *Psalms* of David, which in the *Vulgate* begins with these words, signifying 'Out of the depths'. It is sung when the bodies of the dead are committed to the grave.

Deptford (det'ford), a parliamentary and municipal borough, England, in the counties of Kent and Surrey, on the right bank of the Thames, now forming part of London. It has some manufactures of pottery, chemicals, and soap. The old naval dockyard was shut up in 1869, but the royal victualling yard is still the largest establishment of its kind. Deptford sends one member to Parliament. Pop. 109,496.

Deputies, Chamber of, the lower of the two legislative chambers in France and in Italy, elected by popular suffrage, and corresponding in some respects to the House of Commons in Britain.

De Quincey, Thomas, English author, was the son of a Manchester merchant, and born at Greenhay, near Manchester, on 15th Aug., 1785, died

at Edinburgh, 8th Dec., 1859. In 1793 his father died, leaving the family a fortune of £30,000. After attending some time the Bath and Manchester grammar schools, where he showed precocious ability, especially in classical studies, he importuned his guardian to send him to Oxford University, and on being refused he ran away from school, ultimately arriving in London in an absolutely destitute condition. His sufferings at this time he has described in his *Confessions of an English Opium Eater*. At length, in 1803, he matriculated at Oxford, and it was in the second year of his course here that he began to take opium in order to alleviate severe neuralgic pains. On leaving college he settled at Grasmere, Westmoreland, in the vicinity of Wordsworth and Southey, and devoted himself to literary work. Here or in London he remained till 1828, reading voraciously, and writing for the *London Magazine*, *Knight's Quarterly Magazine*, and *Blackwood's Magazine*. From 1828 to 1840 he lived in Edinburgh, then removed with his family to Lasswade, which continued to be his headquarters. His writings, nearly all contributions to magazines, are distinguished by power of expression, subtle thought, and an encyclopædic abundance of curious information. His work belongs to that class of literature which he himself called "the literature of power", as distinguished from "the literature of knowledge". He was eccentric in his habits, incapable of managing money matters, but amiable and polite.—BIBLIOGRAPHY: A. H. Japp, *Thomas De Quincey: his Life and Writings*; H. S. Salt, *De Quincey*; J. Hogg, *De Quincey and His Friends*; Sir L. Stephen, *Hours in a Library*; Arvede Barine, *Poètes et Névrosés*.

Dera Ghazi Khan, a district and town in the Punjab, Hindustan. The former, which is in Derajat division, has an area of 5606 sq. miles, and a population of 445,000. The town has a population of 18,466, half Hindus and half Mohammedans. It has extensive manufactures of silk, cotton, and coarse cutlery.

Dera Ismail Khan, a town in Hindustan, in the North-West Frontier Province, several miles to the west of the Indus, which is here crossed by boat-bridges, or boats, connecting the town with the Indus Valley railway. Dera Ismail Khan is a staple place for cotton goods, has a cantonment, and carries on a trade with Afghanistan. It is well laid out, has several schools and a large bazaar. Pop. 35,131.

Derajat (-jät'), a commissionership of Hindustan, in the west of the Punjab, occupying part of the valley of the Indus. It is well watered and fertile, and contains numerous towns and villages. Pop. 1,800,000, mostly Mohammedans.

Derbend', or **Derbent'**, a fortified town in Daghestan, Transcaucasia, on the west shore of the Caspian, an ancient place formerly belonging to Persia. The manufactures consist of woollen stuffs, copper- and iron-ware, and rose-water; and there is some trade in saffron, largely grown in the vicinity. Pop. 32,718.

Derby, Edward Geoffrey Smith Stanley, fourteenth Earl of, an English statesman, born at Knowsley Park, Lancashire, 29th March, 1799, died there, 23rd Oct., 1869. In 1820 he was returned to the House of Commons as member for Stockbridge. At first inclining to the Whig party, he joined Canning's ministry in 1827, and in 1830 became Chief Secretary for Ireland in Lord Grey's Government, greatly distinguishing himself by his speeches in favour of the Reform Bill in 1831-2. The opposition led by O'Connell in the House of Commons was powerful and violent, but Stanley, while supporting a Bill for the reform of the Irish Church and the reduction of ecclesiastical taxation, was successful in totally defeating the agitation for the repeal of the Union. He warmly advocated the abolition of slavery, and passed the Act for this purpose in 1833; but in the following year a difference of opinion with his party as to the diversion of the surplus revenue of the Irish Church led him to join the Tories. In 1841 he became Colonial Secretary under Sir Robert Peel, but resigned on Peel's motion for repeal of the corn laws. In 1851 and 1858 he formed ministries which held office only for a short period; and again in 1860, when his administration signalized itself by the reform of the government in India, the conduct of the Abyssinian War, and the passing of a Bill for electoral reform (1867). Early in 1868, owing to failing health, he resigned office, recommending Disraeli as his successor. Lord Derby joined to great ability as a statesman, and brilliant oratorical powers, a high degree of scholarly culture and literary ability. Among other works he published a successful translation of Homer's *Iliad* in 1864.—Cf. T. E. Kebbel, *English Statesmen Since the Peace of 1815: Derby*.

Derby, Edward George Villiers Stanley, seventeenth Earl of, British politician, born in April, 1865. He served for some years in the Grenadier Guards, and entered the House of Commons in 1892. In 1895 he became a Lord of the Treasury, in 1900 Financial Secretary to the War Office, and in 1903 Postmaster-General. He succeeded to the earldom in 1908. Lord Derby became quite famous as the originator of the *Derby Scheme*, when he was Director-General of Recruiting in 1915. Under-Secretary for War in 1916, he succeeded Lloyd George as Secretary when the latter became Premier. In 1918 he was appointed British Ambassador in Paris, but resigned his position in Sept., 1920, and left Paris in Nov., greatly to the regret of the French people, to whom he had proved a staunch friend.

Derby, a municipal, parliamentary, and county borough in England, capital of Derbyshire, on the Derwent, here crossed by a graceful bridge of three arches, 115 miles N.N.W. of London. It is pleasantly situated in a wide and fertile valley open to the south, and is well and regularly built in the modern quarter. It has some fine public buildings, amongst which are the churches of All Saints, St. Alkmund, and St. Werburgh, the county hall, school of arts, and infirmary. There is also a very handsome free library and museum. The manufactures include silk, cotton, hosiery, lace, articles in Derbyshire spar, iron castings, and porcelain; and the principal engineering works of the Midland Railway are here. Derby is one of the oldest towns in the kingdom, and is supposed to owe its origin to a Roman station, Derventio. Under the Danes it took the name of Deoraby. Richardson, the novelist, and Herbert Spencer were natives. It returns two members to Parliament. Pop. 123,930 (1919).—The county of *Derby*, in the centre of the kingdom, is about 55 miles long and from 15 to 30 miles broad; area, 650,369 acres, five-sixths being arable or in permanent pasture. It exhibits much varied and romantic scenery, the southern and eastern parts having a fertile soil, while the north-western portion is bleak, with a rocky and irregular surface. Here is the loftiest range of the English Midlands, the mountains of the Peak. The Peak itself is 2000 feet high. The principal rivers are the Derwent, the Trent, the Wye, the Erwash, the Dove, and the Rother. Oats and turnips are important crops, and dairy-husbandry is carried on to a large extent. Coal is abundant in various parts of the county, iron-ore is also plentiful, and lead, gypsum, zinc, fluor-spar, and other minerals are obtained. The manufactures are very considerable, especially of silk, cotton,

and lace, machinery, and agricultural implements. The county is divided into eight parliamentary divisions, each with one member. Pop. 683,423.—Cf. J. C. Cox, *Derbyshire*.

Derby-day, the great annual London holiday, on which is run the horse-race for the stakes instituted by Lord Derby in 1780. It always takes place on a Wednesday, being the second day of the grand race-meeting which falls in the week after Trinity Sunday. The race is run on Epsom Downs, an extensive Surrey course 15 miles from London. The course is 1½ miles, and the time usually about 2 minutes, 42 seconds. The entry money of each subscriber is fifty guineas, and the stakes are run for by three-year-old colts and fillies entered when yearlings. In the first year of the Derby there were only thirty-six entries, but they are now very numerous, and the value of the winner's prize is at least £5000. The race is the most popular of British sporting events. The Ascot races are patronized by royalty; the world of fashion is to be found at Goodwood; but Derby-day draws to Epsom a vast crowd of every class. During the European War the race was suspended, but a substitute Derby was run at Newmarket from 1915 to 1918. In 1919 the Derby was again run at Epsom, and was won by Lord Glanely's *Grand Parade*. In 1920 it was won by Loder's *Spion Kop*, and in 1921 by Joel's *Humorist*.

Derby Scheme, a scheme produced by Lord Derby in 1915 with a view to making a final effort on behalf of voluntary recruiting. The National Register (q.v.), taken in August, 1915, of all persons between the ages of fifteen and sixty-five, had shown that over five million men of military age had not 'joined up'. As the number of recruits each week fell short of the number required by Lord Kitchener, Lord Derby, appointed Director of Recruiting, produced a scheme in which men were divided into 46 groups. Groups 1 to 23 were for single men, and groups 24 to 46 for married men. Thus a married man of eighteen was in group 24, whilst a single man of forty-one was in group 23. The scheme ensured that single men would be called up first, whilst men who had 'attested' could appeal to a tribunal and claim temporary or permanent exemption. Two and a half millions offered themselves in consequence of the Derby Scheme, and two millions were accepted.

Dereham (dēr'am), **East**, a town in England, nearly in the centre of the county of Norfolk, with manufactures of agricultural implements, iron-foundries, and a brisk trade. The poet Cowper was buried in the church there, and George Borrow was born there in 1803. Pop. 5524.

Der'elict, a vessel or anything relinquished or abandoned at sea, but most commonly applied to a ship abandoned by the crew and left floating about.

Derg, Lough: (1) a lake, Ireland, county of Donegal, about 3 miles long by 2½ miles broad at the broadest part, and studded with islets, one of which, called Station Island, is a great resort of Roman Catholic pilgrims; (2) an expansion of the River Shannon between County Tipperary and Counties Clare and Galway, about 24 miles long and averaging 2 miles in breadth.

Derham, William, English philosopher and divine, born in 1657, died 1735. He was long rector of Upminster in Essex. In 1696 he published his *Artificial Clockmaker*. His best-known works are entitled *Physico-Theology*, *Astro-Theology*, and *Christo-Theology*.

Derma, or **Dermis**, is the true skin lying under the epidermis (cuticle), which is known in contrast as the scarf-skin.

Dermes'tes, a genus of beetles, one species of which (*D. lardarius*) is known by the name of bacon-beetle, and is often found in ill-kept ham or pork shops.

Dermot Mac Murragh, the last Irish King of Leinster, attained the throne in 1140. Having carried off the wife of O'Ruare, Prince of Leitrim, he was attacked by the latter, and after a contest of some years driven out of Ireland (1167). He then did homage to the English king, and with the help of Richard, Earl of Pembroke, recovered his kingdom, but died in the same year (1170), and was succeeded by Pembroke, who had married his daughter.

Dernburg, Bernhard, German financier and administrator, born in Darmstadt in 1865. His father was an editor of the *Berliner Tageblatt*. At the age of nineteen he came to New York to study banking methods, stayed there for several years, and on his return to Germany became director of a financial company. In 1906 he succeeded Prince von Hohenlohe Langenburg as Colonial Secretary. During his administration the condition

of German colonies, and especially of German East Africa, was greatly improved. From 1914 to 1915 he was a zealous organizer of German propaganda in the United States, but was subsequently compelled to leave the country.

Déroulède, Paul, French poet, politician, and agitator, born in Paris in 1846, died there in 1914. Called to the Bar in 1870, he served in the Franco-Prussian War, and afterwards carried on an active and passionate agitation for a war of revenge against Germany. He founded the Ligue des Patriotes, which was suppressed by the Government in 1889, was an ardent supporter of General Boulanger, and one of the leaders of the reactionary forces during the Dreyfus case. He sat in the Chamber of Deputies from 1893 to 1895 and from 1898 to 1899. In 1900 he plotted against the Republic, and sought to bring about a nationalist *coup d'état* with a view to overthrowing the parliamentary constitution. Found guilty, he was sentenced to ten years' exile, but was allowed to return in 1905, under the law of amnesty. His works include: *Chants de Soldat* (1872); *Nouveaux Chants de Soldat* (1875); *Chants du paysan* (1894); *Poésies Militaires* (1896); *La Moäbite*, a religious drama (1880), forbidden by the Censor; and *L'Helman*, a patriotic play (1877).

Derrick, a simple kind of crane, chiefly used on board ship, consisting of a stout pole swung from a mast, and carrying hoisting-tackle at its upper end. The name is derived from that of a celebrated hangman.

Derringer, a small pocket pistol with a short barrel and a large calibre (usually .41), very effective at short range. It is named after its inventor, a gun-smith in the United States of America, and would seem to have been first used in about 1850. It is a single-shot weapon, and so has largely been superseded by revolvers and automatic pistols.

Der'vish, or **Dervise** (Pers., 'seeking doors' or beggar, and equivalent to the Ar. *fakir*), a Mohammedan devotee, distinguished by austerity of life and the observance of strict forms. There are many different orders of dervishes, the underlying idea of most of them being the revival and increase of the Moslem faith. Some live in monasteries, others lead an itinerant life, others devote themselves to menial or arduous occupations. They are respected by the common people, and the mendicants among them carry a wooden bowl

into which the pious cast alms. One of their forms of devotion is dancing or whirling about, another is shouting or howling, uttering the name *Allah*, accompanied by violent motions of the body, till they work themselves into a frenzy and sometimes fall down foaming at the mouth. They are credited with miraculous powers, and are consulted for the interpretation of dreams and the cure of diseases. See *Mohammedanism*.—BIBLIOGRAPHY: E. W. Lane, *Modern Egyptians*; J. P. Brown, *The Dervishes, or Oriental Spiritualism*; S. M. Zwemer, *Arabia, the Cradle of Islam*.

Der'went, the name of four rivers in England, in Derbyshire, Yorkshire, Durham, and Cumberland respectively, the last draining Derwentwater Lake. Also a river in Tasmania.

Derwentwater, James Radcliffe, third and last Earl of, one of the leaders in the rebellion of 1715, born in London 28th June, 1689. The standard of revolt having been raised in Scotland, Lord Derwentwater commenced the movement in England on 6th Oct., 1715, but was forced, along with the other Jacobite nobles, to surrender at discretion on 13th of Nov. He was executed on Tower Hill 24th Feb., 1716, his estates being confiscated, and in 1735 granted to Greenwich Hospital.

Der'wentwater, or **Keswick Lake**, a beautiful lake in Cumberland, England, in the vale of Keswick. It is about 3 miles in length and 1½ miles in breadth, and stretches from Skiddaw on the north to the hills of Borrowdale. Near the south-east corner is the celebrated cascade of Lodore. Its waters are carried to the sea by the Derwent.

Derzhavin (der-zhä'vin), Gabriel Romanovitsh, a Russian lyric poet, born in 1743, died in 1816. He entered the army as a private soldier, distinguished himself highly, and was eventually transferred to the Civil Service, in which he obtained the highest offices. In 1803 he retired from public life and devoted himself entirely to poetry. One of his most beautiful poems is *Oda Bog*, or *The Address to the Deity*.

Desaguadero (des-a' -gwa' -dā'rō), a river of Bolivia, in a valley of the same name, issuing from Lake Titicaca, and carrying its waters into Lake Aullagas. Also a river in the Argentine Confederation flowing into Lake Bevedero Grande, and separating the provinces of San Juan and Mendoza. Desaguadero signifies in Spanish 'a channel of outlet'.

Desaix de Veygoux (dè-sā dè vā-gö), Louis Charles Antoine, a distinguished French general, born in 1768 at St. Hilaire d'Ayat, in Auvergne, died 1800. He was of noble family, and entered the army as a sub-lieutenant. He distinguished himself greatly in 1794 under Pichegru, and two years later with the army of the Rhine under Moreau. In 1797 he accompanied Bonaparte to Egypt, and was very successful in reducing Upper Egypt. After the Treaty of El Arish he followed Bonaparte to Italy, took command of the corps of reserve, and, arriving on the field of Marengo at a critical moment, decided the victory by a brilliant charge, 14th June, 1800. He himself fell, mortally wounded by a musket bullet.

Desault (dè-sö), Pierre Joseph, one of the most celebrated surgeons of France, was born in 1744, and died in 1795. After some experience in the military hospital at Bèfort, he went to Paris in 1764, studied under Petit, and two years afterwards became a lecturer on his own account. His reputation soon increased, and he became principal surgeon in the Hospital de la Charité, and in 1788 was put at the head of the great Hôtel Dieu in Paris. Here he founded a surgical school, in which many of the most eminent surgeons of Europe were educated. A digest of his surgical works was published by Bichat in 1798 (*Œuvres chirurgicales de Desault*).

Des´cant, in music, an addition of a part or parts to a subject or melody, a branch of musical composition which preceded the more modern counterpoint and harmony, coming into existence at the end of the eleventh or beginning of the twelfth century.

Descartes (dā-kärt), René, a great French philosopher and mathematician, with whom the modern or new philosophy is often considered as commencing. He was born 31st March, 1596, at La Haye, in Touraine, died 11th Feb., 1650. Educated at the Jesuit College of La Flèche, where he showed great talent, he entered the military profession and served in Holland and in Bavaria. In 1621 he left the army, and after a variety of travels finally settled in Holland, and devoted himself to philosophical inquiries. Descartes, seeing the errors and inconsistencies in which other philosophers had involved themselves, determined to build up a system anew for himself, divesting himself first of all of the beliefs he had acquired by education or otherwise, and resolving to accept as true only what could stand the test of reason. Proceeding in this way, he found (*Meditationes de*

Prima Philosophia) that there was one thing that he could not doubt or divest himself of the belief of, and that was the existence of himself as a thinking being, and this ultimate certainty he expressed in the celebrated phrase "*Cogito, ergo sum*" (I think, therefore I am). Here, then, he believed he had found the test of truth. Starting from this point Descartes found the same kind of certainty in such propositions as these: that the thinking being or soul differs from the body (whose existence consists in space and extension) by its simplicity and immateriality and by the freedom that pertains to it; that every perception of the soul is not distinct; that it is so far an imperfect finite being; that this imperfection of its own leads it to the idea of an absolutely perfect being; and from this last idea he deduces all further knowledge of truth. Descartes has been rightly called the father of modern philosophy, for in him the modern spirit came into existence. In an age of unrest and ferment his work was that of a great systematizer. He also contributed greatly to the advancement of mathematics and physics, the method of co-ordinates, which has revolutionized geometry, and is of fundamental importance in modern mathematical physics, being due to him. His system of the universe attracted great attention in his time. One of his fundamental doctrines was that the universe is full of matter, there being no such thing as empty space. On this basis he developed the hypothesis of celestial vortices, immense currents of ethereal matter, by which he accounted for the motion of the planets (*Principia Philosophiæ*, 1644). His works effected a great revolution in the principles and methods of philosophy. In 1647 the French court granted him a pension of 3000 livres, and two years later, on the invitation of Christina of Sweden, he went to Stockholm, where he died.—BIBLIOGRAPHY: Sir J. P. Mahaffy, *Descartes*; A. Fouillée, *Descartes*; E. Boutroux, *L'Imagination et les mathématiques selon Descartes*; N. Smith, *Studies in Cartesian Philosophy*; E. S. Haldane, *Descartes: his Life and Times*.

Descent', in law, is the transmission of the right and title to lands to the heir, on the decease of the proprietor, by the mere operation of law. The rule determining to whom an estate belongs, on the decease of the proprietor, is that of consanguinity, or relationship by blood, though with some exceptions, as in the case of the portion, or the use of a portion, of a man's property given by the laws of Britain to his widow. The rules of descent, designating what relations shall inherit, and their respective shares, will be

determined by the genius and policy of the Government and institutions. Hence the practice of entailments in the feudal system. And wherever the Government is founded in family privileges, or very intimately connected with them, as is the case in all Governments where the hereditarily aristocratical part of the community has a great preponderance, the sustaining of families will very probably be a characteristic feature in the code of laws. Thus, in Britain, all the lands of the father, unless otherwise directed by will, go to the eldest son. (In England, however, this privilege may shortly be abolished, or the succession to real estate assimilated to the succession to personal estate.) In the United States of America this distinction in favour of the eldest son has been abolished, and the laws there are founded upon the principle of equal distribution both of real and personal estate among heirs of the nearest surviving degree. Kindred in blood are divided into three general classes, viz. 1, descendants; 2, ancestors; 3, collateral relatives, that is, those who have descended from the same common ancestor. The civil law computes the degrees by counting the generations up to the common ancestor, as father, grandfather, great-grandfather; or mother, grandmother, great-grandmother; and from him or her down to the collateral relative, as brother, cousin, &c., making the degree of relationship the sum of these two series of generations. Every person has two sets of ancestors, the paternal and maternal, and therefore two sets of collateral relatives. There is also a distinction of collateral kindred, into those of the whole blood and those of the half blood.—Cf. Pollock and Maitland, *History of English Law*.

Deschanel, Paul Eugène Louis, French statesman and tenth President of the French Republic, born at Brussels on 13th Feb., 1856, where his father was in exile for opposition to Napoleon III. Educated at the Collège Sainte Barbe and the Lycée Condorcet, Paris, he was private secretary first to De Marcère, the Home Secretary, in 1876, and then to the Premier, Jules Simon, in 1877. Sub-prefect at Dreux, Brest, and Meaux, he was elected to the Chamber of Deputies in Oct., 1885, for Nogent le Rotrou. In the Chamber of Deputies he soon displayed his power of eloquence, and eventually became the leader of the Progressive Republicans and exponent of the separation of Church and State. His authority, which he owed not only to his eloquence but also to his great tact and correct and dignified conduct during debates, gradually increased in the Chamber of Deputies,

and even his political enemies admired his courage and his superiority of intellect. He was consequently elected Vice-President of the Chamber in 1896 and in 1897, and succeeded Henri Brisson as President of the Chamber in 1898, retaining his office till 1902, when M. Léon Bourgeois became President of the Chamber. Re-elected President of the Chamber in 1912, and withdrawing his candidature to the presidency of the Republic in favour of Poincaré in 1913, Deschanel retained his post until 1920, when he was elected President of the Republic by 734 votes. In the meantime, however, his health had given way, and on 23rd May, 1920, he fell out of a moving train, an accident resulting in a prolonged illness. A prompt recovery was at first expected, but this hope soon proved vain. Unable to attend to the duties of his office, he resigned on 16th Sept., and was succeeded by Alexandre Millerand. He was elected to the Académie Française on 18th May, 1899, and his works include: *La Question du Tonkin* (1883), *La Politique Française en Océanie* (1884), *Orateurs et Hommes d'État* (1888), *Figures des Femmes* (1889), *Figures Littéraires* (1889), *La Question Sociale* (1898), *Quatre ans de Présidence* (1902), *Politique Intérieure et étrangère* (1906), *Paroles Françaises* (1911), *La France Victorieuse* (1919), and *Gambetta* (1920).

Deseada (de-se-ä´da´), or **Désirade**, one of the Leeward Islands, belonging to the French, in the Caribbean Sea, about 10 miles long and hardly 5 miles broad. The soil is in some places black and good, in others sandy and unproductive. Pop. about 1500.

Deser´tion. In the military sense this implies the act of an officer or soldier who, either with the intention of not returning at all or of returning when by reason of his absence it has become impossible for him to perform a given duty, absents himself from His Majesty's service. The essence of the offence is intention, and this must be proved to the satisfaction of the court martial by which the offender is tried. The distance travelled is no criterion, and a man arrested while leaving the barracks in disguise may be tried for attempted desertion. In this connection it should be noted that an attempt to desert, or an attempt to procure another person to desert, is also an offence, and is punishable under Section 12 of the Army Act equally with the completed act. As far back as the middle of the fifteenth century the crime of desertion by a soldier was made a felony triable by the ordinary law of the land, and with certain interludes death remained the maximum penalty

for desertion for some 200 years. The present existing Army Act recognizes two degrees of desertion, i.e. that committed on active service or when under orders for active service, and that committed under other circumstances. Under the former heading the maximum punishment is death; under the latter, imprisonment or penal servitude, according to the number of times the offence has been committed. A soldier convicted of desertion forfeits the whole of his service prior to conviction.

Desertion, in law, the term applied to the act by which a man abandons his wife or a wife her husband. Such desertion without due cause is, in England, ground for a judicial separation. Property a wife may have acquired since desertion is protected against her husband, and he may be forced to provide for her maintenance. By Scottish law, where either the husband or wife has deserted and remained separate without due cause for four years, divorce may be obtained.

Deshoulières (dā-söl-yār), Antoinette du Ligier de Lagarde, a French woman of much literary reputation in the seventeenth century, born 1638, died 1694. She was the centre of attraction in the best circles of the period, and was elected a member of several learned societies. She wrote in the *Mercurie Galant* under the name of Amaryllis, and Boileau has described her in his tenth *Satire*. Among her works are odes, eclogues, idyls, and a tragedy (*Genseric*).

Desiccation, a process of dispelling moisture by the use of air, heat, or chemical agents such as chloride of calcium, quicklime, oil of vitriol, and fused carbonate of potash.—*Desiccation cracks*, in geology, are the fissures caused in clayey beds by the sun's heat, and seen in various rock strata.

Design, thought, arrangement, or grouping, imagination or invention in works of art. A design is a composition or invention, pictorial, architectural, or decorative. It may be simply an imperfect sketch, as a record of a first thought; or it may be a fully matured work, as a cartoon in preparation for fresco painting, or a drawing to illustrate a book.

Desmidia'ceæ, or **Desmidie'æ**, a family of microscopic, fresh-water Green Algæ, group Conjugatæ. They are green gelatinous plants composed of variously formed cells having a bilateral symmetry, which are either free, or in linear series, or collected into bundles or into star-like groups, and

embedded in a common gelatinous coat. Desmidiaceæ differ from Diatomaceæ in their green colour and absence of silica.

Desmo'dium, a large genus of leguminous plants, sub-ord. Papilionaceæ, natives of tropical and sub-tropical countries. *D. gyrans* is the telegraph plant, so called from the curious movements performed by the two small lateral leaflets of each leaf, which, at high temperatures, wave up and down like the arms of a semaphore.

Des Moines (dè moin), a city of the United States, capital of the state of Iowa and of Polk County, on the Des Moines River, about 350 miles west of Chicago. Among its chief buildings are the new State house, the State arsenal, university, and opera-houses. There are coal-mines in the vicinity, and the manufactures and other industries are varied and rapidly increasing. Pop. 104,052 (1917).

Des Moines, the largest river in the state of Iowa, rises in the S.W. of Minnesota, and flows in a south-easterly direction till it falls into the Mississippi about 4 miles below Keokuk, after a course of 300 miles.

Desmoncus, a genus of tropical American climbing palms, scrambling by means of stout hooks which are modified leaflets.

Desmoulins (dā-mö-lan), Benoît Camille, born in 1760, was conspicuous during the first period of the French Revolution. He was amongst the most notable of the pamphleteers and orators who urged the multitude forward in the path of revolution. He, along with others, prepared the plan for the taking of the Bastille (July, 1789), was one of the founders of the club of Cordeliers, and the promoter of the assembly in the Champ de Mars. In 1789 he began his career as a journalist by the issue of a weekly publication (*Les Révolutions de France et de Brabant*). In 1793 he gave his vote for the death of the king. Having become closely connected with Danton and the party of opposition to Robespierre, and inveighing against the reign of blood and terror, he was arrested on the order of the latter on 30th March, 1794, tried on the 2nd April, and executed on the 5th. He met his fate in an agony of despair.—Cf. F. A. Aulard, *Les Orateurs de la Législative et de la Convention*.

Desna, a river in the Ukraine, which rises in the government of, and about 50 miles east of the town of Smolensk, flows through the governments of Orel and Tshernigov till it joins the Dnieper near Kiev. It is 500 miles in length and navigable nearly throughout.

De Soto, Hernando, a Spanish explorer and discoverer of the Mississippi, born about 1496, died in 1542. He accompanied expeditions to the New World under Davila and Pizarro, and played a distinguished part in the conquest of Peru. In 1536 he led an expedition to Florida, whence after many difficulties he penetrated to the Mississippi, where he was attacked with fever and died.—The name De Soto has been given to a county in the N.W. of Mississippi, and to several places in the United States.

Des'pot (Gr. *despotēs*), originally a master, a lord; at a later period it became an honorary title which the Greek emperors gave to their sons and sons-in-law when governors of provinces. At present *despot* means an absolute ruler, and in a narrower sense a tyrannous one.

Dessalines (dā-sa' -lēn), Jean Jacques, Emperor of Hayti, born in Africa about 1760. He was a slave in 1791, when the insurrection of the blacks occurred in that island, but was set free along with the other slaves in St. Domingo in 1794. His talents for war, his courage, and unscrupulous conduct raised him to command in the insurrections of the coloured people, and after the deportation of Toussaint-L'Ouverture, and the subsequent evacuation of the island by the French, Dessalines was appointed Governor-General for life with absolute power; and the year following (1804) was declared Emperor with the title of Jacques I. But his rule was savage and oppressive, and both the troops and the people, sick of his atrocities, entered into a conspiracy against him, and on 17th Oct., 1806, he was slain by one of his soldiers.

Dessau (des'ou), a town in Germany, capital of the former Duchy of Anhalt, in a beautiful valley on the left bank of the Mulde, mostly well built, with fine squares and many handsome buildings. The manufactures consist of woollens, woollen yarn, carpets, machinery, and tobacco. The former ducal palace has a picture-gallery and interesting relics and antiquities. Pop. 56,606.

De Stendhal. See *Beyle, Marie-Henri*.

Dester'ro, now **Florianopolis**, a seaport of Brazil, capital of the state of Santa-Catharina. The harbour, next to Rio de Janeiro, is the best in Brazil. Pop. 9000.

Destructors, Refuse, the apparatus or plant used in the cremation of house and factory refuse. Formerly refuse from large towns and populous areas was either disposed of for manurial purposes or spread over waste land, where its presence speedily became a nuisance. The consequent danger to the public health, coupled with the increasing difficulty and cost of its disposal, led to attempts being made to deal with it by more sanitary methods. The first attempts to cremate it in a closed furnace on a large scale, made in London in 1870, were not successful, primarily owing to the unscientific design of the furnace. In 1877, however, there was built in Manchester the forerunner of the modern destructor, to the designs of the late Mr. Alfred Fryer. This consisted of two simple cells, the primary feature of the design being 'the charging of the refuse into the back of the furnace, and drawing out the resulting clinker from the front'. With certain modifications, this design is in general use at the present time. The basic principles underlying their scientific design may be summed up in the following points: (a) Charging at regular intervals, the refuse being dumped into a hopper, and fed into the back of the furnace as a charge, means being taken to prevent undue escape of gases. The moisture having evaporated, the material may then be raked forward onto the fire-bars, where combustion takes place. (b) To avoid nuisance, the resulting gases must be inodorous, which requires a temperature of about 2000°F. This is obtained by passing the products of combustion over the hottest portion of the fire, prior to their passage into the main flue. (c) Removal of the fine dust and particles in suspension in the flue gases by means of spiral chambers at the base of chimney. (d) Provision of forced draught. A well-designed plant should show little, if any, suspended matter in the gases on emission from the chimney. Formerly no use was made of the heat generated, but modern installations now invariably have steam-raising plant incorporated in the design. This power is commonly used to generate electricity. Some portion may be utilized in crushing and screening the clinker, which is frequently used in making slab-paving, or ground fine and used as a substitute for sand in mortar. The following data may be found of service: (a) 60 per cent of the average refuse is combustible. (b) The calorific value may be taken as

being one-seventh that of good steam-coal. (c) 100 electrical units may be obtained from each ton of refuse consumed. (d) With forced draught, as much as 100 lb. of refuse has been burnt per square foot of grate area per hour. (e) 250 tons of refuse per annum per 1000 of population may be estimated for.—BIBLIOGRAPHY: W. H. Maxwell, *Removal and Disposal of Town Refuse*; Kempe, *Engineer's Year Book*.

Destutt de Tracy, Antoine Louis Claude, French philosophical writer, born in 1754 of a family of Scottish extraction, died in 1836. As a philosopher he belonged to the Sensationalist school, and considered all our knowledge to be derived originally from sensation. He has been called the logician and metaphysician of the school of Condillac. Among his chief works are: *Idéologie* (1801), *Logique* (1805), and *Traité de la Volonté* (1815).

Deter'minant, a mathematical expression which appears in the solution of a system of equations of the first degree.

The solution of the equations

$$a_1x + b_1y = c_1,$$

$$a_2x + b_2y = c_2,$$

is given by

$$x(a_1b_2 - a_2b_1) = c_1b_2 - c_2b_1,$$

$$y(a_1b_2 - a_2b_1) = a_1c_2 - a_2c_1.$$

Here x and y have the same coefficient $a_1b_2 - a_2b_1$. This is called a determinant of the second order, and is written

$$\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}.$$

We may now write

$$x \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} = \begin{vmatrix} c_1 & b_1 \\ c_2 & b_2 \end{vmatrix}.$$

Similarly with three equations,

$$a_1x + b_1y + c_1z = d_1,$$

$$a_2x + b_2y + c_2z = d_2,$$

$$a_3x + b_3y + c_3z = d_3,$$

we find

$$\begin{aligned} & x\{a_1(b_2c_3 - b_3c_2) + a_2(b_3c_1 - b_1c_3) + a_3(b_1c_2 - b_2c_1)\} \\ &= d_1(b_2c_3 - b_3c_2) + d_2(b_3c_1 - b_1c_3) + d_3(b_1c_2 - b_2c_1) \end{aligned}$$

The coefficient of x , and the right-hand member, are here determinants of the third order, and the result is written

$$x \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}.$$

When the coefficient of x is expanded fully, it consists of six terms, half of them positive and half negative, each term being the product of three letters, one from each row and one from each column. Thus in every term, such as the leading term $a_1b_2c_3$, the three letters a, b, c , and the three suffixes 1, 2, 3, all occur once and once only. With regard to the signs, we note that if the letters a, b, c are always kept in this order, an interchange of two suffixes changes the sign. Thus, e.g. we may start from $+ a_1b_2c_3$, interchange the suffixes 2 and 3, and thus find $- a_1b_3c_2$. Again, if in the latter term $- a_1b_3c_2$ we interchange the suffixes 1 and 3, we find $+ a_3b_1c_2$. It is easy to verify that after any number of interchanges of this kind a particular term will always come up with the same sign.

Thus the interchange of two rows (and similarly of two columns) changes the sign of the determinant. It follows that if two rows (or two columns) are identical, the determinant is zero. It is now easy to define functions of n^2 letters, arranged in a square array of n rows and n columns, with similar properties to those observed above in the case of determinants of the third order. Determinants are very extensively used in higher algebra, co-ordinate

geometry, and other branches of mathematics. See also *Elimination*.—
BIBLIOGRAPHY: C. Smith, *Algebra*; T. Muir, *Theory of Determinants*.

Determinants, in biology, the name applied by Weismann to hypothetical particles contained in the nuclei of germ-cells, which determine the existence and nature of the various parts of the body of the embryo. The theory of determinants is part of an elaborate attempt to explain the facts of heredity on a mechanical basis. Although it may serve as a working hypothesis, it is regarded with disfavour by many experts, and may be regarded as an elaboration of the 'provisional theory of pangenesis' advanced by Darwin.

Deter'minism, a term employed by recent writers, especially since J. Stuart Mill, to denote a philosophical theory which holds that the will is not free, but is invincibly determined either—according to the older form of the theory—by a motive furnished by Providence, or—according to the modern form—by the aggregation of inherited qualities and tendencies. Biological determinism maintains that each of our voluntary acts finds its sufficient and complete cause in the physiological conditions of the organism. Psychological determinism ascribes efficiency to the psychical antecedents. Opposed to determinism is the doctrine of indeterminism or indifferentism. See *Free-will*.

Det'inue, in law, the form of action whereby a plaintiff seeks to recover a chattel personal unlawfully detained.

Det'mold, a town, Germany, capital of Lippe-Detmold, on the left bank of the Werra, 50 miles south-west of Hanover, with a new and an old palace (or castle), good public library, and museum. In the vicinity a colossal statue has been erected to the Hermann or Arminius who overthrew the Roman general Varus and his legions in a battle which was fought near this place. The Senner race of horses is bred near Detmold. Pop. 14,295.

Det'onating Powders, certain chemical compounds which, on being exposed to heat or suddenly struck, explode with a loud report, owing to one or more of the constituent parts suddenly assuming the gaseous state. The chloride and iodide of nitrogen are very powerful detonating substances. Mercuric fulminate or fulminating mercury ($C_2HgN_2O_2$) explodes violently when forcibly struck or when heated to 180°F. It is used

for making percussion caps, and in detonators for exploding guncotton and nitroglycerin preparations. Silver fulminate ($C_2Ag_2N_2O_2$) explodes even more violently.

Detonating Tube, a species of eudiometer, being a stout glass tube used in chemical analysis for detonating gaseous bodies. It is generally graduated into centesimal parts, and perforated by two opposed wires for the purpose of passing an electric spark through the gases which are introduced into it, and which are confined within it over mercury and water.

Detroit (de-troit'; Fr. *détroit*, a strait or channel), a flourishing port and city of the United States, the largest town in Michigan, situated on the Detroit River, connecting Lakes Erie and St. Clair. The site rises gradually from the river, and the city is generally well built. Among the chief edifices are the city hall, the house of correction, post office, and opera-house. Detroit has increased very rapidly, a fact which is due to its admirable position for trade, and to its connections with a region into which a constant tide of emigration is flowing. Among the industrial establishments are saw-mills, flour-mills, building-yards for ships and boats, foundries, tanneries, blast-furnaces, pork-packing establishments, tobacco and cigar manufactories, and locomotive works. The harbour is one of the finest in the United States, and has a depth of water sufficient for the largest vessels. Detroit owes its origin to the French, who visited the site in 1648 and erected the Fort Pontchartrain in 1701. Pop. 688,028 (1919).

Detroit River, or **Strait of St. Clair**, a river or strait of North America which runs from Lake St. Clair to Lake Erie. It is 28 miles long, and of sufficient depth for the navigation of large vessels. It is about $\frac{3}{4}$ mile wide opposite Detroit and enlarges as it descends.

Dettingen (det'ing-en), the name of several places in Germany, amongst which is a village of Bavaria, on the right bank of the Main, famous for the victory gained by the English and Austrians under George II of England over the French in 1743.

Deuca'lion, in Greek mythology, the son of Prometheus and father of Hellen, ancestor of the Hellenes. According to tradition he saved himself and his wife, Pyrrha, from a deluge which Zeus had sent upon the earth, by building a ship which rested upon Mount Parnassus. To repair the loss of

mankind they were directed by an oracle to throw stones behind them; the stones thrown by Deucalion became men, those thrown by Pyrrha women.

Deus ex Machinâ (mak'i-na; Lat., 'a god out of the machine'), a phrase used to designate the resorting to supernatural causes to explain phenomena that one is not able to account for by natural means. The phrase is taken from the practice on the classical stage of introducing a god from above by means of some mechanical contrivance in order to effect a speedy *dénouement* of the plot.

Deuteronomy (Gr. *deuteronomion*, the second law), the name of the fifth book of the *Pentateuch* (q.v.). Until the seventeenth century it was believed to have been written by Moses, but now it is generally held to be a compilation, the bulk of it having been written in the reign of Manasseh, or according to other scholars in the reign of Josiah. At any rate the book was discovered or rediscovered while Josiah was king (2 *Kings*, xxii). It represents the latest phase in the development of the teaching of Moses. Its chief aim is to combat idolatry, and to concentrate the religious life of the country at Jerusalem. It has a lofty moral tone. Of one passage (vi, 4 and 5) Christ said: "On these two commandments hang all the law and the prophets" (*Matt.* xxii, 40), and all His answers to the Tempter in the Wilderness are taken from the book of *Deuteronomy*.

Deutz (doits), a town in Prussia, on the right bank of the River Rhine, opposite the city of Cologne, with which it communicates by a bridge. It is strongly fortified as part of the defences of Cologne, in which it is now incorporated. There are some manufactories of porcelain and glass, also an iron-foundry and machine-works. Pop. 17,060.

Deut'zia, a genus of plants, nat. ord. Saxifragaceæ, containing seven or eight species, all of which are interesting from the beauty of their flowers, some of them favourite garden and greenhouse plants. They are small shrubs indigenous to China and Japan, and Northern India.

De Valera, Eamon, Irish Republican, born at New York in 1883, his father being a Spaniard, and his mother an Irishwoman. Educated at the Royal University of Ireland, he early became known for his revolutionary activities. In 1917 he was elected president of the Gaelic League and was arrested as an agitator. Elected to Parliament in 1918, while he was in

prison, he refused to take his seat. He was elected 'President' of the so-called 'Irish Republic' soon afterwards, and in Feb., 1919, escaped from prison and reached New York, where he started an active propaganda and began to raise funds for the Irish cause. He returned to Ireland in 1921.

Dev'enter, an old town in Holland, province of Overijssel, 8 miles north from Zutphen, at the confluence of the Schipbeek and Ijssel. Its industries embrace carpets, cast-iron goods, printed cottons, hosiery, and a kind of cake called *Deventer Koek*. It has a large export trade in butter. Pop. 32,483 (1918).

Dev'eron, a river of Scotland belonging to Aberdeenshire and Banffshire, 60 miles long. It flows into the Moray Firth at Banff.

Deviation of the Compass, the deflection of a ship's compass needle from the magnetic meridian, caused by adjacent iron. Hard iron is very retentive of a magnetic state, and is specially liable to become magnetized during a hammering process, as in the building of the ship. Soft iron easily receives or loses magnetism, and its magnetic state varies with every shifting of the ship's head. The effect of the former can be counteracted by magnets suitably placed near the compass, that of the latter by spheres of soft iron. The ship is swung, and the compass errors found in the various positions. The effects of the several contributing causes can then be separated, and the nature of the correctors necessary inferred with considerable accuracy. When these have been provided, the small residual errors for different positions of the ship are determined, and a table is constructed from which the navigator may read the slight correction to apply to the indication of his compass in steering any desired course.

Device', a name common to all figures, ciphers, characters, rebuses, and mottoes which are adopted by a person or a family by way of badge or distinctive emblem, often a representation of some natural body, with a motto or sentence applied in a figurative sense.

Devil (Gr., *diabolos*, a slanderer or accuser), in theology, the name given to a fallen angel, who is the instigator of evil, and the ruler of darkness. Most of the old religions of the East acknowledge a host of devils. The doctrine of Zoroaster, who adopted an evil principle called Ahriman, opposed to the good principle and served by several orders of inferior spirits, spread the

belief in such spirits among the people. The Greek mythology did not distinguish with the same precision between good and bad spirits. With the Mohammedans *Eblis*, or the devil, was an archangel whom God employed to destroy a pre-Adamite race of *jinns*, or genii, and who was so filled with pride at his victory that he refused to obey God. The Satan of the New Testament is also a rebel against God. He uses his intellect to entangle men in sin and to obtain power over them. But he is not an independent self-existent principle like the evil principle of Zoroaster, but a creature subject to omnipotent control. The doctrine of Scripture on this subject soon became blended with numerous fictions of human imagination, with the various superstitions of different countries, and the mythology of the pagans. The excited imaginations of hermits in their lonely retreats, sunk as they were in ignorance and unable to account for natural appearances, frequently led them to suppose Satan visibly present; and innumerable stories were told of his appearance, and his attributes—the horns, the tail, the cloven foot,—distinctly described. Theology has always treated the devil from a psychological or ethical standpoint. From the New Testament we hardly learn more regarding the devil than that he has a distinct personality; that he is a spirit or angel who in some way fell; that he is devoid of truth and of all moral goodness, always warring against the soul of man and leading him towards evil; that he has demons, spirits, or angels under him who work his will, and enter into or 'possess' men; but of his or their origin, original state, or fall, we really learn nothing.—BIBLIOGRAPHY: Mayer, *Historia Diaboli*; Lecanu, *Histoire de Satan, sa chute, son culte, ses manifestations, ses œuvres*; Carus, *History of the Devil*.

Devil-fish, the popular name of various fishes, one of them being the angler. Among others the name is given to several large species of ray (especially *Ceratoptera Vampyrus* which attains the breadth of 20 feet) occasionally captured on the Atlantic and Pacific coasts of America, and much dreaded by divers, whom they are said to devour after enveloping them in their vast wings. During gales of wind or owing to strong currents these immense fish are driven into shoal water, and, being unable to extricate themselves, fall an easy prey to the fishermen, who obtain considerable quantities of oil from their livers. The name is also applied to the larger eight-armed cephalopod molluscs belonging to *Octopus* and

allied genera. A combat with one of these is described in Victor Hugo's *Toilers of the Sea*.

Devil's Bit, the common name of a British species of scabious (*Scabiōsa succīsa*), nat. ord. Dipsacæ. It has heads of blue flowers nearly globular, and a fleshy root, which is, as it were, cut or bitten off abruptly. It flowers from June to October, and is common in meadows and pastures.

Devil's Bridge, a name for several bridges in wild situations; one being in Cardiganshire, spanning a gorge of the Mynach; another in Switzerland, over which the St. Gothard Railway crosses the Reuss.

Devil's Parliament, the name given to a Parliament convened by Henry VI, which met at Coventry in 1459 and unjustly accused the Duke of York of high treason.

Devil's Punch-bowl, a small lake of Ireland, near the Lakes of Killarney, between 2000 and 3000 feet above the sea, supposed to be the crater of an ancient volcano. The name is also given to Highcomb Bottom, a glen in Surrey.

Devil's Wall, in the south of Germany, a structure which was originally a Roman rampart, intended to protect the Roman settlements on the left bank of the Danube and on the right bank of the Rhine against the inroads of the Teutonic and other tribes. Remains of it are found from the Danube, in Bavaria, to Bonn, on the Rhine.

Devil-worship, the worship paid to the devil, an evil spirit, a malignant deity, or the personified evil principle in nature, by many of the primitive tribes of Asia, Africa, and America, under the assumption that the good deity does not trouble himself about the world; or that the powers of evil are as mighty as the powers of good, and have in consequence to be bribed and reconciled. There is a sect called devil-worshippers inhabiting Armenia and the valley of the Tigris, who pay respect to the devil, to Christ, and to Allah or the supreme being, and also worship the sun.

Devise, in law, usually the disposition of real estate by will, but also sometimes applied to any gift by will, whether of real or personal estate.

Devitrification. Glass which has been kept for a considerable time at a temperature just below its fusion-point gradually becomes opaque or crystalline in appearance; this phenomenon is spoken of as *devitrification*. Poor glass, badly prepared window-glass, and glass which has been subjected to strain tend to devitrify on exposure to air, some of the ingredients separating in a crystalline form.

Devi'zes, a town in England, county of Wilts, giving name to a parliamentary division, finely situated on a commanding eminence, 82 miles west by south of London. The name is derived from the Lat. *divisæ* (*terræ*, divided lands), because the ancient castle of Devizes was built at the meeting-place of three different manors. Agricultural engines and implements are made, and malting is carried on. Pop. 6740.

Devon, or **Devonshire**, a maritime county in the s.w. of England, its northern coast being on the Bristol Channel and its southern on the English Channel; area, 1,671,364 acres, the county being the third largest of England. Its principal rivers are the Torridge and the Taw, flowing north into the Bristol Channel; and the Exe, Axe, Teign, Dart, and Tamar, flowing into the English Channel. From Exeter to the confines of Cornwall extends the wide and barren tract called Dartmoor; but the vale of Exeter, comprising from 120,000 to 130,000 acres, and the south extremity of the county called *South Hams*, limited by a line drawn from Torbay to Plymouth Sound, are amongst the most fertile districts of England. Tin, lead, iron, copper, manganese, granite, and the clay used by potters and pipe-makers are the chief mineral products. The geological formation of the Old Red Sandstone is so largely developed that the term Devonian has to some extent become its synonym. Agriculture is in a somewhat backward state, owing, probably, to the general preference given to dairy husbandry, for which the extent and richness of its grass-lands make the county most suitable. Wheat, barley, beans, pease, and potatoes are the principal crops. About three-fourths of the county is under crops or in pasture. There is a large trade in butter, cheese, and live-stock, and the 'clotted' cream and cider of Devonshire are well known as specialties of the county. There are seven parliamentary divisions, each with one member. Pop. 699,703.—**BIBLIOGRAPHY:** *Victoria County History, Devonshire*; F. J. Snell, *Devonshire, Historical, Descriptive, and Biographical*.

Devo'nian System, in geology, a name originally given to rocks in Devonshire and Cornwall, intermediate between the Silurian and Carboniferous strata, and consisting of sandstones, calcareous slates and limestones, &c. They are divided into lower, middle, and upper series, the middle most abounding in fossils, including corals, crinoids, brachiopods, and molluscs. Devonian rocks occupy a large area in Central Europe, as well as in the United States, Eastern Canada, and Nova Scotia. The terrestrial and lacustrine equivalents are known as the Old Red Sandstone.

Dev'onport, formerly a county borough and port of England, county of Devon, contiguous to Plymouth. It is the seat of one of the royal dockyards, and an important naval and military station. A bastioned wall and fosse defend the town on the north-east and south sides, while the sea entrance is protected by heavy batteries on Mount Wise. Connected with the dockyards and fortifications are the gun wharf, foundries, machine-works, rope-walks, store-houses, and naval and military barracks. It has no special trade beyond that connected with the dockyards and Government works. Formerly a parliamentary borough, returning two members to Parliament, Devonport now gives its name to a parliamentary division of Plymouth, with which it is amalgamated since Nov., 1914. Pop. 84,370.

Devonshire, Spencer Compton Cavendish, eighth Duke of, long known as Marquess of Hartington, born in 1833, died 1908. He was the eldest surviving son of the seventh Duke of Devonshire, whom he succeeded in the dukedom in 1891. He was educated at Trinity College, Cambridge, where he graduated as M.A. in 1854. He was attached in 1856 to Earl Granville's Russian mission, and in 1857 was elected as a Liberal one of the members for North Lancashire. In 1863 he was for a short time a Lord of the Admiralty, and he then became Under-Secretary for War, being raised to Cabinet rank as War Secretary in 1866. In 1868 he lost his seat for North Lancashire, but became Postmaster-General under Gladstone, and was returned for the Radnor boroughs. In 1871 he was appointed Chief Secretary for Ireland. He went out with the Gladstone ministry in 1874, and on Gladstone's retirement he became the leader of the Liberal party. On the fall of the Conservative Government in 1880 he was elected for North-East Lancashire, and became Secretary for India under Gladstone, being transferred to the War Office in 1882. In the general election of 1885 he was returned for the Rossendale division of Lancashire. He strenuously opposed

Gladstone's Home Rule Scheme of 1886, and became the leader of the Liberal Unionists, but long declined to take office in the Cabinet. In 1895, however, he became Lord President of the Council, and from 1900 to 1902 he was President of the newly-instituted Board of Education. In 1903 he withdrew from co-operating with Mr. Balfour as Premier, mainly because he disliked the fiscal changes proposed by Chamberlain. He then accepted the position of head of the Free-Trade Unionists.

Dew is the name given to the minute drops of water which at certain times appear by night upon grass, flowers, foliage, and other surfaces which readily radiate heat. The first attempts at a scientific explanation of dew were made early in the nineteenth century, when Dr. Wells propounded his theory on the subject. According to this, the origin is to be found in the moisture previously contained in a vaporous state in the atmosphere. During the day the earth both absorbs and emits heat, but by night its supply of warmth is cut off, while it continues, under favouring circumstances, to lose heat by radiation into surrounding space. For any given state of the atmosphere, in respect of the amount of aqueous vapour it contains in a specified volume, there is a certain temperature at which it can hold just that amount and no more in suspension. If it be cooled to that temperature, it begins to deposit its moisture in the liquid form, and if the cooling proceeds further, more and more is deposited. This particular temperature is the *dew-point* appropriate to the given state of humidity of the air. If the dew-point is below 32°F., the vapour will, when the dew-point is reached, pass directly into the solid form and be deposited as hoar-frost. While there is a certain amount of truth in the foregoing theory, it does not contain the whole facts of the case. Dr. John Aitken, of Falkirk, Stirlingshire, established by experiments that in most cases probably the dew found on plants does not come mainly from the atmosphere. To some extent it exudes from the plants themselves. They derive moisture from the soil, and in the process of supplying their tissues it passes to their outer surfaces, whence in the daytime it is evaporated into the air. By night the fall of temperature checks evaporation of this moisture, and when it reaches their surfaces, it may remain there in the form of dew-drops. Also, moisture is condensed out of the atmosphere upon the cooler plant surfaces, but even this has more often been yielded not long before to the atmosphere through the medium of the plant tissues, so that upon the whole the plants play a much more

important part in the process than had been supposed. Dew is most copiously produced when there is a large difference between the day and night temperatures. Hence it is favoured by the absence of clouds, which would throw back much of the heat radiated from the earth. Other favouring circumstances are a still condition of the atmosphere, good radiating surfaces, and an abundant supply of moisture in the soil.

Dewar (dū´a`r), Sir James, scientist, born at Kincardine-on-Forth, 1842, educated at Dollar Academy, Edinburgh University—where he was assistant to Lord Playfair when professor of chemistry—and Ghent. In 1873 he was elected Jacksonian professor of experimental philosophy at Cambridge, an office which he still holds, and in 1879 a professorial fellow of St. Peter's College. In the latter year he also became Fullerian professor of chemistry at the Royal Institution, London. He is a fellow of the Royal Society, has been awarded several medals and prizes for his scientific researches, including the Rumford medal in 1894 for his investigations into the properties of matter at its lowest temperatures, this branch of science, with which the liquefaction of air and gases is connected, being peculiarly his own. He was the first to reduce hydrogen gas to the liquid and solid form. Together with Sir F. Abel, he was the discoverer of cordite. He is an honorary graduate of several universities, a late president of the Chemical Society and vice-president of the Royal Society, and was president of the British Association in 1902. He has contributed numerous papers to the proceedings of the learned societies of Great Britain, and was knighted in 1904. His collected papers on spectroscopy appeared in 1915.

Dewar Flask, a flask devised by Dewar for the examination of liquefied gases. It consists of an inner vessel surrounded by a vacuum, which much reduces loss or gain of heat by the liquid. The principle now finds practical application in the 'Thermos Flask'.

Dewás, a native state of Central India consisting of two combined states with two chiefs. Total pop. 250,000. Dewás, the chief town, has a pop. of 5200.

Dewberry (*Rubus cæsius*), a European plant belonging to the ord. of the Rosaceæ, and to the same genus as the bramble, from which it is distinguished by its smaller berries, with fewer grains, and by the bloom,

resembling dew, with which they are covered, and from which the plant derives its name. It is common in some parts of England. The Canadian dewberry (*R. procumbens*) yields a much superior fruit.

De Wet, Christian Rudolf, Boer general, born 1854 in the Orange Free State. His father removing into the Transvaal, he fought as a field-cornet at Majuba. As member of the Volksraad (1889-97) he helped to draw the two Dutch republics together, and in the South African War commanded first in Natal, and then in the west under Cronje, whose rescue at Paardeberg he attempted, but unsuccessfully. After March, 1900, he distinguished himself by his attacks on the British lines of communication, and by his skill in evading capture. He became commander-in-chief of the Free State forces, and was the only undefeated Boer general at the end of the war, after which he, with Louis Botha and Delarey, came to Europe to collect funds for his countrymen. A member of the Legislative Assembly and Minister of Agriculture, Orange Free State, from 1907 to 1914, he joined the rebellion at the outbreak of the European War. Captured at Waterburg on 1st Dec., he was sentenced to a fine of £2000 and six years' imprisonment, but was soon released. He published *The Three Years' War* in 1902.

Dewey, George, American naval officer, born at Montpelier, Vermont, 26th Dec., 1837, died 16th Jan., 1917. He began active service in the Mediterranean squadron, and in 1862, under Farragut, he was present at the passage of Forts Jackson and St. Philip. Lieutenant-commander in 1865, commander in 1872, he was in command of the Asiatic squadron in 1898, when he destroyed the Spanish fleet at Manila on 1st May. He was made Admiral of the Navy and was thanked by Congress. From 1900 till his death he was President of the General Board of the Navy.

Dewey, John, American philosopher and psychologist, born at Burlington, Vermont, 20th Oct., 1859. Educated at the University of Vermont, he received his Ph.D. degree from Johns Hopkins University. Professor of philosophy at the Universities of Minnesota, Michigan, and Chicago, where he was also director of the School of Education, he became professor of philosophy at Columbia University in 1904. He is one of the two (William James being the other) American leaders of *pragmatism*, a philosophical conception according to which questions that have no bearing on experience and on life have no significance whatever, and are meaningless.

(See *Pragmatism*.) His works include: *Outlines of a Critical Theory of Ethics* (1881), *Study of Ethics* (1894), *My Pedagogical Creed* (1897), *The School and Society* (1900), *Studies in Logical Theory* (1903), *How we Think* (1909), *Interest and Effort in Education* (1913), *Democracy and Education* (1916), *Letters from China and Japan* (1920).

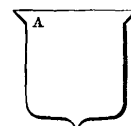
De Wint, Peter, English landscape painter in water-colours, born 1784, died 1849. He studied in the schools of the Royal Academy, where he occasionally exhibited; but most of his pictures were shown in the exhibitions of the Water-colour Society. English scenery was his favourite subject. He occasionally painted in oil with marked success. Several of his pictures are in the National Gallery and the Victoria and Albert Museum.

De Witt, Jan, Grand-Pensionary of Holland, celebrated as a statesman and for his tragical end, was the son of Jacob de Witt, burgomaster of Dort, and was born in 1625. He became the leader of the political party opposed to the Prince of Orange, and in 1652, two years after the death of William II, was made Grand-Pensionary. In 1665 the war with England was renewed, and conducted by De Witt with great ability till its termination in 1665. In 1672 Louis XIV invaded the Spanish Netherlands and involved Holland in war. De Witt's popularity, already on the decline, suffered still further in the troubles thus occasioned, and he felt it necessary to resign his office of Grand-Pensionary. At this time his brother Cornelius, who had been tried and put to torture for conspiring against the life of the young Prince of Orange, lay in prison. Jan de Witt went to visit him, when a tumult suddenly arose amongst the people, and both brothers were murdered, 20th Aug., 1672. De Witt was a man of high character, simple and modest in all his relations.—Cf. Motley, *History of the United Netherlands*.

Dew-point, the temperature at which the air is saturated with the water-vapour which it contains. If the temperature of the air falls to the dew-point, dew is deposited. The dew-point is determined by means of an instrument called a hygrometer. When the air is 'dry', the dew-point is low, and evaporation proceeds rapidly, whilst a 'moist' atmosphere is one whose temperature is near the dew-point, and in which evaporation takes place slowly.

Dewsbury, a town, England, in the West Riding of Yorkshire, and 30 miles south-west of the town of York, chiefly engaged in the manufacture of heavy woollen cloths, including blankets, carpets, rugs, flannels, and baizes. In 1862 it was made a municipal borough; in 1867 a parliamentary borough, returning one member. Pop. of municipal borough, 53,351.

Dexter, a term meaning on the right-hand side, chiefly used in heraldry. The *dexter chief point* is a point in the right-hand upper corner of the shield, being in the dexter extremity of the chief.



A,
Dexter
chief
point.

Dex'trine, or **British Gum**, $(C_6H_{10}O_5)_x$, a generic name applied to soluble gummy substances intermediate between starch and glucose. They are prepared from starch by the aid of dilute mineral acids or of enzymes, and are usually named according to the colour they give with iodine, e.g. erythro-dextrine, &c. When heated with dilute acids, they are transformed into glucose. They are white, odourless substances, and are good substitutes for gum-arabic. Dextrine is used in calico printing for thickening colours; for the preparation of gums, and for stiffening cloth.

Dextro-compounds, bodies which cause the plane of a ray of polarized light to rotate to the right. Dextrine itself, dextro-glucose, naturally occurring tartaric acid, malic acid, cinchonine, and many other bodies have this property; while others, which have the opposite effect, of causing the plane to rotate to the left, are called *lævo-compounds*.

Dey, an honorary title formerly bestowed by the Turks on elderly men, and assumed by the rulers (under the Turkish Sultan) of Algiers, Tripoli, and Tunis.

Dhalak (dha'-la'k'), an archipelago of the Red Sea, belonging to the Italian territory of Eritrea. It consists of nearly 100 islets, mostly uninhabited, clustering round the Island of Dhalak-el-Kebir, which is about 35 miles long by 30 miles broad. This island possesses a pearl-fishery.

Dhar (dhär), a small native state in Central India, with an area of about 1740 sq. miles. The soil is fertile, and yields wheat, rice, and opium. Pop.

169,474.—The capital is of the same name, is surrounded by a mud wall, and has some striking buildings. Pop. 19,000.

Dhárangaon, a town of Hindustan, in Khandesh district, Bombay. Pop. 15,000.

Dharmsála, a hill station with military cantonments, in Kangra district, Punjab, India. Pop. 6170.

Dharwar, the chief town of Dharwar district, in the Bombay Presidency, Hindustan, a straggling place with some trade. There is a fort well planned and strongly situated, but now falling into ruins, and military cantonments at 2 miles' distance. Pop. 32,000.—The Dharwar district has an area of 4535 sq. miles; pop. 1,051,314.

Dhawala'giri, or **Dhaulágiri**, one of the highest peaks of the Himalayas, in Nepál; height, 26,828 feet.

Dhole (dōl), the Cingalese name for the wild dog of India (*Cyon dukhunensis*). It is distinguished from the genus *Canis* or dog proper by its having one molar fewer in either side of the lower jaw. It is of a fox-red or rufous-fawn colour, in size between a wolf and a jackal, and hunts always in packs.

Dholera (dhō-lā'ra), a town of Hindustan, Bombay Presidency, on a stream entering the Gulf of Cambay, an important cotton-mart. Pop. 10,190.

Dholka, a town of Hindustan, Bombay Presidency, probably one of the oldest towns in Gujarat. Pop. 16,700.

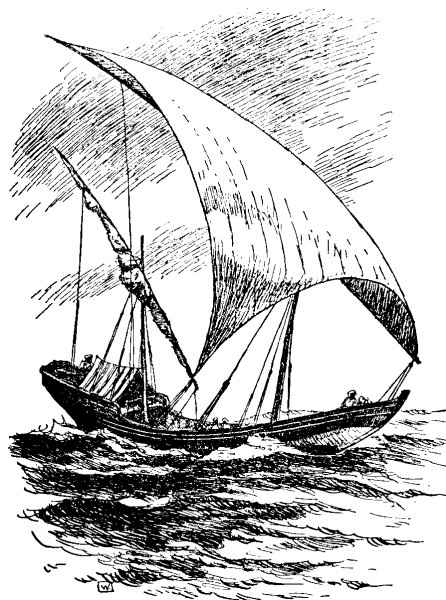
Dholpur, native state of Central India, Rájputána; area, 1155 sq. miles; pop. 263,188.—The capital is also called Dholpur. Pop. 9750.

Dhow (dou), an Arab sea-going vessel, ranging from a comparatively small size up to 200 tons burden, with one mast and a large triangular (*lateen*) sail. It is used for merchandise and is often employed in carrying slaves from the east coast of Africa to Arabia.

Dhúliá, a town of Hindustan, Khandesh district, Bombay Presidency. Pop. 22,000.

Diabase, originally an equivalent of diorite; then used for chloritic igneous rocks of the Intermediate series of various grain; and now usually for a type of dolerite in which the felspar is embedded in augite.

Diabétes is a disease characterized by great thirst, a voracious appetite, and the passage of large quantities of saccharine urine, while there is usually marked emaciation and debility. As a rule the skin is dry and the patient does not perspire. Skin irritations (pruritus) of an intense type occur frequently and cause much discomfort. Constipation is the rule, but the digestion usually remains good, and enormous quantities of food are taken without causing disturbance.



Arab Dhow

Acute and chronic forms are recognized, but there is no essential difference, except that in the former the patients are younger, the course more rapid, and the emaciation more marked. Beyond the large quantity, the outstanding feature of the urine is the presence of sugar, varying from 2 per cent in mild cases to 10 per cent in severe cases.

The disease is due to disturbance in the carbohydrate metabolism, with the result that these carbohydrates are not properly assimilated, but passed as sugar in the urine. Much research has been undertaken to find what organ or organs of the body cause this defect in metabolism, and recent work in connection with the pancreas has established a definite relationship between cells in that organ and the disease.

In treatment the main consideration is to reduce the carbohydrates in the dietary, and many diets have been produced for this purpose.

Diabetic patients may take:—liquids: clear soups, lemonade, coffee, tea, cocoa (without sugar), soda-water, and such like waters or milk (in moderation); animal foods: fish of all sorts, fresh meat, poultry, game, eggs, butter, cream-cheese; vegetables: lettuce, tomatoes, spinach, radishes,

asparagus, water-cress, cucumbers, chicory, mustard; fruits: lemons, oranges, and in moderation currants, plums, cherries, pears, apples (tart), melons, raspberries, strawberries, nuts: bread: gluten bread, almond or coco-nut biscuits. A substitute for bread is one of the greatest difficulties, as many gluten foods are very unpalatable.

Further, it is important to observe strict personal hygiene and to take moderate exercise, lead a regular, quiet life in an equable climate, and above all to avoid worries of any sort. In acute cases the disease may run a very rapid course, coma frequently supervening a few days before death, but chronic cases may live for ten to twenty years.

Diabetic Sugar. Sugar is present in normal urine in very small amount—so small that it cannot be detected by the ordinary tests. In diabetes the percentage of sugar in the urine may rise, in mild cases, to 1½ to 2 per cent, and in severe cases it may reach to 10 per cent.

Diablerets (dē-a ´ b-lè-rā), **Les**, a mountain group of the Bernese Alps, Switzerland, between the cantons Vaud and Valais. The highest peak has a height of 10,620 feet.

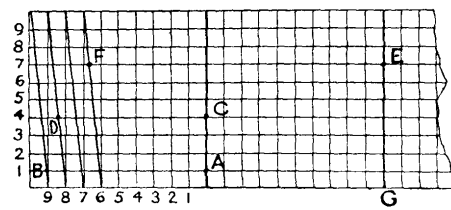
Diachylon (dī-ak´i-lon), a substance prepared by heating together oxide of lead or litharge, olive-oil, and water, until the combination is complete, and replacing the water as it evaporates. It is used for curing ulcers, and is the basis of many plasters.

Di´adem (Gr. *diadein*, to bind round), an ancient ornament of royalty. It was originally a head-band or fillet made of silk, linen, or wool, worn round the temples and forehead, the ends being tied behind and let fall on the neck, as seen in old representations of the diadem of the Indian Bacchus. In later times it was usually set with pearls and other precious stones. The term is also used as equivalent to crown or coronet.

Diæ´resis, a separation of one syllable into two, also the mark (¨) by which this separation is distinguished, as in *aërial*.

Diagno´sis, in medicine, the recognition of diseases by their distinctive signs or symptoms; the discovery of the true nature and seat of a disease.

Diag'onal Scale, a scale which consists of a set of parallel lines drawn on a ruler, with lines crossing them at right angles and at equal distances. One of these equal divisions, namely, that at the extremity of the ruler, is subdivided into a number of equal parts, and lines are drawn through the points of division obliquely across the parallels. With the help of the compasses such a scale facilitates the laying down of lines of any required length to the 200th part of an inch. The length 1·67 inches, for example, is given by EF in the figure. Similarly AB = ·91 inch, CD = ·84 inch.

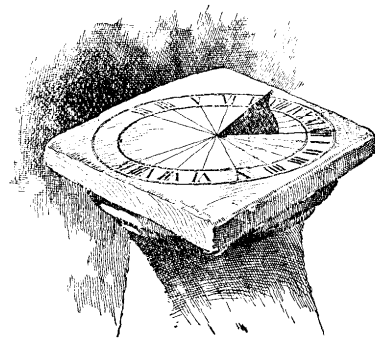


Diagonal Scale

Diag'oras, ancient Greek poet and philosopher, born in Melos, an island of the Cyclades, and flourished about 425 B.C. He spent a great part of his life in Athens. Like his teacher, Democritus, he attacked the prevailing polytheism, and sought to substitute the active powers of nature for the divinities of the Greeks. On this account he had to leave Athens.

Di'agram (Gr. *diagraphein*, to describe), a figure or geometrical delineation applied to the illustration or solution of geometrical problems, or any illustrative figure in which outlines are chiefly presented, and the details more or less omitted.

Dial, or **Sun-dial**, an instrument for showing the hour of the day from the shadow thrown by a *stile* or *gnomon* upon a graduated surface while the sun is shining. This instrument was known from the earliest times amongst Egyptians, Chaldeans, and Hebrews. From those Eastern nations it came to the Greeks. It was introduced into Rome during the first Punic War. Dials are of various construction, horizontal, inclined, or upright, the principle in every case being to show the sun's distance from the meridian by means of the shadow cast by the stile or gnomon. The stile is made parallel with the earth's axis, and may be considered as coinciding with the axis of the sun's apparent diurnal motion. Consequently, as the sun moves westwards the shadow of the stile moves round opposite to it, in the same direction, falling



Sun-dial

successively on lines drawn to represent the hours of the day. The dial, of course, gives true solar or *apparent* time, which, except on four days of the year, is somewhat different from *mean* time. Dials are now rather articles of curiosity or ornament than of use.—BIBLIOGRAPHY: Leybourn, *The Art of Dialling*; Dawbarn, *The Sun-dial*.

Di´alect, the language of a part of a country, or a distant colony, deviating either in its grammar, vocabulary, or pronunciation, from the language of that part of the common country whose idiom has been adopted as the literary language, and the medium of intercourse between well-educated people. Although the use of provincial dialects becomes inconvenient after a language has acquired a fixed literary standard, the study of such dialects is always valuable to the philologist for the light they throw on the history of the language. The diffusion of education and of printed books has much relaxed the hold which the provincial dialects of various countries once had on the people, and in general it may be said that the educated classes of any country now speak each of them a uniform language.

Dialec´tics (Gr. *dialektice*, from *dialektos*, discourse, dialogue), a philosophic term originally signifying *investigation by dialogue*. It was first used by Plato to designate the Socratic method. Afterwards it came to denote the art of inference or argument, and in this sense was synonymous with *logic*. The term is used in Kant's philosophy to mean the logic of appearance, or that logic which treats of inevitable tendencies towards error and illusion in the very nature of reason.

Di´allage, an altered form of the mineral augite, with a lamellar structure, and a submetallic lustre on its planes of separation. Schillerstein, or schiller spar is a similar product of the allied but rhombic mineral hypersthene. It forms diallage rock, and enters into serpentine rock.

Dialling, the art, of making sundials; also the art and practice of mine-surveying, in which the theodolite and magnetic needle are employed.

Di´alogue, a conversation or discourse between two or more persons. The word is used more particularly for a formal conversation in theatrical performances, and for a written conversation or composition, in which two or more persons carry on a discourse. This form was much in favour amongst the ancient philosophers as a medium for expressing their thoughts

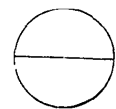
on subjects. The *Dialogues* of Plato are the finest example. Many of the great French and Italian writers have used this form. In the seventeenth century Fontenelle and Fénelon both wrote *Dialogues des Morts*, a title borrowed from Lucian. Landor's *Imaginary Conversations* (1821-8) is the best production of this kind in English.

Dial'ysis, the process by means of which a crystalline substance may be separated from a colloidal body. Certain substances are capable of passing through parchment, others are not, e.g. a solution containing sugar and silicic acid may be separated by placing the solution in a parchment-paper tube suspended in water; the silicic acid remains in the parchment tube, and the sugar passes through into the surrounding water. The solution is said to be dialyzed.

Diamagnet'ic, a term applied to substances which, when under the influence of magnetism and freely suspended, take a position at right angles to the lines of magnetic force. From the experiments of Faraday it appears that all matter is subject to the magnetic force as universally as it is to the gravitating force, arranging itself into three divisions, the *ferromagnetic*, *paramagnetic*, and *diamagnetic*. Among the former are iron, nickel, cobalt, magnetic oxide of iron, and Heusler's alloy. The more feebly magnetic bodies are classed as paramagnetics, and those which behave as described above are called diamagnetic substances. Among the latter are bismuth, antimony, cadmium, copper, gold, lead, mercury, silver, tin, zinc, and most solid, liquid, and gaseous substances. A diamagnetic body is one which is not so magnetic as the medium in which it is suspended. The action of bismuth, the strongest diamagnetic substance, is weak when compared with the magnetic action of iron.

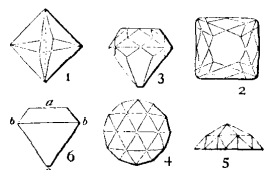
Diamanti'na, a town, Brazil, in the diamond-mining district in the state of Minas Geraes, the inhabitants of which are almost all engaged in the gold and diamond trade. Pop. about 14,000.

Diam'eter (Gr. *dia*, through, and *metron*, measure), the straight line drawn through the centre of a circle and terminated by the circumference. It thus divides the circle into two equal parts, and is the greatest chord. The length of the diameter is to the length of the circumference of the circle as 1 to 3.14159265..., the latter



Diamete
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number being an interminable decimal. The name is also given to any chord of a conic which passes through its centre.

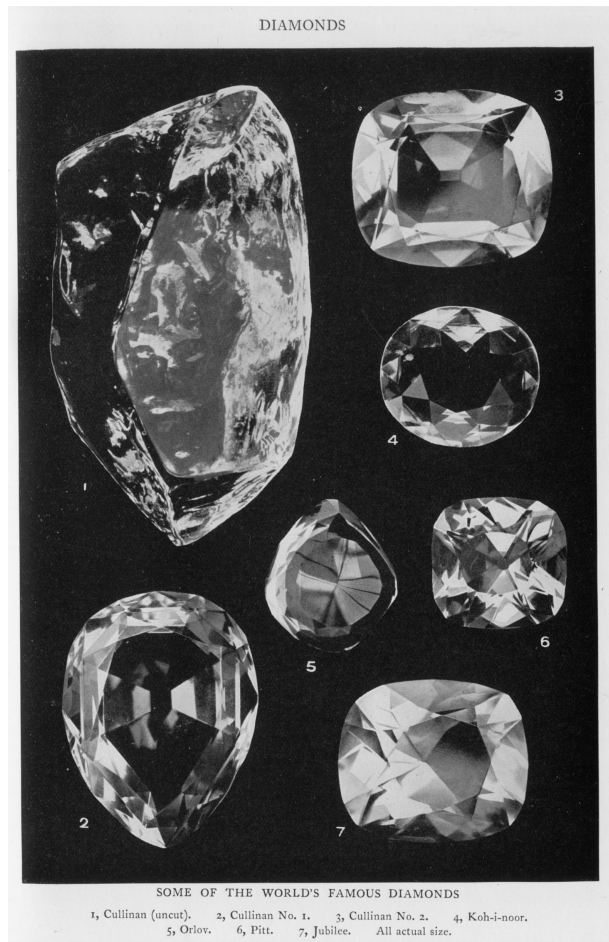


Diamonds, rough and variously cut

Di'amond, the hardest and one of the most valuable of gems, and the purest form in which the element carbon is found. (See *Carbon*.) It crystallizes in forms belonging to the regular or cubic system, the most common being the regular octahedron and rhombic dodecahedron (twelve faces). The finest diamonds are colourless, perfectly clear, and pellucid. Such are said

to be of the finest *water*. But diamonds are often blue, pink, green, or yellow, and such are highly prized if of a decided and equal tint throughout. The hardness of the diamond is such that nothing will scratch it, nor can it be cut but by itself. The value of a diamond is much enhanced by cutting facets upon it inclined at certain angles to each other so as to produce the greatest possible play of colour and lustre. What is called the *brilliant* cut best brings out the beauty of the stone. Its upper or principal face is octagonal, surrounded by many facets. But this form of cutting requires an originally well-shaped stone. For other diamonds the *rose* cut is used. In this form six triangles are cut on the top so that their apices meet in a point called the summit. Round this are disposed other facets. Stones which are too thin to cut as rose-diamonds are cut as *table*-diamonds, which have a very slight play of colour. In the cut, fig. 1 is the diamond in its rough state; fig. 2 is the vertical, and fig. 3 the lateral appearance of a brilliant; fig. 4 the vertical, and fig. 5 the lateral appearance of a rose-cut diamond; in fig. 6 the flat portion *a* in a cut stone is called the *table*; the part *a b b*, which projects from the setting, is the *front*, the part *b b c*, sunk in the setting, is the *back* or *culasse*, while the line *b b* is the *girdle*. The art of cutting and polishing the diamond was unknown in Europe till the fifteenth century, and the stone itself was not nearly so highly valued in the Middle Ages as the ruby. Diamonds are valuable for many purposes. Their powder is the best for the lapidary, and they are used for jewelling watches, and in the cutting of window- and plate-glass. When used as a glazier's tool the diamond must be uncut. Inferior kinds of diamonds are also extensively used by engineers in rock-boring, and by copperplate engravers as etching-points. Diamonds are obtained from deposits of various kinds, mostly alluvial (sands, clays, &c.), being separated by washing. They have been found in India, Borneo, and

other parts of the East; sometimes in N. America and Australia; Brazil has produced large numbers; but the chief diamond-field of to-day is in Cape Province, the centre being Kimberley. Diamonds were discovered here in 1867, and since then the output has amounted to over £183,000,000 in value. The diamonds are no longer obtained by mere surface workings, but the excavations have been carried down to a depth of 2000 feet. 'River diggings' are also carried on on the banks of some of the rivers. Some of the S. African diamonds are very large. One of them, the Cullinan diamond, discovered in 1905, is a monster of 3025 carats, of very good colour, being by far the largest diamond known. A celebrated diamond is the Koh-i-noor (Mountain of Light), an Indian stone belonging to the British crown. Its history extends over five or six centuries. It weighed at one time 280 carats, but by cutting has been reduced to about 106 carats. The Orlov diamond, which belonged to the Emperor of Russia, weighed 194 carats; the Pitt diamond, among the French crown jewels, weighs 136½ carats.—
BIBLIOGRAPHY: A. Jeffries, *A Treatise on Diamonds and Pearls*; H. Emanuel, *Diamonds and Precious Stones*; E. W. Streeter, *Precious Stones and Gems*; *idem*, *The Great Diamonds of the World*; G. F. H. Smith, *Gem-Stones*; P. A. Wagner, *The Diamond Fields of Southern Africa*.



Diamond-beetle, *Entimus imperialis*, a handsome South American insect belonging to the family Curculionidæ or weevils. It is spangled with golden-green on a black background.

Diamond Harbour, a port on the left bank of the Hugli River, about 38 miles by the railway from Calcutta, formerly much used as an anchorage for ships waiting for the tide.

Diamond Lore. In Hellenic, Arabian, Chinese, and other literature the diamond is connected with the eagle and snakes. Diamonds, according to ancient belief, lie in deep valleys infested by snakes, or entirely surrounded by straight, high cliffs. Pieces of flesh are thrown down and eagles seize them. The birds are followed to their nests, where the diamonds that adhered to the flesh are found. Mixed with this legend is the older one regarding the 'eagle stone', which assists parturition. It was believed a

woman was easily delivered if the 'eagle stone' were placed on her abdomen. The Chinese legend was imported with the diamond from Fu-lin (Syria). Indian diamond lore is mixed with pearl lore. According to the Buddha birth stories, diamonds are found in the sea. The ancients asserted that the diamond could not be injured by iron, fire, or smoke. Before it could be broken it had to be steeped in ram's blood. The alchemists used lead as a substitute for ram's blood. In Chinese lore diamonds are rulers of gold and have their origin in gold. A similar belief prevailed in mediæval Europe, adamantine gold being credited with the same virtues as the diamond. Both gold and the diamond were sacred. The diamond is a mediæval symbol of Christ; in the Far East it is connected with Buddha. The association of the diamond with snakes gave origin to the belief that it was poisonous, the saliva of the snakes clinging to it. Diamond dust is regarded in India as a deadly poison. Like the sacred pearl, the diamond has been credited with nocturnal luminosity. Certain varieties of diamonds when heated, rubbed, or exposed in bright sunshine emit slight rays of light for a short time in darkness. The belief in 'night shining gems', however, had origin in pearl lore, the pearl having been connected with the moon ('the pearl of heaven'). Coral, rhinoceros-horn, fern seed, the mandrake, &c., were likewise connected with the moon-goddess and credited with nocturnal luminosity.



Diana. From a statue in the British Museum

Diamond Necklace, an affair of some note in French history immediately preceding the Revolution. See *Marie Antoinette*; *La Motte*; and *Rohan, Louis*.

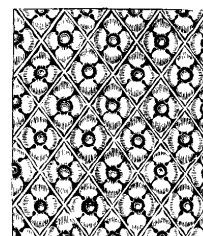
Dian'a, in Roman mythology, an ancient Italian goddess, in later times identified with the Greek Artēmis, with whom she had various attributes in common, being the virgin goddess of the moon, and of the chase, and having as attributes the crescent moon, bow, arrows, and quiver. The name is a feminine form of Janus. She seems to have been originally the patron divinity of the Sabines and Latins. She was worshipped especially by women, as presiding over births, no man being allowed to enter her temple.

Diana-monkey (*Cercopithēcus Diana*), a species of monkey found in West Africa, and so named from the crescent-shaped band on the forehead resembling the crescent moon, which was the symbol of Diana. Another characteristic feature is the possession of a pointed white beard.

Diana of Poitiers, Duchess of Valentinois, born in 1499. She was the mistress of King Henry II of France, and descended from the noble family of Poitiers, in Dauphiny. At an early age she married the Grand-Seneschal of Normandy, Louis de Brézé, became a widow at thirty-one, and some time after the mistress of the young Duke of Orleans. On his accession to the throne in 1547, as Henry II, Diana continued to exercise an absolute empire over him till his death in 1559. After that event she retired to her castle of Anet, where she died in 1566.—Cf. Capefigue, *Diane de Poitiers*.

Diapa'son, in music, the concord of the first and last notes of an octave. The word is also used for the most important foundation-stops of an organ. They are of several kinds, as *open diapason*, *stopped diapason*, *double diapason*. The French use the term as equivalent to pitch in music.

Di'aper, a kind of textile fabric much used for towels and napkins, and formed either of linen or cotton, or a mixture of the two, upon the surface of which a flowered or figured pattern is produced by a peculiar mode of twilling.—As a term in ornamentation diaper is applied to a surface covered with a flowered pattern sculptured in low relief, or to a similar pattern in painting or gilding covering a panel or flat surface.



Diaper
Ornamentation.
Westminster
Abbey

Diaphanoscope, (1) an apparatus by means of which transparent positive photographs may be viewed. The name is also given to (2) an instrument employed in obstetrical surgery; electric light, contained in a glass tube or bulb, is introduced into the female internal organs, and, thus illumined, their condition can be examined through the translucent walls of the abdomen.

Diaphoret'ics (Gr. *diaphorein*, to carry through) are agents used in medical practice to produce perspiration. The Turkish bath, hydropathic treatment, diluent drinks, &c., are employed for this purpose. The degree of

perspiration produced is more than normal, but less than in sweating. See *Sudorifics*.

Diaphragm (dī'a-fram), in anatomy, a muscular membrane placed transversely in the trunk, and dividing the chest from the abdominal cavity. In its natural situation the diaphragm is convex on the upper side and concave on its lower, but when the lungs are filled with air it becomes almost flat. It is the principal agent in respiration, particularly in inspiration. A complete diaphragm is found only in Mammalia.

Diarbek'ir, a town in Armenia, formerly in Asiatic Turkey, capital of the vilayet of same name, on a high bank overlooking the Tigris, and surrounded by a lofty massive wall. It has manufactures of iron- and copper-ware, leather, silk, woollen, and cotton goods, and a considerable trade. Pop. about 38,000.—The province of Diarbekir has an area of 14,480 sq. miles, and a pop. of 471,500.

Diarrhœ'a is morbidly frequent evacuation of the bowels. Several forms are recognized. *Choleraic diarrhœa*; this form is acute, and is marked by great frequency, with serous stools, and accompanied by vomiting and collapse. *Critical diarrhœa* occurs at the crisis of a disease. *Lienteric diarrhœa* is marked by the passage of fluid stools containing scraps of undigested food. *Mucous diarrhœa* is marked by the presence of mucous in the stools. *Summer diarrhœa* occurs chiefly among young children and infants, most frequently in late summer. It is usually acute in type and associated with marked prostration. In epidemic form it may give rise to a high mortality in crowded districts and in institutions. *Nervous diarrhœa* is produced by some emotional cause. The treatment, whatever the type, is to get rid of the cause of the irritation, and to avoid further irritation in the intestinal tract. For this purpose purgatives are given, and all solid food forbidden. This is followed by gastro-intestinal sedatives and a gradual return to normal diet—substances causing least digestive difficulty being first given.

Diastase is an unorganized ferment or enzyme produced in the germination of barley, oats, &c. It is soluble in water, and the solution has the property of inducing fermentation or hydrolysis of starch into dextrine and glucose. To prepare diastase, barley is allowed to germinate; germination is then

interrupted by raising the temperature, and the grain is treated with a mixture of water and alcohol under pressure, and filtered. Diastase, being soluble, is obtained in the filtrate.

Diathermancy, the property that is possessed in various degrees by different substances, of transmitting radiant heat. Bodies that are equally transparent, that is, bodies which have equal power of transmitting rays of light, are very different in their power of transmitting heat-rays. Thus a thin plate of glass and a thin plate of rock-salt may be nearly equally transparent, but the plate of rock-salt has far superior power of transmitting rays of heat. The latter, it has been found, allows 92 per cent of the total heat from most sources to pass; glass and other substances transmit a much smaller proportion, and the amount varies with the source. Rock-salt is diathermanous to heat from nearly all sources. It has been shown that rock-salt is extremely opaque or athermanous to the radiations from a piece of heated rock-salt. The diathermancy of the plates in every case decreases very rapidly as their thickness is increased. See *Radiation*.

Diathesis is the term given in medicine to a constitutional predisposition to a disease; thus uratic diathesis is a tendency to gout; aneurysmal diathesis is an inherent predisposition to aneurysms.

Diatomaceæ, a family of Algæ, consisting of microscopic unicellular plants with brown chromatophores found in fresh, brackish, and salt water, and on damp ground. The cell wall contains a very large quantity of silica, and is formed in each cell into three portions, viz. two generally symmetrical valves and a connecting hoop. The species consist of single free cells, or the cells remain connected so as to form usually linear colonies, sometimes enclosed in a transparent gelatinous sheath. The ordinary method of increase is by cell division. A sexual process resembling that of the conjugatæ also occurs. Diatoms constitute an important source of food for the lower marine animals, and thus indirectly for the food-fishes. Diatomaceæ are found fossil, forming considerable deposits of tertiary age, as at Bilin, Richmond in the United States, &c. Fossil polishing-powders, as tripoli and bergmehl, are composed of them; also kieselguhr, which, impregnated with nitroglycerine, forms dynamite. They are abundant in guano.

Diatomite (Ger. *kieselguhr*), a diatomaceous earth (see *Diatomaceæ*) generally found underlying peat. In Skye, at Loch Quire, it is found about 18 inches below the surface, and extends downward for about 7 feet, and in some places to a much greater depth. Another important area is north of Toome Bridge in the county of Antrim. Diatomite is principally used for the manufacture of dynamite on account of its value as an absorbent. It is described also as extremely well adapted for the manufacture of silicate paints, siliceous glazings, porcelain, boiler-coatings, and for isolating felt and bricks for cold-storage buildings.

Diatonic, a term originally applied by the Greeks to one of their three genera of music. In modern music it is applied to the natural scale, and to the intervals, chords, melodies, or harmony characteristic of it. A diatonic chord is a chord having no note chromatically altered. A diatonic interval is an interval formed by two notes of the diatonic scale unaltered by accidentals. A diatonic melody is a melody composed of notes belonging to one scale only.

Diaz, Bartolommeo, a celebrated Portuguese navigator of the fifteenth century, named in 1486 commander of one of that long succession of exploratory expeditions which the Portuguese court had during this century become distinguished for promoting. The two vessels composing the expedition sailed along the African coast till they reached Cape Negro (lat. 15° 50' S.), where Diego Cam, a previous explorer, had stopped. At 29° S. they anchored at a point to which they gave the name of Angra das Voltas (Bay of Detours). In sailing south from this point they doubled the Cape of Good Hope without knowing it, and landed at a bay on the east coast. Diaz now wished to continue his voyage in order to discover the country of Prester John, but the sailors refused to accompany him. In again doubling the Cape he gave it the name of Cabo Tormentoso (Cape of Storms), which the king changed to its present designation. In 1500 Diaz had command of a vessel in the expedition of Cabral which discovered Brazil. In returning home the vessel which he commanded was lost, 29th May, 1500.

Diazo Compounds, or **Diazonium Compounds**, a name given to substances containing the chemical group - N:N -; thus diazo-benzene chloride, $C_6H_5 - N:N \cdot Cl$, or diazo-toluene sulphate, $C_6H_4(CH_3) - N:NH \cdot SO_4$, &c.

These substances are formed from the aromatic amines by treatment with nitrous acid at low temperatures. Primary amines all react with nitrous acid at moderate temperatures when the amino group (NH_2) is replaced by a hydroxyl group (OH); thus ethylamine ($\text{C}_2\text{H}_5\text{NH}_2$) reacts with nitrous acid (HNO_2), yielding alcohol ($\text{C}_2\text{H}_5\text{OH}$), $\text{C}_2\text{H}_5\text{NH}_2 + \text{HNO}_2 \rightarrow \text{C}_2\text{H}_5\text{OH} + \text{N}_2 + \text{H}_2\text{O}$. The aromatic amines, however, if treated below 0°C . with nitrous acid, yield diazo compounds, and not hydroxy compounds, e.g. aniline treated with nitrous acid in hydrochloric acid solution yields diazo-benzene chloride, $\text{C}_6\text{H}_5\text{NH}_2 \cdot \text{HCl} + \text{HONO} = \text{C}_6\text{H}_5 - \text{N}:\text{N} \cdot \text{Cl} + 2(\text{H}_2\text{O})$. Diazo salts are crystalline compounds soluble in water, sparingly soluble in alcohol, and are unstable, decomposing explosively if struck or suddenly heated. In solution in water they decompose as the temperature rises, liberating nitrogen, and forming hydroxy compounds, $\text{C}_6\text{H}_5\text{N}:\text{NCl} + \text{H}_2\text{O} = \text{C}_6\text{H}_5\text{OH} + \text{N}_2 + \text{HCl}$. Diazo salts are valuable in the synthesis of different classes of compounds, as the $-\text{N}:\text{N}-$ group reacts readily with other groups. For this purpose it is usually only necessary to prepare a solution containing the diazo compound. In the preparation of the azo dyes the starting-point is a primary amine; the amino group is 'diazotized', i.e. treated with a solution of sodium nitrite and dilute mineral acid at low temperature. A diazo salt is formed, and is then made to react in solution with a hydroxy compound or an amino compound, &c., with the formation of a highly coloured azo compound. The diazo compound is then said to be coupled. Diazo compounds are therefore important intermediate substances in the manufacture of azo dyes.

The diazo group may also be exchanged for the hydroxyl group by warming the solution with water, or for the cyanogen group by warming with a solution of potassium cyanide, e.g. diazo-benzene chloride warmed with potassium cyanide solution is converted into the nitrile of benzoic acid, $\text{C}_6\text{H}_5 - \text{N}:\text{NCl} + \text{KCN} = \text{C}_6\text{H}_5\text{CN} + \text{KCl} + \text{N}_2$; or transformed into halogen derivatives of hydrocarbons by warming with cuprous chloride, e.g. diazo-benzene chloride is transformed into chlorobenzene, $\text{C}_6\text{H}_5 - \text{N}:\text{NCl} + \text{Cu}_2\text{Cl}_2 = \text{C}_6\text{H}_5\text{Cl} + \text{N}_2 + \text{Cu}_2\text{Cl}_2$. These reactions, where nitrogen is eliminated from the compound, and a group or element replaces the two atoms of nitrogen, afford a means of synthesizing a variety of compounds.

Dib'din, Charles, an English dramatic manager and poet, composer and actor, born in 1745, died in 1814. At the age of fifteen he made his appearance on the stage, and was early distinguished as a composer. He invented a new kind of entertainment, consisting of music, songs, and public declamations, which he wrote, sang, composed, and performed himself, and by this means succeeded in amusing the public for twenty years. In 1769 he composed some of the music for the Shakespeare jubilee at Stratford-on-Avon. His patriotic songs were very popular, and his sea-songs, amongst which are *Tom Bowling*, *Poor Jack*, and *The Trim-built Wherry*, are still favourites in the British navy. He also wrote a *History of the Stage*, and the novels *The Devil* and *Hannah Hewitt*.—His son, Charles Dibdin, composed and wrote many small pieces and occasional songs.—Another son, Thomas, early displayed the same dramatic tastes as his father, was connected with various theatres, and wrote a great many songs and a number of dramas.

Dibdin, Thomas Frognall, an English bibliographer, born in 1776, died in 1847, was the son of the elder brother of Charles Dibdin the celebrated naval song-writer. After studying law and practising as a provincial counsel, he took orders and became a popular preacher in London. Here his bibliographical tastes developed themselves, and the Roxburghe Club being established in 1812, he became its first vice-president. Among his numerous writings may be noted: *Bibliomania*, *Bibliographical Decameron*, *Typographical Antiquities of Great Britain*, and *The Library Companion*.

Dibranchia'ta. See *Cephalopoda*.

Dice, cubical pieces of bone or ivory, marked with dots on each of their six faces, from one to six, according to the number of faces. They are shaken in a small box and then thrown on the table. Dice are often loaded or falsified in some way so as to make the high or the low sides turn down. The origin of dice is ascribed to Palamedes of Greece (1244 B.C.), although Herodotus attributes the invention of knuckle-bones and of dice to the Lydians. Dice were well known amongst the ancient Egyptians and Greeks, and are still very popular in Japan, China, India, and other Asiatic countries.

Dicen'tra. See *Diëlytra*.

Dichasium. See *Cyme*.

Dichlamydeous (dī-klam-id'ī-us), in botany, said of plants that have both calyx and corolla.

Dichogamy, in flowers, the condition in which anthers and stigmas ripen at different times, whereby self-pollination is effectually prevented. If the anthers ripen first, as in borage, columbine, crocus, Caryophyllaceæ, Compositæ, Labiataë, &c., the flower is said to be protandrous; protogynous flowers, with stigmas ripening and withering before the pollen is shed, occur, e.g. in Christmas rose, Colchicum, horse-chestnut, and in the majority of wind-pollinated plants.

Dichotomy (dī-kot'o-mi), a cutting in two; a division by pairs. Hence, in botany, a mode of branching by constant forking, each branch dividing into two others. See *Branching*.

Dichroic, or more generally **Pleochroic**, **Crystals** (dī-krō'ik), crystals that have the property of exhibiting different colours, according to the direction in which they are traversed by rays of light. When polarized light is passed through a transparent plate of a pleochroic mineral, the colour will vary with the direction in which the light-vibrations take place. Hence, *face-pleochroism*, the colour of the plate, may be distinguished from the colours given by *axis-pleochroism*, the colours given by light vibrating parallel with certain optical directions in the crystal.

Dichroite (dī'kro-īt), or **Iolite**, a mineral, a silicate of magnesium, iron, and aluminium, which readily undergoes modifications and passes into hydrous silicate. It exhibits marked pleochroism, whence the name.

Dick, Thomas, LL.D., a Scottish author of popular scientific works, born at Dundee in 1774, died 29th July, 1857. He was for many years a teacher at Perth, but subsequently resided at Broughty-Ferry, where he devoted himself to astronomical science, especially in its relations to religion. Some years before his death a small pension was granted to him by the Government. Amongst his works are: *The Christian Philosopher* (1823), *The Philosophy of Religion* (1825), *The Philosophy of a Future State* (1828), and *Celestial Scenery* (1838).

Dickens, Charles, one of the greatest English novelists, born 7th Feb., 1812, at Landport, Portsmouth, died 9th June, 1870. His father, John Dickens, was

then in the employment of the Navy Pay Department, but subsequently became a newspaper reporter in London. Young Dickens received a somewhat scanty education, was for a time a mere drudge in a blacking warehouse, and subsequently a clerk in an attorney's office. Having perfected himself in shorthand, however, he became a newspaper critic and reporter, was engaged on *The Mirror of Parliament*, and *The True Sun*, and in 1835 on *The Morning Chronicle*. For some time previously he had been contributing humorous pieces to *The Monthly Magazine*; but at length, in 1835, appeared in *The Morning Chronicle* the first of that series of *Sketches by Boz* which brought Dickens into fame. It was followed in quick succession by a pamphlet entitled *Sunday under Three Heads*, by Timothy Spark (1836); *The Tuggs at Ramsgate* (1836); *The Village Coquette*, a comic opera (1836); and a farce called *The Strange Gentleman* (1836). In the same year Chapman & Hall engaged the new writer to prepare the letterpress for a series of comic sketches on sporting subjects by Seymour, an artist who had already achieved fame, and suggested as a subject the adventures of an eccentric club. Seymour committed suicide soon after, and H. K. Browne joined Dickens as illustrator, the result being the immortal *Pickwick Papers*. The great characteristics of Dickens's genius were now fully apparent, and his fame rose at once to the highest point it was possible for a writer of fiction to reach. A new class of characters, eccentric indeed, but vital representations of the humours and oddities of life, such as Mr. Pickwick, Sam Weller and his father, Mr. Winkle, and others, was made familiar to the public. Under the name of *The Posthumous Papers of the Pickwick Club* this work was published in 2 vols. 8vo, in 1837. In the same year Dickens was engaged as editor of *Bentley's Magazine*, to which he contributed *Oliver Twist*, a work which opened up that vein of philanthropic pathos and indignant satire upon institutions which became a distinguishing feature of his works. Before the completion of *Oliver Twist*, *Nicholas Nickleby* was begun, being issued complete in 1839. As the special object of *Oliver Twist* was to expose the conduct of workhouses, that of *Nicholas Nickleby* was to denounce the management of cheap boarding-schools. *Master Humphrey's Clock*, issued in weekly numbers, contained among other matter two other leading tales, *The Old Curiosity Shop* and *Barnaby Rudge*, the latter a historical tale, going back to the times of the Gordon riots. It was published complete between 1840 and 1841. In 1841 Dickens visited America, and on his return he wrote *American Notes for General*

Circulation (1842). His next novel, *Martin Chuzzlewit* (1844), dwelt again on his American experiences. This work also added a number of typical figures—Mr. Pecksniff, Mark Tapley, Sarah Gamp, and others—to English literature. The series of *Christmas Tales*, in which a new element of his genius, the power of handling the weird machinery of ghostly legend in subordination to his own peculiar humour, excited a new sensation of wonder and delight. These, enumerated consecutively, were: *A Christmas Carol* (1843), *The Chimes* (1844), *The Cricket on the Hearth* (1845), *The Battle of Life* (1846), *The Haunted Man* and *The Ghost's Bargain* (1847). The extraordinary popularity of these tales created for a time a new department in literature, that of the sensational tale for the Christmas season. In 1845 Dickens went to Italy, and on his return *The Daily News*, started on 1st Jan., 1846, was entrusted to his editorial management; but, despite his early training, this was an occupation uncongenial to his mind, and in a few months the experiment was abandoned. His *Pictures from Italy* were published the same year. Next followed his novel of *Dombey and Son* (1848), and *David Copperfield*, a work which has a strong autobiographical element in it (1849-50). In 1850 Dickens became editor of the weekly serial *Household Words*, in which various original contributions from his own pen appeared. In 1853 his *Bleak House* came out. *A Child's History of England*, commenced in *Household Words*, was published between 1852 and 1854. *Hard Times* appeared in *Household Words*, and was published in 1854. *Little Dorrit*, commenced in 1856, dealt with imprisonment for debt, the contrasts of character developed by wealth and poverty, and executive imbecility, idealized in the Circumlocution Office. In 1859, in consequence of a disagreement with his publishers, *All the Year Round* superseded *Household Words*; and in the first number of this periodical, 28th May, was begun *A Tale of Two Cities*. *Great Expectations* followed in the same paper, on 1st Dec., 1860. In *All the Year Round* also appeared a series of disconnected sketches called *The Uncommercial Traveller*, published in 1868. *Our Mutual Friend*, completed in 1865, and published in the usual monthly numbers, with illustrations by Marcus Stone, was the last great serial work which Dickens lived to finish. It contained some studies of characters of a breadth and depth unusual with Dickens, and is distinguished among his works by its elaborate plot. The first number of his last work, *The Mystery of Edwin Drood*, was issued on 1st April, 1870, and only three numbers had appeared when he died somewhat suddenly, at his

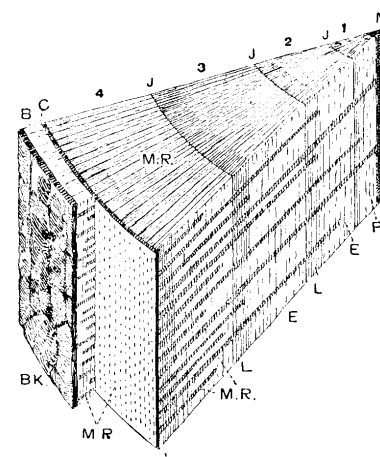
residence, Gad's Hill Place, near Rochester, on 9th June. He had considerably overtaxed his strength during his later years, more especially by his successive series of public readings from his own works, one series being delivered in America between 1867 and 1868. He was buried in Westminster Abbey. Dickens's work as a novelist is firmly based upon a wide and keen observation of men. The essence of his art was caricature, and for comic effect he, therefore, often exaggerated the abuses he attacked. His characters exhibit little more than one trait or quality, but the single trait or quality which they embody is truly conceived, and exhibited with great vitality and humour. His creative power was immense, and his great humour is admitted by all, even by those who consider his pathos as overdrawn. In spite, therefore, of all that is grotesque and overstrained in his work, he has been rightly placed amongst the great artists.—BIBLIOGRAPHY: J. Forster, *Life of Charles Dickens*; George Gissing, *Charles Dickens: a Critical Study*; G. K. Chesterton, *Charles Dickens*; R. H. Shepherd, *The Bibliography of Dickens*; Sir F. T. Marzials, *Life of Charles Dickens*; E. Pugh, *The Charles Dickens Originals*; P. H. Fitzgerald, *Memoirs of Charles Dickens*.

Dicksonia, a genus of Leptosporangiate ferns, section Gradatæ, mostly large tree-ferns, such as *D. antarctica*, a native of Australia and New Zealand, often grown in greenhouses.

Dicotyledon (dī-kot-i-le'don), a plant whose seeds are readily recognized by the embryo containing a pair of cotyledons or seed-leaves, which are always opposite to each other. Dicotyledons are further characterized by their netted-veined leaves and their 'open' vascular bundles containing a cambium; the parts of the flower are commonly in fours or fives. In Bentham and Hooker's system the class is divided into four subclasses—Thalamifloræ, Calycifloræ, Corollifloræ, and Monochlamydeæ. Engler's system recognizes only two subclasses, viz. Archichlamydeæ and Sympetalæ.

Dictaphone, an adaptation of the gramophone, in which the principle of that invention is applied to the requirements of modern business. Letters or memoranda are spoken into the machine, which 'records' them on waxen cylinders. The machine is then passed on to a shorthand writer or typist (or the cylinder may be transferred to a duplicate machine), and the recorded matter is dictated. The motive power is electricity; the speed of dictation is capable of adjustment to that of the writer; and by means of an accessory machine the records can be scraped and re-used.

Dicta'tor, an extraordinary magistrate of the Roman Republic, first instituted B.C. 501. The power of naming a dictator, when an emergency arose requiring a concentration of the powers of the State in a single superior officer, was vested by a resolution of the Senate in one of the Consuls. The dictatorship was limited to six months, and the person who held it could not go out of Italy. This rule was laid aside during the first Punic War. The dictator was also forbidden to appear in Rome on horseback without the permission of the people, and he had no control over the public funds without the permission of the Senate. He had the power of life and death, and could punish without appeal to the Senate or people. All the



Portion of a Four-year-old
Dicotyledonous Woody Stem cut
in Winter

B, Bast. B.K., Bark external to the first periderm layer, corresponding to the primary cortex. C, Cambium ring. E, Early wood. J, J, J, Junction of the wood of successive years. L, Late wood. M, Medulla. M.R., Medullary rays, various views. P, Protoxylem. 1, 2, 3, 4, The four successive annual rings.

other magistrates were under his orders. Originally the dictator was a patrician, but in 356 B.C. the plebeian Marcius Rutilus was called upon to fill the office of dictator. The term is now often applied to rulers enjoying or exercising extra-constitutional power.—Cf. A. H. J. Greenidge, *Roman Public Life*.

Dictionary (from the Lat., *dictio*, a saying, expression, word), a book containing the words, or subjects, which it treats, arranged in alphabetical order. It may be either a vocabulary, or collection of the words in a language, with their definitions; or a special work on one or more branches of science or art prepared on the principle of alphabetical arrangement, such as dictionaries of biography, law, music, medicine, history, or philosophy. Amongst dictionaries of the English language, the earliest seem to have been those of Bullokar (1616) and Cockeram (1623). That of Dr. Johnson, published in 1755, made an epoch in this department of literature. Previous to this the chief English dictionary was that of Bailey, a useful work in its way. An enlarged edition of Johnson's dictionary, by the Rev. H. J. Todd, appeared in 1818; and this, again enlarged and modified, was issued under the editorship of Dr. R. G. Latham (1864-72). The best-known American dictionary of the English language is that by Noah Webster, published in 1828, and since entirely recast. Richardson's dictionary, published during 1836 and 1837, was valuable chiefly for its quotations. Ogilvie's *Imperial English Dictionary*, based on Webster, and first published between 1847 and 1850, has been issued in a remodelled and greatly enlarged form (4 vols. 1881-2 and subsequently, Charles Annandale, LL.D., editor). It is one of the encyclopædic dictionaries. Cassell's *Encyclopædic Dictionary* is another extensive work (1879-88). The largest completed English dictionary is the *Century Dictionary* (New York, 1889-91, 6 vols. quarto). *The Standard Dictionary* is another American work. A new English dictionary 'on historical principles', first edited by Sir J. A. H. Murray, LL.D., with the assistance of many scholars, and now edited by Dr. Henry Bradley, is being published at the Clarendon Press since 1884. Among foreign dictionaries are the polyglot dictionary of Calepino (1502), the Latin and Greek *Thesaurus* of Robert and Henry Stephanus, the Italian *Vocabulario della Crusca* (1612), &c. The chief etymological dictionary of English words is that by Professor Skeat (1882). Among French dictionaries (for French

people) the chief is that of Littré; among German, the dictionary begun by the brothers Grimm.

Dictyotaceæ, a family of Brown Algæ, section Cyclosporeæ. *Dictyota dichotoma*, with a delicate, flattened, repeatedly forked thallus, is not uncommon in sandy pools on our coasts. The plants are of three kinds, viz. ♂, bearing antheridia; ♀, bearing oogonia; and neuter, producing tetraspores. The oospores give rise to neuter plants, the tetraspores to ♂ or ♀ plants. This is one of the best instances of 'homologous' alternation of generations, i.e. that type in which the different generations are identical in form, differing only in their reproductive organs and in the number of chromosomes in their nuclei. Another genus is Padina.

Didactic Poetry, that kind of poetry which professes to give a kind of systematized instruction on a definite subject or range of subjects. Thus the *Georgics* of Virgil and the *De Rerum Natura* of Lucretius profess to give, the one a complete account of agriculture and kindred arts, the other a philosophical explanation of the world. Other examples of purely didactic poetry are Horace's *Ars Poetica*, and Pope's *Essay on Criticism*. In a larger sense of the word most great poems might be called didactic, since they contain a didactic element in the shape of history or moral teaching, Dante's *Divina Commedia*, Milton's *Paradise Lost*, or Goethe's *Faust*, for example. The difference may be said to be this, that in the one case the materials are limited and controlled by nothing but the creative fancy of the poet, while in the other they are much more determined by the actual nature of the subject treated of.

Didel'phia, one of the three subclasses of the Mammalia (the others being Monodelphia and Ornithodelphia), comprising only one order, that of Marsupials or pouched mammals.

Diderot (dēd-rō), Denis, a French writer and philosopher, born in 1713, at Langres, in Champagne, died in 1784. He was educated in the school of the Jesuits, and afterwards at Paris, at the College of Harcourt, but declined to study law, preferring to earn his living by teaching mathematics. His first works were the *Essai sur le Mérite et la Vertu* (1745); and the *Pensées Philosophiques* (1746), a pamphlet against the Christian religion. His *Lettre sur les Aveugles à l'Usage de Ceux qui Voyent* is in the same strain. These

heterodox publications cost him an imprisonment for some time at Vincennes. Diderot now tried writing for the stage, but his pieces were failures. In 1749 he had begun, along with D'Alembert and some others, the *Encyclopædia*. At first it was intended to be mainly a translation of one already published in English by Chambers. Diderot and D'Alembert, however, enlarged upon this project, and made the new *Encyclopædia* a magnificently comprehensive and bold account of all the thought and science of the time. Diderot, besides revising the whole, undertook at first the mechanical arts, and subsequently made contributions in history, philosophy, and art criticism. But the profits of all his labour were small, and it was only the liberality of the Empress Catherine, who purchased his library for 50,000 livres and made him a yearly allowance of 1000 livres, that saved Diderot from indigence. In 1773 he visited St. Petersburg to thank his benefactress and was received with great honour. On his return to France he lived in retirement, passing the last ten years of his life in writing and conversations, wherein, as Marmontel said, he was at his best. Besides his articles in the *Encyclopædia* he wrote numerous works, some of which were published after his death. Among the best known are *Le Neveu de Rameau*, a kind of philosophical dialogue which Goethe thought worthy of translation; *Essai sur la Peinture*, and *Paradoxe sur le Comédien*, suggestive essays on the principles of painting and acting; two lively tales, *La Religieuse* and *Jacques le Fataliste*. On account of his great interest in almost every branch of human knowledge, Voltaire nicknamed him 'Pantophile Diderot'.—BIBLIOGRAPHY: T. Carlyle, *Essay on Diderot*; F. Brunetière, *Études Critiques*; R. L. Cru, *Diderot as a Disciple of English Thought*.

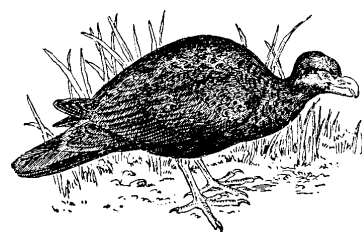
Dido, or **Elissa**, the reputed founder of Carthage. She was the daughter of a king of Tyre, called by some Belus, by others Metten or Matgenus. After her father's death, her brother Pygmalion murdered her husband Sicharbas, or as Virgil calls him Sychæus, with the view of obtaining his wealth. But Dido, accompanied by many Tyrians of her party, fled with all the treasure over sea, and, landing on the coast of Africa, founded Carthage about 860 B.C. The story is told by Virgil, with many inventions of his own, in the *Æneid* (Books i and ii).

Didot (dē-dō), a famous house of printers, booksellers, and typefounders at Paris. The founder was *François Didot*, born in 1689, died 1757. Of his

sons *François-Ambroise* (born 1720, died 1804) and *Pierre-François* (born 1732, died 1795) the first distinguished himself in the type-founding art as an inventor of new processes and machines, the second was equally eminent for his bibliographical knowledge, and contributed much also to the advancement of printing.—*Pierre* (born 1761, died 1853) succeeded his father *François-Ambroise* in the printing business. He made himself famous by his magnificent editions of classic authors in folio, amongst which his *Virgil* (1798) and his *Racine* (1801) may be particularly mentioned. He did much also for the improvement of types. He is known also as an author.—*Firmin* (born 1764, died 1836), the brother of *Pierre*, took charge of the type-founding, was the inventor of a new sort of script, and an improver of the stereotype process.—*Ambroise-Firmin* (born 1790, died 1876), and *Hyacinthe-Firmin* (born 1794, died 1880) occupied a distinguished position amongst the publishers of Paris. The former left a collection of MSS. which was worth, at the time of his death, about two million francs. The house has now extended its trade into everything connected with bookselling, paper-making, and bookbinding.

Didsbury, a district of Manchester, on the Midland Railway, 4 miles south by east of Manchester, a place of residence of many Manchester business men. There is an important Wesleyan Training College there. Pop. 9234.

Didun'culus, a genus of birds allied to the pigeons, and comprising only the one species, *D. strigirostris*, native to some of the Samoan Islands. This bird is of special interest as being the nearest living ally of the extinct dodo. It has a length of about 14 inches, with a glossy plumage verging from a velvety black on the back to greenish black on the head, breast, and abdomen. The large beak, which is nearly as long as the head, is greatly arched on the upper half, while the lower is furnished with two or three tooth-like indentations.



Didunculus strigirostris

Didym'ium, a rare metallic element, occurring along with *lanthanum* in the mineral cerite as discovered by Mosander in 1842. It has been resolved into two new elements: *Praseodymium* (Pr, 140.9) and *Neodymium* (Nd, 144.3).

Die, a metallic stamp for impressing a design or figure upon coins or other metallic objects. See *Dies* and *Die-sinking*.

Die (dē), an ancient town, France, department of Drôme, 26 miles south-east of Valence; with an ancient cathedral and Roman remains. Pop. 4000.

Dié (di-ā), **St.**, a town, France, department of Vosges, on the Meurthe, 25 miles E.N.E. of Epinal. Both iron and copper are worked; there are marble quarries; and numerous manufactures are carried on. Pop. 19,029.

Diebitsch-Sabalkanski, Hans Karl, a Russian general, born at Grossleippe, in Silesia, in 1785, died 9th June, 1831. He was educated at the military school of Berlin, but in 1801 quitted the Prussian service for that of Russia. He was present at the battles of Austerlitz and Friedland, served with distinction in the campaign of 1812, took part in the battles of Dresden and Leipzig, and was made lieutenant-general at the age of twenty-eight. He had the chief command in the Turkish War of 1828-9, stormed Varna, and made the famous passage of the Balkans, for which the surname of Sabalkanski was conferred on him. In 1830 he commanded the army sent against the revolted Poles, but did not distinguish himself in this service.—Cf. Bantych-Kamenski, *Biographies of Russian Field-m Marshals*.

Dieffenbach (dē'fen-ba' h), Johann Friedrich, German surgeon, born at Königsberg in 1792, died in 1847. After having studied at Bonn and Paris, he settled in Berlin, where his talent as an operator soon attracted notice. Surgery is particularly indebted to him for new methods of forming artificial noses, eyelids, and lips, and curing squinting and stammering.

Diego Garcia. See *Chagos*.

Di'electric, in electricity, a name applied by Faraday to any medium through or across which electrostatic induction can take place. See *Electricity* (Electrostatics). Faraday first showed that electrostatic induction was not action at a distance, but took place by means of the insulating medium separating the two conductors. The medium he named a *dielectric*, and measured its specific inductive capacity by taking that of common air as unity.

Diëly'tra, or **Dicentra**, a genus of plants, of the nat. ord. Fumariaceæ or Fumitories. The best known is *D. spectabilis*, a native of Northern China

and Siberia, now common in European and other gardens. It blossoms in April and May, and its long drooping racemes of purplish-red blossoms present a very graceful appearance. It grows freely in the open air. It is sometimes called bleeding heart or virgin's heart from the shape of the blossoms.

Diemen (dē'men), Anton van, Dutch administrator, was born in 1593, died in 1645. Having gone to India, he speedily rose to the highest dignities, and was at length, in 1636, made Governor-General. He administered the government with much ability, and contributed much to the establishment of the Dutch commerce in India. Abel Tasman, whom he sent with a vessel to the South Seas in 1642, gave the name of *Van Diemen's Land* to the island now called Tasmania.

Dieppe (dē-ep'), a seaport town, France, department of Seine-Inférieure, on the English Channel, at the embouchure of the Arques, 93 miles N.N.W. of Paris. Almost the only public edifices worth special notice are the two Gothic churches, St. Jacques, begun in the thirteenth century, and St. Rémi, founded in 1522, and the old castle (1433), now a barracks. To the west of Dieppe proper is the suburb La Barre; and on the opposite side of the harbour Le Pollet, inhabited chiefly by sailors and fishermen. The old port is spacious, but a new channel with its own harbour system has been added, and vessels of 20-foot draft can now enter. Dieppe is one of the chief watering-places of France, and is much frequented by visitors in summer and autumn. The great bathing establishment forms a luxurious retreat for bathers and invalids, and includes a ballroom. The manufactures include works in ivory, horn, and bone, lace-making, sugar-refining, and shipbuilding. There is a busy fishery, and the foreign trade is still considerable. There is constant steam intercourse between this port and Newhaven. In early times Dieppe was the chief port of France, but its prosperity diminished after the revocation of the Edict of Nantes (1685). During the European War Dieppe was an important Allied base. Pop. 23,973.

Dierx, Léon, French poet, born in the Island of Réunion in 1838. Educated in Paris, he became one of the foremost of the Parnassians. His works include: *Aspirations*, *Poèmes et Poésies*, *La rencontre*, and *Les Amants*. His

collected poems (1889-90) were crowned by the Académie. He died in Paris, 11th June, 1912.

Dies and Die-sinking. Die-sinking is the art of preparing dies, a die being a plate or block, usually of metal, so cut or shaped as to be capable, by means of stamping or pressure, of transferring a given design to some article which is to be manufactured in quantity. The pressure may be applied by any sort of power, from hand to hydraulic.

The steel for the manufacture of steel dies is carefully selected, forged at a high heat into the rough die, softened by careful annealing, and then handed over to the engraver. After the engraver has worked out the design in intaglio, the die is put through the operation of hardening, after which, being cleaned and polished, it is called a *matrix*. This is not, however, generally employed in multiplying impressions, but is used for making a *punch* or steel impression for relief. For this purpose another block of steel of the same quality is selected, and, being carefully annealed or softened, is compressed by proper machinery upon the matrix till it receives the impression. When this process is complete, the impression is retouched by the engraver, and hardened and collared like the matrix. Any number of dies may now be made from this punch by impressing upon it plugs of soft steel.

There is hardly any article which does not in the course of its manufacture require the use of a die of some kind. For all sorts of metal-work, seals, rings, silverware, moulds and shapes of sheet steel or tin, dies are employed. For this class of work they are usually of steel. For embossing articles of leather, wood, celluloid, rubber, cloth, or clay, dies of brass and phosphor-bronze are commonly used, these being easier media to work in and yet sufficiently strong. The dies for letter headings and company seals are cut, in reverse to the design required, in steel; those for sealing-wax seals in steel or brass; the lettering being usually punched in by hand by separate letter punches, which themselves have been cut in relief on steel. Designs which are unusually ornate may be engraved by hand. Dies for embossing designs on leather, catalogue covers, cardboard articles, cards, and soft materials are usually modelled in brass. The design in reverse is cut out to a depth corresponding to the relief wanted. These dies are usually worked up by hand by the engraver. Dies for the reproduction of rubber stamps for printing on clay are cut in phosphor-bronze or hard brass in

relief and reverse, and with an extreme bevel. The dies or blocks are then struck deeply into lead, and melted rubber is poured into the moulds so formed. When set, the rubber is removed and mounted as a hand-stamp ready to impress the clay in ink. Dies for wallpapers are cut on rollers. Steel dies for flower-shapes have a cutting edge, so that they can stamp out and emboss in one action. Of late years machinery has come much into use for relieving the engraver of some of his labour, but the designs are generally kept secret. One machine, called the pantograph or engraving machine, reproduces engravings in all metals and many shapes from patterns. Many of the stamp-duty steel dies made and issued by the Royal Mint are reproduced from this machine in reductions from brass patterns.—
BIBLIOGRAPHY: Lucas, *Dies and Die-making*; J. V. Woodworth, *Dies: their Construction and Use*.

Diesel, Rudolph, German inventor, born in Paris in 1858, died in 1913. Educated in England and at Munich, he proposed in 1893 to utilize directly the energy created by the combustion of fuel, a proposal which led to his invention of the Diesel engine. (See *Internal Combustion Engines*.) In 1913 he was called to England to consult with the Admiralty on the application of his motor, but was drowned in crossing the Channel. In 1894 he published a monograph entitled *Theory and Construction of a Rational Heat Motor*.

Dies Fasti et Nefasti, a Roman division of days, with reference to judicial business, into working-days and holidays. A *dies fastus* was a day on which courts and assemblies could be held and judgments pronounced; a *dies nefastus*, a day on which courts could not be held nor judgments pronounced.

Dies Iræ (dī'es ī'rē), one of the great Latin hymns of the Mediæval Church, generally used as part of the requiem or mass for the souls of the dead. It describes, as its name ('the day of wrath') denotes, the Last Judgment of the world, and seems to have been suggested by the description in *Zephaniah*, i, 15 and 16. It is supposed to have been written by Thomas da Celano, a Franciscan friar of the thirteenth century. It was translated by Crashaw and Dryden in the seventeenth century, and by Macaulay and others in the nineteenth, but none of these translations conveys the solemn force of the original.

Diest (dēst), a town, Belgium, province of Brabant, 32 miles E.N.E. of Brussels. It has some manufactures, but the chief products of the place are beer and gin, the former being largely exported. The town was occupied by the Germans in 1914, and re-entered by the Belgians in 1918. Pop. 8800.

Diesterweg, Friedrich Adolf Wilhelm, German educator, born in 1790, died in 1866. In 1820 he became director of the new Teachers' Seminary at Mörs, and soon gained a reputation as teacher and educator. He was a follower of Pestalozzi, and aimed at making every subject of instruction a means of education. In 1827 he founded the *Rheinische Blätter für Erziehung und Unterricht*, wherein he advocated his pedagogical views.

Di'et, a meeting of some body of men held for deliberation or other purposes; a term especially applied to the legislative or administrative assemblies of Austria, Germany, and Poland.

Dietet'ics (Gr. *diaita*, daily regimen), that part of medicine which relates to the regulation of diet. The ideal diet is clearly that which, without burdening the viscera uselessly, furnishes all necessary nutritive elements, with due consideration for special physiological conditions in any given case. Under the head of *Aliment* the physiological properties of various foods have already been considered theoretically in respect of their capacity to supply physical waste in nitrogenous and non-nitrogenous matter. (See *Aliment*.) No single substance contains the elements needed to replace this waste in their requisite proportions, and a mixed diet is therefore necessary. For instance, to secure the required amount of carbon a man would need to eat about 4 lb. of lean beef, while 1 lb. would yield all the nitrogen required; thus, apart from the labour of digesting 4 lb. of beef, the body would be compelled to get rid of the excess of nitrogen. Bread, on the other hand, has carbon in abundance, but is deficient in nitrogen; so that by uniting 2 lb. of bread with $\frac{3}{4}$ lb. of lean meat, the due proportion of carbon and nitrogen is satisfactorily supplied. Milk and oatmeal taken together also contain nitrogenous and non-nitrogenous substances in nearly the required proportions. A certain proportion of saline matter is also necessary. The nature of the food most suitable for a healthy man is dependent in part upon general conditions, such as climate and season, and in part upon special conditions of individual habit. The inhabitants of the Arctic regions need large quantities of oleaginous food; those of the Tropics live chiefly on

starchy products. With increased activity and exertion, as in training, an increase in the nitrogenous foods becomes necessary. In a state of health we need not draw hairbreadth distinctions as to the superior salubrity of the several sorts of diet, the quantity rather than the quality of food being the main consideration. Those persons who have been most remarkable for health and long life have generally been contented with two moderate meals a day, which are certainly quite sufficient during a state of health. In various countries the breakfast generally consists of tea, coffee, or cocoa, with a certain proportion of bread and butter; persons with delicate digestive powers, or who lead a sedentary life, cannot with safety or comfort eat animal food *constantly* to breakfast. At dinner all made-dishes highly spiced, such as curries, turtle-soup, &c., as provoking appetite, are hurtful; and the custom of late dining is not to be commended. Stewed and boiled meats are more difficult to digest than meat cooked by fire alone. The flesh of young animals seems to be more difficult of digestion than that of old; and the flesh of tame than that of wild animals. All sorts of fat meat must be taken in smaller quantities. Hence, also, ham, bacon, and salted meats cannot be eaten in such quantities as the tender flesh of poultry. Fish has the advantage of being easily soluble. All boiled vegetables are in general easy of digestion; raw vegetables and salads are rather more difficult. Fruit should be taken in the forenoon rather than after a hearty meal. The moderate use of fermented liquors is far from being invariably an evil, but the smaller the quantity habitually used the better in the majority of cases.

In all diseases attended with much fever or quickness of pulse the stomach loathes animal food, and there is generally a great increase of thirst, to quench which water, either quite cold, or iced, or tepid, or rendered acid, may be freely indulged. Infusions, too, of barley, sage, balm, &c., may be taken. In chronic diseases attended with hectic fever, milk is the most proper diet. The best food for infants is, of course, their mother's milk; but whenever they begin to cut teeth a little animal food, such as soft-boiled eggs, beef-tea, and even chicken minced very fine, may be given. Many infants suffer from having too much sugar given them in their food.—
BIBLIOGRAPHY: R. H. Chittenden, *Physiological Economy in Nutrition*, and *Nutrition of Man*; Hutchison, *Food and Dietetics*; Lusk, *The Science of Nutrition*.

Dietrich (dē'trih), Christian Wilhelm Ernst, a German painter and engraver, born in 1712, died in 1774. He studied under his father, and afterwards under Alexander Thiele at Dresden, where he became court-painter and professor in the academy. He adopted several different manners, successfully imitating Raphael and Mieris, Correggio, and Ostade.

Dietrich of Bern (dē'trih), the name under which Theodoric the Great, King of the Ostrogoths, appears in the old German legends. Bern stands for Verona, his capital.

Dieu, or **D'Yeu** (dyeu; ancient **Insula Dei**), an island off the west coast of France, department of Vendée. It is inaccessible on the west side, but on the east has a tolerable harbour defended by batteries. The chief industry is fishing. There are four lighthouses on the island. Pop. 3809.

Dieu et Mon Droit (dyeu e mon' drwä; 'God and my right'), the battle-cry of Richard I at the battle of Gisors (1198), signifying that he was not subject to France, but owed his power to God alone. The battle-cry was then adopted as the motto of the arms of England, and revived by Edward III in 1340, when he claimed the crown of France. Except during the reigns of Elizabeth and Anne, who used the motto *Semper eadem*, and of William III, who personally used *Je maintiendray*, it has ever since been the royal motto of England.

Diez (dêts), Friedrich Christian, German philologist of the Romance languages, born in 1794, died in 1876. Having qualified himself as a lecturer at Bonn, he was appointed professor of the Romance languages there in 1830. His work stands in much the same relation to the Romance dialects which the researches of Grimm occupy with respect to German dialects. In addition to various works on the poetry of the Troubadours, he published a very valuable *Grammatik der Romanischen Sprachen* (1836-42, translated into English by Cayley in 1863), and an *Etymologisches Wörterbuch der Romanischen Sprachen* (1853).

Difference, a stock-exchange term. When stock is bought or sold merely as a speculation for the rise or fall, with no intention of the buyer to 'take up' the stock, or of the seller to deliver it, the 'difference' is the movement in price which may take place between the date of the transaction and the following 'settling-day'. If the price falls, the buyer has to pay the difference

upon 'carrying over' his purchase to the next account; if it rises, the seller is at the loss. Since the first weeks of the European War all stock-exchange transactions have been made, in theory at least, for cash, and speculative business of this nature has been consequently much reduced.

Differences, Finite, a calculus much used in actuarial work, which deals with a series of numbers by considering the differences of the successive terms.

If $u_1, u_2, u_3, \dots$ are the terms of the series, then $u_2 - u_1, u_3 - u_2, u_4 - u_3, \dots$ form another series called the series of first differences. The notation used is $u_2 - u_1 = \Delta u_1, u_3 - u_2 = \Delta u_2, \dots$

These first differences may themselves be differenced, giving the second differences $\Delta u_2 - \Delta u_1, \Delta u_3 - \Delta u_2, \dots$, which are written $\Delta^2 u_1, \Delta^2 u_2, \dots$

Similarly, we form the third differences $\Delta^3 u_1 = \Delta^2 u_2 - \Delta^2 u_1$, and so on.

As an example, let the original series be the cubes of the natural numbers.

1	8	27	64	125	216	343	512
7	19	37	61	91	127	169	
12	18	24	30	36	42		
6	6	6	6	6			
0	0	0	0				

Here we begin by writing down the series of cubes as far, say, as 216; beneath these we write the first differences $8 - 1 = 7, 27 - 8 = 19$, &c. We thus obtain the part of the table to the left of the diagonal line.

We observe that the third differences are constant, each being 6. (It is easy to prove generally that the n th differences of the series, $1^n, 2^n, 3^n, \dots$, are constant.) Knowing the third differences, we can now extend the table as far

as we wish to the right of the diagonal line. We get first $6 + 30 = 36$, $36 + 91 = 127$, $127 + 216 = 343$. We infer that $7^3 = 343$.

Since $u_1 - u_0 = \Delta u_0$, we have $u_1 = u_0 + \Delta u_0 = (1 + \Delta)u_0$.

Similarly, $u_2 = (1 + \Delta)u_1 = (1 + \Delta)^2 u_0$; and, generally,

$$u^x = (1 + \Delta)^x u_0$$

$$= \left\{ 1 + x\Delta + \frac{x(x-1)}{1 \times 2} \Delta^2 + \dots \right\} u_0$$

$$= u_0 + x\Delta u_0 + \frac{x(x-1)}{1 \times 2} \Delta^2 u_0 + \dots$$

a formula much used by calculators, and known as Newton's interpolation formula.

The above symbolic method of proof only applies when x is a positive integer, but the result is used in practice even for fractional values of x , as in most cases the high differences become negligible.

If n is a positive integer, it is easy to prove that

$$\Delta^n u_x = u_{x+n} - nu_{x+n-1} + \frac{n(n-1)}{1 \times 2} u_{x+n-2} - \dots$$

If the n th differences vanish, or are negligible, this gives

$$0 = u_{x+n} - nu_{x+n-1} + \frac{n(n-1)}{1 \times 2} u_{x+n-2} - \dots + (-1)^n u_x,$$

another useful interpolation formula, by which we can calculate any missing term of a series.—BIBLIOGRAPHY: G. Boole, *Finite Differences; Textbook of the Institute of Actuaries*.

Differential Equation, an algebraical relation involving derivatives or differentials. Examples:

$$\frac{d^2z}{dt^2} = g: ydx + xdy + zdz = 0.$$

An *ordinary* differential equation involves only one independent variable, a *partial* differential equation involves more than one. Examples of ordinary equations:

$$\frac{d^2y}{dx^2} + \frac{1}{x} \frac{dy}{dx} + y = 0; \frac{d^2x}{dt^2} + a \frac{dy}{dt} + px + qy = 0.$$

Examples of partial differential equations:

$$x \frac{dz}{dx} + y \frac{dz}{dy} = nz; \frac{d^2u}{dx^2} + \frac{d^2u}{dy^2} + \frac{d^2u}{dz^2} = 0.$$

Equations, whether ordinary or partial, can also be classified as *linear* or *non-linear*. A linear equation is a rational integral equation of the first degree in the dependent variable or variables and their derivatives. The equation

$$x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} + (x^4 + 1)y = 0$$

is linear, but

$$\left(\frac{dy}{dx} \right)^2 = xy \text{ and } y \frac{dy}{dx} = x^2$$

are non-linear. The *order* of an equation is the order of the highest derivative or differential which it contains. Of the three equations last written, the first is *linear of the second order*, the other two are of the *first order and second degree*. To *integrate* a differential equation or system of equations is to find a relation or relations among the variables, equivalent to the given equation or equations. Thus the integral of

$$\frac{d^2z}{dt^2} = g \text{ is } z = \frac{1}{2}gt^2 + At + B,$$

where A and B are *arbitrary constants*. An ordinary equation of the n th order with one dependent variable has exactly n arbitrary constants in its complete integral, or solution. In a practical problem the arbitrary constants are determined by the *initial, or boundary, conditions*. The solution of $d^2z/dt^2 = g$, e.g. is completely determinate if the values of z and dz/dt when $t = 0$ are given. The solution of *partial* equations may involve arbitrary *functions*, which become definite when proper initial or boundary conditions are assigned. Thus the equation $du/dx = du/dt$ has for its complete solution $u = \varphi(x + t)$, where φ may be a function of any form whatever; if now we are given that, when $t = 0$, $u =$ a given function $f(x)$, we obtain $f(x) = \varphi(x)$, so that the solution required is $u = f(x + t)$. Certain ordinary linear equations of the second order are specially important, both from the beauty of their theory and from their usefulness in Mathematical Physics. Some of these equations are: Bessel's equation, Legendre's equation, the hypergeometric equation, Mathieu's equation, Lamé's equation. Linear partial equations of the second order are fundamental in Physics. Such are: Laplace's equation,

$$\frac{d^2V}{dx^2} + \frac{d^2V}{dy^2} + \frac{d^2V}{dz^2} = 0;$$

the wave equation,

$$\frac{d^2V}{dt^2} = c^2 \left(\frac{d^2V}{dx^2} + \frac{d^2V}{dy^2} + \frac{d^2V}{dz^2} \right);$$

the equation of conduction of heat,

$$\frac{dV}{dt} = k \left(\frac{d^2V}{dx^2} + \frac{d^2V}{dy^2} + \frac{d^2V}{dz^2} \right).$$

These involve one dependent variable only. Equations with several dependent variables occur in Elasticity, Electrodynamics, and Hydrodynamics. A notable feature of the hydrodynamical equations is that they are not linear.

No general rules exist enabling us to deal with a differential equation taken at random, and only a few types have been completely solved. Of soluble equations, the most important are those which are *linear with constant coefficients*.

Example 1. $d^2x/dt^2 - 7dx/dt + 12 = 0$. To solve this, try $x = e^{mt}$. We find $e^{mt}(m^2 - 7m + 12) = 0$. Thus $m = 3$ or 4 . It is now easy to show that $x = Ae^{3t} + Be^{4t}$ is a solution, where A and B are arbitrary constants. This is the general solution. We can determine A and B if the values of x and dx/dt are given for a definite value of t , say $t = 0$.

Example 2. $d^2y/dt^2 = c^2 d^2y/dx^2$. Try $y = e^{lx + mt}$. We find $m^2 = c^2l^2$, or $m = \pm cl$. Hence $y = Ae^{l(x + ct)} + Be^{l(x - ct)}$ is a solution for all values of A , B , l ; so, also, is the sum of any number of terms of similar forms. We may infer that the general solution is

$$y = f(x + ct) + F(x - ct),$$

where f and F are arbitrary functions. It is only in exceptional cases that an equation can be solved, as in these two examples, by an analytical formula; indeed, differential equations are the most fertile source of new functions in analysis. But, as in the analogous cases of algebraic equations and definite integrals, it may be quite possible to find, by methods of approximation, an arithmetical solution which is sufficient for the purpose in hand.—
BIBLIOGRAPHY: H. T. H. Piaggio, *Differential Equations*; J. M. Page, *Ordinary Differential Equations*; A. R. Forsyth, *A Treatise on Differential Equations*; E. T. Whittaker and G. N. Watson, *Modern Analysis*.

Diffraction, a term applied to the bending that rays of light undergo in passing close to the edge of an opaque body. Thus when a beam of direct sunlight is admitted into a dark room through a narrow slit, and falls upon a screen placed to receive it, there appears a line of white light bordered by coloured fringes; these fringes are produced by diffraction, and in the case given it may be seen that the red or long-wave rays are diffracted more than the blue rays. See *Interference*.

Diffusion, the gradual mixing of gases, liquids, or solids when brought into direct contact. When a block of lead is placed on a block of gold, with

their smooth surfaces in close contact, it is found that, after several weeks, gold has diffused into the lead, and lead into the gold. In the case of gases, when a jar of oxygen and a jar of hydrogen are connected together by a tube or opening of any kind, they rapidly become mixed; and their mixture does not depend on gravity, but takes place in opposition to that force, as may be shown by placing the jar of hydrogen gas above the other. Oxygen is sixteen times heavier than hydrogen, bulk for bulk, but the heavier gas moves upwards and the lighter downwards, and the process of intermixture, or *diffusion*, goes on till the two gases are apparently equably distributed throughout the whole space. After that they have no tendency whatever to separate. Similarly, if two vessels, one containing oxygen and the other hydrogen, be connected by a tube which is stuffed with a plug of porous material, such as plaster of Paris, the gases gradually diffuse one into the other through the porous plug. The two gases, however, do not pass through the porous separator at equal rates, but in *inverse proportion to the square roots of the densities of the gases*. Thus in the case of two vessels, one containing hydrogen and the other oxygen, which is sixteen times as heavy as hydrogen, the hydrogen will pass towards the oxygen jar four times as quickly as the oxygen will pass towards the hydrogen jar. Kindred phenomena occur when two liquids that are capable of mixing, such as alcohol and water, are put in contact, the two gradually diffusing one into the other in spite of the action of gravity. In some cases, however, as where ether and water are employed, the diffusion is only partial, this result arising from the fact that these two liquids are not miscible in all proportions. When solutions of various solid bodies are placed in contact, interdiffusion also takes place. On the results of his examination of the phenomena of diffusion of liquids and salts across porous membranes or *septa*, Graham founded a method of separating *colloid* from *crystalloid* bodies, which he called *dialysis*.

Digam´ma, a letter which once belonged to the Greek alphabet, and which remained longest in use among the Æolians. It resembled our letter F, and hence was called *digamma*, that is, double Γ. It appears to have had the force of *f* or *v*. Its existence was first pointed out by Richard Bentley.

Digby, Sir Everard, an English gentleman, born of a Roman Catholic family in 1578. He enjoyed some consideration at the court of Elizabeth and

James I, by whom he was knighted. Having contributed money to the Guy Fawkes conspiracy, he was tried and hanged in 1606.

Digby, Sir Kenelm, eldest son of the preceding, born in 1603, died in 1665. He studied at Oxford, was knighted in 1623, and on the accession of Charles I was created a gentleman of the bedchamber, a Commissioner of the Navy, and a governor of Trinity House. He soon after fitted out at his own expense a small but successful squadron against the French and Venetians. In 1636 he became a Roman Catholic, and was imprisoned as a Royalist during 1642-3, when he was allowed to retire to the Continent. At the Restoration he returned to England, became a member of the Royal Society, and was much visited by men of science. He wrote numerous works: a *Treatise on the Nature of Bodies*, a *Treatise on the Nature and Operation of the Soul*, and *Of the Cure of Wounds by the Powder of Sympathy*.

Di'gest, a name originally given to a collection or body of Roman laws, digested or arranged under proper titles by order of the Emperor Justinian. Hence applied to any somewhat similar collection.

Diges'ter, a strong vessel of copper or iron, on which is screwed an air-tight cover with a safety-valve, the object being to prevent loss of heat by evaporation, and to enable boiling to take place at a high pressure. Water may be thus heated to 400°F.; at which temperature its solvent power is so greatly increased that bones are converted into a jelly.

Diges'tion is that process in the animal body by which the aliments are so acted upon that the nutritive parts are prepared to enter the circulation, and separated from those which cannot afford nourishment to the body. The organs effecting this process are called the *digestive* organs, and consist of the stomach, the great and small intestines, &c. (see *Intestine*, *Stomach*), the liver, and pancreas. When the aliments, after being properly prepared and mixed with saliva by mastication, have reached the stomach, they are intimately united with a liquid substance called the *gastric juice*, by the motion of the stomach. By this motion the aliments are mechanically separated into their smallest parts, penetrated by the gastric juice, and transformed into a uniform pulpy or fluid mass. The gastric juice acts upon the albuminous parts of the food, converting them into peptones, which can

pass through organic membranes and thus enter the blood. This action is aided by the warmth of the stomach. The pulpy mass, called *chyme*, proceeds from the stomach, through the pylorus, into that part of the intestinal canal called the small intestine, where it is mixed with the pancreatic juice, bile, and intestinal juice. The pancreatic juice converts starch into sugar, albumins into peptones, and emulsionizes fats, so that all these kinds of food are rendered capable of absorption. The process is aided by the intestinal juice. The bile also acts upon fats, and thus the food is formed into the *chyle*, which is absorbed into the system by the capillary vessels called *lacteals* (see *Chyle*; *Chyme*), while the non-nutritious matters pass down the intestinal canal and are carried off.—BIBLIOGRAPHY: F. Hare, *Food Factor in Disease*; Taylor, *Digestion and Metabolism*.

Digit (dij'it; Lat. *digitus*, a finger), in arithmetic, any one of the ten numerals, 1, 2, 3, 4, 5, 6, 7, 8, 9, 0. Digit is also a measure of a finger's breadth, equal to $\frac{3}{4}$ inch.

Digit, in astronomy, is the measure by which we estimate the quantity of an eclipse. The diameter of the sun or moon's disk is conceived to be divided into twelve equal parts, called *digits*; and according to the number of those parts or digits which are obscured, so many digits are said to be eclipsed.

Digitalin (dij-i-tā'lin), a vegetable alkaloid, the active principle of the *Digitālis purpurēa* or foxglove. It has a bitter taste, and is a strong poison, but is used medicinally, especially for the heart. See next article.

Digitalis (dij-), a genus of plants, nat. ord. Scrophulariaceæ, containing about twenty species of tall herbs, natives of Europe and Western Asia. The purple foxglove (*D. purpurēa*) is a common wild flower in Britain, and several species are grown in gardens. Various preparations from the foxglove receive this name, and are used in medicine, principally in cases of heart disease.

Digitigra'da (*digitus*, finger, toe, and *gradi*, to walk), a section of the Carnivora, so called from their walking on the ends of their toes; as the dog, cat, and their allies. See *Plantigrade*.

Digito'rium, a small portable dumb instrument having a short keyboard with five keys like those of a piano, used by piano-players for practice, to

give strength and flexibility to the fingers.

Digne (dēny), a town, France, capital of the department of Basses-Alpes, picturesquely situated on a mountain slope, 60 miles north-east of Marseilles. In 1629 a plague reduced the population from 20,000 to 1500. Pop. 7317.

Dijon (dē-zhōn; Lat. *Castrum Divonense*), a town in Eastern France, capital of the department of Côte-d'Or, in a fertile plain, at the foot of a range of vine-clad slopes, formerly surrounded by ramparts, which now furnish beautiful promenades. At some distance it is surrounded by a series of forts. Some of the buildings belong to the period when Dijon was capital of the dukedom of Burgundy, the chief being the cathedral of St. Bénigne, a building of vast extent with a lofty wooden spire above 300 feet high; the churches of Nôtre Dame and St. Michael; the ancient palace of the Dukes of Burgundy, now used as the hôtel de ville and museum; and the *palais de justice*, formerly the Parliament House of Burgundy. Dijon is the birth-place of Bossuet. It has important educational institutions and a valuable library. Industries: woollens, hosiery, candles, mustard, vinegar, chemicals, paper-hangings, tanneries, foundries, machine factories, cotton- and oil-mills. The trade is considerable, particularly in the wines of Burgundy. Pop. 76,847.

Dike, or **Dyke** (connected with the Gr. *teichos*, wall), a word variously used in different localities to represent a ditch or trench, and also an embankment, rampart, or wall. It is specially applied to an embankment raised to oppose the incursions of the sea or of a river, the dikes of Holland being notable examples of work of this kind. These are often raised 40 feet above the high-water mark, and are wide enough at the top for a common roadway or canal, sometimes for both. The Helder Dike, one of the largest, is about 6 miles in length and costly in upkeep. See *Embankment*.

Dike, or **Dyke**, in geology, a term applied to intrusive igneous masses, such as basalt, which fill up veins and fissures in the other rocks, and sometimes project on the surface like walls through their superior resistance to weathering.

Dilapidation, in English ecclesiastical law, is where an incumbent of a church living suffers the parsonage-house or outhouses to fall down, or be in decay for want of necessary repairs; or it is the pulling down or

destroying any of the houses or buildings belonging to a spiritual living, or destroying of the woods, trees, &c., appertaining to the same. An outgoing incumbent (or his heirs) is liable for dilapidation to his successor. In general, the term is applied to the act of allowing or causing any lands, houses, &c., to become waste or to decay.

Dilem'ma (from Gr. *di*-, double, and *lēmma*, proposition, assumption), in logic, a form of argument used to prove the falsehood or absurdity of some assertion, as in the following instance: If he did so he must be either foolish or wicked; but we know he is neither foolish nor wicked; therefore he cannot have done so. The two suppositions, which are equally untenable, are called the 'horns' of the dilemma.

Dilettante (di-let-ta' n'tā), an Italian expression, signifying a lover of the arts and sciences, who devotes his leisure to them as a means of amusement and gratification, being thus nearly equivalent to *amateur*. It is also used in reference to the trifler and dabbler in art and science. In 1734 a number of gentlemen founded in London a Dilettanti Society, which published a splendid work on *Ionian Antiquities*, 1769, 1881 (4 vols.); *Specimens of Ancient Sculpture*, 1809, 1835.

Dilke, Sir Charles Wentworth, English writer and politician, son and grandson of men well known in their day, was born in 1843, died in 1911. He graduated at Cambridge, and was called to the Bar. His first work, *Greater Britain*, the result of a tour round the world from 1866 to 1867, became very popular. In 1868 he was elected member of Parliament for Chelsea, and he remained so up to 1885. After a few years' retirement (due to a divorce case) he became member of Parliament for Forest of Dean. He was Under-Secretary for Foreign Affairs, President of the Local Government Board, &c. He succeeded his father as owner of the *Athenæum*, and became the proprietor of *Notes and Queries*. *The Present Position of European Politics*, and *Problems of Greater Britain*, are among his works.

Dill, an umbelliferous plant, *Anēthum graveōlens*, a native of the southern countries of Europe, the fruits, commonly but erroneously called seeds, of which are moderately warming, pungent, and aromatic, and are employed medicinally as a carminative.

Dillenia´ceæ, an order of plants, chiefly fine trees, inhabiting the East Indies, allied to Ranunculaceæ and Magnoliaceæ.

Dillingen (dil´ing-en), an old town, Bavaria, on the Danube, formerly the seat of a Jesuit university. Pop. 6291.

Dillon, John, Irish politician and agitator, born in Dublin in 1851, the son of John Blake Dillon (1816-66), a leader of the Young Ireland party. Educated at the Catholic University of Dublin and at the Royal College of Surgeons, he became a doctor of medicine. He identified himself with the Parnellite movement, and entered Parliament for Tipperary in 1880. An ardent Nationalist, not hesitating to incite his compatriots to lawlessness, he was sent to prison in 1888. Without a seat in Parliament from 1883 to 1885, he was returned in the latter year for East Mayo, which he represented thereafter. In 1918, after the death of John Redmond, he was elected chairman of the Irish Nationalist party, which, however, owing to the rise of the Sinn Féin party, was a nominal distinction only.

Dilman´, a town, Persia, province of Azerbaijan, 75 miles west of Tabreez. Pop. estimated at 15,000.

Dilo´lo, a small lake in Angola, near the southern boundary of Belgian Congo, lat. 11° 22' S.; long. 22° 34' E.: regarded as the source of the Zambesi.

Dil´uents (Lat. *diluere*, to wash away), in medicine, are those substances which are taken to increase the proportion of fluid in the blood. They consist of water and watery liquors.

Dilu´vium, the name formerly given by geologists to certain gravels and comparatively recent deposits, which seemed to have been the result of a rush of water or deluge.

Dime (Fr. *dîme*, Lat. *decimus*, tenth), the term for the tenth part of a dollar or ten-cent piece in the United States of America, a silver coin whose English equivalent is about 5d. Hence the phrases *dime novels*, *dime museums*, &c.

Dimensions, Algebraical. There are three dimensions in space: length, breadth, and height or depth. An area is said to be of two dimensions

because it has length and breadth only; a volume is of three dimensions. In algebra terms like x^2 , xy are said to be of two dimensions because there are two letters multiplied together, and their product would measure an area if each letter denoted a length. Similarly, x^3 , xyz are said to be of three dimensions, and the meaning is extended to cover the product of any number of letters. An expression of more than one term is said to be of the same degree as its term of highest dimensions. For example, $3x^2y^2z^2 + 5xyz + 6x^3 + 3x^2y^2$ is said to be of the sixth degree because $x^2y^2z^2 = x \times x \times y \times y \times z \times z$ is of six dimensions.

Dimensions, Physical. One of the aims of physical science is to express all its measurements in terms of the three fundamental units of length, mass, and time. A velocity, for example, is specified by the number of units of length traversed in the unit of time, so that we may write $v = l \div t$, or $v = lt^{-1}$. On this account velocity is said to have the dimensions LT^{-1} . Similarly, acceleration, being velocity added per unit time, has the dimensions of velocity $\div$ time, or LT^{-2} ; and force, being proportional to mass and acceleration jointly, has the dimensions MLT^{-2} .

When a physical law is expressed as an equation connecting the numbers of units of the quantities involved, every term in this equation must be of the same dimensions in any one of the fundamental units. This is the *Principle of Dimensions*, first stated by Joseph Fourier, founder of the theory of the conduction of heat. In order to see its truth, we have only to observe that an equation containing terms of different dimensions would give inconsistent results if the unit of length were varied. Suppose it to be suggested, for example, that the period of vibration t of a simple pendulum of length l is given by the formula $t = 2\pi l/g$, where g denotes the acceleration of a falling body. The dimensions of the expression on the right are $L \div (LT^{-2})$, or T^2 , whereas the term t on the left has dimensions T^1 . Suppose the unit of length is the foot and the unit of time the second, so that $g = 32$, and let $l = 3$. We find in this case $t = 6\pi/32$, so that the period is $3\pi/16$ seconds. But if we change the unit of time to one minute, g becomes $32 \times 60 \times 60$, and the formula gives $t = 6\pi/(32 \times 60 \times 60)$, so that the period is $3\pi/(16 \times 60 \times 60)$ minutes.

The two results are obviously inconsistent. If, however, we take the correct formula, namely $t = 2\pi\sqrt{l/g}$, we find on trial that we obtain the same value for the period however we change the unit of time or the unit of length. Both sides are in this case of dimensions T^1 .

The principle of dimensions provides therefore a useful check on the accuracy of formulæ. But it does much more than this. It often gives very valuable information about the relations of physical phenomena in cases where these relations are far too complicated to be completely worked out by mathematical analysis. To mention but one example, it is by the use of this principle that modern naval architecture is able to predict the behaviour of ocean-going ships from experiments in ponds on small-scale models.

Dimin'utive, in grammar, a word having a special affix which conveys the idea of littleness, and all other ideas connected with this, as tenderness, affection, or contempt. The opposite of *diminutive* is *augmentative*. In Latin, diminutives almost always ended in *-lus*, *-la*, or *-lum*; as *Tulliola*, *meum corculum*, little Tullia my dear, or little heart; *homunculus*, a manikin. The Italian is particularly rich in diminutives and augmentatives, such compound diminutives as *fratellinucciettinetto* (a diminutive of *frate*, brother) being sometimes employed. Among English diminutive affixes are *-kin*, as in *manikin*, a little man; *pipkin*, a little pipe; *-ling*, as in *gosling*, a little goose; *darling*, that is, *dearling*, or little dear; and *-et*, as in *pocket*, from *poke*, a bag or pouch; *tablet*, a little table. Diminutives are not confined to nouns, and *dandle*, *scribble*, *tipple*, are examples of diminutive verbs, and *greenish*, *whitish*, are diminutives of adjectives. Diminutives are also formed, in colloquial and familiar language, by adding *-y* or *-ie* to the names, as *Charley*, *Mousie*, &c.

Dim'ity, a stout cotton fabric, ornamented in the loom either by raised stripes or fancy figures. It is usually employed white, as for bed and bedroom furniture.

Dimorph'ism, in crystallography, the crystallization of a body in forms belonging to two different systems, or in incompatible forms of the same system, a peculiarity exhibited by sulphur, carbon, &c.

Dimorphism, in botany. See *Heterostyly*.

Dinajpur, a town, Hindustan, Bengal, capital of a district of same name, 205 miles north of Calcutta; pop. 12,500.—The district covers an area of about 4118 sq. miles; pop. 1,687,860.

Dinan (dē-năn), a town, France, department of Côtes-du-Nord (Brittany), on the Rance, 14 miles south of St. Malo. It was besieged and captured by the English under the Duke of Lancaster in 1359, but retaken by Du Guesclin. It stands on a steep hill nearly 200 feet above the river, is surrounded by high old walls pierced with four gates, and is a picturesque and interesting old place. In the cathedral of St. Sauveur the heart of Bertrand du Guesclin is buried. Pop. 11,410.

Dinant (dē-năn), a town, Belgium, in the province and 14 miles S. of Namur; picturesquely and strongly situated on the Meuse; a place of antique appearance. The town house was once the palace of the Princes of Liège. The town was destroyed by the Germans in 1914. It is one of the most popular Belgian summer resorts. Pop. 7690.

Dina'pur, a town, Hindustan, Patna district, Bengal, on the right bank of the Ganges, about 12 miles north-west of Patna, cantonment and military head-quarters of the district, with extensive barracks. The environs are studded with handsome bungalows. Pop. 31,025.

Dinar (Lat. *denarius*), formerly an Arab gold coin, also a Persian coin; at present the chief Serbian coin, value one franc.

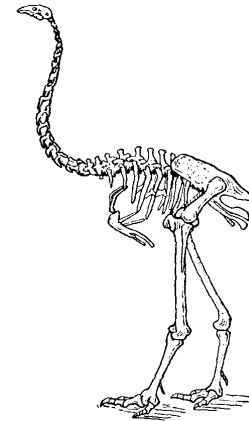
Di'nas Bricks, an infusible kind of brick made of a peculiar rock, containing 98 per cent of silica, with a little alumina, which occurs at Dinas, in the Vale of Neath, in Glamorganshire, S. Wales. The rock is crushed, moistened with water, and moulded by a machine.

Dindigul, a town of India, Madura district, Madras, with a fort on a rocky height; manufactures cigars. Pop. 21,000.

Dindings, The, properly two small islands, also called **Pangkor Islands**, in the Straits of Malacca, belonging to the Straits Settlements, off the coast of Perak (British). The name now includes a strip of territory on the Malay Peninsula opposite; total area about 265 sq. miles, two-thirds of which is covered by dense forests. Coco-nuts, coffee, and pepper are grown with success. Lumut, on the mainland, has a fine natural harbour.

Din'dorf, Karl Wilhelm, German classical scholar, born 1802, lived most of his life at Leipzig, and died 1883. His chief publications were editions of the Greek dramatists (*Poetæ Scenici Græci*) and works elucidative of them and other Greek writers.

Dingo, the native wild dog of Australia (*Canis Dingo*), of a wolf-like appearance and extremely fierce. The ears are short and erect, the head elongated, the tail rather bushy, and the hair of a reddish-dun colour. In habit the dingo is rather fox-like, usually lying concealed throughout the day and making predatory expeditions at night. It is very destructive to sheep, killing more than it eats. It was probably introduced by prehistoric man.

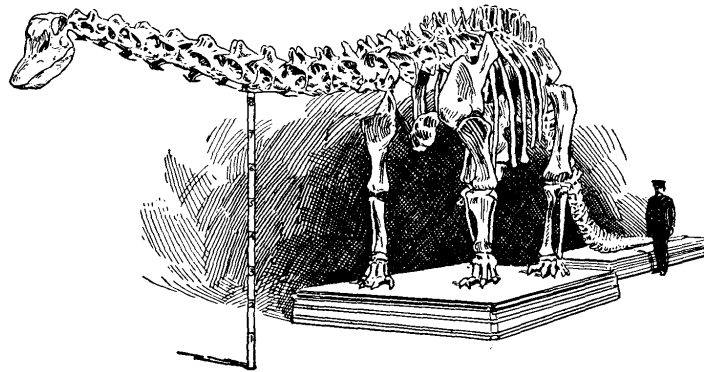


Dinornis maximus
(the Moa)

Ding'wall, a royal and parliamentary burgh and seaport of Scotland, county town of Ross and Cromarty, situated at the head of Cromarty Firth. This town, erected into a royal burgh in 1227, unites with Wick and other places in returning a member to Parliament. Pop. 2590.

Dino'ceras (Gr. *deinos*, terrible, *keras*, a horn), a fossil mammal found in the Eocene strata of North America, in some respects akin to the elephant and of equal size, but without a proboscis. Its bones were very massive; it had two vertical tusks in the upper jaw, three pairs of horns, and the smallest brain, proportionally, of any known mammal.

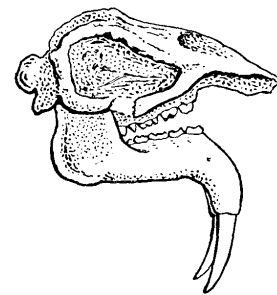
Dinor'nis (Gr. *deinos*, terrible, *ornis*, a bird), an extinct genus of large wingless birds—classed with the small existing Apteryx. The bones of several species have been found in New Zealand. The largest must have stood 12 feet in height, several of its bones being at least twice the size of those of the ostrich. The body seems to have been even more bulky in proportion, the tarsus being short and stout in order to sustain its weight. They do not appear to have become extinct until the seventeenth or eighteenth century, and are spoken of as *moas* by the natives, who buried the eggs (more than 1 foot long) with their dead as provision for their journey to the other world.



Diplodocus Carnegii, a gigantic Dinosaur. Length, 84 feet 9 inches; height at middle of back, 11 feet 5 inches

Dinosauria (Gr. *deinos*, terrible, and *sauros*, a lizard), a group of extinct reptiles, allied in skeletal structure both to the lizards and the birds. While some were only 3 feet long, a large number attained gigantic size. *Atlantosaurus* being 115 feet long. Many were carnivorous, but some of the large heavy forms were herbivorous, and protected by bony spines or plates. The Dinosaurs were the dominant land animals of the Jurassic and Cretaceous periods.

Dinotherium (Gr. *deinos*, terrible, *thērion*, beast), a genus of extinct gigantic proboscidean mammals, precursors of the elephants, the remains of which occur in Miocene formations in several parts of Europe. The type-species (*D. giganteum*) is calculated to have attained the length of 18 feet. It had a proboscis and also two tusks placed at the anterior extremity of the lower jaw, and curved downwards somewhat after the manner of those in the upper jaw of the walrus. The skull is best known from that found in 1835 at Eppelsheim; but the skeleton can now be pieced together from remains in various localities. The vertebræ resemble those of mastodon. *Dinotherium* may have inhabited rivers or estuaries.



Skull of *Dinotherium giganteum*

Diocese (Gr. *dioikēsis*, administration), the circuit or extent of a bishop's jurisdiction. Each English diocese is divided into archdeaconries, each archdeaconry (nominally) into rural deaneries, and each deanery into

parishes. In the Eastern Churches the term *eparchy* is used for diocese. See *Bishop*.

Diocle'tian (Gaius Valerius Diocletianus, surnamed *Jovius*), a man of mean birth, a native of Dalmatia, proclaimed Emperor of Rome by the army A.D. 284. He defeated Carinus in Mœsia (286), conquered the Allemanni, and was generally beloved for the goodness of his disposition, but was compelled by the dangers threatening Rome to share the government with M. Aurelius Valerius Maximian. In 292 Galerius and Constantius Chlorus were also raised to a share in the empire, which was thus divided into four parts, of which Diocletian administered Thrace, Egypt, Syria, and Asia. As the result of his reconstitution of the empire there followed a period of brilliant successes in which the barbarians were driven back from all the frontiers, and Roman power restored from Britain to Egypt. In 305, in conjunction with Maximian, he resigned the Imperial dignity at Nicomedia, and retired to Salona, in Dalmatia, where he cultivated his garden in tranquillity until his death in 313. In the latter part of his reign he was induced to sanction a persecution of the Christians.—BIBLIOGRAPHY: Gibbon, *Decline and Fall*; P. Allard, *La Persécution de Dioclétien*; A. J. Mason, *The Persecution of Diocletian*.

Dioda'ti, Giovanni, Italian Protestant divine, born at Lucca, about 1576, of a noble Catholic family. He was for some time professor, first of Hebrew, then of theology, at Geneva, and in 1619 represented the Genevan clergy at the Synod of Dort, and aided in drawing up the Belgic confession of faith. He is most celebrated for a translation of the Bible into Italian (1607), which is superior to his translation of it into French. He died at Geneva in 1649.

Diodo'rus of Agyrium, in Sicily, and therefore called **Siculus**; a Greek historian in the time of Julius Cæsar and Augustus. His universal history, in the composition of which he travelled through a great part of Europe and Asia, occupied him thirty years, and consisted of 40 books, but only books 1-5 and 11-20, with certain fragments, are now extant.

Diœcious (Gr. *di*, double, *oikos*, a house), in botany, a term applied to plants which have flowers with stamens on one individual and those with pistils

on another; as opposed to *monœcious*. The willow, the yew, the poplar, &c., are diœcious.

Diogenes Laërtius, author of a sort of history of philosophy in Greek, appears to have been born at Laerte, in Cilicia, and to have lived towards the close of the second century after Christ; but no certain information exists either as to his life, studies, or age. The work is divided into ten books, and bears in MSS. the title, *On the Lives, Doctrines, and Apothegms of those who have distinguished themselves in Philosophy*. It is full of absurd and improbable anecdotes, but contains valuable information regarding the private life of the Greeks, and many fragments of works now lost. It was the foundation of the earlier modern histories of philosophy. A translation of his work by C. D. Yonge was published in Bohn's Classical Library.

Diogenes (di-øj'ē-nēz) **of Apollonia** (Crete), known also as the *Physicist*, a Greek philosopher of the fifth century B.C., who belonged to the Ionian school, and considered air as the element of all things. He was a pupil of Anaximenes and a contemporary of Anaxagoras.

Diogenes of Sinopē (on the Black Sea), the most famous of the Cynic philosophers, born about 412 B.C., died about 323 B.C. Having been banished from his native place with his father, who had been accused of coining false money, he went to Athens, and thrust himself upon Antisthenes as a disciple. Like Antisthenes he despised all philosophical speculations, and opposed the corrupt morals of his time; but while the stern austerity of Antisthenes was repulsive, Diogenes exposed the follies of his contemporaries with wit and good humour. As an exemplar of Cynic virtue he satisfied his appetite with the coarsest food, practised the most rigid temperance, walked through the streets of Athens barefoot, without any coat, with a long beard, a stick in his hand, and a wallet on his shoulders, and by night, according to the popular story, slept in a tub (or large earthenware vessel). On a voyage to the Island of Ægina he fell into the hands of pirates, who sold him as a slave to the Corinthian Xenocrates. The latter emancipated him, and entrusted him with the education of his children. He attended to the duties of his new employment with the greatest care, commonly living in summer at Corinth and in winter at Athens. It is at Corinth that he is said to have had his famous interview with Alexander the

Great. The Macedonian conqueror was so struck with the philosopher's self-possession that he went away remarking: "If I were not Alexander, I should wish to be Diogenes". Of the many stories related of him the majority are probably fictions; many, indeed, are chronologically impossible. Concerned with practical wisdom, Diogenes established no system of philosophy. To gain virtue, he maintained, man must avoid physical pleasure, despise the conventions of society, and adopt a simple and natural life. His enemies accused him of various scandalous offences, but there is no ground for supposing him guilty of any worse fault than that of elevating impertinence to the rank of a fine art. See *Cynics*.

Diomedes (dī-o-mē'dēz), in Greek mythology, (1) the son of Mars and Cyrene, and King of the Bistōnes in Thrace, who fed his horses on human flesh, and used to throw all strangers who entered his territories to those animals to be devoured. He was killed by Hercules, who carried off the horses. (2) One of the heroes at the siege of Troy, the son of Tydeus and Deipyle, and King of Argos, one of the suitors of Helen. After she was carried off, Diomedes engaged in the expedition against Troy, in which his courage and the protection of Pallas rendered him one of the most distinguished heroes. He wounded Aphrodite and Ares, and thrice assailed Apollo; and by carrying off the horses of Rhœsus from the enemies' tents, and aiding Ulysses in the removal of Philoctetes from Lemnos, he fulfilled two of the conditions on which alone Troy could be conquered. Finally he was one of the heroes concealed in the wooden horse by which the capture of Troy was at length accomplished. Different accounts were given of his after-life. He is often called *Diomede*.

Dionæ'a, a genus of plants, nat. ord. Droseraceæ. Only one species is known, *D. muscipūla* (Venus's fly-trap), a native of the sandy savannas of Carolina and Florida. It has a rosette of root-leaves, from which rises a naked scape bearing a corymb of largish white flowers. The leaves have a dilated petiole and a slightly stalked 2-lobed lamina, with three short stiff bristles on each lobe. The bristles are remarkably irritable, and when touched by a fly, or other insect, the lobes of the leaf suddenly close on and capture the insect. It dissolves the food thus captured by means of digestive fluid similar to ordinary gastric juice.

Dion Cassius, or **Dio Cassius**, a Greek historian, born about A.D. 155 at Nicæa, in Bithynia. After accompanying his father to Cilicia, of which he held the administration, he came to Rome about 180, and obtained the rank of a Roman Senator. On the accession of Pertinax Dion was appointed Prætor, and in the reign of Caracalla he was one of the Senators whom it had become customary to select to accompany the emperor in his expeditions, of which he complains bitterly. In 219 he was raised to the consulship, and about 224 became Proconsul of Africa. In 229 he was again appointed Consul; but feeling his life precarious under Alexander Severus, he obtained permission to retire to his native town of Nicæa. The period of his death is unknown. The most important of his writings, though only a small part is extant, is a *History of Rome*, written in Greek and divided into eighty books, from the arrival of Æneas in Italy and the foundation of Alba and Rome to A.D. 229.

Dion Chrysostom, a Greek sophist and rhetorician and a favourite of Trajan; born A.D. 50, died about A.D. 110. Eighty of his orations (in excellent Attic) have been preserved.

Dion of Syracuse, in Greek history, a connection by marriage of the elder and the younger Dionysius, tyrants of Syracuse, over whom he long exercised great influence. He attempted to reform the younger Dionysius, but his enemies succeeded in effecting his banishment. He afterwards returned and made himself ruler of the city, but became unpopular, and in 353 B.C. one of his followers, Callipus of Athens, caused him to be assassinated.

Dionys'ia. See *Bacchanalia*.

Dionys'ius, St., a disciple of Origen, and Patriarch of Alexandria in A.D. 248. He was driven from the city in 250, and in 257 was banished to Libya, but was restored in 260. He died in A.D. 265.

Dionysius of Halicarnassus, in Caria, a Greek critic and teacher of eloquence, born about 70 B.C. He went to Rome about 30 B.C., where he wrote his *Roman Antiquities*, in twenty books, in which he relates (in Greek) the early history of Rome and its government up to the times of the first Punic War. We have the first nine books of this work entire, the tenth and eleventh nearly so, and some fragments of the others. His rhetorical

writings are of greater value, especially his essays on the Greek orators. He died about 6 B.C.—Cf. Sir J. E. Sandys, *A History of Classical Scholarship*.

Dionysius the Areopagite, that is, one of the judges of the Areopagus, at Athens, a convert to Christianity by the Apostle Paul about the middle of the first century, and the first Bishop of Athens, where he suffered martyrdom. Certain writings formerly ascribed to him consist of obscurely written treatises on mystical subjects. Scotus Erigena translated them into Latin. In France, where a certain Dionysius (see *Denis, St.*) established the first Christian community at Paris in the third century, they were readily received, this Dionysius being without further inquiry taken for the Areopagite, because the origin of the Gallican Church could thus be carried back to the first century; and France gained a patron who was a martyr and the immediate disciple of an apostle.—Cf. article in Smith's *Dictionary of Christian Biography*.

Dionysius the Elder, in Greek history, tyrant or absolute ruler of Syracuse, born about 430 B.C. of obscure parentage. He obtained the rank of general, and afterwards of commander-in-chief; and, gaining the support of the army, he seized the supreme power in Syracuse, though only twenty-five years of age. He extended his rule over other cities in Sicily; and after some successes and reverses in the struggle with the Carthaginians, he gained a complete victory over them under the walls of Syracuse. In his expeditions into Lower Italy he reduced the city of Rhegium by famine (387). After another short war with Carthage he lived some time in peace, occupied with writing poems and tragedies, with which he contended for the Olympian prize. In 368 he commenced a new war against the Carthaginians, but failed to drive them entirely out of Sicily. He is said to have died from a potion administered at the instigation of his son Dionysius the Younger (367 B.C.).

Dionysius the Little (so called on account of his short stature), a Scythian monk who was abbot of a monastery at Rome in the beginning of the sixth century, and died about the year A.D. 530, according to others about 545, celebrated as the first to introduce the computation of time from the Christian era. This mode of computation, however, was not publicly used until the eighth century.

Dionysius the Younger, a tyrant of Syracuse, who succeeded his father, Dionysius the Elder, in 367 B.C. For the purpose of recalling him from the excesses to which he was addicted, his kinsman Dion persuaded him to invite Plato to his court, but the influence of the philosopher effected no permanent change. Becoming suspicious of Dion, the tyrant banished him and confiscated his property, but in 357 B.C. Dion made himself master of Syracuse. Dionysius fled to Locri, but after the murder of Dion recovered his power in Syracuse. His misfortunes, however, had rendered him more cruel, and Timoleon, who came to Syracuse with aid from Corinth against the Carthaginians, deposed him in 344 B.C. He was carried to Corinth, where he is said to have gained a living by giving lessons in grammar, or as one of the attendants on the rites of Cybele.

Diony'sus, the original Greek name of the god of wine, the name Bacchus, by which he was also called both by the Greeks and the Romans, being at first a mere epithet or surname.—Cf. R. Brown, *The Great Dionysiak Myth*.

Dioon, a genus of Cycads, natives of Mexico, where meal is prepared from the starchy seeds.

Diophan'tus of Alexandria, the first Greek writer on algebra, flourished, according to some authorities, about the middle of the fourth century after Christ. He is called the Father of Algebra, and left behind him thirteen books of *Arithmetical Questions*, of which only six are extant; and a work on *Polygonal Numbers*.

Diop'side, a calcium magnesium silicate, of the pyroxene series, occurring in igneous rocks and altered limestones, with a vitreous lustre, and of a pale-green, or a greenish- or yellowish-white colour.

Diop'sis, a genus of two-winged flies of which the species are native to India and Africa. Each side of the head is drawn out into a long lateral horn, which bears the eye and antenna at its extremity.

Di'optase, emerald copper ore, hydrated silicate of copper, a translucent mineral, occurring crystallized in six-sided prisms. It occurs in Siberia, Hungary, and Chile. It has been used as a gem-stone, notably in Persia and Russia.

Diop̄ter, the unit in terms of which the power of a lens or curved mirror can be expressed. It is obtained by taking the reciprocal of the focal length of the lens or mirror in metres. Thus, a lens with focal length 1 metre has a power of 1 diop̄ter. If the focal length is 2 metres, the power is $\frac{1}{2}$ diop̄ter. The power may also be obtained by dividing 39.37 by the focal length in inches. The unit is employed in classifying spectacle lenses.

Di'orite, a coarsely crystalline igneous rock, sometimes of a whitish colour speckled with black or greenish-black, sometimes very dark in colour, consisting of hornblende and calcium sodium felspar. Dark mica sometimes takes the place of hornblende (mica-diorite). The 'greenstones' of older authors are mostly diorites.

Dioscoreaceæ, a nat. ord. of monocotyledons, with alternate reticulate-veined leaves, tuberous root-stocks, and twining stems. The flowers are small and unisexual. There are 6 genera, with about 100 species. The typical genus is Dioscorea, which includes the yam. Black bryony is the only British representative.

Dioscor'ides, Pedanius, a Greek physician, born in Cilicia in the first century of the Christian era. He was the author of a celebrated work on *materia medica*, in five books, particularly valuable in regard to botany.

Dioscu'ri. See *Castor and Pollux*.

Dios'pyros, a large genus of trees or shrubs, natives of the warmer regions of the world, nat. ord. Ebenaceæ. The trees of this genus supply ebony wood. That from Ceylon is the wood of *D. Eběnum*; from India, of *D. melanoxylon* and other species; and that from Mauritius, *D. tessellaria*. The Chinese date-plum (*D. kaki*) is an apple-like tree which produces large red fruits resembling tomatoes. In China and Japan this tree is as important as the apple is in Northern Europe.

Dip, of the horizon, the angle of depression of the visible horizon at sea below the true horizontal direction, due to the height of the eye above the level of the sea. The dip in minutes of arc is approximately equal to the square root of the height in feet.—*Dip, magnetic*, or *Inclination*, is the angle which a magnetic needle free to move in a vertical circle in the magnetic meridian makes with the horizon. See *Dipping Needle*.

Dip, in geology, the inclination or angle at which strata slope or *dip* downwards into the earth. The degree of inclination or amount of the dip, which is easily measured by a *clinometer*, is the steepest angle made with a horizontal plane by a line drawn in the surface of the stratum. The line of dip is hence perpendicular to the intersection of the stratum with the horizontal, which is called the *strike*.

Dip Circle. See *Dipping Needle*.

Diphthe´ria is an acute infectious disease characterized by the formation of membrane in the throat and air-passages, and associated with severe disturbances affecting especially the heart and nervous system. It is due to a bacillus described by Koch in 1883. It is essentially a disease of the early years of life, and the period between two and twelve years covers the vast majority of cases. The commonest modes of infection are direct and indirect contact, infected milk, and defective drains. Of late years, the domestic cat has been held to be a source of infection. The disease runs a rapid course, beginning with fever, headache, chilliness, lassitude, and occasionally vomiting, while usually there is early complaint of sore throat. The membrane, which appears on the side of the throat, is usually of a dirty yellowish-white colour. It may be limited to a small area, but usually, if untreated, it would spread extensively over the throat, involving the palate and uvula. Diphtheria beginning in the larynx (windpipe) is what is popularly called croup (q.v.). There is danger of death in severe cases during the first few days from early heart failure, and almost any time during convalescence late heart failure may occur. Late heart failure is one of the forms of post-diphtheritic paralysis which arise from disturbances of the nervous system. The other common varieties are paralysis of the palate, of the pharynx, of the eye muscles, and of the respiratory muscles. An effective treatment is found in diphtheric antitoxin, which should be administered as early as possible in the disease. It is given under the skin, and the dose is regulated by the severity of the attack.—BIBLIOGRAPHY: W. F. Litchfield, *Diphtheria in Practice*; W. R. Smith, *Harben Lectures*.

Diphthong (Gr. *di-*, double, and *phthongos*, sound), a coalition or union of two vowels pronounced in one syllable. In uttering a proper diphthong both vowels are pronounced; the sound is not simple, but the two sounds are so blended as to be considered as forming one syllable, as in *void*, *bough*. The

term improper diphthong is applied to the union in one syllable of two or more vowels of which only one is sounded, as in *bean*.

Di'phyodont, a term applied to those animals which develop two sets of teeth, a deciduous or milk set, and a permanent set—as distinct from the monophyodonts, which develop only one set. The majority of mammals are diphyodont, though the number of teeth replaced may vary: thus in man twenty teeth of the adult are preceded by a milk set.

Diplacan'thus, a genus of ganoid fishes, found only in the Old Red Sandstone. They have small scales, a heterocercal tail, and two dorsal fins with a strong spine in front.

Diplei'doscope, an instrument for indicating the passage of the sun or a star over the meridian, by the coincidence of two images of the object, the one formed by single and the other by double reflection. It consists of an equilateral hollow prism, two of whose sides are silvered on the inside so as to be mirrors, while the third is formed of glass. The prism is adjusted so that one of the silvered sides shall be exactly in the plane of the meridian, and the transparent side towards the object.

Diploid Phase, in botany. See *Generations, Alternation of*.

Diplo'ma (Gr. *diplōma*, from *diploō*, to double or fold), literally a document folded but once, and therefore divided into two parts. It is used to signify a document signed and sealed, in which certain rights, privileges, or dignities are conferred, especially a university degree.

Diplo'macy, the science or art of foreign politics. In a more restricted sense the term denotes the science or art of conducting negotiations and arranging treaties between states and nations; the branch of knowledge which deals with the relations of independent states to one another; the agency or management of envoys accredited to a foreign court; the forms of international negotiations. The word, borrowed from the French, was first used in England in 1796 by Burke. The Cardinal de Richelieu is generally considered as the founder of that regular and uninterrupted intercourse between Governments which exists at present between almost all the Christian powers; though the instructions given by Machiavelli to one of his friends, who was sent by the Florentine Republic to Charles V (Charles I of

Spain) show that Richelieu was not the first to conceive the advantages that might be derived from the correspondence of an intelligent agent accredited at the seat of a foreign Government. As a uniform system, however, with a fixed international status, diplomacy was only established in the nineteenth century at the Congresses of Vienna (1815) and Aix-la-Chapelle (1818). Amongst the European powers it is agreed that of ministers of the same rank he who arrives first shall have the precedence over his colleagues.—
BIBLIOGRAPHY: D. J. Hill, *History of Diplomacy in the International Development of Europe*; E. C. Grenville-Murray, *Embassies and Foreign Courts: a History of Diplomacy*; P. Pradier-Fodéré, *Cours de droit diplomatique*; L. Oppenheim, *International Law*; D. P. Heatley, *Diplomacy and the Study of International Relations*.

Diplomatics, originally the science of deciphering ancient MSS. It laid down certain principles for the systematic examination of public documents, and taught the forms and styles adopted in them, and the titles and rank of public officers subscribing them. Among the earliest exponents of diplomatics were Daniel van Papenbroeck, an Antwerp Jesuit (1675), and Mabillon (*De Re Diplomatica*, 1681).

Diplomatic Service, The, as now existing, may be said to have originated in the Venetian Republic, which employed ambassadors as early as the thirteenth century. At first these officials had a very brief term of office, rarely remaining at their post in a foreign country for more than two or three months. By the middle of the fifteenth century, however, a permanent Milanese embassy had been established at Genoa, followed by one at Paris in 1494; while two years later Venice was officially represented at London. The clergy, who for the most part alone possessed the requisite accomplishments for such work, were the usual ambassadors of the Middle Ages; but by the sixteenth century lawyers, or not seldom merchants, were employed. It was not till two hundred years later that the modern attachés, junior officials of an embassy, came into being. The diplomatic service of Great Britain, controlled by the Foreign Office, includes (1) ambassadors, and (2) envoys and ministers plenipotentiary, both of which ranks represent the person of their sovereign and enjoy numerous special privileges in the country to which they are sent. Of lower standing are (3) ministers resident and (4) chargés d'affaires; the last-named are accredited, not to a sovereign, but to his foreign minister, and frequently act merely as temporary

substitutes for an ambassador. Secretaries of more than one grade, with naval, military, and, of late years, commercial attachés, also form members of an embassy. Candidates for the British diplomatic service require a nomination from the Foreign Secretary, must be between the ages of nineteen and twenty-five, are subjected to an examination, and are almost invariably young men of good birth and position. The service is distinguished and affords a pleasant, if to some extent an idle life; but it does not offer any prospect of financial fortune. Lord Stratford de Redcliffe (the 'Great Elchi') and Lord Lyons rank high among distinguished British ambassadors. See *Civil Service*.

Diplozo'on, a parasitic trematode worm which infests the gills of the bream, and which appears to be formed of two distinct bodies united in the middle, and resembling an **X** or St. Andrew's cross, two sexually mature individuals being thus united.

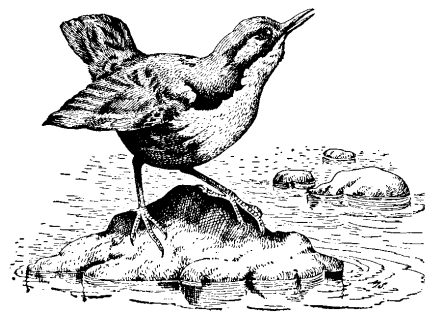
Dip'noi, mud-fishes or lung-fishes, an ancient order now represented by three genera—Neoceratodus, Protopterus, and Lepidosiren. Like some adult Amphibia they possess both gills and lungs, the latter corresponding to a specialized swim-bladder. The heart has two auricles instead of one only, as in all other fishes. The single species of Neoceratodus (*N. forsteri*) is a large form with paddle-like fins and large overlapping cycloid scales. It is native to the Burnett and Mary Rivers of Queensland. Protopterus is represented by three African species, which inhabit rivers and swamps from the Senegal to the Zambezi. It is smaller than Neoceratodus, somewhat eel-shaped, with very narrow fins, and small cycloid scales embedded in the skin. It spends the summer in a torpid condition, buried in the mud, and is dug up by the natives as an article of food. Lepidosiren includes a single South American species (*L. paradoxa*) ranging from the Amazon to Paraguay. It is closely related to Protopterus, which it resembles in shape and the character of the fins.

Dippel, Johann Conrad, German theologian and alchemist, born 1672, died 1734. He studied theology, defended the orthodox party against the Pietists, led a turbulent life at Strasbourg, and then joined the Pietists until an unfortunate tractate placed him in disfavour with both parties. He then turned his attention to alchemy, and during a residence at Berlin produced the oil called after him, from which indirectly followed the discovery of

Prussian or Berlin blue. After various adventures and wanderings in Sweden, Denmark, and Germany, he died at Berleburg.

Dippel's Oil, a pharmaceutical preparation obtained by the destructive distillation of animal matter, such as horn, ivory, blood. The crude form was refined by Dippel, and at one time was a good deal used in medicine as a diaphoretic and hypnotic. It is a form of bone oil, a product obtained in the manufacture of bone-black, or animal charcoal, by the distillation of bones. Crude bone oil has a most offensive smell. It contains the carbonate and other salts of ammonium, and a large variety of organic substances.

Dipper, a bird of the genus *Cinclus*, allied to the wrens. The common dipper, water-ouzel, or water-crow (*Cinclus aquaticus*), is a familiar European bird; it is about 7 inches in length, with a very short tail, small rounded wings, and large powerful feet; the bill is of moderate length, straight, and slender. The male has the upper part of the body dark brown, the throat and breast white, belly rusty. The dipper frequents streams, and feeds largely on water-insects and larvæ. It can dive and walk under water, effecting its progress by grasping the stones with its feet. The song is sweet and lively. Other species are found in North Asia, America, and North Africa.



Dipper (*Cinclus aquaticus*)

Dipping-needle, or **Inclination Compass**, an instrument for showing the direction of the earth's magnetic force. In essentials the instrument consists of a light magnetized steel bar supported on a horizontal axis which passes, as nearly as possible, through the centre of inertia of the bar. When a needle thus mounted is placed anywhere not in the magnetic equator, it dips or points downward; and if the vertical plane, in which it moves, coincides with the magnetic meridian the position of the needle shows at once the direction of the magnetic force. The angle between the magnetic axis of the dipping-needle and the horizontal is called the dip or inclination. This varies from 90° at the magnetic poles to 0° at the magnetic equator. The dip is 70° at Glasgow, and varies slowly with the passage of time. In the northern

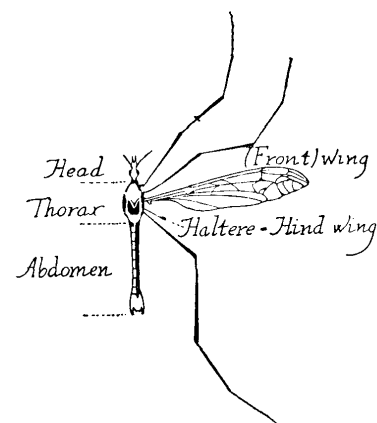
hemisphere, the north-seeking pole of the dipping-needle dips downwards, the reverse being the case south of the magnetic equator.

Dipro'todon, a gigantic fossil marsupial from Pleistocene beds in Australia, allied to the kangaroos. The skull is 3 feet long.

Dipsas, a genus of tree-snakes of which the species are native to tropical South America. Some of the hinder teeth in the upper jaw are grooved for the conduction of poison, but this is not sufficiently deadly to be dangerous to human beings. The related genus *Dipsadomorphus* includes a number of Indian species.

Dipsoma'nia (Gr. *dipsa*, thirst, and *mania*, madness), a term used to denote an insane craving for intoxicating liquors, when occurring in a confirmed or habitual form. It is a form of acute alcoholism seen in persons with a strong hereditary tendency to drink. The only remedy appears to be seclusion, with enforced abstinence and healthy occupation. Homes for this purpose have been established in Britain under the Habitual Drunkards Act of 1879 and Inebriates Act of 1888. There are corresponding institutions in the United States.

Dip'tera, two-winged flies, an order of insects embracing a vast number of species, of which about 40,000 have been named. The two transparent wings correspond to the fore-wings of other insects, the hind-wings being often represented by small club-shaped structures (halteres or balancers). See diagram, p. 42. There are two large compound eyes, and the mouth-parts are often modified for piercing and sucking. There is a well-marked metamorphosis, the larvæ being usually limbless maggots. The Diptera include many agricultural and horticultural pests, and a number are notorious as disease carriers. See *Blow-fly*; *Bot-fly*; *Crane-fly*; *Gnat*; *House-fly*; *Hover-fly*; *Midge*; *Mosquito*; *Tsetse-fly*.



Diptera—Diagram of a two-winged insect—a Daddy-long-legs. Appendages of left side omitted

Dipteris, a genus of Leptosporangiate ferns, section Mixtæ, formerly included in Polypodium, but now recognized as the sole living genus of the Dipteridineæ, a family which was largely developed in Mesozoic times. They are Indo-Malayan ferns with creeping rhizomes and long-stalked, fan-shaped, forked, leathery fronds.

Dipterocarpaceæ, an important order of Asiatic dicotyledonous trees, allied to the mallows (Malvaceæ). The different species produce a number of resinous, oily, and other substances; one, a sort of camphor; another, a fragrant resin used in temples; and others, varnishes; while some of the commonest produce pitches, and sal (*Shorea robusta*) yields valuable timber.

Diptych (dip'tik), in Greek originally signified the same as *diploma*, something folded; the double tablets of metal, ivory, &c., used by the Greeks and Romans. Diptychs became important in the Christian Church, in them being written the names of Popes, and other distinguished persons, who had deserved well of the Church, to be mentioned in the church prayers. Diptychs also often contained pictures of biblical scenes.—Cf. Sir W. Smith, *A Dictionary of Greek and Roman Antiquities*.

Dipyre (dī'pīr), or **Mizzonite**, a mineral—aluminium calcium sodium silicate—of the scapolite series. Its name indicates the double effect of fire upon it (Gr. *di*, double, *pyr*, fire) in producing first phosphorescence, and then fusion.

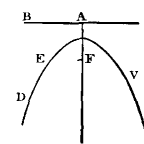
Diræ, one of the names under which the Eumenides were known to the Romans. See *Furies*.

Directors, persons elected to meet together at short fixed intervals and consult about the affairs of corporations or joint-stock companies, and to advise and assist the manager. These are termed *Ordinary Directors*, as in many companies there is a body called *Extraordinary Directors*, who have little or no business functions, and are chosen as a rule on account of their social position imparting a degree of distinction to the concern. Directors are appointed by a general meeting of the shareholders in the undertaking, and a certain number of them, usually a third, retire every year. Ordinary directors are granted a certain remuneration for their services. The duties and responsibilities of directors are defined by the constitution of the

company, or by the various Acts of Parliament affecting joint-stock and other companies.

Directory, the name given to a body of five officers to whom the executive authority in France was committed by the Constitution of the year III (1795). The two legislative bodies, called the *councils*, elected the members of the Directory: one member was obliged to retire yearly, and his place was supplied by election. This body was invested with the authority which, by the Constitution of 1791, had been granted to the king. By the Revolution of the 18th Brumaire the Directory and the Constitution of the year III were abolished. It was succeeded by the Consulate, with Napoleon as First Consul.

Directrix, a fixed line that is required for the description of a curve. The term is chiefly used in connection with the parabola, ellipse, and hyperbola, which are the loci of points that move so that their distances from a fixed point (the focus) are in a constant ratio to their distances from the directrix. The directrix of a parabola is a line perpendicular to the axis produced, and at a distance from the vertex equal to the distance of the vertex from the focus. Thus AB is the directrix of the parabola VED, of which F is the focus.



Directrix of a Parabola

Dirk, a kind of dagger formerly used as a weapon of offence by the Highlanders of Scotland. Dirks are worn by midshipmen and cadets of the Royal Navy, and still form part of the full Highland costume.

Dirk-Hartog Island, on the west coast of Australia, 45 miles long, north to south, and 10 miles broad.

Dirt-beds, in geology, layers of ancient soil, such as those in the Oolitic strata of the Isle of Purbeck (Dorset), which contain the stumps of trees that once grew in them.

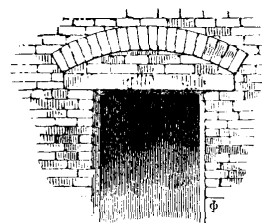
Disability, in law, incapacity to do any legal act. It is either *absolute*, which wholly disables the person, such as outlawry or excommunication—or *partial*, such as infancy, coverture, insanity, or drunkenness.

Disbarring, or **Disbarments**, expelling a barrister from the Bar, a prerogative which, in England, is possessed by the benchers of each of the

four Inns of Court. The party disbarred may lodge an appeal with the judges in their capacity of visitors.

Disc, or **Disk**, the central part of the capitulum of *compositæ*, surrounded by the ray. Also a part of a flower, sometimes cup-shaped, at the base of the stamens, consisting in some cases of rudimentary stamens, in others of the modified receptacle.—In astronomy the term is applied to the visible face or figure exhibited by the sun, moon, or a planet. In the case of the moon and certain planets it may be of gibbous, semicircular, or crescent form.

Discharging Arch, an arch formed in the substance of a wall to relieve the part which is below it from the superincumbent weight. Such arches are commonly used over lintels and flat-headed openings.



Discharging Arch

Discipline, Books of, two books connected with the Church of Scotland. The *First Book of Discipline* was drawn up by John Knox and four other ministers, and laid before the General Assembly in 1560. Though not formally ratified by the Privy Council, it was secretly subscribed by the greater part of the nobility and barons who were members of the Council. Another similar document, the *Second Book of Discipline*, was prepared and sanctioned by the General Assembly of 1578, and has from that time been recognized as the authorized standard of the Church of Scotland in respect of government and discipline.

Disclaim'er, in its stricter legal sense, a plea containing renunciation or a denial of some claim alleged to have been made by the party pleading.

Discomycetes, a large section of the ascomycetous Fungi, distinguished by the fact that the hymenium covers the surface of an open, disc-like or cup-shaped fruit-body called an *apothecium*. It includes many important genera, such as *Dasyscypha*, *Peziza*, and *Sclerotinia*.

Discopho'ra, (1) a sub-class of the Hydrozoa, comprising most of the organisms known as sea-jellies, jelly-fishes, and sea-nettles; (2) leeches (q.v.).

Dis'count, the charge made by a banker for interest of money advanced by him on a bill or other document not presently due. In advancing money on

such a security the banker deducts the charge for interest on his advance from the total amount represented on the security, pays the difference, which is called the *proceeds* of the bill, to the person parting with it, and collects the full amount to reimburse himself for outlay and interest at maturity. Popularly the term *discount* is applied to any deduction from the full amount of an account made by the party to whom it is paid, especially on prompt or early payment. When a bill which has been *discounted* is paid by the acceptor before it is due, the discount allowed for prepayment is called *rebate*.

Discov'ery, in law, the act of revealing or making known any matter by a defendant in his answer to a bill in chancery. The word is also used in reference to the disclosure by a bankrupt of his property for the benefit of his creditors, and to the right of a party to a lawsuit to obtain from his adversary, on oath, full disclosure of the facts within his knowledge, and production of the documents in his possession, pertinent to the action.

Discus, Disc, or Disk, among the Greeks and Romans a quoit of stone or metal, convex on both its sides, sometimes perforated in the middle. The players aimed at no mark, but simply tried to throw the quoit to the greatest possible distance. It was sometimes furnished with a thong of leather to assist in the throwing. The thrower of the discus was called *discobolus*.

Disease, any morbid state of the body, or of any organ or part of the body. Diseases are described as local or constitutional, epidemic, endemic, contagious, acute, and chronic. As to their classification, see *Nosology*. The influence of the parents on the organization of the child is so great that not only peculiarities of external form, but the peculiar constitution, the greater or less activity and development of the organs, are found to pass from parent to child. As it is in the particular state of the several tissues and functions that certain diseases have their foundation, the liability to such affections is inherited with the organic structure, and children are not infrequently attacked by ailments from which one of their parents or grandparents had previously suffered. In spite of the emphasis that has recently been put on the facts of heredity, the present tendency of preventive medicine holds out a larger hope by impressing upon us the fact that in such affections, for example, as tuberculosis and insanity, the individual's

environment and personal history are perhaps more potent than hereditary influence in the causation or the prevention of disease.

Diseases of Plants. See *Plant Pathology*.

Disestablishment, the severance of connection between Church and State, with the resultant emancipation of the Church from civil control, is a movement in which there has been considerable growth during the last half-century. The Church of the West Indies was disestablished in 1868, and all Colonial Churches, with the exception of the Church in India, are now free from State authority. In 1869 an Act, taking effect two years later, was passed for the disestablishment of the Church of Ireland, in which country the mass of the inhabitants are Roman Catholics. In France all recognition of Roman Catholicism as the State religion ceased in 1905, the French Government now tolerating all religions. In 1914 was passed a Bill for disestablishing the Church in Wales, though its effect was, owing to the war, deferred till 1920 by the Suspensory Bill. The Church of Scotland has been practically free from State control since 1874, in which year patronage was abolished; and the presence of the Lord High Commissioner, who represents the sovereign at the opening of the General Assembly, and the precedence enjoyed by the Moderator during his term of office, are little more than decorative courtesies. Among a section of the English public there has been from time to time an agitation for the disestablishment of the Church of England; and the movement is even said to be viewed with favour by some advanced High Churchmen, who believe that it would result in increased freedom for the adoption of their special views; but, though a Disestablishment Society exists, there seems no immediate prospect of a decisive step in this direction.

Dishonour of a Bill, the refusal or neglect to accept or pay when due a bill of exchange, or promissory note, or draft on a banker. It is absolutely necessary that the holder of a dishonoured bill should give immediate notice of the non-payment to the drawer or endorsers.

Disinfection, the means employed for killing the germs of infectious or contagious disease by physical or chemical agencies. The former are the more important, and consist in applying water or steam at the boiling-point, or hot air at 160°C. Ten minutes' boiling, or half an hour in hot air, kills all

ordinary disease germs, but a longer exposure is necessary to kill germs (especially those of putrefaction), which form spores (see *Bacteria*). The most important chemical agents are chlorine, iodine, carbolic acid, bleaching powder, Condy's red fluid (containing permanganate of potash), perchloride of mercury, formalin, and flavine. Carbolic acid is one of the most effective, needing, however, care in the handling, as it is very poisonous and in strong solution causes severe burns. It does not in its common form mix with water, but solutions can be made by using hot water. A greater dilution than 1 part in 40 of water is useless as a disinfectant. For application to the skin, tincture of iodine is one of the readiest preparations. In cases of infectious disease the most important points are the immediate disinfection of all the excretions of the patient. Expectoration should be received into a sputum-cup containing 1-20 carbolic acid, and all handkerchiefs when soiled should be similarly treated. The personal linen and sheets of the patient should be placed in carbolic acid (1-40) in a slop-pail, and should be boiled before being sent to the laundry. All plates, spoons, &c., used for the patient's food, should be boiled or scalded immediately after use. The excretions of the bowels or kidneys should be treated with bleaching powder. Those in attendance should wear an overall when in the sickroom, and should wash the hands and face before coming into contact with anyone outside. They should wash out the mouth frequently with Condy as strong as can be tolerated. Their linen should be treated in the same way as that of the patient. At the close of the illness all bedding should be baked in the hot-air oven which most local authorities now provide for the purpose. Everything washable in the sickroom should be washed with soft soap, and it is better that the room should be repapered.

Disin'tegrator, a machine for pulverizing and sometimes for mixing various materials, such as rock, asphalt, ore, artificial manures, sugars, corn, and the ingredients of mortar.

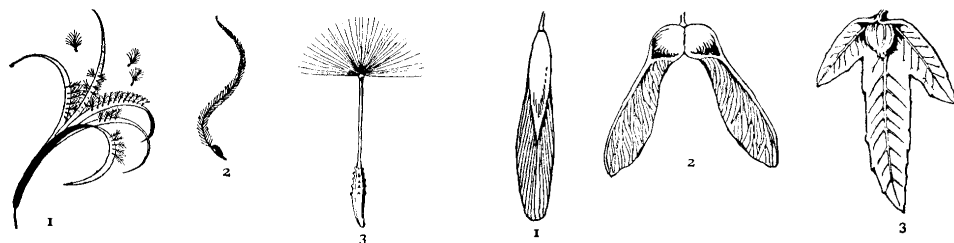
Disloca'tion, a surgical term applied to cases in which the articulating surfaces of the bones have been forced out of their proper places. The particular dislocation takes its name either from the joint itself or its farthest bone, and is called *compound* when accompanied by an external wound. The most common dislocations are those of the hip, shoulder, elbow, knee, and ankle, and the chief obstacle to their reduction is the spasmodic and violent contraction of the muscles consequent upon them, the application of

considerable force being often necessary to set the joint. Chloroform is of great use, not only in preventing pain but in relaxing the muscles. The most dangerous dislocations are those of the bones of the spine.—In geology it signifies the displacement of parts of rocks or portions of strata from the situations they originally occupied.

Dismal Swamp, a large tract of marshy land in America, beginning a little south of Norfolk, in Virginia, and extending into North Carolina, containing 150,000 acres; 30 miles long, from north to south, and 10 miles broad. This tract was entirely covered with trees, with almost impervious brushwood between them, but it has now in part been cleared and drained. In the midst of the swamp is a lake, called *Drummond's Pond*, 7 miles in length. A navigable canal through the swamp connects Chesapeake Bay and Albemarle Sound.

Dispensa'tion is the act by which an exception is made to the rigour of the law in favour of some person. The Pope may release from all oaths or vows, and may sanction a marriage within the prohibited degrees of the Mosaic law, or exempt from obedience to the disciplinary enactments of the canon law. In England the monarch claimed, in former times, a similar dispensing power in civil law, but the prerogative was so much abused by James II that it was abolished by the Bill of Rights. The power of commuting sentences in capital cases is the only form in which the dispensing power of the Crown still exists. In ecclesiastical matters a bishop may grant a dispensation allowing a clergyman to hold more than one benefice, or to absent himself from his parish.—Cf. F. W. Maitland, *Constitutional History of England*.

Dispersal of Seeds and Fruits serves (a) to scatter these reproductive structures and so reduce



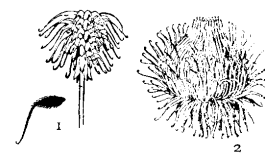
Parachute Fruits

1, Ripe fruit of willow-herb, dehiscing. 2, Single fruit of clematis. 3, Single fruit of dandelion.

Winged Fruit

1, Ash. 2, Sycamore. 3, Hornbeam.

internecine competition; (b) to bring the seeds into new surroundings, which may be more favourable than those of the parent plant. The chief agents of dispersal are wind and animals. Very minute seeds, like those of orchids, are carried away by the gentlest air-currents. Larger wind-borne seeds are *winged*, as in the pine, most Bignoniaceæ, &c.; or provided with a tuft of hairs acting as a *parachute*, as in willow, willow-herb, cotton, &c. Winged *fruits* are exemplified by ash, elm, sycamore, many docks, &c.; parachute *fruits* by Compositæ, Clematis, cotton-grass, &c. In the case of animal dispersal, the whole fruit is usually involved, being either *edible*, with hard indigestible seeds which are cast up or voided with excreta (fleshy fruits), or *hooked* so as to adhere to fur or wool, as in bidens, cleavers, enchanters' nightshade, and other 'burs'. A small number of fruits and seeds, such as the coco-nut and the seeds of water-lilies, are adapted for water transport. In certain cases seeds are scattered for short distances by an 'explosive' mechanism, as in wood-sorrel, impatiens, sand-box tree, squirting cucumber, and other 'sling' fruits.



Hooked Seeds
1, Geum and single fruit. 2, Burdock.

Dispersion, in optics, the angular separation of light rays of different colour, that is, of different wave-length. Dispersion may be caused either by refraction or by diffraction. When a beam of composite light passes obliquely from air into a second transparent medium, each constituent of the light is bent or refracted through a different angle from the original direction of the beam, with the result that the different colours are separated fanwise, or dispersed at the surface of the second medium. In the refraction spectrum of white light, when caused by passage through a glass prism, the red rays are least deviated and the violet rays most deviated, if we consider only the visible spectrum. The difference



Explosive or Sling Fruits
1, *Ecbalium Elaterium*, flowers and fruit, one fruit detached from its stalk and with its seeds squirting out. 2, *Oxalis Acetosella*, entire plant, with one unripe fruit on a hooked stalk, and one ripe fruit on an erect stalk ejecting its seeds. 3, Ripe fruit of *Oxalis Acetosella* ejecting the seeds (enlarged).

of the angles of deviation for two selected rays measures their dispersion, and if this angle is divided by the deviation of the mean ray, we obtain the dispersive power of the prism. Transparent media vary in their dispersive powers; for example, carbon disulphide has more than three times the dispersive power of crown glass. The true nature of dispersion was first demonstrated by Newton, who concluded that the colours of the spectrum were homogeneous and caused by simple vibrations of definite wave-length, the different colours being unequally refrangible. Newton was, however, led to the erroneous view that the dispersion was proportional to the refraction. This was later disproved by the construction of achromatic lenses, or lenses which caused deviation without dispersion, and of direct-vision spectroscopes, or instruments which caused dispersion with no deviation of the central part of the spectrum. The dispersive power is not the same for all parts of a refraction spectrum; besides, the same colours do not occupy the same positions in spectra formed by prisms of different material. This arises from the fact that there is no simple relation between the deviation of a ray and its wave-length; consequently, such spectra are called irrational, and the property is known as the irrationality of dispersion. In the diffraction spectrum, the order of the colours is reversed, red undergoing the greatest deviation; also, the deviation for a given colour is nearly proportional to the wave-length. The diffraction spectrum is therefore termed a normal spectrum.

All substances do not give the same order of colours in their spectra; certain exceptions are known in which the usual order of the colours is changed. Christiansen showed that an alcoholic solution of fuchsine gave a spectrum containing only violet, red, and yellow; the violet is least refracted, and the yellow most, and a dark band lies between the violet and the red. This has been called anomalous dispersion, and similar effects have been observed in iodine and sodium vapours, and in solutions of colours derived from aniline which exhibit surface colour.

The theory of dispersion now generally accepted is that of Sellmeier, which was published in 1871. Sellmeier assumed that when light waves pass through a material substance, they set the particles of the substance in vibration, and these resonant vibrations react in such a way as to modify the velocity with which the waves are transmitted. Applying the dynamical principles of wave motion to the case of an elastic solid in which heavy

particles are embedded, Sellmeier obtained an equation which connected the refractive index of the substance with the wave-length of the incident light. Equations of similar form were subsequently derived by Ketteler and Helmholtz. The consideration of Sellmeier's equation leads to important conclusions. If the period of vibration of the incident waves is very short, as compared with those of the particles forming the solid, no refraction will take place, and the rays will travel through the solid without deviation and without change of velocity. This is verified in the case of X-rays, which consist of extremely short waves and which are not deviated on passing through light-opaque solids. Sellmeier's equation may also be modified to apply to the case of anomalous dispersion. The phenomenon is always associated with absorption of light of a particular wave-length or range of wave-lengths, and the conclusion is drawn that the medium will possess an abnormally high refractive index for waves slightly longer than those which it absorbs, and an abnormally low index for waves slightly shorter than those which it absorbs. This result has been verified by various investigators. Rubens has determined the values of the constants in Sellmeier's equation for rock-salt, sylvine, fluorspar, and quartz, and has shown that the equation gives correct values for the refractive indices of these substances over the entire range of wave-lengths to which they are transparent.—BIBLIOGRAPHY: T. Preston, *Theory of Light*; E. Edser, *Light for Students*; P. Drude, *Theory of Optics*.

Displacement. The position of a point in space is fixed by means of its distances x , y , z , from three mutually rectangular planes. If the point moves to another position, it is said to be displaced, and the rates of displacement parallel to these planes measure the velocities dx/dt , dy/dt , dz/dt parallel to these planes. If the acting forces be resolved in directions parallel to these planes, relations may be found between the forces, the accelerations parallel to the planes and the mass of the body. These relations are called the equations of motion. In hydrostatics a body immersed in a liquid displaces a certain volume of the liquid, and the upthrust of the liquid in the body is, by the principle of Archimedes, equal to the weight of liquid displaced. It follows that, in the case of a floating ship, the weight of the ship is equal to the weight of water displaced. This weight is called the displacement of the ship, and is measured in tons.

Disposition, in Scots law, is, in its general acceptation, a deed by which a person provides for the general disposal of his property heritable and movable, after his death, equivalent to a will or testament; also a conveyance of property.

Disraeli, Benjamin. See *Beaconsfield*.

D'Israeli (diz-rā'e-li), Isaac, man of letters, and father of the well-known statesman, was born at Enfield, Middlesex, in 1766, died in 1848. His father, Benjamin D'Israeli, a descendant of a family of Spanish Jews which had settled at Venice in the fifteenth century to escape the Inquisition, came over to England in 1748 and made a large fortune. Isaac D'Israeli, however, showed a strong repugnance to commerce, and was finally permitted to follow his literary bent. An anonymous reply to Peter Pindar, entitled *On the Abuse of Satire*, was followed during 1791 to 1793 by the appearance of his *Curiosities of Literature*, the success of which determined much of his afterwork. His *Essay on the Literary Character* was published in 1795, and some time afterwards a volume of romantic tales, *The Loves of Mejnoun and Leila*. Between 1812 and 1822 appeared his *Calamities of Authors*, *Quarrels of Authors*, and *Inquiry into the Literary and Political Character of James I*; the three being afterwards published collectively under the title of *Miscellanies of Literature*. In 1828 appeared the commencement of his *Life and Reign of Charles I*, a work completed in 1831. An affection of the eyes put an end to a projected *Life of Pope* and a *History of English Freethinkers*, but in 1841 he published a selection from his MSS. under the title of *Amenities of Literature*. The greater part of his life was passed in his library. For his son, see *Beaconsfield*.

Disrup'tion, the name commonly applied in Scotland to the act by which, in 1843, 474 ministers and professors of the Established Church gave up their livings to vindicate principles which they held to be essential to the purity of the Church, and in harmony with its earlier history. See *Free Church*.

Diss, a town, England, Norfolk, on the slope of a hill 18 miles south by west of Norwich. It was formerly noted for the manufacture of 'Suffolk hempen cloth', worsted yarn, and knit hosiery. Pop. 3763.

Dissection, a word, derived from Latin, that is etymologically equivalent to the word *anatomy*, derived from Greek. Its literal meaning is 'cutting up', and it is used to define the technical procedures for acquiring a practical knowledge of the anatomy or structure of the body. Dissection of the human body is an essential part of the education of a medical practitioner, for it is the only means whereby he can acquire a thorough and practical familiarity with the geography of the territories in which all his professional activities lie. Therefore the medical student is required thoroughly to explore every part of the human body, to examine all its constituent parts, to learn to recognize their properties, positions, and relationships, and to train his eyes and fingers to appreciate their distinctive qualities. This process of exploration usually occupies about eighteen months or more of the student's time; but it represents the foundation upon which all his professional knowledge and experience are built up. For this purpose it is of the utmost importance that he be provided with ample facilities for acquiring the training which is essential to the medical practitioner. But the supply of subjects for dissection is difficult to acquire. Until a century ago teachers in medical schools, being unable to get an adequate supply of bodies for dissection by legal means, were forced to deal with 'body snatchers' who plundered cemeteries. Eventually the terrible scandals associated with the names of Burke and Hare forced the Government to pass an Anatomy Act to make better provision for this necessary part of medical education. Within recent years the action of Boards of Guardians has so hampered the administration of the Act that teachers in medical schools are threatened with the same dilemma as their colleagues a century ago had to face. The Guardians of the Poor in some localities prefer to bury the unclaimed bodies of the dead at the ratepayers' expense rather than allow them to be used for the necessary instruction of surgeons and physicians. Offers are repeatedly made by men and women, often well-known and distinguished people like the late Miss Florence Nightingale, to place their bodies at the service of medical education; but in accordance with the law of the land such bequests are invalid, because once a person is dead the corpse is not his property but belongs to his relatives. Hence it is only the unclaimed bodies that are legally available for dissection.

Dissei'zin, or **Disseisin**, in law, is the dispossessing one of a freehold estate, or interrupting his *seisin*. Of freeholds only can a *seizin* be had, or a

disseizin done. Whether an entry upon lands is or is not a disseizin, will depend partly upon the circumstances of the entry, and partly upon the intention of the party as made known by his words or acts.

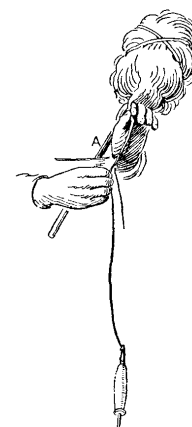
Dissent'ers, the common name by which in Britain all Christian denominations, excepting those of the Established Churches, are usually designated, though in Acts of Parliament it generally includes only Protestant dissenters, Roman Catholics being referred to under their specific name. The most important bodies of English dissenters are the different bodies of Methodists, the Congregationalists, and the Baptists; and of Scottish dissenters, the United Free Church and the Free Church. The Nonconformists were dissenters from the English Church, and the name is sometimes used as meaning simply dissenters, though it has properly a wider meaning.

Dissentis', a Swiss town, canton of Grisons, 3800 feet above the sea, at the junction of the Middle and Vorder Rhine, with a Benedictine abbey established so long ago as A.D. 614. Pop. 1420.

Dissociation. Certain substances tend to break down into simpler substances with change of temperature; thus ammonium chloride on heating gives a mixture of hydrochloric acid (HCl) and ammonia (NH₃), and on cooling these substances recombine to give ammonium chloride (NH₄Cl) again. Dissociation is therefore a particular case of decomposition, where the products of decomposition recombine on obtaining the original conditions.

Dis'sonance, in music, that effect which, results from the union of two sounds not in accord with each other. The ancients considered thirds and sixths as dissonances; and, in fact, every chord except the perfect concord is a dissonant chord. The old theories include an infinity of dissonances, but the present received system reduces them to a comparatively small number. The most common are those of the tonic against the second, the fifth against the sixth, or (the most frequent of all) the fourth against the fifth.

Dis'taff, the first instrument employed in spinning. It consisted of a staff, on one end of which the wool or flax was rolled. The spinner held it in the left hand, and drew out the fibres with the right, at the same time twisting them. A small piece of wood called a spindle was attached to the thread, the weight of which carried it down as it was formed. When the spindle reached the ground, the thread which had been spun was wound round it, and it was then again fastened near the beginning of the new thread. In ancient and modern art the Fates are usually represented with the distaff, engaged in spinning the thread of life.



A, Distaff. It is shown held by the waist-belt of the spinner. The spindle is revolving and twisting the wool.

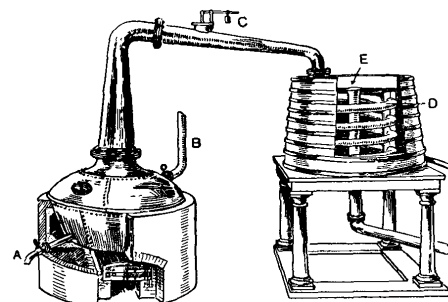
Distemp'er, a disease of the dog commonly considered as of a catarrhal nature. In most cases a running from the nose and eyes is one of the first and chief symptoms, the defluxion becoming after some time mucous and purulent. The animal is subject to violent fits of coughing combined with vomiting, loses its appetite, its flesh begins to waste, and if the disease be virulent, symptoms of affection of the brain manifest themselves, accompanied by fits, paralysis, or convulsive twitchings. In the first stage of the disease laxatives, emetics, and occasional bleeding are the principal remedies; diarrhœa should be checked by astringents, and to reduce the violence of the fits warm bathing and antispasmodics should be resorted to. The distemper is generally contagious, and occurs but once in a lifetime.

Distemper (It. *tempera*), in painting, a preparation of colour mixed with size, yolk of egg or white of egg. Prepared with size, it is used chiefly in scene painting and household decorations, but in other forms it is much used for easel and mural paintings. Before the introduction of oil as a medium in the fifteenth century, fresco and distemper were the principal methods of painting. Distemper is usually but not necessarily applied to a dry ground, fresco always to a wet.

Distich (dis'tik), a couplet of verses, especially one consisting of a Latin or Greek hexameter and pentameter, making complete sense. Distichs have been frequently made use of by the modern German poets.

Distillation, the volatilization and subsequent condensation of a liquid in an apparatus known as a *still* and heated by a fire or flame. The operation is

performed by heating the crude liquid or mixture in a retort or vessel known as the *body of the still*. This is made of various shapes and materials, and is closed, with the exception of a slender neck which opens into the condenser, a long tube through which the hot vapour from the still is passed. The tube is kept at a sufficiently low temperature to cause the vapour to condense, the common method of securing this being to surround the tube with a constantly renewed stream of cold water. In some cases ice or a freezing



Distillation of Alcohol

A, Emptying pipe. B, Wash inlet. C, Vacuum safety valve. D, Worm. E, Cooling water inlet.

mixture may be required to effect condensation. In a large-scale apparatus the condensing tube is coiled round and round in a tub or box, and is known as a *worm*. From the end of it the vapour condensed into a liquid drops into a receiver. The simplest case of distillation is that of water containing solid matter in solution, the solid matter remaining behind in the still or retort while the water trickles pure into the receiver, through a worm made of block-tin, as most other metals are attacked by distilled water. When the mixture to be distilled consists of two or more liquids of different boiling-points, such as alcohol and water, the more volatile comes off first, accompanied by a certain proportion of the vapour of the other, so that it is hardly possible completely to separate bodies by one distillation. This is effected by repeated successive distillations of the liquid with or without the addition of substances to retain the impurities. When the production of one of the ingredients only is aimed at by this process, it is called *rectification*, but when it is desired to separate and collect all the liquids present, or to divide a mixture into portions which volatilize within certain ranges of temperature, the process is called *fractional distillation*. In the laboratory, distillation is employed for purifying water, for recovering alcohol and ether, and for the preparation, purification, and separation of a great number of bodies. Substances which decompose at their boiling-points can be distilled under reduced pressure. On the large scale distillation is employed in the preparation of potassium, sodium, zinc, mercury; of sulphuric acid, ether, chloroform, carbon bisulphide, essential oils and perfumes; in the purification of coal and wood tar, and the products obtained from them; and on an extensive scale in the manufacture of whisky, brandy, or other spirit. The distillation of whisky has long been familiar in Britain, especially in Scotland and Ireland, and, when performed by means of the old *pot-still*, is a simple operation indeed, and one that even yet is practised

surreptitiously in out-of-the-way localities. On the large scale a more elaborate apparatus is employed, and for alcohol of a cheap class Coffey's or other patent still is much used. Copper is the metal that suits best as the material for the stills used in distilling whisky. Sea-water is distilled in many cases for drinking or cooking purposes. This water is, of course, very pure, but its taste is far from agreeable. *Destructive distillation*, or *dry distillation*, differs from the preceding in this respect, that the original substance is not merely broken up into bodies by the mixture of which it is formed, but is so treated that it is further decomposed, and products are obtained which were not present uncombined in the original material. (See *Coal-tar*.) The term is restricted to the action of heat upon complex organic substances out of contact with the air. The products of destructive distillation are numerous and varied. On the manufacturing scale the process is conducted sometimes for the sake of one part of the products, sometimes for the sake of another. Coal, for example, may be distilled not solely for the gas, but also for ammoniacal water, benzene, anthracene, as well as for the sake of the fixed carbon or coke, the volatile portions being too often neglected and practically wasted. But much more economical methods of making coke are now practised than formerly. Wood is distilled partly for the sake of the pyroligneous acid and the tar, partly for the charcoal. Bones are distilled for the sake of the charcoal, though the oil is also collected. Shale is distilled both for the oil and for the paraffin wax, ammonia, &c., obtained.

Distinguished Conduct Medal, a medal instituted in 1854 under the name of the Meritorious Service Medal as "a mark of the Sovereign's sense of the distinguished service and gallant conduct in the field of the army then serving in the Crimea". The regulations concerning this medal were revised in 1862, when it received its present name. It is given to warrant-officers, non-commissioned officers, and privates. As from the 1st Aug., 1918, the medal is awarded for services in action only. The medal is silver, 1.4 inches in diameter; on the obverse is the Sovereign's head, and on the reverse the inscription "For Distinguished Conduct in the Field". It is suspended from a ribbon 1¼ inches wide, crimson, blue, and crimson, the colours being of equal width. Bars may be awarded for additional deeds of gallantry; in service uniform the possession of a bar is indicated by a silver rosette worn on the ribbon. The letters D.C.M. are placed



Distinguished
Conduct
Medal

after the name of the recipient, who receives either a gratuity of £20 on discharge, or an increase in pay of 6*d.* a day.



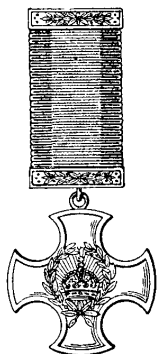
On Left, Distinguished Flying Cross (obverse) On Right, Distinguished Flying Medal (obverse)

Distinguished Flying Cross, a decoration instituted during the European War, and awarded to officers and warrant-officers of the Air Force for acts of gallantry when flying in active operations against the enemy. The ribbon is 1¼ inches wide, and has narrow purple and white alternate diagonal stripes. The letters D.F.C. are placed after the name of the recipient. A corresponding medal, the Distinguished Flying Medal, is awarded to non-commissioned officers and men.

Distinguished Service Cross, a naval decoration formerly known as the Conspicuous Service Cross, and instituted in 1901. It is awarded to naval officers below the rank of lieutenant-commander, and to warrant-officers, for services before the enemy. The ribbon is dark-blue, white, and dark-blue, in stripes of equal width. The letters D.S.C. are placed after the name of the recipient.



Distinguished Service Cross (obverse)



Distinguished Service Order (obverse)

Distinguished Service Order, The, was instituted by Royal Warrant on 6th Sept., 1886. All commissioned officers of the Navy, Army, or Air Force are eligible to be appointed Companions of this Order. It is conferred upon officers who have been specially mentioned in dispatches for meritorious or distinguished services. Towards the end of the European War (1st Aug., 1918) it was decided that this decoration was only to be given for services in action, i.e. (1) for service under fire, or (2) distinguished individual service in connection with air-raids, bombardments, or other enemy action. Bars may be awarded for additional deeds of gallantry; in undress uniform the possession

of a bar is indicated by a silver rosette worn on the ribbon. The ribbon is red, edged with blue, and is 1 inch wide. It is the same as the ribbon of the Waterloo and Peninsular medals, only narrower. The badge of the order is a gold cross *patée* enamelled white, edged gold, having on one side thereof, in

the centre, within a wreath of laurels enamelled green, the Imperial Crown in gold, upon a red enamelled ground, and on the reverse, within a similar wreath and on a similar red ground, the Imperial and Royal cipher (G.R.I.) The letters D.S.O. are placed after the name of the recipient, who ranks between Commanders of the Order of the British Empire and Members of the Royal Victorian Order (4th class).

Dis'tomum, a genus of trematode or suckorial parasitic worms or flukes, infesting various parts in different animals. *D. hepaticum*, the common liver fluke, when adult inhabits the gall-bladder or ducts of the liver in sheep, and is the cause of the disease known as the rot. It may also occur in ox and man. In form it is ovate, flattened, and presents two suckers (whence the name), of which the anterior is perforated by the aperture of the mouth. The digestive tube divides into two branching halves, the excretory organs consist of delicate branching tubes, and there is a complex set of hermaphrodite reproductive organs. The minute larvæ live in a small water-snail (*Limnæa truncatula*), from which they ultimately escape to encyst on grass, &c., by which sheep may become infested. The adults of other species of *Distomum* live within the bodies of various fishes, amphibia, reptiles, birds, and mammals.

Distress' (Lat. *distringere*, pull asunder), in law, is the taking of a personal chattel of a wrong-doer or a tenant, in order to obtain satisfaction for the wrong done, or for rent or service due. If the party whose goods are seized disputes the injury, service, duty, or rent, on account of which the distress is taken, he may replevy the things taken, giving bonds to return them or pay damage in case the party making the distress shows that the wrong has been done, or the service or rent is due. Wrongful distress is actionable. Another kind of distress is that of *attachment*, to compel a party to appear before a court when summoned. The distresses most frequently made are on account of rent and taxes and *damage-feasance*.—Cf. Blackstone, *Commentaries on the Laws of England*.

Distribution is that part of the subject-matter of economics which deals with the division, among the individuals composing a community, of the product of their labour and of land and capital at their command. It is the most contentious part of economics, as well as that which has most social interest and importance, for in it there is a sharper conflict than anywhere else between economic considerations and ideals of social justice, and the problem is to secure such a distribution as will satisfy social justice, without abandoning the incentive to effort provided by the possibility of a greater reward.

The present position is one of unresolved discord between the economic and the equalitarian schools. The economic analysis is based upon a division of the factors of production into three, viz. Labour, Capital, and Land. These are represented as competing among themselves for employment, which is given to each in accordance with the return which will be secured as a result of an addition to the amount already employed; the rewards of each fall roughly under the headings of Wages and Salaries, Interest and Rent, respectively. The demarcation cannot be exact, as much that in common life is regarded as interest on capital partakes really of the nature of wages, as being earnings of management or reward for risk-taking, while certain forms of capital, whose quantity is temporarily fixed, may receive a return which is more of the nature of rent.

The fundamental assumption of the economists is that competitive distribution is indispensable, both in order to secure a fruitful distribution of the resources and efforts of society between the three factors of production, and also owing to the need of an incentive to keep individuals working hard enough to ensure a supply of goods sufficient to maintain and improve the standard of life of the community. Economists do not assume that men would not work at all if they were guaranteed a living wage whether they did or not, but that men would not work enough, and would be guided in the direction of their efforts by personal idiosyncrasy rather than by the needs of the community. The only alternatives to a great falling off in national wealth are, in their view, either competitive distribution or enforcement of work by a rigid discipline, which would be as hard as the present system, while being less adaptable.

Against this analysis idealist thinkers rebel, on the ground that it fails to justify the great inequality in the distribution of wealth between different individuals and classes of society. Their protest dates from the harsh dogmatism of the economists who propounded the now discredited 'Iron Law of Wages' and 'Wages Fund Theory', which both consigned wage-earners to lives of unremitting toil for a reward inevitably limited to a very small amount. Against these views appeal is made to justice and equality, and the result is summed up in the phrase 'to each according to his needs, from each according to his ability', i.e. the community should be so organized that each member receives that which he needs, while giving in return work according to the best of his ability. An alternative and cruder theory is that which demands absolute equality in the distribution of wealth. The economic basis for these theories is the claim that all production is the result of labour, and that the workers are

entitled to the 'whole produce of their labour', interest and rent being 'surpluses' appropriated by the capitalist owing to the exploited workers being kept at the level of subsistence. The economic statement of the case is generally associated with the name of Marx, who was, however, greatly influenced by English writers; but the driving force of the movement is moral and emotional.

District Courts, an important series of courts in the United States, each under a single judge, and having original jurisdiction in civil, criminal, and admiralty causes. Generally there is one for each state. By the Judiciary Code, enacted by Congress in 1911, the circuit courts, which had hitherto shared original jurisdiction with the district courts, became only appellate tribunals.

Distringas (Lat., that you distrain), a notice proceeding upon an affidavit filed in the High Court by a party not the registered holder of shares or stock but beneficially interested therein, and served upon the particular company or public body, whereby it is precluded from registering any transfer of the shares or stock or any mandate for payment of the dividends without previous intimation to such party. The latter has thus the opportunity, if desired, to apply for an interim injunction, and, should he not do so within eight days from the date upon which the transfer or mandate was lodged, the restraint flies off.

Dithyram'bus, or **Dith'yramb**, in Greek literature, a poem sung in honour of the god Bacchus or Dionysus, at his festivals. The choral portion of Greek tragedy arose out of the dithyramb. It was composed in a lofty and often inflated style: hence the term is applied to any poem of an impetuous and irregular character.

Ditmarshes (Ger. *Dithmarschen*), a district of Holstein, in Germany, consisting of a monotonous flat stretching along the North Sea, between the mouths of the Elbe and the Eider, and so little raised above the sea as to require the protection of strong embankments. Ditmarshes was incorporated in Prussia in 1866. The area is 500 sq. miles, and the total pop. 96,373.

Dit'tany, the popular name of the plants of the genus *Dictamnus*, an herb of the rue family (Rutaceæ), found in the Mediterranean region. The leaves are pinnate, the large white or rose-coloured flowers are in terminal racemes. The whole plant is covered with oily glands, and the secreted oil is so volatile that in hot weather the air round the plant becomes inflammable. *D. Fraxinella* and *D. albus* are found in gardens.

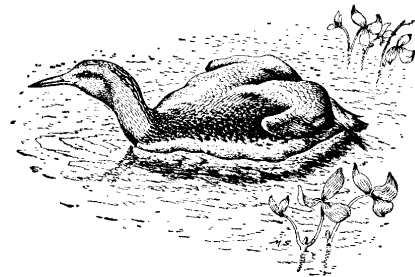
Dittay, in Scots law, a technical term signifying the matter of charge or ground of indictment against a person accused of a crime; also, the charge itself.

Diuret'ics are agents used to increase the flow of urine. Many drugs are used for this purpose: caffeine and theobromine, digitalis and squills, potassium salts, carbonates, calomel and blue pill.

Divan', a Persian word having several significations. It is used in Turkey for the highest Council of State, the Turkish ministry; and for a large hall for the reception of visitors. Low couches, covered with rich carpets and cushions, are ranged along the walls of the room. Hence in Western Europe the term is applied to a café, and to a kind of cushioned seat. In India the term is applied to the Prime Minister of a native State. Among several Oriental nations this name is given to certain collections of lyric poems by one author. The *divans* of Hafiz and Saadi, the Persian poets, are among the most important.

Divergent, in algebra, opposed to convergent, a term applied to an infinite series which cannot be said to have a sum because there is no definite limit towards which the sum of its terms tends as the number of terms is increased indefinitely.

Divers, birds remarkable for the habit of diving. The divers (Colymbidæ) are a family of swimming birds, characterized by a strong, straight, rather compressed pointed bill about as long as the head; a short and rounded tail; short wings; thin, compressed legs, placed very far back, and the toes completely webbed. They prey upon fish, which they pursue under water, making use partly of their wings, but chiefly of their legs and webbed feet in their subaqueous progression. The leading species are the great northern diver (*Colymbus glaciālis*), the red-throated diver (*C. septentrionālis*), and the black-throated diver (*C. arcticus*). These birds inhabit the Arctic seas of the New and Old Worlds; they are abundant in the Hebrides, Norway, Sweden, and Russia. The great northern diver, loon, or ember goose is about $2\frac{3}{4}$ feet long, and is of handsome plumage.



Red-throated Diver (*Colymbus septentrionalis*)

Div'idend, literally what is to be divided, a term used in arithmetic and in reference to stocks. In the latter sense it is the interest or profit of stocks

divided among, and paid to, the proprietors. No dividend must be paid except out of profit (Companies Act of 1862). The term also signifies the payment made to creditors out of the estate of a bankrupt.

Dividing Engine, a machine for marking the divisions on the scales of scientific, mathematical, or other instruments. Some of these perform work of extraordinary fineness and accuracy. It may also be employed to measure lengths accurately, or to divide a given length into any number of equal or unequal parts. See *Graduation*.—Cf. Stewart and Gee, *Practical Physics* (vol. i).

Dividing Range, Great, an Australian chain of mountains, forming the watershed between the rivers flowing into the Pacific and those running westward. It is situated at an average distance of 30 miles from the sea, though in some places it recedes as much as 60 miles, and stretches from Cape York on the north to Wilson's Promontory on the south. The culminating point is Mount Townshend (7353 feet).

Divi-divi, the pods of *Cæsalpinia coriaria*, a tree which grows in tropical America, and a member of the family which yields sapan, brazil, and other red woods. The pods are about 1 inch broad and 3 inches long, but are generally bent or curled up; are excessively astringent, containing a large proportion of tannic and gallic acid, for which reason they are used by tanners and dyers.

Divination, the act of divining; a foretelling future events, or discovering things secret or obscure, by the aid of superior beings, or by other than human means. Cicero divided it into two kinds, *natural* and *artificial*, or *intuitive* and *inductive*. *Natural* divination was supposed to be effected by a kind of inspiration or divine afflatus; this method of divination is familiar as represented by oracles; *artificial* divination was effected by certain rites, experiments, or observations, as by sacrifices, observation of entrails and flight of birds (ornithomancy), of the behaviour of fishes (ichthyomancy), lots, omens, and position of the stars. Among modes of divination were: *axinomancy*, by axes; *belomancy*, by arrows; *bibliomancy*, by the Bible; *oneiromancy*, by dreams; *pyromancy*, by fire; *hydromancy*, by water; *coscinomancy*, by observing the results of the turning of a sieve hung on a thread. Cf. Bouché-Leclercq, *Histoire de la divination dans l'antiquité*.

Divine Right, the claim set up by some sovereigns or their supporters to the absolute obedience of subjects as ruling by appointment of God, insomuch

that, although they may themselves submit to restrictions on their authority, yet subjects endeavouring to enforce those restrictions by resistance to their sovereign's acts are considered guilty of a sin. This doctrine, which came into general use in the seventeenth century, and is so celebrated in English constitutional history, especially in the time of the Stuarts, may now be considered to be exploded. Hobbes was one of the chief supporters of the theory of Divine Right, whilst Milton was a strong opponent. Cf. J. N. Figgis, *Theory of the Divine Right of Kings*.

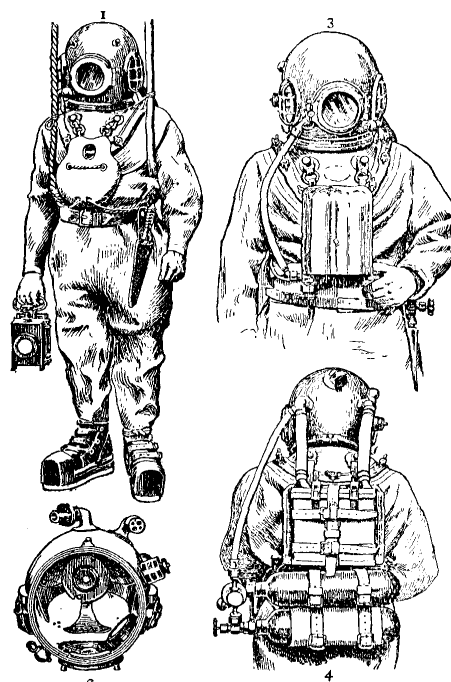
Divine Service, Tenure by, a species of tenure, now obsolete, by which the tenant held the land on condition of performing some divine service, such as saying so many masses or distributing a certain amount in alms.

Diving, the art or act of descending into the water to considerable depths, and remaining there for a time. The uses of diving are important, particularly in searching for pearls, corals, sponges, examining foundations of bridges, salvage of wrecked ships, recovering valuables, clearing propellers, valves, and cleaning bottoms of ships when no dry docks are available. Without the aid of artificial appliances a skilful diver may remain under water for two, or even three minutes; accounts of longer periods are doubtful or absurd. Various methods have been proposed, and engines contrived, to render diving more safe and easy. The great object in all these is to furnish the diver with fresh air, without which he must either make but a short stay under water or perish.

Diving-bell, a contrivance for the purpose of enabling persons to descend and to remain below the surface of water for a length of time, for various purposes, such as laying foundations of bridges, blasting rocks, and recovering treasure from sunken ships. Diving-bells have been made of various forms, more especially in that of a bell or hollow truncated cone, with the smaller end closed, and the larger one, which is placed lowermost, open. The air contained within these vessels prevents them from being filled with water on submersion, so that the diver may descend in them and breathe freely for a long time, provided he can be furnished with a new supply of fresh air when the contained air becomes vitiated by respiration. The diving-bell is usually made of cast iron, and weighted, and has several strong convex lenses set in the sides or roof, to admit light to the persons inside. It is suspended by chains from a barge or lighter, and can be raised or lowered at pleasure upon signals being given by the persons within, who are supplied with fresh air pumped into a flexible pipe by means of force-pumps carried in the lighter, while the heated air escapes by a cock in the upper part of the bell. Modern diving-bells are

usually rectangular in shape, and have a trunk or tube on top reaching to the surface of the water, and fitted with an air-lock to enable men to go into or out of the bell without moving it from the bottom; they are fitted with telephones and electric lights. A constant flow of fresh air is kept up, and all excess of air escapes from the lower part of the bell, the pressure of the air being kept slightly above that of the water outside. The diving-bell has long been found highly useful for carrying on work under water, a steam-crane being usually employed for the movements required. A form, called the *nautilus*, has been invented which enables the occupants, and not the attendants above, to raise or lower the bell, move it about at pleasure, or raise great weights with it and deposit them in any desired spot.

Diving-dress, a waterproof dress of india-rubber cloth used by professional divers, and covering the entire body except the head. The dress has a neck-piece or breastplate, fitted with a segmental screw bayonet joint, to which the head-piece or helmet, the neck of which has a corresponding screw, can be attached when wanted. The helmet has usually three eyeholes, covered with strong glass, and protected with guards. Air is supplied by means of a flexible air-pipe which screws on to a non-return valve on the helmet, and is connected with an air-pump on the surface. To allow of the escape of excess air a valve is fitted in the helmet, so constructed as to prevent water getting in, though it lets the air out. It can be adjusted by the diver to suit his convenience, no matter at what depth he may be. Leaden weights are attached to the diver, and his boots are weighted, so that he can descend a ladder and walk about on the bottom. Communication can be carried on with those above by signals on the breast-rope between the diver and his attendant, or he may converse with them through a speaking-tube or by telephone, which is usually fitted in the breast-rope. One form of diving-dress makes the diver independent of any connection with persons above water except by breast-



Diving Dress

1. Ordinary diving-dress with (2) helmet. 3. Front view and (4) back view of self-contained diving apparatus.

Reproduced by courtesy of Messrs. Siebe, Gorman, & Co

rope. It is elastic and hermetically closed. A reservoir containing highly compressed air is fixed on the diver's back. This supplies him with air by a self-regulating apparatus at a pressure corresponding to his depth. When he wishes to ascend, he simply inflates his dress from the reservoir. Another form, known as the Fleuss dress, also makes the diver independent of exterior aid. The helmet contains a supply of compressed oxygen, and the exhaled breath is passed through a filter in the breast-piece which deprives it of its carbonic acid, while the nitrogen goes back into the helmet to be mixed with the oxygen, the supply of which is under the diver's own control, and to be breathed over again. A diver has remained for an hour and a half under 35 feet of water in this dress. The safe limit for diving is 200 to 300 feet, the deepest dive in this country being 210 feet; but great care must be exercised in bringing the diver to the surface. Diving for pearls, sponges, valuables, &c., is now to a great extent carried on by means of diving-dresses.—BIBLIOGRAPHY: C. W. Domville-Fife, *Submarine Engineering of To-day*; G. W. M. Boycott, *Compressed Air Work and Diving*.

Divining Rod, a rod, usually of hazel, with two forked branches, used by persons who profess to discover minerals or water under ground. The rod, if carried slowly along by the forked ends, dips and points downwards, it is affirmed, when brought over the spot where the concealed mineral or water is to be found. Divination by means of rods is of great antiquity, and has been described by Cicero and Tacitus; their rods, however, were short bits of stick, and the forked hazel twig does not seem to have come into use before the early sixteenth century. Dr. H. Mayo gave a collection of discoveries made by it in his work *On the Truth contained in Popular Superstitions* (1847-51). The use of the divining rod is still common in many parts.—Cf. P. L. L. de Vallemont, *La Physique Occulte: ou Traité de la baguette divinatoire*.

Divisibil'ity, that general property of bodies by which their parts or component particles are capable of separation. The study of radioactivity has shown that larger atoms may be broken up into smaller ones, and the old conception of the atom as an absolutely indivisible unit is no longer entertained by physicists. (See *Matter*.) Numerous examples of the division of matter to a degree almost exceeding belief may be easily instanced. Thus glass test-plates for microscopes have been ruled so fine as to have 225,000 spaces to the inch. Cotton yarn has been spun so fine that one pound of it extended upwards of 1000 miles, and a Manchester spinner is said to have attained such a marvellous fineness that one pound would extend 4770 miles. One grain of

gold has been beaten out to a surface of 52 sq. inches, and leaves have been made 367,500 of which would go to the inch of thickness. Iron has been reduced to wonderfully thin sheets. Fine tissue paper is about the 1200th part of an inch in thickness, but sheets of iron have been rolled much thinner than this, and as fine as the 4800th part of an inch in thickness. Wires of platinum have been drawn out so fine as to be only the 30,000th part of an inch in diameter. Human hair varies in thickness from the 250th to the 600th part of an inch. The fibre of the coarsest wool is about the 500th part of an inch in diameter, and that of the finest only the 1500th part. The silk line, as spun by the worm, is about the 5000th part of an inch thick; but a spider's line is only the 30,000th part of an inch in diameter; insomuch that a single pound of this attenuated substance might be sufficient to encompass our globe. The trituration and levigation of powders, and the perennial abrasion and waste of the surface of solid bodies, occasion a disintegration of particles almost exceeding the powers of computation. The solutions of certain saline bodies, and of other coloured substances, also exhibit a prodigious subdivision of matter. A single grain of sulphate of copper, or blue vitriol, will communicate a fine azure tint to five gallons of water. In this case the sulphate must be attenuated at least 10,000,000 times. Odours are capable of a much wider diffusion. A single grain of musk has been known to perfume a large room for the space of twenty years. At the lowest computation the musk had been subdivided into 320 quadrillions of particles, each of them capable of affecting the olfactory organs.

Division, in arithmetic, the dividing of a number or quantity into any parts assigned; one of the four fundamental rules, the object of which is to find how often one number is contained in another. The number to be divided is the *dividend*, the number which divides is the *divisor*, and the result of the division is the *quotient*. Division is the converse of multiplication.

Division, in the army, the smallest formation of troops which is a mixed force. A division, besides three brigades of infantry, includes artillery, engineers, and administrative troops. It is commanded by a major-general, and if at full strength consists of about 20,000 men.

Division, in Parliament, the mode of determining a question at the end of a debate. In the House of Commons the Speaker puts the question, and declares whether in his opinion the 'Ayes' or the 'Noes' have it. Should his opinion not be acquiesced in by the minority, the House is cleared, and the 'Ayes' directed to go into the right lobby and the 'Noes' into the left, where they are counted by

two tellers appointed for each party. In the House of Lords the two sides in a division are called 'Contents' and 'Not-contents'.

Division of Labour, a method employed in productive undertakings for the simplifying of the work to be done by each of the workmen engaged therein. The separation of the process of production into a series of simple operations means that less ability on the part of the workman is required in order that he may acquire the necessary skill in performing any particular operation, and saves much time, partly because practice leads to each operation being more quickly performed, and partly by avoiding the waste which takes place in workmen moving from one operation to another. Owing to both of these causes, the cost of producing complicated articles may be immensely reduced. By standardizing operations division of labour tends to the invention of machinery; increases the skill and dexterity of the individual workman in the particular operation in which he is engaged; enables semi-skilled or unskilled labour to replace skilled; makes a more continuous and economical use of capital possible; and conduces to the more economical distribution of labour by classing work-people according to their capacity. It has, however, a deteriorating effect on the labourer's usefulness as an all-round workman, and is liable to kill his interest in his work, thereby reducing the incentive to industry. What is called division of labour in English economics has been sometimes termed *co-operation* by foreign economists.

Divorce (Lat. *divortium*, from *divertere*, to turn apart, separate), a separation, by law, of husband and wife, which is either a divorce *a vinculo matrimonii*, that is, a complete dissolution of the marriage bonds, or a divorce *a mensâ et thoro* (from bed and board), whereby the parties are legally separated, but not unmarried. The causes admitted by different codes of laws as grounds for the modification or entire dissolution of the marriage contract, as well as the description of tribunal which has jurisdiction of the proceedings, and the form of the proceedings, are various. Divorce was permitted by the law of Moses, but forbidden in the New Testament, except for adultery. The early laws of Rome permitted the husband to divorce his wife for adultery and many other alleged offences. The facility of divorce continued, without restriction, under the Roman emperors, but as the modern nations of Europe emerged from the ruins of the Roman Empire, they adopted the doctrine of the New Testament. Marriage, under the Roman Church, instead of a civil contract, came to be considered a sacrament of the Church, which it was unlawful to dissolve. The ecclesiastical courts could indeed annul a marriage, but only for a cause that

existed at the time the marriage was contracted, such as prior contracts or impotency. For any cause arising after marriage they could only pronounce a divorce *a mensâ et thoro*, which did not leave either party free to marry again, except by Papal dispensation. A divorce *a vinculo matrimonii*, for any cause arising subsequent to marriage, could formerly be obtained in England only by an Act of Parliament, and the ecclesiastical courts must have previously pronounced a divorce *a mensâ et thoro*. The Act passed in 1857, however, established a new court for trying divorce causes, called the Court for Divorce and Matrimonial Causes, since absorbed into the Probate, Divorce, and Admiralty Division of the High Court of Justice. According to present practice the husband may obtain a divorce for simple adultery; but if the wife is the petitioner, she must show that her husband has been guilty of certain kinds of adultery, or of adultery coupled with desertion or gross cruelty. Either party may marry again after divorce. A divorce cannot be obtained if it appear that the petitioner has been guilty of the same offence, or has been accessory to or has connived at the offence, or if there has been collusion between the parties to obtain a divorce, or if they have condoned the offence by living together as man and wife after discovery. The husband may claim damages from the adulterer, and the court may also order the adulterer to pay the costs of the proceedings, in whole or in part. The Act also abolished divorces *a mensâ et thoro*, substituting, however, judicial separations. (See *Judicial Separation*.) A decree for a divorce is always in the first instance a *decree nisi* (q.v.). A Matrimonial Causes Bill, introduced in the House of Lords in 1920, proposes to abolish the legal disabilities of a wife in this matter, and to grant divorce to husband or wife on any of the following grounds: (a) adultery; (b) desertion for at least three years; (c) cruelty; (d) incurable insanity coupled with at least five years' confinement under the lunacy laws; (e) incurable habitual drunkenness coupled with three years' separation under a temporary separation order granted on the ground of habitual drunkenness; and (f) imprisonment under a commuted death sentence. In Scotland, from the time of the Reformation, divorce might be obtained by either party on the ground of adultery, marriage being held to be only a civil contract, and as such under the jurisdiction of the civil courts. Condonation, or connivance, or collusion is sufficient to prevent a divorce from being obtained on the ground of adultery, but not recrimination, that is, a counter charge of adultery. Wilful desertion for at least four years is also held a valid reason for divorce. The action is carried on before the Court of Session. In other countries, including British colonies, the law relating to divorce varies greatly. In the United States of America, marriage, though it may be celebrated before clergymen as well as civil

magistrates, is considered to be a civil contract, and the laws as to divorce, and the facility or difficulty of obtaining it, differ greatly in the several states. In France divorce was legalized in 1884, with conditions, after having been prohibited for many years.—BIBLIOGRAPHY: Lehr, *Le mariage, le divorce et la séparation de corps dans les principaux peuples civilisés*; Stephen, *Commentaries on the Laws of England*; Bryce, *Marriage and Divorce*.

Dixmude, a town of Belgium, in Flanders. It was the scene of severe fighting during the European War. Captured by the Germans, it was retaken by the Belgians on 29th Sept., 1918. Pop. 4040 (1911); 450 (1919).

Dixon, William Hepworth, miscellaneous writer, born at Manchester, 1821, died in London, 1879. In 1849 he published a memoir of Howard the philanthropist, which was followed by the *Life of William Penn* (1851), and by a work on Admiral Blake (1852). In 1853, after having been a contributor, he became chief editor of *The Athenæum*, a post which he retained till 1869. During this period he published several very popular works, including the *Personal History of Lord Bacon*, *The Holy Land*, and *New America*, the last being followed by *Spiritual Wives*. After his retirement from *The Athenæum*, and in the last ten years of his life, he gave to the world about twenty-five volumes of history, travel, and fiction, among others, *Free Russia*; *Her Majesty's Tower*; *The Switzers*; *History of Two Queens*, *Catherine of Aragon and Anne Boleyn*; *Diana Lady Lyle*, and *Ruby Grey* (both novels); and his last work, *Royal Windsor*.

Dizful, a town of Persia, near the western boundary, on the River Dizful; a place of great trade and manufactures. Pop. about 25,000.

Djavid Bey, Hussein, a Turkish Jew, one of the leaders of the Committee of Union and Progress, and Minister of Finance from 1914 to 1918. His advice that Turkey should adhere to strict neutrality during the European War was, unfortunately for his country, not followed by his colleagues, who decided to join Germany.

Djemal Pasha, Turkish soldier and politician, and one of the leaders of the Committee of Union and Progress. Minister of Marine in 1914, he commanded the Turkish forces in Palestine from 1915 to 1917, and became notorious on account of his oppression of and cruelties against the inhabitants of Syria and Palestine, and by his hatred of Zionism. Accused in 1918 of various crimes, such as misappropriation of funds, and compelled to flee from Constantinople,

he was condemned to death in his absence in 1919. He was nevertheless one of the Turkish delegates who signed the Peace Treaty on 11th May, 1920.

Djokdjokarta, a Dutch residency in the Island of Java, on the south coast, with a capital of the same name. Its forests abound in teak. Its natural fertility is great, and rice, coffee, and tobacco are extensively cultivated. It is ruled by a sultan who is dependent on the Dutch. Pop. 441,800. The town is large and regular, and contains the sultan's water-palace, and the seat of the Dutch Resident, which is a fort commanding both the palace and the town. Pop. 97,058.

Dnieper (nē'pèr; Russ. *Dnjepr*, dnyepr; anciently **Borysthēnēs**, or **Danapris**), a great river of Russia which rises in the government of Smolensk, flows first south-west, then south-east, and again south-west to the Black Sea. It begins to be navigable a little above Smolensk, and has a total length, including windings, of 1230 miles. Among its tributaries are the Beresina, the Pripet, the Desna, and the Psiol. In its lower course there are important fisheries. The region watered by the river in its lower course is famous for its great fertility, and is known as the *black-soil*. Between Kiev and Alexandrovsk it forms a series of cataracts. Since 1838 there have been steamboats on the river, and the trade carried by it is considerable. Through the Beresina Canal the Dnieper communicates with the Baltic Sea.

Dniester (nēs'tèr; Russ. *Dnjestr*; ancient **Tyras**), a large river of Europe (Poland, Roumania, and Ukraine), which has its source in the Carpathian Mountains, in Austrian Galicia, enters Russia at Chotin, and empties itself into the Black Sea, after a course of about 750 miles. Its navigation is difficult on account of frequent shallows and rapids.

Doab' (that is, *Two Waters*), a name in Hindustan applied indiscriminately to any tract of country between two rivers, like the Greek *Mesopotamia*. The tract between the Ganges and the Jumna is usually called *the Doab*; other similar tracts have their distinctive names, the Punjab being divided into five districts of this kind.

Dobell', Sydney, English poet and man of letters, born in 1824, died 22nd Aug., 1874. His first poem, *The Roman*, appeared in 1850, and was favourably received by the critics. Among his other works are: *Balder*, *Sonnets on the War*, and *England in Time of War*.

Döbeln (deu'beln), a town of Saxony, about 40 miles south-east of Leipzig, with a great trade in grain, and manufactures of cloth, yarn, leather, and lacquered wares. Pop. 17,920.

Döbereiner's Lamp, a contrivance for producing an instantaneous light, invented by Professor Döbereiner, of Jena, in 1824. The light is produced by throwing a jet of hydrogen or a mixture of oxygen and hydrogen gases upon recently prepared spongy platinum, when the metal instantly becomes red-hot, and then sets fire to the gas. The action depends upon the readiness with which spongy platinum absorbs both oxygen and hydrogen. The intimate contact of the two gases leads to chemical combination with evolution of heat. Coal-gas may be employed instead of hydrogen, and this has led to the application of the principle to self-lighting devices for igniting gas flames.

Dobritch, or **Bazarjik**, a town in Roumania, was, till the end of the second Balkan War, the chief town in Bulgarian Dobrudja. It is situated on a small tributary of the Danube, and is on the Cernavoda-Constanza railway. At one time it was famous for its panair or great annual fair, but it is now no longer an important centre. Pop. 10,000.

Dobrudja, or **Dobrud'scha**, **The** (anciently **Scythia Minor**), a territory forming part of the kingdom of Roumania, included between the Danube, which forms its boundary on the west and north, the Black Sea on the east, and Bulgaria (to which it belonged before 1878) on the south. Its area, formerly 6000 sq. miles, was increased by the Treaty of Bucharest of 10th Aug., 1913, and is now 8969 sq. miles. During the European War the Dobrudja was invaded in 1916 by an army of Germans, Bulgarians, and Turks, under General von Mackensen. In 1918 Roumania was compelled by the Treaty of Bucharest to cede the Dobrudja to Bulgaria, but it was restored to her in 1919. There are some fertile spots, but on the whole it is marshy and unhealthy. The population is of various nationalities, Roumanians, Bulgars, Greeks, Turks, and Jews. The inhabitants engage in tillage and stock-rearing. The principal towns are Kustendje and Tultcha. See *Roumania*. Pop. 381,306.

Dobson, Henry Austin, poet, born at Plymouth in 1840. He was educated at Beaumaris, Coventry, and Strasbourg; in 1856 obtained a clerkship under the Board of Trade, where he rose to be one of the officials known as principals. His earliest verses first appeared in book form under the title *Vignettes in Rhyme and Vers de Société* (1873). His other volumes of verse include: *Proverbs in Porcelain* (1877); *Old World Idylls* (1883); and *At the Sign of the*

Lyre (1885), which *The Athenæum* pronounced to be "of its kind as nearly as possible perfect". Among his prose works may be mentioned his lives of *Hogarth*, *Fielding*, *Steele*, *Goldsmith*, *Horace Walpole*, *Richardson*, and *Fanny Burney*; *Thomas Bewick and his Pupils*; *Four Frenchwomen*, a study on Charlotte Corday, the Princesse de Lamballe, and Mesdames Roland and de Genlis; three series of *Eighteenth Century Vignettes*; *A Paladin of Philanthropy*; *Sidewalk Studies*; *Old Kensington Palace*; *At Prior Park*; *A Bookman's Budget*; and several editions of standard works. His collected poems were published in one volume in 1897. Many of Mr. Dobson's poems are written in various French forms, such as the rondeau and ballade, and all are marked by gracefulness and ease. He died in Sept., 1921.



Curly or Yellow Dock
(*Rumex crispus*)

Docē'tæ (from Gr. *dokein*, to seem or appear), the name given, in the earlier ages of the Church, to those who denied the reality of the human form of Christ, maintaining it to be merely a phantom or shadow. In the sense of regarding Christ's body as a heavenly and ethereal, instead of a human one, docetism had its partisans even among the orthodox.

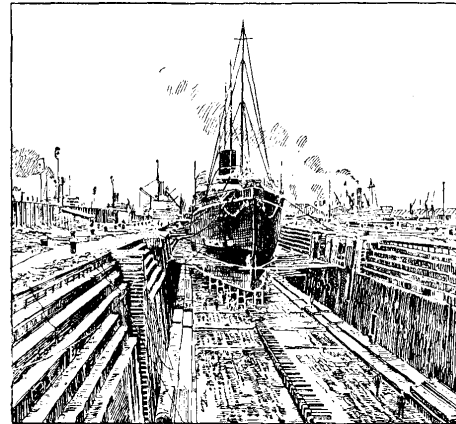
Dock, a name applied to different plants of the genus *Rumex*, belonging to the rhubarb family (Polygonaceæ). These are large herbaceous plants, with stout roots, alternate and often entire leaves, and bearing panicles of small greenish flowers. They are very troublesome as weeds, but the roots of some of them are used medicinally as astringents.

Docket, or **Docquet** (dok'et; from *dock*, to shorten; Icel. *dokr*, stumpy tail), in law, a term variously used, as for a summary of a larger writing; a small piece of paper or parchment containing the heads of a writing; an alphabetical list of cases in a court, or a catalogue of the names of the parties who have suits depending in a court.

Docks, are artificial enclosures for the reception of shipping, for the purpose of loading, discharging, or repairs. They may be divided into four types, viz. tidal docks or basins, wet docks, dry or graving docks, and floating docks.

Tidal Docks or Basins are open permanently to the main channel or river, and the water-level therefore varies with the rise and fall of the tide. This variation of level is a serious hindrance in the work of loading or discharging cargo, unless the tidal range be small. It should be noted that this form is more properly termed a tidal basin.

Wet Docks have a water entrance normally closed by gates or caissons, which permit the water-level of the enclosed area being maintained at high-water level. This uniformity of level is of great service in dealing with cargo, but this type has the disadvantage of only permitting traffic in and out of the dock at high water. This disadvantage may be modified by the provision of a lock at the dock entrance.



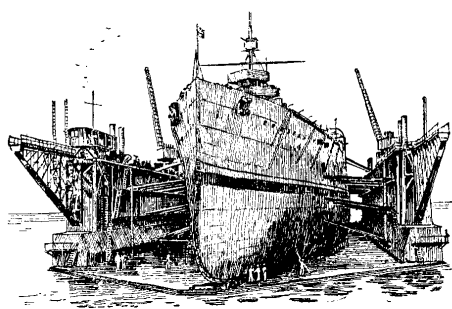
Dry Dock, Tilbury, Essex

Dry or Graving Docks are used for the purpose of examining and repairing ships.

The entrance is controlled by either gates or caissons. The ship having entered, the gates are closed, and the water pumped out, allowing the vessel to settle down gradually upon a row of keel blocks, running up the centre line of the dock. The sides of the docks are built in the form of large continuous steps or 'altars', which support the ends of the timber shores which serve to keep the ship in an upright position. The floor is graded to channels, leading to powerful pumps, usually of the centrifugal type, and capable of emptying a large dock in one hour.

Floating Docks fulfil the same functions as dry docks. In their modern form, they consist of a hollow steel box or pontoon, carrying hollow longitudinal walls at each side. These walls contain the pumps and controlling machinery, the pontoon portion being capable of being filled with or emptied of water, thus raising or sinking the dock. This lower portion is subdivided into a great number of compartments, all of which may be filled separately, so that errors of trim can be corrected. In making use of this type the dock is lowered by flooding the lower compartments. The ship is then floated into position and shored.

The initial cost of these two latter types is in favour of the floating dock, the annual maintenance charges of which, however, may be five to ten times those



The Medway Floating Dock lifting
H.M.S. *Lion*, 30,000 tons

of an ordinary masonry dock. The mobility of a floating dock may be considered an advantage, allowing it to be easily removed to another locality to meet changing conditions, but this adaptability should not be overvalued.

The average life of a steel dock may be assumed to be 40 years, whereas a masonry dock may be said to last indefinitely, subject only to the fact that it may outlive its

usefulness.

The first wet docks constructed in England were those now called the Commercial Docks, in London, which existed in a much less extensive form so early as 1660. In 1800 the West India Docks were constructed, and were followed by the East India Docks, Millwall Docks, London Docks, the St. Katharine Docks, and the Victoria Docks, affording, together with those at Tilbury, more than 600 acres of water accommodation, besides wharf and warehouse grounds, where all kinds of appliances and machinery for the speedy and convenient transfer of goods and cargoes are in use. Some of the warehouses are extremely capacious, the tobacco warehouse of the London Docks being itself nearly 5 acres in extent. Next after the London docks come those of Liverpool, which extend more than 6 miles along the north bank of the Mersey, and cover, together with the Birkenhead docks, nearly as large a total acreage as those of London. The other important British docks are those at Southampton, Bristol, Cardiff, Hull, Great Grimbsy, Newcastle, Shields, Barrow, Leith, Glasgow, Dundee, &c. A floating dock at Hamburg has a length of 728 feet, inside width of 123 feet, and a lifting capacity of 46,000 tons.—

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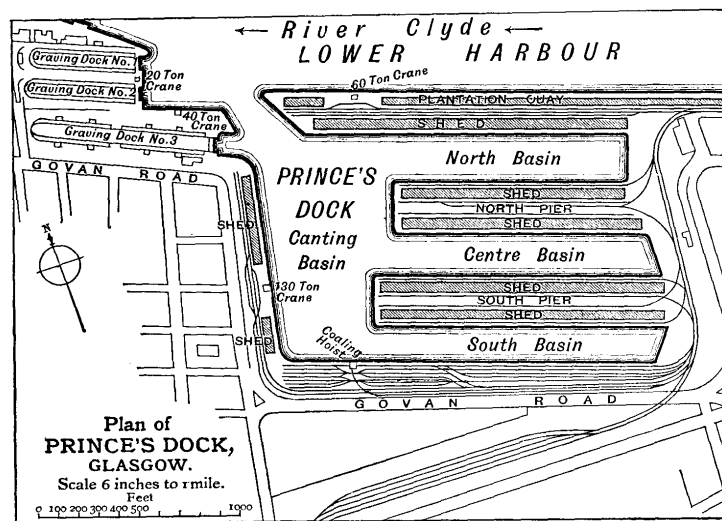
Dock-warrants, orders for goods kept in the warehouses connected with a dock. They are granted by the proper officer at the dock to the importer in favour of any one that he may name. These warrants are held to be negotiable, so that they may pass from one holder to another, the property of them being always vested in the holder.

Dock-yards, establishments supplied with all sorts of naval stores, materials and conveniences for the construction, repairs, and equipment of ships of war.

In England the royal dock-yards are at Chatham, Sheerness, Portsmouth, Devonport, and Pembroke, besides the Deptford and Woolwich store-yards. There are also royal dock-yards or naval victualling yards at Haulbowline in Cork Harbour, Rosyth, Invergordon, at the Cape of Good Hope, Gibraltar, Malta, Bermuda, Bombay, Calcutta, Hong-

Kong, Sydney, and Wei-hai-wei. Others in the colonies have been given up. The dock-yards are under the direct control of the Admiralty. The chief officer of the greater or home dock-yards is generally an admiral, with a considerable staff of officials under him, professional and other. Since the introduction of steam the engineering department has become an important one in such establishments.

Doctor, a term literally signifying teacher. In the Middle Ages, from the twelfth century, it came into use as a title of honour for men of great learning, such as Thomas Aquinas (Doctor Angelicus) and Duns Scotus (Doctor Subtilis). It was first made an academical title by the University of Bologna, and emperors and Popes soon afterwards assumed the right of granting universities the power of conferring the degree in law. The University of Paris followed in the footsteps of Bologna, and in 1145 the title was bestowed on Peter Lombard. The faculties of theology and medicine were soon included, but for a long time the faculty of arts retained the older title of *Magister*, till the German universities substituted that of Doctor. In England the Doctor's degree was introduced into the universities during the reign of John or Henry III. The title of Doctor is in some cases an honorary degree, and in other cases (as in medicine and science) conferred after examination. The title of D.C.L. (Doctor of Civil Law), for example, at the Universities of Oxford and Durham, is frequently an honorary degree, and so also are those of D.D. (Doctor of Divinity) and LL.D. (Doctor of Laws) at various universities. The Popes and the Archbishops of Canterbury exercise the right of conferring the degree of



Plan of PRINCE'S DOCK, GLASGOW.

Doctor both in law and divinity. Oxford and Cambridge, and many other universities, create doctors of music (Mus.D.).

Doctors' Commons was a college founded in 1567 for the Doctors of the Civil Law in London, and was at one time the seat of the Court of Arches, the Archdeacon's Court, and the Court of Admiralty. The practitioners in these courts were called advocates and proctors. In 1857 an Act was passed empowering the college to sell its property and dissolve, and making the privileges of the proctors common to all solicitors.

Doctors of the Church, a name given to four of the Greek Fathers (Athanasius, Basil, Gregory Nazianzen, and Chrysostom) and five of the Latin Fathers (Hilary, Ambrose, Jerome, Augustine, and Gregory the Great). The Roman Catholic Church, however, recognizes eighteen 'Doctors of the Church', including, besides those already mentioned, Chrysologus, Leo, Isidore, Peter Damian, Anselm, Bernard of Clairvaux, Thomas Aquinas, Bonaventura, and Alphonsus of Liguori. The title is conferred only after death.

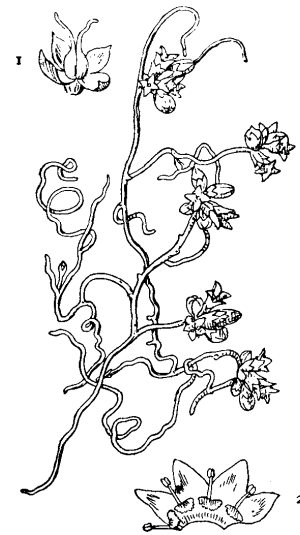
Doctrinaires, a section of French politicians, represented by the Duc de Broglie, Royer-Collard, Guizot, and others, who became prominent after the Restoration in 1815. They favoured a constitutional monarchy similar to that which then existed in Britain. In the Chambers they thus occupied a place between Radicals and ultra-Royalists. They received the name of *doctrinaires* because they were looked upon more as theoretical constitution-makers than practical politicians, and the term is now used with a wider application to political theorists generally.

Dodder, the common name of the plants of the genus *Cuscuta*, a group of slender, branched, twining, leafless pink or white annual parasites, nat. ord. Convolvulaceæ. The seeds germinate on the ground, but the young plant speedily attaches itself to some other plant, from which it derives all its nourishment. Four species are common in England—*C. europæa*, found on nettles and vetches; *C. Epithymum*, on furze, thyme, and heather (these two being natives); *C. trifolii*, on clover; *C. Epilinum*, on cultivated flax (both introduced from abroad).

Doddridge, Philip, D.D., an English Dissenting divine, born in London in 1702, died in 1751. He was an earnest pastor, and the author of many hymns and devotional treatises. *The Rise and Progress of Religion in the Soul* and *The Family Expositor* are amongst his best known works. Many of his hymns are

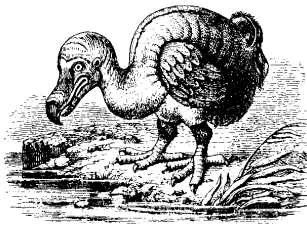
still sung, such as *O God of Bethel, by whose hand;*
and *O happy day that fixed my choice.*

Dodecanese (Gr., twelve islands), a name applied since the Turko-Italian War of 1912-3 to a group of islands lying off the south-west coast of Asia Minor, the base of the Italian troops. The group, consisting of the Southern Sporades, includes the Islands of Patmos, Cos, Lipsos, and others near Rhodes. Italy agreed to restore the islands to Turkey, but was prevented from carrying out the agreement by the breaking out of the Balkan War. The Treaty of Peace with Turkey (11th May, 1920) allotted these islands to Italy, who ceded them again to Greece (with the exception of Rhodes).



Dodder
1, Pistil. 2, Corolla.

Dodgson, Charles Lutwidge. See *Carroll, Lewis*.



Dodo, from painting in the
Belvedere, Vienna

Dodo (*Didus ineptus*), an extinct genus of birds once abundant on the Island of Mauritius, and assigned by naturalists to the ord. Columbæ or pigeons, though an extreme modification of the type. It was a massive clumsy bird, larger than a swan, covered with down instead of feathers, with short ill-shaped legs, a strong bulky hooked beak, and wings and tail so short as to be useless for flight. Its extinction was due to its organization not being adapted to the new conditions which colonization and cultivation introduced. It probably became extinct soon after 1681.

Dodo'na, a celebrated locality of ancient Greece, in Epirus, where was one of the most ancient Greek oracles. It was a seat of Zeus (surnamed the Pelasgian), whose communications were announced to the priestesses in the rustling of the leaves on its oak tree, and the murmuring of water which gushed forth from the earth.

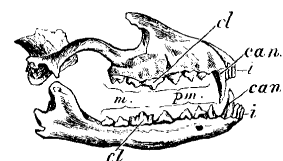
Dods, Marcus, theologian, born at Belford, Northumberland, in 1834, his father being minister of the Scottish Church there, died in 1909. He was educated in Edinburgh, where he took his M.A. degree at the age of twenty. In 1858 he was licensed as a minister of the Free Church of Scotland, and eight years afterwards was ordained to Renfield Free Church, Glasgow, where he

remained until his appointment in 1889 to the chair of New Testament Exegesis in New College, Edinburgh. Of his works some of the most important are: *The Prayer that Teaches to Pray* (1863, 6th edition, 1889); *Epistles to the Seven Churches* (1865); *Israel's Iron Age* (1874); *Mohammed, Buddha, and Christ* (1877); *The Book of Genesis* (1882); *Parables of Our Lord* (1883 and 1885); *How to become like Christ* (1897); *Forerunners of Dante* (1903); *The Bible: its Origin and Nature* (1905); and articles in the *Encyclopædia Britannica*.

Dodsley, Robert, English poet, dramatist, and publisher, born in 1703, died in 1764. His first volume of verses, *The Muse in Livery*, appeared in 1732. *The Toy Shop* was performed at Covent Garden in 1735. In 1737 his *King and the Miller of Mansfield* was performed at Drury Lane, and met with an enthusiastic reception. He also wrote a tragedy, entitled *Cleone*, which had an extraordinary success on the stage. A selection of *Fables in Prose*, with an *Essay on Fables* prefixed, was one of his latest productions. He planned the *Annual Register* (commenced in 1758); the *Collection of Old Plays* (12 vols. 12mo), which now chiefly sustains his name as a publisher; and the *Collection of Poems by Different Hands* (6 vols. 12mo).

Doe, John, and Richard Roe, two fictitious personages of the English law who formerly appeared in a suit of ejectment. This fictitious form of procedure was abolished in 1852.

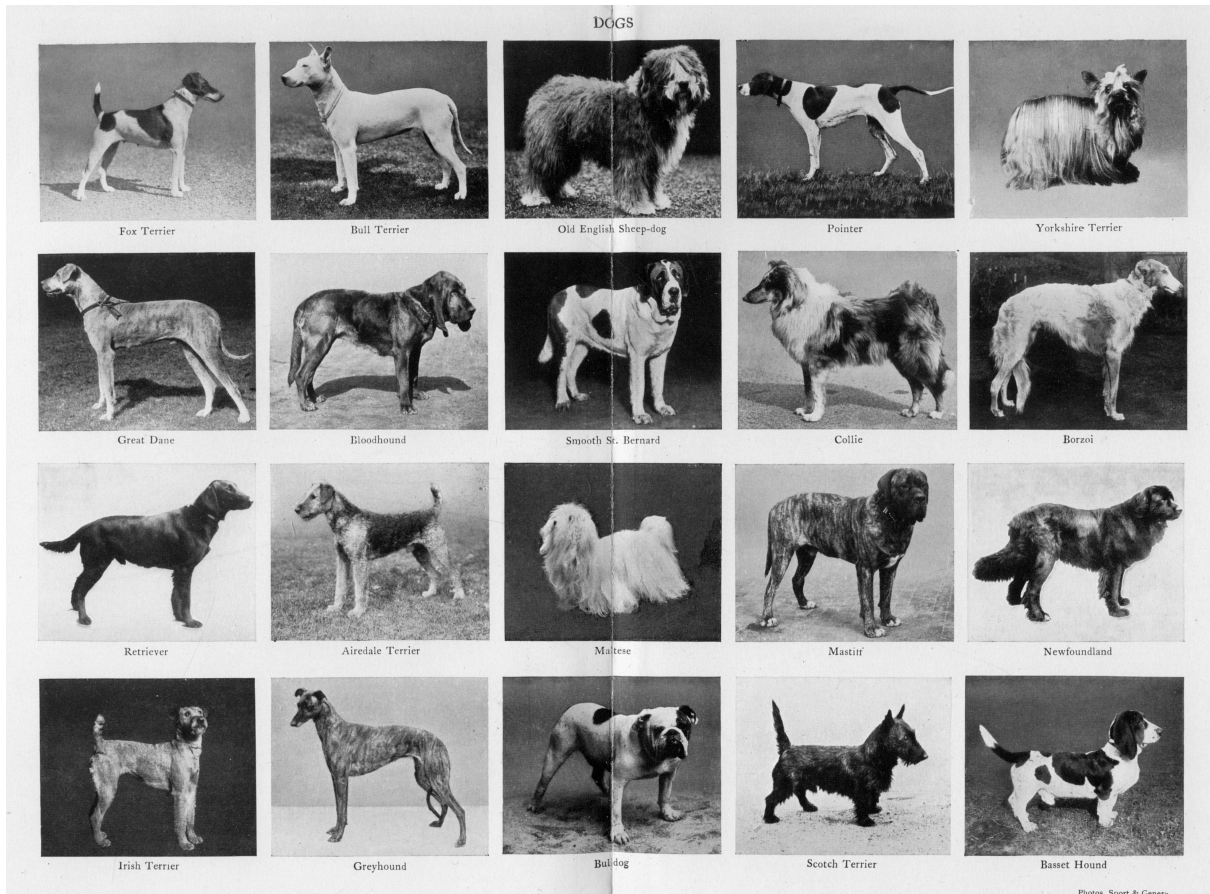
Dog (*Canis familiaris*), a digitigrade, carnivorous animal, forming the type of the genus *Canis*, which includes also the wolf, the jackal, and the fox. The origin of the dog is a much-debated question. The original stock is unknown, but various species of wolf and jackal have been suggested as ancestors. Probably a number of wild types were domesticated by prehistoric man, and there has been a good deal of crossing between these different stocks. It is generally agreed that no trace of the



Jaws of Dog
m, Molars. *pm*,
 Premolars. *cl*,
 Carnassial. *can*,
 Canines. *i*, Incisors.

dog is to be found in a primitive state, the dhole of India and dingo of Australia being believed to be wild descendants from domesticated ancestors. Several attempts to make a systematic classification of the varieties of dogs have been made, but without much success, it being difficult in many cases to determine what are to be regarded as types, and what as merely mongrels and cross-breeds. Colonel Hamilton Smith divides dogs into six groups as follows: (1) *Wolf-dogs*, including the Newfoundland, Esquimaux, St. Bernard, shepherd's

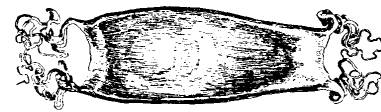
dog, &c.; (2) *Watch-dogs and Cattle-dogs*, including the German boar-hound, the Danish dog, the matin dog, &c.; (3) *Greyhounds*, the lurcher, Irish hound, &c.; (4) *Hounds*, the bloodhound, staghound, foxhound, setter, pointer, spaniel, cocker, poodle, &c.; (5) *Terriers* and their allies; (6) *Mastiffs*, including the different kinds of mastiffs, bull-dog, pug-dog, &c. (See the different articles.) On each side of the upper jaw are three incisors, one canine, four premolars, and three or two molars: on each side of the lower jaw the same number, except that the molars are four or three. The fore-feet have five toes, the hind-feet four or five; the claws are strong, blunt, and formed for digging, and are not retractile. The tail is generally long, and is curled upwards. The female has six to ten mammæ; she goes with young nine weeks as a rule. The young are born blind, their eyes opening in ten to twelve days; their growth ceases at two years of age. The dog commonly lives about ten or twelve years, at the most twenty. By English law it is prohibited to use dogs for purposes of draught.—
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Dog-days, the name applied by the ancients to a period of about forty days, the hottest season of the year, at the time of the heliacal rising of Sirius, the dog-star. The time of the rising is now, owing to the precession of the equinoxes, different from what it was to the ancients (1st July); and the dog-days are now counted from 3rd July to 11th Aug., that is, twenty days before and twenty days after the heliacal rising.

Doge (dōj; from Lat. *dux*, a leader, later a duke), formerly the title of the first magistrates in the Italian Republics of Venice and Genoa. The first doge of Venice elected for life was Paolo Anafesto, in 697; and in Genoa, Simone Boccanera, in 1339. In 1437 the Doge of Venice obtained from the emperor a diploma creating him 'Duke of Treviso, Feltre, Belluno, Padua, Brescia, Bergamo, &c.'. In Venice the dignity was always held for life; in Genoa, in later times, only for two years. In both cities the office was abolished by the French in 1797. The title was re-established in 1802 for the Ligurian Republic, but was abolished in 1805.

Dog-fish, a name given to several species of small shark, common around the British Isles. The rough skin of one of the species (*Scyllium canicūla*), the lesser spotted dog-fish, is used by joiners and other artificers in polishing various substances, particularly wood. This species is rarely 3 feet long, *S. catūlus*, the greater spotted dog-fish, is in length from 3 to 5 feet. It is blackish-brown in colour, marked with numerous small dark spots. Both species are very voracious and destructive. Their flesh is hard, dry, and unpalatable. The common or picked dog-fish (*Acanthias vulgāris*) is common in British and N. American seas, and is sometimes used as food. It is fierce and voracious. *S. profundorum* has been brought up from 816 fathoms in the North Atlantic. The tiger or zebra-shark (*Stegostoma tigrinum*) is a handsome dog-fish native to the Indian Ocean. It is marked with dark stripes on a yellow ground, and may attain the length of 15 feet.



Egg Capsule of Lesser Spotted Dog-fish

Dogger, a Dutch vessel equipped with two masts and somewhat resembling a ketch. It is used particularly in the North Sea for the cod and herring fisheries.

Dogger Bank, an extensive sand-bank, near the middle of the North Sea, between Denmark on the east and England on the west, celebrated for its cod-fishery. It commences about 36 miles east of Flamborough Head and extends E.N.E. to within 60 miles of Jutland, in some places attaining a breadth of about 60 miles, though it terminates merely in a point. Where shallowest the water over it is 9 fathoms. In Oct., 1904, the Russian Baltic Squadron fired upon a British fishing-fleet on the Dogger, killing two men. The incident was settled by arbitration. During the European War a naval battle was fought off the Dogger Bank on 24th Jan., 1915, in which three powerful German cruisers were seriously injured by a British fleet under Admiral Beatty, but made their escape to Heligoland.

Doggett's Coat and Badge, the prize for a rowing-match which is held annually on the 1st Aug., the course being on the Thames from London Bridge to Chelsea. The match is open to six young watermen whose apprenticeship ends the same year, and the prize is a waterman's red coat bearing a badge which represents the white horse of Hanover. It was instituted—in celebration of the accession of George I—by the actor Thomas Doggett, born in Dublin; but though first rowed in 1716, its winners have only been recorded since 1791. The match, like other events of the same kind, suspended during the War, was held again on 3rd and 4th Aug., 1920, for the years 1915 to 1920.

Dog-gods. The dog was the first domesticated animal, and at an early period was deified. In ancient Egypt the god Anubis, who guided souls to the Otherworld, was dog-headed, as Herodotus says, or jackal-headed. Yama, the Indian god of death, has a dog form, and in the *Mahábhárata*, as Dhárma, god of justice (one of his forms), leads the Pandava brothers to Paradise as a dog. Indra had a dog form, and the custom still prevails among Northern Indian hill tribes of pouring hot oil in a dog's ear to bring rain; The Big Dog (Indra) is supposed to hear the howl of the tortured dog. The dog of Hades figures in Greek, Scandinavian, Celtic, and other mythologies. Cuchullin slew the dog of Hades, and was called 'Hound (*cu*) of Culann'. The widespread belief that dogs howl when a sudden death is about to take place has a long history. Dog ancestors figure in American, Javan, and other myths. In Japanese temples are images of 'Ama-Inu' (Heavenly Dog). Red Indian myths refer to a Dog Creator and to the eclipse-causing dog. Eskimos thrash dogs during an eclipse.

Dogma (Gr. *dogma*, from *dokein*, to seem), an article of religious belief, one of the doctrines of the Christian faith. The history of dogmas, as a branch of theology, exhibits in a historical way the origin and the changes of the various Christian systems of belief, showing what opinions were received by the various sects in different ages of Christianity, the sources of the different creeds, by what arguments they were attacked and supported, what degrees of importance were attached to them in different ages, the circumstances by which they were affected, and the mode in which the dogmas were combined into systems. Lectures on this subject are common in the German universities. In English dogma and dogmatism have come to be frequently used for assertion without proof.

Dogmat'ics, a systematic arrangement of the articles of Christian faith (dogmas), or the branch of theology that deals with them. The first attempt to furnish a complete and coherent system of Christian dogmas was made by Origen in the third century.

Dōgra. The Dōgras are a race of Indian hillmen, descended from Rajputs, who in almost prehistoric times overran the country to the east of Jammu, in Southern Kashmir, and founded principalities. They are now found in and recruited from the country comprised within a rough triangle having Jammu, Simla, and Chamba at its angles; or more generally still, between the upper waters of the Rivers Chenab and Sutlej, two of the five rivers from which the Punjab takes its name. By religion the Dōgras are, as becomes their descent, strict Hindus; indeed, their religion may be considered Hinduism in its purer

and more original form, uncontaminated by outside influences. As is the case with other high-caste Hindus, the Dōgra's life is largely governed by ceremonial observances, those relating to birth, initiation, marriage, and death being the more important. He is, however, rather less fastidious about ceremonial cooking than either the sacerdotal Brahman or his distant Rajput cousin of the plains.

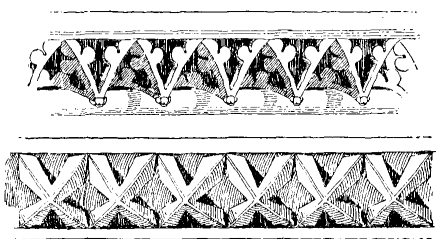
The Dōgra is not a big man, his average height being barely 5 feet 5 inches, but he is sturdy, and develops well with training, and, being fond of sport and games, makes an excellent soldier, especially in his native hills. Being Rajputs by descent, they are essentially a race of soldiers, and, enlisted in the Indian army, give excellent service as such, being law-abiding, hardy and enduring, and quietly courageous. It has already been said that Dōgras are not so ceremonious in their cooking arrangements as either Brahmans or down-country Rajputs, who require separate cooking-places for each man; the Dōgra, on the other hand, though remaining very particular about his drinking-water utensils, will yet agree to have his food prepared for him in messes of five to ten men; this trait makes his value on service and under service conditions considerably greater than that of either of the two above-mentioned classes. In the Indian army there are three first-line battalions of Dōgras: the 37th, 38th, and 41st. These are what are known as class regiments; that is, they consist entirely of the one class. In addition to the three regiments, Dōgras also enlist in the cavalry and in class-company regiments, i.e. regiments enlisting a variety of classes while keeping them separate inside the regiment.

Dog-rose, the *Rosa canīna*, or wild brier, nat. ord. Rosaceæ. It is a common British plant, growing in thickets and hedges. The fruit is known as the hip.

Dog's-mercury, *Mercuriālis perennis*, nat. ord. Euphorbiaceæ, a woodland herb common in Britain. It has poisonous properties, and may be made to yield a fugitive blue dye.

Dog's-tail Grass (*Cynosūrus*), a genus of grasses. *Cynosūrus cristātus* is a perennial found wild all over Great Britain in pastures, lawns, and parks. Its roots are long and wiry, and, descending deep into the ground, ensure the herbage against suffering from drought. Its stem is from 1 to 2 feet high, and its leaves are slightly hairy.

Dog's-tooth Ornament, an architectural ornament or moulding consisting usually of four leaves radiating from a raised point at the centre. It is the



Dog's-tooth Ornaments. Early English Style

characteristic decorated moulding of Early English architecture, as the zigzag is of the Norman.

Dog-tooth Spar, a form of mineral calcium carbonate or calc-spar found in Derbyshire and other parts of England, and named from a supposed resemblance of its pointed crystals to a dog's tooth.

Dog-watch, a nautical term distinguishing two watches of two hours each (4 to 6 p.m. and 6 to 8 p.m.). All the other watches count four hours each, and without the introduction of the dog-watches the same hours would always fall to be kept as watch by the same portion of the crew.

Dogwood, a common name of trees of the genus *Cornus*, but specifically applied in Britain to *C. sanguinea*. It is a common shrub in copses and hedges in England; the small cream-white flowers are borne in dense roundish clusters. The branchlets and leaves become red in autumn. The wood is used for skewers and for charcoal for gunpowder.

Doit, an ancient Scottish coin, of which eight or twelve were equal to a penny sterling. In the Netherlands and Lower Germany there was a coin of similar name and value.

Dol, a town of France, department of Ille-et-Vilaine, 14 miles south-east by east of St. Malo. The old cathedral of St. Samson mostly dates back to the thirteenth century. To the north of the town stretches a salt-marsh, protected from inroads of the sea by a twelfth-century dyke, and in the centre of the marsh Mont-Dol rises to a height of 213 feet. Pop. 3540.

Dolci (dol'chē), Carlo, celebrated painter of the Florentine school, was born at Florence in 1616, and died there in 1686. His works, principally heads of madonnas and saints, have a character of sweetness and melancholy. Among his chief productions are: *Archduchess Claudia*, in the Uffizi (Florence); *St.*



Dog's Mercury (*Mercurialis perennis*)

1, Male flower. 3, Same, enlarged. 2, Female flower.

Cecilia at the Organ and *Herodias with the Head of John the Baptist*, both in the Dresden Gallery; *Ecce Homo* and *St. Andrew in Prayer*, at the Pitti Gallery; and *Magdalene*, at Munich.

Dolcinites, a Christian sect of Piedmont, so named from their leader Dolcino. They arose in 1304 as a protest against Papacy, but were suppressed by the troops of the Inquisition in 1307.

Dol'drums, among seamen, the parts of the ocean near the equator that abound in calms, squalls, and light baffling winds; otherwise known as the horse-latitudes.

Dôle, a town in France, Jura, 26 miles southeast of Dijon. It is of Roman origin, was long the capital of Franche Comté, and has some interesting antiquities. The manufactures are Prussian blue, hosiery, ironware, and leather. Pop. 16,294.

Dol'erite, compact rock of the Basaltic series, but crystalline throughout, composed of augite and labradorite with some titaniferous iron ore and often olivine. It makes, when unaltered, an excellent road-metal.

Dolgel'ly, a town of Wales, capital of Merionethshire, near the foot of Cader Idris. It was there that Owen Glendower held his Parliament in 1404 and signed his treaty with Charles VI of France. It has manufactories of woollens, flannels, and cloths. Pop. 2160.

Dolichocephalic (dol-i-ko-se-fal'ik), long-headed: a term used in anthropology to denote those skulls in which the diameter from side to side is less in proportion to the longitudinal diameter (i.e. from front to back) than 8 to 10.

Dol'ichos (-kos), a genus of leguminous plants, sub-ord. Papilionaceæ. They are found in the tropical and temperate regions of Asia, Africa, and America, and all produce edible legumes. *D. Lablab* is one of the most common kidney beans in India, and *D. biflorus* (horse-gram) is used as cattle-food in the same country. *D. Pachyrrhizus tuberōsus* of Martinique has a fleshy tuberous root which is an article of food.

Dol'ichosaurus ('long lizard'), an extinct snake-like reptile found in the English chalk, whose remains indicate a creature of aquatic habits from 2 to 3 feet in length.

Dollar, a silver or gold coin of the United States, of the value of 100 cents, or rather above 4s. sterling. The same name is also given to coins of the same general weight and value, though differing somewhat in different countries, current in Mexico, a great part of South America, Singapore, and the Philippine Islands. The name is from the Dutch (also Danish and Swedish) *daler*, from Ger. *thaler*, so named from Ger. *thal*, a dale, because first coined in Joachimsthal, in Bohemia, in 1518. By the Act of 14th March, 1900, the gold dollar was declared to be the standard of value in the United States, but no provision was made for the issue of a coin corresponding to the unit.

Dollar, a town and police burgh, Scotland, Clackmannanshire, 10 miles E. by N. of Stirling, noted for its academy, founded by John Macnab, who left £90,000 for this purpose. The building, a handsome structure in the Grecian style, was erected in 1819. The population of the village is 1497.

Dollart, The, a gulf of the North Sea, at the mouth of the Ems, between the Dutch province of Groningen and Hanover. It was originally dry land, and was formed by irruptions of the sea which took place in 1277 and 1530, overwhelming thirty-four large villages and numerous hamlets.

Döllinger (deul'ing-ër), Johann Joseph Ignaz, a celebrated German theologian and leader of the Old Catholic party, was born at Bamberg, in Bavaria, in 1799, died in 1890. In 1822 he entered the Church, and soon after published *The Doctrine of the Eucharist during the First Three Centuries*, a work which won him the position of lecturer on Church history at the University of Munich. In later years he took an active part in the political struggles of the time as representative of the university in the Bavarian Parliament, and as delegate at the Diet of Frankfurt voted for the total separation of Church and State. In 1861 he delivered a course of lectures, in which he attacked the temporal power of the Papacy. But it was first at the Œcumenical Council of 1869-70 that Dr. Döllinger became famous over Europe by his opposition to the doctrine of Papal infallibility. In consequence of his opposition to the Vatican decrees, he was excommunicated in 1871 by the Archbishop of Munich. A few months later he was elected rector of the University of Munich, where he remained until his death. When the sentence of his excommunication was pronounced, he received honorary degrees from the Universities of Oxford and Edinburgh. Among his numerous works are: *Origins of Christianity*, *A Sketch of Luther*, *The Papacy*, *Lectures on the Reunion of the Churches*, and *Papal Legends of the Middle Ages*.

Dol'fond, John, an English optician of French descent, born in 1706, died in 1761. He devoted his attention to the improvement of refracting telescopes, and succeeded in constructing object-glasses in which the refrangibility of the rays of light was corrected.

Dolls, representing more or less realistically the human form, have, for more than fifty centuries, been the common playthings of children, more especially of girls, whose maternal instinct impels them to lavish upon these often crude surrogates all the affection and devotion which their elders display towards real babies. But in ancient times, and even in the ritual of many modern religions, worship is not infrequently paid to human images, which in ancient times, and among the less cultured modern peoples, are hardly distinguishable from dolls, such as children regard as playthings and their fancy endows with a crude animism. In the earliest times in which members of our own species, *Homo sapiens*, are known to have lived in Europe, i.e. at the latter part of the so-called Old Stone (palæolithic) Age, it was the custom to make grotesque representations of the female form as small figurines of clay or stone, which were regarded as amulets identified with the Great Mother, the giver of birth or life to mankind. As 'givers of life' such amulets were believed to be able to protect their possessors against the risk of death, because they were regarded in the most literal sense of the term as life-giving. But it was not merely against the risk of death that such amulets were believed to be potent: they could add 'vital substance' to the living and the dead, rejuvenating and reinvigorating the former, and enhancing the chances of continued existence and survival to the latter. Enormous respect was naturally paid to figurines supposed to possess such far-reaching powers; and when the Great Mother came to be identified with various animals, such as the cow, pig, &c., the amulet was identified with these 'givers of life' and sometimes represented in their shape. This is intimately associated with the origin of *totemism* (q.v.). It is probable that the modern doll is in part at least the survivor of these primitive images of the deities of early peoples. The fact that modern dolls are usually of the female sex may also be due to the fact of the earliest prototype of the doll being an amulet representing the Great Mother.

Dol'man, a long robe worn by the Turks as an upper garment. It is open in front, and has narrow sleeves. It has given its name to a kind of loose jacket worn by ladies, and to the jacket worn by hussars.

Dol'men, a Celtic name meaning 'table-stone'. Although some apply the name to prehistoric stone chambers covered with more than one slab (really 'corridor

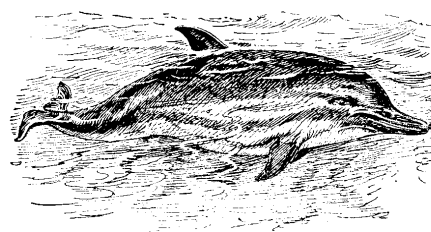
tombs'), the Dolmen proper, whether round or square, has a single cover-slab, and three, four, or even more stones supporting it. Some authorities consider the name *dolmen* as simply a French equivalent for *cromlech* (q.v.).

Dolomieu (dol-o-myeu), Déodat Guy Silvain Tancred Gratet de, a French geologist and mineralogist, born in 1750 at Dolomieu (Isère), died in 1801. After some years of military service, he devoted himself to geological researches. He accompanied the French expedition to Egypt, but was shipwrecked on his return off the coast of Taranto, and imprisoned and harshly treated by the Neapolitan Government. Among his works are: *Voyages aux Îles de Lipari* (1783), *Sur le Tremblement de Terre de la Calabre* (1784), *Philosophie minéralogique* (1802).

Dolomite, a mineral, the main constituent of magnesian limestone. It is composed of carbonate of calcium and carbonate of magnesium in equal molecular quantities, and varies from grey or yellowish-white to yellowish-brown. Dolomite is easily scratched with the knife, and is semi-transparent. It effervesces only slightly in *cold* hydrochloric acid. Its rhombohedral crystals are sometimes called *bitter-spar*. A variety is *pearl spar*, which has crystals with curvilinear faces and a pearly lustre.

Dolomites, a group of European mountains, a division of the Alps, in the Trentino, North Italy, and having the Piave and Rienz on the east, the Adige and Eisack on the west. They are named from the prevalence of the mineral dolomite, and present most interesting and picturesque scenery, the peaks being endlessly varied in form. The highest summits are Palle di San Martino (10,968 feet); Sorapiss (10,798 feet), and Monte Tofana (10,715 feet).

Dolphin (*Delphīnus*), a cetaceous animal, forming the type of a family (Delphinidæ) which includes also the beluga or white whale (*Delphinapterus leucas*), the narwhal (*Monodon*), porpoises (*Phocæna*), the ca'ing whale (*Globicephalus melas*), and the killer whale or grampus (*Orca gladiator*). Dolphins inhabit every sea from the equator to the



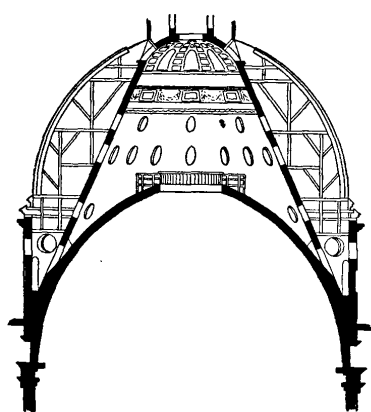
Common Dolphin (*Delphīnus delphis*)

poles; they are gregarious, and swim with extraordinary velocity. The common dolphin (*D. delphis*) measures from 6 to 10 feet in length, has a long, sharp snout with numerous nearly conical teeth in both jaws; its flesh is coarse, rank, and disagreeable, but is used by the Laplanders as food. It lives on fish,

molluscs, &c., and often may be seen in numbers round shoals of herring. The animal has to come to the surface at short intervals to breathe. The blow-hole is of a semilunar form, with a kind of valvular apparatus, and opens on the vertex, nearly over the eyes. The structure of the ear renders the sense of hearing very acute, and the animal is observed to be attracted by regular or harmonious sounds. A single young one is produced by the female, who suckles and watches it with great care and anxiety, long after it has acquired considerable size. Dolphins are associated with many legends, and they figure in armorial bearings.—The name is also commonly but improperly given to fishes (species of *Coryphæna*) belonging to the mackerel family. They abound within the tropics, are about 4 or 5 feet long, very swift in swimming, and are used as food, though said sometimes to be poisonous. The corn aphid (*Siphonophora granaria*) is locally known as the dolphin.

Domain', same as *Demesne*; also applied especially to Crown lands or Government lands.—*Right of eminent domain*, the dominion of the sovereign power over all the property within the State, by which it is entitled to appropriate any part necessary to the public good, compensation being given.

Dombrowski (-brov'skē), Jan Henryk, a Polish general, distinguished in the wars of Napoleon, born in 1755, died in 1818. He supported the rising of the Poles under Kosciusko in 1794. In 1796 he entered the service of France, and at the head of a Polish legion rendered signal services in Italy from 1796 to 1801. He took a distinguished part in the invasion of Russia in 1812, and also in the campaign of 1813. After Napoleon's abdication he returned to Poland, and the year following was made a Polish Senator by Alexander I.



Dome of St. Paul's
Section showing the inner and
outer domes with the conical



The Pantheon, Rome

Dome, a vaulted roof of spherical or other curvature, covering a building or part of it, and forming a common

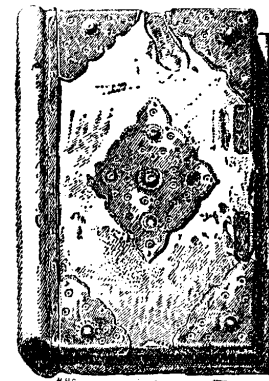
feature in Byzantine and also in Renaissance architecture. Cupola is also used as a synonym, or is applied to the interior, dome being applied to

wall. Diameter inside dome at base 102 feet.

the exterior. (See *Cupola*.) Most modern domes are semi-elliptical in vertical section, and are constructed of timber; but the ancient domes were nearly hemispherical and constructed of stone. Of domes the finest, without any comparison, ancient or modern, is that of the Rotunda or Pantheon at Rome (142½ feet internal diameter and 143 feet internal height), erected in the reign of Augustus, and still perfect. Among others the most noteworthy are St. Sophia at Constantinople (104 × 201 feet), the cathedral of Santa Maria del Fiore at Florence (139 × 310 feet), St. Peter's at Rome (139 × 330 feet), St. Paul's, London (112 × 215 feet), the Hôtel des Invalides (80 × 173 feet) and the Panthéon (formerly the church of St. Geneviève) at Paris (67 × 190 feet). The figures represent the internal diameter and height in English feet. The finest dome in America is that of the Capitol at Washington, built of cast iron.

Domenichino (dō-men-i-kē'nō) (or **Domenico**) **Zampieri**, a distinguished Italian painter, of the Lombard school, born at Bologna on 21st Oct., 1581. He studied under Annibal Carracci, and afterwards went to Rome, where he became painter to Pope Gregory XV. Among his best works are the *Communion of St. Jerome* in the Vatican Museum, the *History of Apollo*, the *Martyrdom of St. Agnes*, and the *Triumph of David*. He died at Naples, 15th April, 1641.

Domesday (or **Doomsday**) **Book**, a book containing a survey of all the lands in England, compiled in the reign and by the order of William the Conqueror. The survey was made by commissioners, who collected the information in each district from a sworn jury consisting of sheriffs, lords of manors, presbyters, bailiffs, villeins—all the classes, in short, interested in the matter. The extent, tenure, value, and proprietorship of the land in each district, the state of culture, and in some cases the number of tenants, villeins, serfs, &c., were the matters chiefly recorded. The survey was completed within a year. Northumberland, Durham, Cumberland, and Westmoreland were not included in the survey, probably for the reason that William's authority was not then (1086) settled in those parts. The *Domesday Book* first appeared in print in 1783 in two folios. In 1816 two supplementary volumes were published. These contained four other records of the same nature. It has been twice republished,



Domesday Book
From the original in
the Public Record
Office, London.

the last time (1861-5) in perfect facsimiles of the original.—BIBLIOGRAPHY: *The Victoria County History*; F. W. Maitland, *Domesday Book and Beyond*; H. Ellis, *General Introduction to Domesday Book*; Birch, *Domesday: a Popular Account*.

Dom'icile, in law, the place where a person has a home or established residence. Domicile is often an important question in determining the efficacy of legal citations, the validity of marriage, the right of succession to property, &c. For some purposes what is called a temporary domicile is sufficient, but in questions of marriage and succession it is the permanent domicile that determines the decision. A permanent domicile may be constituted by birth, by choice, or by operation of the law. To constitute a domicile by choice both actual residence and the intention to make it the permanent home are required. It is a legal principle that the wife takes the domicile of her husband. As a general rule the old domicile, and especially the domicile of origin, continues till a new one has been acquired.

Dom'inant, in music, the fifth tone of the diatonic scale. This tone assumes the character of a key-note itself when there is a modulation into the first sharp remove. Thus G is the dominant of the scale of C, and D the dominant of the scale of G.—*Dominant chord*, in music, that which is formed by grouping three tones, rising gradually by intervals of a third from the dominant or fifth tone of the scale. It occurs almost invariably immediately before the tonic chord which closes the perfect cadence.

Dom'inic, Saint, the founder of the order of the Dominicans, was born in 1170 at Calahorra, in Old Castile. He early distinguished himself by his zeal for the reform of canonical life and by his success as a missionary amongst the Mohammedans. His attention having been directed to the Albigenses in the south of France, he organized a mission of preachers against heresy in Languedoc. In 1215 he went to Rome to obtain the sanction of Pope Innocent III to erect the mission into a new order of preaching friars. His request was only partially granted, and it was the succeeding Pope, Honorius III, who first recognized the importance of a preaching order, and conferred full privileges on the Dominicans. He also appointed Dominic Master of the Sacred Palace or court preacher to the Vatican, an office which is still held by one of the order. Dominic died at Bologna in 1221, and was canonized in 1234 by Pope Gregory IX. St. Dominic is usually considered the founder of the Inquisition, which is supposed to have originated with his mission to the Albigenses; but

his claim is denied, on the ground that two Cistercian monks were appointed inquisitors in 1198.—Cf. A. T. Drane, *The Life of St. Dominic*.

Dominica (dom-i-nē'ka'), a British West India island, so named because discovered by Columbus on a Sunday (Sp. *dominica*), constituting a presidency of the united colony of the Leeward Islands between Martinique on the south, and Guadeloupe on the north. It is about 29 miles in length, north to south, and 12 miles in breadth, east to west; area, 186,436 acres, about 65,000 being under cultivation. It is rugged and mountainous, but it contains many fertile valleys and is well watered. The shores are but little indented, and are entirely without harbours; but on the west side there are several good anchorages and bays. The principal exports consist of sugar, molasses, cocoa, and lime-juice. The imports and exports amount to about £160,000 and £120,000 annually. Dominica was ceded by France to Great Britain in 1763. Roseau is the capital. Pop. 33,863 (including about 420 aboriginal Caribs).

Domin'ical Letter, in chronology, properly called Sunday letter, one of the seven letters of the alphabet, A B C D E F G, used in almanacs, ephemerides, &c., to mark the first seven days of the year and all consecutive sets of seven days to the end of the year, so that the letter for Sunday will always be the same. If the number of days in the year were divisible by seven without remainder, then the year would constantly begin with the same day of the week; but as it is the year begins and ends on the same day, and therefore the next year will begin on the day following, and on leap years two days following, so that the same series is not repeated till after four times seven or twenty-eight years.

Dominican Republic, or **San Domingo**. See *Santo Domingo*.

Domin'icans, called also *predicants*, or *preaching friars* (*prædicatores*), derived their name from their founder, St. Dominic. At their origin (1215, at Toulouse) they were governed by the rule of St. Augustine, perpetual silence, poverty, and fasting being enjoined upon them; and the principal object of their institution was to preach against heretics. Their distinctive dress consists of a white habit and scapular with a large black mantle, and hence they have been commonly known as *Black Friars*. They were almost from the first a mendicant order. They spread rapidly not only in Europe, but in Asia, Africa, and America. In England, where they founded their first house at Oxford, there were fifty-eight Dominican houses at the dissolution of the monasteries, and the Blackfriars locality in London took its name from one of their

establishments. Four Popes, Innocent V, Benedict XI, Pius V, and Benedict XIII, were Dominicans, and the order produced some famous scholars, such as Albertus Magnus and Thomas Aquinas. As fierce opponents and strenuous combatants against any departure from the teaching of the Catholic Church, the Dominicans were entrusted with the conduct of the Inquisition, and became formidable as managers of this ecclesiastical institution, which was committed exclusively to them in Spain, Portugal, and Italy. In 1425 they obtained permission to receive donations, and ceased to belong to the mendicant orders, paying more attention to politics and theological science. With the Franciscans, their great rivals, they divided the honour of ruling in Church and State till the sixteenth century, when the Jesuits gradually superseded them in the schools and courts. They obtained new importance in 1620 by being appointed to the censorship of books for the Church. Amongst notable Dominicans we may mention Savonarola, Las Casas, and Lacordaire, through whose efforts the order was revived in France in the nineteenth century. There are still establishments of the Dominicans both in England and Ireland, twenty-one houses for men and thirteen for women.—
BIBLIOGRAPHY: Caro, *Saint Dominique et les Dominicains*; A. T. Drane, *The Life of St. Dominic, with a Sketch of the Dominican Order*.



Dominican



Dominoes.

Left, Priest wearing domino over surplice, from brass plate at Bury St.

Dom'ino, formerly a dress worn by priests in the winter, which, reaching no lower than the shoulders, served to protect the face and head from the weather. At present it is a masquerade dress worn by gentlemen and ladies, consisting of a long silk mantle with wide sleeves and a masquing hood. The name is also given to a half-mask formerly worn on the face by ladies when travelling or at masquerades.

Dom'inoes, a game played with small flat rectangular pieces of ivory, about twice as long as they are broad. They are marked with spots varying in number. When one player

Edmunds, 1514. Right, Lady wearing domino.

leads by laying down a domino, the next must follow by placing alongside of it another which has the same number of spots on one of its sides. Thus if the first player lays down 6-4, the second may reply with 4-8, or 6-7, &c.; in the former case he must turn in the 4, placing it beside the 4 of the first domino, so that the numbers remaining out will be 6-8; in the latter case he must turn in the 6 to the 6 in like manner, leaving 4-7, to which his opponent must now respond. The player who cannot follow suit loses his turn, and the object of the game is to get rid of all the dominoes in hand, or to hold fewer spots than your opponent when the game is exhausted by neither being able to play. The game was introduced into Europe about the middle of the eighteenth century.

Domitian, or in full Titus Flavius Domitianus Augustus, Roman emperor, son of Vespasian, and younger brother of Titus, was born A.D. 51, and in 81 succeeded to the throne. At first he ruled with a show of moderation and justice, but soon returned to the cruelty and excesses for which his youth had been notorious. He was as vain as he was cruel, and after an ineffective expedition against the Catti, carried a multitude of his slaves, dressed like Germans, in triumph to the city. He executed great numbers of the chief citizens, and assumed the titles of Lord and God. He established the most stringent laws against high treason, which enabled almost anything to be construed into this crime. At length a conspiracy, in which his wife Domitia took part, was formed against him, and he was assassinated in his bedroom A.D. 96.



Domitian

Domrémy la Pucelle (don'-rè-mi la' pu'-sāl), the birth-place of Joan of Arc, a small French village, department of the Vosges, 7 miles N. of Neufchâteau. The house is still shown here in which the heroine was born, and in the neighbourhood is the monument erected to her memory.

Don (ancient **Tanaïs**), a river of Russia, which issues from Lake Ivan-Ozero, in the government of Tula; and flows S.E. through the governments of Riazan, Tambov, Voronej, and the Don Republic, to within 37 miles of the Volga, where a railway connects the rivers. It now turns abruptly S.W. for 236 miles, and falls into the Sea of Azov; whole course nearly 900 miles. The chief tributaries are: right bank, the Donetz and Voronej; left, the Khoper and Manych. Although not admitting vessels of much draught, the Don carries a

large traffic, especially during the spring-floods, and a canal connects it with the Volga system of navigation. It has productive fisheries. The principal port is Rostov.

Don, a river, Scotland, Aberdeenshire, rising near the Banffshire border. It flows tortuously east through the whole breadth of Aberdeenshire, and falls into the North Sea a little to the north of Aberdeen, after a total course of 82 miles. Its salmon fisheries are of considerable value.—Also, a river of Yorkshire, England, which rises near Cheshire, and joins the Ouse after a course of about 70 miles. It is navigable for small craft as far as Sheffield.

Don (Lat. *dominus*, a lord or master), a Spanish title of honour, originally given only to the highest nobility, afterwards to all the nobles, and finally used indiscriminately as a title of courtesy. It corresponds with the Portuguese Dom. During the Spanish occupation it was introduced and became naturalized in some parts of Italy, and was particularly applied to the priests.

Donaghadee (don-ah-a-dē'), a seaport and market town, Ireland, County Down, on the Irish Channel, 16 miles east by north of Belfast. Pop. 4878.

Don'aldson, Sir James, Scottish scholar, born in 1831 at Aberdeen, died in 1915. He was educated at Aberdeen University, and also at Manchester New College, London, and Berlin University. After being rector of Stirling High School, a classical master and rector of Edinburgh High School, he was appointed in 1881 to the Chair of Humanity (Latin) in Aberdeen University. In 1886 he became principal of the United College of St. Salvator and St. Leonard in St. Andrews University, and in 1890 principal of the university. He published a *Modern Greek Grammar for the use of Classical Students* (1853); *Lyra Græca: Specimens of Greek Lyric Poets, with Introduction and Notes* (1854); *History of Christian Literature and Doctrine from the Death of the Apostles to the Nicene Council* (1864-6); *The Ante-Nicene Christian Library* (24 vols., 1867-72, edited jointly with Professor A. Roberts); *The Apostolical Fathers* (1874); *Lectures on the History of Education in Prussia and England* (1874); *The Westminster Confession of Faith* (1905); *Woman: her Position and Influence in Ancient Greece and Rome, and among the Early Christians* (1907); *Addresses Delivered in the University of St. Andrews from 1886 to 1910* (1911). He was knighted in 1907.

Donaldson, John William, a distinguished English philologist, was born in London in 1811, died in 1861. He studied at London University, and at Trinity

College, Cambridge, where he was elected a fellow in 1835. His first work was *The Theatre of the Greeks*, a work showing much erudition. In 1839 he published *The New Cratylus*, which was amongst the earliest attempts to bring the philological literature of the Continent within the reach of the English student. In 1844 appeared the first edition of *Varronianus*, a work on Latin similar in scope to the *Cratylus*. Amongst his other writings are: *Christian Orthodoxy* (1855), *History of the Literature of Ancient Greece*, and grammars of the Hebrew, Greek, and Latin languages.

Donatello (properly, *Donato di Betto Bardi*), one of the revivers of the art of sculpture in Italy, was born at Florence between 1382 and 1387, died at Florence in 1466. His first great works in marble were statues of St. Peter and St. Mark, in the church of St. Michael in his native town, in an outside niche of which is also his famous statue of St. George. Along with his friend Brunelleschi he made a journey to Rome to study its art treasures. On his return he executed for his patrons, Cosmo and Lorenzo de' Medici, a marble monument to their father and mother, which is of high merit. Statues of St. John, St. George, Judith, David, John the Evangelist, and St. Cecilia are amongst his leading works.

Dona'ti's Comet, so called from the Italian astronomer Donati, who first observed it on 2nd June, 1858. Next to the comet of 1811 it was the most brilliant that appeared last century. It was nearest the earth on the 10th Oct., 1858, and was seen until March, 1859. See *Comets*.

Don'atists, one of a body of African schismatics of the fourth century, so named from their founder Donatus, Bishop of Casæ Nigræ, in Numidia, who taught that though Christ was of the same substance with the Father yet that he was less than the Father, that the Catholic Church was not infallible, but had erred in his time and become practically extinct, and that he was to be the restorer of it. All joining the sect required to be rebaptized, baptism by the impure Church being invalid.

Dona'tus, Ælius, a Roman grammarian and commentator, born A.D. 333. He was the preceptor of St. Jerome, wrote notes on Virgil and Terence, and a grammar of the Latin language so universally used in the Middle Ages that 'Donat' became a common term for grammar or primer of instruction.

Donauwörth (don'ou-veurt), a town, Bavaria, at the confluence of the Wörnitz and Danube. It was formerly a free Imperial town, and was stormed by the

Swedes under Gustavus Adolphus in the Thirty Years' War, 1632. Pop. 5500.

Don Beni'to, a town, Spain, province of Badajoz. It has manufactures of woollens, and a trade in cattle, grain, and melons. Pop. 19,212.

Don'caster (*Danum* of the Romans, and *Dona Ceaster* of the Saxons), a municipal borough and market town, England, West Riding of Yorkshire, on the River Don, well built, with straight, broad streets. The parish church with its tower 170 feet high, Christ Church, the town hall, the theatre, are amongst the chief public buildings. It has railway workshops, manufactures of ropes, canvas, and machinery. It has been long celebrated for its annual races, begun in 1615, and held in September. Doncaster was originally a Roman station on the line of Watling Street. Pop. 30,520. It gives its name to a parliamentary division.

Dondrah Head, the southern extremity of the Island of Ceylon. It was the site of the Singhalese capital during part of the seventh century, numerous remains of which are still to be found.

Don'egal (*Dun-na-n Gal*, Fort of the Stranger), a maritime county, Ireland, province of Ulster, bounded north and west by the Atlantic Ocean; area, 1,197,154 acres, of which about a fifth is under crops. The coast is indented with numerous bays; the most remarkable being Lough Swilly. It is the most mountainous county in Ireland, but has some fine fertile valleys. Mount Errigal, the loftiest summit, is about 2460 feet high. The streams and lakes are small, but numerous and abounding in fish. The climate is moist, the subsoil chiefly granite, mica-slate, and limestone, and the principal crops oats, potatoes, and flax. Agriculture generally is in a very backward state. The manufactures are limited, and consist chiefly of linen cloth, woollen stockings, and worked muslin. The fisheries are extensive and valuable, and form the chief employment of the inhabitants of the coast and islands. Grain, butter, and eggs are exported. The minerals include marble, lead, and copper, but are not wrought to advantage. Donegal returns four members to Parliament. Pop. 168,530.

Donetz', a Russian river which rises in the government of Kursk, flows south and east, forming the boundary of several governments, and joins the Don; length, 400 miles. Its basin is rich in coal.

Don'gola, a province of the Anglo-Egyptian Sudan extending on both sides of the Nile from about lat. 18° to lat. 20° N. After having belonged to Egypt, the

rebellion under the Mahdi caused its evacuation by the Egyptian Government, leaving it in the hands of the dervishes, but the Anglo-Egyptian forces under Lord Kitchener again occupied it in 1896. Its chief products are dates, cotton, indigo, and maize. The population is a mixture of Arabs and indigenous Nubians. Its chief town is New Dongola, or El-Ordi, on the Nile. Pop. 20,000.

Doni, a clumsy kind of boat used on the coast of Coromandel and Ceylon; sometimes decked, and occasionally furnished with an outrigger. The donis are about 70 feet long, 20 feet broad, and 12 feet deep, have one mast and a lug-sail, and are navigated in fine weather only.

Donizet'ti, Gaetano, Italian composer, born in 1797 at Bergamo, died 8th April, 1848. He studied music at Bologna under the distinguished Abbé Mattei. His first opera, *Enrico di Borgogna*, was represented at Venice in 1818. In 1822 his *Zoraïde di Granata* gained him the honour of being crowned on the Capitol. In 1830 appeared his *Anna Bolena*, which first, along with *Lucrezia Borgia* and *Lucia di Lammermoor*—the latter his masterpiece—acquired for him a European fame. In 1835 Donizetti was appointed professor of counterpoint at the Royal College of Naples, but removed in 1840 to Paris, bringing with him three new operas, *Les Martyrs*, *La Favorita*, and *La Fille du Régiment*, of which the last two are amongst his most popular productions. Of his other operas none except *Linda di Chamounix* (1842) and *Don Pasquale* (1843) achieved any special triumph. He wrote in all as many as seventy operas.

Donjon, the principal tower of a castle, situated in the innermost court or bailey, which the garrison could make the last line of defence. Its lower part was commonly used as a prison.

Don Juan (Sp. pron. hu'-a'n'), the hero of a Spanish legend which seems to have had some historical basis in the history of a member of the noble family of Tenorio at Seville. According to the legend, Don Juan was a libertine of the most reckless character. An attempt to seduce the daughter of the Governor of Seville brought the indignant father and the profligate don into deadly conflict, in which the former was slain. Don Juan afterwards, in a spirit of wild mockery, goes to the grave of the murdered man and invites the statue of him erected there to a revel. To the terror of Don Juan the 'stony guest' actually appears at the table to bear him away to the infernal world. The legend has furnished the subject for many dramas and operas. The most famous of the latter is Mozart's *Don Giovanni*, which has made the story familiar to

everybody. Amongst the former are *Burlador de Sevilla* by Tellez, *Don Juan, ou Le Festin de Pierre* by Molière (1665), and *The Libertine* by Shadwell. The *Don Juan* of Byron bears no relation to the old story but in name and in the libertine character of the hero. Among the modern authors who have made use of the theme are: Prosper Mérimée, in *Les âmes du purgatoire*; A. Dumas, in *Don Juan de Marana*; Bernard Shaw, in *Man and Superman*; and José Zorilla, in *Don Juan Tenorio*.—Cf. Gendarme de Bévette, *La Légende de Don Juan*.

Donkey-engine, a small engine used in various operations where no great power is required. Thus a donkey-engine is often stationed on the deck of a ship to work a crane for loading and unloading.

Donnay, Maurice Charles, French dramatist, born in Paris in 1859. His first work, *Phryné*, was produced in 1891, and was followed by an adaptation of Aristophanes' *Lysistrata*. His later works include: *Douloureuse* (1897), *Le Torrent* (1898), *L'Affranchie* (1898), *L'Autre Danger* (1901), *Le Retour de Jérusalem* (1903), *Le Ménage de Molière* (1912). He was elected to the Académie Française in 1907.

Donne, John, a celebrated poet and dean of St. Paul's, was the son of a merchant of London, in which city he was born in 1573. He studied both at Oxford and Cambridge. In his nineteenth year he abjured the Catholic religion, and became secretary to the Lord Chancellor Ellesmere, but finally lost his office by a clandestine marriage with his patron's niece, a daughter of Sir George More. The young couple were in consequence reduced to great distress, till his father-in-law relented so far as to give his daughter a moderate portion. In 1610 he wrote for the king the *Pseudo-Martyr*, and in 1612 he published a philosophical poem, *The Progress of the Soul*. By the desire of King James, Donne took orders, and, settling in London, was made preacher of Lincoln's Inn. In 1621 he was appointed Dean of St. Paul's. He was chosen prolocutor to the Convocation from 1623-4. He died in March, 1631, and was interred in St. Paul's. As a poet, Donne stands extremely high, and may be deemed the founder of what Dr. Johnson calls the *metaphysical* class of poets. Abounding in thought, this school generally neglected versification, and that of Dr. Donne was peculiarly harsh and unmusical. His style is quaint and pedantic; but he displays sound learning, deep thinking, and originality of manner. A collection of his poems appeared in 1633, and they include: *The Storm*, *The Calm*, *The Blossom*, and *Upon Parting with his Mistress*. He also wrote *Letters*, *Sermons*, *Essays on Divinity*, and other pieces.—Cf. E. Gosse, *Life and Letters of John Donne*.

Donnybrook, formerly a village, Ireland, now a suburb of Dublin. Its famous fair, instituted under King John in 1204, and which seldom passed off without riot and bloodshed, was abolished in 1855.

Don Republic, a republic in south-eastern Russia, formerly a government of imperial Russia, and known as the Don Cossacks' Territory. The republic was proclaimed in Jan., 1918; its capital is Novo-Tscherkask.

Doombook, a code of laws compiled by King Alfred from the West-Saxon collection of his ancestor Ina, and comprising small portions of the Mercian laws of Offa, and of the Kentish collection of Ethelbert. The code begins with the words: "The Lord spake these words to Moses, and thus said: I am the Lord thy God".—Cf. B. Thorpe, *Ancient Laws and Institutes of England*.

Doon, a river in Ayrshire, Scotland, which issues from the long, narrow Loch Doon, and falls into the Firth of Clyde; length, 27 miles.

Doppler's Principle, the principle by which the motions of heavenly bodies in the line of sight can be determined spectroscopically. When the distance between a body and the observer is being diminished, the number of waves of light of any definite colour received from it per second is increased, just as a swimmer meets more waves in the same time if he swims to meet them. Each colour of light is, therefore, slightly shifted towards the violet end of the spectrum. Thus any radiation or absorption lines seen are shifted towards the violet. If the body's distance is increasing, they are shifted towards the red. By investigation of such displacements, the motions of stars, nebulae, &c., in the line of sight can be measured, the dimensions of the orbits of double-stars gauged, the motions of gases in the solar atmosphere determined, and the rotational velocity of the sun's surface found for latitudes where sun-spots are rarely or never visible. The speed and direction of the sun's motion through space, viz. 12 miles per second towards a point in the constellation Hercules, has also been determined by this means.

Dor, or **Dorr**, the black-beetle, *Geotrūpes stercorarius*, one of the most common British beetles, of a stout form, less than 1 inch long, black with metallic reflections, not to be confounded with the cockroach, commonly but erroneously called black-beetle. It may often be heard droning through the air towards the close of the summer twilight.

Dora, the name of two rivers in Northern Italy, both tributaries of the Po. The Dora Baltea rises on the southern slopes of the Mont Blanc group, and falls,

after a course of about 100 miles, into the Po below Chivasso; the Dora Riparia, about 75 miles long, rises in the Cottian Alps, and joins the Po below Turin.

Do´ran, John, English writer, born 1807, died 1878. He began writing when a mere youth, and produced a great number of books, among them being *Lives of the Queens of England of the House of Hanover*, *Monarchs retired from Business*, *History of Court Fools*, *The Book of the Princes of Wales, Their Majesties' Servants* (a history of the English stage from Betterton to Kean), *Saints and Sinners*, *A Lady of the Last Century* (Mrs. Montague), *London in Jacobite Times*, and *Memories of Our Great Towns*. *In and about Drury Lane* was published after his death. He was editor of *Notes and Queries* at the time of his death.

Dorcas Society (from *Dorcas* mentioned in *Acts*, ix, 36-42), an association generally composed of ladies for supplying clothes to the poor. Frequently the members of the society meet at stated times and work in common. Partial payment is generally required from all recipients except the very poor. Most of these societies are connected with churches of one denomination or another.

Dor´chester, a municipal borough of England, chief town of Dorsetshire, 118 miles s.w. of London. There are large cavalry and infantry barracks a little to the west of the town. The trade consists chiefly in agricultural produce. Dorchester was an important Roman station (*Durnovaria*), and many interesting Roman remains, such as the 'Maumbury Rings' (a Roman amphitheatre), are still to be found in the vicinity. It was a parliamentary borough till 1885, when it was merged in the county. Pop. 9842.

Dordogne (dor-dony), a department of France, which includes the greater part of the ancient province of Périgord, and small portions of Limousin, Angoumois, and Saintonge. Area, 3544 sq. miles, of which about a third is fit for the plough. The chief minerals are iron, which is abundant, slate, limestone, marble, and other stone. Mining, iron manufacture, &c., are carried on to a considerable extent, and there are a number of vineyards. The climate is mild but somewhat changeable. Pop. 396,702.—The river *Dordogne*, principal river of the department, rises on the flanks of the Puy-de-Sancy, flows w.s.w., and, after a course of 290 miles, unites with the Garonne in forming the Gironde.

Dordrecht. See *Dort*.

Doré (dō-rā), Gustave Paul, a prolific French draughtsman and painter, born at Strasbourg, 6th Jan., 1833, died 23rd June, 1883. He studied at Paris, contributing, when only sixteen years of age, comic sketches to the *Journal pour Rire*. He distinguished himself greatly as an illustrator of books. His illustrations of Rabelais (*Rabelais Illustré*), of Perrault's *Tales*, Sue's *Wandering Jew*, Dante's *Divina Commedia*, and Cervantes' *Don Quixote* displayed great fertility of invention, and the fine fantasy of his landscapes and the dramatic effectiveness of his groups acquired for him a European reputation. His illustrations of the Bible, of Ariosto's *Orlando Furioso*, and Milton's *Paradise Lost* are also of high excellence. As a painter he has grandeur of conception and a bold expressive style. Amongst his chief works are: *Christ leaving the Prætorium*, *Paolo and Francesca di Rimini*, *The Flight into Egypt*, and *Mont Blanc*. In later years Doré also won fame as a sculptor.

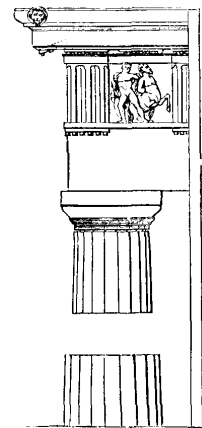
Dore´ma, a genus of plants, nat. ord. Umbelliferæ. *D. ammoniäcum*, a Persian species, yields the ammoniacum of commerce, a milky juice that exudes from punctures on the stem and dries in little 'tears'.

Doria, one of the most powerful families of Genoa, became distinguished about the beginning of the twelfth century, and shared with three other leading families, the Fieschi, Grimaldi, and Spinola, the early government of the Republic. Amongst the older heroes of this family are Oberto Doria, who in 1284 commanded the Genoese fleet which at Meloria annihilated the power of Pisa; Lamba Doria, who in 1298 defeated the Venetian Dandolo at the naval battle of Curzola; Paganino Doria, who in the middle of the fourteenth century distinguished himself by great victories over the Venetians. But the greatest name of the Dorias is that of Andrea, born at Oneglia in 1466, of a younger branch of the family. After serving some time as a condottiere with the princes of Southern Italy, he was entrusted by the Genoese with the reconstruction of their fleet. Disagreement with the Genoese factions drove him to take service with Francis I of France, in which he highly distinguished himself, and in 1527 he took Genoa in the name of the French king. But being displeased with the projects of Francis for reducing Genoa to a place of secondary importance, he went over to the service of Charles V, carrying with him the whole influence and resources of Genoa, and hastened the deliverance of Italy from French domination. He entered Genoa in 1528, re-established order, reorganized the Government, and, although refusing the title of doge, practically controlled its affairs to the end of his life. The country bestowed on him the title of *Father of Peace*. As Imperial Admiral he performed many services for Charles, clearing

the seas of Moorish pirates and assisting the emperor in his expeditions to Tunis and Algiers. In 1547 his authority was threatened by the conspiracy of Fieschi, and he narrowly escaped assassination in the tumult. He died in 1560.

Dorians, one of the three great branches of the Greek nation who migrated from Thessaly southwards, settling for a time in the mountainous district of Doris in Northern Greece and finally in Peloponnesus. Their migration to the latter was said to have taken place in 1104 B.C.; and as among their leaders were certain men reputed to be descendants of Hercules (or Herakles), it was known as the return of the Heraclidæ. The Dorians ruled in Sparta with great renown as a strong and warlike people, though less cultivated than the other Greeks in arts and letters. Their laws were severe and rigid, as typified in the codes of the great Doric legislators Minos and Lycurgus.—The *Doric dialect* was characterized by its broadness and hardness, yet on account of its venerable and antique style was often used in solemn odes and choruses.

Doric Order, in architecture, is the oldest, strongest, and simplest of the three Grecian orders, and the one that is best represented among the remains of ancient Greek architecture. The Doric column is distinguished by its want of a base (in the more ancient examples, at least), by the small number of its flutings, and by its massive proportions, the true Grecian Doric having the height of its pillars six times that of the diameter. The capital was small and simple, and the architrave, frieze, and cornice were rather plain and massive.



Grecian Doric
Order

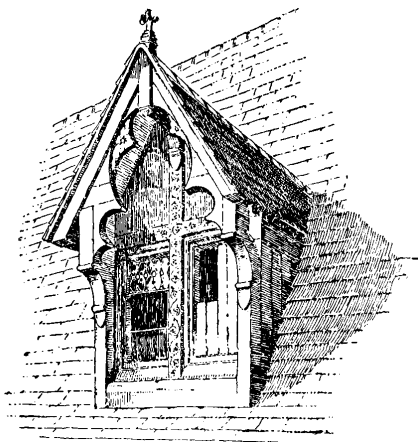
Dorigny (do-rē-nyē), the name of several French painters and engravers. Michael, born in 1617, became professor in the Academy at Paris, and died in 1665. Louis, son of the preceding, was born in 1654, settled in Italy, and died in 1742. Sir Nicholas, brother of Louis, born in 1658 at Paris, was the most celebrated of the three. He spent eight years in engraving the famous cartoons of Raphael at Hampton Court, and was knighted by George I. He died in 1746.

Doris, anciently a small and mountainous region of Northern Greece, at one time the abode of the Dorians. It now forms an eparchy in the nomarchy of Phocis.

Dorking, a town of England, county of Surrey, 22 miles s.s.w. of London, largely consisting of villa residences. Large numbers of fowls, known as

Dorkings, of an excellent breed, having five claws on the foot, are reared here, and sent to the London markets. The breed was introduced both into Gaul and Britain when those countries were subject to the Roman power. Pop. (urban district), 7850.

Dormant State, a state of torpidity in which certain animals pass a portion of the year. In cold and temperate climates this period of long sleep takes place during the winter months, and is properly called *hibernation*. It commences when the food of the animal begins to get scarce, continues for a longer or shorter period, and is deeper or lighter according to the habits and constitution of the animal. Bats, bears, some animals of the rodent order, such as the porcupine, the dormouse, and the squirrel, all the animals belonging to the classes of Amphibia and Reptilia, such as tortoises, lizards, and snakes (frogs, &c.), also many species of molluscs and insects, hibernate more or less completely, retiring to suitable places of concealment—the bat to dark caves, the hedgehog to fern-brakes, and snakes to holes in trees. During hibernation there is a great decrease of heat in the bodies of the animals, the temperature sometimes sinking to 40° or even 20°F., or in general to a point a little above that of the surrounding atmosphere. The respiration as well as the pulsation of the heart is exceedingly slow, and the irritability of the animal often so low that in some cases it can be awakened only by strong electric shocks. With frogs and other amphibia the dormant state is very common, and if the temperature is kept low by artificial means, they may remain dormant for years. The term *æstivation* has been used to describe a similar condition into which certain animals, such as serpents and crocodiles, in tropical countries pass during the hottest months of the year. Plants also present many interesting examples of the dormant state, by which unfavourable periods or conditions are tided over. A seed, for instance, contains a dormant embryo or plantlet, which resumes growth (i.e. germinates) when the temperature rises above a certain level, provided sufficient moisture and air are present. Many lower forms, notably bacteria, are able to form thick-walled cells (spores) that can retain their vitality for a considerable time.



Dormer Window. Abbeville,
Fifteenth Century

Dormer Windows (Fr. *dormir*, to sleep), are windows inserted in the inclined plane of a sloping roof, on a frame rising vertically above the rafters. They are named dormer windows because they are found chiefly in attic bedrooms.

Dormouse
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Dormouse (*Muscardinus avellanarius*)

mammals constituting a special family (Gliridæ or Myoxidæ) allied to rats and mice. They inhabit temperate and warm countries, and subsist entirely on vegetable food. Their pace is a kind of leap, but they have not the activity of squirrels. Whilst feeding they sit upright and carry the food to their mouths with their paws. Dormice pass the winter in a lethargic or torpid state, reviving only for a short time on a warm sunny day, when they take a little of their hoarded stores and then relapse into the dormant state. The squirrel-tailed or 'fat' dormouse (*Myoxus glis*) of the continent was esteemed as an article of diet by the ancient Romans. The common British dormouse (*Muscardinus avellanarius*) is a graceful little creature about 3 inches in length, with a tail 2½ inches long. It feeds on hazel-nuts, eggs, and insects, and constructs a spherical nest. It is not known to occur in Scotland or Ireland.

Dorn'birn, a manufacturing town in Tyrol, about 6 miles from the Lake of Constance. Pop. 28,550.

Dornick, a kind of stout figured linen fabric used for table-cloths, and generally chequered. It derives its name from Doornik, or Tournai, in the Netherlands. The manufacture was brought into Norfolk by the Dutch.

Dornoch (-noh), a seaport and royal burgh of Scotland, county of Sutherland, at the entrance of the Dornoch Firth, the seat of the extinct

bishopric of Caithness. It is one of the Wick district of parliamentary burghs. Pop. 740.—The *Firth* runs inland for about 16 miles between Ross-shire and Sutherlandshire.

Dor'ohoi, a town of Roumania in N.W. Moldavia, near the Austrian frontier. Pop. 11,140.—The department of *Dorohoi* has an area of 1090 sq. miles, and a pop. of 189,789.

Dor'pat (Tartu), a town, formerly in the Russian province of Livonia, now belonging to the Republic of Esthonia. It is situated on the Embach, about 135 miles N.E. of Riga. Dorpat is chiefly remarkable for its university and other educational establishments. It is an ancient town, and was once a member of the Hanseatic Union. Captured by the Russians in 1559, it was ceded to Poland in 1582, was subsequently taken by the Swedes, and in 1704 passed to the Russians, who called it *Yuryev*. The town was occupied by the Germans on 18th Feb., 1918. The vernacular language is Esthonian, but the upper classes speak German. Pop. 44,140.

D'Orsay, Alfred, Count, a dilettante artist and man of fashion, born at Paris 1801, died 1852. When a young man he visited England, and became acquainted with Byron and other literary and fashionable celebrities. He married a daughter of the Earl of Blessington, but after the earl's death a separation took place, and D'Orsay became an inmate of Gore House, which the Countess of Blessington had made the centre of a famous literary coterie. A zealous Bonapartist, he followed Prince Louis Napoleon to Paris in 1849, and enjoyed his favour till his death. Disraeli has described him in his novel *Henrietta Temple*, under the name of 'Count Mirabel'.—Cf. Richard Madden, *Life of Lady Blessington*.

Dorset, Earls of. See *Sackville*.

Dorset, or **Dorsetshire**, a maritime county in the south of England, having on the south the English Channel; area, 627,265 acres, over 490,000 being under crop. The general surface of the county is undulating; its principal elevations being chalk hills known as the North and South Downs, upon which immense flocks of sheep are pastured. On the south, on the borders of Hampshire and along part of the sea-coast, is a heathy common. A great part of the county is in grass, and dairy husbandry is extensively carried on. Neither coal nor ores of any kind are found, but the quarries yield the well-

known Portland stone. Pipe-clay, plastic clay, and potter's clay also abound. The principal manufactures are those of flax, canvas, duck, &c., also silk and woollens. The fish frequenting the coast are of various kinds, but mackerel is the most abundant. Near the mouth of Poole harbour is a prolific oyster bank. The principal rivers are the Stour, the Frome, and the Piddle. The county has four parliamentary divisions, with a member for each. Dorchester is the county town. Other towns are Bridport, Poole, and Weymouth. Pop. 223,274.

Dorsetshire Regiment, The, once known as the East Middlesex, dates from 1702, and is intimately associated with Gibraltar, which it twice defended during siege (1727 and 1779-82). For its services in India under Clive it bears the motto *Primus in Indis*. It later took part in the relief of Ladysmith, and during the European War suffered heavily on the Western front, being also represented at Kut and Ramadie.

Dorste'nia, a genus of plants, nat. ord. Moraceæ, found in tropical America. They have their naked flowers buried in a flat, fleshy, somewhat concave receptacle. *D. Contrayerva* and other species have a stimulant and tonic rhizome, which is used medicinally under the name of *contrayerva*.

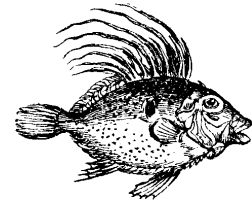
Dort, or **Dordrecht**, a town, Holland, province of South Holland, 14 miles S.E. of Rotterdam, on the Merwede, an arm or part of the Maas, and on an island separated from the mainland by an inundation in 1421. It is an old town, founded in 1018 by Count Dietrich III of Holland, with a fine Gothic Church (Groote Kerk, 'Great Church'), a good town-house and museum. It was formerly of more importance than now, but it still carries on an extensive trade, being not only near the sea, but by the Rhine, the Maas, and other water communications, connected with an immense extent of inland territory. Pop. 47,300.

Dort, Synod of, an assembly of Protestant divines convoked at Dort on 13th Nov., 1618, dissolved on 9th May, 1619. Besides the Dutch and Walloon divines, it included representatives from England, Scotland, Switzerland, and part of Germany, in all about sixty-two native and twenty-four foreign deputies. The Synod was convoked principally for the sake of crushing the Arminian party, and extreme measures were taken to prevent that party being represented in the assembly or having a free voice there.

The result was the condemnation of the Arminians and the dogmatic establishment of Calvinism in the Reformed Church. The Synod also set on foot the Dutch translation of the Bible known as the Dort Bible.

Dort'mund, a city of Prussia, province of Westphalia, on the Emscher, 47 miles N.N.E. of Cologne, starting-point of an important canal to the lower Ems. It has rapidly increased in recent years, being the centre of important railway systems, having extensive coal-mines in the vicinity, and active manufactures of iron, steel, machinery, and railway plant. There are also a number of breweries, potteries, tobacco factories, and chemical works. It was once a free Imperial Hanseatic town, and the seat of the chief tribunal of the Vehme. Pop. 214,226.

Dory, or **John Dory** (*Zeus faber*), a bony fish which is the type of a special family (Zeidæ), and is celebrated for the delicacy of its flesh. It seldom exceeds 18 inches in length, and is yellowish-green in colour with a blackish spot on each side, which, according to an old superstition, is the mark of St. Peter's forefinger and thumb; another claimant for this honour is the haddock.



Dory (*Zeus faber*)

The dory is found on the Atlantic shores of Europe and in the Mediterranean. The name John Dory is supposed to be derived from the Fr. *jaune dorée*, golden yellow.

Dosith'eans, an ancient sect among the Samaritans, so called from their founder Dositheus, who was a contemporary and associate of Simon Magus, and lived in the first century of the Christian era. They rejected the authority of the prophets, believed in the divine inspiration of their founder, and had many superstitious practices.

Dosso Dossi, Giovanni di Lutero, Italian painter of the Ferrara school; born 1479, died 1542. He was much honoured by Duke Alfonso of Ferrara, and immortalized by Ariosto (whose portrait he executed in a masterly manner) in his *Orlando*. Modena, Ferrara, and Dresden possess most of his works. *Circe in the Woods* is in the Borghese Gallery, and *St. Sebastian* in the Brera at Milan.

Dost Mohammed Khan, born about 1790, a successful usurper who obtained possession of the throne of Afghanistan after the flight of Mahmud

Shah in 1818. He ruled with very great ability, and although driven from his throne by a British army, was ultimately restored and became a steady supporter of British power in the East. He died in 1863.

Dostoiev'sky, Feodor Mikhailovitsh, a famous Russian novelist, born 1821, died 1881. After serving as an officer of engineers he devoted himself to literature, but becoming connected with the communistic schemes of Petrashevsky, he was condemned to death. At the last moment, when Dostoievsky was already on the scaffold, the sentence was commuted, and he was banished to the mines of Siberia. Pardoned by Alexander II, he returned in 1856 to resume his literary activity. His first novel, *Poor People*, came out in 1846. Among his works that have appeared in English are: *Crime and Punishment*; *Injury and Insult*; *The Friend of the Family*; *The Gambler*; *The Idiot*; *Prison Life in Siberia*; *Uncle's Dream*; *The Permanent Husband*; *The Brothers Karamzov*; *Letters from the Underworld and Other Tales*. There is a complete edition of his novels by C. Garnett, 1912.—**BIBLIOGRAPHY**: J. A. T. Lloyd, *A Great Russian Realist*; J. M. Murray, *F. Dostoievsky: a Critical Study*.

Dot'terel (*Eudromias morinellus*), a species of plover which breeds in the north of Europe, and returns to the south for the winter. In Scotland it appears in April and leaves in August, the young being hatched in July, but comparatively few breed in the British Islands. It is found all over Europe and Northern Asia. The dotterel is about 8 inches long. Contrary to the general rule the hen is larger and more brightly coloured than the cock, and the latter performs most of the duties of incubation.

Douai (dö-ā), town, France, department of Nord, on the Scarpe, 18 miles south of Lille. It is one of the oldest towns in France, of which it became part by the Treaty of Utrecht. It is strongly fortified, has a fine town-house, several handsome churches, an academy of arts and law, a lyceum, museum and public library, Benedictine college, and hospital; a cannon foundry, linen manufactories, machine-works, and tanneries. There was long here a college for British Roman Catholic priests, the most celebrated of its kind, founded by Cardinal Allen in 1568. Douai was captured by the Germans during the European War, and retaken by the Allies in Oct., 1918. It received the Cross of the Legion of Honour in Sept., 1919. Pop. 36,314.

Douai Bible, the English translation of the Bible used among English-speaking Roman Catholics, and executed by divines connected with the English College at Douai. The New Testament was published in 1582 at Rheims, the Old during 1609-10 at Douai, the translation being based on the *Vulgate*. Various revisions have since materially altered it.

Douarnenez (du''-a' r-nè-nā), a seaport, France, Finistère, on a beautiful bay of the same name, 13 miles north-west of Quimper. It depends chiefly on the sardine fishery. Pop. 13,753.

Double Fertilization. See *Embryo-sac*.

Double-flowering, the development, often by cultivation, of the stamens and pistils of flowers into petals, by which the beauty of the flower is enhanced and its reproductive powers sacrificed.

Double-insurance, the effecting of two insurances upon the same goods. In marine insurance it is lawful for a shipper to insure his goods twice, but only to give an additional security in the event of the failure of the first underwriters. In the event of a loss it is ultimately divided among the underwriters in the ratio of the risks they have taken.

Double-stars, or **Binary Stars**, stars which are so close together that they appear as one to the naked eye, but are seen to be double when viewed through a telescope. One of these stars may revolve about the other, or, more accurately speaking, both revolve round the common centre of gravity.

Doublet, a close-fitting garment, covering the body from the neck to a little below the waist. It was introduced from France into England in the fourteenth century, and was worn by both sexes and all ranks until the time of Charles II, when it was superseded, as far as men were concerned, by the coat and waistcoat. The garment got its name from being originally lined or wadded for defence.

Doublet, in lapidary work, a counterfeit stone composed of two pieces of crystal, with a colour



1, Doublet, time of Edward

between them, so that they have the same appearance as if the whole substance of the crystal were coloured.

IV. 2,3, Doublets, time of Elizabeth. 4, Doublet, time of Charles I.

Double-vault, in architecture, one vault built over another so that a space is left between the two. It is used in domes or vaulted roofs when the external and internal arrangements require vaults differing in size or shape, the outer and upper vault being made to harmonize with the exterior of the building, the inner or lower with the interior.

Doubloon, a gold coin of Spain and of the Spanish American States, originally double the value of the pistole. The doubloon of Spain was subsequently equivalent to about a guinea sterling. The doubloon of Chile was worth about 18s. 9d. sterling; that of Mexico, £3, 4s. 8d.

Doubs (dö), a department of France, having Switzerland on its eastern frontier. Its surface is traversed by four chains of the Jura. The temperature is variable, and the climate somewhat rigorous. About a third of the land is arable, but much the greater part is covered with forests. Maize, potatoes, hemp, flax are the principal crops. Much dairy produce is made into Gruyère cheese. The minerals include iron, lead, and marble. Pop. 284,975. —The River Doubs rises in the department to which it gives its name, flows first north-east, then north-west till it joins the Saône at Verdun-sur-Saône; length, 250 miles.

Douche (dösh), a jet or current of water or vapour directed upon some part of the body; employed in bathing establishments. When water is applied, it is called the *liquid douche*, and when a current of vapour, the *vapour douche*.

Douglas (dug'las), a family distinguished in the annals of Scotland. Their origin is unknown. They were already territorial magnates at the time when Bruce and Baliol were competitors for the crown. As their estates lay on the borders they early became guardians of the kingdom against the encroachments of the English, and acquired in this way power, habits, and experience which frequently made them formidable to the Crown. We notice in chronological succession the most distinguished members of the family. James, son of the William Douglas who had been a companion of Wallace, and is commonly known as the Good Sir James, early joined

Bruce, and was one of his chief supporters throughout his career, and one of the most distinguished leaders at the battle of Bannockburn. He was called 'Black Douglas' from his swarthy complexion. He fell in battle with the Moors while on his way to the Holy Land with the heart of his master, in 1331.—Archibald, youngest brother of Sir James, succeeded to the regency of Scotland in the infancy of David. He was defeated and killed at Halidon Hill by Edward III in 1333.—William, son of the preceding, was created first earl in 1357. He recovered Douglasdale from the English, and was frequently engaged in wars with them. He fought at the battle of Poitiers and died in 1384.—James, the second earl, who, like his ancestors, was constantly engaged in border warfare, was killed at the battle of Otterburn in 1388. After his death the earldom passed to an illegitimate son of the Good Sir James, Archibald the Grim, Lord of Galloway.—Archibald, son of Archibald the Grim and fourth earl, was the Douglas who was defeated and taken prisoner by Percy (Hotspur) at Homildon 14th Sept., 1402. He was also taken prisoner at Shrewsbury 23rd July, 1403, and did not recover his liberty till 1407. He was killed at the battle of Verneuil, in Normandy, in 1427. Charles VII created him Duke of Touraine, which title descended to his successors. He was surnamed 'The Tyneman', or loser, on account of his many misfortunes in battle.—William, sixth earl, born 1422, together with his only brother David was assassinated by Crichton and Livingstone at a banquet to which he had been invited in the name of the king, in Edinburgh Castle, on 24th Nov., 1440. Jealousy of the great power which the Douglasses had acquired from their possessions in Scotland and France was the cause of this deed.—William, the eighth earl, a descendant of the third earl, restored the power of the Douglasses by a marriage with his cousin, heiress of another branch of the family; was appointed Lord-Lieutenant of the kingdom, and defeated the English at Sark. Having entered into treasonous league, he was invited by James II to Stirling and there murdered by the king's own hand, 22nd Feb., 1452.—James, the ninth and last earl, brother of the preceding, took up arms with his allies to avenge his death, but was finally driven to England, where he continued an exile for nearly thirty years. He entered Scotland on a raid in 1484, but was taken prisoner and confined in the abbey of Lindores, where he died in 1488. His estates, which had been forfeited in 1455, were bestowed on the fourth Earl of Angus, the 'Red Douglas', the representative of a younger branch of the Douglas family, which continued long after to flourish. The fifth Earl of

Angus, Archibald Douglas, was the celebrated 'Bell-the-Cat', one of whose sons was Gawin Douglas the poet. He died in a monastery in 1514. Archibald, the sixth earl, married Queen Margaret, widow of James IV, attained the dignity of regent of the kingdom, and after various vicissitudes of fortune, having at one time been attainted and forced to flee from the kingdom, died about 1560. He left no son, and the title of Earl of Angus passed to his nephew David. James Douglas, brother of David, married the heiress of the Earl of Morton, which title he received on the death of his father-in-law. His nephew, Archibald, eighth Earl of Angus and Earl of Morton, died childless, and the earldom of Angus then passed to Sir William Douglas of Glenbervie, his cousin, whose son William was raised to the rank of Marquess of Douglas. Archibald, the great-grandson of William, was raised in 1703 to the dignity of Duke of Douglas, but died unmarried in 1761, when the ducal title became extinct, and the marquessate passed to the Duke of Hamilton, the descendant of a younger son of the first marquess. The line of Angus or the Red Douglas is now represented by the Houses of Hamilton and Home, who both claim the title of Earl of Angus.—BIBLIOGRAPHY: David Hume of Godscroft, *A History of the House of Douglas and Angus*; Sir H. Maxwell, *A History of the House of Douglas*.

Douglas, Gawin, an early Scottish poet of eminence. He was the son of Archibald, fifth Earl of Angus, and was born at Brechin about 1474. He received a liberal education, commenced at home and completed at the University of Paris. On returning to Scotland he took orders in the Church, and ultimately became Bishop of Dunkeld, through the influence of his nephew, the sixth Earl of Angus, who married Queen Margaret, widow of James IV. He died of the plague in 1522 in London, where he had been obliged to take refuge on account of political commotions. He translated Virgil's *Æneid* into verse with much spirit and elegance, prefixing original prologues to the different books of the original. This was the first poetical translation into English of any classical author. It was written about 1512, and first published in 1553. He also wrote *The Palace of Honour* and *King Hart*, both allegorical poems.—Cf. J. H. Millar, *Literary History of Scotland*.

Douglas, Sir Howard, Baronet, G.C.B., a British general, born in 1776, the son of Admiral Sir Charles Douglas. He served in Spain in the Peninsular

War, and acquired much reputation by his writings on military subjects, especially by his *Military Bridges and the Passage of Rivers* (1816), and *Treatise on Naval Gunnery* (1819). From 1823 to 1829 he was Governor of New Brunswick, and from 1835 to 1840 Lord High Commissioner of the Ionian Islands. He attained the rank of general in 1851, and died in Nov., 1861.

Douglas, Stephen Arnold, American politician, born in Vermont, 1813, died 1861. Having gone to Jacksonville, Illinois, he became an attorney, was appointed Attorney-General for the State, and in 1843 was elected a member of the United States House of Representatives. In 1847 he was elected to the Senate, and by re-election was a member of this body till his death. He was especially prominent in connection with the question as to the extension of slavery into new states and territories, which he maintained was a matter to be settled by the people of the respective states or territories, and not by Congress. He was a presidential candidate in 1860, when Lincoln was elected.

Douglas, Sir William Fettes, painter, born in Edinburgh 1822, died in 1891. He was educated at the High School in that city, spent ten years in a bank before finally deciding (in 1847) upon the artist's profession. In 1851 he was elected an Associate of the Royal Scottish Academy, and three years later a full member. In 1877 he became Curator of the National Gallery of Scotland, resigning the post in 1882 on his election as president of the Royal Scottish Academy. Among the finest of his early pictures are: *Bibliomania* (1852, in the National Gallery), *The Ruby Ring* (1853), *The Alchemist* (1855), *Hudibras and Ralph visiting the Astrologer* (1856), and *The Rosicrucian* (1856), many of these showing much of the Pre-Raphaelite spirit, with abundance of detail. After 1870 he devoted himself rather to landscape, and his *Stonehaven Harbour* and *A Fishing Village* (1874-5) are perhaps his masterpieces. He was knighted in 1882.

Douglas (dug'las), capital of the Isle of Man, is situated on the south-east coast, on a beautiful semicircular bay. It is frequented by immense numbers of visitors during the summer. Among the objects of interest are the House of Keys, the custom-house, the extensive breakwater, and the promenade. Pop. 21,192.

Douglass, Frederick, American lecturer and journalist, was born at Tuckahoe, in Maryland, about 1817. His father was a white man, but his mother being a negro slave, he was, according to the law, reared as a slave. In 1832 he was purchased by a Baltimore shipbuilder, but made his escape in 1838. As he had taught himself to read and write, and showed talent as an orator, he was employed by the Anti-slavery Society as one of their lecturers. In 1845 he published his autobiography, and afterwards made a successful lecturing tour in England. In 1870 he started a journal entitled *The New National Era*; in 1871 he was appointed secretary of the Commission to Santo Domingo; in 1877 Marshal for the district of Columbia, then Commissioner of Deeds, and eventually Minister to Hayti. He died in 1895.

Doulton (dōl'tun), Sir Henry, 'the greatest potter of the nineteenth century', born in Lambeth in 1820, died in 1897. On leaving University College School, in 1835, he joined his father, who had carried on a small pottery since 1815, and began by perfecting himself in all the mechanical processes then used by potters. He scored his first distinct success in 1846 with glazed drain-pipes, and in 1851 and 1862 the firm obtained medals for stoneware vessels and chemical apparatus. At the South Kensington Exhibition in 1871 a striking display was made of the new Doulton artistic ware. Doulton exhibited at Vienna in 1873, and at Paris five years later, when he was made a chevalier of the Legion of Honour. He then established a school of artists in connection with his manufactory, with the object of promoting originality in design. He received the Albert gold medal of the Society of Arts in 1885, and was knighted at the Jubilee two years later.

Doum Palm (dōm), a palm tree, *Hyphæne thebaica*. It is remarkable, like the other species of the genus, for having a repeatedly branched stem. Each branch terminates in a tuft of large fan-shaped leaves. The fruit is about the size of an apple; it has a fibrous mealy rind, which tastes like gingerbread (whence the name *gingerbread tree* sometimes applied to this palm), and is eaten by the poorer inhabitants of Upper Egypt, where it grows. An infusion of the rind is also used as a cooling beverage in fevers. The seed is horny, and is made into small ornaments. Ropes are made of the fibres of the leaf-stalks.

Doune (dön), a police borough in W. Perthshire, Scotland, on the River Teith, 9 miles north-west of Stirling, once famous for its manufacture of Highland pistols and sporrans. The old ruined castle—an imposing structure now partially repaired—is described in Scott's *Waverley*. Pop. 890.

Douro (dö' rō), one of the largest rivers of the Spanish Peninsula, which, flowing west, traverses about one-half of Spain and the whole of Portugal, and, after a course of 500 miles, falls into the Atlantic 3 miles below Oporto. It is navigable for small vessels for about 70 miles.

Dove. See *Turtle-dove* and *Pigeon*.

Dove (dōv), a river, England, Derbyshire, which, after a course of 39 miles through highly picturesque scenery, falls into the Trent.

Dove-cotes. Pigeon-keeping to provide a food-supply is a practice of considerable antiquity, and dove-cotes are found in many quarters of the world. Those introduced into Britain by the Normans were modelled on the Roman *columbarium*, a massive circular structure, lined with nest-holes, and having a domed roof. A fine example of this type, built 1326, survives at Garway, Herefordshire. Till towards the end of the sixteenth century, these buildings, numbering some 26,000, formed items of manorial privilege in England, and were long confined to Scottish baronies. About this period square and octagonal forms became common, a fine brick specimen of the latter style remaining at Whitehall, Shrewsbury. In Scotland typical 'doo'-cots' exist in the Edinburgh suburbs of Liberton and Corstorphine. That at Liberton, a type common in Scotland but rare in England, is oblong, with lean-to roof and two compartments—probably to avoid disturbing the whole flock when 'squabs' were taken from the nests. Dove-cotes fell generally into disuse when the introduction of 'roots' insured the winter feeding of farm-stock and a consequent steady supply of



Doum Palm (*Hyphaene thebaica*)

fresh meat; but their antiquarian interest and frequent beauty call for the careful preservation of existing specimens.

Dove Deities. The cult of the dove is of great antiquity. In Crete and at Mycenæ, and in the area of Hittite control in Asia Minor, it was connected with the Mother-goddess. The bird appears in archaic clay figurines from Phœnicia, Rhodes, Delos, Athens, and Etruria. Whether or not the dove cult originated in Crete or Asia Minor is uncertain. Some think it is of Egyptian origin, but there is no trace of a dove goddess in Nilotic art. In the love poems found in Egyptian tombs, however, the dove is referred to, being in one case addressed by a lover, who asks it if love is to be denied to her; she then tells the dove that she has found her chosen one and is happy by his side. The pigeon was protected in Egypt, and is still regarded as a 'luck bird', and it may have been connected with ancient folk religion. In Babylonia and Assyria the dove was associated with the goddess Ishtar, but not specially during the earlier periods. The Allatu bird is, however, referred to in the Gilgamesh epic, and Pinches has translated the suggestive reference in an Ishtar hymn, *Like a lonely dove I rest*. In another hymn the worshipper moans like a dove. According to Diodorus, the famous Assyrian Queen Semiramis, who was abandoned after birth, was protected and fed by doves. In Crete two forms of the Great Mother-goddess, who was an Aphrodite in one of her phases, were the snake goddess and the dove goddess. Two doves appear on a model of a Mycenæan shrine. The dove is associated with the Hittite goddess at Marash, Yarre, and Fraktin. It is sometimes found with the nude Syrian goddess. Lucian states, in reference to a Syrian cult, that the dove is the holiest bird (*De Dea Syria*, chapter liv), and that there was a golden dove in the temple, but nothing was known regarding its origin, some referring it to Dionysus, some to Deukalion, and some to Semiramis (a name said to mean mountain dove) (chapter xxxiii). Ælian tells that the dove was the sacred companion of Astarte (*Hist. Nat.*, iv, 2). Like the Egyptians, the Semites regarded the pigeon with veneration. The Hebrews sacrificed it on special occasions (*Num.* vi, 10; *Lev.* xiv, 4, 49). Noah sent out a dove from the ark (*Gen.* viii, 8). A prophet mourns as a dove (*Is.* xxxviii, 14). The dove is 'silly' (*Hos.* vii, 11). Doves were sold in the temple (*Mark*, xi, 15). The Spirit of God appears as a dove (*Matt.* iii, 16). According to Herodotus, the Persians drove away white pigeons, connecting them with leprosy (*Book I*, 139). At Dodona, the famous

sanctuary in Epirus, auguries were taken from the moaning of doves in the tree-tops, and the priestesses of Zeus were called doves (*Peleiai*). Doves and pigeons were mystical birds in the British Isles. In England it was believed that one could not die on a bed of pigeon feathers, and the dying person had, therefore, to be removed from one so that the suffering might not be prolonged. The early Christian saints revered the dove. St. Gregory the Great is shown with a dove on his shoulder, and the emblem of St. Remigius is a dove with an oil-cruse in its beak. A snow-white dove with golden bill was wont to sit on the head of St. Mungo, the patron saint of Glasgow. Stories used to be told of sick persons recovering on seeing a white dove on an altar. In the folk lore attached to the memory of Michael Scott, the dove is connected with heaven, and the raven with hell. St. Columba's name signifies 'Dove'. Dove-cotes were connected with old churches, especially in England.

Dō'ver, a municipal borough of England, county of Kent, 67 miles south-east of London. It lies on the coast of the Straits of Dover, and is 21 miles distant from Calais on the French coast. It is an important railway terminus, and as a port for mail and packet service with the Continent has a large passenger traffic. Ship-building, sail-making, and fisheries are carried on. There are two docks and a tidal harbour; an outer harbour of 70 acres, enclosed by a new pier and the extended Admiralty Pier, completed in 1871. Very extensive harbour improvements, begun in 1893, were carried out in subsequent years. The celebrated castle stands on a high chalk cliff. Dover is the chief of the Cinque Ports, and has extensive barracks. A parliamentary borough till 1918, Dover now gives its name to a parliamentary division of Kent. Dover was frequently raided by German aviators during the European War. Pop. 43,645.

Dover, a city of the United States, in New Hampshire. It is situated on both sides of the Cochecho, which has here a fall of over 30 feet, affording abundant water-power for the large iron and cotton manufactories. Pop. 13,247.

Dover, Straits of, the narrow channel between Dover and Calais which separates Great Britain from the French coast. At the narrowest part it is only 21 miles wide. The depth of the channel at a medium in the highest spring-tides is about 25 fathoms. On both the French and English sides the

chalky cliffs show a correspondence of strata which leaves no room for doubt that they were once united, a fact which is clearly shown by many other proofs.

Dover's Powder, a preparation frequently used in medical practice to produce perspiration. It consists of 1 grain of opium, 1 of ipecacuanha, and 8 of sulphate of potash in every 10 grains, which constitute a full dose. It is named after Thomas Dover, an English physician of the eighteenth century.

Dovre-Fjeld (dō-vre-fyel), an assemblage of mountain masses in Norway, forming the central part of the Scandinavian system, and extending as a plateau 2000 feet high E.N.E. from lat. 62° N. to lat. 63°. It is generally composed of gneiss and mica schist. One of the mountains belonging to it is Snehaetta, 7620 feet.

Dow, Gerard, an eminent painter of the Dutch school, was the son of a glazier, and born at Leyden in 1613. He studied under Rembrandt, and united his master's manner in chiaroscuro with the most minute finish and delicacy. Among his pictures, generally of small size and mostly scenes of family life, are: *The Evening School*, *Young Mother*, *Woman Sick with Dropsy*, and *The Bible Reader*. Dow died in 1675.

Dowden, Edward, English critic, historian, and educator, was born at Cork in 1843, died in 1913. He studied at Queen's College, Cork, and Trinity College, Dublin, where he gained great distinction, especially in English and Philosophy; and in 1867 he was elected to the professorship of English literature in the university. He was the first Taylorian lecturer at Oxford University in 1889, and held the Clark lectureship in English literature at Trinity College, Cambridge, from 1893 to 1896. Besides contributing to periodicals, Professor Dowden published various works on literary subjects, in particular: *Shakspeare: his Mind and Art* (1875); *Shakspeare Primer*; *Studies in Literature*; *Southey*; *Southey's Correspondence with Caroline Bowles*; *Life of Shelley* (2 vols., 1886), the chief authority on the poet's life, being founded on papers in the possession of the Shelley family; *Wordsworth's Poetical Works* (1892-3); *Introduction to Shakspeare* (1893); *New Studies in Literature* (1895); *The French Revolution and English Literature* (lectures delivered at Princeton College, New Jersey, in 1896); *History of French Literature* (1897); *Robert Browning* (1904); *Michel de*

Montaigne (1905); *Essays, Modern and Elizabethan* (1910). A volume of poems by him appeared in 1876, and his collected *Poetical Works and Letters* appeared in 1914.

Dower (Fr. *douaire*, Lat. *dos*, dower), in English law, is the right which a wife (not being an alien) has in the freehold lands and tenements of which her husband dies possessed and undisposed of by will. By common law this right amounts to one-third of his estate during her life; by local custom it is frequently greater. Where the custom of *gavelkind* prevails, the widow's share is a half, and that of *free-bench* gives her the whole or a portion of a copyhold, according to the custom of the manor. The term is also applied to the property which a woman brings to her husband in marriage, but this is more correctly *dowry*.

Dowie, John Alexander, religious impostor, born at Edinburgh in 1847, died in 1907. Educated at the university of his native town, he joined his family in Sydney, Australia, and entered the ministry as clergyman of the Congregational denomination. In 1878 he started evangelistic work, maintaining that it was wrong to take a minister's salary. In 1882 he established a tabernacle at Melbourne, and began to practise faith-healing. He then came to the United States, where he organized his own Church, establishing it in 1901 at Zion City, 42 miles from Chicago, on the shore of Lake Michigan. He styled himself 'Elijah II', and 'the First Apostle of the Lord Jesus Christ, and General Overseer of the Catholic Apostolic Church'. Suspended in 1906 by his congregation of Zion City, who accused him of tyranny, polygamy, and misuse of funds, he died shortly afterwards.

Dowlais, a town of South Wales, Glamorganshire, included in the parliamentary borough of Merthyr Tydfil, from which it is distant 1½ miles north-east, with important iron- and steelworks. Pop. 18,112.

Dowlas, a kind of coarse linen formerly much used by working people for shirts; this use of it is now generally superseded by calico.—Cf. Shakespeare, *1st Henry IV*, iii, 3.

Dowletabad. See *Daulatabad*.

Down, a county of Ireland, in Ulster, bounded on the north by Belfast Lough and on the east by the Irish Sea; area, 610,730 acres, of which over

five-sixths are productive. Down is copiously watered by the Rivers Bann, Lagan, and Newry, and has numerous small lakes. The surface is very irregular, and in parts mountainous, Slieve Donard, in the Mourne Mountains, being 2796 feet high. Agriculture is in a flourishing condition, oats, wheat, flax, turnips, and potatoes being the principal crops. The native breed of sheep is small, but valued for the delicacy of its mutton and the fine texture of its wool. The principal manufactures are linen and muslin. The fisheries on the coast, principally cod, haddock, and herring, are considerable. The county has five parliamentary divisions, each returning a member. The county town is Downpatrick; others are Newry, Newtownards, Bangor, and Banbridge. Pop. 204,303.

Downing College, one of the colleges of the University of Cambridge, chartered in 1800 and opened in 1821. Its founder was Sir George Downing, a Cambridgeshire gentleman.

Downing Street, a street in London, leading from Whitehall. The name is used as a synonym for the British Government, the Foreign Office and Colonial Office being located in it. No. 10 is the official residence of the Prime Minister, and No. 11 that of the Chancellor of the Exchequer.

Downpatrick, a market town of Ireland, county town of Down, 21 miles S.E. of Belfast. It is the seat of the diocese of Down, Connor, and Dromore, has a cathedral, and is celebrated as the supposed burial-place of St. Patrick. Pop. 3200.

Downs, a term given to undulating grassy hills or uplands, specially applied to two ranges of undulating chalk hills in England, extending through Surrey, Kent, and Hampshire, known as the North and South Downs. The word is sometimes used as equivalent to *dunes* or sand-hills.

Downs, The, a celebrated roadstead for ships, extending 6 miles along the east coast of Kent in England, protected on the seaward side by the Goodwin Sands.

Downton, a town of England, in Wilts., on the Avon, 6½ miles S.S.E. of Salisbury; an ancient place, with a large cruciform church in the Norman and later styles, an old earthwork mound called 'the Moat', and an agricultural college. Pop. 1933.

Doxology (from Gr. *doxa*, praise, glory, and *logos*), a set form of words giving glory to God, and especially a name given to two short hymns distinguished by the title of *greater* (Glory be to God on high, &c.) and *lesser* (Glory be to the Father, &c.). Both the doxologies have a place in the Church of England liturgy, the latter being repeated after every psalm, and the former used in the communion service.

Doyen, Eugène Louis, famous French surgeon, born at Rheims in 1859, died at Paris in 1916. He made numerous discoveries in gynæcological surgery, and in 1895 established a private clinic, where many French and foreign surgeons came to study under him. His surgical methods were adopted, although his claim to have discovered the germ of cancer has been disputed. In 1898 he received the degree of LL.D. from the University of Edinburgh, where he introduced the method of teaching surgery by means of the cinematograph. His works include: *La maladie et le médecin* and *Le Cancer*.

Doyle, Sir Arthur Conan, English novelist, a nephew of Richard Doyle, born at Edinburgh, 1859, studied medicine, and for some years practised, but gave up the profession for that of literature. In 1887 he produced *A Study in Scarlet*, in which he created the detective Sherlock Holmes. Among his other books are: *Micah Clarke*, *The Sign of Four*, *The White Company*, *The Adventures of Sherlock Holmes*, *Brigadier Gerard*, *The Great Boer War*, *The Crime of the Congo*, *The Lost World*, *The Poison Belt*, *The British Campaign in France and Flanders*, *The New Revelation*, and *The Vital Message*.

Doyle, Sir Francis Hastings Charles, English poet, born 21st Aug., 1810, died 8th June, 1888, was the son of Major-General Sir Francis Hastings Doyle, first baronet, succeeding his father in the title in 1839. Educated at Eton and Oxford, he graduated with first-class honours in classics in 1832, and held a fellowship at All Souls' from 1835 to 1844. After some years' work as a barrister, he became Receiver-General, and in 1869 Commissioner of Customs, having two years previously been elected professor of poetry at Oxford in succession to Matthew Arnold, a position to which he was re-elected for a second term five years later. He had already published *Miscellaneous Verses* (1840); *The Two Destinies* (1844); *The Return of the Guards and other Poems* (1866); and subsequently

printed his *Oxford Lectures* (1869 and 1877) and *Reminiscences and Opinions, 1813-85* (1886).

Doyle, Richard, an artist, born in London in 1824, died in 1883. He was long well known as a constant contributor of satirical designs to *Punch*, and also showed much talent in illustrations to Leigh Hunt's *Jar of Honey*, Thackeray's *Newcomes* and his *Rebecca and Rowena*, and Ruskin's *King of the Golden River*. Afterwards he devoted himself to water-colour painting.

Dozy (dō'zi), Reinhart, Dutch Orientalist and historian, born 1820, died 1883. He was thoroughly versed in most of the Semitic tongues, and spoke and wrote almost all the European languages with facility. Among his works (sometimes in Dutch, sometimes in French) are: *Histoire des Musulmans d'Espagne de 711-1110*, *Géographie d'Edrisi*, *De Israelieten te Mekka*, *Het Islamisme*], *Supplements aux Dictionnaires Arabes*.

Draa, or **Wady Draa**, a river, or rather water-course, of Morocco, rising in the Atlas Mountains and flowing generally south-east, until, after penetrating the Anti-Atlas range and passing several oases, it suddenly turns westwards, and forms the shallow lagoon El Debaia. From this point until it enters the ocean it is a wady, and forms the southern boundary of Morocco.

Dracæ'na, a genus of endogenous evergreen plants, nat. ord. Liliaceæ. It includes the dragon tree of Teneriffe (*D. Draco*), celebrated for producing the resin called dragon's blood. Several species of *Dracæna* are cultivated in greenhouses for the beauty of their foliage, but many of the fine plants known by this name belong strictly to other genera.

Drachenfels (drä'hen-fels; 'dragon rock'), "the castled crag of Drachenfels", as Byron calls it, a hill in Rhenish Prussia, about 8 miles south-east of Bonn, rising 900 feet above the Rhine, and crowned by the old castle of Drachenfels.

Drachma (drak'ma), the unit of weight and of money among the ancient Greeks. It was the principal Greek coin, was made of silver, and was worth (the Attic drachma) about $9\frac{3}{4}d$. As a weight amongst the Greeks it was about 2 dwt. 7 grains troy. The monetary unit of modern Greece is also called a drachma. Since 1867 its value has been equivalent to that of the franc of the Latin Monetary Union. It is divided into 100 *lepta*.

Draco, a legislator of Athens, about 620 B.C., whose name has become proverbial as an inexorable and bloodthirsty lawgiver, and whose laws were said to have been written in blood, not ink. Suidas says that he met his death at Ægina, being unintentionally suffocated by the caps and cloaks thrown at him by some of his enthusiastic supporters.

Draco, the Dragon, a constellation of the northern hemisphere, consisting of a long and straggling line of stars, coiled about Ursa Minor, the Lesser Bear. The Pole of the Ecliptic, or earth's orbital plane, is in this constellation, and round that point the Pole of the Equator, at present close to α Ursæ Minoris (the 'Pole Star'), travels in a circle in about 26,000 years. Some 4000 years ago α Draconis was pole-star.

Drag, (1) a long coach or carriage, generally uncovered and seated round the sides; (2) an apparatus for retarding or stopping the rotation of one wheel or of several wheels, in carriages especially; (3) an apparatus, consisting of a frame of iron with a bag-net attached, used to recover articles lost in the water.

Drag-net, a net drawn along the bottom of a river or pond to catch fish. The use of drag-nets is usually prohibited in rivers where fish breed, as it takes all indiscriminately.

Drago Doctrine, a doctrine formulated by L. M. Drago, an Argentinian jurist and Minister for Foreign Affairs, and asserting the principle that no power had a right to impose itself by force of arms upon any of the Spanish American nationalities. Drago first advanced his doctrine in 1902, when the British, German, and Italian fleets were blockading the Venezuelan coast to compel President Castro to pay certain claims made upon his government.

Dragomirov, Mikhail Ivanovitch, Russian general, born in 1830, died in 1905. He became known as lecturer on military tactics, and was appointed chief of the Russian general staff at Kiev. During the Russo-Turkish War he distinguished himself at the crossing of the Danube at Sistova, and was wounded at the Shipka Pass. Retired from active service, he was director of the War Academy at St. Petersburg, Governor-General of Kiev from 1898 to 1902, and member of the Council of the Empire. His works include: *The Austro-Prussian War*, *A Study on the Novel 'War and Peace'*, *The French Soldier*, *War is an Inevitable Evil*, and *Duels*.

Dragomirov, Vladimir, son of the former, was prominent as a commander during the European War, and took part in the offensive in Galicia in 1916. In 1919 he was president of General Denikin's Political Council, and Governor of Kiev.

Dragon (Gr. *drakon*, 'the seeing one', a serpent). This 'composite wonder beast' is prominent not only in fairy lore and mediæval romances, but in ancient religious systems. In the mythical history of the East the dragon is the symbol of anarchy and destruction, and the idea was taken over by Christianity, which looked upon the dragon as an emblem of the devil. In Ancient Egypt certain of the deities had serpentine forms, as have still some of the dragons of India, China, and Japan. The Egyptian 'fiery flying serpent' is a dragon, as is also the Apep serpent of night and death, through which the sun-barque of Ra was supposed to pass each night. Biblical references to it as the 'worm' include: "Their worm shall not die" (*Is.* lxvi, 24); "The worm shall eat them like wool" (*Is.* li, 8); "In that day the Lord, with his sore, and great, and strong sword, shall punish leviathan the piercing serpent," &c. (*Is.* xxvii, 1); "The great dragon was cast out, that old serpent, he that is called the Devil and Satan" (*Rev.* xii, 9). See also *Ps.* lxxxvii, 4, and lxxxix, 10; *Amos*, ix, 3; and *Is.* li, 9. Osiris, as the Nile, was a dragon on which were human heads. In the Pyramid Texts he is addressed as "the Great Green" (the Mediterranean Sea), and the one who is "round as the Great Circle (Okeanos)." On the sarcophagus of Seti I he is "Osiris encircling the Nether World." Set, who slew him, had a 'roaring serpent' form and hid in a hole; he resembled Typhon. The Babylonian dragon Tiamat was the Great Mother of all the deities, and was slain by her descendant Marduk (Merodach), who formed the earth and sky from her body; her blood ran as the flooded rivers to the sea. In India the drought-demon is a water confiner. When slain by Indra with the thunderbolt, the rainy season ensues. The Naga serpent-gods are dragons who may assume human or half-human, half-reptile forms. They guard treasure and chiefly pearls. Early pearl-fishers believed that the shark was the owner and guardian of pearls. Among the Chinese dragons is the lion-headed shark. All the Chinese dragons have pearls in their mouths, and are supposed to spit out pearls. Dragon deities are connected with the moon, which is 'the night-shining pearl,' and in Mexico 'the pearl of heaven'. The Mexican dragon resembles the Chinese, Indian, and Babylonian dragons. The Indian

wonder-beast, the *Makara*, the vehicle of the sea-god Varuna, is similar to the dragon of the Babylonian mother-goddess Ishtar seen on the famous Ishtar gate of Babylon. *Makara* forms include the lion-headed dolphin, the crocodile-headed fish, and the ram-headed fish so like the 'goat fish' or 'antelope fish' of the Babylonian sea-god Ea, and resembling the Greek horse-headed, dog-headed, and man-headed fish (Tritons). Japanese dragons are serpentine 'water fathers', which are prayed to in time of drought. The Chinese dragons are rain-bringers which sleep during the winter (the season of drought) in pools and rise to fight and thunder in spring. They are hatched from stones as snakes, or from sea-plants, or are transformed fish, or are born from aged pine trees. They are coloured according to their attributes, and may assume human forms or be horse-headed with a snake's tail. That the composite dragon-god is a mixture of several ancient animal-gods is evident by the following description of one class of dragon by a Chinese writer: "His horns resemble those of a stag, his head that of a camel, his eyes those of a demon, his neck that of a snake, his belly that of a clam, his scales those of a carp, his claws those of an eagle, his soles those of a tiger, his ears those of a cow." Dragons may transform themselves into fishes, snakes, crows, dogs, rats, cows, sharks, whales, crocodiles, &c., as well as human beings. The 'will o' wisp' is the 'dragon lantern', and the dragon pearl is the 'jewel that grants all desires' in India, China, and Japan. Dragon herbs cure diseases and prolong life. Dragons carry souls to the Celestial regions, or draw vehicles in which souls stand. This Far Eastern belief existed in Ancient Crete too. On a Cretan sarcophagus is a chariot drawn by two griffins (forms of the dragon) in which stands a woman, probably a goddess, and a swathed pale figure, the deceased. Shakespeare has interesting dragon references, including:

The dragon wing of night o'erspreads the earth,...

(*Troilus and Cressida*, V, 8, 17.)

Swift, swift, you dragons of the night, that dawning
May bare the raven's eye.

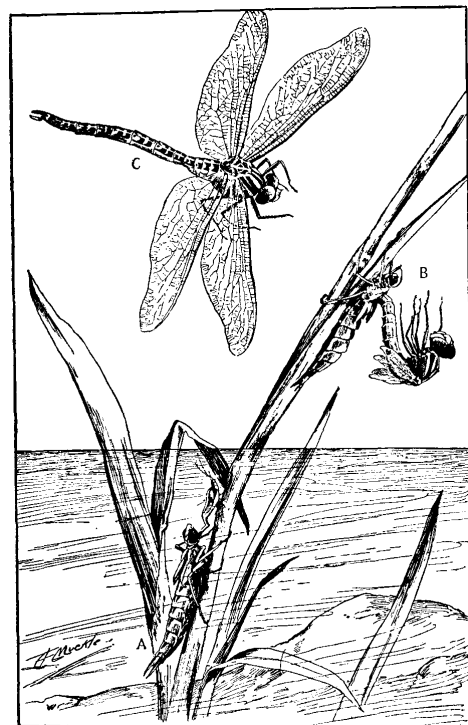
(*Cymbeline*, ii, 2, 49.)

It is believed that the treasure-guarding dragon of the romances had origin in mixed memories regarding the pearl-guarding shark, the fiery flying serpent, and the ancient serpent and crocodile demons of destruction, flood, darkness, and death. The whole idea of dragons may have originated from traditions about the pterodactyls which lived in the Mesozoic period.

The dragon being a symbol of destruction and a power of evil, the slaying of a dragon was considered a great achievement of mediæval heroes, such as King Arthur, Beowulf, Siegmund, and Tristram.—Cf. G. Elliot Smith, *The Evolution of the Dragon*; M. W. de Visser, *The Dragon in China and Japan*.

Dragon, or **Dragon-lizard**, a name for several species of lizards inhabiting South-East Asia. The common flying lizard (*Draco volans*), the best type of the genus, is about 10 or 12 inches in length, the tail being extremely long in proportion to the body. The sides are furnished with peculiar extensions of the skin, resembling wings, which help to support it in the air as it springs from branch to branch. These wing-like processes are borne by prolongations of five or six of the hindermost ribs, and can be folded up. Its food consists almost exclusively of insects.

Drag'onet, the common name of small marine fishes constituting a special family (Callionymidæ). The gemmeous dragonet (*Callionymus lyra*) is found in the British seas. The female is dull brown and much smaller than the male, which is brilliantly coloured with spots and bars of blue on a yellow ground. His first dorsal fin is large and drawn out into a long filament.



Dragon-fly

A, Larva of dragon-fly. B, Dragon-fly escaping from chrysalis. C, The perfect insect.

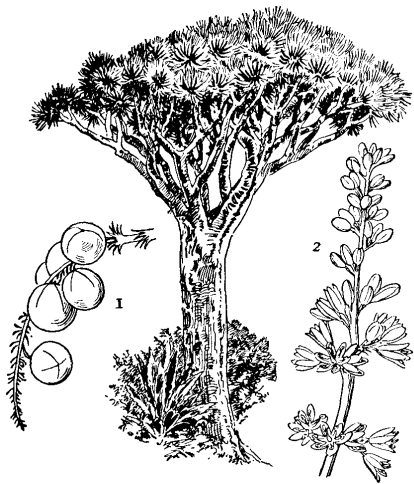
Dragon-fly, the common name of members of a family (Odonata or Libellulidæ) of neuropterous insects. They have a large head, large eyes, and strong horny mandibles. They are beautiful in form and colour, and are of very powerful flight. The great dragon-fly (*Æschna grandis*) is about 4 inches long, and the largest of the British species. They live on insects, and are remarkable for their voracity. The dragon-fly deposits its eggs in the water, where the wingless nymphs live on aquatic insects. The nymph stage lasts for a year. The family is of very wide distribution. The small blue Agrion is a common European form, but the familiar Libellula is the most extensively distributed. See *Demoiselle*.

Dragonnades, or **Dragonades**, the name given to the persecutions directed against the Protestants, chiefly in the south of France, during the reign of Louis XIV, shortly before the revocation of the Edict of Nantes in 1685. Bands of soldiers, headed by priests, marched through the villages, giving the Protestant inhabitants the alternative of renouncing their faith or being given over to the extortions and violence of the soldiery. The dragoons were conspicuous in these expeditions, to which they gave their name. The Dragnonnades drove thousands of French Protestants out of France.—Cf. Tylor, *The Huguenots in the Seventeenth Century*.

Dragon's Blood, or **Gum Dragon**, a resinous juice, usually obtained by incision from various tropical plants, as *Calāmus Draco*, *Dracæna Draco*, *Pterocarpus Draco*, &c. It varies in composition, and is often much adulterated. It is opaque, of a reddish-brown colour, brittle, and has a smooth shining conchoidal fracture. It is soluble in alcohol and oil, but scarcely so in water. It is used for colouring varnishes, for staining marble, leather, and wood, and for tooth tinctures.

Dragon Tree (*Dracæna Draco*), a tree-like liliaceous plant, with a stem simple or divided at top, and in old age often much branched. It is a native of the Canaries, and yields the resin known as dragon's blood. It is often grown in greenhouses.

Dragoon, a kind of mounted soldier, so called originally from his musket (*dragon*) having on the muzzle of it the head of a dragon. At one time dragoons served both as mounted and foot soldiers, but now only as the



Dragon Tree (*Dracæna Draco*)
1, Fruiting branchlet. 2, Flowers.

former. In the British army there are *heavy* and *light dragoons*. The first dragoon regiment, the Scots Greys, was formed in 1681.

Draguignan (dra ' -gē-nyän), a town of Southern France since 1793, capital of the department of Var, in a beautiful valley, 41 miles north-east of Toulon. It has some interesting buildings, and manufactures of silk, soap, and leather. Pop. 9974.

Drainage. The term comprises the drainage of areas of country by rivers and streams, the reclamation of areas from the sea, and country formerly marshy, and the provision of culverts and pipe-drains to buildings and towns.

The Fens in Lincolnshire are a notable example of a comprehensive system of drainage by means of open ditches (locally called 'drains'), into which the surplus water is lifted by means of wind- and steam-pumps.

Low-lying or flat country often requires a considerable amount of drainage, which is carried out by means of a regular system of earthenware pipes, laid 2 to 3 feet deep, and from 15 to 35 feet apart. These pipes are porous, from 2 to 3 inches in diameter, laid with butt joints, and lead into larger mains, and thence by open ditches to streams.

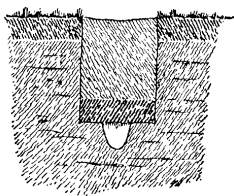
A method recently introduced consists in drawing a pointed cylindrical tool, 2 inches in diameter, through the ground at the required depth. This tool is dependent from a thin steel plate, which connects it with the carriage above, so that it can be drawn underground in any desired direction. This system is economical in first cost, but its useful life is considerably less than that of a piped drain, and its use is obviously confined to soils of the heavier variety.

In considering the provision of drainage to water-logged or low-lying land, every care should first be taken to improve the existing natural means of drainage, such as deepening and cleaning out streams and ditches, and removing obstructions.

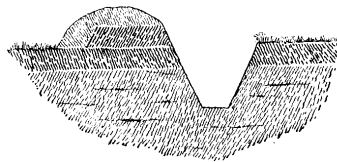
It should be borne in mind that the object of land drainage is not only to remove the surplus water, but to promote a free and natural circulation of water in the soil, and to allow the mineral constituents of the water to reach the plant roots.

In the drainage of buildings, glazed socketed stoneware pipes are used, varying in diameter from 3 to 9 inches, laid straight in plan and in longitudinal section, and laid to falls, calculated to give a minimum velocity of 3 feet per second when flowing half full. These are laid in trenches, with inspection chambers at all changes in direction, and should be laid on and surrounded with concrete. In bad ground, or under dwelling houses, cast-iron pipes are employed, with special turned and bored joints. In towns these drains lead into the public sewers, which are similar stoneware pipes from 6 to 18 inches in diameter, larger sizes being constructed in brickwork or concrete, and either circular or egg-shaped form. In populous areas these sewers attain very large dimensions, the northern outfall sewers of London consisting of five parallel sewers, each 9 feet in diameter. See *Draining*.—
BIBLIOGRAPHY: G. S. Mitchell, *Handbook of Land Drainage*; Moore and Silcock, *Sanitary Engineering*; Gilbert Thomson, *Modern Sanitary Engineering*.

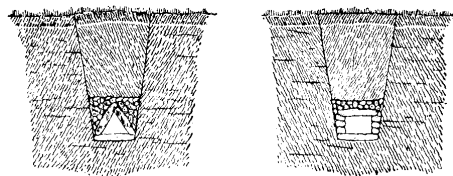
Drainage Tubes are fenestrated india-rubber tubes used in surgery to effect the gradual removal of the contents of a suppurating cavity. The inner end of the tube is in the cavity, and the outer end projects above the skin surface, and is usually fixed by a stitch or safety-pin, and covered with suitable dressings.



Covered Drain
without Pipes



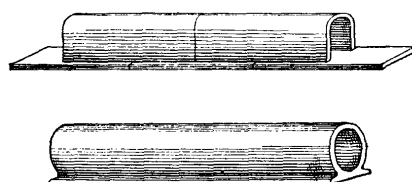
Open Drain



Stone-laid Drains

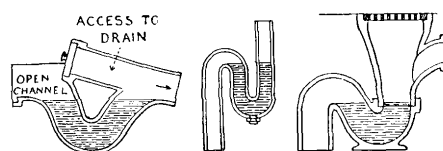
Draining, in agriculture, a method of improving the soil by withdrawing the superfluous water from it by means of channels that are generally covered over. Plants cannot thrive unless there is free circulation of air and water round their roots. The successful practice of draining in a great measure depends on a proper knowledge of the superficial strata, of their situation, relative degrees of porosity, &c. Some strata allow water to pass through them, while others more impervious force it to run or filtrate along their surfaces till it reaches more level ground below. In general, where the grounds are in a great measure flat and the soils of materials which retain the excess of moisture, they require artificial means of drainage to render them capable of yielding good crops whether of grain or grass. The wetness of land, which makes it inferior for agricultural purposes, may appear not only as surface-water but as water which flows through the lower strata, and to draw off these there are the two distinct operations of surface-draining and under-draining. The rudest form of open drains are the deep furrows lying between high-backed ridges, and meant to carry off the surplus water after the soil is completely saturated, but in doing so they generally carry off also much of the best of the soil and of the manure which has been spread upon it. The ordinary ditch is a common form of water-course useful in certain cases, as in hill pastures. But covered drains at a depth of 4 feet or so are the common forms in draining agricultural lands. They are generally either *stone-drains* or *tile-drains*. Stone-drains are either formed on the plan of open culverts of various forms, or of small stones in sufficient quantity to permit a free and speedy filtration of the water through them. The box-drain, for instance, is formed of flat stones neatly arranged in the bottom of the trench, the whole forming an open tube. In tile-drains, tiles or pipes of burnt clay are used for forming the conduits. They possess all the qualities which are required in the formation of drains, affording a free ingress to water, while they effectually exclude earth, as well as other injurious substances, and vermin. Drainage tiles and pipes have been made in a great variety of forms, the earliest of which, since the introduction of thorough draining, was the horse-shoe tile, so called from its shape. These should always rest on soles, or flats of burned clay. Pipe tiles, which combine the sole and cover in one piece, have been made of various shapes, but the best form appears to be the cylinder. An important department of draining is the draining off of the waters which are the sources of springs. The judicious

application of a few simple drains, made to communicate with the watery layers, will often dry swamps of great extent, where large sums of money, expended in forming open drains in the swamp itself, would leave it but little improved. In the laying out of drains the first point to be determined is the place of outfall, which should always afford a free and clear outlet to the drains, and must necessarily be at the lowest point of the land to be drained. The next point to be determined is the position of the minor drains. In the laying out of these the surface of each field must be regarded as being made up of one or more planes, as the case may be, for each of which the drains should be laid out separately. Level lines are to be set out a little below the upper edge of each of these planes, and the drains must then be made to cross these lines at right angles. By this means the drains will run in the line of the greatest slope, no matter how distorted the surface of the field may be. All the minor drains should be made to discharge obliquely into mains or submains, and not directly into an open ditch or water-course. As a general rule, there should be a main to receive the waters of the minor drains from every 5 acres. The advantages of drainage are obvious. In the first place it brings the soil into a more suitable condition for the growth of plants, aiding in producing the finely divided and porous state which allows the roots and rootlets to spread themselves at will in order to obtain the needed supplies of food, air, and moisture. It also allows the sun's rays to produce their full effect on the soil and plants. In the presence of stagnant water a great part of this effect would be lost. See *Drainage*.



Horse-shoe and Cylindrical Drain Tiles

Drain-trap, a contrivance to prevent the escape of foul air from drains, while allowing the passage of water into them. They are of various forms. In the traps represented below it will be seen that there must always be a certain quantity of water



Drain-traps

maintained to bar the way against the escape of the gas from the drain or sewer. When additional liquid is conveyed to the trap, there is, of course, an overflow into the drain. In older types of drains the gas was prevented from escaping by a metal plate thrown obliquely over the drain mouth and dipping into the water in the vessel beyond it.

Drake, Sir Francis, an English navigator, born at Tavistock, in Devonshire, in 1539, or according to some authorities in 1545. He served as a sailor in a coasting vessel, and afterwards joined Sir John Hawkins in his last expedition against the Spaniards (1567), losing nearly all he possessed in that unfortunate enterprise. Having gathered a number of adventurers round him, he contrived to fit out a vessel in which he made two successful cruises to the West Indies in 1570 and 1571. Next year, with two small ships, he again sailed for the Spanish Main, captured the cities of Nombre de Dios and Vera Cruz, and took a rich booty which he brought safely home. In 1577 Drake made another expedition to the Spanish Main, having this time command of five ships. On this the most famous of his voyages Drake passed the Straits of Magellan, plundered all along the coasts of Chile and Peru, sacked several ports, and captured a galleon laden with silver, gold, and jewels, to the value of perhaps £200,000. He then ran north as far as 48° N. lat., seeking a passage to the Atlantic, but was compelled to return to Port San Francisco on account of the cold. He then steered for the Moluccas, and holding straight across the Indian Ocean doubled the Cape of Good Hope, and arrived at Plymouth 3rd Nov., 1580, being thus the first of the English circumnavigators. As there was no war between England and Spain, the proceedings of Drake had a somewhat dubious character, but the queen maintained that they were lawful reprisals for the action of the Spaniards, and showed her favour to Drake by knighting him on board his own ship. Five years afterwards Drake was again attacking the Spaniards in the Cape Verde Islands and in the West Indies, and in 1588 particularly distinguished himself as vice-admiral in the conflict with the Spanish Armada. In 1593 he represented Plymouth in Parliament. His later expeditions, that in 1595 against the Spanish West Indies and that to Panama, were not so successful, and his death, which took place on 28th Jan., 1596, at sea off Porto Bello, was hastened by disappointment.—BIBLIOGRAPHY: Sir J. S. Corbett, *Sir Francis Drake* (English Men of Action Series), and *Drake and the Tudor Navy*.

Drakenberg Mountains, a range of South Africa forming the western frontier of Natal, and rising to the height of 11,000 feet, a continuation of the Quathlamba range.

Drama (Gr. *drama*, action, from *drān*, to act or do), a form of art which imitates action by introducing real persons to represent the fictitious characters, and to carry on the story by means of action and dialogue.

Man is naturally an imitative animal, and some crude form of drama must have been in existence in very early times. We can see the origins of drama in many of the games played by children, where important events such as war, marriage, and sacrifice are represented in song and dance. In Greece, the cradle of drama as of everything that is good, there must have been in prehistoric times war-dances which formed the basis of tragedy, and rough vintage revel dances which formed the basis of comedy.

Greek Drama.—The Greek drama was religious in its origin. It arose from the dithyrambs or songs composed in honour of Dionysus, the god of all vegetation, though identified most closely with the vine. When vegetation died in winter, this was considered to be the death of Dionysus; when it bloomed anew in the spring, this was thought to be the god's resurrection. The one event was celebrated with gloomy song and dance, and the other with merry revels and crude indecency. The history of Greek drama is the history of the decline and fall of the chorus. At first the chorus was the whole play, and in the *Supplices* of Æschylus, the earliest extant tragedy, the chorus played a predominating part. According to tradition, Thespis (about 535 B.C.) introduced for the first time a masked actor, who carried on a dialogue with the leader of the chorus. Æschylus introduced a second actor, and Sophocles a third. It is thought that there never were more than three actors, but, of course, duplication of parts was permitted. There were also frequently mute characters (*kōpha prosōpa*) on the stage. Dialogue became more important in the later plays of Æschylus, and chorus became less important; Sophocles developed his dialogue in masterly style, though his choruses are among the most beautiful things in all Greek poetry; in Euripides the choruses, however lovely in themselves, are less an integral part of the drama than they were in the plays of his predecessors. In fact the chorus acted as a clog on the freedom of the dramatist, who wished to develop exciting situations and depict realistic characters. In comedy the

same decline of the chorus is to be found; in the *Acharnians*, the earliest comedy, the chorus is very prominent; in the *Plutus*, the last comedy extant, it is comparatively unimportant. Sumptuary laws had something to do with this, and there is a vast difference between the magnificently apparelled chorus in the *Birds*, and the chorus in the *Lysistrata*, which represented elderly Athenian men and women in their everyday costume.

Greek tragedies were usually presented in the form of trilogies, that is, in sets of three plays all dealing with the same subject. To these was added, as a rule, a fourth play, known as a satyr-play. The *Cyclops* of Euripides is the only example extant of this kind of play. It is not very amusing, though it contains a certain amount of horse-play and high spirits. The satyr-play was intended to lighten the gloom of the three preceding tragedies. We have one complete trilogy preserved—the magnificent *Oresteia* of Æschylus, consisting of the *Agamemnon*, *Choephoræ*, and *Eumenides*—plays which are bracketed with *Lear* and *Othello* as the highest and most majestic of all tragedies. In later times the three plays of the trilogy dealt with different subjects. The tragedies to be performed were carefully selected by some of the Athenian magistrates, and at the festival prizes were given for the best tragedy, on the recommendation of a carefully chosen jury. Comedies were presented one at a time; prizes were offered for the best of them also. Greek tragic actors wore long flowing robes, and added to their height by means of the *cothurnus* or thick-soled boot; it is believed that they wore masks with some sort of speaking-trumpet in the mouth, so that their words would be audible to the vast audience which assembled in the theatre, a huge circular open-air amphitheatre.

Each of the three Greek tragic writers whose work has been preserved is supreme in his own way. Æschylus's lyric dramas are among the greatest writings of all time; the plays of Sophocles are masterpieces of deft construction, of well-woven plot, and ironic dialogue; and his choruses are lyrics of the greatest beauty. Æschylus is more titanic; Sophocles is more humane. Euripides, the latest of the three, is a great poet and a champion of the weak, such as women and slaves; moreover, he sees deeply into men's hearts. He is really the founder of romantic drama, through the Roman Seneca, who imitated him. Of Greek comic poets we only possess one, but he is a host in himself. Aristophanes is a Gargantuan mirth-maker; he bestrides the narrow world like a Colossus. He plays with a master's hand

upon every note in the whole comic gamut. His works, owing to the conditions of the old comedy, were very frequently political and highly personal in their tone. The later plays are less so. The *Birds*, *Clouds*, and *Frogs* are among the very greatest comic creations; only a little less great is the *Lysistrata*, where a serious purpose is veiled by intense indecency. The old comedy, however, was essentially the product of its own age; it did not invite, or even permit, imitation. The new comedy, of which Philemon, Menander, and Diphilus were the principal writers, gradually supplanted it. Their plays were more or less romantic comedies with carefully constructed plots. They are all lost, but we may gain some idea of them from Plautus and Terence, and from the fragments which have been found, some of them fairly recently.

Roman Drama.—Roman drama is not intrinsically good; it is in many respects a weak imitation of Greek drama, but it has been very much more important in its influence. Early English, French, and Italian dramatists all turned to Seneca as a model for tragedy, and to Plautus and Terence as models for comedy. This was partly due to the fact that though most of them had small Latin, they had less Greek; but it was partly because the Latin writers were easier to imitate. Italy had native farces, known as Atellan Fables, which were not without their influence on the development of comedy. These plays were broadly farcical, and dealt almost entirely with country life. The two great Roman comedy writers, Plautus and Terence, based their work, however, upon the new comedy of Greece, especially upon the plays of Menander and Diphilus. Plautus is decidedly coarse at times, and sometimes his fun is too much like that of a fourth-form boy at a public school, but his work is wholesome and vigorous, and he is a more creative and virile writer than Terence. Terence's plays are somewhat weak dramatically, but are written in a style of great beauty. He was a careful literary craftsman. Seneca, the only Roman tragic writer, had an immense influence on later dramatists. It is hard to account for this. He based his work upon Euripides, but he suppressed everything that makes Euripides tender and human. Senecan tragedy abounded in bloodshed and horrors; the

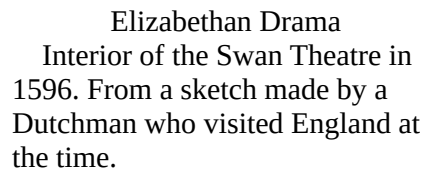


A type of mask worn by the characters in a Roman play after Terence's time. The audience knew from the mask the particular character the actor represented.

speeches are full of pompous rant, and their metre is most monotonous. Some of the choruses are good rhetorical writing, though scarcely great poetry. Seneca's influence pervades all our early tragedy; it is clearly seen in *Gorboduc* and in Jonson's *Sejanus* and *Catiline*; even Shakespeare is not without traces of it. As the Roman Empire declined so did the Roman stage; finally nothing was performed save pantomime, in the proper sense of the word, where everything was done in dumb-show. This appeared to content the populace of the Roman Empire, even as the cinema seems to satisfy the citizens of a later and perhaps greater empire.

Mediæval Drama.—There is no drama between the death of Seneca and the Renaissance, unless we except the six curious 'comedies' of the nun Hrosvitha of Gandersheim (born about A.D. 935). These plays are based upon Terence, though they do not follow their model closely. They are, of course, written in Latin. They have some vivid dramatic touches, and frequent felicities of expression. They were probably intended for recitation, not for representation on the stage. They must be regarded as an isolated phenomenon. The Church for long discouraged drama, but ended by adapting it to its own purposes. As in Greece, therefore, drama originated in England from religion. The priests impressed certain events in sacred history upon the minds of their congregation by means of dramatic performances which at first took place actually in the church. Thus the removal of the stone from the mouth of the sepulchre and the discovery of the empty tomb was performed at Easter, and the finding of the Babe in the manger by the three Magi was represented at Epiphany. It is easy to understand how performances of this sort arose from the singing of suitable anthems on festival days. The Oberammergau passion-play is a somewhat sophisticated representative of these liturgical plays; it cannot be called a survival, as it only dates back to 1633. These mystery-plays, so called because they were produced by the trade-guilds (Lat. *ministerium*, a trade), were eventually brought out into the market-place on wagons, and were moved round to various 'stations' in the town, different plays being performed at each station. A distinction is sometimes made between mystery- and miracle-plays, the former being defined as dealing with gospel events only, while the latter deal with incidents derived from the legends of the saints. Several collections of these plays survive—the Wakefield, Chester, and Coventry plays. They are written in a lively fashion, and are

Elizabethan Drama.—The first English comedy, *Ralph Roister Doister*, appeared in 1551. It is by Nicolas Udall, and is based upon the *Miles Gloriosus* of Plautus. *Gammer Gurton's Needle*, a more native production, thought to have been by John Still, who was master of St. John's College, Cambridge, and Bishop of Bath and Wells, appeared about 1566. Drama now improved rapidly, and was soon to attain perfection in Shakespeare. Marlowe, Kyd, Greene, Lyly, Nash, Lodge, and Peele all helped to prepare the way. The greatest of these is Marlowe, who died at the age of twenty-nine, leaving behind him the great plays *Tamburlaine* (1588), *Faustus*, and *Edward II.* In his development of blank verse he contributed greatly to the success of the drama. The earliest tragedy, *Gorboduc* (1562), is in blank verse. Marlowe made of blank verse an instrument of great power and pathos and sublimity. In the plays



drama reached its greatest height. In comedy, tragedy, history, in handling dramatic situations, and in liquid perfection of verse, he is supreme. Like the very greatest masters, he founded no school, and his contemporaries owe little to him. While they are all put in the shade by his myriad-minded genius, they are all partakers with him in the glory of their age, and are all great in themselves. Jonson (1573-1637) is one of the most important, as he to some extent founded a school and exercised considerable influence over later writers. He was a scholarly and laborious playwright, who over-elaborated some of his work, but who was a masterly adept at constructing a play, and a vigorous realist. Chapman (1559-1634), Dekker (1570-1641), and Marston (1575-1634) were all good workmanlike dramatists. Beaumont and Fletcher produced between them a great body of work, some of inferior quality, but all of great power. In some respects their work is less unlike that of Shakespeare than the work of other Elizabethans. Middleton, Webster, Tourneur, Thomas Heywood, and Massinger are all excellent in their way, Massinger in particular being a master of stage-craft. Shirley and Ford conclude the list of the great Jacobean dramatists. The Puritans caused the theatres to be closed in 1642.

Spanish and French Drama.—Meanwhile a similar outburst of dramatic activity was taking place on the Continent. In Spain, Lope de Vega (1562-1634) wrote a prodigious quantity of plays, and wrote them with much brilliance. Calderon wrote some beautiful plays, several of which have been translated by Edward Fitzgerald. Cervantes, though much better known as a novelist, wrote many good plays. The Spanish school directly inspired Corneille (1606-1684) to write his play *Le Cid*, and so begin the great age of classical French tragedy. Racine (1639-1700) is the other great name. French classical drama, though somewhat fettered by its observance of laws that were wrongly considered essential, is extremely dignified and beautiful. In Molière (1622-73) France possesses the greatest of all writers of society comedies. He is as supreme in his kingdom as Shakespeare is in his empire. He borrowed from his predecessors with all the licence of genius, but he paid usurious interest on his borrowings.

Restoration Drama.—When the theatres were reopened after the Restoration, many dramatists began to write. Restoration comedy was largely based on Molière, who was brutalized by Wycherley, and adapted but not improved by Congreve. Congreve was, however, a master of

sparkling dialogue, and in one play, *The Way of the World*, he has shown himself not unworthy of comparison with his master. Vanbrugh and Farquhar are the other two important writers of comedies; all their comedies are more or less disfigured by cynicism and immorality, the reaction after the Puritan restraint. Restoration tragedy is much less important than Restoration comedy. Otway, Lee, and Southerne are its chief exponents.

Eighteenth Century Drama.—Some of these dramatists bring us into the eighteenth century, which was not on the whole prolific in good plays. Fielding wrote many amusing farces, but all were more or less hack-work. At a later period Foote, Cumberland, and the two Colmans wrote good acting plays, which have not lived. The two plays of Goldsmith and several of the plays of Sheridan still hold the stage. Sheridan owed much to the Restoration dramatists, especially Vanbrugh, but as he improved his originals in many respects, and made them much more presentable in decent society, he is entitled to most of the reputation he long enjoyed.

In France, Marivaux (1678-1763) wrote sentimental comedies, while Beaumarchais, whose own life was more exciting and varied than most plays, wrote comedies with brilliant plots. In Italy, Maffei, Goldoni, and Alfieri are notable dramatists; the last named wrote propaganda in the disguise of tragedy. In Germany, Lessing by precept and example inaugurated the 'romantic movement'; Schiller and Goethe are the two greatest names associated with the stage. *Wallenstein* in particular is a good chronicle-play, while *Faust* is considered one of the greatest of all German plays.

Nineteenth and Twentieth Century Drama.—Victor Hugo led the Romantic movement in France, and wrote many great plays, such as *Hernani* and *Ruy Blas*. De Musset wrote his plays, which he called *proverbes*, under the same influence, and later followers of this school are Rostand and the Belgian M. Maurice Maeterlinck. The French dramatists Augier, Scribe, and Sardou had an overwhelming influence on the English stage, not altogether for its good. English drama was at a low ebb in the middle of the nineteenth century. Lytton's plays, though sometimes performed still, are extremely theatrical. Boucicault, who made a great success by dramatizing 'the pathos of Paddy', is not a great writer. H. J. Byron was an inveterate punster and writer of burlesques of no value. One of his plays, *Our Boys*, was acted for

many years. Robertson is the most outstanding author of what is known as 'the cup and saucer' school of comedy. His plays are very much acting plays; they are not literature, and are quite removed from real life. Gilbert was a man of great gifts, but though some of his farces and comedies are good, he was not a master of drama as he was of *libretti* writing. He did little to improve the drama of his day. Sir A. W. Pinero began his career as a dramatist under the ægis of Robertson, but continued it under that of Ibsen. Ibsen (1828-1906) exercised a not altogether wholesome influence upon English drama for a considerable time. His plays are extremely well-constructed, and he refused to tolerate many conventions, such as asides and soliloquies. In many of his plays he adopted the retrospective method, where the plot consists not so much in anything being done as in the gradual discovery of what has been done long before the rise of the curtain. Sophocles had done this most skilfully in *Ædipus Tyrannus*, but Ibsen carried the method to perfection in *The Wild Duck* and *Rosmersholm*. All Ibsen's plays are more or less unpleasant, and he did not make many of his characters sympathetic. Pinero, after writing several farces, wrote *The Second Mrs. Tanqueray* (1893), a masterpiece after the style of Ibsen. *His House in Order* is a cleverly constructed example of the retrospective method. H. A. Jones (born 1851) has written many excellent and extremely powerful plays, of which the best known are *The Liars* and *The Case of Rebellious Susan*. G. Bernard Shaw (born 1856), who combines some of the qualities of a Greek sophist with some of the foibles of a modern Irishman, has written some amusing plays, though others have been spoilt by his tendency to turn them into propaganda. Galsworthy has written plays of great earnestness; in some he has neglected the Aristotelian maxim that every play must have a beginning, a middle, and an end. Sir James Barrie has accumulated a large fortune by means of his plays, and in one at least, *Peter Pan*, he has made a bid for immortality.

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Contemporary Drama of England; F. Brunetière, *Les Époques du théâtre français 1636-1850*; A. Filon, *The Modern French Drama*; A. d'Ancona, *Origini del teatro italiano*; G. H. Lewes, *The Spanish Drama*.

Drammen, or **Dram**, a seaport of Norway, in a valley on both sides of the Drammen, at its mouth in the Drammenfiord, 25 miles s.s.w. of Christiania. It has manufactures of leather, soap, ropes, sail-cloth, earthenware, and tobacco; and is the second port in the kingdom for the export of timber.

Draper, John William, American chemist and physiologist, born at Liverpool 1811, died 1882. He went to America in 1833, and was successively professor of physical science in Hampden-Sidney College, Virginia, and of natural history, chemistry, and physiology in the University of New York. He made many contributions to scientific literature, and devoted much attention to the chemical action of light, in connection with which he effected some discoveries. Among his chief works is his *History of the Intellectual Development of Europe* (2 vols., 1863). His son, Henry Draper (born 1837, died 1882), chemist and astronomer, made some valuable researches on the spectra of the heavenly bodies.

Draughts, a game resembling chess played on a board divided into sixty-four checkered squares. Each of the two players is provided with twelve pieces or 'men' placed on every alternate square at each end of the board. The men are moved forward diagonally to the right or left one square at a time, the object of each player being to capture all his opponent's men, or to hem them in so that they cannot move. A piece can be captured only when the square on the diagonal line behind it is unoccupied. When a player succeeds in moving a piece to the farther end of the board (the crown-head), that piece becomes a king, and has the power of moving or capturing diagonally backwards or forwards. When it so happens that neither of the players has sufficient advantage in force or position to enable him to win, the game is drawn. *Checkers* is the common American name of the game.

The game does not offer the same scope for brilliance and originality as the sister game of chess, but still is much more profound than is generally supposed. It has been cultivated in Britain, and especially in Scotland, certainly for over two hundred years, and has served as a field of exercise for some extremely able intellects, which but for it might hardly have been

exercised at all. Among famous players are: Andrew Anderson, of Carlisle, who published a celebrated work on the game in 1852; James Wyllie, the 'Herd Laddie', who travelled over the world playing exhibition matches, and was for many years world's champion; Robert Martins, English champion about 1870, who played several matches with Wyllie; R. D. Yates, a young American player, who defeated Wyllie for the championship, but shortly afterwards gave up the game; James Ferrie, of Coatbridge, who in 1894 defeated Wyllie and became champion, to be defeated in turn by Richard Jordan of Edinburgh in 1896; Robert Stewart, of Fifeshire, many times Scottish champion, and probably the strongest player now living.

The Scottish tourney, held annually in Glasgow since 1893, except for a few years on account of the War, has done much to stimulate interest in the game. The English Draughts Association also holds a biennial tourney. Several international matches have taken place between Scotland and England, the first in 1884. This, like nearly all the other matches, was won by Scotland. In 1905 a very strong British team visited America and decisively defeated a side representing the United States.—BIBLIOGRAPHY: James Lees, *A Guide to the Game of Draughts*. Early works by Payne, Sturges, Drummond, Hay, Anderson, and Bowen are now very scarce. There are several periodicals devoted to the game; and some newspapers, notably *The Glasgow Weekly Herald*, give it a column weekly.

Drave, or **Drau** (drä've, drou), a European river which rises in Tyrol, flows E.S.E. across the north of Illyria and the south of Styria, and between Hungary on the left and Croatia and Slavonia on the right, and, after a course of nearly 400 miles, joins the Danube 14 miles east of Essek. It is navigable for about 200 miles.

Dravid'ian, a term applied to the vernacular tongues of the great majority of the inhabitants of Southern India, and to the people themselves who inhabited India previous to the advent of the Aryans. The affinities of the Dravidian languages are uncertain. The family consists of the Tamil, Telugu, Canarese, Malayâlam, Tulu, Tuda, Gond, Rajmahal, Oraon, &c. Only the first four mentioned have a literature, that of the Tamil being the oldest and the most important. Originally the word Dravidian was a purely philological term, but it is now used in an ethnological sense as well.—Cf.

R. Caldwell, *Comparative Grammar of the Dravidian or South Indian Family of Languages*.

Drawback, usually a certain amount of duties or customs dues paid back or remitted to an importer when he exports goods that he has previously imported and paid duty on, as, for instance, tobacco, &c.; or a certain amount of excise paid back or allowed on the exportation of home manufactures.

Drawbridge, a bridge with a lifting floor, such as was formerly used for crossing the ditches of fortresses, or any movable bridge over a navigable channel where the height of the roadway is insufficient to allow vessels to pass underneath. Modern drawbridges across rivers, canals, the entrances of docks, &c., are generally made to open horizontally, and the movable portion is called a bascule, balance, or lifting bridge, a turning, swivel, or swing bridge, or a rolling bridge, in accordance with the mode in which it is made to open. Swing-bridges are usually divided into two parts meeting in the middle, and each moved on pivots on the opposite sides of the channel, or they may move as a whole on a pivot in the middle of the channel. Rolling bridges are suspended from a structure high above the water, and are propelled backwards and forwards by means of rollers.

Drawing is the art of representing upon a flat surface the forms of objects, and their positions in relation to each other. The idea of nearness or distance is given by the aid of perspective, foreshortening, and gradation, and in the same way the three-dimensional quality of objects is expressed. The term drawing is sometimes limited to the representing of the forms of objects in outline, with or without the shading necessary to develop roundness or *modelling*. But the term has a wider significance. Any arrangement of colours or tones which serve to express form, and the relation of one form to another, is really drawing; and thus a painting may show fine draughtsmanship without line being used at all. Drawing is not a matter of the medium employed, but of the manner in which it is used. It is, however, convenient to classify drawings according to mediums, each of which produces its effect in a different way. Thus drawing may be divided into (1) pen drawing; (2) chalk, pencil, or charcoal drawing; (3) crayon and pastel drawing; (4) drawing shaded with the brush; (5) architectural or mechanical drawing. *Pen drawings* are often confined to pure outlines; an appearance

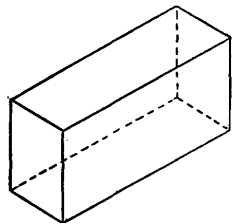
of *relief* or projection being given by thickening or doubling the lines on the shadow side, or they may be shaded by combination of lines. *Chalk, pencil, and charcoal drawings* may be in line or tone. When the chalk is powdered and rubbed in with a stump, large masses and broad effects can be produced with much rapidity. The best draughtsmen, however, rarely use a stump. In *crayon and pastel drawings* the colours of the objects represented are more or less completely represented in the medium. *Drawings shaded with the brush* are outlined with the pencil or pen, the shading being laid on or *washed in* with the brush in tints of Indian ink, sepia, or colour. This was the method of the early water-colour painters in England. *Architectural and mechanical drawings* are those in which the proportions of a building, machine, &c., are accurately set out for the guidance of the constructor; objects are, in general, delineated by geometric or orthographic projection.

The great schools of painting all show excellent drawing, though differing in character. In Italy the Florentine school combined study of the antique with anatomical research, and produced many vigorous and expressive draughtsmen, notably Michael Angelo and Leonardo da Vinci. The Roman school, under the influence of Raphael, sacrificed vigour and expressiveness to elegance and the representation of ideal form. In the Lombard school a severe style of drawing is seen through harmonious colouring, and in the Venetian school the drawing is often veiled in the richness of the colour. The German and Dutch schools excel in a careful and minute style of naturalistic drawing, combined with good colour. The French school in the time of Poussin was very accurate in its drawing; at a later period its style betrayed a tendency to mannerism. David introduced, again, a purer taste in drawing and a close study of the antique, and these are qualities which distinguish his school (the so-called Classical school), of which Ingres is the leading representative, from the Romantic and Eclectic schools of a later period. The drawing of the British school is naturalistic rather than academic, but the work of Gainsborough and Alfred Stevens is comparable with that of earlier masters.—BIBLIOGRAPHY: Ruskin, *Elements of Drawing*; Spiers, *Architectural Drawing*; R. S. Bowers, *Drawing and Design for Craftsmen*; J. H. Brown, *Sketching without a Master*; Harold Speed, *Drawing*.

Drawing-room, a room appropriated for the reception of company; a room in which distinguished personages hold levees, or private persons receive

parties. *Court drawing-rooms* are those assemblies held from time to time for the reception or presentation to the sovereign of such ladies as by custom, right, or courtesy are admissible. Receptions at which men are presented are known as *levees*. The sovereign sometimes deposes a member of the royal family to receive, in which case presentations are equivalent to those made to the sovereign in person.

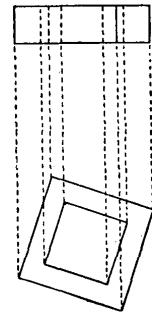
Drawings. The term 'drawings' is usually taken to mean drawings of an architectural or engineering nature, such as the plans of a new building prepared by an architect, or the designs for engineering works, or for machinery produced by an engineer.



Isometric projection
of a Brick $8\frac{3}{4} \times 4\frac{1}{4}$
 $\times 2\frac{3}{4}$

Three methods are commonly made use of in preparing drawings. (1) Orthographic, which represents the subject under consideration in one plane only, and from which dimensions may be scaled off, and which is the normal method of preparing an engineering drawing. (2) Perspective or radial projection is made use of by an architect for displaying the elevations of a

building, and gives a truer appreciation of the actual appearance of the building than can be obtained by orthographic projection. (3) Isometric projection enables one to show the length, breadth, and thickness of an object drawn to scale on the one drawing. Such a drawing is really composed of three sets of parallel straight lines, and is not strictly a true representation of the object as it would appear to the eye. It has the great advantage, however, that measurements may be directly scaled from it, and lines which are parallel in the object are also parallel in the drawing.



Plan and
Elevation of
a Square
Slab pierced
by a square
hole
obtained by
orthographic
projection

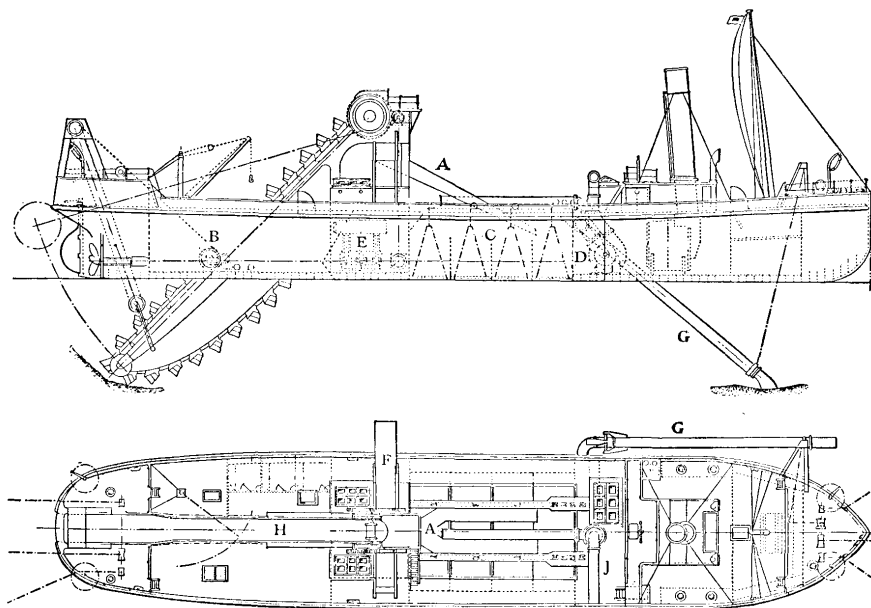
Drayton, Michael, an English poet, born in 1563, is said to have studied at Oxford, and afterwards held a commission in the army. The poem by which his name is chiefly remembered is his *Polyolbion* (1622, reprinted in 1890), a sort of topographical description of England. It is generally extremely accurate in its details, with, at the same time, many passages of true poetic fire and beauty. Other works are his *Nymphidia*, *the Court of Fairy*; *The*

Barons' Wars; The Legend of Great Cromwell; The Battle of Agincourt; besides numerous legends, sonnets, and other pieces. He died in 1631, and was buried in Westminster Abbey.—Cf. O. Elton, *Michael Drayton: a Critical Study*.

Drayton, Market, or **Drayton-in-Hales**, a town, England, county Salop, 18 miles northeast of Shrewsbury. It has a church, supposed to have been erected, with exception of the steeple, in the reign of William I. There are paper and hair-cloth manufactories. Pop. 5167.

Dreams, trains of ideas which present themselves to the mind during sleep. The principal feature of the state of dreaming is the absence of conscious control over the current of thought, so that all kinds of fantastic notions, which in the waking state would at once be put aside, are woven into the texture of the dream. The usual content of dreams consists of aspirations or dreads, which the dreamer had recently entertained or experienced, mixed up with incidents which excited intense emotion at some earlier period of the individual's history, and especially in early childhood. The memory of unpleasant experiences, such, for example, as the horrors of trench warfare, which is repressed in the waking state, tends to force itself on the individual's attention when the conscious control is relaxed in sleep, and to give rise to disturbing dreams which may become so intense as to interfere with sleep and cause insomnia. The only rational remedy for this distressing trouble is to discover the painful incident and persuade the patient frankly to face it and not 'try to forget'. In recent years S. Freud has placed the study of dreams upon a scientific basis. He maintains that dreams represent the fulfilment of wishes. There is usually an utter want of coherency in the images that appear before the mental eye, but this excites no surprise in the dreamer. Occasionally, however, intellectual efforts are made during sleep which would be difficult to surpass in the waking state. It is said that Condillac often brought to a conclusion in his dreams reasonings on which he had been employed during the day; and that Franklin believed that he had been often instructed in his dreams concerning the issue of events which at that time occupied his mind. Coleridge composed from 200 to 300 lines during a dream: the beautiful fragment of *Kubla Khan*, which was all he was able to commit to paper when he awoke, remains a specimen of that dream-poem. Dreams are subjective phenomena dependent on natural causes. They are retrospective and resultant instead of being prospective or

prophetic. The latter opinion has, however, prevailed in all ages and among all nations; and hence the common practice of divination or prophesying by dreams, that is, interpreting them as presages of coming events. Some authorities declare that all our dreams take place when we are in process of going to sleep or becoming awake, and that during deep sleep the mind is totally inactive. This is denied by the majority of philosophers, and with apparent reason.—BIBLIOGRAPHY: Havelock Ellis, *The World of Dreams*; S. Freud, *The Interpretation of Dreams*; W. H. R. Rivers, *Instinct and the Unconscious*.



Plan and Elevation of an 800-ton Dredger

A, Shoot. B, Hoist-gear. C, Hopper. D, Sand-pump and its engines. E, Engines. F, Overboard shoot. G, Suction-pipe. H, Bucket-well. J, Overboard discharge.

Dredging, a term applied to the operation of removing mud, silt, and other deposits from the bottom of harbours, canals, rivers, docks, &c. The most simple dredging apparatus is the spoon apparatus, which consists of a strong iron ring or hoop, properly formed for making an impression upon the soft matter at the bottom, so as to scoop it into a large leather bag attached to the ring and perforated with a number of small holes. The means for working it is a long handle, a suspending rope, and a crane or sweep-pole planted in a boat. Much more effective is the steam dredging-machine now in common use. It has a succession of strong iron buckets on an

endless chain, which travels on a frame whose lower end is vertically adjustable so as to regulate the depth at which it works. It is worked by steam, and the buckets tear up the matter at the bottom, raise it, and discharge it into punts or hoppers close to the dredging vessel. Various forms of steam-pump dredgers, in which suction-pipes are the chief features, are also used. The River Clyde, from being a shallow stream, has been converted, mainly by dredging, into a waterway carrying large vessels up to Glasgow. Dredging rivers for gold is now largely carried on; and the gold-dredge may even be floated in water artificially supplied. The operation of dragging the bottom of the sea for molluscs, plants, and other objects, usually for scientific observation, is also called dredging. The oyster-dredge is a light iron frame with a scraper like a narrow hoe on one side, and a bag attached to receive the oysters. The dredges used by naturalists are mostly modifications of or somewhat similar to the oyster-dredge. Scientific dredging has of late assumed great importance as making us acquainted with the life of deep-sea areas.

Dreisenssia, a genus of bivalve mollusc allied to the mussels. One species (*D. polymorpha*) is a native of the streams which flow into the Caspian, but has been accidentally introduced into most rivers and estuaries of Europe, including those of Britain, where it is now abundant.

Drelincourt (drè-lan'-kör), Charles, a French Calvinistic minister, born at Sedan 1595, died at Paris 1669. He was the author of many controversial works, and of *Consolations against the fear of Death*. To promote the sale of the English translation of this work, De Foe wrote his *Apparition of Mrs. Veal*.

Drenthe (dren'te), a province of Holland, bounded by Hanover, Overijssel, Friesland, and Groningen; area, 948 sq. miles. It is in general more elevated than the surrounding provinces, especially in the centre. The soil is generally poor, and the surface largely consists of heath and morass, but the province is famed for its horses and cattle. Drenthe is remarkable for the great number of so-called 'giants' graves' or barrows scattered over the country. Its capital is Assen. Pop. 200,951.

Dres'den, the capital of the Republic (former kingdom) of Saxony, is situated in a beautiful valley on both sides of the River Elbe, which is here

spanned by four stone bridges and an iron railway bridge. It is first mentioned in history in 1206, and has been the residence of the sovereigns since 1485; was greatly extended and embellished by Augustus the Strong (1694-1736), and rapidly increased during the nineteenth century. Among the chief sights are the museum (joined on to an older range of buildings called the Zwinger), a beautiful building containing a famous picture-gallery and other treasures; the Japanese palace (Augusteum), containing the royal library (founded by the Elector Augustus in the sixteenth century) of 570,000 volumes, besides a rich collection of manuscripts; the Johanneum, containing the collection of porcelain and the historical museum, a valuable collection of arms, armour, and domestic utensils, belonging to the Middle Ages. The palace, built about 1530, restored and remodelled externally between 1890 and 1902, and until 1918 the residence of the kings of Saxony, has also a fine interior, and contains (in what is called the Green Vault) a valuable collection of curiosities, jewels, trinkets, and works of art. The theatre is one of the finest structures of the kind in the world. The city is distinguished for its excellent educational, literary, and artistic institutions, among which are the Technical High School, much on the plan and scale of a university; the Conservatory and School of Music; and the Academy of Fine Arts. The manufactures are not unimportant, and are various in character; the china, however, for which the city is famed, is made chiefly at Meissen, 14 miles distant. The commerce is considerable, and has greatly increased since the development of the railway system. The chief glory of Dresden is the gallery of pictures, one of the finest in the world, which first became of importance under Augustus II, King of Poland and Elector of Saxony, but owes its most valuable treasures to Augustus III, who purchased the greater portion of the gallery of the Duke of Modena for £180,000. The pictures number about 3000, and in particular comprise many fine specimens of the Italian, Dutch, and Flemish schools. Besides this fine collection the museum contains also engravings and drawings amounting to upwards of 350,000. There is also a sculpture-gallery, the Albertinum, where by casts and otherwise the progress of sculpture is exemplified from the earliest times, all the most important antiques being shown. Dresden, being thus rich in treasures of art and favoured by a beautiful natural situation, is the summer resort of many foreigners. It suffered severely in the Thirty Years' War, and also in 1813, when it was the

head-quarters of Napoleon's army. It was occupied by the Prussians in 1866, but was evacuated in the following spring. Pop. 548,308.

Dresden, Battle of, a battle fought in 1813 (26-27th Aug.), between the French under Napoleon and the Allies under Schwarzenberg. Napoleon had come to the relief of the city, which was occupied by the French. The Allies assaulted and bombarded the city, and soon after a great pitched battle was fought (27th Aug.), the Allies being defeated.

Dresden China, a delicate, semi-transparent, highly finished china made at Meissen, 14 miles from Dresden. The manufacture resulted from an accidental discovery made by Böttger, a young chemist, in 1710, and the vases, statuettes, groups of figures, candelabra, and clocks, manufactured during the eighteenth century are highly prized.



Dresden China.
Candelabrum

Dreux (*dreu*; **Durocassis** or **Drocæ** of the Romans), a French town, department of Eure-et-Loir, on the Blaise, near to where it joins the Eure, 20 miles N.N.W. of Chartres. It is built at the foot of a hill crowned by a dilapidated castle, which contains a chapel, founded in 1142; to which has been added the costly mausoleum of the Orleans family. A battle took place near the town in 1562 between the Royalists under Montmorency and the Huguenots under Condé, in which the latter were defeated. Pop. 10,692.

Dreyfus, Alfred, captain of artillery and general staff-officer in the French army, was born of a Jewish family in Mulhouse, Alsace, in 1859. In Oct., 1894, he was arrested on a charge of communicating military documents to a foreign Government, supposed to be Germany; and at a secret court-martial, which sat in December, he was condemned to public degradation and lifelong imprisonment. Early in 1895 he was sent to the Île du Diable (Devil's Island), near Cayenne, to undergo his sentence. About the middle of the same year Colonel Picquart became head of the Intelligence Department, and in the course of his official duties discovered various circumstances tending to throw doubt on the correctness of the court-

martial's decision, and pointing to another officer, of the name of Esterhazy, as the real traitor. Picquart was superseded by a Colonel Henry in Nov., 1897, and in the following January Esterhazy, charged by a brother of the condemned man with having written the *bordereau*, or memorandum, which was the chief document relied on by the prosecutors of Dreyfus, was acquitted by a court-martial. Two days later M. Zola, the eminent novelist, in a letter headed *J'accuse* published in the *Aurore*, made serious charges against the general staff and the Government in connection with the Esterhazy court-martial. He was prosecuted, and condemned to pay a heavy fine and undergo a term of imprisonment. In June, 1898, M. Brisson succeeded M. Méline as Prime Minister, and next month M. Cavaignac, his War Minister, read to the Chamber several documents which he regarded as conclusive proof of the guilt of Dreyfus. The chief of these was soon admitted by Colonel Henry to have been forged by him, and M. Cavaignac at once resigned. In June, 1899, the Cour de Cassation ordered a fresh court-martial. The court-martial, which sat at Rennes, found Dreyfus guilty with extenuating circumstances. He was sentenced to ten years' imprisonment, but was pardoned by President Loubet almost immediately. In 1906, when Clemenceau was Prime Minister, the sentence was annulled, and Dreyfus was reinstated in the army (as major). He was shot at by a reactionary journalist in 1908, but escaped without serious injury. In Sept., 1919, Lieutenant-Colonel Dreyfus was publicly presented with the insignia of an officer of the Legion of Honour. Several times during the progress of the case France seemed on the verge of revolution.—Cf. J. Reinach, *Histoire de l'affaire Dreyfus*.

Driffield, Great, a town, England, Yorkshire, at the head of a navigable canal communicating with the Humber at Hull. It lies in a fertile district, has an ancient parish church, and manufactures linseed-cake and manures. Pop. 5676.

Drift, in geology, a term applied to earth and rocks which have been conveyed by flood-action, glaciers, or floating ice and deposited over the surface of a country. It is sometimes used in a wider sense to denote all post-Pliocene sands, gravels, and clays, such as the superficial deposits shown on the 'drift' maps of the Geological Survey.

Drift, in mining, a horizontal tunnel or passage excavated underground that follows the course of a vein or stratum. Drift, in musketry, is the lateral deviation of the bullet after it has left the barrel of the rifle; it is due to the spin of the bullet and the resistance of the air.

Drift Sand, sand thrown up by the waves of the sea, and blown when dry some distance inland until arrested by obstacles, round which it gradually accumulates until the heaps attain considerable dimensions, often forming dunes or sand-hills. Coast-land sometimes requires artificial protection from encroachment by drift-sand.

Drill, a tool used for boring holes in wood, metal, stone, ivory, &c. It consists of a sharp spindle to which a circular motion is communicated by various contrivances. Drills are of various designs. For rock-boring the diamond rock-drill, an instrument with cutting edges made of bort or black diamond, is now generally adopted. See *Boring*.—Cf. Dana and Saunders, *Rock Drilling*.

Drill is the A B C of all military movements. In the Training Manuals of the British army the word is defined as "the training of the soldier to perform certain movements as a second nature". It follows, therefore, that drill is an essential part of the training of every soldier, more especially in the early days of his training, in that without it, and without the power of movement in obedience to the expressed will of a superior given by it, a body of soldiers would be merely a collection of armed men who, however willing individually, would be incapable of carrying out collectively an order given for the general good. In the early days of our history, when fighting was largely individual, and the whole duty of a soldier was 'to do unto the other fellow as he would do unto you—and do it first' (with a club), drill, as we know it, was unknown; each man armed himself as he thought fit, and, beyond getting into some formation for the actual purpose of the assault, a battle was largely a go-as-you-please affair. In Saxon days the normal formation for the battle was the wedge; that is two men at the point followed by three, and so on till the available number was used up. This, of course, formed a solid pointed mass with considerable weight, and was used both for attack and defence. But once this formation was broken it was next to impossible to reform it. An instance of this weakness occurred at the battle of Hastings. The English were in this one and only wedge formation,

officers and the better armed men at the point, and the less skilful and more indifferently armed at the base. Doubtless the troops had been got into this formation after much exertion in the way of pushing and vituperation, and, once in it, had been told on no account to break it. At a certain stage in the battle the heavily armoured Normans pretended flight; this was too much for the English, who broke their ranks and gave chase, each after his own particular source of ransom. This ended the battle; the Normans turned, and, owing to the entire inability of the English to re-form their ranks, the wedge, and with it the English army, ceased to exist: the result of want of discipline and absence of drill.

Drill and discipline are complementary to each other. In one of the battles of the Peninsular War, the 28th (now the Gloucestershire Regiment) were being hotly attacked in front by a French column. The regiment was firing in two ranks—the front rank kneeling and the rear rank standing—when suddenly a fresh attack developed from the rear. It was a matter of seconds for the commanding officer of the 28th to order the rear rank to turn about; drill and discipline did the rest, and the rear rank turned round, knelt down, and beat off the new attack. Since then the 28th has worn its badges both front and rear of its head-dresses.

Drill is an aid to discipline in that it teaches men that there is a right and a wrong way of doing a thing. Drill and the spirit arising from it has a great steadying effect on the nerves, as when in the European War the Guards Division, after being almost decimated during a German 'push', was brought out of the line and kept to steady drill for a week. To those who did not understand this appeared harsh and futile; to those who did it appeared, as it was, the best means of steadying men tried beyond endurance, and of preparing them for further efforts. Our English drill has passed through many phases in its time; but from the days when large bodies of men performed complicated manœuvres at the executive command of one man, through the times when drill, perhaps, was considered to be the be-all and end-all of the soldier, to modern days when it is recognized as a means to an end, the guiding principle remains the same, viz. that one of the first essentials for a soldier is that he shall be so trained by drill that he shall know instinctively how to do the right thing at the right time and in the right way. Even now, when drill movements are no longer performed in face of an enemy, accuracy and attention to detail are insisted on in all parade-

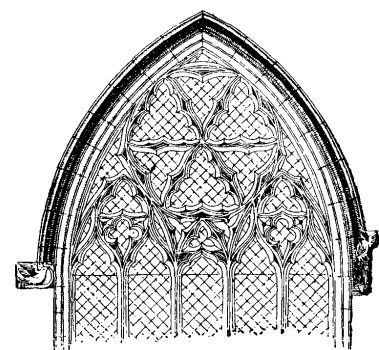
ground movements as part of the education of the soldier and as an aid to discipline. Drill for the soldier takes the place of the five-finger exercises for the musician. Neither of them, in itself, is of any particular value, but each adds to the efficiency of those who practise it.

Drill (*Papio leucophæus*), a large variety of baboon, smaller and less fierce than the mandrill, and like it a native of the coast of Guinea. The face and ears are bare and of a glossy black colour, the palms of the hands and soles of the feet are also naked and of a deep copper colour.

Drilling, the plan of sowing in parallel rows as distinguished from sowing broadcast. It was introduced into England by Jethro Tull, who invented the first implement for drilling, and published a work on the subject in 1731. The crops which are now generally drilled are turnips and flax. The first form of drill was of very simple construction, and was only adapted for potatoes, beans, peas, carrots, clover, cereals, sowing one row at a time, but now a great variety of improved implements is in use, some of which distribute artificial manure with the seed. Among the principal advantages of drilling over broadcast sowing we may mention that a considerable saving of seed is effected in the sowing of grain crops, but the great advantage is that in the case of green crops it enables the farmer more readily to clean the land both by the hand- and by the horse-hoe. To keep the soil stirred and pulverized, which can only be properly done when the crops have been drilled, favours the retention and absorption of the moisture.

Dripstone, a projecting tablet or moulding over the head of a Gothic doorway, window, archway, or niche to throw off the rain. It is also called a *weather moulding*, and *label* when it is turned square. It is of various forms; sometimes a head is used as a termination or support, in others an ornament or simple moulding is adopted.

Driver, Rev. Samuel Rolles, D.D., professor of Hebrew and Biblical critic, born at Southampton 2nd Oct., 1846, died in 1914. He



Dripstone over window,
decorated Gothic, mixed tracery.
Two different forms of
termination.

was educated at Winchester and at New College, Oxford, where he graduated with first-class honours in classics in 1869. In 1866 he gained the Pusey and Ellerton Hebrew Scholarship, and the Kennicott Scholarship (also for Hebrew) four years later, besides prizes for Septuagint Greek and Syriac. He was for some years a fellow and tutor of his college, and from 1876 to 1884 a member of the Old Testament Revision Committee. In 1883, on the death of Dr. Pusey, he became Regius professor of Hebrew at Oxford and (*ipso facto*) a Canon of Christ Church. Of his numerous works we may mention: *A Treatise on the Use of the Tenses in Hebrew* (1874); *Isaiah: his Life and Times* (1888); *Introduction to the Literature of the Old Testament* (1891), a work suited for popular reading, which has passed through many editions; *Sermons on Subjects connected with the Old Testament* (1892); *The Parallel Psalter* (1904); *Commentaries* on various books of the Bible; and articles in *Bible Dictionaries* and in periodicals. He was a joint editor of the new *Hebrew-English Lexicon of the Old Testament* published by the Clarendon Press.

Drogheda (droh'e-da), an ancient town and seaport, formerly a parliamentary borough, Ireland, in the county of Louth, on both sides of the Boyne, about 4 miles from the sea, 26 miles north of Dublin. The Boyne is here crossed by a railway viaduct of 18 arches and 95 feet high. Flax- and cotton-spinning are carried on; there are also salt-works, breweries, and tanneries; and the fisheries are increasing. There is a good export trade in cattle, sheep, grain, butter, and eggs. In 1412 a Parliament assembled here which enacted Poynings' Law. The town was for a long time strongly fortified, and was taken by Cromwell with great slaughter in 1649; it surrendered to William III immediately after the battle of the Boyne. Pop. 12,500.

Drohobycz (dro'ho-bich), a Polish town in Galicia, formerly belonging to Austria, 41 miles s.s.w. of Lemberg. Its Catholic church is one of the handsomest in the country. It has an important trade, particularly in salt, obtained from springs in the vicinity. Pop. 40,000.

Droit d'Aubaine (drwä-dō-bān), an old rule in some European countries, by which the property of a foreigner who died was claimed by the State, unless the person had a special exemption. In France, where it was not

abolished till 1819, the Scots, Savoyards, Swiss, and Portuguese were exempted.

Droitwich (droit'ich), a town of England, in the county and 6 miles N.N.E. of Worcester, on the Salwarp. It is famous for its brine springs, from which salt has been manufactured for more than 1000 years. Pop. 4146.

Drôme, a south-east department of France, covered almost throughout by ramifications of the Alps, the average height of which, however, does not exceed 4000 feet; area, 2508 sq. miles, of which about one-fourth is waste, one-third under wood, and a great part of the remainder under tillage and pasture. A considerable extent of the area is occupied by vineyards, and several of the wines produced have a high reputation, especially Hermitage. Olives, chestnuts, and silks are staple productions. Valence is the capital. Pop. 263,509.

Dromedary. See *Camel*.

Dromore', an episcopal city, Ireland, County Down, on the Lagan, here crossed by two bridges, 16 miles south-west of Belfast. Its cathedral contains the tomb of Jeremy Taylor. Pop. 2307.

Dropsy (Edema) is a condition usually marked by enlargement and swelling of the affected parts, and due to an accumulation of serous fluid in the tissue spaces and cavities of the body. Different names are given to such accumulations in particular areas, thus *anasarca* refers to accumulations in the limbs and body generally; *ascites* to an accumulation of fluid in the peritoneal cavity (abdomen); *hydrothorax* to an accumulation in the pleural cavity (lungs); *hydrocephalus* to an accumulation in the brain. The commonest cause of dropsy is heart disease, where first the lower limbs, and then the trunk, are affected. It also appears in diseases of the kidneys and liver, and it may be produced in a limb by any obstruction of the veins of the part.

Drop-wort (from the small tubers on the fibrous roots), *Spiræa filipendŭla*, nat. ord. Rosaceæ, a British plant of the same genus as queen-of-the-meadow, found in dry pastures. The hemlock drop-wort, or water drop-wort, is *Cenanthē fistulōsa*.

Drosera'ceæ, a nat. ord. of polypetalous Dicotyledons, consisting of insectivorous marsh herbs, whose leaves are usually covered with glands or glandular hairs. It contains six genera, including the sundew (*Drosëra*), and Venus's fly-trap (*Dionæa*). (See *Sundew* and *Dionæa*.) They have no known qualities except that they are slightly bitter. The leaves are generally circinate in the bud, as in ferns.



Russian Droshky

Droshky, a kind of light, four-wheeled carriage used by the

Russians. It is not covered, and in some types there is in the middle a sort of bench placed lengthways on which the passengers ride as on a saddle; but the name is now applied to various kinds of vehicles, as to the common cabs plying in the streets of German cities.

Drouais (drö-ā), Jean Germain, French historical painter of considerable repute, born at Paris in 1763, died at Rome 1788. His chief pictures are: *The Canaanitish Woman at the Feet of Jesus*, *Dying Gladiator*, and *Marius at Minturno*.

Drouet (drö-ā), Jean Baptiste, Comte d'Erlon, French general, born 1765, died 1844. He served in the campaigns of the Moselle, Meuse, and Sambre (1793-6), in the Peninsula, and at Waterloo, where he commanded the first *corps d'armée*. In 1834-5 he was Governor-General of Algeria, and in 1843 was made a marshal.

Drouyn de Lhuys (drö-an ' dé lwēs), Édouard, French statesman and diplomatist, born 1805, died 1881. He entered the diplomatic service in 1831, and was chargé d'affaires at the Hague during the events which led to



Common Water Drop-wort
(*Enanthe fistulōsa*)
a, Cluster of florets. b, Single floret.

the separation of Belgium from Holland. In 1840 he was head of the commercial department under the Minister of Foreign Affairs. Opposition to Guizot caused his dismissal in 1845. He became Minister for Foreign Affairs in 1848, Ambassador to London in 1849; and again Foreign Minister in 1851, and in 1863. On the fall of the empire he fled to Jersey, but subsequently returned to France.

Drowning means death by the air being prevented from entering the lungs owing to the mouth and nostrils being immersed in a liquid, the liquid being commonly water. Death may, therefore, occur by drowning in a small quantity of water. Thus a child may fall head downwards into a tub and be drowned, though the tub is not half full of water, sufficient to cover the mouth and nostrils being all that is necessary, and a man overcome by a fit or by drunkenness may fall on a road with his head in a ditch or pool of water, and thus meet death. Death is thus due to suffocation, to the stoppage of breathing, and to the entrance of water into the lungs. When death has been caused by drowning, the skin presents the appearance called goose-skin (*cutis anserīna*), the face and surface of the body generally are usually pale, a frothy liquid is found in the lungs and air-passages, and about the lips and nostrils; water may be found in the stomach, and clenched fingers, holding substances grasped at, may serve to show that a struggle has taken place in the water, and that the body was alive at the time of immersion. Complete insensibility arises, it is probable, in from one or two minutes after submersion, recovery, however, being still possible, and death occurs in from two to five minutes. So long as the heart continues to beat, recovery is possible; after it has ceased it is impossible. Newly-born children and young puppies stand submersion longer than the more fully grown.

For the restoration of the apparently drowned several methods are suggested. Those of Dr. Silvester, recommended by the English Humane Society, and Dr. Benjamin Howard, of New York, will be described.

Whichever method is adopted, the following steps must first and immediately be taken: Pull the body up on to dry ground. Send immediately for medical assistance, warm blankets, dry clothing, brandy and hot water, if any one is at hand to send. No delay must be permitted, however, in treating the drowned, so that if only one person is on the spot he must begin to treat the victim *instantly*, without seeking assistance. Remove all clothing

from the neck and chest. Fold the articles of dress removed so as to make a firm pillow, which is to be placed under the shoulders, so that the upper part of the body is slightly raised and the head slightly thrown back. Cleanse the mouth and nostrils, open the mouth and pull forward the tongue. If natural efforts to breathe are made, try to stimulate them by brisk rubbing of the sides of the chest and of the face. If no effort to breathe is made, proceed to produce the entrance and outflow of air from the lungs by Silvester's or Howard's method.

Silvester's method: Stand or kneel behind the person's head, grasp each arm at the elbow, draw both arms simultaneously upwards till they are extended in line with the body, as a man places them when he stretches himself. Let this movement occupy about two seconds. This enlarges the chest and causes the entrance of air to the lungs. Without a pause carry the arms down to the sides, making them overlap the chest a little, and firmly press them on the chest. This movement should occupy other two seconds. It expels air from the lungs. Repeat the movements, and maintain them steadily and patiently at the rate of fifteen times a minute, until breathing has been fully restored, or until medical aid arrives, or until death is certain. An hour is not too long a time to persist, and so long as there seems the least effort to breathe the movements must be persevered in.

Howard's method: Place the body on its face, with the roll of clothing under the stomach; the head being supported on the hand as shown in fig. 1. Pull the body over the roll of clothing to expel water from the chest. Then turn the body on the back, the shoulders being supported as shown in fig. 2. Kneel over the body. Place both hands on the lower part of the chest, so that the thumbs hook in under the lowest ribs and the fingers are spread out on the chest. Steadily press forwards, raising the ribs, your own body being thus thrown leaning forward. This enlarges the cavity of the chest and causes air to enter. When the ribs have been raised to the utmost extent, with a slight effort push yourself back to the more erect position, allowing the ribs to recoil to their former position. This expels the air. Repeat the process fifteen times a minute. One person will find it more easy to maintain this method for a prolonged period than Silvester's, especially if the patient be big and heavy.



Fig. 1



Fig. 2

Howard's Method for restoring the apparently Drowned

Meanwhile, if other persons are present they should be occupied rubbing the body and limbs (*always upwards*) with hands or warm flannel, applying hot flannels, bottles, &c., to the limbs, feet, arm-pits, &c. As soon as the person is sufficiently restored to be able to swallow, give small quantities of hot brandy and water, hot wine and water, hot coffee, &c., and use every effort to restore and maintain warmth.

Drowning was formerly a mode of capital punishment in Europe. The last person executed by drowning in Scotland suffered death in 1685. It survived in Switzerland until 1652, and in Austria until 1776. In Russia the punishment was abolished early in the eighteenth century.

Droylsden, a town of England, Lancashire, $3\frac{1}{2}$ miles E. of Manchester, of which it is practically a suburb. Pop. 13,259.

Droz (drō), François Xavier Joseph, French moralist and historian, born at Besançon 1773, died 1850. In 1806 he published an *Essai sur l'Art d'être Heureux*, which was very popular; and in 1823 *De la Philosophie Morale, ou des Différents Systèmes sur la Science de la Vie*, which procured his admission into the Academy. His reputation is, however, founded chiefly on his *Histoire du Règne de Louis XVI*.

Drugget, a coarse kind of woollen felt or cloth, formerly used by the lower classes for purposes of clothing, but now chiefly used as a covering for carpets.

Druids, the priests of the Celts of Gaul and Britain. According to Julius Cæsar they possessed the greatest authority among the Celtic nations. They had some knowledge of geometry, natural philosophy, &c., superintended the affairs of religion and morality, and performed the office of judges. They were also well versed in the knowledge of the mysterious powers of plants and animals, and were adepts in the magic arts. They venerated the mistletoe when growing on the oak, a tree which they likewise esteemed sacred. They had a common superior, who was elected by a majority of votes from their own number, and who was appointed for life. They took unusual care to fence themselves round with mysteries, and it is probable that they cherished doctrines unknown to the common people; but that they had a great secret philosophy which was handed down by oral tradition is very unlikely. Of their religious doctrines little is known.—BIBLIOGRAPHY: J. Rhys, *Lectures on the Origin and Growth of Religion as Illustrated by Celtic Heathenism*; D'Arbois de Jubainville, *Les Druides et les dieux celtiques à forme d'animaux*.

Druids, Order of, a secret organization, founded in London in 1781, for the purpose of mutual aid and protection. Their rites somewhat resemble those of freemasons; their lodges are known as 'groves'.

Drum, a musical instrument of percussion, of Eastern origin, either cylindrical or hemispherical in shape, with the end or ends covered with tightened parchment, which is stretched or slackened at pleasure by means of cords with sliding knots or screws. Drums are of three kinds: (1) the long or bass drum played with stuffed-nob drumsticks, and used only in large orchestras or military bands; (2) the side-drum, having two heads, the upper

one only being played upon by two sticks of wood; (3) the kettle-drum, a hemisphere of brass or copper, the end of which is covered with parchment, always used in pairs, one drum being tuned to the key-note, and the other to the fifth of the key, the compass of the two together being an octave. The use of drums was introduced into Europe either by the Moors or the Crusaders.

Drumclog', a moorland tract in Lanarkshire, Scotland, 6 miles s.w. of Strathaven, the scene of a skirmish between Claverhouse and the Covenanters, in which the former was defeated (1679). A graphic description of the battle is given by Scott in his *Old Mortality*.

Drum-fish, or **Drum**, *Pogonias chromis*, and other species of the same genus, fishes found on the Atlantic coast of N. America, and so named from the deep drumming sound they make, by means of the swim-bladder and its muscles, during the spawning season in April. It is the most powerful sound-producing apparatus known among fishes. They often weigh about 20 lb.

Drum-major, in the British army, a warrant or non-commissioned officer whose duty it is to teach and direct the drummers. He marches at the head of the band carrying the regimental baton.

Drum'mond, Professor Henry, was born at Stirling in 1851, died in 1897. Educated at the Universities of Edinburgh and Tübingen, he entered the ministry of the Free Church, and having devoted much attention to science, in 1877 was appointed lecturer on natural science in the Free Church College (or divinity hall), Glasgow, becoming professor in 1884. He travelled much, and wrote a popular book on *Tropical Africa* (1888). His most remarkable work is *Natural Law in the Spiritual World* (1883), which has passed through many editions and been translated into various languages. He is author, also, of *Travel Sketches in Our New Protectorate*, *The Greatest Thing in the World*, and *The Ascent of Man* (1894).

Drummond, Rev. James, Unitarian theologian, was born at Dublin in 1835, died 13th June, 1918. After receiving his early education at a private school, he entered Trinity College, Dublin, where he graduated and obtained the first gold medal for classics in 1855. In 1859 he became colleague of the Rev. W. Gaskell in Cross Street Chapel, Manchester, and ten years later was

appointed professor of theology at Manchester New College, London, of which institution (now known, since its removal to Oxford in 1889, simply as Manchester College) he was principal from 1885 to 1906, when he retired from his post. His works include: *Spiritual Religion: Sermons on Christian Faith and Life* (1870); *The Jewish Messiah: a Critical History of the Messianic Idea among the Jews* (1877); *Introduction to the Study of Theology* (1884); *Philo-Judæus* (2 vols., 1888); *Via, Veritas, Vita* (the Hibbert lectures for 1894); *The Pauline Benediction* (1897); *The Epistles to the Thessalonians, &c.* (International Handbooks, 1899); *Some Thoughts on Christology* (1902); *The Character and Authorship of the Fourth Gospel* (1904); *Johannine Thoughts* (1909); *Paul: his Life and Teaching* (1911).

Drummond, William, of Hawthornden, a Scottish poet distinguished for the elegance and tenderness of his verses, was born at Hawthornden House, 7 miles from Edinburgh, 1585, died 1649. He was educated at the University of Edinburgh; after which he spent four years in foreign travels, residing for a part of the time at Bourges, to study the civil law. On his return to Scotland he retired to Hawthornden and gave himself up to the cultivation of poetry and polite literature, and here he spent the most of his life. He entertained Ben Jonson on the occasion of a visit which the English dramatist made to Scotland in the winter of 1618-9, and took notes of Jonson's conversation, first published in entirety in 1842 (*Notes of Ben Jonson's Conversations with William Drummond*). He was the first Scottish writer to abandon the native dialect for the language raised to supremacy by the Elizabethan writers. His chief productions are: *The Cypress Grove*, in prose, containing reflections upon death; *Flowers of Zion, or Spiritual Poems*; *Tears on the Death of Mæliades* (that is, Prince Henry); *Poems, Amorous, Funeral, Divine, Pastoral, in Sonnets, Songs, Sextains, Madrigals*; *The River Forth Feasting* (on King James's visit to Scotland in 1617); *Polemo-Middinia, or the Battle of the Dunghill: a Macaronic Poem*; and *History of the Lives and Reigns of the Five Jameses, Kings of Scotland*. As an historian he is chiefly remarkable for an ornate style, and a strong attachment to the High Church principles of the Jacobites.

Drumont, Édouard, French journalist and anti-Semitic agitator, born at Paris in 1844, died there in 1917. His work *Mon vieux Paris* (1879) was crowned by the Académie Française. In 1886 he published *La France Juive devant l'opinion*. He thus began a violent campaign against the Jews which

he continued until his death, especially in his organ *La Libre Parole*, founded in 1892. In 1898 he was elected to the Chamber of Deputies, retaining his seat till 1902. His other works include: *La fin d'un monde* (1888), *Testament d'un Anti-Sémite* (1891), *De l'or et de la boue du Sang* (1896), and *Les Juifs et l'affaire Dreyfus* (1899).

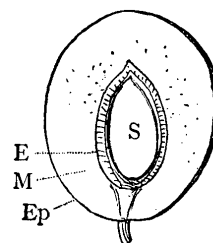
Drunkards, Habitual. The Habitual Drunkards Act of 1879 provided for the licensing of retreats for receiving habitual drunkards, and for the regular inspection of such retreats. An habitual drunkard desiring admission to a retreat had to make a written application, accompanied by a declaration of two persons that the applicant was an habitual drunkard, and attested by two justices of the peace. No patient in a retreat was permitted to leave before the expiration of the term stated in the application, such term not to exceed one year. This Act was to expire in ten years; but another Act, passed in 1888, made it permanent, with some modifications. The Inebriates Act of 1898 introduced several important changes. It transferred the licensing power in counties from justices of the peace to County Councils and their committees, and in boroughs from magistrates to town councillors or police commissioners. The maximum period of detention was extended to two years, and the attestation of one justice was made sufficient for a valid application. It also gave power to the Secretary of State to establish State inebriate reformatories, or to grant certificates to reformatories suitable for such a purpose. If an habitual drunkard, when drunk, commit an offence punishable by imprisonment or penal servitude, the court may, in addition to, or in substitution for, the ordinary sentence, order him to be detained three years in a State or certified inebriate reformatory. The proof that the accused is an habitual drunkard may consist either in his own admission or in the jury's verdict after inquiry. The Inebriates' Act of 1899 was very short, and made no important change in the law. In Scotland habitual drunkenness is now a ground justifying judicial separation of spouses, while under the Matrimonial Causes Bill, 1920, it is proposed that in England, where one spouse has been granted a temporary separation order on the ground of the incurable habitual drunkenness of the other, and such order has been in force for at least three years, a divorce should be granted.

Drunkenness, the state of being drunk or overpowered by alcoholic liquor, or the habit of indulging in intoxication. A similar condition may be produced by numerous agents, but the term is always applied to the act or

habit of drinking alcoholics to excess. By the law of Britain drunkenness is an offence against the public economy, and those found drunk are liable to fine or imprisonment. Drunkenness is no excuse for any crime, but it renders a contract invalid if either of the parties was in a state of complete drunkenness when the contract was signed.

Drunken Parliament, in Scottish history, a name given to the Privy Council who, under their powers as representing the estates between sessions, met at Glasgow and passed an Act (1st Oct., 1662) to remove the recusant ministers from their parishes within a month. All the members were said to have been drunk except Lockhart of Lee, who opposed the measure.

Drupe, in botany, a stone fruit; a fruit in which the outer part of the pericarp becomes fleshy or softens like a berry while the inner hardens like a nut, forming a stone with a kernel, as the plum, cherry, apricot, and peach. The stone enclosing the kernel is called the *endocarp*, while the pulpy or succulent part is called the *mesocarp*. In some fruits, as those of the almond, the horse-chestnut, and coco-nut, the mesocarp is not succulent, yet, from their possessing the other qualities of the drupe, they receive the name. See *Berry*.



Longitudinal
Section of Plum
S, Seed. E,
Endocarp, or shell.
M, Mesocarp, or
intermediate layer.
Ep, Epicarp, or
skin.

Drury Lane Theatre, one of the principal theatres in London, was established by Thomas Killigrew in the reign of James I. In 1671 it was burned down, and was rebuilt by Sir Christopher Wren between 1672 and 1674, but again destroyed by fire in 1809. On this occasion it was rebuilt by B. Wyatt, and was reopened on 10th Oct., 1812, with an address composed by Lord Byron. It was in connection with this opening that James and Horace Smith wrote the *Rejected Addresses*. Nearly all the great English actors from Betterton and Garrick have been more or less connected with Drury Lane. —Cf. J. Doran, *In and About Drury Lane*.

Druses, a curious people of mixed Syrian and Arabian origin, inhabiting the mountains of Lebanon and Anti-Lebanon, and also the Hauran (south-west of Damascus). In their faith are combined certain Jewish, Christian, and

Mohammedan doctrines. They describe themselves as followers of Khalif Hakim Biamr-Allah, whom they regard as an incarnation of deity, the last prophet, and the founder of the true religion. They are nearly all taught to read and write. They maintain a semi-independence, and between 1840 and 1860 they engaged in bloody conflicts with their neighbours the Maronites. Their total number is estimated at 100,000. They are very friendly to the English, and some of them have been converted to Christianity. Cf. E. Sell, *The Druses*.

Drusus, the name of several distinguished Romans, among whom were: *Marcus Livius*, orator and politician; became tribune of the people in 122 B.C. He opposed the policy of Gaius Gracchus, and became popular by planting colonies.—*Marcus Livius*, son of the above, was early a strong champion of the senate or aristocratic party, but showed great skill in manipulating the mob. He rose to be tribune of the people, and was assassinated 91 B.C.—*Nero Claudius*, brother of the Emperor Tiberius, born 38 B.C. By a series of brilliant campaigns he extended the Roman Empire to the German Ocean and the River Elbe, and was hence called *Germanicus*. By his wife Antonia, daughter of Mark Antony, he had a daughter, Livia, and two sons, Germanicus and Claudius, the latter of whom afterwards became emperor. He died in 9 B.C.

Dry'ads, wood nymphs, in the Greek mythology; supposed to be the tutelary deities of trees. Each particular tree or wood was the habitation of its own special dryad.

Dryas. See *Mountain Avens*.

Dry'burgh Abbey, a monastic ruin in Scotland, consisting of the nave's western gable, the gable of the south transept, and a fragment of choir and north transept of an abbey founded in 1150 on the banks of the Tweed, about 5 miles E.S.E. of Melrose. It is celebrated as the burial-place of Sir Walter Scott and his family.

Dry Cell, originally a cell of the Leclanché type, in which the solution of sal-ammoniac was replaced by a paste containing this substance. The formulæ or recipes from which dry cells are now made up are numerous, although the electrodes, as a rule, remain the same as in the Leclanché cell. The E.M.F. of the cell is about 1.5 volts, and three dry cells are used to light

up a small 'flash' lamp. A battery of dry cells forms a convenient and portable means of supplying a small current at voltages up to 100.

Dryden, John, English poet, was descended from an ancient family, his grandfather being Sir Erasmus Dryden of Canons Ashby, Northamptonshire. Born near Aldwinkle, Northamptonshire, in 1631, he was admitted a King's scholar at Westminster under the celebrated Dr. Busby, whence he went to Trinity College, Cambridge, being elected to a scholarship there. After leaving the university, he went to London, where he acted as secretary to his cousin Sir Gilbert Pickering, a favourite of Cromwell; and on the death of the Protector he wrote his *Heroic Stanzas* on that event. At the Restoration, however, he hailed the return of Charles II in *Astræa Redux*, and from that time his devotion to the Stuarts knew no decay. In 1661 he produced his first play, *The Duke of Guise*; but the first that was performed was *The Wild Gallant*, which appeared in 1663 and was not a success. This was followed by *The Rival Ladies*, and *The Indian Queen*, a tragedy on Montezuma in heroic verse, written in collaboration with Sir Robert Howard, whose sister, Lady Elizabeth Howard, Dryden married in 1663. He followed up *The Indian Queen* with *The Indian Emperor*, which at once raised Dryden to the highest pitch of public estimation, an elevation which he retained till his death. The great fire of London put a stop for some time to theatrical exhibitions. In the interval Dryden published the *Annus Mirabilis*, an historical account of the events of the year 1666, one of the most elaborate of his productions. In 1668 he also published his celebrated *Essay on Dramatic Poesy*—the first attempt to regulate dramatic writing. In 1668 *The Maiden Queen*, a tragi-comedy, was represented. This was followed in 1670 by *The Tempest*, an alteration from Shakespeare, in which he was assisted by Sir William Davenant. It was received with general applause, notwithstanding the very questionable taste and propriety of the added characters. Dryden was shortly afterwards appointed to the offices of Historiographer Royal and Poet Laureate, with a salary of £200 a year. He now became professionally a writer for the stage, and produced many pieces, some of which have been strongly censured for their licentiousness and want of good taste. The first of his political and poetical satires, *Absalom and Achitophel* (Monmouth and Shaftesbury), was produced in 1681, and was followed by *The Medal*, a satire against sedition; and *Mac Flecknoe*, a satire on the poet Shadwell. In 1682 he published a

poem called *Religio Laici*, wherein he maintained the doctrines of the Church of England. On the accession of James in 1685 Dryden became a Roman Catholic, a conversion the sincerity of which has been not unreasonably regarded with suspicion, considering the time at which it occurred. At court the new convert was received with open arms, a considerable addition was made to his pension, and he defended his new religion at the expense of the old one in a poem, *The Hind and the Panther*. Among his other services to the new king were a savage reply to an attack by Stillingfleet, and panegyrics on Charles and James under the title of *Britannia Rediviva*. At the Revolution Dryden was deprived of the offices of Poet Laureate and Historiographer, and of the certain income which these offices secured him. During the remaining ten years of his life he produced some of his best work, including his admirable translations from the classics. He published, in conjunction with Congreve, Creech, and others, a translation of Juvenal, and one of Persius entirely by himself. About a third part of Juvenal was translated by Dryden, who wrote an essay on satire which was prefixed to the whole. His poetic translation of Virgil appeared in 1697, and, soon after, the well-known lyric *Alexander's Feast*, and his *Fables*. He died 1st May, 1700, in the sixty-ninth year of his age, and was buried in Westminster Abbey. Dryden is unequalled as a satirist among English poets, and the best of his tragedies are unsurpassed by any since written. His poetry as a whole is more remarkable for vigour and energy than beauty, but he did much to improve English verse. He was also an admirable prose writer. Personally he was modest and kindly. The whole of his works, edited by Sir W. Scott, were published in 1818 (18 vols. 8vo); they were republished with additional notes, &c., by Professor Saintsbury (1882-93).—BIBLIOGRAPHY: R. Garnett, *Age of Dryden*; Sir A. W. Ward, *History of English Dramatic Literature*; *Cambridge History of English Literature* (vol. viii).

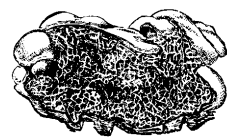
Drying-machine, a machine consisting of any number of steam-heated cylinders up to thirty or even more, each about 22 inches in diameter, and used in bleachworks, dye-houses, and in cloth-finishing departments; used as a separate machine to dry fabrics which contain a certain amount of moisture left in at some previous operation, but often used in conjunction with a starching-mangle or similar apparatus. All the cylinders are in a horizontal plane, and usually in two rows, but such rows may be disposed

either in horizontal or vertical frames. Floor-space is economized in the latter arrangement, and two or more groups of two rows per group may be provided for. Each cylinder is provided with some type of safety air-valve, which yields to allow air to enter in proportion as the steam is condensed in the cylinder, and so prevents the latter from collapsing. The condensed steam is withdrawn either by means of siphons or revolving scoops, so that the interior may be as dry as possible. The long length of cloth, either from the squeezing-rollers of the starch-mangle or from a loose or rolled state of cloth from some other machine, is conducted over guide-rollers, then under and over the two rows of steam-heated cylinders, and finally led from the last cylinder to the roller of a plaiting-down apparatus, or otherwise delivered. Both sides of the cloth thus come into direct contact with half the number of cylinders as it is drawn through the machine, and the dried cloth is ultimately delivered by the plaiting-down apparatus in folds ready for the subsequent operations.

Drying-oils, linseed and other oils, which are the bases of many paints and varnishes. When exposed to the air, they absorb oxygen, and are converted into a transparent, tough, dry mass or varnish.

Dry-point, a method of engraving generally regarded as part of etching, but more closely allied to line engraving. Instead of the copper being covered with etching ground and the lines bitten with acid, a pointed instrument is drawn across it, which incises a fine line with a more distinct burr on each side than that raised by a graver. This burr helps to give a characteristic quality to the line, but is rapidly worn away by printings. Dry-point may be used by itself, but is frequently combined with etching proper.

Dry-rot, a well-known disease affecting timber, occasioned by various species of Fungi, the mycelium of which penetrates the timber, destroying it. *Merulius lacrymans*, which is found chiefly in fir-wood, is the most common and most formidable dry-rot fungus in Britain; while *Polyporus destructor* is equally destructive in Germany. *P. vaporarius* may also cause dry-rot. Damp, unventilated situations are most favourable to the development of dry-rot Fungi. Various methods have been proposed for the prevention of dry-rot; that most in favour is thoroughly saturating the wood with creosote, which



Dry-rot Fungus
(*Merulius
lacrymans*)

makes the wood unfit for vegetation, but proper ventilation is the surest safeguard.

Dual, in grammar, that number which is used, in some languages, to designate two things, whilst another number (the plural) exists to express many. The Greek, Sanskrit, and Gothic among ancient languages, and the Lithuanian and Arabic among modern, possess forms of the verb and noun in which two persons or things are denoted, called the *dual* numbers.

Du'alism, the philosophical exposition of the nature of things by the hypothesis of two dissimilar primitive principles not derived from each other. Dualism in religion is chiefly confined to the adoption of a belief in two fundamental beings, a good and an evil one, as is done in some Oriental religions, especially that of Zoroaster. In metaphysics, dualism is the doctrine of those who maintain the existence of matter and form, or mind and matter, as distinct substances, in opposition to idealism, which maintains that we have no knowledge or assurance of the existence of anything but our own ideas or sensations. Dualism may correspond with realism in maintaining that our ideas of things are true transcripts of the originals, or rather of the qualities inherent in them, the spirit acting as a mirror and reflecting their true images; or it may hold that, although produced by outward objects, we have no assurance that in reality these at all correspond to our ideas of them, or even that they produce the same idea in two different minds. Among modern philosophers Professor W. M'Dougall and Bergson have defended the doctrine of dualism. See *Monism*.—BIBLIOGRAPHY: J. Ward, *The Realm of Ends*; W. James, *Essays in Radical Empiricism*; H. Bergson, *Matter and Memory*.

Dubail, Augustin Edmond, French general, born at Belfort 15th April, 1851. He served in the Franco-Prussian War, and was for many years colonel of a Zouave regiment in Algeria. Chief of the Staff of the French army in 1914, he commanded the First Army operating in Alsace-Lorraine, and successfully defended Nancy. Appointed Military Governor of Paris in 1915, he held this post until June, 1918.

Du Barry, Marie Jeanne Bécu, Comtesse, mistress of Louis XV, was born at Vaucouleurs in 1743. She came young to Paris, and was presented to the king in 1769, who had her married for form's sake to the Comte du Barry.

She exercised a powerful influence at court, and with some of her confidants completely ruled the king. Important offices and privileges were in her gift, and the courtiers abased themselves before her. After the death of Louis she was dismissed from court and sent to live in a convent near Meaux. She received a pension from Louis XVI. During the reign of terror she was arrested as a Royalist and executed, Nov., 1793.—Cf. N. Williams, *Madame du Barry*.

Dubit'za, a fortified town of Bosnia, in Yugo-Slavia, on the right bank of the Unna, about 10 miles from its confluence with the Save. In the sixteenth and seventeenth centuries it was a frequent point of contention between Austria and Turkey. In 1878, with the rest of Bosnia, it passed under Austrian administration. Pop. 3260.—*Dubitza*, in Croatia, on the opposite bank of the Unna, has 6660 inhabitants.

Dub'lin, the metropolis of Ireland, is situated in County Dublin, on the east coast of the island, at the mouth of the Liffey, the banks of which for more than 2 miles from the sea are lined with quays. The river, which divides the city into two unequal parts, is crossed by numerous bridges. In the old part of the city the streets are irregular, narrow, and filthy; in the more modern and aristocratic quarters there are fine streets, squares, and terraces, but with little pretension to architectural merit. The public buildings, however, are especially numerous and handsome. The main thoroughfare, east to west, is by the magnificent quays along the Liffey. The principal street at right angles to the river is Sackville Street, a splendid street 650 yards long and 40 yards wide, forming a thoroughfare which is continued across the river by O'Connell Bridge, a magnificent structure the same width as Sackville Street. The principal public secular buildings are the castle, the official residence of the viceroy; the Bank of Ireland, formerly the Irish Parliament House; Trinity College; the custom-house, destroyed in 1921; the King's Inns; the post office; rotunda; corn exchange; commercial buildings; the mansion house; and the city hall or corporation buildings. The most important literary and scientific institutions are Trinity College (Dublin University); the National University of Ireland; the Royal College of Science; the Catholic University; the College of Surgeons; the Royal Dublin Society; the Royal Hibernian Academy of Painting, Sculpture, and Architecture; the Royal Irish Academy for Promoting the Study of Science, Literature, and Antiquities; the Archæological Society; and the Royal

Zoological Society. Dublin contains two Protestant Episcopal cathedrals—St. Patrick's Cathedral, erected in 1190, and thoroughly restored between 1860 and 1865, through the munificence of Sir Benjamin Lee Guinness; and Christ's Church, built in 1038 and restored between 1870 and 1877, the restoration being carried out at the expense of Henry Roe. The Roman Catholic Cathedral is a very large edifice. The charitable institutions are numerous, and some of them possess splendid buildings. There are several extensive military and constabulary barracks in the city and vicinity. A little north-west of the city, up the Liffey, is the Phoenix Park, with an area of 1759 acres. In it are the Viceregal Lodge, the usual residence of the King's representative; the Chief Secretary's and Under-Secretary's official residences; the Royal Hibernian Military School; and the depot of the Royal Irish Constabulary; as also the gardens of the Royal Zoological Society. The manufactures carried on are of little note: poplins, for which Dublin has been long celebrated, are still in some request, and brewing and distilling are largely carried on. Since 1918 Dublin returns seven members to the House of Commons. The Sinn Fein members, however, elected in 1918, never attended the Imperial Parliament. Serious risings occurred in Dublin at Easter 1916, in 1919, and 1920, and there were also disorders in May 1921. Dublin is an ancient town, but its early history is obscure. It was held by the Danes for more than three centuries from 836. Pop. 399,000 (1919). The county, which is in the province of Leinster, on the east coast of the island, has an area of 218,873 acres, about a third of it under crops of various kinds, chiefly grass and clover. The surface on the whole is flat, but the ground rises at its southern boundary into a range of hills, the highest of which—Kippure—is 2473 feet above the sea. There is about 70 miles of sea-coast, the chief indentation being Dublin Bay. The principal stream is the Liffey, which intersects the county west to east. Important water communications are the Royal and the Grand Canals, both centering in Dublin, and uniting the Liffey with the Shannon. The manufactures are unimportant, but the fisheries are extensive. Since 1918 the county returns four members to the House of Commons. Pop. 172,394; Dublin (county borough), 304,802.—Cf. D. A. Chart, *The Story of Dublin*.

Dublin, University of, an institution founded in 1591, when a charter, or letters-patent, was granted by Queen Elizabeth for the incorporation of the "College of the Holy and Undivided Trinity", the University and Trinity

College being practically the same. The corporation now consists of a provost, seven senior fellows, twenty-six junior fellows, and seventy foundation scholars. The Senate of the university consists of "the chancellor of the university, or in his absence, of the vice-chancellor, and such doctors or masters of the university as shall have and keep their names on the books of Trinity College". The Senate possesses the right of electing the chancellor of the university; it is also the body which grants degrees. The fellows are appointed for life, after an examination. The scholars are chosen from among the undergraduates, after an examination in mathematics and logic, or in Greek, Latin, and logic. The scholarships are tenable for five years, or till the degree of M.A. is attained. The course of general instruction extends over four years. The academical year is divided into three terms—Michaelmas, Hilary, and Trinity—and every student must keep at least two terms in each year in order to obtain a degree. The system of instruction is superintended by the fellows, both junior and senior, together with a large staff of professors in the various departments of science and literature. Eighteen of the junior fellows act as tutors, and every student must place himself under one of these on entering the college. The B.A. degree is given after examination in the usual subjects, and may be a pass or honours degree; the M.A., as at Oxford and Cambridge, is gained by the payment of a fee after a certain time has elapsed. There are also a law school, a medical school, and a school of engineering, and degrees are granted in these subjects, as well as in arts and divinity. The college possesses a library of about 285,000 printed volumes and 1700 manuscripts. It has also a botanic garden and museum. In 1613 James I granted to the university the right of returning two members to Parliament. One was taken away at the Union, but was restored by the Reform Bill of 1832. The number of students in 1920 was 1350.—Cf. *Dublin University Calendar*.

Dubno, a town of the Ukraine, government of Volhynia. It was a place of some importance before the annexation of Western Poland by Russia. During the European War it was recaptured in the Russian advance in June, 1916. Pop. 14,000.

Dubois (du ´ -bwä), Guillaume, a French cardinal, was the son of an apothecary, born in 1656, died 1723. He became tutor to the Duke of Chartres, afterwards Duke of Orleans and Regent, and maintained his influence by pandering to the vices of his pupil. He became Privy

Councillor and overseer of the duke's household, and Minister for Foreign Affairs under the regency. The archbishopric of Cambrai having become vacant, Dubois ventured to request it of the regent, although he was not even a priest. The regent was astonished at his boldness; but he obtained the post, having in one morning received all the clerical orders, and, a few days after, the archbishopric. By his consummate address he obtained a cardinal's hat, and in 1721 was appointed Prime Minister. Dubois was an avaricious, lying, licentious creature, yet clever and industrious, and able to make himself very agreeable where it suited his interest.

Dubois (du´-bwä), Paul, French sculptor, born 1829, died in 1905. He first studied law, but from 1856 to 1858 gave himself up to sculpture under Toussaint at Paris, and then went to Italy, where the sculptors of the early Renaissance, Donatello and Luca Della Robbia, had a decided influence upon him. Among his works are a *St. John*, a *Narcissus*, a *Madonna and Child*, *Eve Awakening to Life*, a figure of *Song* for the opera-house at Paris, and numerous busts; but his greatest work is the monument of General Lamoricière in the Cathedral of Nantes, with figures of *Military Courage*, *Charity*, *Faith*, and *Meditation*, which rank among the best products of French plastic art. He is also distinguished as a painter of portraits. He was director of the École des Beaux Arts from 1878 until his death, and received the grand cross of the Legion of Honour.

Du Bois-Reymond (du´ bwä-rā-mōn´), Emil, German physiologist, and an especial authority on animal electricity, born at Berlin 1818, died in 1896. He studied theology, geology, and afterwards anatomy and physiology, and became professor of physiology in the University of Berlin in 1858. His principal publication is *Researches in Animal Electricity*.

Dubov´ka, a town of South Russia, government of Saratov, on the Volga; it has an extensive river trade in wool, iron, oil, and grain. Pop. 16,530.

Dubuque (du-būk´), a city of Iowa, United States, on the right bank of the Mississippi. It occupies an important commercial position as a railway centre and entrepôt for the agricultural and mineral products of the northern half of Iowa, and the timber of Wisconsin, and also from the valuable lead-mines in its vicinity. Pop. 39,428.

Ducange (du´-känzh), Charles Dufresne, Sieur, a French historian and linguist, was born in 1610 near Amiens, died at Paris 1688. He studied in the Jesuits' College at Amiens, afterwards at Orleans and Paris. At this last place he became Parliamentary Advocate in 1631, and in 1645 Royal Treasurer at Amiens, from which place he was driven by a pestilence, in 1668, to Paris. Here he devoted himself entirely to literature, and published his great works, viz. his *Glossaries of the Greek and Latin peculiar to the Middle Ages and the Moderns*, his *Historia Byzantina*, the *Annals of Zonaras*, the *Numismatics of the Middle Ages*, and other important works.

Ducas, Michael, Byzantine historian, flourished in the fifteenth century. His *Historia Byzantina*, which contains a reliable account of the siege and sack of Constantinople, was largely used by Gibbon.

Duc´at (Lat. *ducātus*, a duchy), a coin formerly common in several European states. They were either of silver or gold: value of the former, 3s. to 4s., of the latter about 9s. 4d. They were named from being first coined in one of the Italian duchies.

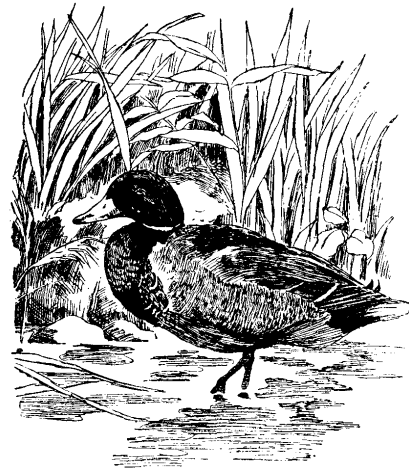
Ducatoon´, formerly a Dutch silver coin worth 3 gulden 3 stivers, or 5s. 3d. sterling. There were coins of the same name in Italy. In Tuscany its value was about 5s. 5d., in Savoy slightly more, and in Venice about 4s. 9d.

Du Chaillu (du´-shā-yü), Paul Belloni, traveller, born in Paris 1835, died 1903. He spent his youth in the French settlement at the Gaboon, on the west coast of Africa, where his father was a merchant. In 1852 he went to the United States, of which he afterwards became a naturalized citizen. In 1855 he began his first journey through Western Africa, and stayed till 1859 alone among the different tribes, travelling on foot upwards of 8000 miles. He collected several gorillas, never before hunted, and rarely, if ever, before seen by any European. An account of this journey, *Explorations and Adventures in Equatorial Africa*, was published in 1861. A second expedition was made in 1863, an account of which, under the title *A Journey to Ashango Land*, appeared in 1867. *The Land of the Midnight Sun*, an account of a tour in Northern Europe (1881), had a considerable success. He published a number of books intended for boys, and based on his travels. One of his works is *The Viking Age* (1889), on the ancestors of the English-speaking peoples.

Duchesne, or **Du Chesne** (du ´ -shān), André, French historian, born in 1584, died in 1640. His most important works are his collection of French historians—*Historiæ Francorum Scriptores*; *Historiæ Normanorum Scriptores 838-1220*; *Histoire d'Angleterre, d'Écosse, et d'Irlande*; *Histoire des Papes*.

Ducis (dü-sēs), Jean François, French dramatic writer, born at Versailles 1733, died 1816. Of his original works, the tragedy *Abufar* was much admired; but he is now best known for his adaptations of Shakespeare to the Parisian stage.

Duck, the name given to web-footed birds constituting the sub-family Anatinae of the family Anatidae, which also includes swans and geese. The ducks are very numerous as species, and are met with all over the world. They are often migratory, going northward in summer to their breeding-places. Their food is partly vegetable, partly animal. The common mallard or wild-duck (*Anas boschas*) is the original of the domestic duck. In its wild state the male is characterized by the deep green of the plumage of the head and neck, by a white collar separating the green from the dark chestnut of the lower part of the neck, and by having the four middle feathers of the tail recurved. The wild-duck is taken in large quantities by decoys and other means. Some tame ducks have nearly the same plumage as the wild ones; others vary greatly, being generally duller or pure white, but all the males have the four recurved tail-feathers. There are several favourite varieties of the domestic duck, those of Normandy and Picardy in France, and the Aylesbury ducks in England, being remarkable for their great size and delicacy of flesh. Other species of the sub-family are: shoveller (*Spatula clypeata*), garganey (*Querquedula circia*), pintail or sea-pheasant (*Dafila acuta*), teal (*Nettion crecca*), widgeon (*Mareca penelope*), gadwall (*Chaulelasmus streperus*), sheldrake (*Tadorna cornuta*), tree-ducks (species of *Dendrocygna*). In a wider sense the name 'duck' is applied to species of other sub-families of the Anatidae as follows: Merganettinae: blue duck (*Hymenolæmus malacorhynchus*) of New Zealand. Erismaturinae:



Mallard or Wild-duck

musk duck (*Biziura lobata*) of Tasmania and Australia. Lake ducks (species of *Erismatura*). Fuligulinæ: eider duck (*Somateria mollissima*), q.v.; scoter or black duck (*Edemia nigra*); harlequin duck (*Cosmonetta histrionica*); logger-head or steamer duck (*Tachyeres cinereus*) of South America: scaup (*Fuligula marila*); canvas back (*F. vallisneria*), q.v.; pochard (*Nyroca ferina*). Plectopterinae: summer duck (*Aix sponsa*) of N. America and Cuba; mandarin duck (*A. galericulata*) of E. Asia; Muscovy or musk duck (*Cairina moschata*), ranging from Mexico to the Argentine.—BIBLIOGRAPHY: Nourse, *Turkeys, Ducks, and Geese*; Rankin, *Natural and Artificial Duck Culture*; J. G. Millais, *British Diving Ducks*.

Ducking-stool, a stool or chair in which common scolds were formerly tied and plunged into water. They were of different forms, but that most commonly in use consisted of an upright post and a transverse movable beam on which the seat was fitted or from which it was suspended by a chain. The ducking-stool is mentioned in *Domesday Book* (Chester): it was extensively in use throughout the country from the fifteenth till the beginning of the eighteenth century, and in one case—at Leominster—was used as late as 1809.

Duckweed, the popular name of several species of *Lemna*, nat. ord. Lemnaceæ, plants growing in ditches and shallow water, floating on the surface, and serving as food for ducks and geese. Five species are known in Britain, and others are common in America. They consist of small fronds bearing naked unisexual flowers.

Duckworth, Sir John Thomas, a British admiral, born in 1748, died 1817. He joined the navy when eleven years of age; and was post-captain in 1780. In 1793, on the breaking out of the French war, he was appointed to the command of the *Orion*, 74, forming part of the Channel fleet under Lord Howe, and distinguished himself in 1794 in the great naval victory on 1st June. In 1798 he aided in the capture of Minorca. From 1800 to 1806 he rendered important services on the West India station, in particular gaining a complete victory over a French squadron, for which he received a pension of £1000 a year and the thanks of both Houses of Parliament. In 1807, having been ordered to Constantinople, he forced the passage of the Dardanelles, but suffered severely from the Turkish batteries in returning. From 1810 to 1814 he commanded the Newfoundland fleet, and in 1817 he

was appointed to the chief command at Plymouth. In 1813 he was created a baronet.

Duclos (du´-klō), Charles Pinot, a French novelist, writer of memoirs, and grammarian, born in 1704 at Dinant, died at Paris 1772. He became secretary of the French Academy, and on the resignation of Voltaire he was appointed to the office of historiographer of France. His writings are lively and satirical. Among the best are: *Confessions du Comte de * * ** (1741), *Considérations sur les Mœurs de ce Siècle*, *Mémoires secrets sur les Règnes de Louis XIV et XV*, and *Remarques sur la Grammaire générale de Port-Royal*.

Ductil´ity, the property of solid bodies, particularly metals, which renders them capable of being extended by drawing, while their thickness or diameter is diminished, without any actual fracture or separation of their parts. The following is nearly the order of ductility of the metals which possess the property in the highest degree, that of the first mentioned being the greatest: gold, silver, platinum, iron, copper, nickel, palladium, cadmium, zinc, tin, lead. Dr. Wollaston succeeded in obtaining a wire of platinum only $\frac{1}{30000}$ th of an inch in diameter. The ductility of glass at high temperatures seems to be unlimited, while its flexibility increases in proportion to the fineness to which its threads are drawn.

Duddon, an English river which flows 20 miles along the boundaries of Cumberland and Lancashire to the Irish Sea, and is the subject of a series of sonnets by Wordsworth, written in 1820.

Du Deffand, Madame. See *Deffand*.

Duderstadt (dö´dér-sta´t), an old German town, province of Hanover, 10 miles east of Göttingen, formerly a member of the Hanseatic League and a place of some importance. Pop. 5380.

Dudevant, Madame. See *Sand, George*.

Dud´ley, Sir Edmund, born 1462, executed 1510, noted in English history as an instrument of Henry VII in the arbitrary acts of extortion by the revival of obsolete statutes and other unjust measures practised during the latter years of his reign. On the accession of Henry VIII he was arrested for

high treason, and perished on the scaffold with his associate Sir Richard Empson.

Dudley, Lord Guildford, son of John, Duke of Northumberland, was married in 1553 to Lady Jane Grey, whose claim to the throne the duke intended to assert on the death of Edward VI. On the failure of the plot Lord Guildford was condemned to death, but the sentence was not carried into effect till the insurrection of Wyatt induced Mary to order his immediate execution (1554).

Dudley, John, Duke of Northumberland, son of Sir Edmund Dudley, minister of Henry VII, was born in 1502, beheaded 1553. He was left by Henry VIII one of the executors named in his will, as a kind of joint-regent during the minority of Edward VI. Under that prince he manifested the most insatiable ambition, and obtained vast accessions of honours, power, and emoluments. The illness of the king, over whom he had gained complete ascendancy, aroused his fears, and he endeavoured to strengthen his interest by marrying his son Lord Guildford Dudley to Lady Jane Grey, descended from the younger sister of Henry VIII, and persuaded Edward to settle the crown on his kinswoman by will, to the exclusion of his two sisters, the Princesses Mary and Elizabeth. The attempt to place Lady Jane Grey on the throne failed, and many of the conspirators were executed.

Dudley, Robert, Earl of Leicester. See *Leicester*.

Dudley, a town and parliamentary borough of England, in an isolated part of Worcestershire enclosed by Staffordshire, 8 miles west by north of Birmingham. It is situated in the midst of the 'black country', and has extensive coal-mines, iron-mines, ironworks, and limestone quarries. It produces nails, chain-cables, anchors, vices, boilers, fire-irons, and has also glassworks, brickworks, and brass-foundries. There are the remains of a castle, said to have been founded in the eighth century by a Saxon prince called Dud, who has given the town its name. Dudley returns one member to Parliament. Pop. of municipal borough, 51,079.

Dudley Limestone, a highly fossiliferous limestone belonging to the Silurian system, occurring near Dudley, and equivalent to the Wenlock limestone. It abounds in beautiful masses of coral, shells, and trilobites.

Duel (Lat. *duellum*, from *duo*, two), a premeditated and prearranged combat between two persons with deadly weapons, for the purpose of deciding some private difference or quarrel. The combat generally takes place in the presence of witnesses called seconds, who make arrangements as to the mode of fighting, place the weapons in the hands of the combatants, and see that the laws they have laid down are carried out. The origin of the practice of duelling is referred to the trial by 'wager of battle' which obtained in early ages. This form of duel arose among the Germanic peoples, and a judicial combat of the kind was authorized by Gundebald, King of the Burgundians, as early as A.D. 501. When the judicial combat declined, the modern duel arose, being probably to some extent an independent outcome of the spirit and institutions of chivalry. France was the country in which it arose, the sixteenth century being the time at which it first became common, especially after the challenge of Francis I to Charles V in 1528. Upon every insult or injury which seemed to touch his honour, a gentleman thought himself entitled to draw his sword, and to call on his adversary to give him satisfaction, and it is calculated that 6000 persons fell in duels during ten years of the reign of Henri IV. His minister, Sully, remonstrated against the practice; but the king connived at it, supposing that it tended to maintain a military spirit among his people. In 1602, however, he issued a decree against it, and declared it to be punishable with death. Many subsequent prohibitions were issued, but they were all powerless to stop the practice. During the minority of Louis XIV, more than 4000 nobles are said to have lost their lives in duels. The practice of duelling was introduced into England from France in the reign of James I; but it was never so common as in the latter country. Cromwell was an enemy of the duel, and during the Protectorate there was a cessation of the practice. It came again into vogue, however, after the Restoration, thanks chiefly to the French ideas that then inundated the court. As society became more polished duels became more frequent, and they were never more numerous than in the reign of George III. Among the principals in the chief duels of this period were Charles James Fox, Sheridan, Pitt, Canning, Castlereagh, the Duke of York, the Duke of Richmond, and Lord Camelford. The last mentioned was the most notorious duellist of his time, and was himself killed in a duel in 1804. A duel was fought between the Duke of Wellington and Lord Winchelsea in 1829, but the practice was dying out. It lasted longest in the army. By English law fatal duelling is considered murder, no matter how fair the

combat may have been, and the seconds are liable to the same penalty as the principals. In 1813 the principal and seconds in a fatal duel were sentenced to death, though afterwards pardoned. An officer in the army having anything to do with a duel renders himself liable to be cashiered. In France duelling still prevails to a certain extent; but the combats are usually very bloodless and ridiculous affairs. In the German army until 1918 it was common, and was recognized by law. The duels of German students, so often spoken of, seldom cause serious bloodshed.—BIBLIOGRAPHY: Millingen, *History of Duelling*; Steinmetz, *Romance of Duelling*; G. Letainturier-Fradin, *Le duel à travers les âges*; C. A. Thimm, *Bibliography of Fencing and Duelling*; A. Hutton, *The Sword and the Centuries*.

Duen'na, the chief lady-in-waiting on the Queen of Spain. In a more general sense, an elderly woman holding a middle station between a governess and companion, appointed to take charge of the young daughters of Spanish and Portuguese families.

Dufaure (du' -fōr), Jules Armand Stanislas, French orator and statesman, born 1798, died 1881. He practised law at Bordeaux; entered the Chamber of Deputies in 1834, and became an influential leader of the Liberal party. Under the Republic he was Minister of the Interior, but was driven from the public service by the *coup d'état* of 1851, and for the next twenty years devoted himself closely to his Bar practice and pamphlet writing. Under the Government of Thiers he acted as Minister of Justice; and in 1876, and again from 1877 to 1879, he was head of the Cabinet.

Duff, Alexander, Scottish missionary, born 1806, died 1878. Educated at St. Andrews University, in 1829 he set out for India as the first Church of Scotland missionary to that country, and reached Calcutta after being twice shipwrecked. He opened a school in which he taught successfully the doctrines of Christianity, as well as general knowledge, but on his secession (along with the other missionaries of the Church) from the Church of Scotland in 1843, he had to give up the school and begin again. In 1849 he visited Scotland, where he remained until 1856. He assisted in founding the University of Calcutta, and, having been obliged to return home for reasons of health, he raised £10,000 to endow a missionary chair in the New College, Edinburgh, becoming himself its first occupant. His chief writings are: *The Church of Scotland's India Mission* (1835); *Vindication of the*

Church of Scotland's India Mission (1837); *India and India Missions* (1840); *The Jesuits* (1845); and *The Indian Mutiny: its Causes and Results* (a series of letters published in 1858).

Duff, Sir Mountstuart Elphinstone Grant, writer on political and other subjects, born in Aberdeenshire in 1829, died in 1906. He was educated at Edinburgh Academy, The Grange, Bishop Wearmouth, and Balliol College, Oxford, was called to the Bar at the Inner Temple in 1854, and in 1857 entered the House of Commons as Liberal member for the Elgin Burghs, which constituency he continued to represent until 1881. He was Under-Secretary for India in W. E. Gladstone's ministry from 1868 to 1874, and Under-Secretary for the Colonies from 1880 to 1881, in which latter year he was appointed Governor of Madras. His Indian administration was most successful, and on his retirement in 1886 he was made a G.C.S.I. He was president of the Royal Geographical Society from 1889 to 1893, and of the Royal Historical Society from 1892 to 1899, and was also a trustee of the British Museum. His published works include: *Studies in European Politics* (1866); *A Political Survey* (1868); *Elgin Speeches* (1871); *Notes of an Indian Journey* (1876); *Miscellanies, Political and Literary* (1879); *Memoir of Sir H. S. Maine* (1892); *Ernest Renan* (1893); and *Notes from a Diary* (7 vols., 1897-1905).

Duff'erin and Ava, Frederick Temple Hamilton-Blackwood, Marquess of, British statesman and author, son of the fourth Baron Dufferin and a granddaughter of R. B. Sheridan, born at Florence 1826, died in 1902. He began his public services in 1855, when he was attached to Earl Russell's mission to Vienna. Subsequently he was sent as Commissioner to Syria in connection with the massacre of the Christians (1860); was Under-Secretary of State for India (1864-6); Under-Secretary for War (1866); Chancellor of the Duchy of Lancaster (1868-72); Governor-General of Canada (1872-78); Ambassador at St. Petersburg (1879-81); at Constantinople (1882); sent to Cairo to settle the affairs of the country after Arabi Pasha's rebellion (1882-3); Viceroy of India (1884-8); Ambassador to Italy (1889-91); to France (1891-96). He was elected president of the Royal Geographical Society in 1878, and Lord Rector of Glasgow University in 1891. Besides being a noted diplomatist, he was also a popular author. In 1847 he published *Narrative of a Journey from Oxford to Skibbereen during the year of the Irish Famine*; in 1860, *Letters from High Latitudes*; also various pamphlets

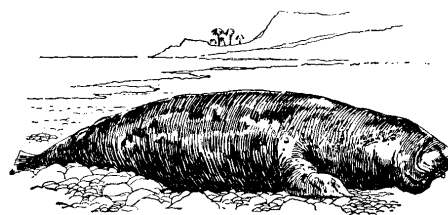
on Irish questions. In 1888 he was made Marquess of Dufferin and Ava.— Cf. Sir A. Lyall, *Life of the Marquess of Dufferin and Ava*.

Dufrénoy (du' -frē-nwä) Pierre Armand, French geologist and mineralogist, born in 1792, died in 1857. He became director of the school of mines, and published a great variety of papers on geology and mineralogy. In 1841 he published, in collaboration with Élie de Beaumont, a great geological map of France with three volumes of text, and this was followed by his *Traité de Minéralogie*. He introduced a new classification of minerals, based on crystallography.

Dufresny (du' -frā-nē), Charles Rivière, a French comic poet, born in 1648, died in 1724. He was clever and versatile, and had great skill as a landscape-gardener and an architectural designer. Among his dramatic pieces may be mentioned *L'Esprit de Contradiction*, *Le Mariage Fait et Rompu*, and *Le Double Veuvage*. He also wrote *Amusements sérieux et comiques d'un Siamois*, which suggested to Montesquieu his *Lettres Persanes*.

Dugdale, Sir William, an English antiquary, born in 1605 of a good family in Warwickshire, died 1686. He was made Chester herald in 1644; accompanied Charles I through the civil war; and after the Restoration received knighthood, and was appointed garter king-at-arms. In concert with Roger Dodsworth he produced an important work on English monasteries entitled *Monasticon Anglicanum*. Among his other works are: *Antiquities of Warwickshire*; *The Baronage or Peerage of England*; *Origines Judiciales, or Historical Memoirs of the English Law, Courts of Justice, &c.*; a *History of St. Paul's Cathedral*; and various minor writings. He also completed and published the second volume of Spelman's *Concilia*.

Dugong, a herbivorous mammal, the *Halicōrē dugong*, belonging to the ord. Sirenia, which also includes the manatees. It is a native of the Indian and Australian seas; possesses a tapering body ending in a crescent-shaped fin, and is said sometimes to attain a length of 20 feet, though generally it is about 7 or 8 feet in length. The fore-limbs are in the form of



Dugong (*Halicōrē dugong*)

flippers; hind-limbs are absent. The skin is thick and smooth, with a few scattered bristles; the colour bluish above and white beneath. Its food consists of marine plants; it yields little or no oil, but is hunted by the Malays for its flesh, which resembles veal, and is tender and palatable. It has been suggested that the appearance of this animal has given rise to the legends of mermaids and mermen.

Duguay-Trouin (dü-gā-trö-an), René, a distinguished French seaman, born at St. Malo in 1673, died at Paris 1736. As commander of a privateer he took many prizes from the British between 1690 and 1697. He then entered the royal marine as a captain, and signalized himself so much in the Spanish War that the king granted him letters of nobility, in which it was stated that he had captured more than 300 merchant ships and twenty ships of war. By the capture of Rio de Janeiro (1711) he brought the Crown more than 25,000,000 francs. Under Louis XV he rendered important services in the Levant and the Mediterranean.

Du Guesclin (dü-gā-klan), Bertrand, Constable of France, born about 1314, died 1380. Mainly to him must be attributed the expulsion of the English from Normandy, Guienne, and Poitou. He was captured by Chandos at the battle of Auray in 1364, and ransomed for 100,000 francs. While serving in Spain against Peter the Cruel, he was made prisoner by the English Black Prince, but was soon liberated. For his services in Spain he was made Constable of Castile, Count of Trastamare, and Duke of Molinas; and in 1370 he was made Constable of France.

Duikerbok, species of *Cephalophus*, small South African antelopes with short horns (none in the female), and a tuft of stiff hairs between them.

Duisburg (dö'is-bu`rh), a flourishing town in Rhenish Prussia, 13 miles north of Düsseldorf. It is an ancient place, believed to be of Roman origin. It early rose to be a free town, and became a member of the Hanseatic League. It possesses a beautiful church of the fifteenth century, and has iron manufactories, engineering works, chemical works, and cotton and woollen mills; and a large trade greatly facilitated by a canal communicating with the Rhine, which is about 2 miles distant. Pop. 229,483.

Dujardin (du'-zha`r-dan'), Karel, a Dutch artist, who excelled in painting landscapes, animals, and scenes in low life, born in 1640 at Amsterdam,

died at Venice 1678. His paintings are rare, and command high prices. His masterpiece, *The Charlatans* (1657), is in the Louvre.

Duke (Fr. *duc*, Sp. *duque*, It. *duca*, all from Lat. *dux*, leader, commander), a title belonging originally to a military leader. In Britain it is the highest rank in the peerage. Royal dukes have a special status and precedence. The first hereditary duke in England was the Black Prince, created by his father, Edward III, in 1336. The duchy of Cornwall was bestowed upon him, and was thenceforward attached to the eldest son of the king, who is considered a duke by birth. The duchy of Lancaster was soon after conferred on Edward's third son, John of Gaunt, and hence arose the special privileges which these two duchies still in part retain. A duke in the British peerage, not of royal rank, is styled 'your grace', or 'my Lord Duke'; his wife is a duchess. (See *Address, Forms of*.) The coronet consists of a richly-chased gold circle, having on its upper edge eight golden leaves of a conventional type called strawberry leaves; the cap of crimson velvet is closed at the top with a gold tassel, lined with sarsenet, and turned up with ermine. (See *Coronet*.) At various periods and in different continental countries the title duke (*Herzog* in Germany) was given to the actual sovereigns of small states. The titles 'grand-duke' and 'grand-duchess', 'archduke' and 'archduchess', were in use also on the European continent, especially in Russia and Austria until 1918. In the Bible the word *dukes* is used (*Gen.* xxxvi) for the *duces* of the *Vulgate*.

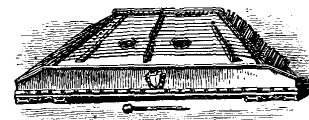
Dukhoborzi (duˈh-o-borˈtsē), a Russian sect of religious mystics which arose in the eighteenth century. The name means 'spirit-fighters', as the sect was accused by the orthodox priests of fighting against the spirit of God. They reject the doctrine of the Trinity, of the deity of Christ, hold property in common, and refuse oaths and military service, thus resembling Quakers. In 1899 a body of several thousands emigrated to Canada, where they received territory in Assiniboia and Saskatchewan.

Dukinfield, or **Duckinfield**, a municipal borough, England, county Cheshire, separated by the Tame from Ashton-under-Lyne, and mostly within Stalybridge parliamentary borough. Collieries, cotton-factories, brickworks, and tileworks give employment to the population. Pop. (municipal borough), 19,426.

Dulce (duˈlˈsā), a lake of Guatemala, on the east coast, communicating with the Gulf of Honduras by the lakelet el Golfete. It is about 30 miles long by 12 broad, and affords profitable turtle hunting.

Dulcigno (duˈl-chēnˈyō), a small seaport town, formerly in Albania, now in Montenegro, on the Adriatic, the seat of a Roman Catholic Bishop. It was captured by the Austrians in 1916, and retaken by Italian troops in 1918. Pop. 5000.

Dulcimer, one of the most ancient musical instruments, used in almost all parts of the world. The modern instrument consists of a shallow trapezium-shaped box without a top, across which runs a series of wires, tuned by pegs at the sides, and played on by being struck by two cork-headed hammers. It is in much less common use in Europe now than it was a century or two ago, and is interesting chiefly as being the prototype of the piano. It is still, however, occasionally to be met with on the Continent at fairs in the country, and in England in the hands of street musicians. It was known in Persia and Arabia under the name of *santir*, and was introduced into Europe by the Crusaders. The Hebrew psaltery is supposed to have been a variety of the dulcimer.



Italian Dulcimer

Dulcinists, followers of Dulcinus, a layman of Lombardy, in the fourteenth century, who preached the reign of the Holy Ghost, affirming that the Father had reigned till Christ's incarnation, and that the Son's reign terminated in 1300. He was followed by a great many people to the Alps, where he and his wife were taken and burned by order of Clement IV.

Dulcitol, or **Dulcite**, is an alcohol closely allied to the sugars. It is found in Madagascar manna, from which it is extracted by boiling water.

Dulī'a (Gr. *douleia*, service, from *doulos*, a slave), an inferior kind of worship or adoration, as that paid to saints and angels in the Roman Catholic Church. The Catholics recognize different degrees of worship. The lowest degree is the *dulia*, which is given to saints and angels. *Hyperdulia* is reserved for the Virgin alone; and *latria* is given to God and to each person of the Trinity.

Dulse, a red sea-weed, the *Rhodymenia palmāta*, used in some parts of Scotland as an edible. It has a reddish-brown, or purple, leathery, veinless frond, several inches long, and is found at low water adhering to the rocks. It is an important plant to the Icelanders, and is stored by them in casks to be eaten with fish. In Kamchatka a fermented liquor is made from it. In the south of England the name is given to the *Iridæa edūlis*, also an edible red sea-weed.

Duluth (du-luth'), a town of the United States, capital of St. Louis county, Minnesota, at the south-west extremity of Lake Superior. The Northern Pacific and Lake Superior and Mississippi railways terminate here; and extensive docks and other works have been constructed, affording a convenient outlet for the surrounding wheat region. Pop. 97,077.

Dulwich (dul'ich), a suburb of London, in County Surrey, about 5 miles south of London Bridge, giving name to a parliamentary division of the borough of Camberwell; noticeable on account of its school, *Dulwich College*, called the 'College of God's Gift,' founded as a charitable institution in 1619 by the actor Edward Allen or Alleyn. Four parishes were benefited by the charity: St. Luke's, Middlesex; St. Botolph's, Bishopsgate; St. Saviour's, Southwark; and St. Giles', Camberwell. Having outlived its usefulness, in 1857 an Act was passed by which the college was reconstituted. It now consists of two branches, the educational and the eleemosynary, between which the surplus revenue is divided in the proportion of three-fourths to the former and one-fourth to the latter. The educational branch comprises two schools, the upper and the lower; the former giving boys a high-class education (lower fees for those of the privileged parishes), and having a number of scholarships and exhibitions. The eleemosynary branch maintains a certain number of resident and non-resident poor people. The original revenues were only £800, but now amount to £20,000. Dulwich College is celebrated for its pictures, many of which were bequeathed by the founder; but the greater and more valuable portion of them was the bequest of Sir Francis Bourgeois, a landscape-painter, who died in 1810. The collection includes many fine pictures of the Dutch school.

Duma, or **Douma**, the Lower House of the former Russian Parliament, the Upper House being the Council of the Empire. In 1905 Tsar Nicholas II

granted his country a Constitution, promising that responsible Government would be established, and that no law would be made effective without the consent of the Duma. The first Duma accordingly met in 1906, and was to have had the power of a Parliament in Constitutional countries. The Legislative Assembly could make new laws, modify existing ones, issue the national Budget, &c., but had no right to alter the fundamental laws of the empire. In spite of the promises, however, given by the Tsar, the Imperial Government paid no attention to the demands of the Assembly, and when the criticisms of the Deputies became too loud, the first Duma was dissolved. A second Duma assembled the next year, but its members, in consequence of governmental restrictions on elections, were mostly Conservatives. In the opinion of the Government, however, even the second Duma was too Liberal in its tendencies, and it was promptly dissolved. The third Duma, which met in 1907, and whose members were mostly landed proprietors, retired officers, and priests, was absolutely subservient to the autocratic Government, and, from that date to the outbreak of the Russian Revolution in 1917, the power and influence of the Russian Duma were almost nil. The Duma ceased to exist on 7th Nov., 1917, when the Bolsheviks came into power, and the Government of Commissaries of the People was set up.

Dumas (du ' -mä), Alexandre (called *Dumas Père*), French novelist and dramatist, born at Villers-Cotterets 1803, died at Puys, near Dieppe, 1870. He was the son of a republican general, and grandson of the Marquis de la Paillette and a negress, Tiennette Dumas. In 1823 he went to Paris, and obtained an assistant-secretaryship from the Duke of Orleans, afterwards Louis Philippe. He soon began to write for the stage, and in 1829 scored his first success with his drama *Henri III et sa cour*. It was produced when the battle between the Romanticists and the Classicists was at its height, and hailed as a triumph by the former school. The same year appeared his *Christine*, and in quick succession *Antony*, *Richard d'Arlington*, *Térésa*, *La Tour de Nesle*, *Catharine Howard*, and *Mlle de Belle-Isle*. Dumas had now become a noted Parisian character. The critics fought over the merits of his pieces, and the scandal-mongers over his prodigality and *galanteries*. Turning his attention to romance, he produced a series of historical romances, among which may be mentioned: *Le bâtard de Mauléon*; *Isabelle de Bavière*; *Les Deux Dianas*; *La Reine Margot*; *Les Trois Mousquetaires*,

with its continuations *Vingt Ans Après* and the later *Vicomte de Bragelonne*. His *Monte-Cristo* and several others are also well known to English readers through translations. Several historical works were also written by him: *Louis XIV et son Siècle*, *Le Regent et Louis XV*, *Le Drame de '93*, *Florence et les Médicis*, &c. The works which bear his name amount to some 1200 volumes, including about 60 dramas; but the only claim he could lay to a great number of the productions issued under his name was that he either sketched the plot or revised them before going to press. He earned vast sums of money, but his recklessness and extravagance eventually reduced him to the adoption of a shifty, scheming mode of living. His *Mémoires*, begun in 1852, present interesting sketches of literary life during the Restoration, but display intense egotism. In 1860 he accompanied Garibaldi in the expedition which freed Naples from the Bourbons. He died at the residence of his son, and was buried in Villers-Cotterets in 1872. Dumas was remarkable for his creative rather than for his artistic genius, and although he frequently squandered his gifts, he was admired even by the highly cultured, such as Thackeray and others.—BIBLIOGRAPHY: H. Blaze de Bury, *Alexandre Dumas: sa vie, son temps, son œuvre*; A. B. Davidson, *Alexandre Dumas père: his Life and Works*.

Dumas, Alexandre, son of the above, born 1824, died in 1895, novelist and dramatist. His works treat mostly of the relations between vice and morals. His first novels, *La Dame aux Camélias* and *Diane de Lys*, were very successful, as were also the plays which were founded on them. His dramas, which are much superior to his novels, deal satirically with the characters, follies, and manners of French society. He was thus a pioneer in the 'comedy of manners'. His plays, besides his dramatized novel *La Dame aux Camélias*, which marked a date in the history of the French stage, and which supplied Verdi with the plot for *La Traviata*, are: *Le demi-monde*, *Le fils naturel*, *L'Ami des femmes*, *La princesse Georges*, and *L'Étrangère*.

Dumas, Matthieu, French soldier and military writer, born in 1753, died in Paris 1837. He early entered the French cavalry, took part in the War of American Independence, and was employed in the Levant and in Holland. At the commencement of the Revolution he assisted Lafayette in organizing the National Guard. On the triumph of the extreme party in 1797 Dumas was proscribed, but made his escape to Holstein, where he wrote the first part of his *Précis des Événements Militaires*, a valuable source for the

history of the period of which it treats (1798-1807). He was recalled from exile by Napoleon, who had become First Consul. His first employment was to organize the reserve for the army of Italy. In 1802 he was appointed State Councillor; in 1805 he became general of division, and was shortly afterwards Neapolitan minister in the service of Joseph Bonaparte. In 1808 he was actively employed in the arrangements for the war against Austria, fought in the battles of Essling and Wagram, and arranged the terms of the armistice of Znaim. He held the office of General Intendant of the army in the campaign of 1812. After the Restoration Louis XVIII appointed him Councillor of State, and gave him several important appointments connected with the army. In 1830 he aided in bringing on the revolution of July, and after the fall of Charles X he obtained the chief command of all the national guards of France, together with a peerage. He published a translation of Napier's *History of the Peninsular War*.

Du Maurier (du ´ -mō´ri-ā), George Louis Palmella Busson, artist and writer, was born in Paris 1834, died in 1896. He was the son of an English mother and a Frenchman who had been naturalized as a British subject. At the age of seventeen he took up the study of chemistry in London, but soon adopted art as a profession. After studying in Belgium and France, he returned to London, and soon began to contribute drawings to *Punch*, *Once a Week*, *Cornhill Magazine*, &c. He succeeded Leech on *Punch*, and became famous chiefly through his drawings for that publication. He also illustrated various books, and wrote three novels, *Trilby*, *Peter Ibbetson*, and *The Martian*. His elder son, Guy Du Maurier, born in 1865, killed in France in 1915, was the author of *An Englishman's Home* (1909).

Dumba, Konstantin Theodor, Austro-Hungarian diplomatist, who became conspicuous as an agent of German propaganda in the United States during the European War. He went as Ambassador to the United States in 1913, and at the outbreak of the war endeavoured to organize a vast conspiracy there, with a view to hampering the productions of munitions for the Allies. Abusing his position of Ambassador, he planned explosions and strikes in American factories. On 1st Sept., 1915, J. F. J. Archibald, an American newspaper correspondent, was arrested at Falmouth, and among the papers found upon him was a letter from Dr. Dumba to Baron Burian, Austrian Minister for Foreign Affairs, wherein the former suggested a plan of crippling the munition factories in America by creating strikes. President

Wilson then demanded his recall, and he left the United States in Oct., 1915, Great Britain granting him a safe conduct.

Dumbar'ton, a royal and parliamentary burgh and seaport, Scotland, chief town of Dumbarton county, stands on the Leven near its junction with the Clyde, 16 miles W.N.W. of Glasgow. Shipbuilding is carried on to a great extent, and there are foundries and engine-works. Dumbarton unites with Port-Glasgow, Renfrew, Rutherglen, and Kilmarnock in sending a member to Parliament. Originally it was called Alcluyd, and it was the chief town of Cumbria or Strathclyde. Pop. 21,989. A little to the south is the famous rock and castle of Dumbarton, rising above the Clyde. The rock, which is of basalt, is 240 feet in height, and about 1 mile in circumference at the base. It is one of the fortresses stipulated to be kept in repair by the Act of Union, and the barracks contain accommodation for 150 men. There has been a stronghold here from the earliest times, and the fortress of Dumbarton occupied an important place in Scottish history.—*Dumbartonshire*, the county of Dumbarton, is partly maritime, partly inland, consisting of two detached portions, the larger and most westerly lying between the Clyde, Loch Long, and Loch Lomond, and the far smaller portion being about 4 miles east of the former, and comprising only two parishes. More than half the area of the county is occupied by mountains, some of them attaining a height of upwards of 3000 feet. The lower lands are fertile, and in general well cultivated. More than one-half of Loch Lomond and fully two-thirds of the islands in it belong to Dumbartonshire. The Gareloch, an arm of the Firth of Clyde, forms a part of the county into a peninsula. The principal rivers are the Leven, from Loch Lomond, and the Kelvin, both belonging to the Clyde system. The chief minerals are coal, limestone, ironstone, and slate, all of which are wrought more or less. On the banks of the Leven and elsewhere are extensive cotton printing and bleaching establishments; and there are extensive shipbuilding yards along the Clyde. Besides Dumbarton, the chief town, the county contains the towns of Helensburgh and Kirkintilloch, and the manufacturing villages of Alexandria, Renton, and Bonhill. Vestiges of the Roman wall of Antoninus still exist. The county returns one member to the House of Commons. Pop. 139,831.

Dumb-cane, a plant of the ord. Araceæ, the *Dieffenbachia seguina*, of the West Indies, so called from its acridity causing swelling of the tongue when chewed, and destroying the power of speech.

Dumdum, a military village and extensive cantonment, Hindustan, province of Bengal, 4½ miles E.N.E. of Calcutta. The village is famous as being the scene of the first open manifestation of the sepoys against the greased cartridges, which led to the mutiny of 1857. Pop. 12,000.

Dumdum Bullet (so called from the arsenal at Dumdum, a small village 4½ miles from Calcutta), a hollow-nosed bullet which expands on impact, and so causes an ugly wound. It was used in Indian frontier fighting to stop the rushes of fanatical tribesmen. While the term 'Dumdum' bullet should strictly only be applied to hollow-nosed bullets, it is popularly applied to any kind of expanding bullet. Ordinary bullets can be converted into expanding ones by means of filing the cupro-nickel envelope until the lead core is exposed, by means of slitting the envelope at the shoulders, or simply by reversing the bullet in its socket. Expanding bullets are considered legitimate in big-game shooting, but in the Declaration signed at the Hague, 29th July, 1899, Germany expressly promised not to use such bullets in warfare. In spite of this the Germans freely used bullets of this kind in the European War.

Dumfries (dum-frēs'), a river port, railway centre, and until 1918 a parliamentary burgh, Scotland, capital of the county of same name, and the chief place in the south of Scotland; situated on the left bank of the Nith, about 6 miles from its junction with the Solway Firth. It is connected with the suburb Maxwelltown (in Kirkcudbright) by three bridges, one dating from the thirteenth century. It is a pleasing, well-built town, with various handsome public edifices. There are iron-foundries, hosiery and tweed factories, tanneries, and coach-building works. The River Nith is navigable to the town for vessels of under 60 tons, but the port has decreased in importance since the development of the railway system. Dumfries is a place of great antiquity. The church of the Minorites which once stood here was the scene of the murder of the Red Comyn by Bruce in 1306. Burns spent his closing years here, and the street in which he lived now bears his name. His remains rest under a handsome mausoleum, and a statue of him was erected in 1882. Dumfries was the head-quarters of the Young Pretender in 1745. Until 1918 Dumfries united with Annan, Sanquhar, Lochmaben, and Kirkcudbright (the Dumfries burghs) in sending a member to Parliament. Pop. 19,076.—*Dumfriesshire*, the county of Dumfries, abuts on the Solway Firth, having on its borders the counties of Lanark, Peebles,

Selkirk, Roxburgh, Ayr, and Kirkcudbright; area about 1100 sq. miles or 702,946 acres, of which about a third is under cultivation. The surface is irregular, but for the most part mountainous, especially in the north and north-west districts, where the hills attain a considerable elevation, some of them exceeding 2000 feet. The dales of the Nith, Annan, and Esk, the chief rivers of the county, contain fine pasture holms and good arable land. Oats, potatoes, and turnips are the most common products. Good cattle are reared, and are much in request for the English market. The sheep on the hill pastures are mostly Cheviots; on the lower and arable lands the Leicester breed prevails. The minerals most abundant are coal, lead, iron, antimony, and gypsum. Coal and lead are worked to a small extent. Limestone and freestone abound in various parts. There are no manufactures worth mentioning. The county returns one member to the House of Commons. Its principal towns are Dumfries, Annan, Sanquhar, Lockerbie, Moffat, Langholm, and Lochmaben. Pop. 72,825.

Dumont (du´-mōn), Pierre Étienne Louis, the friend and literary assistant of Mirabeau and Jeremy Bentham, was born at Geneva in 1759, died at Milan 1829. Ordained a minister of the Protestant Church in 1781, he attached himself to the democratic party in Geneva, and when the opposite party gained the ascendancy he went to St. Petersburg, in 1782, where he was appointed pastor of the French Reformed Church. Soon after he accepted an offer to act as tutor to the sons of Lord Shelburne, afterwards Marquess of Lansdowne, which brought him to London, where he became intimate with Jeremy Bentham and Sir Samuel Romilly. Visiting Paris during the first years of the Revolution, he gained the friendship of Mirabeau, whom he assisted in the composition of speeches and reports, and of whom he wrote some interesting *Recollections*. On his return to London he formed that connection with Bentham which fixed his career as a writer; recasting, popularizing, and editing Bentham's works in a form suitable for the reading public. He returned to Geneva in 1814 and became a Senator.

Dumont d'Urville (du´-mōn du´r-vēl), Jules Sebastien César, French navigator, was born in 1790, killed in a railway accident between Paris and Versailles 1842. After completing his studies at Caen, he entered the French navy, in which he ultimately rose to be rear-admiral. From 1826 to 1829 he commanded the corvette *Astrolabe*, which was sent to obtain tidings of La Pérouse, and to make hydrographic observations. He made surveys of the

coasts of Australia and New Zealand, and found remains of the shipwreck of La Pérouse on one of the Pacific islands. The result of this voyage was the publication of *Voyage de Découverte autour du Monde*. In 1837 he sailed with the *Astrolabe* and *Zélée* on a voyage of Antarctic discovery, and after many dangers, and having visited many parts of Oceania, he returned in 1840. On his return he began the publication of *Voyage au Pôle sud et dans l'Océanie*, which was finished by one of his companions.

Dumouriez (du'-mö-ri-ā), Charles François Duperrier, a French general of great military talent, was born at Cambrai in 1739 of a noble family of Provence, died near Henley-on-Thames 1823. He served as an officer in the Seven Years' War. In 1768 he went to Corsica as quartermaster-general of the small army which was sent for the conquest of that island, and was afterwards made colonel. In 1778 he was appointed Governor of Cherbourg. At the Revolution he joined the Jacobins, and subsequently the Girondists, and in 1792 he was Minister of Foreign Affairs. War breaking out between France and Austria, he resigned in order to take command of the army; invaded Flanders, and defeated the Austrians at Jemappes and conquered Belgium. Instead of prosecuting the war vigorously, he now entered upon measures for the overthrow of the Revolutionary Government, issued a proclamation, in which he promised the restoration of the constitutional monarchy in the person of the heir to the crown, but was attacked by the Versailles volunteers, and compelled to flee (4th April, 1793). The Convention set a price of 300,000 livres upon his head. At first he retired to Brussels, and after various wanderings found a final refuge in England. His *Memoirs*, written by himself, appeared in 1794; an enlarged edition in 1822. He was also the author of a large number of political pamphlets.—BIBLIOGRAPHY: J. Holland Rose and A. M. Broadley, *Dumouriez and the Defence of England against Napoleon*; H. Welschinger, *Le Roman de Dumouriez*.

Düna (dü'na'), or **Western Dvina**, a river of Russia, which rises in the government of Tver, about 15 miles w. of the source of the Volga, falls into the Gulf of Riga, has a course of about 650 miles, and waters the seven governments of Tver, Pskov, Vitebsk, Mogilev, &c., draining an area of about 65,000 sq. miles. It is navigable for a considerable distance, but is frozen for about four months each year.

Dünaburg (dü´na-bu`rg), **Dvinaburg**, or **Dvinsk**, a fortified town in Latvia, formerly belonging to Russia, in the government of Vitebsk on the right bank of the Düna, or Dvina, 112 miles south-east of Riga. It carries on various industries, a considerable trade, and has three yearly fairs. The official name is Dvinsk, or Daugavpils. It was captured by the Germans in Feb., 1918. Pop. 110,912.

Dünamünde (dü´na-mu´n-de; 'Dünamouth'), a fortress and port on the Gulf of Riga, at the mouth of the Düna, having a large winter harbour for the shipping of Riga. Pop. 2500.

Dunbar´, William, the most eminent of all the old Scottish poets, was born, probably in East Lothian, about 1460-5. In 1475 he went to St. Andrews, where, in 1477, he took the degree of B.A., and two years later that of M.A. After this he seems to have become a begging friar of the Franciscan order, and made journeys in England and France, but he returned to Scotland about 1490, and attached himself to the court of James IV, from whom he received a pension of £10. On the marriage of James IV to Margaret of England Dunbar celebrated the event in a poem of great beauty, entitled *The Thrissil and the Rois*. His pension was ultimately raised to £80 a year, and he was the recipient of various additional gratuities, though he appears frequently to have addressed both the king and the queen for a benefice, but always without success. After the defeat at Flodden his name disappears from the royal accounts, and he probably died about 1520. His works, which consist of elaborate allegories, satirical and grimly humorous pieces, and poems full of brilliant description and luxuriant imagination, first collected by David Laing (Edinburgh, 1834), were edited by John Small and Æ. J. G. Mackay, for the Scottish Text Society, between 1884 and 1893. —Cf. *The Cambridge History of English Literature* (vol. ii).

Dunbar´ (Gael., Castle Point), a town of Scotland; a royal and municipal (formerly parliamentary) burgh and seaport in Haddingtonshire, at the mouth of the Firth of Forth. It is a place of great antiquity, having originated in a castle, once of great strength and importance, which underwent several memorable sieges, on one occasion being successfully defended (in 1338) against the English for nineteen weeks by Black Agnes of Dunbar, Countess of March. In 1650, at the 'Race of Dunbar', Cromwell totally defeated the

Scottish army under David Leslie near the town. The harbour is not very commodious, but the town is an important fishing-station. Pop. 4830.

Dunblane', an old episcopal city, Scotland, in Perthshire, 6 miles north-east of Stirling, on the Allan. The ancient cathedral, partly in ruins, dates from the twelfth century. The nave is 130 feet by 58 feet, and the choir, now the parish church, is 80 feet by 30 feet. The building was restored in 1893. Bishop Leighton held the see from 1662 to 1670. About two miles from the town the indecisive battle of Sheriffmuir was fought in 1715, between the Royal forces under the Duke of Argyle, and the Jacobites under the Earl of Mar. Pop. 4600.

Dun'can, Adam, Viscount, a British naval officer, was born in Dundee in 1731, died 1804. He went to sea when young, and was a post-captain in 1761. In the following year he served at the taking of Havana; and in 1779 he shared in the victory of Admiral Rodney over the Spaniards. In 1789 he became rear-admiral of the blue, and in 1794 vice-admiral of the white squadron. The following year he was appointed commander of the North Sea fleet, and in Oct., 1797, won a brilliant victory over the Dutch fleet off Camperdown, for which he was rewarded with the title of Viscount Duncan of Camperdown and a pension of £2000 a year.

Duncan, Thomas, an eminent Scottish painter, was born in 1807, died at Edinburgh 1845. He studied under Sir W. Allan, and was elected an associate of the Royal Academy in 1843. His principal works were illustrative of Scottish history and character. Among the best known of them are: *The Abdication of Mary Queen of Scots*, *Anne Page and Slender*, *Prince Charles Edward and the Highlanders entering Edinburgh after Prestonpans*, *Charles Edward asleep in a Cave after Culloden*, and *The Martyrdom of John Brown of Priesthill*.

Dun'cansby Head, a promontory in Caithness-shire, Scotland, forming the N.E. extremity of the Scottish mainland, $1\frac{3}{4}$ miles E. of John o' Groat's House, and $18\frac{1}{2}$ miles N. by E. of Wick. Close by the promontory are two rocks, of fantastic form and great height, called the Stacks of Duncansby, which in spring and summer are covered with sea-fowl.

Dundalk (dun-da'k'), a seaport and former parliamentary borough, Ireland, capital of the county of Louth, on Castletown River, about 2 miles above its

mouth in Dundalk Bay. It has railway workshops, tanyards, and a spinning-mill; the trade, chiefly in cattle and agricultural produce, is extensive. It was the seat of the court of Edward Bruce from 1315 to 1318. In 1649 it was captured by Cromwell. Pop. 13,128.

Dundas' of Arniston, the name of a family several members of which held a conspicuous place in the legal and political history of Scotland.—Sir James Dundas, the first of Arniston, knighted by James VI, was the third son of George Dundas of Dundas, a descendant of the Dunbars, Earls of March.—His eldest son, Sir James, was member of Parliament for Mid-Lothian, and was appointed one of the judges of the Court of Session (1662).—His eldest son Robert was also raised to the bench of the Court of Session, and filled that station for thirty-seven years. He died in 1727.—His eldest son Robert (1685-1753) was successively Solicitor-General for Scotland, Lord-Advocate, member of Parliament for the county of Edinburgh, and Dean of the Faculty of Advocates. In 1737 he was raised to the bench, and on the death of Lord-President Forbes of Culloden, in 1748, he was appointed his successor.—His eldest son Robert (1713-87) also attained to the position of Lord-Advocate, and Lord-President of the Court of Session.—His brother Henry Dundas, Viscount Melville, distinguished statesman, was born in 1741 and died 1811. He obtained the post of Solicitor-General in 1773, that of Lord-Advocate in 1775, and was made joint keeper of the signet for Scotland in 1777. In 1782 he was appointed Treasurer of the Navy and member of the Privy Council; and from that time took a leading part in all the Pitt measures, and had supreme influence in Scotland. Among other offices he held that of First Lord of the Admiralty; and in 1805 he was impeached before the House of Lords of high crimes and misdemeanours in his former office of Treasurer of the Navy, but was finally acquitted. He was created Viscount Melville in 1801, a title still borne by his direct descendant.

Dundee', John Graham of Claverhouse, Viscount. See *Graham*.

Dundee (Gael. *Dun Taw*, fort on the Tay), a city, royal and parliamentary burgh, and seaport, Scotland, in the county of Forfar, on the north shore of the Firth of Tay, about 8 miles from the open sea, 37 miles N.N.E. of Edinburgh; in population the third town in Scotland. It stretches along the Tay, or east to west, and of late years has been greatly extended in both

directions. The more recently made streets are spacious and handsome, but most of those of more ancient date are narrow, and irregularly built. The most conspicuous building is St. Mary's Tower, or the Old Steeple as it is popularly called, 156 feet high, erected in the middle of the fourteenth century. Three modern parochial churches have been built on to it in form of a cathedral, the nave, choir, and transept respectively forming a separate church. Among public buildings are: the town hall, several public halls, the high school, exchange, infirmary, lunatic asylum, Albert Institute and free library. University College, for men and women, was opened in 1883, and affiliated to the University of St. Andrews in 1897. It was founded by private munificence, receiving an endowment of £140,000, and has ten professorships. Dundee has several public parks and recreation grounds and a good supply of water. The town has long been celebrated for its textile manufactures, particularly those of the coarser descriptions of linen, and it is now the chief seat of the linen trade in Scotland and of the jute trade in Great Britain, there being a great number of mills and factories engaged in the spinning and weaving of flax, jute, and hemp. Shipbuilding is extensively carried on, and there are large engineering establishments. Another branch of business is the northern seal and whale fishery. Dundee is also famous for its marmalade and other preserves and confectionery. The shipping accommodation includes five large wet-docks, with a connected tidal harbour and graving-docks. The chief foreign trade is with the Baltic and Archangel in the importation of flax and hemp, with Norway, Sweden, and Canada in timber, and with Calcutta in jute. The railway facilities of Dundee were greatly increased in 1878 by the opening of a bridge across the Tay; but on the 28th of Dec., 1879, the bridge was destroyed by a violent storm, when about 100 people in a train in the act of crossing lost their lives. A new bridge, to replace the one destroyed, was opened for traffic in June, 1887; it is a very substantial structure about 2 miles in length. Dundee was made a royal burgh by William the Lion about 1200, was twice in the possession of the English under Edward I, and was as often retaken by Wallace and Bruce. In 1645 it was besieged, taken, and sacked by the Duke of Montrose; and six years afterwards it was stormed by Monk, when a great number of its inhabitants were put to death. Since 1868 the town has returned two members to Parliament. In 1888 it was raised by royal grant to the rank of a city. In 1914 the burgh of Broughty Ferry was

annexed to Dundee, increasing its area to 5964 acres. Pop. 176,062; population in 1921, 168,217.

Dundon'ald, Thomas Cochrane, tenth Earl of, British admiral, was born in Lanarkshire 1775, died 1860. At the age of eighteen he embarked with his uncle, then Captain, and afterwards Sir Alexander Cochrane, in the *Hind*, of twenty-eight guns, and soon distinguished himself by his daring and gallantry. In 1800 he was appointed to the *Speedy* sloop-of-war of fourteen guns, and in the course of thirteen months captured over fifty vessels, but was at last captured himself. In 1805, while in command of the *Pallas* frigate, he took some rich prizes, and for the next four years in the *Impérieuse* performed remarkable exploits in cutting out vessels, storming batteries, and destroying signals. On his return to England he entered Parliament, and by his attacks on the abuses of the naval administration made himself obnoxious to the authorities. He gave further offence by charging Lord Gambier, his superior officer, with neglect of duty (which was true); by denouncing the abuses of the prize-court, and the treatment of the prisoners of war. His enemies succeeded in 1814 in convicting him on a charge—since proved to be false—of originating a rumour, for speculative purposes, that Napoleon had abdicated. He was expelled from Parliament, ignominiously ejected from the Order of the Bath, imprisoned for a year, and fined £1000. The electors of Westminster immediately paid his fine and re-elected him, but he had to remain in prison till the expiration of his sentence. In 1818 he took service in the Chilian navy, his exploits greatly aiding the national independence of that country, as well as soon after of Brazil. In 1832 he was restored to his rank in the British navy. In 1831, by the death of his father, he had succeeded to the title of Earl of Dundonald; in 1841 he became vice-admiral of the blue; in 1848 he was appointed commander-in-chief on the North America and West India station; and in 1851 and 1854 respectively he became vice-admiral of the white, and rear-admiral of the United Kingdom. He was reinstated in the Order of the Bath (G.C.B.) on 25th May, 1847. He did much to promote the adoption of steam and the screw propeller in war-ships. He wrote an autobiography, which, though left incomplete, is a most interesting work.—Cf. J. B. Atlay, *The Trial of Lord Cochrane before Lord Ellenborough*.

Dunedin (dun-ē'din), capital of the provincial district of Otago, New Zealand, and the most important commercial town in the colony, stands at

the upper extremity of an arm of the sea, about 9 miles from its port, Port Chalmers, with which it is connected by railway. Though founded in 1848, its more rapid progress dates only from 1861, when extensive gold-fields discovered in Otago attracted a large influx of population. There are many handsome buildings, both public and private: the municipal buildings, the post office, hospital, lunatic asylum, Government offices, the university, high schools (boys' and girls'), the new museum, several banks (especially the Bank of New Zealand), the athenæum and mechanics' institute, the freemasons' hall, and two theatres. Wool is the staple export. Several woollen and other manufactories are now in existence. There is a regular line of steamers between this port and Melbourne, and communication is frequent with all parts of New Zealand. Through the opening of the new Victoria Channel from Port Chalmers vessels drawing 16 feet can now ascend to Dunedin at low water. Pop. 68,716.

Dunes, low hills of sand accumulated on the sea-coasts of Holland, Britain, Spain, and other countries, in some places encroaching on and covering what once was cultivated land, but in others serving as a natural barrier to protect the country from the destructive encroachments of the sea.

Dunferm'line, a royal and police burgh of Scotland, county of Fife, 3 miles N. of the Firth of Forth, and 13 miles north-west of Edinburgh. The streets though narrow are well built. Dunfermline was early a favourite residence of the kings of Scotland, and at it were born David II, James I, Charles I, and his sister Elizabeth. The Benedictine abbey founded by Malcolm Canmore (1070) is now represented chiefly by the Abbey Church, in which are the remains of Queen Margaret and Canmore, Alexander I and his Queen, David I, Malcolm IV, and Robert Bruce. Dunfermline was made a royal burgh in 1588. The town has greatly benefited through the munificence of the late Andrew Carnegie, a native, who, besides other benefactions, settled on it the sum of £500,000. In the manufacture of table-linen it is unrivalled by any town in the kingdom. There are collieries adjacent. The Dunfermline burghs return one member to Parliament. Pop. 28,103.

Dungan'on, a town of Ireland, County Tyrone, 35 miles west by south of Belfast. It has manufactures of linen and earthenware. Till 1885 it returned a member to the House of Commons. Pop. 3830.

Dungarpur (dön-gar-pör'), an Indian native state in Rájputána; area, 1000 sq. miles; pop. 153,381.—*Dungarpur* is also the name of the chief town and residence of the Maharawal of the state.

Dungar'van, a seaport of Ireland, County Waterford, on the Bay of Dungarvan, much resorted to for sea-bathing. The harbour is shallow, and the trade depends almost entirely on agricultural produce. Till 1885 it returned a member to Parliament. Pop. 4977.

Dung Beetle, a name applied to a large number of lamellicorn beetles (in which the antennæ terminate usually in lateral leaflets) from their habit of burying their eggs in dung. The *Geotrūpes stercorarius*, 'dor' or 'shard-borne' beetle, and the *Scarabæus sacer*, or sacred beetle of the Egyptians, are examples.

Dungeness (dunj-nes'), a low headland on the s. coast of Kent, 102 miles S.E. of Rye; has a lighthouse with fixed light.

Dunkeld', a small town of Scotland, on the Tay, about 14 miles north by west of Perth; pop. 613. It is a very ancient place, and from 850, when Kenneth I removed the remains of St. Columba from Iona to a church which he had built here, became the metropolitan see of Scotland, till supplanted by St. Andrews. The choir of the ancient cathedral is still used as the parish church. Near it is Dunkeld House, the seat of the Duke of Atholl, the grounds of which are the finest and most extensive in Scotland.

Dunkers, or **Tunkers**, also called **Dippers**, a religious sect in America, founded by Conrad Peysel, a German, in 1724, and which takes its name from the Ger. *tunken*, to dip, from their mode of baptizing converts. They reject infant baptism; use great plainness of dress and language; refuse to take oaths or to fight; and anoint the sick with oil in order to hasten their recovery, depending on this unction and prayer, and rejecting the use of medicine. Every brother is allowed to speak in the congregation, and their best speaker is usually set apart as their minister.

Dunkirk' (Fr. *Dunkerque*), a seaport town, France, department of Nord, at the entrance of the Straits of Dover, surrounded by walls, and otherwise defended by forts and outworks. It has several fine churches, a college, a public library, and a gallery of paintings; manufactures of earthenware,

leather, soap, starch, ropes; sugar-refineries, breweries, and distilleries, and a large trade. In 1658 Dunkirk was given up to the English by Turenne, and continued with them till 1662, when Charles II sold it to Louis XIV. It is one of the chief French torpedo stations, and during the European War was a British base and frequently bombed by the enemy. Pop. 38,891.

Dun'lin, a British bird (*Tringa alpina*), a species of sandpiper, occurring in vast flocks along sandy shores. It is about 8 inches in length from the point of the bill to the extremity of the tail, and its plumage undergoes marked variations in summer and winter, the back passing from black with reddish edges to each feather, to an ashen grey, and the breast from mottled black to pure white. During the winter it migrates to Asia, Africa, the Canaries, West Indies, and California.

Dunmow', Great and Little, two villages, England, county of Essex. The latter is remarkable for the ancient custom, revived in 1855, of giving a flitch of bacon to any couple who, a year and a day after their marriage, could swear that they had neither quarrelled nor repented. The prize, instituted in 1244 by Robert de Fitzwalter, was first claimed in 1445.

Dun'nage, faggots, boughs, or loose wood laid in the hold of a ship to raise heavy goods above the bottom to prevent injury from water; also loose articles wedged between parts of the cargo to hold them steady.

Dunne, Finley Peter, American humorist, born in Chicago in 1867. After serving as reporter on various papers, he became editor of the *Evening Journal* (1897-1900). Dunne first attracted attention by a series of sketches in the *Times-Herald*, where he humorously commented upon all sorts of subjects in the name of one Martin Dooley, publican of Archey Road. His works include: *Mr. Dooley in Peace and War* (1898), *Mr. Dooley's Philosophy* (1900), *Observations by Mr. Dooley* (1902), and *Mr. Dooley Says* (1910).

Dunnet Head, a bold rock promontory in Caithness, with sandstone cliffs 100 to 300 feet high, the most northerly point of the mainland of Scotland, crowned by a lighthouse visible at a distance of 25 miles.

Dunnot'tar Castle, an extensive ruin on the coast of Kincardineshire, Scotland, on a precipitous rock rising from the sea. It dates from the close

of the fourteenth century, and was long the stronghold of the Keiths, earls marischal. During the Commonwealth this castle was selected for the preservation of the Scottish regalia; and in 1685 it was used as a State prison for Covenanters. It was dismantled in 1720.

Dunois (du' -nwä), Jean, Count of Orleans and of Longueville; a French hero, natural son of Louis, Duke of Orleans, born 1402, died 1468. Dunois made the name 'Bastard of Orleans' illustrious by his military exploits. He began his career with the defeat of Warwick and Suffolk, whom he pursued to Paris. Being besieged by the English, he defended Orleans until relieved by the Maid of Orleans. In 1450 he had completely freed France from the English, and was rewarded by the title of 'deliverer of his country', the county of Longueville, and the dignity of High Chamberlain of France.

Dunoon', a town, police burgh, and watering-place of Scotland, in Argyleshire, on the shore of the Firth of Clyde, 27 miles by river from Glasgow. It extends for about 3 miles s.s.w. from the Holy Loch, and consists of Hunter's Quay to the N., Kirn and Dunoon proper to the S.; each with its separate steamboat pier. On a green rocky knoll are remains of the castle of Dunoon, once a residence of the family of Argyll. Pop. 9860.

Duns, John, commonly called *Duns Scotus*, an eminent scholastic divine, born 1265 or 1274, but whether in England, Scotland, or Ireland is uncertain. He was admitted when young into an institution belonging to the Franciscan friars at Newcastle, whence he was sent to Merton College, Oxford. In 1301 he was appointed divinity professor at Oxford, and the fame of his learning and talents drew crowds of scholars from all parts. In 1304 he went to Paris, and was appointed professor and regent in the theological schools, in which situation he acquired the title of *Doctor Subtilis*, 'the subtle doctor'. He opposed Thomas Aquinas on the subject of grace and free-will; and hence the *Scotists* are opposed to the *Thomists*. Duns Scotus was the apostle of *realism*, which was opposed to the systems of *nominalism* and *conceptualism* promulgated by the other sections into which the schoolmen were divided. He died, it is said, at Cologne in 1308, leaving behind him numerous works. He was a genuine scholastic philosopher, who worked out ideas taken from Aristotle, St. Augustine, and the preceding scholastics.—BIBLIOGRAPHY: W. J. Townsend, *The Great Schoolmen*; C. R. Hagenbach, *History of Doctrines*; E. Pluzanski, *Essai sur*

la philosophie de Duns Scot; and article in *Dictionary of National Biography*.

Duns, or **Dunse** (dunz, duns), police burgh and county town of Berwickshire, Scotland, on the Whitadder; has manufactures of linen, and paper-mills. Pop. 3040. On Duns Law (700 feet) are traces of a camp formed by Leslie's Covenanters in 1639.

Dunsin'ane, a hill in Scotland, one of the Sidlaws, 1012 feet high, about 7 miles N.E. of Perth, with vestiges of a hill-fort locally called Macbeth's Castle, and immortalized by Shakespeare in *Macbeth*.

Dun'stable, a town, England, county of Bedford, 32 miles north-west of London. It was an important Roman station, and had a palace and a priory founded by Henry I in 1131. Part of the latter is used as the parish church. It has a grammar school, founded in 1715. Dunstable is famous for its manufactures of straw-plait. Pop. 8057.

Dun'stan, St., an English archbishop and statesman, was born at Glastonbury in 925, died at Canterbury 988. As a youth he was remarkable for his learning and his skill in music, painting, carving, and working in metals. He entered the Benedictine order, became an anchorite at Glastonbury, and in 945 was made abbot by King Edmund. After the death of Edmund, Edred, the next king, made him his Prime Minister and principal director in civil and ecclesiastical affairs. In the reign of Edwy he was banished, but was recalled by Edgar, and made Archbishop of Canterbury. He was again deprived of power on the accession of Ethelred in 978, and devoted the last years of his life to his diocese and the literary and artistic pursuits of his earlier days. He did much to improve education and to raise the standing and character of the priesthood. The old biographies of him have all a large legendary element.—Cf. W. Stubbs (editor), *Memorials of St. Dunstan* (Rolls Series, 1874).

Duodecimal System, in numeration, a system of numbers the scale of which is twelve. *Duodecimals* is a term applied to an arithmetical method of ascertaining the number of square feet, &c., in a rectangular area or surface, whose sides are given in feet, inches, and lines. The method is similar to that of ordinary decimals, the scale being twelve instead of ten.

Duodec'imo (often contracted 12mo) is that form of volume in which each leaf forms a twelfth part of the sheet.

Duode'num (Lat. *duodeni*, by twelve), the commencement of the intestinal canal, the first of the smaller intestines, so called because its length is about twelve fingers' breadth.

Dupanloup (du'-pän-lō), Félix Antoine Philibert, French prelate, born at St. Félix, in Savoy, 1802; became a French subject by naturalization in 1838; died at Paris 1878. He was ordained in 1825, appointed professor of theology at the Sorbonne in 1841, and Bishop of Orleans in 1849. From that time he took a prominent part in all the political and religious discussions in France. He belonged to the Gallican party, but submitted to the decisions of the council of the Vatican; and was a strenuous advocate of free education. He wrote: *La Pacification Religieuse*, *De l'éducation*, and *De la haute éducation*.

Dupleix (du'-plā), Joseph, a French leader in India, born 1697, died 1763. He accumulated a fortune by commercial operations in India, and in 1742 was appointed Governor of Pondicherry for the French East India Company. He formed the project of founding a French Empire in India, and soon made himself master of the Carnatic partly by conquest and partly by political intrigue. He was opposed by Clive, and a long string of British successes caused the complete overthrow of all his plans. Recalled in 1753, he died in want and obscurity in Paris.

Dupont (du'-pōn), Pierre, French poet and song-writer, born at Lyons 1821, died at St. Étienne (Loire) 1870. He was educated by his godfather, a priest, and began to write and compose songs at an early age. After issuing a volume of poems in 1844, he went to Paris and obtained a place in the office of the secretary of the Institute. Some of his songs, such as *Song of Bread* and *Song of the Workers*, had a Socialistic ring which proved obnoxious to the Government which came into power in Dec., 1852. He was arrested, imprisoned, and condemned to be banished for seven years; but his release was soon procured. His poems have been collected under the titles *Cahiers de Chansons*, *La Muse Populaire*, *Chants et Chansons*, *Poésie et Musique*, and *Études Littéraires*.

Dupont de Nemours (du ´-pōn də nè-mör), Pierre Samuel, French political economist, born at Paris, Dec., 1739; died in America 1817. He early gained a reputation for his writings on commerce, and his exposition of the theories of the physiocrats, and was employed by Turgot and Vergennes in the public service. During the ministry of Calonne he became Councillor of State, and in 1787 was secretary to the Assembly of the Notables. He was twice president of the National Assembly. During the Revolution he opposed the extreme republicans, and narrowly escaped the guillotine at the downfall of Robespierre. From 1798 to 1802 he was in America, and on his return to France he refused all public office. He finally returned to America in 1815. Among his writings are: *Philosophie de l'Univers*, *Vie de Turgot*, and a translation of Ariosto.

Düppel (du ´p´l), a fortified village in Schleswig-Holstein, on the coast of the Little Belt. The place is of considerable strategical importance, and has been the scene of some severe struggles between the Danes, to whom it formerly belonged, and the Germans. It was captured by the Prussians in 1864, after a siege and bombardment which lasted nearly two months.

Dupuy, Charles Alexander, French statesman, born at Le Puy, Haute-Loire, in 1851. Educated at the Lycée of Le Puy and the Lycée Charlemagne in Paris, he was professor of philosophy at the colleges of Nantes and Aurillac, and afterwards vice-rector of the Corsican College at Ajaccio. He entered the Chamber of Deputies in 1885, was Minister of Public Instruction in 1889, and succeeded Ribot as Premier in 1893, but resigned and became President of the Chamber of Deputies. He was again Premier from 1894 to 1895, and from 1898 to 1899, and was elected to the Senate in 1900. He was Minister of Labour from 1912 to 1914.

Dupuytren (du ´-pu ´-i-trän), Guillaume, Baron, French surgeon and anatomist, born in 1777, died at Paris 1835. In 1812 he became professor of surgery, and in 1815 first surgeon to the Hôtel Dieu, Paris. In 1823 he was appointed first physician to Louis XVIII, and retained the same situation under Charles X. He was considered the first French surgeon of his day, made important discoveries in morbid anatomy, and invented several useful surgical instruments.

Duquesne (du´-kān), Abraham, French admiral, born 1610, died 1688. In his seventeenth year he was in the sea-fight off Rochelle, and distinguished himself during and after the year 1637 in the war against Spain. In 1647 he commanded the expedition against Naples. In the Sicilian War he thrice defeated the combined fleets of Holland and Spain, under the renowned De Ruyter. After he had reduced Algiers and Genoa, Louis XIV conferred upon him the fine estate of Bouchet, and made it a marquisate, with the title of Duquesne. He was a Protestant, and the only person exempted from the banishment of his sect, occasioned by the repeal of the Edict of Nantes.

Dura´men, the name given by botanists to the central wood or heart-wood in a tree trunk. It is harder than the newer wood that surrounds it, and is often dark-coloured from being impregnated with tannin and other antiseptic substances.

Durance (du´-räns), a river of France, which rises in the Cottian Alps, and, after a course of about 180 miles, joins the Rhone about 4 miles below Avignon. Marseilles is supplied with water from the Durance.

Duran´go, a town in Mexico, capital of the state of Durango, about 500 miles N.W. of Mexico, on an elevation 6845 feet above the sea. It is well built, has a cathedral, a mint, manufactures of cotton and woollen goods and leather. Pop. 34,085.—The state (area, 42,272 sq. miles) is partly mountainous and unproductive, but has valuable gold-, silver-, and iron-mines, and also fertile tracts. Pop. 436,147 (1910); estimated pop. in 1919, 509,585.

Durante (dö-ra´n´tā), Francesco, Italian musician, born 1684, died 1755. He attained a high degree of eminence in vocal church music, and he trained the most celebrated musical masters of the eighteenth century in Naples—Pergolese, Sacchini, Piccini, Guglielmi, and Jomelli.

Durazzo (dö-rät´sō; ancient **Dyrrhachium**, or **Epidamnus**), a seaport of Albania, on the Adriatic, 50 miles south by west of Scutari. It is fortified, and has a good harbour. For four centuries the town remained under Turkish rule and lost all importance, but in 1912, during the Balkan War, it came again into prominence. It was occupied by the Serbians on 28th Nov., 1912, and in 1913 was incorporated in the newly created Principality of Albania.

During the European War Durazzo was captured by the Austrians in Feb., 1916, but was reoccupied by the Italians on 14th Oct., 1918. Pop. 5000.

Dur'ban, chief port of Natal, on a land-locked bay (Port Natal). It is well laid out, has a fine town hall, handsome churches, post office, hospital, electric tramways, and parks and gardens, and is connected by railway with Maritzburg and the interior. The harbour now admits large vessels. Founded in 1834, it was named after a governor of the Cape. Pop. 48,475.

Durbar (du'r-bär'; Pers. *dar*, door, and *bar*, court, admittance), a term signifying the court, council-chamber, or audience-room in the palaces of the native princes of India; hence, a general reception by a ruler in British India or by any officer of rank. Durbars were held on a ridge at Delhi on the proclamation of Queen Victoria (1877), of King Edward VII (1903), and of King George V (1911).

Düren (dü'ren), a town in the Rhine province, on the right bank of the Roer, 16 miles E. by N. of Aix-la-Chapelle. It has important manufactures of woollens, paper, leather, rails, and hardware, and an extensive trade. Pop. 32,511.

Dürer (dü'rér), Albrecht, German painter, designer, sculptor, and engraver on wood and metal, born at Nürnberg 1471, died there 1528. His father was a skilful goldsmith of Hungary. In 1486 he left his father's trade and became an apprentice of Michael Wohlgemuth, then the best painter in Nürnberg. Having finished his studies, he entered upon his 'wanderjahre', the usual course of travels of a German youth. On his return to Nürnberg he married Agnes, the daughter of Hans Frey, a mechanic, who has been falsely accused for centuries of embittering his life and bringing him to his grave. In 1505 he went to Venice to improve himself in his art. His abilities excited envy and admiration. He painted the *Martyrdom of Bartholomew* for St. Mark's church, which painting was purchased by the Emperor Rudolph and removed to Prague. He also travelled to Bologna, to improve his knowledge of perspective. On his return to Nürnberg his fame spread far and wide. Maximilian I appointed him his court-painter, and Charles V confirmed him in this office. All the artists and learned men of his time honoured and loved him, and for many years he was one of the chief burghers of his native town. Profound application and great facility in the mechanical part of his

art were the characteristics of Dürer, and enabled him to exert a great influence on German art. He was the first in Germany who taught the rules of perspective, and of the proportions of the human figure. He not only made use of the burin, like his predecessors, but was also among the first to practise etching. He also invented the method of printing woodcuts in two colours. Among his masterpieces in painting are a *Crucifixion*, *Adam and Eve*, an *Adoration of the Magi*, and portraits of Raphael, Erasmus, and Melanchthon, who were his friends. Among his best engravings on copper are his *Fortune*, *Melancholy*, *Adam and Eve in Paradise*, *St. Hubert*, *St. Jerome*, and *The Smaller Passion*, in sixteen plates. Among his best engravings on wood are *The Greater Passion* (so called), in thirteen plates; *The Smaller Passion*, with the frontispiece, thirtyseven pieces; *The Revelation of St. John*, with the frontispiece, fifteen plates; *The Life of Mary*, two prints, with the frontispiece. Dürer has also much merit as a writer, and published works on *Human Proportion*, *Fortification*, and the *Use of the Compass and Square*.—BIBLIOGRAPHY: Thausing, *Dürer, Geschichte seines Lebens und seiner Kunst* (2 vols.; English translation by F. A. Eaton); L. Cust, *Albrecht Dürer: a Study of his Life and Works*; F. Nüchter, *Life and Selection from Works*.

Du'ress (Lat. *duritia*, severity, from *durus*, hard), in law, restraint or compulsion, is of two kinds: *duress of imprisonment*, which is imprisonment or restraint of personal liberty; and *duress by menaces or threats* (*per minas*), when a person is threatened with loss of life, or with some kind of injury. An act done under duress is voidable or excusable.

D'Urfey (dur'fi), Thomas, an English poet and wit, the grandson of a French Protestant refugee, was born at Exeter in 1653, and died in 1723. He abandoned law for literature, and wrote a large number of comedies of a licentious character. His bombastic tragedy *The Siege of Memphis* appeared in 1676. D'Urfey's name is now principally remembered in connection with his *Pills to Purge Melancholy*, a collection of songs and ballads, partly his own, and many of them coarse or licentious. His society was generally courted by the witty, and he enjoyed the favour of four successive monarchs.

Durga (du'r'gä), a Hindu divinity, one of the names given to the consort of Siva. She is generally represented with ten arms. In one hand she holds a

spear, with which she is piercing Mahisha, the chief of the demons, the killing of whom was her most famous exploit; in another a sword; in a third the hair of the demon chief; and in others, the trident, discus, axe, club, and shield. A great festival in her honour, the *Durga puja*, is celebrated annually, lasting for ten days.



Durga

Durham, Simeon of, English chronicler of the twelfth century; wrote *Annals of England to the Reign of Henry I*, particularly valuable for events connected with the north of England. They were continued by John of Hexham.

Durham (du´ram), an ancient city, capital of the county of the same name, on the River Wear, which is crossed here by four bridges, 14 miles s. of Newcastle. The principal public buildings are the ancient castle—now appropriated to the uses of the university—the cathedral and other churches, the town hall, county prison, and grammar-school. The educational institutions comprise the university, opened in 1833, the grammar-school, a training-school for school-mistresses, and other schools. There are manufactures of carpeting and mustard. The cathedral occupies a height overlooking the Wear. The larger portion of it is Norman in style, with insertions in all the English styles. Three magnificent and elaborately ornamental towers spring up from the body of the building, one from the centre 212 feet high, and two together from the west end each 143 feet high; the entire length is 420 feet. It was founded by William de St. Carilef, assisted by Malcolm, King of Scotland, in 1093. A parliamentary borough until 1918, Durham returned two members to the House of Commons from 1673 to 1885, and one member from 1885 to 1918. Pop. 17,500.

Durham Book, a Latin text of the gospels written by Bishop Eadfrith of Lindisfarne, with an interlinear Saxon gloss, finished in the year 720; now in the British Museum.

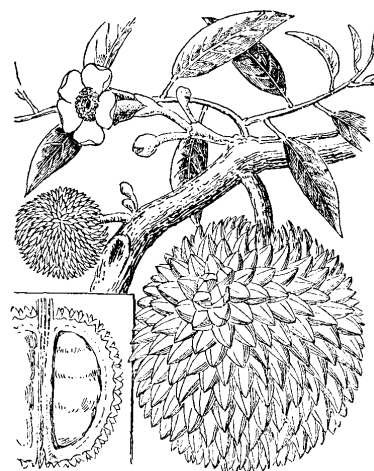
Durham, County of, a county on the N.E. coast of England, having on the E. the North Sea, on the N. Northumberland, from which it is divided by the Rivers Tyne and Derwent, Cumberland on the w., and Yorkshire on the s.,

the River Tees parting the two counties. Its area is 647,592 acres, of which two-thirds are under cultivation. The western portion of the county is hilly, enclosing fertile valleys, the eastern portion is more level, and the centre contains extensive coal-fields. Durham is the chief coal county in England, and also produces fire-clay. The chief crops are wheat, oats, turnips, and potatoes. The cattle are esteemed both for the dairy and for fattening. In connection with the commerce of the county may be noticed its foundries, ironworks, potteries, glass-houses, iron-shipbuilding, engine and machine works, and chemical works. For parliamentary purposes it is divided into eleven divisions, each of which sends one member to the House of Commons. It was formerly one of the three counties called counties *palatine*. The chief towns besides Durham are Sunderland, Gateshead, South Shields, Stockton, Darlington, and Hartlepool. Pop. 1,369,860.— Cf. *Victoria History of the County of Durham*.

Durham University, founded in 1832, opened in 1833, incorporated by royal charter in 1837. It is connected with the bishopric of Durham, the office of warden being annexed to the deanery of Durham, and a canonry in the cathedral being annexed to each of the professors of divinity and classical literature. There are also professors of mathematics, Hebrew, medicine, &c. The students mostly reside within the university buildings, but in 1870 a regulation was passed dispensing with the necessity of residing in any college, hall, or house connected with the university in order to be admitted as a member. The management of the university is entrusted, under the Bishop of Durham as visitor, to the dean and chapter of the cathedral as governors, and to the warden, senate, and convocation, the last including all persons regularly admitted since the opening of the university to the degrees of Doctor in Divinity, Civil Law, and Medicine, and to the degree of Master of Arts. The academical year is divided into three terms—Michaelmas, Epiphany, and Easter. For the degree of B.A., or a licence in theology, a residence of two years (of six months each) is necessary. The M.A. degree may be obtained by a graduate who is of the standing of nine terms since taking his degree of B.A. Armstrong College, founded in 1874, and the College of Medicine, both at Newcastle-on-Tyne, form part of the University of Durham.

Durian, or **Durion** (*Durio zibethinus*), a large and lofty tree growing in the Malayan Archipelago. The largish flowers, of a yellow-green colour, are

produced on the stem or main branches, and are followed by the large fetid fruit, which is of the size of a man's head, and is a favourite food of the natives during the time (May and June) when it is in season. There is usually a second crop in November. The smell is offensive, like putrid animal matter, but with this is associated the most delicious flavour, which places it, notwithstanding the odour, in the opinion of many, in the foremost place among tropical fruits.



Durian and Section of Fruit

Dürkheim (du · rk'hīm), an old town in Rhenish Bavaria (the Palatinate), 14 miles w.s.w. of Mannheim, well known for its mineral water. The town was destroyed by the French in 1689. Pop. 6523.

Durkheim, Émile, French philosopher and sociologist, born at Les Vosges 15th April, 1858, died in Paris 15th Nov., 1917. Educated at the École Normale Supérieure, he travelled in Germany, where he studied social conditions. In 1887 he founded the first French chair in sociology at the University of Bordeaux. Durkheim's merit consists in having separated sociology from mere psychology, and in having made a distinction between individual mental phenomena and Folk-psychology. In 1898 he founded and published annually *L'Année Sociologique*. His other works include: *De la division du travail social* (1893); *Les règles de la méthode sociologique* (1894); *Le Suicide* (1897); *Les formes élémentaires de la vie religieuse*; *Sociologie et sciences sociales* (1910); *Le système totémique en Australie* (1912); *La Sociologie* (in *La Science Française*, 1915); *Qui a voulu la guerre?*; *Les Origines de la guerre, d'après les documents diplomatiques* (1915).

Durlach (du'r'la'h), a town in Baden, 4 miles E.S.E. of Carlsruhe, at the foot of the Turmberg, with manufactures of machinery, chemicals, and leather. Pop. 13,896.

Durmast, a species of oak, *Quercus sessiliflora*, or according to some, *Q. pubescens*, so closely allied to the common oak (*Q. robur*) as to be

reckoned only a variety of it. Its wood is, however, darker, heavier, and more elastic, less easy to split, not so easy to break, yet not so difficult to bend. It is highly valued, therefore, by the builder and cabinet-maker.

Duroc (du'-rok), Michel Géraud Christophe, Duke of Friuli, a distinguished general under Bonaparte, born at Pont-à-Mousson in 1772, killed, 1813, at the battle of Bautzen. He served as aide-de-camp to Napoleon in the Italian and Egyptian campaigns. In 1805 he was made grand-marshal of the palace; and was frequently employed in diplomatic missions, though he still took his full share in the wars of France till the time of his death. He was a great favourite of Napoleon, and was killed by his side.

Durra, or **Dhurra**, Indian millet, the seed of *Sorghum vulgāre*, after wheat the chief cereal crop of the Mediterranean region, and largely used in those countries by the labouring classes for food. Varieties are grown in many parts of Africa, one of them being known as Kaffir corn.

Dürrenstein (du'r'en-stīn), a village in Lower Austria, on the Danube, 41 miles west by north of Vienna. Here are the ruins of the castle in which Leopold, Duke of Austria, imprisoned Richard Cœur-de-Lion on his return from Palestine, 1192.

Duruy (du'-ru'-ē), Victor, French historian and educationist, born at Paris 1811, died in 1894. His father was a workman in the Gobelins tapestry works, and the boy did not begin his education until he was grown up. He was admitted to the École Normale Supérieure in 1830, graduated in 1833, and was appointed successively teacher of history in the Lycée Henri IV, then at the Normal School and the Polytechnic School, inspector of the Academy of Paris, inspector-general of secondary education, and Minister of Public Instruction (1863-9). He is author of *Géographie Politique de la République Romaine et de l'Empire*, *Géographie Historique du Moyen Age*, *Histoire Romaine*, *Histoire de France*, *Histoire Grecque*, *Histoire Populaire Contemporaine*, &c. Some of these are simply school-books, but his *Histoire des Romains* (translated into English) and his *Histoire de la Grèce Ancienne* (translated into English) are extensive and important works, the former especially.

Duse, Eleonora, Italian actress, born at Vigevano, near Venice, in 1859. At the age of thirteen she made her first appearance on the stage, and in 1883

she acted at Rome, when she was recognized as one of the greatest Italian, and even one of the greatest living actresses. From that time her career was one of uninterrupted success, and she gained a world-wide reputation. Duse is one of the dramatic artists who discard the customary mannerisms of the stage, and all that is conventional but unreal in modern acting. Some of d'Annunzio's best plays were specially written for her. Among her most remarkable impersonations are those of Francesca da Rimini, Marguerite Gautier in *La Dame aux Camélias*, Magda, La Tosca, Paula in *The Second Mrs. Tanqueray*, and Nora in Ibsen's *A Doll's House*. She appeared for the first time in London in 1897.

Düsseldorf (du ` s'sel-dorf), a town of Germany, in the Rhine province, beautifully situated among villas and gardens on the right bank of the Rhine, 22 miles N.N.W. of Cologne, one of the handsomest towns in the valley of the Rhine. It is a great focus of railway and steamboat communication, and has a number of handsome public buildings, and several remarkable churches. Among the public institutions particular notice is due to the Academy of Art, founded in 1767, by the Elector Theodore, and afterwards directed by Cornelius, Schadow, and Bendemann. It has the honour of having founded a school of painting, which takes the name of Düsseldorf, and has had a large number of distinguished pupils. The industries embrace iron, machinery, railway plant, cotton, leather, chemicals, and beer, and the trade is large. Pop. 358,728.

Dust, solid matter in a fine state of division. Spores of plants, bacteria, &c., are found in the atmosphere, but in general organic particles are numerous only over thickly populated districts. Inorganic particles are derived from various sources. Where the soil is dry, dust is whirled aloft by the winds, this cause giving rise to the great sand-storms of tropical desert regions. Volcanoes in eruption eject large quantities of dust. It is estimated that millions of meteors are encountered by the earth per day. Most of these are excessively minute. They are speedily disintegrated, and generally entirely reduced to dust at high levels. Evaporation is almost always proceeding over seas and oceans, and from foam thrown up and swept along by the winds the dissolved salts are liberated as solid particles. Again vast quantities of dust are produced in the consumption of fuel.

The late Dr. John Aitken, F.R.S., of Falkirk, Stirlingshire, contrived a means of gauging the dust contents of the atmosphere. This consists of a glass box about a centimetre in thickness. Two pieces of wet filter paper inside serve to keep the contained air damp. The bottom of the box is a micrometer plate, divided rectangularly in millimetres. It can be examined from above by a lens. An air-pump can withdraw definite volumes of air as desired. When the air is partially withdrawn, the expansion of the remainder produces cooling. The dust particles form nuclei for condensation of the vapour. They are thus precipitated on the plate, and counted, leaving the air dust-free. A measured quantity of the air to be tested is next drawn in and shaken up. Further operation of the air-pump causes its expansion, and the deposition of its dust particles, which can then be counted. Dr. Aitken found the proportion of dust on Ben Nevis to vary at different times from under 100 particles to over 14,000 per cubic centimetre. Over oceans the numbers were from about 500 on the Indian to 2000 on the Atlantic. But over cities 100,000 per cubic centimetre are frequently present. A puff of cigarette smoke was estimated to contain 4,000,000,000 particles.

Many phenomena are connected with the existence of dust in the atmosphere. As a result of Dr. Aitken's discoveries the belief largely prevailed that the formation of fog, of rain, and other varieties of precipitation, was necessarily dependent on dust particles as nuclei of condensation. Though they certainly function to a preponderating extent, it has been shown that gaseous particles can act similarly, particularly when air is ionized. Dust is the main cause of the scattering of the sun's rays which produces twilight, the blue of the sky, the gorgeous red and golden hues of sunrise and sunset, and the purple lights of advancing dusk. After the great Krakatoan eruptions of 1883, dust was carried in the upper atmosphere several times round the earth, and caused extraordinary colour effects. To a lesser degree similar phenomena followed the West Indian eruptions of 1902. The unusual sunlessness of the summer of 1912 was attributed to dust expelled in the preceding great eruptions at Katmai, Alaska.

Dutch Clover, *Trifolium repens*, commonly called white clover, a valuable pasture plant. It has a creeping stem; the leaflets are broad, obovate, with a horse-shoe mark in the centre; the white or pinkish flowers are in a globular head.

Dutch East Indies, forming a large and important colonial possession of the Netherlands Government, lie between 6° N. and 11° S., and 95° E. and 141° E. The colony includes Java and Madura, with the 'Outposts', which comprise Sumatra, the south-east and west portions of Borneo, Banca, Billiton, Celebes, the Timor and Riau-Lingga Archipelagos, the smaller Sunda Islands, and the north and west of New Guinea. The total area is about 735,000 sq. miles; the population of about 47,000,000 is composed of 46,000,000 natives of Malay race, 832,000 Arabs, Chinese, and other Orientals, and some 80,000 whites. The origin of the colony may be traced to the treaty made by the Dutch with the Sultan of Bantam in Sumatra (1595), which was followed by the formation of the Dutch East India Company (1602), the establishment of Batavia (1619) on the ruins of the native town of Jacatra, and the settlement in Sumatra (1677). The Dutch East India Company was dissolved in 1798, since which date the colony has been administered from the Netherlands by a Governor-General, who, assisted by a Council of five members nominated by the queen, has the power to pass laws, subject to the general regulations adopted in 1854. Some of the outlying islands are, however, administered by their native princes under the 'advice' of a Netherlands Resident. Batavia (population about 234,000), a town in the province of the same name on the north-west coast of Java, is the administrative capital and an important centre of trade. Java and Sumatra, containing about three-fourths of the total population of the colony, are self-supporting as regards food, besides producing for European consumption large quantities of tobacco, tea, coffee, sugar, cinchona, tin, rubber, and copra. The colonial army numbers some 1200 officers and 40,000 men; compulsory service for white men within certain age-limits was adopted in 1918. There is also a small naval force.—BIBLIOGRAPHY: Bemmelen and Hooyer, *Guide to the Dutch East Indies*; W. Cool, *With the Dutch in the East*; J. M. Brown, *The Dutch East*.

Dutch Metal, an alloy containing 84.5-84.7 per cent of copper and 15.5-15.3 per cent of zinc, with a fine golden-yellow colour, ductile, malleable, and tenacious. When beaten out by a process analogous to that for gold-leaf, until the sheets are less than 1/50,000th part of an inch thick, it constitutes Dutch leaf or Dutch foil, and is used as a cheap substitute for gold-leaf for ornamental purposes.

Dutch Pink, a bright yellow colour used in distemper, for staining paper-hangings, and for other ordinary purposes. It is composed of chalk or whiting coloured with a decoction of birch leaves, French berries, and alum.

Dutch Rush, *Equisetum hyemāle*, one of the plants known as horse-tails, with a firm texture and so large an amount of silica in the cuticle that it is employed as a fine sand-paper for polishing delicate woodwork. The plant is found in marshes and woods in Britain, but for economic use it is imported from Holland, whence its popular name.

Dutrochet (du'-tro-shā), René Joachim Henri, a French physiologist, born in Poitou in 1776, died at Paris in 1847. He served for some time as medical attendant to Joseph Bonaparte during the Spanish campaign 1808-9; but afterwards returned to France, and retired to the estate of Châteaurenault, where he devoted himself exclusively to physical and physiological studies. His chief works have been published in a collective form with the title *Mémoires pour servir à l'Histoire Anatomique et Physiologique des Végétaux et des Animaux* (1837, 2 vols.).

Dvina, Northern, a Russian river formed by the union of two small streams in the government of Vologda. It flows in a north-westerly direction, and falls by four mouths into the White Sea. At Archangel, before it divides, it is 4 miles broad. It is navigable as far as Suchona, and is connected with the Volga and Neva by canal.

Dvořák (dvor-shäk'), Anton, a Bohemian musical composer, born in 1841, died in 1904. He studied at the Prague Conservatoire, and composed several operas on national Bohemian subjects, songs, orchestral arrangements of Bohemian dances, several symphonies, a *Stabat Mater*, a cantata (*The Spectre Bride*), and an oratorio (*St. Ludmila*). In 1892 he was director of the New York National Conservatory, and after 1895 he lived in Prague, where he wrote, in 1889, the opera *Der Teufel und die wilde Kätze*.

Dwarf, a term applied to any animal or plant greatly below the usual size of its kind, particularly to a human being of small dimensions. Strictly speaking, the term should be used with reference to individuals and not to races. When a whole population consists of people of small stature, the proper term to apply to them is pigmies, not dwarfs. Accounts of pigmy tribes have been common from early times, such tribes being located

especially in Africa; and it would appear from the accounts of Du Chaillu, Schweinfurth, and other travellers that there are several pigmy tribes throughout this continent. The Obongo, a race of dwarfs, are described as living in woods near the Okanda River, in wretched huts made of branches. Other races are the Mabongo, and the Akka dwarfs of Central Africa (see *Akkas*); and a race exists in the Congo State, not as a distinct community, however, but mixed with other tribes. Individual dwarfs occur in all races, and were formerly a fashionable appendage to the courts of princes and the families of nobles. Jeffery Hudson, the favourite dwarf of Charles I, at the age of thirty is said to have been only 18 inches high, though he afterwards grew to 3 feet 9 inches. Bébé, the celebrated dwarf of Stanislas of Poland, was 33 inches; Wybrand Lolkes, a Dutch dwarf, when sixty years of age was only 27 inches; Charles H. Stratton, 'General Tom Thumb', was 31 inches high at the age of twenty-five; Francis Flynn, 'General Mite', was only 21 inches at sixteen. In most of the extreme cases the dwarfing is the result of some defect in the ductless glands which regulate the normal growth of the body. Stories of dwarfs and brownies are to be found in the folk-lore of many tribes on earth.—BIBLIOGRAPHY: E. J. Wood, *Giants and Dwarfs*; E. Tyson, *Philological Essay concerning Pygmies of the Ancients*.

Dwarfing, the process of training up trees or shrubs for ornament in houses so as to cause them never to reach more than a very small size, by keeping them in poor soil, giving them little water, pinching off strong shoots, &c. It is much practised among the Chinese and Japanese.

Dwight (dwīt), Timothy, American divine, born in Massachusetts 1752, died 1817. His father was Colonel Timothy Dwight, and his mother was a daughter of Jonathan Edwards. He served as chaplain in the revolutionary army, and ultimately became president of Yale College. His *Theology Explained and Defended* (1818) was for long a standard work both in Britain and in America.

Dy'aks, the aborigines of Borneo, chiefly inhabiting the interior of the island. They are a finely formed race, of a yellow complexion, and are described as docile, industrious, and superior to the Malays. The more advanced of them practise agriculture and dwell in neatly-constructed and tolerably comfortable houses. In Sarawak they have made considerable advances in civilization. The practice of head-hunting (hunting their

enemies to make trophies of their heads) is practised among them, but has been abolished where European influence prevails.—Cf. Hose and M'Dougall, *The Pagan Tribes of Borneo* (2 vols.).

Dyaus (dyous), the god of the sky in the older mythology of the Hindus. His name is etymologically connected with that of the Greek Zeus.

Dyce (dīs), Alexander, Shakespearean editor, born at Edinburgh 30th June, 1798, died 15th May, 1869. He was educated at Edinburgh and Oxford, but in 1827 settled in London, where most of his life was passed. He first became known by his editions of Collins, Peele, Webster, Marlowe, and Skelton, accompanied by notes and biographies of the authors. His chief work was an edition of Shakespeare in six volumes, with notes (1853-8).

Dyce, William, painter, born at Aberdeen in 1806, died near London, 1864. He studied in London and Rome, and practised his art in Edinburgh. In 1840 he became director of the School of Design in London, and in 1844 was appointed professor of fine art in King's College, London. He was elected a Royal Academician in 1848. Amongst his chief works are: *Francesca da Rimini* (1837); *Joash shooting the Arrow of Deliverance* (1844); *King Lear in the Storm* (1851); *Christabel* (1855); *The Good Shepherd* (1856); *The Baptism of Ethelbert*, a large fresco for the Houses of Parliament, and a series of frescoes illustrative of the legends relating to King Arthur, for the same. He also executed for the Prince Consort, at Osborne, the fresco *Neptune giving the Empire of the Sea to Britannia*.

Dyeing is the art of colouring textile and other materials in such a way that the colours are not readily removed by the action of light, washing, &c. Like spinning and weaving, it was originally a home industry, as it still is in many places. The natural dyes formerly employed are now largely displaced by dyes derived from coal-tar products, the first discovery of which was made by Perkin in 1856; a few mineral colours are employed in cotton dyeing. Before dyeing, the materials have generally to be cleansed or bleached to get rid of undesirable colouring matters or impurities; and frequently a textile material is subjected to some subsidiary treatment in order to obtain special effects. For example, cotton yarn may be subjected to the action of strong caustic soda ('mercerizing' process) while in a state of great tension, in order to give it a permanent silky lustre. Dyeing is not

only an art, it is also a branch of applied chemistry. One fundamental principle is, that the colouring matter and other necessary substances must be applied in a state of solution, and while in direct contact with the fibre they must be rendered insoluble, so that they are precipitated within or upon the fibre and fixed permanently. The method of effecting this varies greatly, according to the fibre and the colouring matter employed. As a rule, the vegetable and the animal fibres are dyed by very different methods. The affinity of the animal fibres for certain colouring matters is often so great that they are readily dyed by simple immersion in hot colour solutions; but this simple process is not generally sufficient. According to the method of their application in dyeing, the following groups of dye-stuffs may be distinguished: *Acid Dyes*, *Basic Dyes*, *Direct Dyes*, *Developed Dyes*, *Mordant Dyes*, *Vat Dyes*. A dye is *substantive* to a particular fibre when it dyes that fibre directly, and *adjective* when the presence of a third substance known as a *mordant* is necessary.

The *acid dyes* are so called because they are of an acid character and are applied in an acid dye-bath. As a rule, they are only suitable for dyeing the animal fibres, e.g. wool and silk, also leather, horn, feathers, &c., and they are substantive to these materials. The purple vegetable dyestuff *orchil* belongs to this class. The acid dyes derived from coal-tar are very numerous, and yield a great variety of hues—red, orange, yellow, green, blue, violet, brown, and black. The *basic dyes* are so called because their essential constituents, to which they owe their dyeing power, are organic bases. The bases themselves are colourless and too insoluble in water to be of use, hence they are employed in the form of their soluble coloured salts, usually the hydrochlorides of the colour-bases. Their solutions are precipitated by tannic acid, because it combines with the colour-bases to form insoluble tannates. Wool, silk, and animal substances generally have a direct attraction for colour-bases, and hence these fibres are readily dyed by simple immersion in hot aqueous solutions of the basic dyes. Cotton and linen, on the other hand, are not dyed so readily; they need first to be mordanted or impregnated with the *mordant* tannic acid. Most of the colours of this class are fugitive to light, and all but one, viz. *barberry root*, are derived from coal-tar products.

The *direct dyes* are so called because they dye cotton without the aid of any mordanting process. The first of this class derived from coal-tar was *congo*

red, discovered in 1884; at present this group includes a very great variety of fast colours, and forms, indeed, one of the most important and valuable series of dye-stuffs employed. Cotton, linen, and the vegetable fibres generally are dyed in the simplest possible manner by merely boiling them in a solution of the dye-stuff, with or without the addition of a little soap, carbonate or sulphate of soda, &c. Wool and silk are frequently dyed in the same manner as cotton. Very few vegetable dye-stuffs belong to the direct colours, e.g. *Safflower*, *Turmeric*, *Saffron*, *Annatto* (see the articles). They are all fugitive, and are now of little or no importance to the dyer. The coal-tar colours of this class, on the other hand, are extremely numerous. The *developed dyes* are formed *in situ* upon the fibre by the successive application of two or more substances. They include *aniline black*, a permanent black produced by the oxidation of aniline, and the *ice colours*, which are *azo dyes* derived from certain coal-tar products containing nitrogen.

The *mordant dyes* form one of the most important classes of colouring matters, for they include not only most of the vegetable dye-stuffs, e.g. madder, logwood, fustic, &c., but also many valuable fast coal-tar colours, commonly known as the *alizarin dyes*, after their typical representative, alizarin. These mordant dyes have by themselves very little colouring power, as a rule, and if employed alone in dyeing give little or no result. If applied, however, in conjunction with metallic salts, notably those of chromium, aluminium, iron, tin, and copper, they each yield a variety of colours, according to the metallic salt employed. In employing them two distinct operations are usually involved: first, that of applying the metallic salt or *mordant*, called the *mordanting process*; and second, that of *dyeing* proper, in which the mordanted material is boiled in a solution or decoction of the dye-stuff. During the dyeing operation the colouring principle of the dye-stuff combines with the metallic salt already upon the material, and the colour is thus produced and fixed upon the fibre. The method of mordanting varies with the fibre and the metallic salt employed. The vegetable dye-stuffs of this class include *Madder*, *Sapanwood*, *Camwood*, *Barwood*, *Old Fustic*, *Young Fustic*, *Quercitron Bark*, *Persian Berries*, *Weld*, *Logwood* (see these separate articles). *Madder* was formerly the most important and highly valued of the dye-stuffs of this class, being especially employed to produce the fine 'Turkey-red' dye; but it is now entirely superseded by the

coal-tar colour alizarin. Similarly, the employment of *cochineal* (an insect dye) has also greatly diminished through the introduction of the cheaper colours. *Camwood* and *barwood* are almost entirely used in wool-dyeing, either in conjunction with the indigo-vat or for the purpose of dyeing various shades of brown. *Old fustic* is the most important of the yellow mordant dye-stuffs, and the colours are fast although not very brilliant. *Quercitron bark* is an excellent dye-stuff employed by wool-dyers for the production of bright orange and yellow colours. *Persian berries* and *weld*, a species of wild mignonette, are both excellent dye-stuffs, but their employment is now limited. *Logwood* is largely employed by wool, silk, and cotton dyers for dyeing black and dark-blues, which, although fast to washing, are only moderately so towards light. The important vegetable dye *catechu* (q.v.) is used in dyeing cotton and wool brown. On wool, catechu yields *khaki* browns in single bath by using copper sulphate as the mordant. On silk, it is largely employed for weighting purposes in the process of dyeing black. Although dyewoods are still much employed, they are being steadily replaced by coal-tar colours. The *vat dyes* are insoluble in water, but yield reduction products which are soluble in aqueous alkali, and can be readily reoxidized to the dye-stuff. In this class may be included the *sulphur dyes*, substances of uncertain composition obtained by fusing certain compounds containing nitrogen with sulphur and sodium sulphide, which are now extensively and increasingly employed, especially as direct dyes for cotton. *Indigo*, a typical vat dye, is prepared both artificially and from natural sources. It is a dark-blue powder quite insoluble in water, but when reduced it yields *indigo-white* which dissolves in aqueous alkali, the solution thus obtained being called an *indigo-vat*. Cotton, wool, or silk steeped for some time in the clear yellow solution of such a vat, and then exposed to the oxidizing influence of the air, is dyed a permanent blue. The indigo-white absorbed by the fibre loses its acquired hydrogen, and thus insoluble indigo-blue is regenerated within and upon the fibre.

In the classification adopted above, the following *mineral colours* employed in cotton dyeing belong to the group of developed dyes, since they are formed on the fibre: *chrome yellow*, obtained by immersing cotton successively in solutions of acetate of lead and bichromate of potash; *iron buff* (oxide of iron), produced by the successive application of sulphate of iron and carbonate of soda; *Prussian blue*, developed by passing the buff-

dyed cotton through an acidified solution of potassium ferrocyanide; *manganese brown* (oxide of manganese), deposited similarly to iron buff. The mineral colours are very useful for certain purposes, and are very fast to light.—BIBLIOGRAPHY: Crookes, *A Practical Handbook of Dyeing and Calico-printing*; W. P. Dreaper, *Chemistry and Physics of Dyeing*; Knecht, Rawson, and Loewenthal, *Manual of Dyeing*; Rawson, Gardner, and Laycock, *A Dictionary of Dyes, Mordants, &c.*

Dyer, John, English poet, born in Carmarthenshire about 1700, died in 1758. Educated at Westminster School, he became a painter, but, not succeeding in that vocation, took orders and was appointed to a small living. In 1727 he published his poem of *Grongar Hill*, in 1740 *The Ruins of Rome*, and in 1757 *The Fleece*, a didactic poem in five books.

Dyer's-broom, a European and now also N. American shrub (*Genista tinctoria*), formerly used with woad for dyeing green.

Dyer's-weed, *Resēda Luteōla*, a British plant of the same genus as mignonette, otherwise called *Yellow-weed*, *Weld*, or *Woad*, nat. ord. Resedaceæ. This plant grows in waste ground; it affords a beautiful yellow dye, and is cultivated for that purpose.—*Dyer's Greenweed* is *Genista tinctoria*.

Dying Declaration, a deposition made by one who is in prospect of death. Such declarations are as a general rule admissible as evidence only in criminal and not in civil cases, and must be made, according to English, though not Scottish law, in the full consciousness of the danger of death.

Dynam'ics is the science which deals with the laws of force in their relation to matter at rest or in motion, and as such it is differentiated from *kinematics*, which considers motion mathematically, and apart from the forces producing it. Dynamics is divided into two great branches: *statics*, which treats of solid bodies at rest under the action of forces; and *kinetics*, which treats of the action of forces in producing motion in solid bodies. Formerly the latter alone was called dynamics, and to this, in conjunction with statics, the general name *mechanics* was given. In the wide sense dynamics includes also hydrodynamics. It is to Newton that we owe the clear statement of the three primary laws of force on which the science of dynamics is based. These are: (1) that every body remains in a state of rest,

or of uniform motion along a straight line, unless it is compelled by force to change that state; (2) that rate of change of momentum is in proportion to the force employed, and occurs along the straight line in which the force acts; (3) that, as the result of every action, there is always an equal and opposite reaction. These laws, which were formulated from experiment, involve the conception of force as a primary influence or action expressed in terms of space, time, and matter. Now, in dealing with the laws of force, a standard of measurement is required which shall be applicable to all forces at all times, and we therefore require to begin by establishing units of space, time, and mass. There are two systems of units in use, the one British, the other French. In the British system the foot is taken as the unit of length, and the second as the unit of time. In the French the centimetre is the unit of length, the second the unit of time; the unit velocity in the one case being that of one foot per second, in the other one centimetre per second. The British unit of mass is the pound (the mass of a certain lump of platinum deposited in the Exchequer Office, London); the French the gramme; and accordingly the French units of space, mass, and time are commonly known as the C.G.S. (centimetre, gramme, second) units. As the weight of a pound (or a gramme) is not the same at all parts of the earth's surface, it cannot give us of itself an absolute or dynamical unit of force, that is, an invariable unit; but taking it in conjunction with unit time and unit velocity, we do obtain such a unit. Two absolute units of force are in common use in dynamics, the *poundal* and the *dyne*, the latter being the absolute unit in the C.G.S. system. The former is *that force which, acting on the mass of one pound for one second, generates in that mass a velocity of one foot per second*. The latter is *that force which, acting on the mass of one gramme for one second, generates in that mass a velocity of one centimetre per second*. It is important in dynamics to distinguish between *mass* and *weight*. The *mass* of one pound is the quantity of matter equal to a certain standard quantity (a certain lump of metal) and is quite independent of force. The *weight* of one pound is the force with which the mass of one pound is attracted to the earth's surface by the force of gravity. Another important term is *momentum*: the momentum of a body in motion at any instant is the product of the mass of the body and the velocity at that instant. See *Couple*; *Elasticity*; *Energy*; *Force*; *Hydrodynamics*; *Kinematics*; *Kinetics*; *Statics*; *Thermodynamics*; *Waves*.—BIBLIOGRAPHY: Kelvin and

Tait, *Natural Philosophy*; A. Gray, *Dynamics*; P. G. Tait, *Dynamics*; S. L. Loney, *Mechanics and Hydrostatics for Beginners*.

Dyn'amite. See *Explosives*.

Dy'namo. See *Generator*.

Dynamom'eter, an apparatus for measuring the power or rate of working of a machine. There are two types, the transmission dynamometer, which measures the power of the machine without sensibly diminishing it; and the absorption dynamometer, which measures the power by using it all. The instrument is generally employed to determine the horse-power transmitted by a shaft or by belting.—Cf. Aspinall Parr's *Electrical Engineering Testing*.

Dyrrhachium. See *Durazzo*.

Dy'sart, a royal and municipal burgh of Scotland, in Fife, on the Firth of Forth. It is an old place, and is a member of the Kirkcaldy district of parliamentary burghs. Pop. of royal burgh, 4197.

Dys'entery is a disease of an acute type, due to the action of a bacillus, characterized by pain and frequent passage of blood and mucus. Owing to improved sanitation, dysentery has become less frequent. In temperate countries sporadic cases occur from time to time, and occasional epidemics break out, but in the tropics widespread epidemics occur, and the disease is a serious menace. It is a very frequent camp disease, and has been the scourge of all armies in tropical and semi-tropical regions. The bacilli are widely spread by the fæces of infected persons, and usually the infection takes place by the mouth. The onset is rapid, and marked by fever, pain in the abdomen, and frequent stools. At first mucus only is seen in the stools, but soon blood appears. In very acute cases the patient is seriously ill in forty-eight hours, and may die on the third or fourth day. Moderate cases may go on for several weeks, with resulting convalescence. Some cases become chronic in type, and a person may have chronic dysentery for years. Bismuth in large doses is given, and morphia is a most useful drug to relieve the pain and quieten the bowel. Normal saline solution is given by rectum after the acute stage, whenever possible. Chronic dysentery requires

dietetic treatment for the persistent dyspepsia and irritability of the bowel. Amœbic dysentery, due to the *Amœba dysenteriaë*, is a distinct disease.

Dys'odile, a yellowish or greenish foliated carbonaceous substance found in Sicily originally, and derived from the decay of minute organisms. When ignited, it burns and emits a very unpleasant smell.

Dyson, Sir Frank Watson, F.R.S., LL.D.(Edin.), British astronomer, born at Ashby 8th Jan., 1868, the son of a Baptist minister. Educated at Bradford Grammar School and Trinity College, Cambridge, he became chief assistant at the Royal Observatory, Greenwich, in 1894, and secretary of the Royal Astronomical Society in 1899. Astronomer Royal for Scotland in 1905, he was appointed Astronomer Royal for England in 1910. He was knighted in 1915.

Dyspep'sia, or **Gastritis**, may be either acute or chronic. *Acute dyspepsia* may follow when more food is taken than the stomach can digest, or when unsuitable articles are taken. The symptoms are headache, depression, nausea, vomiting, with pain, varying from a feeling of discomfort in the abdomen to marked tenderness. The tongue is furred, and usually there is diarrhœa, while in the more severe attacks the onset is marked by chill and a rise of temperature. An attack may last from one day to four days. Treatment for mild cases is simply a dose of castor-oil (children) or blue pill (adults), but in severe cases vomiting should be promoted by warm water, or the stomach tube if necessary, and a dose of calomel (3 grains), followed by salts, should be given. Absolute rest to the stomach is necessary, and only small quantities of water allowed. Repeated attacks lead to the establishment of the chronic form. *Chronic dyspepsia* is a condition of disturbed digestion due to the prolonged use of unsuitable, or improperly prepared, foods. Hot cakes, excess of tea, coffee, or alcohol, rapid and irregular meals are all common causes. It may also arise in the course of diseases like anæmia, chronic tuberculosis, gout, Bright's disease, chronic heart disease, cirrhosis, and in diseases of the stomach itself, as cancer, gastric ulcer, and gastric dilatation. The most common symptoms are a feeling of oppression after food, tenderness over the stomach, headache, nausea, flatulence, constipation, and occasionally vomiting. Treatment consists of dietetic measures, regulated exercises, change of air and surroundings, and avoidance of depression. Milk should be used freely, and

in severe cases should be given alone till improvement sets in. Fats and greasy dishes should be avoided. Fruits are sometimes well borne, and at other times the reverse. Drugs do not play so important a rôle, but bitter tonics, like nux vomica, quassia, gentian, &c., are the best. Constipation should be treated when necessary.

Dyspho'nia is difficulty in speaking, and is the result of some forms of laryngitis. The condition is aggravated by attempts to use the voice, and complete rest is necessary to bring about an early and satisfactory recovery. Tonics, moderate exercise, and a holiday hasten recovery.

Dyspnœa (dis-pnē'a) is difficult or laboured breathing. It is a symptom of diseases of the heart or lungs, and is produced by any condition which interferes with normal respiration. It is sometimes present in nervous disturbances.

Dze'ren, or **Dze'ron**, the Chinese antelope, a remarkably swift species of antelope (*Procapra gutturōsa*) inhabiting the dry arid deserts of Central Asia, Tibet, China, and Southern Siberia. It is nearly 4½ feet in length, and 2½ feet high at the shoulder.

Dzig'getai, or **Kiang** (*Equus hemiōnus*), a species of wild ass native to Central Asia, allied both to the horse and ass. Its head is large like that of the ass, but in form resembles that of the horse. The ears also resemble those of the horse. It runs with a rapidity exceeding that of the best Arabian horses.

Dzoungaria, or **Sungaria**, a Chinese territory in Central Asia, stretching from about 43° to 48° N. lat. and from about 82° to 86° E. long. It has an area of 147,950 sq. miles, and pop. 600,000. It is administratively connected with Kuldja, and since the surrender of Kuldja by the Russians in 1880 is again under Chinese rule. Dzoungaria, once the centre of an independent empire, was first conquered by the Chinese in 1757.

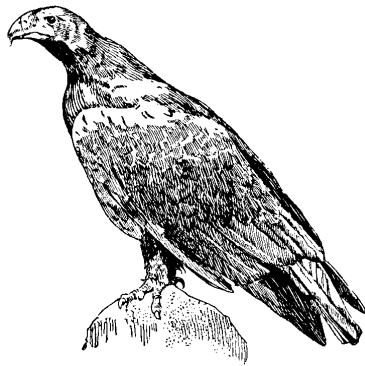
E

E, the second vowel and the fifth letter of the English alphabet. It occurs more frequently in English words than any other letter of the alphabet. Its long or natural sound in English coincides with the sound of *i* in the Italian and French languages, as in *here*, *mere*, *me*. It has also another principal sound, a short one, heard in *met*, *men*. It has besides a sound like *a* in *bare*, as in *there*, *where*, &c., and the obscure sound which is heard in *her*. As a final letter in English it is generally silent, but it serves to indicate that the preceding vowel is to have its long sound, as in *mane*, *cane*, *plume*. When two *e*'s come together the sound is generally the same as that of the single *e* long, as in *deem*, *esteem*, *need* (compare, however, *pre-exist*, &c.).

E, in music, is the third note or degree of the diatonic scale, answering to the *mi* of the Italians and French.

Eadie (ē'di), John, D.D., a Scottish preacher and theologian, born 1810, died 1876. He was educated at Glasgow University, and entered the ministry of the Secession Church, becoming in 1843 professor of biblical literature in the Divinity Hall of the Church, a post which he continued to hold after the Secession body was merged in the United Presbyterian Church (in 1847). Among his works are *Biblical Cyclopædia*; *Analytical Concordance to the Scriptures*; *Ecclesiastical Cyclopædia*; *Commentary on the Greek Text of Ephesians*, and similar works on *Colossians*, *Philippians*, and *Galatians*; and *The English Bible*. He was one of the scholars engaged on the Revised Version of the New Testament.

Ead'mer, an English monk, the friend and biographer of St. Anselm. In 1120 he was chosen Bishop of St. Andrews; but as the Scottish king refused to recognize the right of the Archbishop of Canterbury to consecrate him, he returned to England and died a simple monk about 1124. Besides the life of St. Anselm, Eadmer wrote lives of St. Wilfrid, St. Dunstan, St. Odo, and other English saints, as well as a valuable history (*Historiæ Novorum*) of events in England and the English Church from 1066 to 1122.



Imperial Eagle (*Aquila mogilnik*)

Eagle (Lat. *aquila*, Fr. *aigle*), the general name of raptorial birds that form a group or sub-family (Aquilinæ) of the great family Falconidæ, which includes the eagles, falcons, and hawks. The eagle is popularly regarded



African Sea-eagle (*Haliaeetus vocifer*)

as the noblest and most courageous of the rapacious birds. It soars to a greater height than any other European bird, from which circumstance the ancients considered it as the bird or messenger of Jove. The genus *Aquila*, which includes the most typical eagles, is distinguished by its long and powerful bill, the curve commencing at the cere, by its wings reaching to the tip of the tail, and by its tarsi being feathered to the toes. The imperial eagle (*A. mogilnik*) of Central Europe, North-East Africa, India, and China is probably the species to which the popular belief in the courage, strength, and nobleness of eagles is to be traced. *A. chrysaëtus*, the golden eagle, is the chief British species. It measures over 6 feet from tip to tip of the expanded wings, and 3 feet from the beak to the end of the tail. The adults have the body brownish, becoming darker with age; the feathers of the head and neck pointed, and of a golden-red hue. This species is found all over the northern hemisphere. It was once common in the Highlands of Scotland, but is now becoming rare. The Kirghis and other tribes of Central Asia use the golden eagle to kill antelopes, foxes, and even wolves. Another British eagle is the erne or sea-eagle (*Haliaeetus albicilla*) found near the sea-coast or lakes, and feeding largely on fish. The general colour is greyish-brown, the head pale-coloured, the tail white. The bald eagle (*Haliaeetus leucocephalus*), found in North America and North-East Asia, is the symbol of the United States, though Franklin deplors the selection on account of his mean and dishonest habit of robbing the industrious osprey of the fish caught by him.

Like all members of the genus, his diet is less restricted than that of the true eagles; and he even takes carrion. See also *Harpy Eagle*.

Eagle, as a symbol. The eagle first appears as a war standard amongst the Persians, through whom it reached the Egyptians. As the standard of the Roman armies it was first used by Marius, and afterwards took the place of all the other emblems at the head of the legions. It was first made of wood, then of silver, and finally, under Cæsar and his successors, of gold. In the Mediæval Ages the eagle became the heraldic emblem of the Holy Roman Empire, and was made double-headed in the fourteenth century. When the Holy Roman Empire fell to pieces in 1806, the double-headed eagle was retained by Austria. The double-headed eagle was assumed by Tsar Ivan III in 1472, and became the national military symbol of Russia; the single-headed eagle was assumed by the modern German Empire in 1871, and by the United States of America. The American eagle stands with outspread wings guarding a shield, with the motto *E pluribus unum*. The eagle was also the badge of several orders, the chief of which were the order of the *Black Eagle*, founded by the Elector of Brandenburg in 1701, and the highest order in Prussia; the order of the *Red Eagle*, also a Prussian order, and founded in 1705; and the Russian order of the *White Eagle*, originally Polish, and instituted in 1325.



Roman
Standard

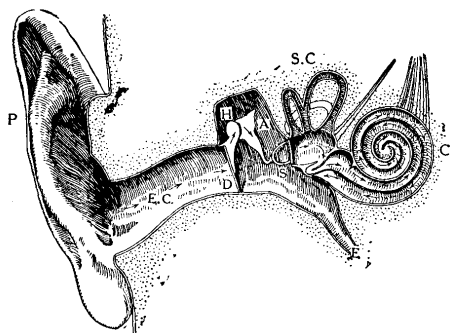
Eagle, a gold coin in the United States of the value of ten dollars, or £2 sterling, first coined in 1795. There are also half-eagles, quarter-eagles, and double-eagles.

Eagle-hawk, a name sometimes applied to small South American eagles (genus *Morphnus*), with short wings and long legs.

Eaglehawk, a gold-mining town in Victoria, Australia, 4 miles from Bendigo. Pop. 8130.

Eagle-owl, a name for several large horned owls, such as *Bubo ignavus* (the great horned owl), little inferior in size to the golden eagle, found in many parts of Europe and sometimes in Britain. An allied species, the Virginian horned owl (*B. virginianus*), is common in the United States.

Ealing, a municipal and parliamentary borough of Middlesex, the former a few miles west of London. Pop. 61,222.



Diagrammatic Section of Outer, Middle, and Inner Ear
P, Concha. E.C., External canal. D, Drum. H, Hammer. A, Anvil. E, Eustachian tube. S, Stirrup, S.C, Semicircular canals. C, Cochlea. Arrows denote the direction of vibration.

Ear, the organ of hearing. In the higher vertebrates it is divided into the outer, middle, and inner ear. The external ear, which is a cartilaginous funnel for collecting the sound waves and directing them inwards, is composed of the *concha*, or projecting part, and of the *auditory canal*, which extends from the *concha* to the membrane of the *tympanum* or *drum*. This membrane is a partition stretched obliquely across the bottom of the auditory canal, which it separates from the middle ear or tympanum; it is semi-transparent and very delicate. It vibrates with the waves of sound which strike against it, and

transmits the vibrations to certain little bones of the cavity of the tympanum. These bones, which have been named respectively the hammer (*malleus*), the anvil (*incus*), and the stirrup (*stapes*), transmit the vibrations to the internal ear, forming a chain communicating at one end with the membrane just mentioned, and at the other with the inner ear. The internal ear consists of a complicated system of tubes known as the *membranous labyrinth*, containing fluid in which waves are set up by the vibrations transmitted to it by the little bones from the drum membrane. The lower part of the labyrinth is coiled like a snail shell, and is called the *cochlea*. It is the real organ of hearing. The upper part consists of three *semicircular canals*, the function of which is to record the position and movements of the body in space. The middle ear communicates with the pharynx by the Eustachian tube, through which air from the mouth may be introduced into the tympanic cavity, so as to permit vibrations of the drum membrane. In the external auditory canal of the ear is produced the cerumen or ear-wax. The cut shows P the concha, E.C. the external canal, D the drum membrane partly removed, S the stirrup, A the anvil and H the hammer, the small bones communicating with the drum and vestibule, C cochlea, S.C semicircular

canals, E Eustachian tube.—Cf. Sir Thomas Wrightson, *An Enquiry into the Analytical Mechanism of the Internal Ear*.

Ear-cockle, a disease in wheat caused by the presence in the grain of worms belonging to the genus *Vibrio*. It is called in some parts of England *purples*.

Earl (A.Sax. *eorl*; Dan. *jarl*), a degree of the British nobility between marquess and viscount, the title of highest antiquity in England. The title was made hereditary by William the Conqueror, and for a time was used interchangeably with that of *count*, the corresponding title on the Continent. The wife of an earl is still called a *countess*. The earl was the highest in rank of the nobility until Edward III created a duke in 1357, and Richard II a marquess in 1385. The first earl of England is the Earl of Arundel. An earl's coronet is composed of eight pearls raised upon points, with small leaves between, above the rim. See *Peer*.

Earle, John, English bishop and writer, born about 1601, died 1665. He was educated at Oxford, and, after writing some short poems, gave to the world anonymously in 1628 *Microcosmographie, or a Piece of the World discovered in Essays and Characters*—a work full of wit, humour, and admirable character-painting. He was tutor to Charles II, accompanied him during his exile, and was held by him in the highest esteem. In 1662 he was consecrated Bishop of Worcester, and next year was translated to Salisbury.

Earle, Rev. John, Anglo-Saxon scholar, was born in 1824, and died in 1903. He studied at Oriel College, Oxford, where he obtained first-class honours in classics, and was elected a fellow (1848). In the following year he took orders, and was appointed for five years university professor of Anglo-Saxon. In 1857 he became rector of Swanswick, near Bath, and in 1871, a prebendary of Wells. The five years' rule having been rescinded, he was re-elected professor of Anglo-Saxon at Oxford in 1876, and continued to hold the position, together with the rectory of Swanswick, until his death. Among his contributions to the study of Anglo-Saxon and modern English are the following: *Two Saxon Chronicles Parallel* (1865); *The Philology of the English Tongue* (1871); *A Book for the Beginner in Anglo-Saxon* (1877); *English Plant Names from the Tenth to the Fifteenth Century* (1880); *Anglo-Saxon Literature* (1884), a very useful and informing little manual;

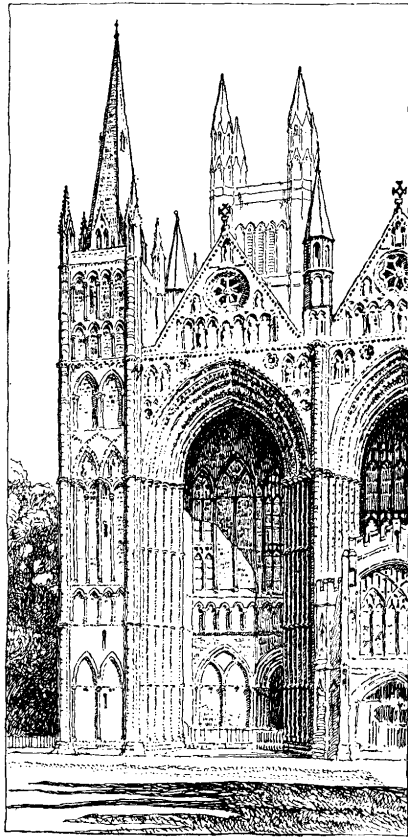
Handbook to the Land Charters and other Saxon Documents (1888); *English Prose: its Elements, History, and Usage* (1890); *The Deeds of Beowulf* (1892), a translation of the well-known Anglo-Saxon epic; *The Psalter of 1539* (1894); and *A Simple Grammar of English now in Use* (1898). He also wrote a book on *Bath, Ancient and Modern* (1864).

Earlestown, a town of Lancashire, England, 14 miles east of Liverpool. There are here engineering-works, sugar-works, and other establishments. Pop. 9020.

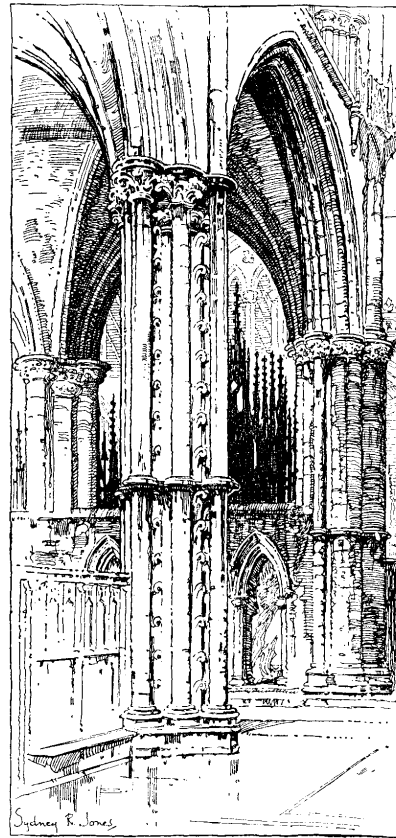
Earl-marshal, a great officer of England, who had, anciently, several courts under his jurisdiction, as the court of chivalry and the court of honour. He is the head of the College of Arms (Heralds' College), grants armorial bearings, and determines all claims in connection with them. Since 1672 the office is hereditary in the family of Howard (Dukes of Norfolk). There was also an earl-marshal of Scotland, the office being hereditary in the Keith family until 1716, when it was abolished.

Earlom, Richard, English mezzotinto engraver, born 1743, died 1822. His engravings from Reynolds, Hogarth, Van Huysum, &c., and from Claude's *Liber Veritatis*, are exceptionally fine, and are standard works in their department.

Earlston (originally **Ercildoun**), a village of Scotland in Berwickshire. Near it are the ruins of the ancient tower, which belonged to the famous Thomas the Rhymer. Pop. 1749.



Early English Architecture
Peterborough Cathedral. Part of the West Front
Cathedral. Early English Pillars in the Eastern Transept



Lincoln

Early English Architecture, the first of the Pointed or Gothic styles of architecture that prevailed in England. It succeeded the Norman in the reign of Richard I (1189), and continued to the end of the reign of Henry II in 1272, a period of 123 years, when it gradually merged into the Decorated style. One of the leading peculiarities in this style is the form of the windows, which are narrow in proportion to their height, and terminate in a pointed arch, resembling the blade of a lancet (and therefore often called the *Lancet* style). Throughout the early period of the style they are very plain, particularly in small churches; but in cathedrals and other large buildings the windows, frequently combined two or more together, are carried to a great height, are richly and deeply moulded, and the jambs ornamented with slender shafts. On the eastern and



Early English
Capital, Salisbury

western fronts of small churches the windows are often combined in this manner, with a circular window above and a richly moulded door below; but in large buildings there is often more than one range of windows, and the combinations are very various. Though separated on the outside, these lancets are in the interior combined into one design, thus giving the first idea of a compound window. The doorways are in general pointed, and in rich buildings sometimes double; they are usually moulded, and enriched with the tooth-ornament. The buttresses are often very bold and prominent, and are frequently carried up to the top of the building with but little diminution, and terminate in acutely pointed pediments, which, when raised above the parapet, produce in some degree the effect of pinnacles. In this style, likewise, flying-buttresses were first introduced (see *Buttresses*), and the buttresses themselves much increased in projection owing to the comparative lightness of the walls, which required some counter-support to resist the outward pressure of the vaulting. The roof in the Early English style appears always to have been high pitched, and the towers surmounted by lofty pointed spires, as at Salisbury Cathedral. In the interior the arches are usually lancet-shaped, and the pillars often reduced to very slender proportions. As if to give still greater lightness of appearance, they are frequently made up of a centre pillar, surrounded by slight detached shafts, only connected with the pillar by their capitals and bases, and bands of metal placed at intervals. These shafts are generally of Purbeck marble, the pillar itself being of stone, and from their extreme slenderness they sometimes appear as if quite inadequate to support the weight above them. The earliest example of Early English architecture is the choir of Canterbury, followed by the choir of Lincoln Cathedral, but some of the best examples are to be seen in Salisbury Cathedral. The architects of this style carried their ideas of lightness to the utmost limits of prudence, and their successors have been afraid to imitate their example. The abacus of the capitals is generally made up of two bold round mouldings, with a deep hollow between. The foliage is peculiar, generally very gracefully drawn, and thrown into elegant curves; it is usually termed *stiff-leaved*, from the circumstance of its rising with a stiff stem from the neck-mould of the capital. The trefoil is commonly imitated, and is very characteristic of the style. The mouldings of this style have great boldness, and produce a striking effect of light and shade. They consist chiefly of rounds separated by deep hollows, in which a peculiar ornament, called the *dog's-tooth*, is

used, whenever ornament can be introduced. This ornament is as characteristic of the Early English as the zigzag is of the Norman. See *Dog's-tooth*.—Cf. F. Bond, *An Introduction to English Church Architecture, from the 11th to the 16th Century*.

Earnest, in law, any sum paid in advance, to bind parties to the performance of a verbal agreement, or something given by a buyer to a seller as a pledge of adherence to a bargain. The party is then obliged to abide by his bargain, and is not discharged upon forfeiting his earnest. In England the general view is that the sum paid as earnest, however small, is part of the price.

Ear-ring, an ornament for the ear, consisting of a ring or hook passing through the lobe, with a pendant of diamonds, pearls, or other jewels frequently attached. Ear-rings were commonly worn amongst the Oriental nations, and by both sexes, especially in Babylonia and Assyria, from the earliest times. Amongst the Greeks and Romans the wearing of ear-rings was usually confined to women. In England the Romanized Britons and the Anglo-Saxons wore them, but the fashion declined in the tenth century, and was again introduced in the sixteenth century, in Queen Elizabeth's time.

Earsdon, an urban district or town of England, South Northumberland, several miles north-west of Tynemouth, with productive collieries. Pop. 10,568.

Earth, the planet which we inhabit, a nearly spherical body which every twenty-four hours rotates from west to east round an imaginary line called its axis—this axis having as its extremities the north and south poles—while in the course of a year it completes a revolution round the sun. To an observer whose view is not obstructed, the visible part of the earth appears as a circular and horizontal expanse, on the circumference of which the heavens appear to rest. Accordingly, in remote antiquity, the earth was regarded as a flat, circular body, floating on the waters. But even in antiquity the spherical form of the earth began to be suspected. It is only on this supposition that we can explain how the horizon of vision grows wider and wider the higher the position we choose, how the tops of towers and mountains at a distance become visible before the bases, how the hull of a ship first disappears as she sails away, and how, as we go from the poles

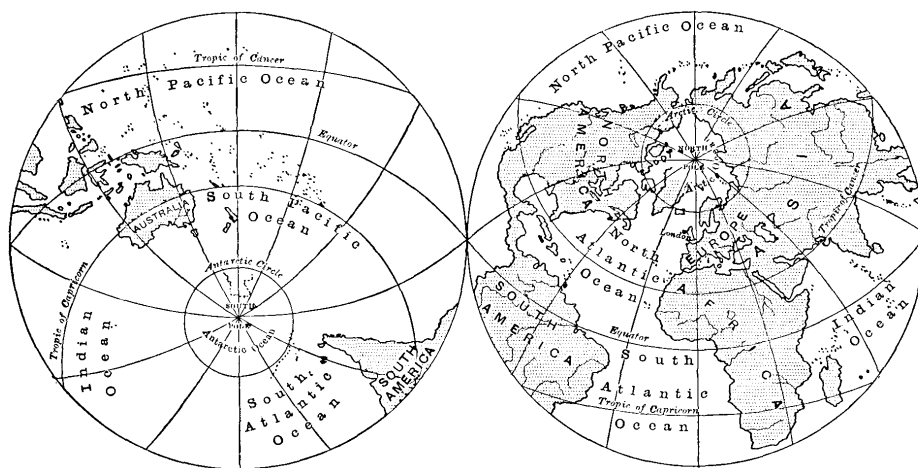
towards the equator, new stars become visible. Besides these proofs there are many others, such as the circular contour of the earth's shadow seen on the moon during an eclipse. The mere fact that the earth can be circumnavigated does not, as is sometimes assumed, prove it to be globular. But its surface, land and ocean, has been almost all explored and accurately mapped, and the relative distances and directions found to obtain between the places on its surface are consistent only with its possessing such a shape.

The earth is not, however, an exact sphere, but is very slightly flattened at the poles, so as to have the form known as an *oblate spheroid*. In this way the *polar diameter*, or diameter from pole to pole, is shorter than the diameter at right angles to this—the *equatorial diameter*. The most accurate measurements make the polar diameter almost 27 miles less than the equatorial, the equatorial diameter being found to be 7926.7 miles, and the polar 7900 miles. The earth is regarded as divided into two halves—the northern and the southern hemisphere—by the *equator*, an imaginary line going right round it midway between the poles. In order to indicate with precision the position of places on the earth additional circles are imagined to be traced upon the surface in such a manner that those of the one set all pass through both poles, while those of the other are drawn parallel to the equator. The former are called *meridians*, the latter *parallels of latitude*, and by reference to them we can state the latitude and longitude, and thus the exact position, of any place.

Many experiments by various methods have been made in order to determine the average density of the earth, and the total quantity of matter it contains. Amongst these methods may be mentioned: (1) that of measuring the deflection of a plumb-line due to a mountain's attraction, and thereby comparing the mass of the earth with that of the mountain; (2) that founded on the difference of oscillation period of a pendulum when placed at the summit of a mountain and when at the sea-level; (3) by the determination of the difference of gravity at the top and the bottom of a deep mine, by pendulum experiments; (4) Cavendish's experiment with the torsion balance, which attempts to compare the attractive force of two large lead balls upon two small lead balls with that exercised by the earth. From these and other experiments it has been calculated that the mean density of the earth is to that of water as about $5\frac{1}{2}$ to 1.

The earth, in common with the other planets, moves round the sun, completing its revolution in about 365 days and 6 hours. The orbit of the earth is an ellipse, with the sun in one of its foci. Hence the earth is not equally distant from the sun throughout the year; it is over 3,000,000 miles nearer at one time than another, its least distance (*perihelion* distance), according to recent calculations, being about 91,340,000 miles; its greatest (*aphelion* distance), 94,450,000; and the mean distance, 92,897,000 miles. From this it may be calculated that the velocity of the earth in its orbit is about $18\frac{1}{2}$ miles a second. About 3rd Jan. the earth is nearest the sun, and about 4th July farthest from it. This position of matters, which is subject to slow alteration in the course of ages, at present tends to moderate the seasonal variations in the northern hemisphere, and to intensify them in the southern. The passage of the earth round its orbit causes the sun to appear as if it described an annual circuit of the heavens; and hence it is that at one time of the year one group of stars is seen in the neighbourhood of the sun near sunrise or sunset, and at another time another group. This apparent path of the sun is the *ecliptic*, and corresponds with what would be the path of the earth as seen from the sun; and the groups of stars through which the sun successively passes form the *zodiac*.

The earth's daily motion about its own axis takes place in twenty-three hours, fifty-six minutes, and four seconds of mean time. This diurnal revolution is



The Hemispheres, showing the Greatest Masses of Land and Water

the occasion of the alternation of day and night. As the axis on which the earth performs its diurnal rotation is inclined towards the plane of its path about the sun at an angle of $66\frac{1}{2}^{\circ}$, and the angle between the plane of the ecliptic and the plane of the earth's equator is therefore $23\frac{1}{2}^{\circ}$, the sun

ascends in the heavens, as seen from our northern latitudes, from 21st March to 21st June (the summer *solstice*), to about $23\frac{1}{2}^{\circ}$ above the celestial equator, and descends again towards the equator from 21st June to 23rd Sept.; it then sinks till 22nd Dec. (the winter *solstice*), when it is about $23\frac{1}{2}^{\circ}$ below the equator, and returns again to the equator by 21st March. This arrangement is the cause of the seasons, and the inequality of day and night attending them. For all places removed from the equator, day and night are equal only twice in the year (at the *equinoxes*). At the summer solstice in the northern hemisphere the north pole of the earth is turned towards the sun, and the south pole away from it, and for places within $23\frac{1}{2}^{\circ}$ of the former there is a period of longer or shorter duration during which the sun is continually above the horizon throughout the twenty-four hours of each day, while round the latter there is an equal extent of surface within which the sun for similar periods is below the horizon. (See *Day*.) The reverse state of matters occurs at the winter solstice. The circles bounding these regions are called respectively the *arctic* and the *antarctic* circle, and the regions themselves the *polar* or *frigid zones*. Throughout a region extending to $23\frac{1}{2}^{\circ}$ on each side of the equator the sun is directly overhead at any place twice in the year. The circles which bound this region are called the *tropics*, that in the northern hemisphere being the tropic of *Cancer*, that in the southern the tropic of *Capricorn*, while the region between is the *torrid zone*. The regions between the tropics and the polar circles are respectively the *north* and *south temperate zones*.

From the evidence furnished by volcanoes, hot springs, sinking of mines, &c., it is known that the earth has a high internal temperature. Taking the average of the various observed rates of increase this temperature seems to increase 1° F. for every 60 feet of descent. Assuming this to continue, the rocks at a depth of 2 miles would be as hot as boiling water, and at a depth of 50 miles the heat would be such as at the surface would melt every known solid. This being so, various theories as to the internal condition of the earth have been proposed: (1) that a thin envelope or crust surrounds a molten interior. It can be shown, however, that as tides must be produced in such a molten mass the cool outer crust would be unable to withstand the enormous force of these unless it were about 2000 miles thick. (2) That the interior is solid, with spaces here and there filled with liquid or gaseous material. This theory assumes that there are within the earth enormous

cavities filled with molten rock, which escapes, when local pressure is removed, in the form of volcanic outbursts. (3) That the earth consists of a thin crust, a large solid nucleus, and a liquid film between the nucleus and the crust. (4) That the earth is solid to the centre, but any part may become liquid if local pressure is removed. On this theory it is supposed that if water should percolate to liquefied rocks, it would be converted into steam, and produce the various volcanic phenomena.

The question of the constitution of the earth's interior has in recent years been much investigated by means of seismographic records. These appear to indicate that there are three distinct divisions. The outer crust has a thickness of from 20 to 40 miles. It possesses a high power of resistance to all kinds of stress. Beneath it is a large shell possessing a density and elasticity resembling fine steel. This shell has a high rigidity against changing forces of shorter duration, like tidal action, but in its outer parts at least yields in time to unvarying long-continued stress. The third or innermost division of the earth is probably molten, as it can transmit compressional waves, but yielding immediately to distortional or twisting forces, is unable to transmit distortional waves. This innermost portion appears to be a sphere of radius approximately one-half that of the earth as a whole. The transition between the crust and intermediate shell is abrupt, but that between the latter and the central portion is more gradual.

The earth (like the other planets) is believed to have condensed and solidified from a gaseous or nebular condition, and to have once had a far higher temperature than now. If such were the case, the outer surface, losing heat by radiation, would be the first part to cool quickly; while the interior, losing its heat by conduction, would not cool so rapidly, and, therefore, would naturally have a higher temperature than the portion at the surface. This is what all observations indicate the condition of the earth to be, and the shape of the earth also indicates that it must once have been in a fluid state. Calculations have been made of the time which has elapsed since solidification commenced, the estimates being in general of the order of hundreds of millions of years. See *Nebular Hypothesis*.

Another feature that the earth as a whole presents is its magnetism. When a magnetic needle is balanced on a point, it remains at rest in one position only, pointing then nearly due north and south. This can be explained only

on the supposition that the earth acts as a great magnet. It has, in fact, two poles—a north and a south magnetic pole—which are not very far from the geographical poles. The magnetic equator, where the vertical force is zero and the dipping needle takes a horizontal position, does not diverge greatly from the geographical equator. The earth acts upon all magnets as they act upon each other, and it is for this reason that they point north and south.

The surface of the earth contains over 196,000,000 sq. miles, of which about two-sevenths is dry land, the remaining five-sevenths being water. The land is arranged into masses of irregular shape and size, the greatest connected mass being in the eastern hemisphere. The chief masses receive the name of continents, detached masses of smaller size being islands. The surface of the land is variously diversified, exhibiting mountains, valleys, plains, plateaus, deserts, &c. The water area of the earth is divided into oceans, seas, bays, gulfs, &c., while rivers and lakes may be regarded as features of the land surface. The great phenomena of the oceans are currents and tides. The population of the whole earth is estimated at from 1600 to 1700 millions. The earth is attended by the moon as a subordinate or secondary planet. See also such articles as *Climate*, *Currents*, *Ocean*, *Earthquake*, and *Seasons*.—BIBLIOGRAPHY: A. von Humboldt, *Cosmos*; E. Reclus, *The Earth and its Inhabitants*; T. G. Bonney, *The Story of our Planet*; T. M. Reade, *The Evolution of Earth Structure*; *Theory of Geomorphic Changes*; A. T. Swaine, *The Earth: its Genesis and Evolution, considered in the Light of the most Recent Scientific Research*.

Earthenware, a name applied to the commoner sorts of pottery-ware. The older kinds of earthenware, such as Majolica, Delft-ware, Faïence, and Palissy-ware, are not only glazed, but are besides elaborately coloured and enamelled and ornamented with raised figures of various kinds. See *Pottery*.

Earth-houses, a name generally given throughout Scotland to underground buildings, also known as 'Picts' houses' or 'Picts' dwellings'. The earth-house in its simplest form consists of a single irregular-shaped chamber, formed of unhewn stones, the side walls gradually converging towards the top until they can be roofed by stones of 4 or 5 feet in width, all covered in by a mound of earth rising slightly above the level of the adjacent ground. In the more advanced form of these structures two or three chambers are found.

Earth-houses are frequent in the north-east of Scotland, occasionally thirty or forty being found in the same locality. Querns, bones, deers' horns, earthen vessels, cups and implements of bone, stone celts, bronze swords, and the like, are occasionally found in connection with them. Very similar structures, known as beehive-houses, occur also in Ireland and Cornwall.

Earth-nut, the *Conopodium denudatum*, an umbelliferous plant common in woods and fields in Britain. The leaves are ternately divided, and the small white flowers are in terminal umbels. The tuber or nut is about 4 or 6 inches below the surface, at the termination of a long slender root. It is brown, the size of a chestnut, of a sweetish farinaceous nature, resembling in taste the common chestnut. Swine are very fond of the nuts, and fatten rapidly where they are abundant. The name is frequently applied to *Carum Bulbocastānum*, which has a similar tuber. See *Ground-nut*.

Earthquake, a shaking of the earth's surface, propagated from place to place by a wave motion. It may vary in intensity from the slightest perceptible tremor to a violent shock which bursts open chasms and changes the appearance of the ground. Earthquakes originate in the crust of the earth, generally at only a very few miles depth, and probably never lower than about 30 miles. The point of origin is called the centre or seismic focus, and the place on the surface vertically over it the epicentre. It is rather difficult to tell the depth of the focus. Mallet estimated this by projecting backward the direction of travel of the wave at different points, as judged from the inclinations of the rents in buildings, &c., assumed to be at right angles to the line of propagation. The accuracy of this method has been improved by substituting evidence of direction as given by seismographs. The focus of an earthquake is often submarine, and subsequent to the shock transmitted through the solid earth a great sea-wave may invade the land and produce far more disastrous effects.

In some cases an earthquake may be caused by a fall of rock in some subterranean cavity. This gives only a minor and local shock. The vast majority of earthquakes are certainly *tectonic*, originating from the snapping of strata under great strain, or the further slipping of portions of the earth's crust along previously existing fault planes. Such dislocations probably arise sometimes from the variations of weight supported by the earth's crust in neighbouring regions, due to the transport by rivers of

material, which they erode at one place and deposit at another. A further cause is the contraction undergone by the earth in its secular cooling. There are also earthquakes of *volcanic* origin, accompanying eruptions, but these are not usually of any great violence, nor do they involve any large area. The coasts of the Pacific Ocean—American, Asiatic, and East Indian—are much visited by earthquakes, in especial the Japanese Islands. The other band of greatest frequency has a direction outlined by the Azores, Alps, Mediterranean, and the Caucasus and Himalaya Mountains. It may be noted that all the regions specially affected are distinguished by steep gradients of the earth's surface.

In recent years much information has been obtained by the investigation of earthquakes by various kinds of seismograph. One single instrument at a particular station, e.g. a Milne seismograph, will enable the distance of the epicentre to be calculated. From the results of three stations, the precise locality can practically always be told. With additional or particular forms of instruments, this may even be possible by

means of the records at one station. Earthquake waves are found to consist of distinctly defined types. The first to arrive are the preliminary tremors or first-phase waves, then the second-phase waves, next the third-phase or large waves, and lastly the concluding waves, consisting largely of 'echoes' or reflected vibrations. The speed of the preliminary tremors is found to be only about 2 miles per second for very short distances, but for a quadrant of the earth's surface they travel at an average of about 7 miles per second, a speed which is only slightly exceeded for still greater distances. The second-phase waves travel with a little under two-thirds of these velocities. These two classes of wave have been proved to travel through the earth, approximately along chords, but with the path slightly bent, convex towards the earth's centre. The first-phase waves are longitudinal, or waves of compression; the second-phase are transverse, or waves of distortion. The greater speed for greater distances is due to the track being more through the earth's interior and less through its outer portions, as the interior

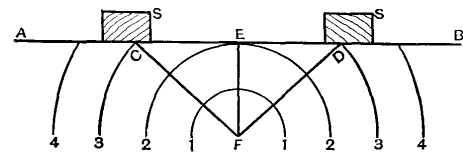


Diagram of Earthquake

AB, Surface of earth. F, Focus. E, Epicentre. 1-4, Successive positions of earthquake wave. S, S, Cracked walls; the cracks being at right angles to CF and DF give some indication of the depth of focus.

transmits wave-motion much more rapidly than the crust. The rigidity at some depth from the surface has been shown to be of the same order of magnitude as the rigidity of steel. The third-phase waves are of much longer vibration period and wide amplitude, and have been compared to a groundswell on the sea. Their time of passage from the epicentre to any place is proportional to the distance measured round the earth's surface, and it is clear that they travel on the surface, and not through the interior. Their speed is nearly 2 miles per second. The difference in time between the arrival of the preliminary tremors at any station and the arrival of the second-phase waves, or between the second-phase and third-phase waves, enables the distance of the epicentre to be easily found, as these differences, of course, become greater with increasing distance.

The number of earthquakes has been found to be enormously greater than was at one time supposed; in fact, small tremors are occurring daily in one part or another of the earth. Great and destructive shocks are generally preceded by minor shocks in the same district, and they are always followed for months afterwards by a series of gradually lessening after-shocks. Among the most remarkable earthquakes of modern times were those which destroyed Lima in 1746, and Lisbon in 1755; more recently destructive earthquakes visited Calabria in 1857, Peru and Ecuador in 1868, the Island of Ischia in 1884, Japan in 1896, North India and Calabria in 1905, San Francisco in 1906, Messina and Reggio in 1908. One of the greatest earthquakes of recent times was that which visited the provinces of Kansu and Shensi in North-West China, on 16th Dec., 1920. (See *Seismograph*.)—
BIBLIOGRAPHY: J. Milne, *Earthquakes and other Earth-Movements*; C. Davison, *A Study of Recent Earthquakes*; C. G. Knott, *The Physics of Earthquake Phenomena*.

Earths, a term applied in geology to certain loosely aggregated siliceous and aluminous materials, the detritus of pre-existing rocks. In chemistry the term earth is given to certain metallic oxides, such as the 'alkaline earths' lime, baryta, and strontia; also to alumina and a series known as the 'rare earths'. The earths were regarded as simple bodies until Sir H. Davy proved them to be compounds of oxygen with metals.

Earth-shine, in astronomy, a name given to the faint light visible on the part of the moon not directly illuminated by the sun, due to the illumination

of that portion by the sunlight which the earth reflects on her. It is most conspicuous when the illuminated part of the disc is small, as soon after new moon. This phenomenon is popularly described as 'the old moon in the new moon's arms'.

Earth-tongue. See *Geoglossum*.

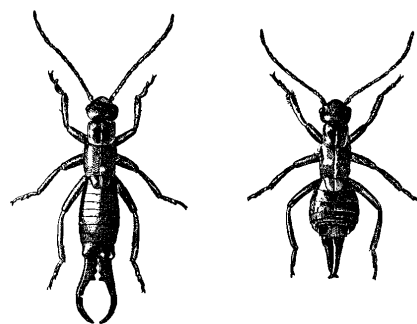
Earthworks (in fortification) are military works formed chiefly of earth and designed either as permanent or temporary defences. They are cheaper, more easily repaired, and expose their defenders to less risk from broken stone than stone-works. See *Entrenchments*.

Earthworm, the name applied to segmented worms (Annelids) that burrow in the soil, and belong to the ord. Oligochæta, a subdivision of the bristle-worms (Chætopoda). They have a long, cylindrical body, divided by transverse furrows into numerous rings. The mouth is destitute of jaws, and they have no eyes, tentacles, or other head appendages. They are hermaphrodite. The commonest British forms are chiefly species of *Lumbricus* and *Allolobophora*. They feed on earth and various kinds of animal and vegetable matter, and move by the contractions of successive parts of the body aided by a double row of bristles. They are of great service to the agriculturist by loosening the soil and increasing its depth. This is chiefly the result of their mode of nourishment, since they deposit the soil they have swallowed, after digestion, in heaps called *worm castings* which bring up rich fine soil to the surface, gradually covering the upper layer sometimes to the extent of several inches.

Ear-trumpet, an artificial instrument for aiding the collection of the vibrations or waves of sound, and carrying them in an intensified form to the internal parts of the ear. They are generally made of tin, vulcanite, or gutta-percha, and are of various forms. A small kind known as ear-cornets or acoustic auricles, attached to the ear by a spring, is sometimes used in slight cases of deafness.

Earwig (*Forficula*), a common orthopterous insect whose name is derived from its supposed habit of insinuating itself into the ears of persons. This is practically impossible, yet the notion is widely spread, as appears from the names given to the earwig in different languages, as in Fr. *perceoreille* (pierce-ear), in Ger. *ohrenhöher* (ear-borer). Much damage is sustained by

gardeners from the depredations of these insects among fruit and tender vegetables, which constitute their proper food. The earwig is about three-quarters of an inch in length, having the wings folded under very short and truncate elytra or wing-cases, and the extremity of the abdomen armed with a horny forceps.



Earwig (*Forficula auricularia*). Male and female

Easement, in law, a right or privilege which one proprietor may have to use the land of another in connection with the needs of his own land, as the use of a way, a water-course, &c. The right to an easement may be acquired either by grant or by uninterrupted enjoyment for a period of years.

East, one of the four cardinal points, being the point in the heavens where the sun is seen to rise at the equinox, or the corresponding point on the earth; that point of the horizon lying on the right hand when one's face is turned towards the north pole. By *the East*, in an indefinite sense, is often meant Syria, Arabia, Persia, India, and the eastern part of the world generally.

Eastbourne, a municipal borough and flourishing watering-place of England, county of Sussex, on the English Channel, near Beachy Head; also a parliamentary division of Sussex. The town is handsomely built, having fine parades and well-planted walks and drives. Pop. 52,544.

East Cape, the most easterly point of Asia, projecting into Behring's Strait nearly opposite Cape Prince of Wales in Alaska.

Easter, the festival commemorating the resurrection of Christ, observed in the Roman Catholic, the Greek, Anglican, Lutheran, and other branches of the Christian Church. By the first Christians it was considered to continue the feast of the *passover*, at which the paschal lamb, a type of Christ, was sacrificed. Hence its name in Greek (*pascha*), French (*pâques*), and other Romance languages is taken from the Hebrew *pesach*, passover. The English name, according to the Venerable Bede, comes from the Anglo-Saxon *Eostre* (from Teutonic *Austrō*), a goddess of light or spring, whose festival was celebrated in April. There was long a dispute in the Christian

Church as to the proper time for holding Easter, the Christians of the East celebrating it on the same day as that on which the Jewish passover fell, that is, the 14th of Nisan (hence they were called *quarto decimani*), while the majority of the Church celebrated it on the Sunday next after this day. The controversy was decided by the Council of Nice (Nicæa) in 325, which settled that it was to be reckoned as at present, namely, that Easter is the first Sunday after the full moon which happens upon or next after the 21st of March; and if the full moon happens on a Sunday, Easter-day is the Sunday after, but, properly speaking, for the 'full moon' in the above the 'fourteenth day of the moon' should be substituted.—Cf. Sir J. G. Frazer, *The Golden Bough*.

Easter Dues, or Offerings, in the Church of England, certain dues paid to the parochial clergy by the parishioners at Easter as a compensation for the tithe for personal labour.

Easter Eggs. The egg was anciently a symbol of the mother goddess and of birth: the sun emerged from the cosmic egg. The Saxon goddess Easter was a life-giver. On Easter Day, the day of Christ's resurrection, eggs (Pasche eggs) were dyed in symbolic colours, and boiled hard to be rolled or used in egg-breaking contests. The Jews have eggs at the Passover Feast.

Easter Island, or Rapanui (discovered by the Dutch Admiral Roggeveen, on Easter, 6th April, 1722), an island, 12 miles long, in the South Pacific Ocean, long. 109° 17' w., lat. 27° 6' s., and utilized for grazing sheep and cattle. It now belongs to Chile, from which it is 2000 miles distant. Pop. 250 in 1916. The Routledge Expedition reported, in 1919, that the inhabitants are of mixed Polynesian and Melanesian origin. Their ancient bird-cult shows very close resemblances to that of the Solomon Islands. Numerous gigantic stone images of a soft 'volcanic ash' were being worshipped when the island was first visited by Europeans in the eighteenth century. Some still lie partly constructed in a crater quarry. These images date back a few centuries, and resemble those made until recently in wood on this island and elsewhere in Oceania, and bear symbols used on these and in tattooing. Local legends of the earliest settlements from distant islands and of local tribal wars still survive. The present inhabitants are undoubtedly descendants of the image-makers and worshippers. Cf. K. Routledge, *The Mystery of Easter Island*.

Eastern Bengal and Assam, a province of India, under a Lieutenant-Governor, formed in 1905 by disjoining from Bengal the three divisions of Chittagong, Dacca, and Rajshahi (with the exception of Darjeeling) and uniting Assam with them, as also the state of Cooch Behar. It was formed in order to provide for the better government both of the area belonging to it and of that of Bengal, which was regarded as having become rather unwieldy, its population being still 54 millions. On 1st April, 1912, however, Assam was separated from Eastern Bengal and reconstituted. Its area is about 53,000 sq. miles, and the pop. nearly 6,750,000.

Eastern Churches, a collective term for the Greek, Armenian, Coptic, Abyssinian, Syrian, and other kindred Churches, as distinguished from the Latin, or Western Church.

Eastern Question, The, an international political problem which occupied the attention of European statesmen during the last two centuries, and even since 1453, when the Turks established their empire and gained sway over the Balkans. It deals with the relations of the Balkan nationalities, Turkey, and the Great Powers to each other. Russia, Germany, Austria, Greece, France, Italy, and Great Britain were all interested in the Near East and in the Eastern Question ever since the Treaty of Kutshuk-Kainardji in 1774. The Levantine commerce and the Mediterranean ports were, and still are, of vital importance not only to Russia, but also to the Balkan States and to the neighbouring European powers. The occupation of Egypt by Great Britain, the Russo-Turkish War of 1878, the proclamation of Bulgaria's independence, the annexation of Bosnia and Herzegovina by Austria in 1908, all contributed to the complication of the Eastern Question. This complication was further increased by Italy's occupation of Tripoli in 1911, by the Balkan Wars (1912-3), and by the construction of the Bagdad Railway with the aid of German capital. It is no exaggeration to say that the Eastern Question was one of the causes which led to the outbreak of the European War of 1914. The Peace Treaties of Versailles, Sèvres, and St. Germain have not yet settled the Eastern Question, and the peace in the Near East is still a problem which occupies the attention of European diplomatists.—BIBLIOGRAPHY: W. A. Phillips, *Modern Europe*; R. W. Seton Watson, *The Rise of Nationality in the Balkans*.

Eastern Rumelia. See *Bulgaria*.

Easter Term, one of the four regular terms of the courts of common law in England, beginning on the 19th April, and continuing till the middle or end of May.—In Oxford University, a term beginning 13th April, ending 27th May; in Cambridge, beginning 18th April, and ending 24th June.

East India Company, a great English company, originally simply a trading association, which played an important part in the history of Hindustan. It was formed in 1599 in London, with a subscribed capital of about £30,000, for the purpose of trade with the East Indies. A charter was granted to it by Queen Elizabeth on



Silver Rupee of East India Company, 1675

31st December, 1600, for fifteen years, renewable for a similar period. In this charter the Company is styled, "The Governor and Company of the Merchants of London trading into the East Indies". The first voyages resulted in large profits. In 1609 the charter was renewed by James I, and made perpetual, reserving power to the Crown to recall it at three years' notice. Additional power was granted to the Company of seizing and confiscating ships and goods of contraband traders, either in the British dominions or in any of the places where they were authorized to trade. Among the motives which had induced the Company to press for this renewal of their charter was the necessity they had experienced from the jealousy of the Dutch and Portuguese to send out vessels fitted not only for trade, but for defence and indeed attack. Accordingly Captain Best, who commanded the eighth expedition, attacked four Portuguese war galleons, convoying 200 sail of merchantmen, off Surat, and gained a complete victory, which so impressed the Great Mogul that he immediately made a treaty with Captain Best, giving the English full liberty to trade in his dominions. This treaty was concluded on 6th Feb., 1613. It was followed at once by a resolution of the Company to trade on a joint-stock. £429,000 was raised as capital, and apportioned in fitting out four voyages for 1613, 1614, 1615, 1617. In 1617 and 1618 the Company was so enlarged as to include 954 proprietors, while a new joint-stock of £1,600,000 was subscribed. In 1619 a treaty was made with the Dutch, by which the two companies were to work in harmony for twenty years; but in 1629 the Dutch massacred the leading members of the English factory at Amboyna. In the feeble reigns of James and Charles I, however, the outrage remained

unredressed, and the English Company, ill supported by the Crown, was often reduced to great straits. Their trade, impeded by the Dutch, became unprofitable, and, to add to their difficulties, Charles I in 1635 gave a licence to a rival company. At length, under Cromwell, the Company received a new charter. A territorial footing had been acquired in Madras in 1640, to which settlement was given the control of all the factories in Bengal and the Coromandel coast, the Supreme Council in India still remaining at Surat. A new charter, granted by Charles II in 1660, enlarged the powers of the Company, giving it political and judicial authority in the factories and colonies established by it, with the right to appoint governors. On the Revolution of 1688 the Company was involved in new difficulties, and in 1692 the Commons presented an address to the Crown praying for their dissolution. At this time, by an accidental failure to pay a tax upon their stock, the Company formally forfeited their charter, and were compelled to accept its renewal with the important proviso of a reservation to the Crown of the right to alter or modify its conditions. The maximum stock to be held by any individual was fixed at £10,000, every £1000 of which was to give a vote, while the right of membership was thrown open to all British subjects. The Scottish Parliament also sanctioned a company, but a war with Spain and the bitter opposition of the English Parliament made difficulties under which this company succumbed. Meantime the misconduct of the English company had so strengthened its enemies that, in spite of all its opposition, a resolution in favour of the formation of a new company passed the House of Commons on 4th May, 1698, and this company was actually constituted by Act 9 William III cap. xlv. This Act provided for the extinction of the old company, but an amalgamation was eventually arranged in 1708. The possessions of the old company at the time of amalgamation, upon which the valuation of £330,000 was placed in 1700, included a large number of places in India, a footing having been by this time acquired in each of the three presidencies, besides possessions in Persia, Cochin-China, and Sumatra. The dividends of the Company rose rapidly after the amalgamation, and finally settled at 8 per cent; and it procured without difficulty, at various periods, a prolongation of its exclusive privileges until 1780, still with three years' notice. In the meantime the French possessions had, as well as the English, been growing in power and importance in the East, and on the outbreak of the war of the Austrian Succession in 1741 commenced those struggles (Clive being the

first great English leader) by which a mercantile company was led on to establish British supremacy over nearly the whole of India. In 1766 the right of the Company to acquire territorial possessions formed a subject of parliamentary inquiry; and the question of the political rights of the Company being thus opened up, the ministry began to act on their view of it by sending out a Crown plenipotentiary to India. A regulating Act was passed in 1773 remodelling the powers of the Company, and placing it completely under the control of Parliament, providing for the establishment by the Crown of courts of judicature in India. The charter, which expired in 1780, was renewed till 1791. The Renewal Act provided that the Company, which was already bound to submit to the Government all dispatches received from India, should submit for approval all dispatches proposed to be transmitted thither. In 1784 another Act established a board, afterwards known as the Board of Control, to superintend, direct, and control all acts, operations, and concerns relating to the civil and military government or revenues of India. The board was to consist of a principal Secretary of State, the Chancellor of the Exchequer, and four Privy Councillors nominated by the Crown. The directors of the Company were bound to submit all their papers except those referring to commercial matters to this board, and obey its instructions. From this time the political power of the Company was little more than nominal. While the right of nominating the servants of the Company was still left to the directors, the absolute right of recall was vested in the Crown. A subsequent Declaratory Bill regulated the power of the Board of Control to send out troops at the expense of the Company. In 1813 the charter was renewed on condition that the right of exclusive trade should be restricted to China, while the India trade should be thrown open to all British subjects. A Church establishment for India was also provided by this Act. The appointment of governors-general, governors, and commanders-in-chief was no longer to be valid without the direct sanction of the Crown. The renewal of the Company's charter in 1834 took place amid continued opposition to their mercantile, and even to their legislative privileges. It continued them in all their possessions except the Island of St. Helena, put an end to the exclusive right of trade with China, and enacted that the Company should with all convenient speed close their commercial business, and make sale of all their property not retained for Government purposes; all their other property was to be held in trust for the Crown, which was to take over their debts and guarantee their dividend out

of the revenues of India. The stock was valued at £6,000,000, which was to bear interest at 10 per cent, and be redeemable after 30th April, 1874, on payment of £12,000,000. The Company was now fairly in liquidation, and on the outbreak of the mutiny of 1857 it was felt indispensable to vest the government of India directly in the Crown, and this was accordingly done in 1858. Henceforth the Company existed only for the purpose of receiving payment of its capital, and of the dividends due upon capital until its repayment.—BIBLIOGRAPHY: J. Bruce, *Annals of the East India Company*; Sir W. W. Hunter, *History of British India*; W. Cunningham, *Growth of English Industry and Trade in Modern Times*; J. Macpherson, *The History and Management of the East India Company*; W. Foster and F. C. Danvers, *Letters received by the East India Company from its Servants in the East* (6 vols.).

East Indies, the name loosely applied to Hindustan, the Indo-Chinese peninsula, and a portion of the Eastern Archipelago, but excluding the Philippine Islands, New Guinea, and Australia.

Eastlake, Sir Charles Lock, English painter, president of the Royal Academy, born at Plymouth 17th Nov., 1793, died at Pisa 23rd Dec., 1865. He studied at the Royal Academy, London, and at Paris. In 1817 he visited Italy and Greece, and painted besides other pictures his *Pilgrims arriving in Sight of Rome*. In 1830 he was elected member of the Royal Academy, and in 1850 became its president, receiving at the same time the honour of knighthood. From 1843 to 1847 he was keeper of the National Gallery, of which he was afterwards director for about ten years. Sir Charles is also known as a writer on art by his *Materials for a History of Oil-painting*. Among his most noteworthy pictures are: *Lord Byron's Dream* (in the Tate Gallery), *Greek Fugitives*, *Escape of the Carrara Family*, *Christ blessing Little Children*, *Christ lamenting over Jerusalem*.

East London, a seaport on the east coast of Cape Province, at the mouth of the Buffalo River, now an important outlet for the produce of this region, connected by railway with Cape Town. Pop. 20,867.

East Main, a considerable river of Canada, having a westward course to James Bay, the southern extension of Hudson Bay, and forming the boundary between Quebec province and Ungava territory.

Easton, a city of Pennsylvania, United States, at the junction of the Delaware and Lehigh Rivers, 75 miles s.w. of New York. It contains iron-foundries, tanneries, and breweries. Pop. 32,000.

East River, a strait in New York State, separating New York from Brooklyn and connecting Long Island Sound with New York Bay, about 20 miles long. The Brooklyn Bridge, and the Williamsburg, Queensboro, and Manhattan Bridges now cross the river.

East St. Louis, a town of the United States, in Illinois, on the east bank of the Mississippi, opposite St. Louis, and connected with it by a great steel arch bridge and another bridge, carrying numerous lines of railway. Pop. 58,547.

Eastwood, a town of England, in Notts, on the Derbyshire border, with coal-mines, 8½ miles w.n.w. of Nottingham. Pop. 4692.

Eau (ō), a French word signifying *water*, and used in English with some other words for several spirituous waters, particularly perfumes, as *eau de Cologne*, and *eau de Luce*.—*Eau de Cologne* is a fragrant water, made originally and in most perfection in Cologne by a manufacturer named Farina, by whose successors the only genuine water is said still to be manufactured. It consists of spirits of wine flavoured by different essential oils blended so as to yield a fine fragrant scent. It was invented in Northern Italy by one of the Farina family, who afterwards settled in Cologne.—*Eau Créole*, a highly esteemed liqueur made in Martinique by distilling the flowers of the mammee apple with spirit of wine.—*Eau de Luce* ('water of Luce'), so called from the name of its inventor, is made by dissolving white soap in spirit of wine, and adding oil of amber and sal ammoniac. It is a milky fluid, antispasmodic and stimulant.—*Eau de Vie* ('water of life'), a term used by the French for the coarser kinds of brandy, *cognac* being the name of the best.

Eau Claire (ō klār), a city of Wisconsin, United States, at the junction of the Eau Claire and Chippewa Rivers, a great lumbering centre. In 1910 Eau Claire adopted the commission form of government, being the first city in the state to do so. Pop. 18,310.

Eaux-bonnes (ō-bon), a watering-place, France, department of Basses Pyrénées, about 25 miles south of Pau. The hot sulphur springs are said to have great efficacy in affections of the chest. Pop. 622.—Near it is *Eaux Chaudes*, also with warm springs.

Eʿbal, a mountain of Western Palestine about half-way between Jerusalem and Nazareth, on the north side of a narrow valley, on the south side of which and directly opposite stands Mt. Gerizim with Nablous almost between. Here the Israelites set up an altar on their entrance into the Holy Land and had the law solemnly read to them by Joshua (*Jos.* viii 30-35). At the east end of the valley are Jacob's Well and Joseph's Tomb.

Ebbsfleet, a hamlet in the Isle of Thanet, county Kent, memorable as the place where the first Anglo-Saxon invaders landed.

Ebbw-vale, a town of England, in Monmouthshire, with ironworks, steelworks, and collieries. Pop. 30,540.

Ebenaʿceæ, a nat. ord. of gamopetalous Dicotyledons, consisting of trees and shrubs, of which the wood is very hard, and frequently of very dark colour in the centre, as ebony. The leaves are alternate, and generally coriaceous and shining; calyx gamosepalous and persistent, with three or six equal divisions; corolla with imbricated divisions. The fruit is a globular berry containing a small number of compressed seeds. The principal genus is *Diospýros*, which yields ebony and iron-wood.

Ebers (āʿbèrz), Georg Moritz, German Egyptologist and novelist, born 1st March, 1837, at Berlin, died in 1898. He studied at Göttingen, and afterwards at Berlin, where he devoted himself to Egyptology. In 1870 he was made professor at the University of Leipzig, but he had to resign in 1889. In 1869 and 1870 he travelled extensively in Egypt and Nubia. Two years later he again visited Egypt, where he discovered the medical papyrus, known as the *Papyrus Ebers*. His most important works have been translated into English, such as *Egypt, Descriptive, Historical, and Picturesque*; and the novels *An Egyptian Princess*, *Uarda*, *Homo Sum*, *The Emperor*, *The Sisters*, all dealing with old Egyptian life; *The Burgomaster's Wife*, and *Only a Word*.

Eberswalde (ā'berz-va ' l-de), a town in Prussia, in the province of Brandenburg, on the Finow Canal, 27 miles north-east of Berlin. It has a school of forestry, piscicultural establishment, botanic gardens, well-frequented mineral springs, and industrial works of various kinds. Pop. 26,064.

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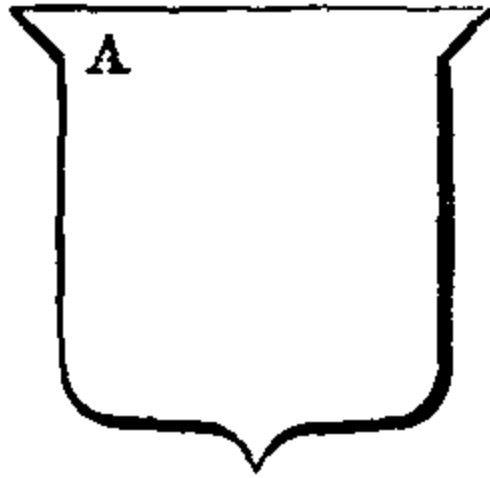
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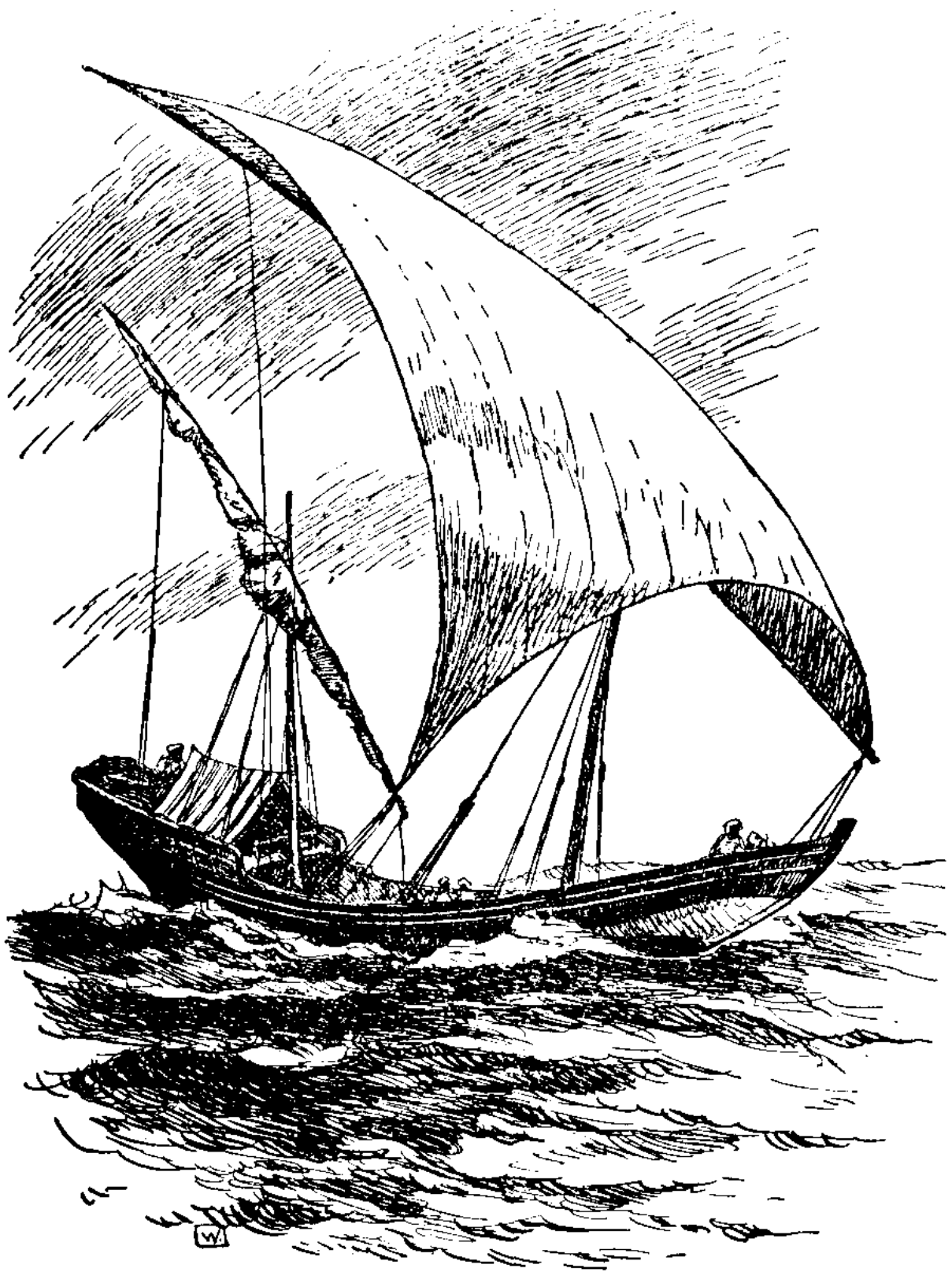
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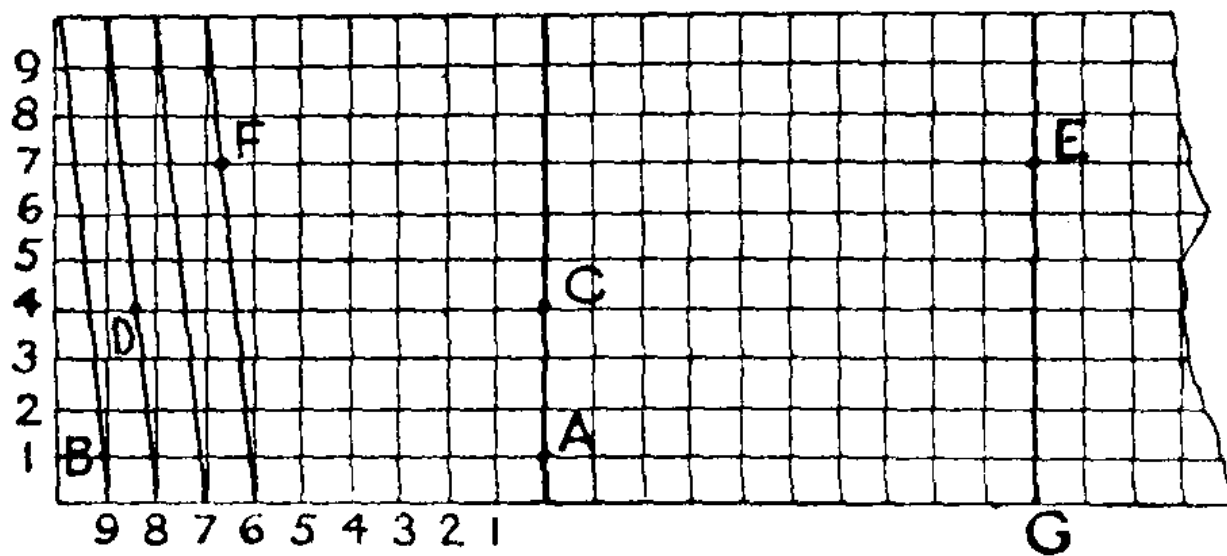
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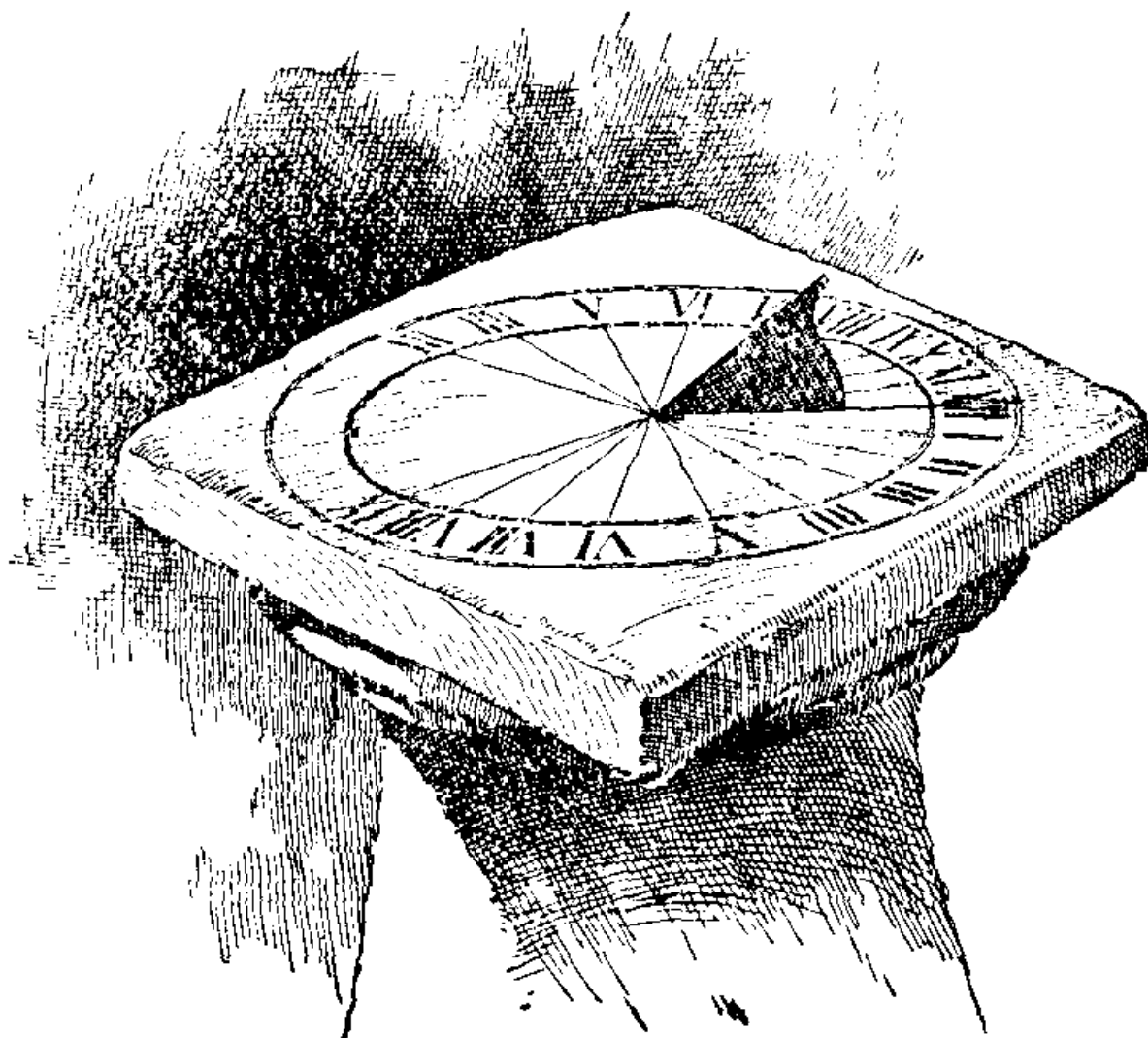
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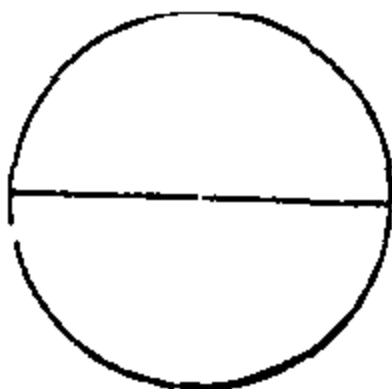
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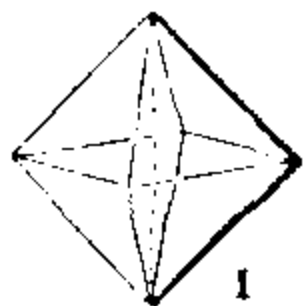
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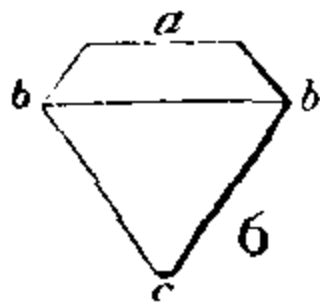
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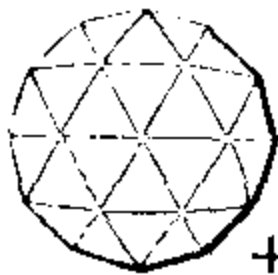
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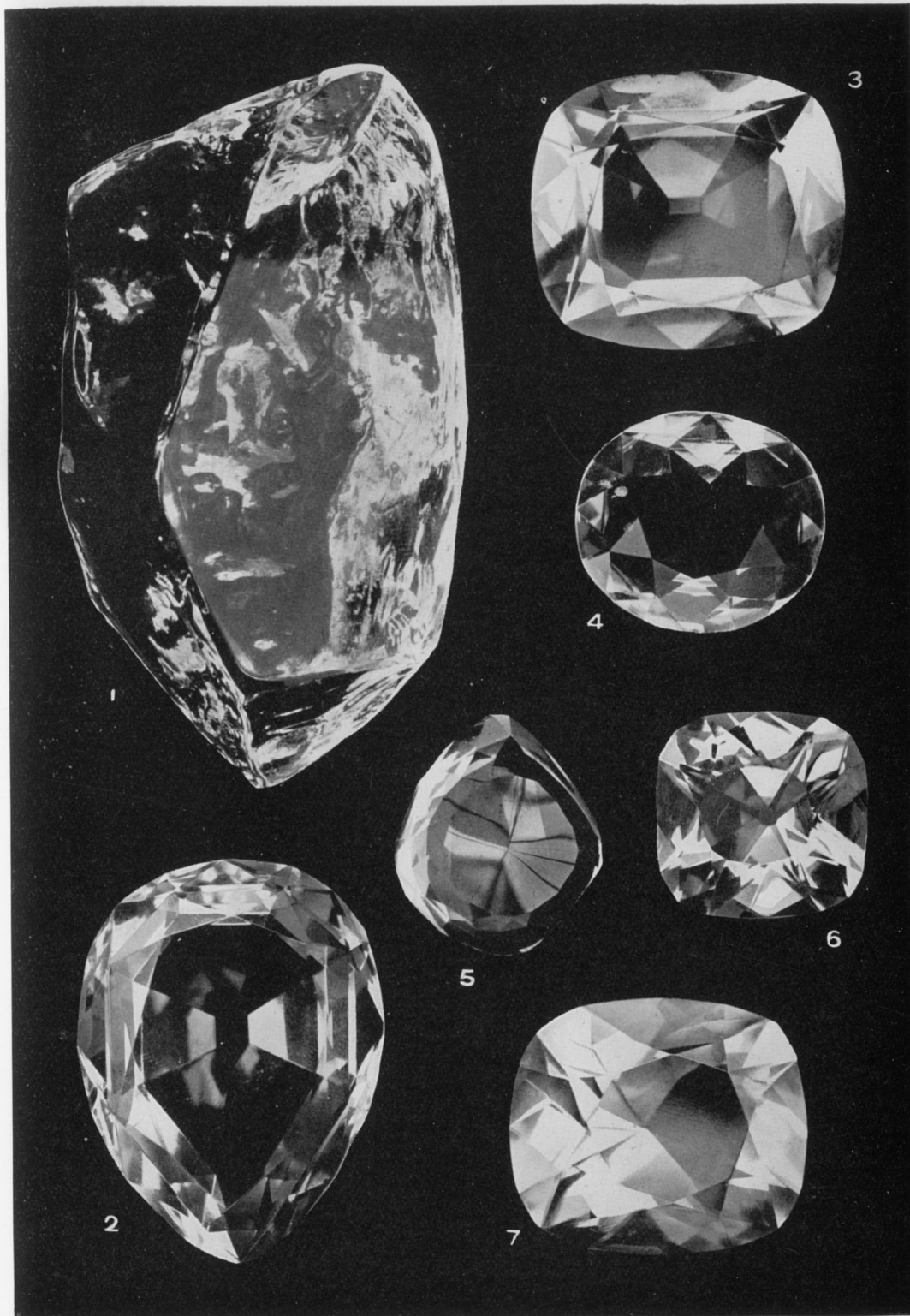
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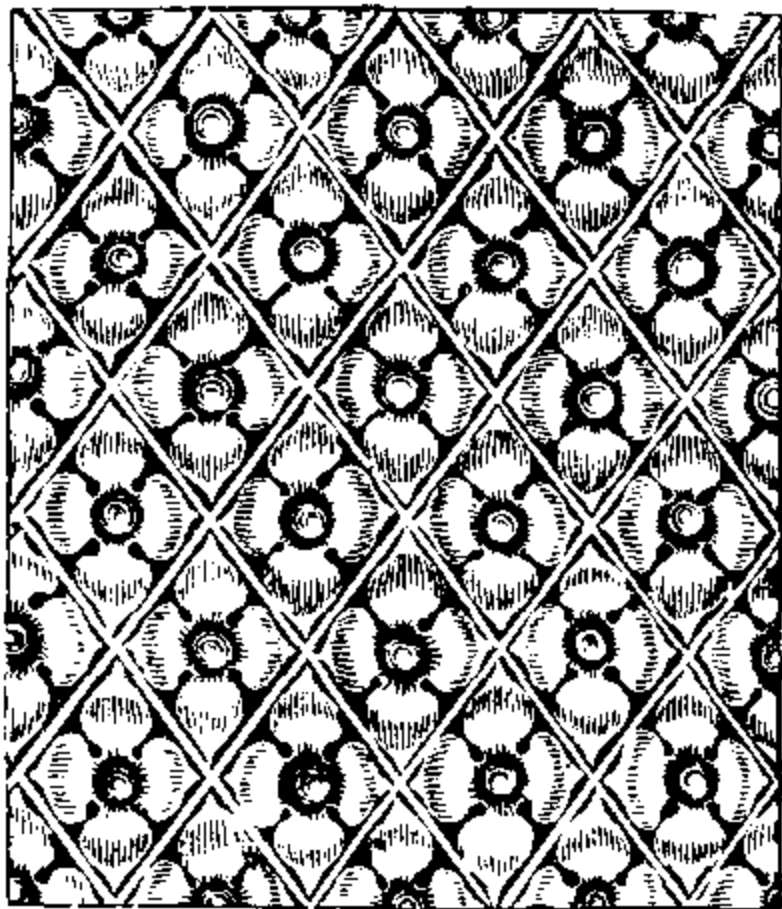
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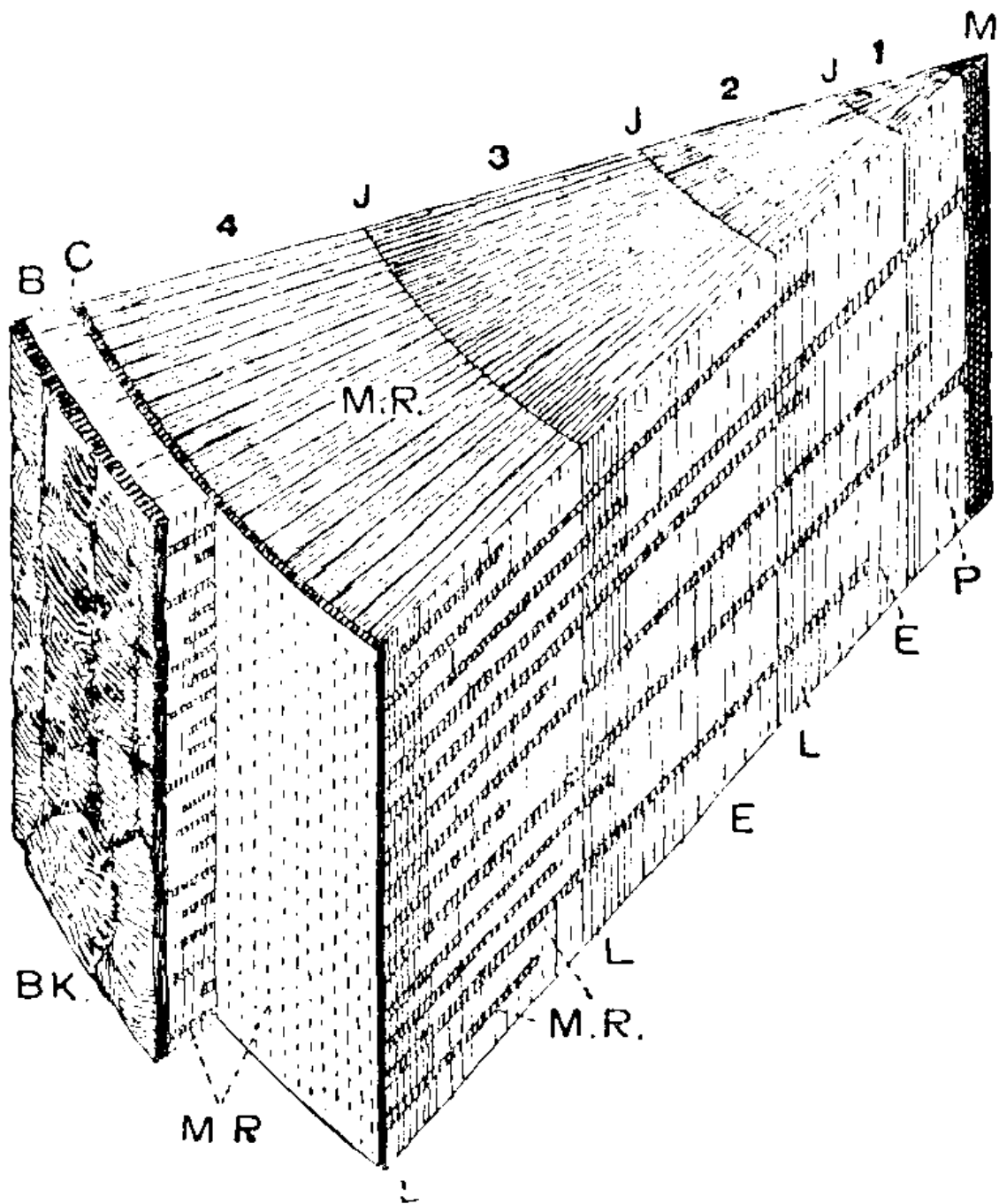
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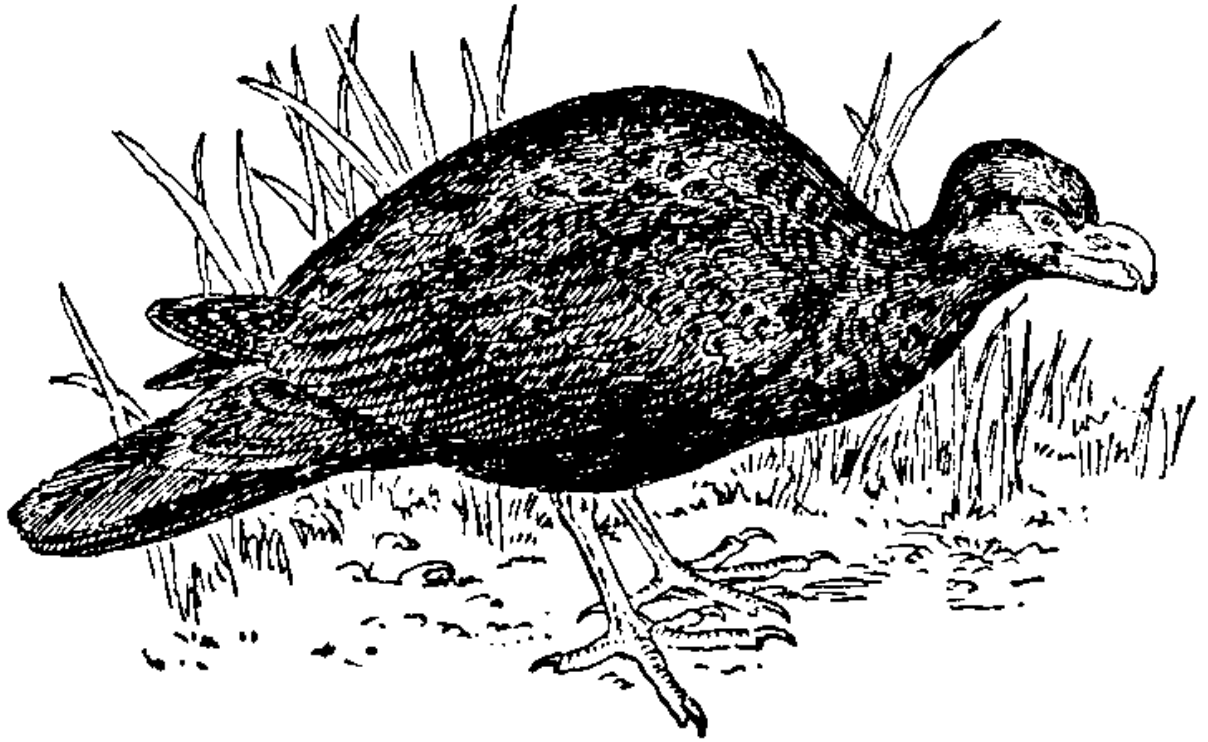
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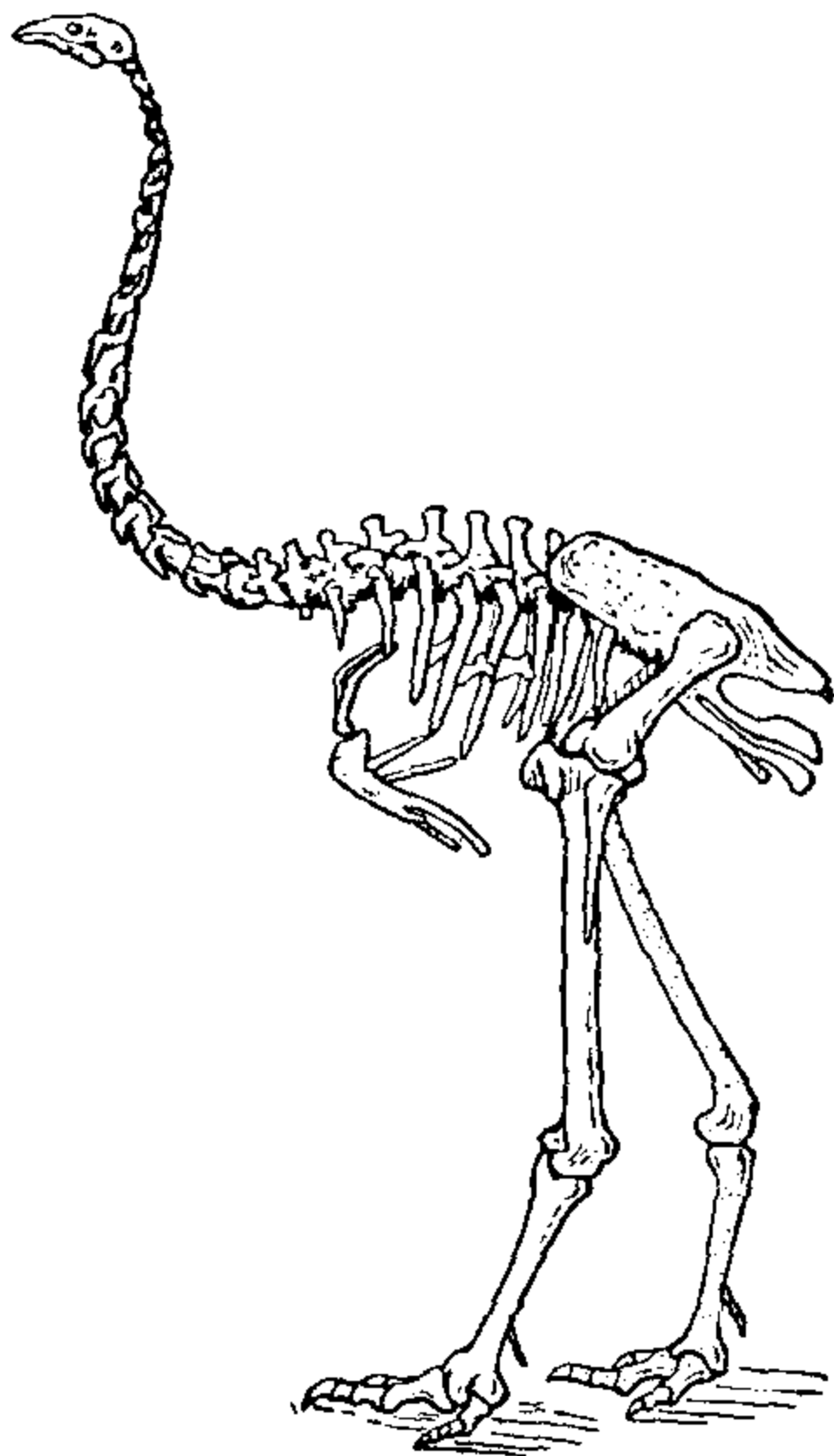
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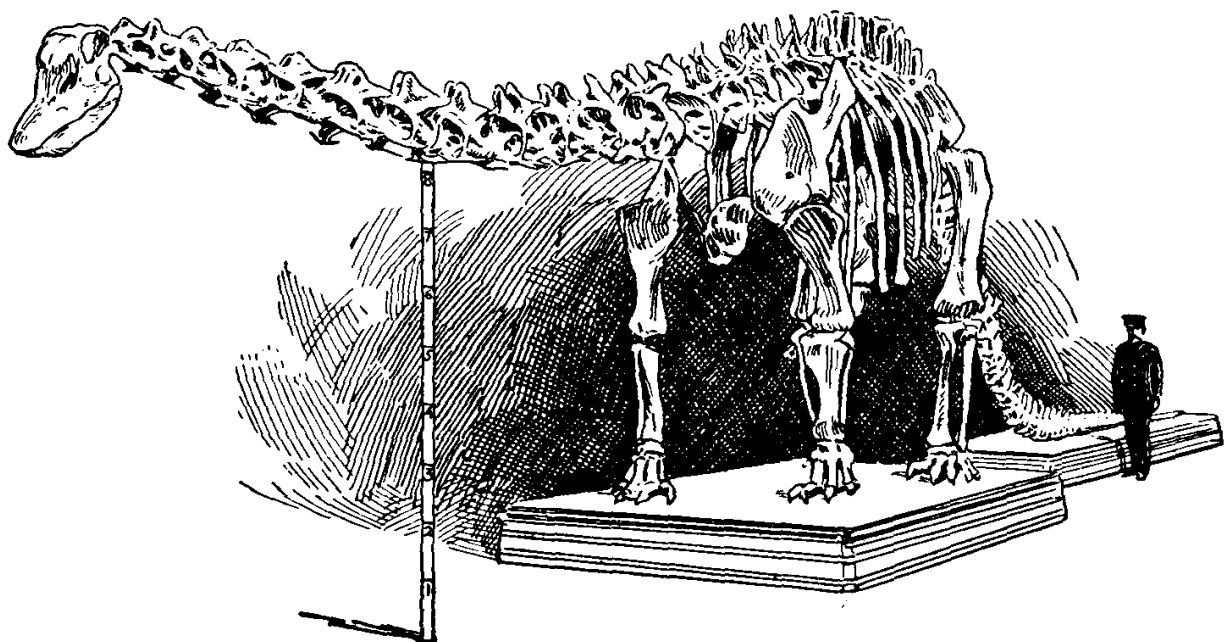
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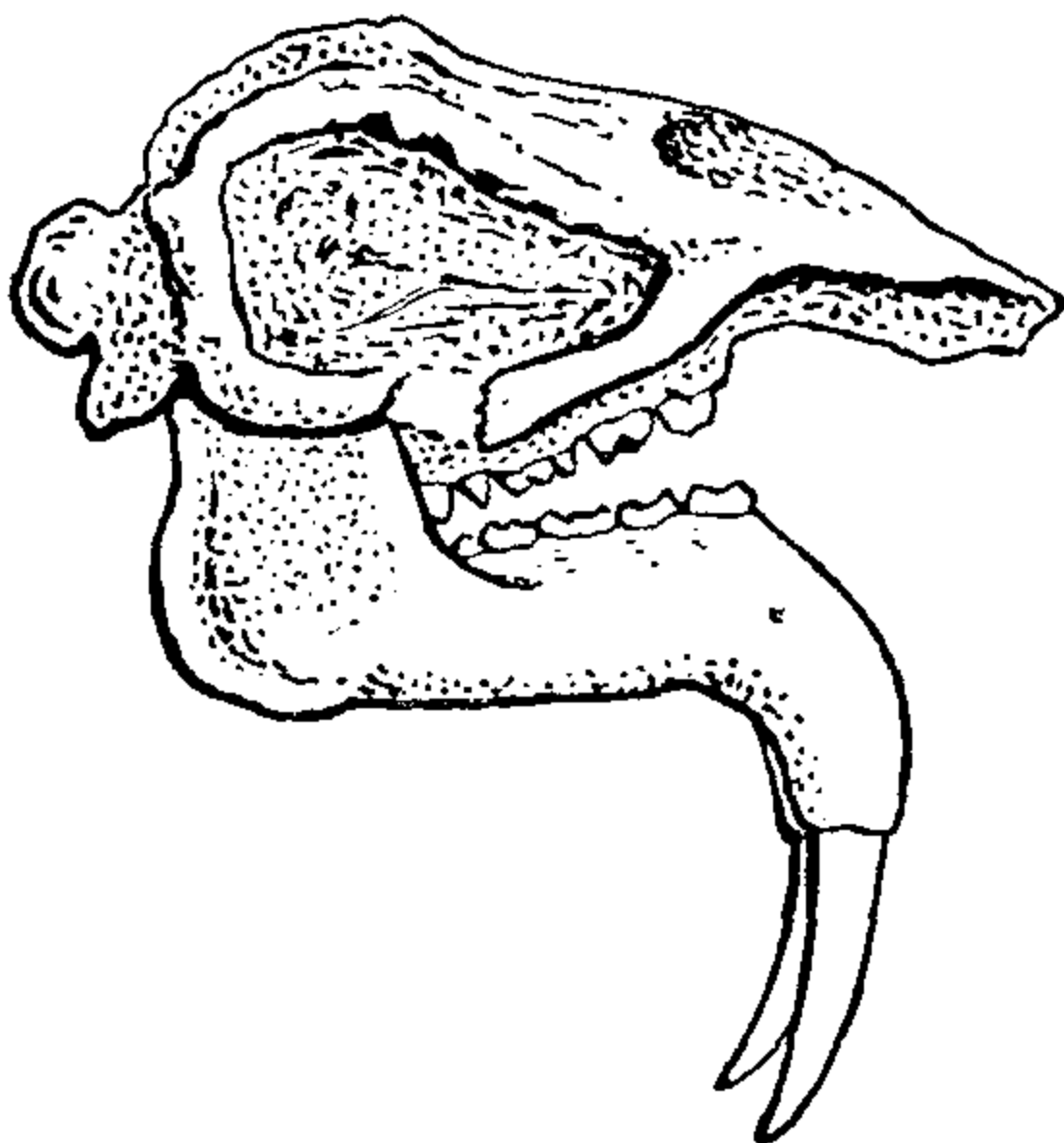
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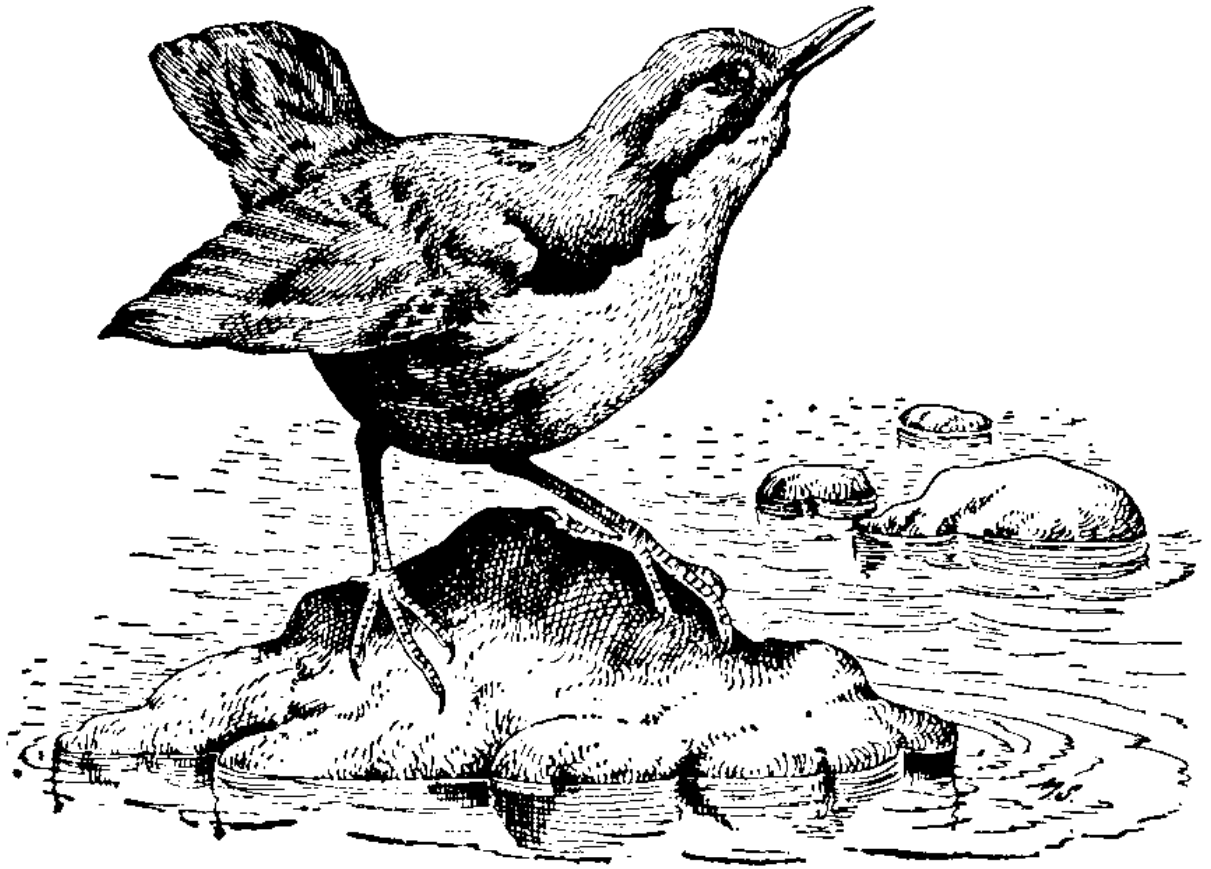
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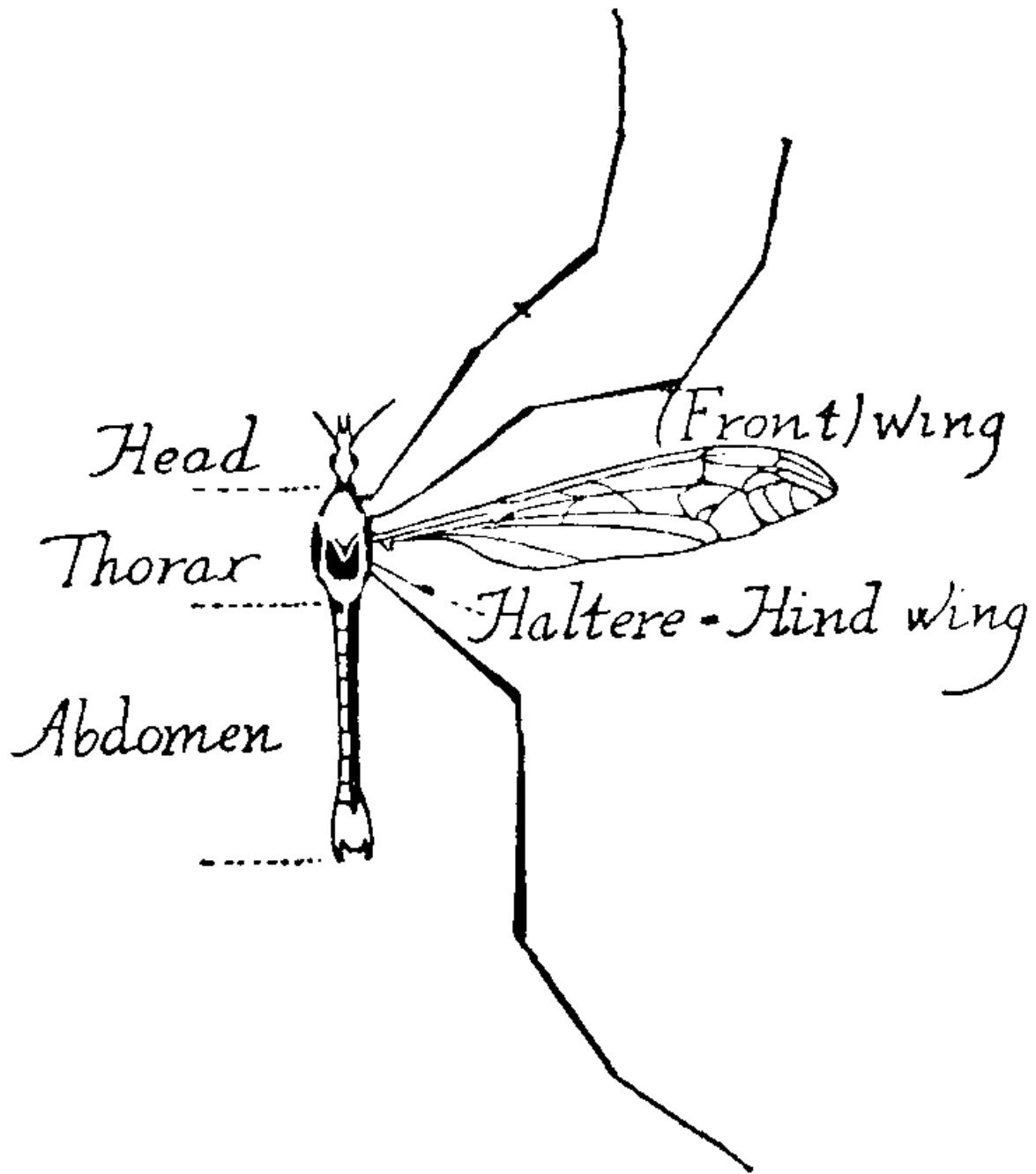
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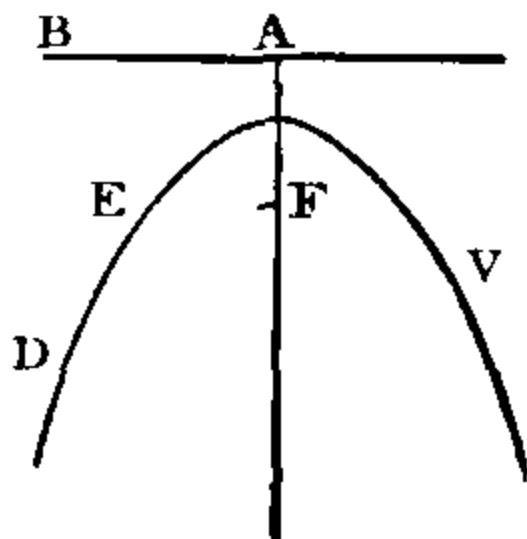
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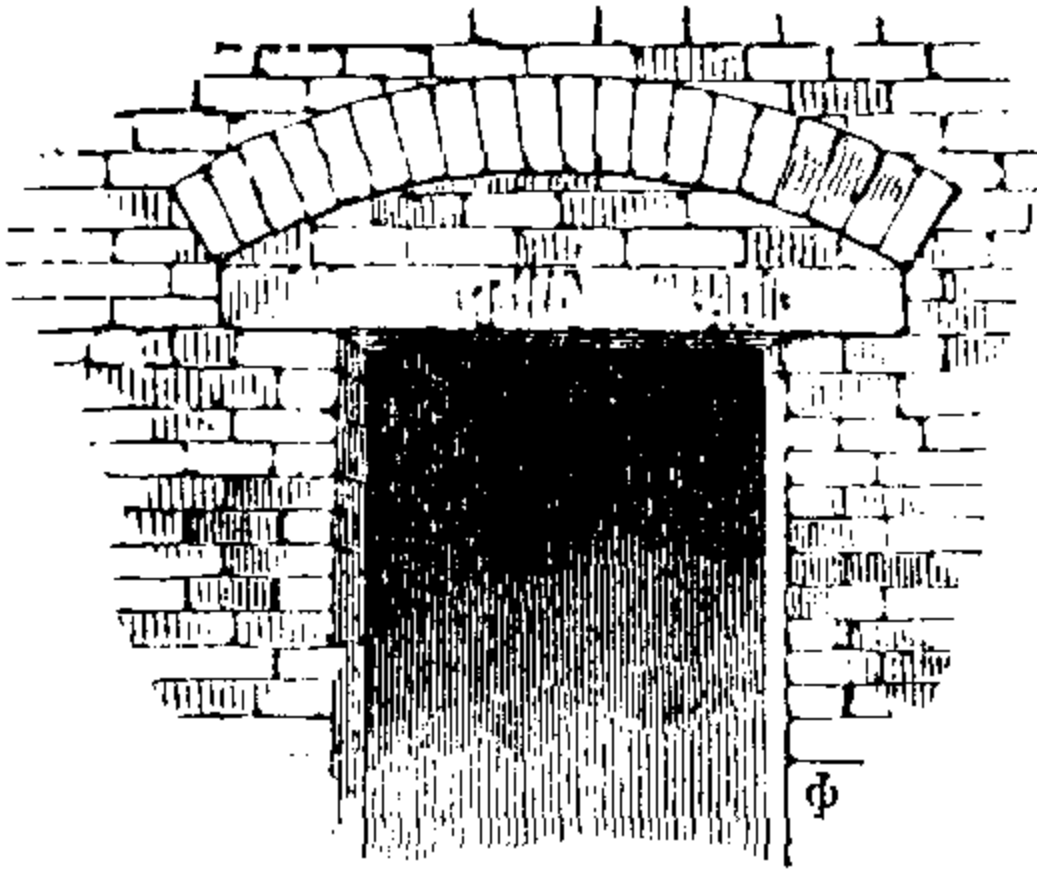
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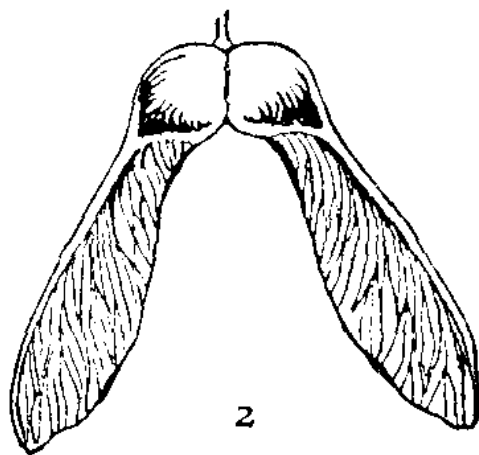
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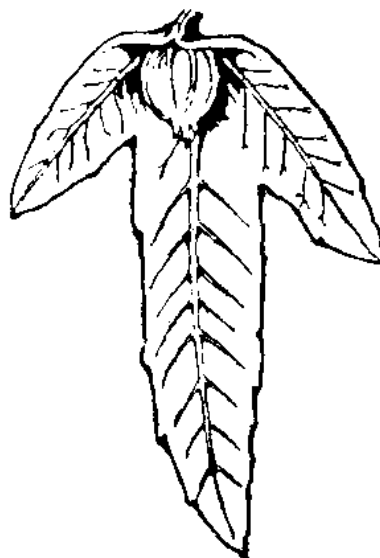
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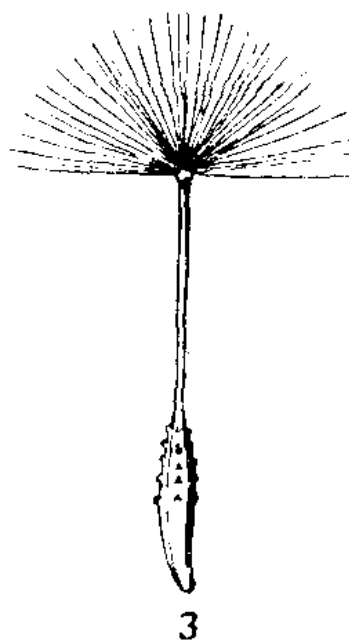


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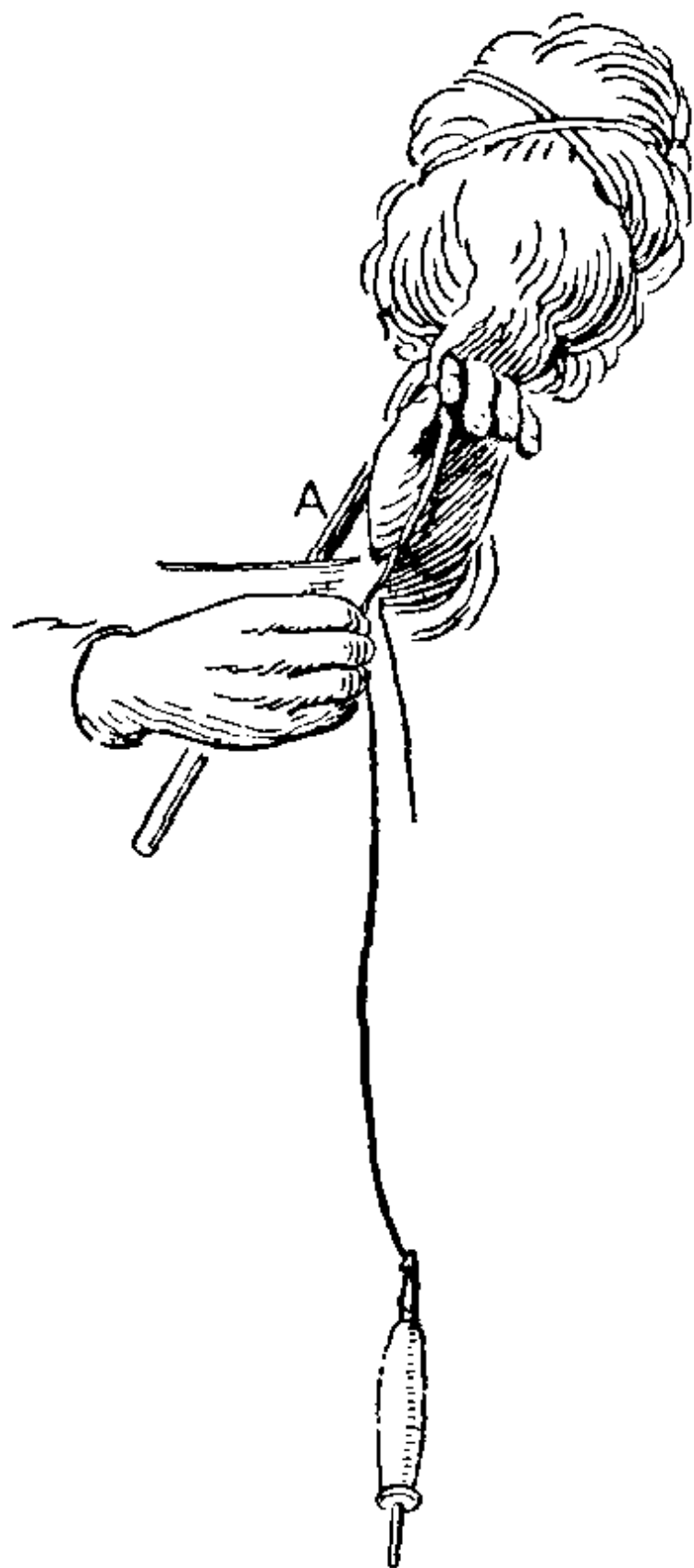
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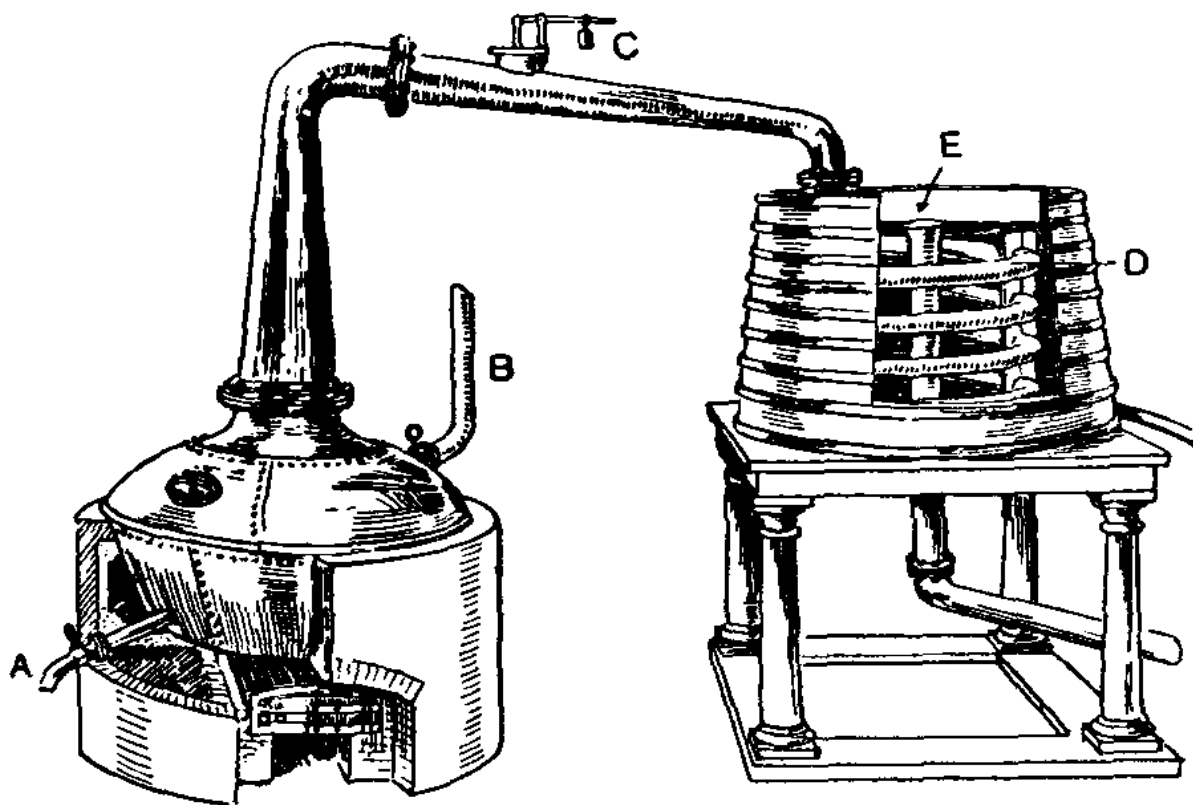
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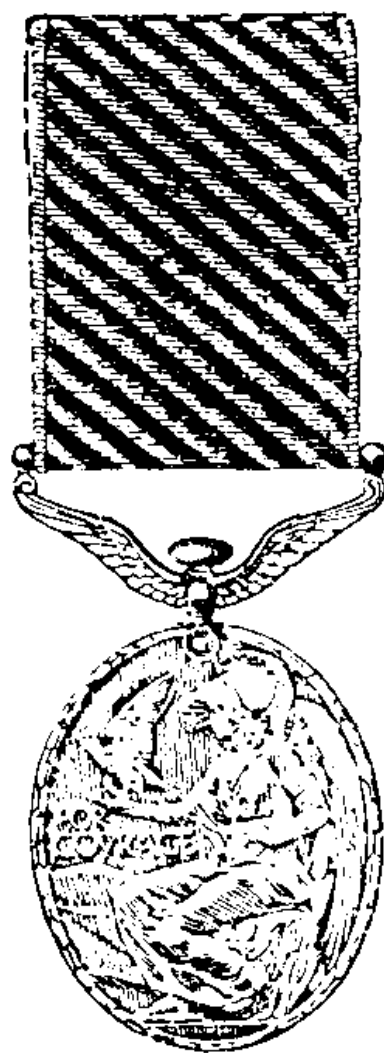
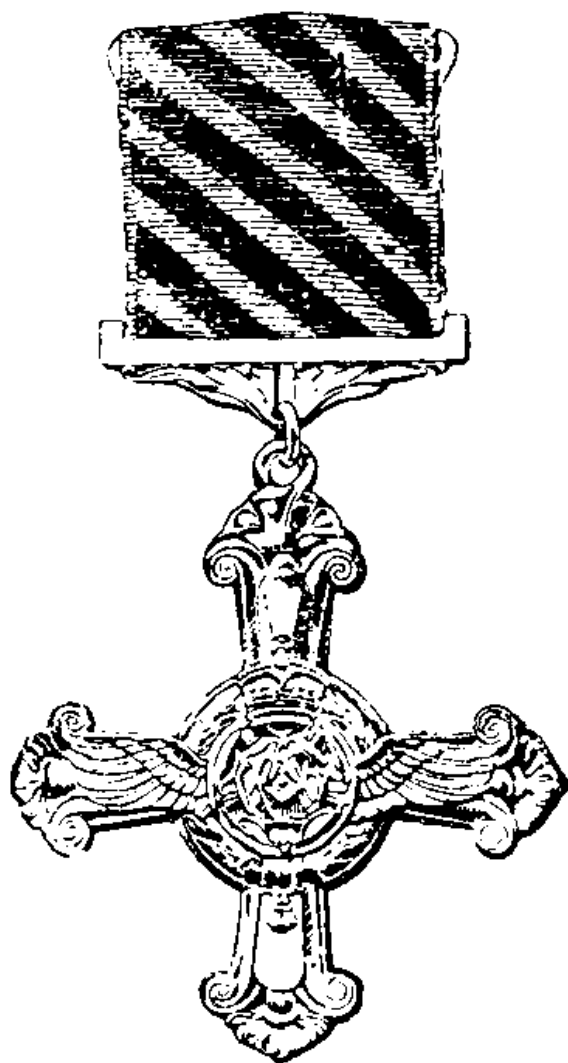
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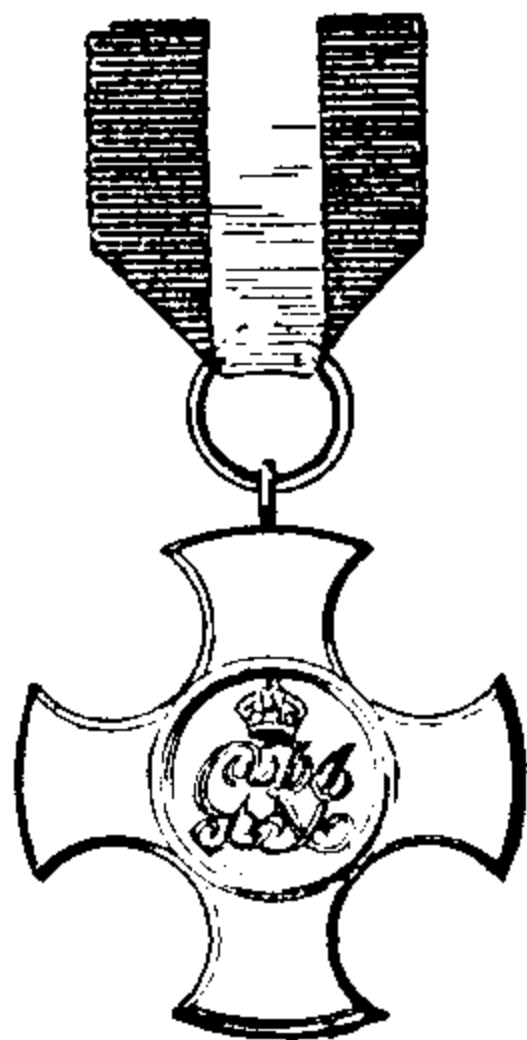
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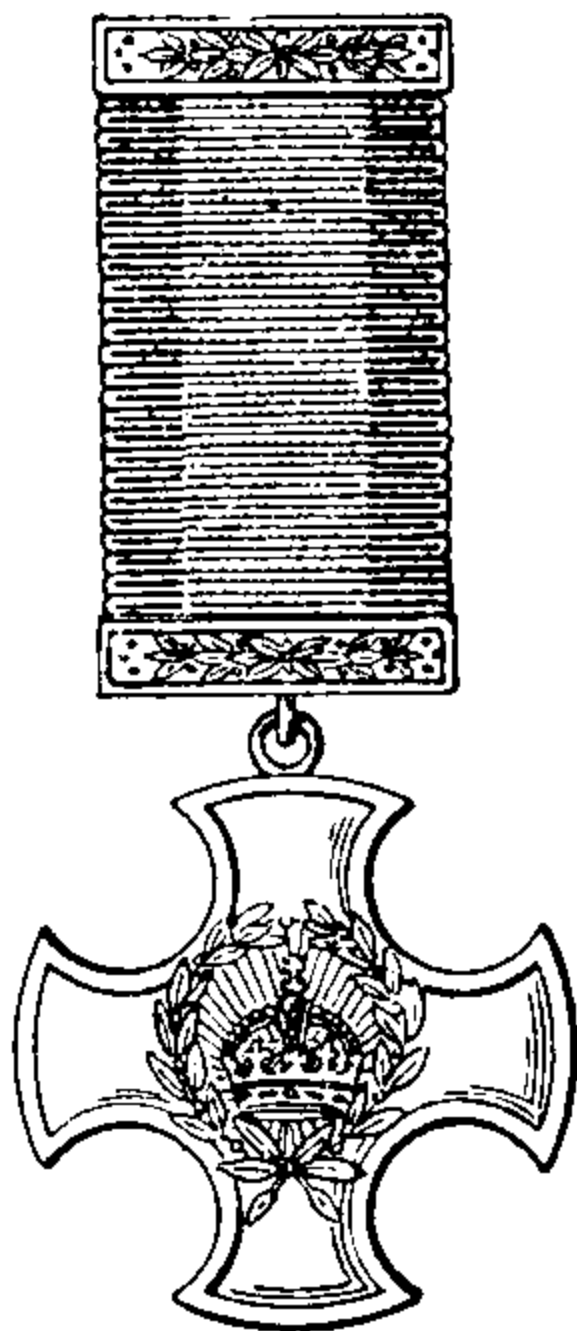
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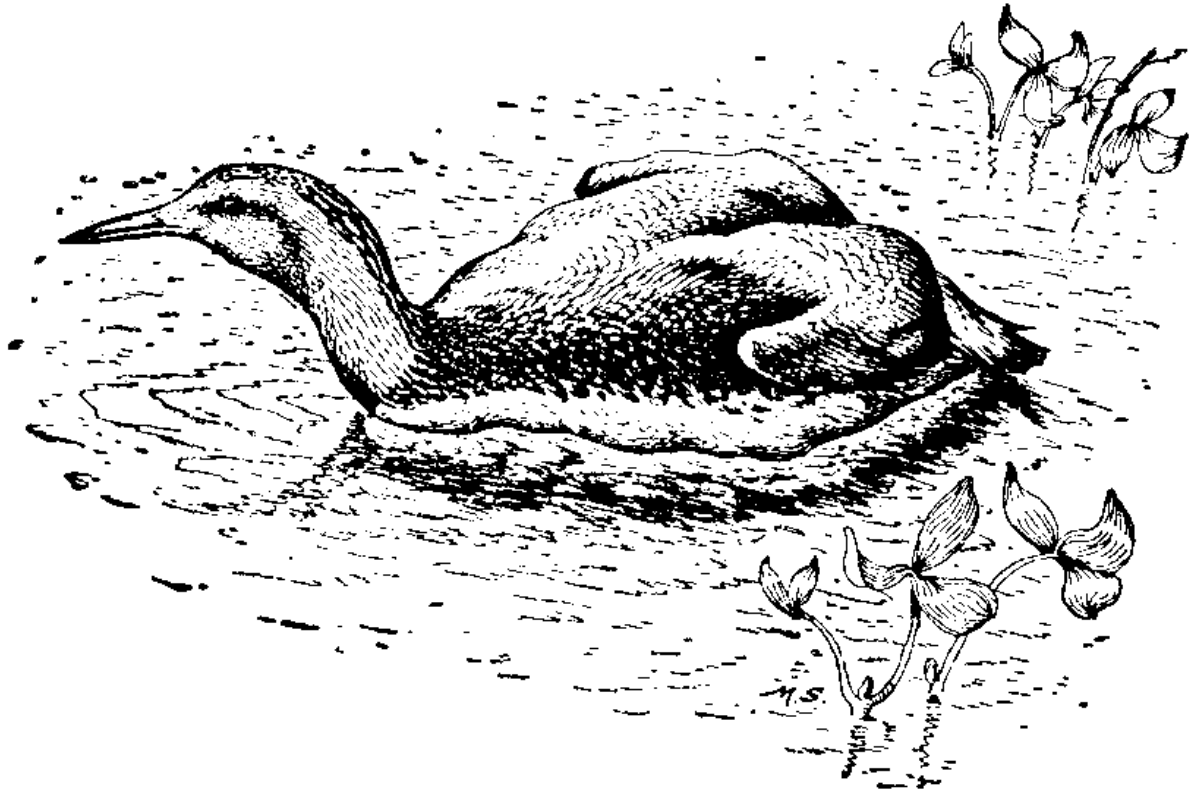
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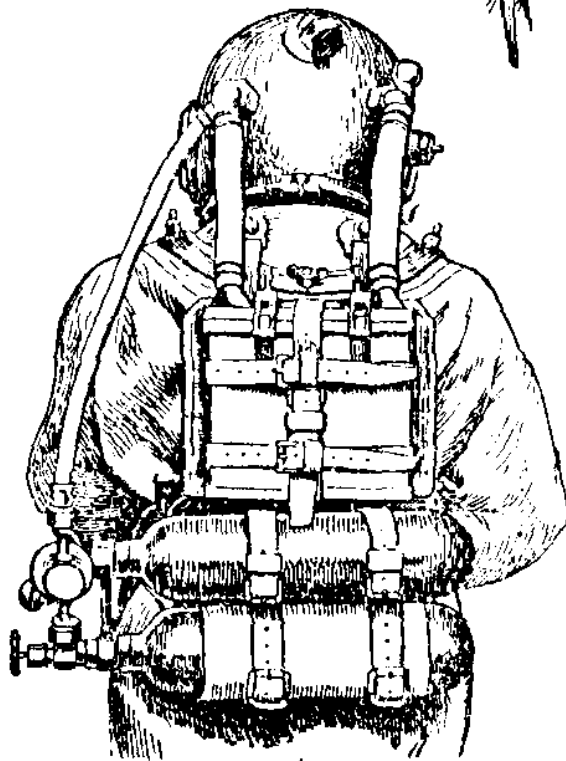
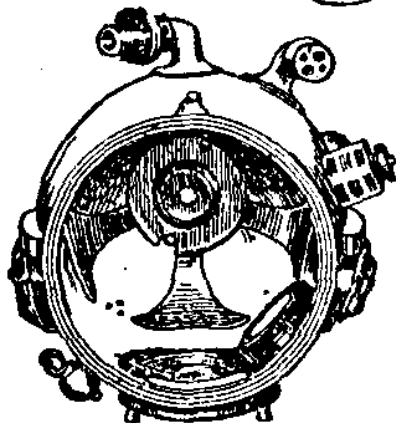
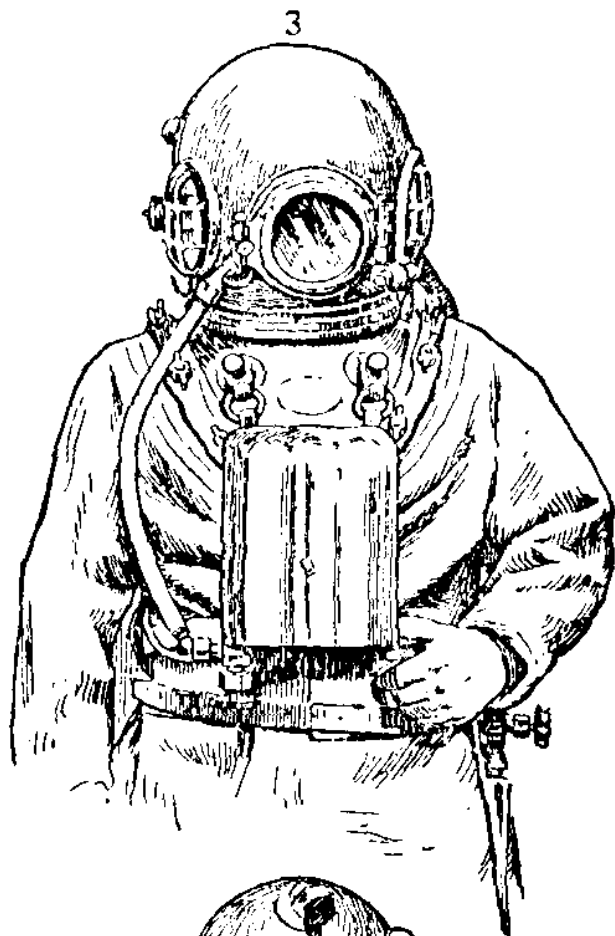
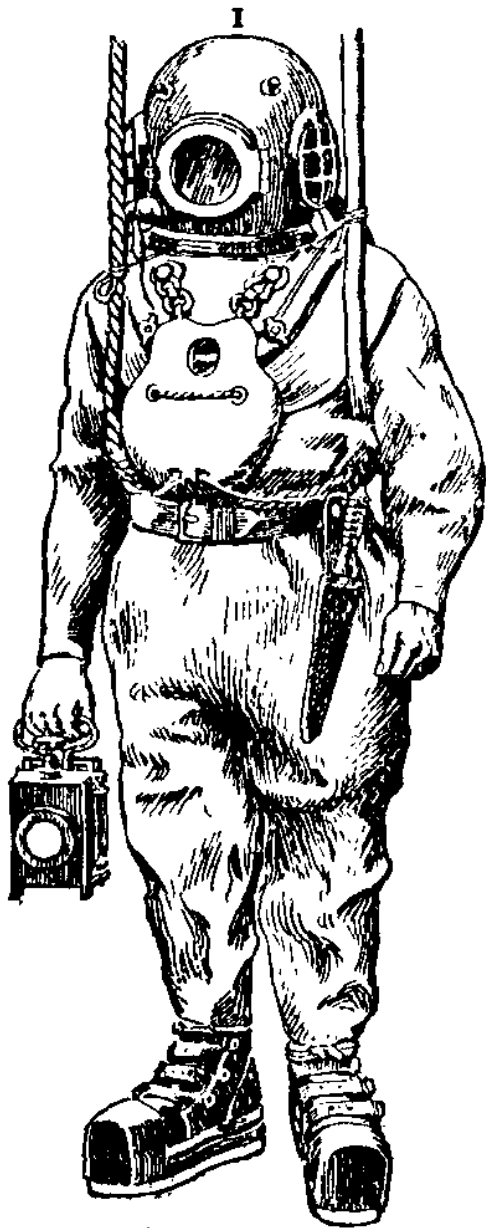
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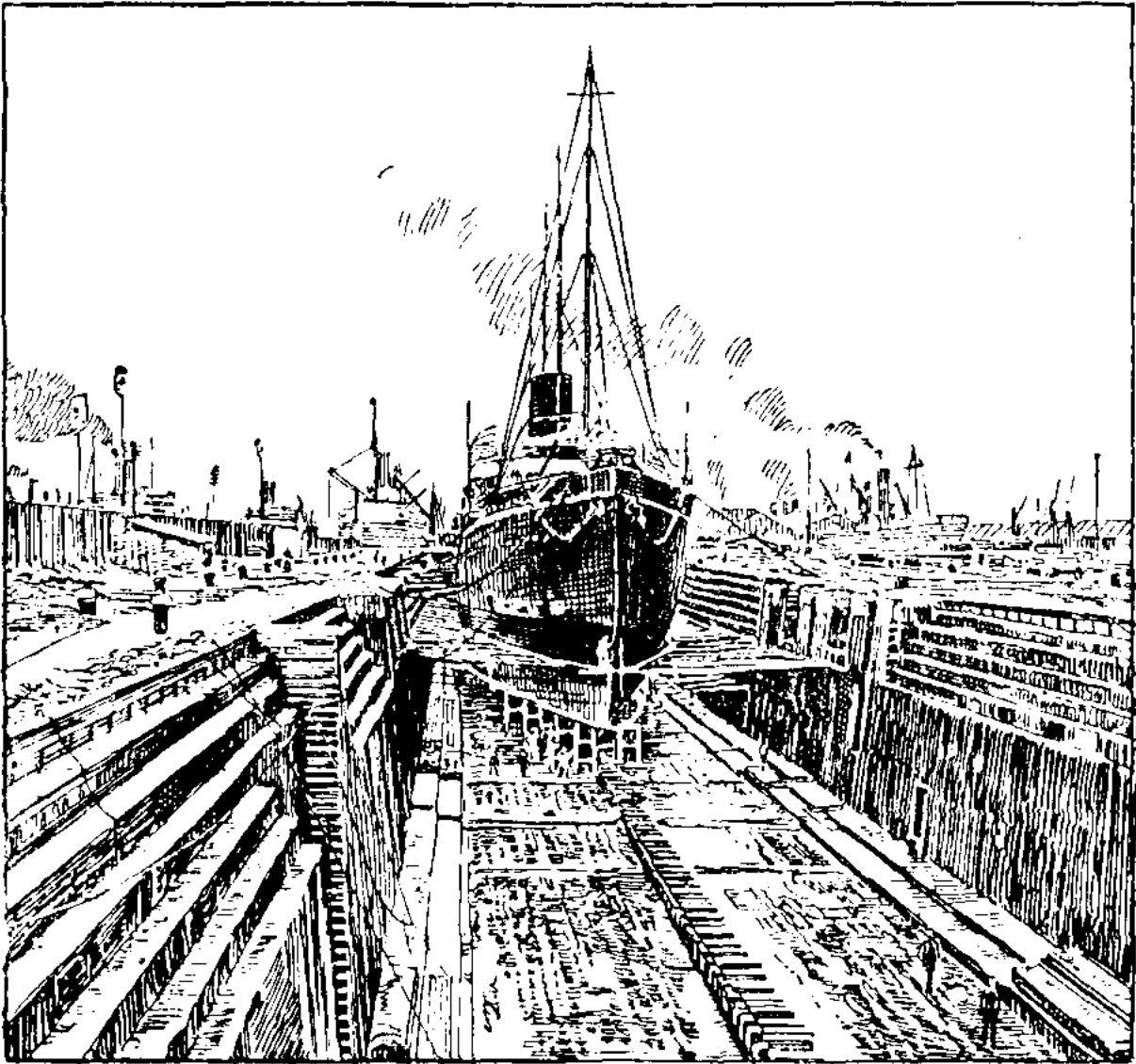
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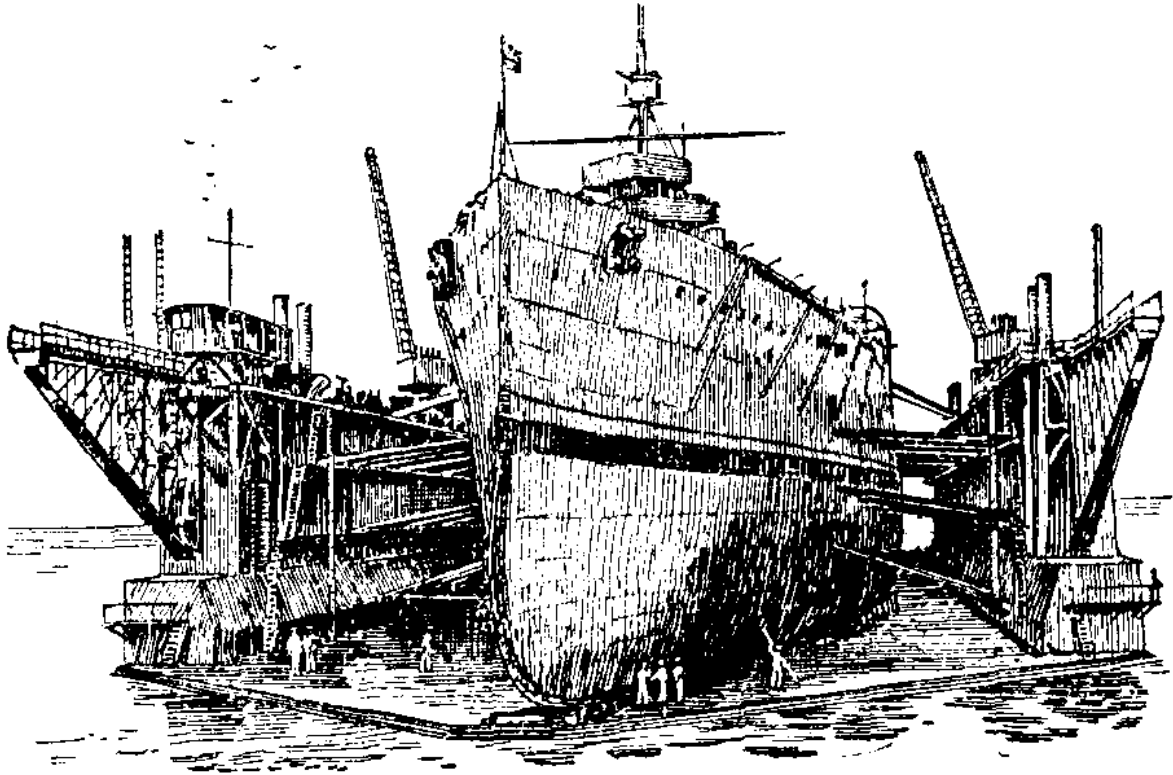
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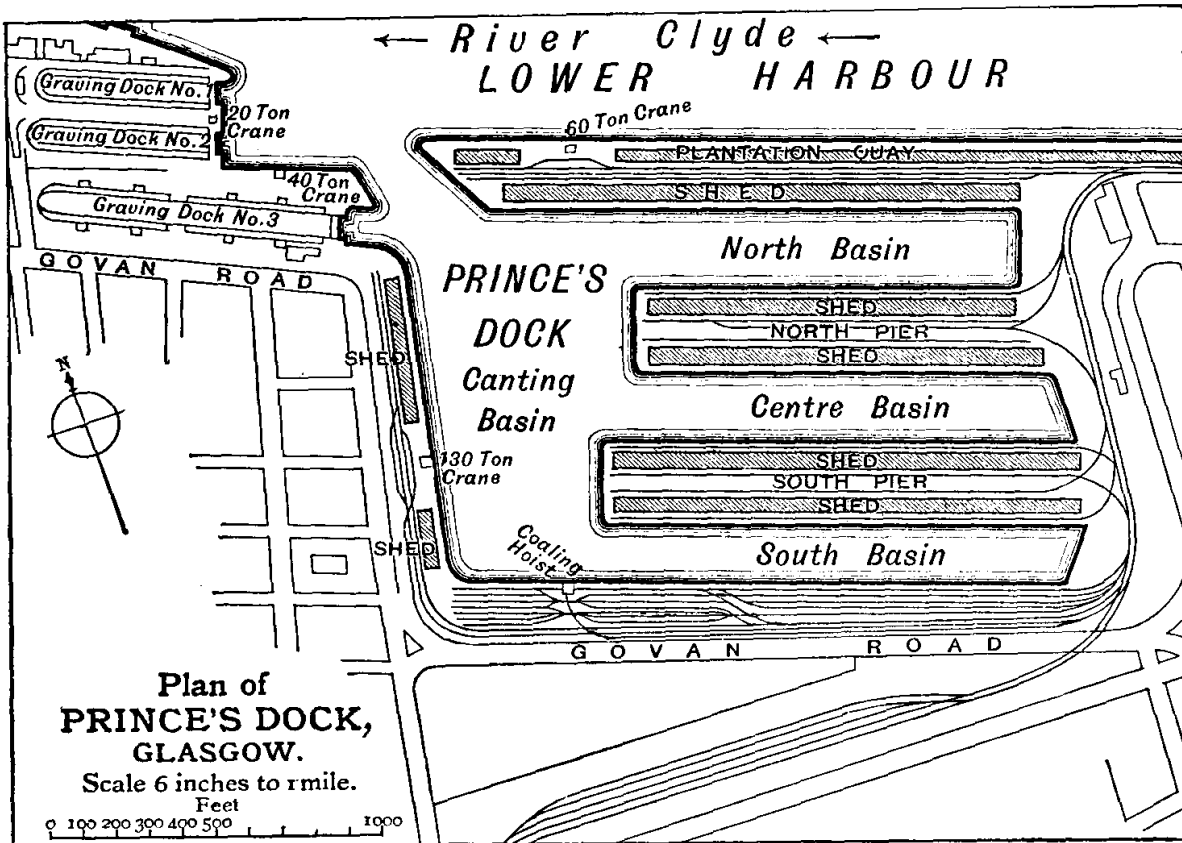
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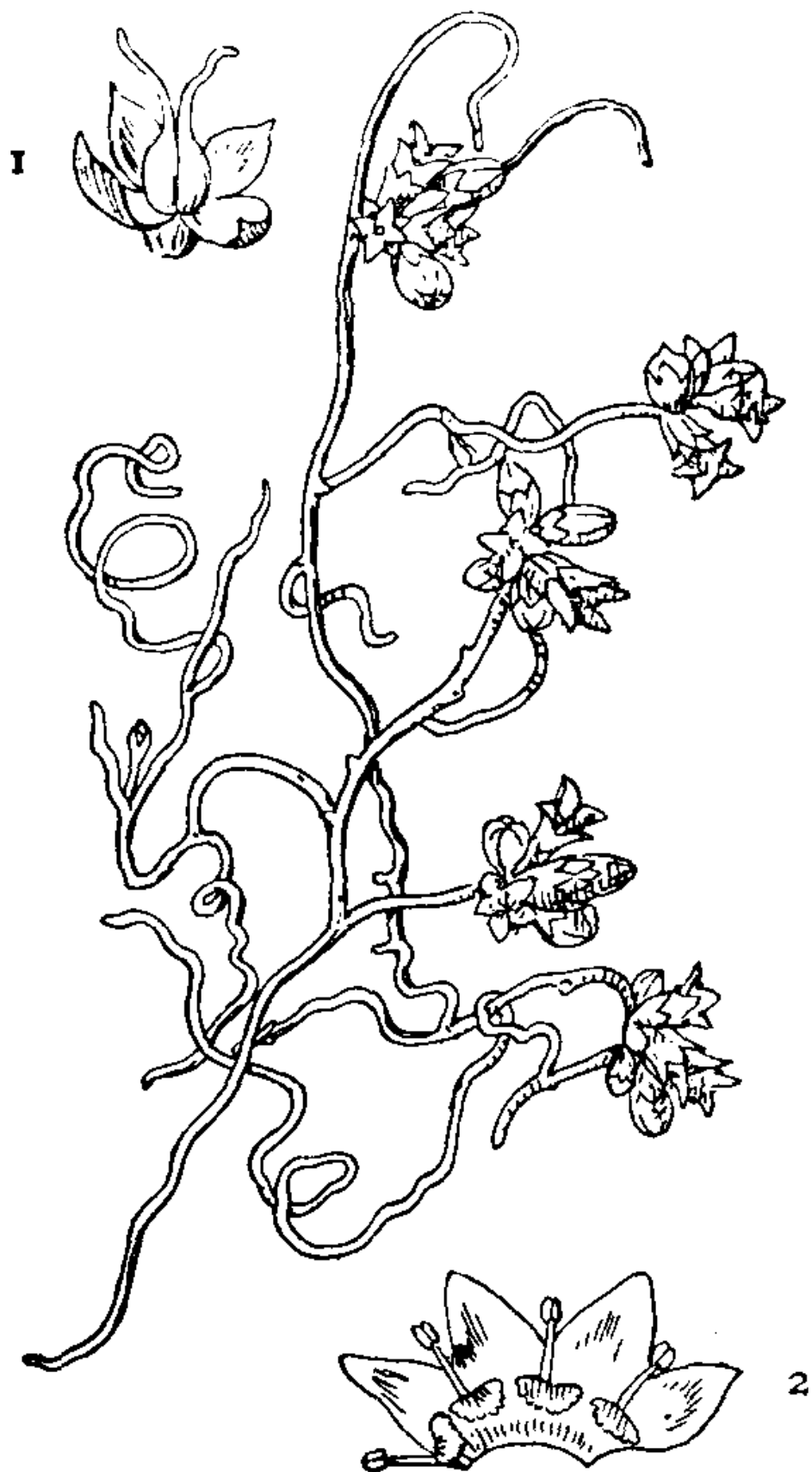
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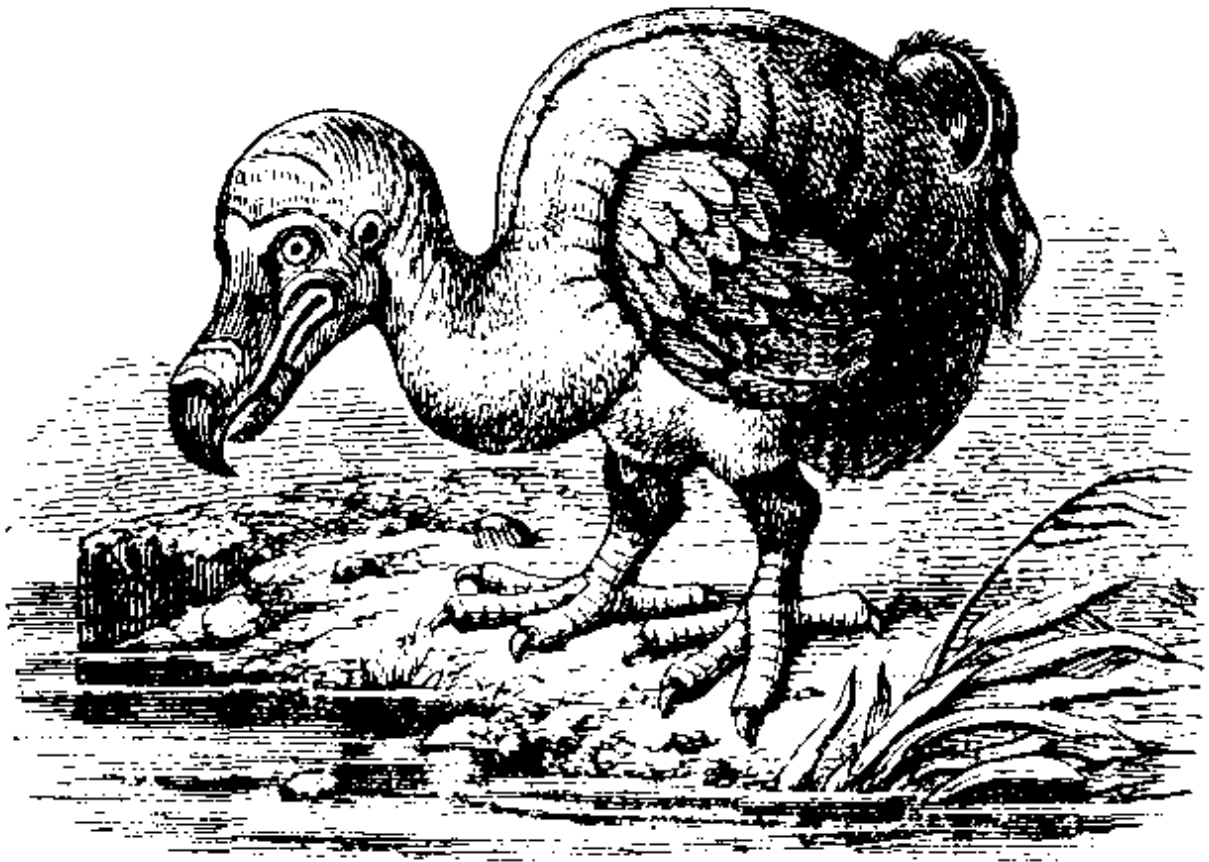
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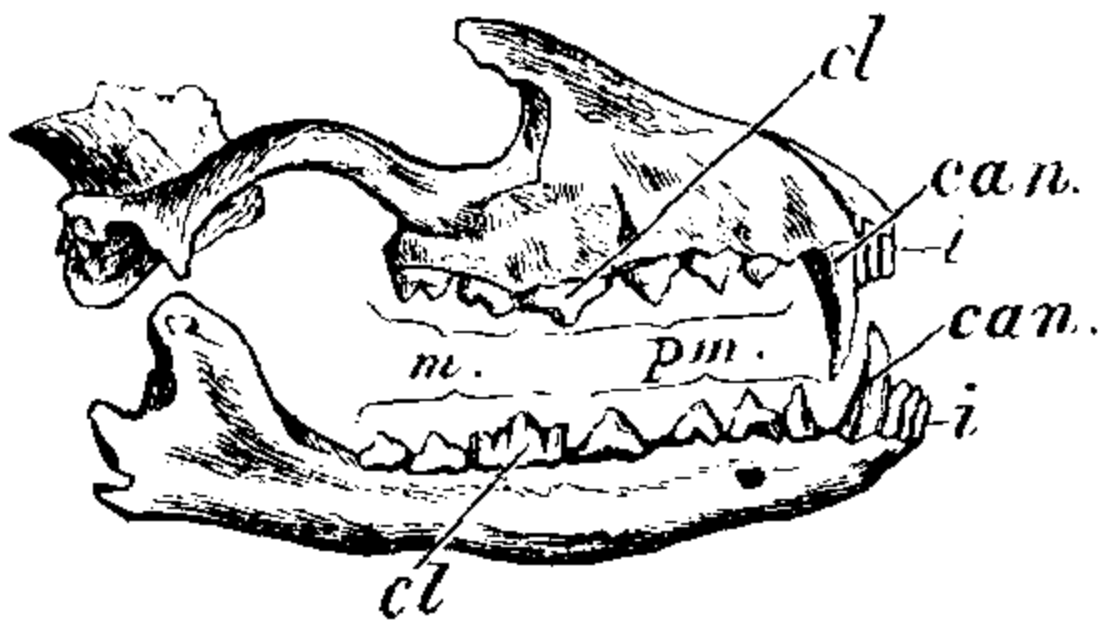
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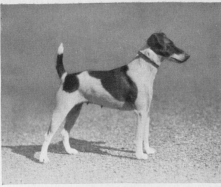


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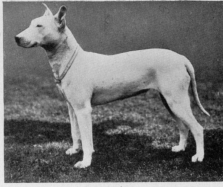


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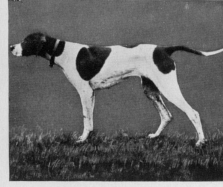
Fox Terrier



Bull Terrier



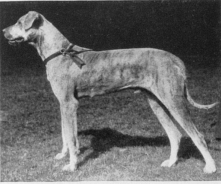
Old English Sheep-dog



Pointer



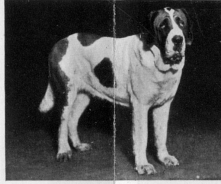
Yorkshire Terrier



Great Dane



Bloodhound



Smooth St. Bernard



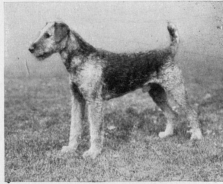
Collie



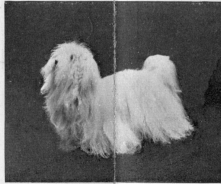
Borzoi



Retriever



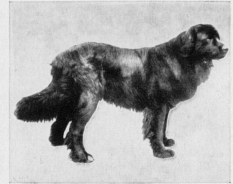
Airedale Terrier



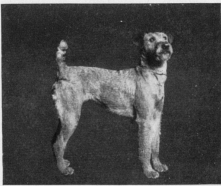
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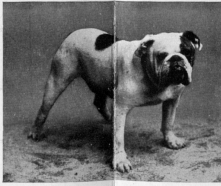
Newfoundland



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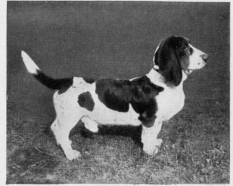
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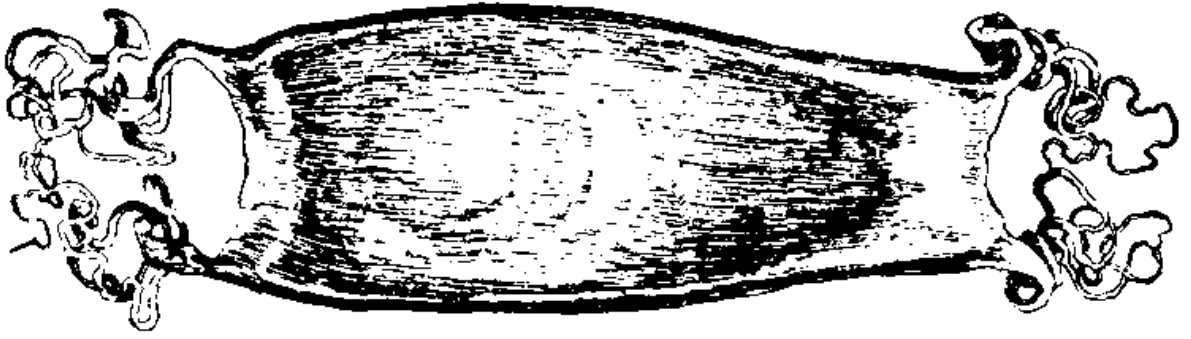
Scotch Terrier



Basset Hound

Photos. Sport & General

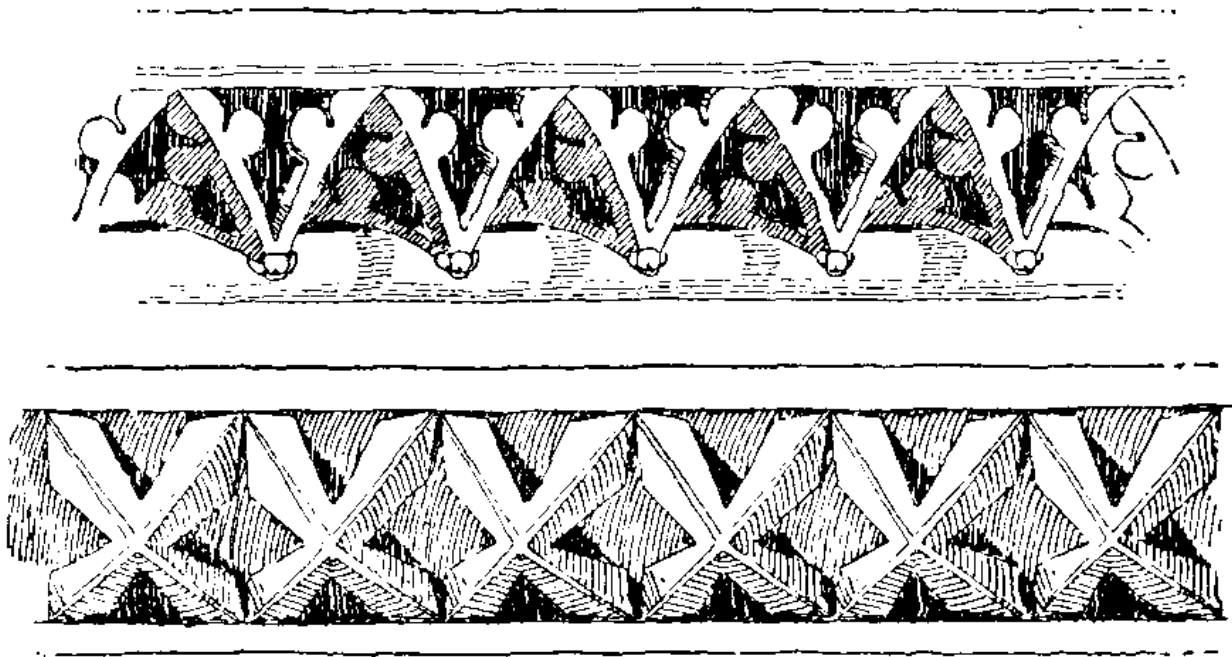
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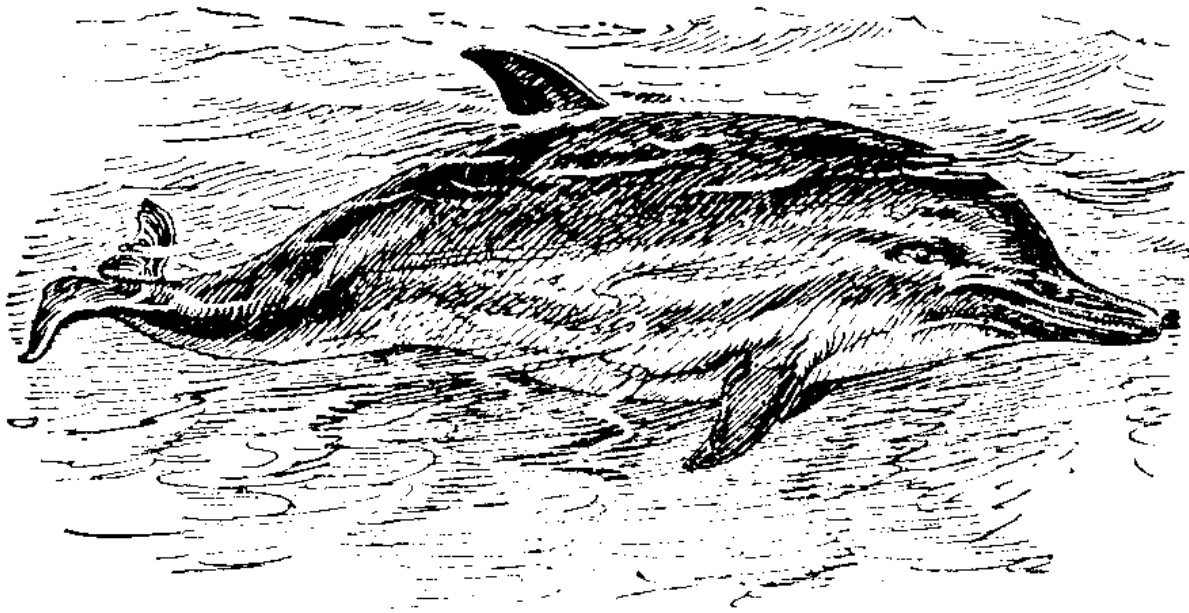
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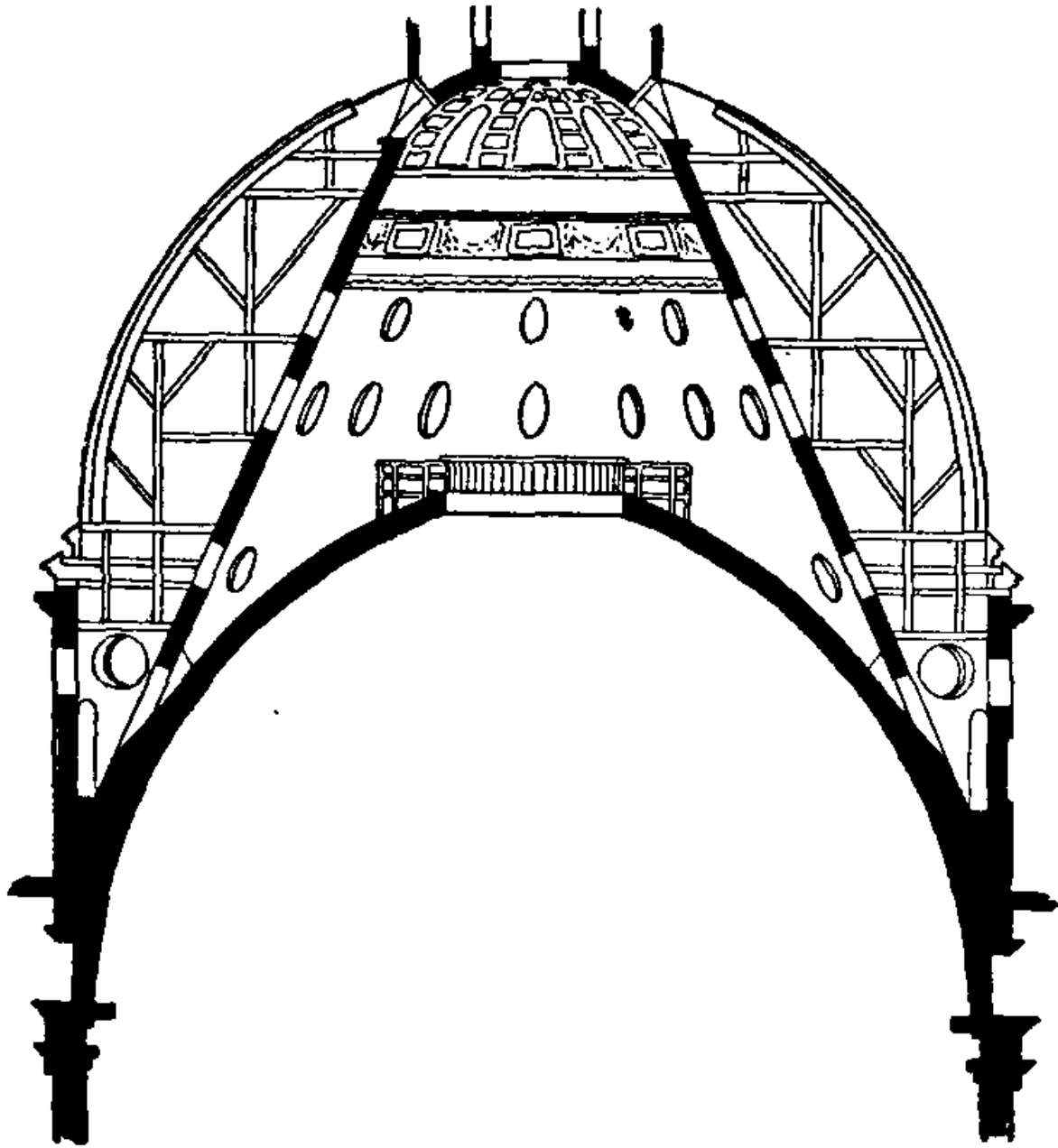
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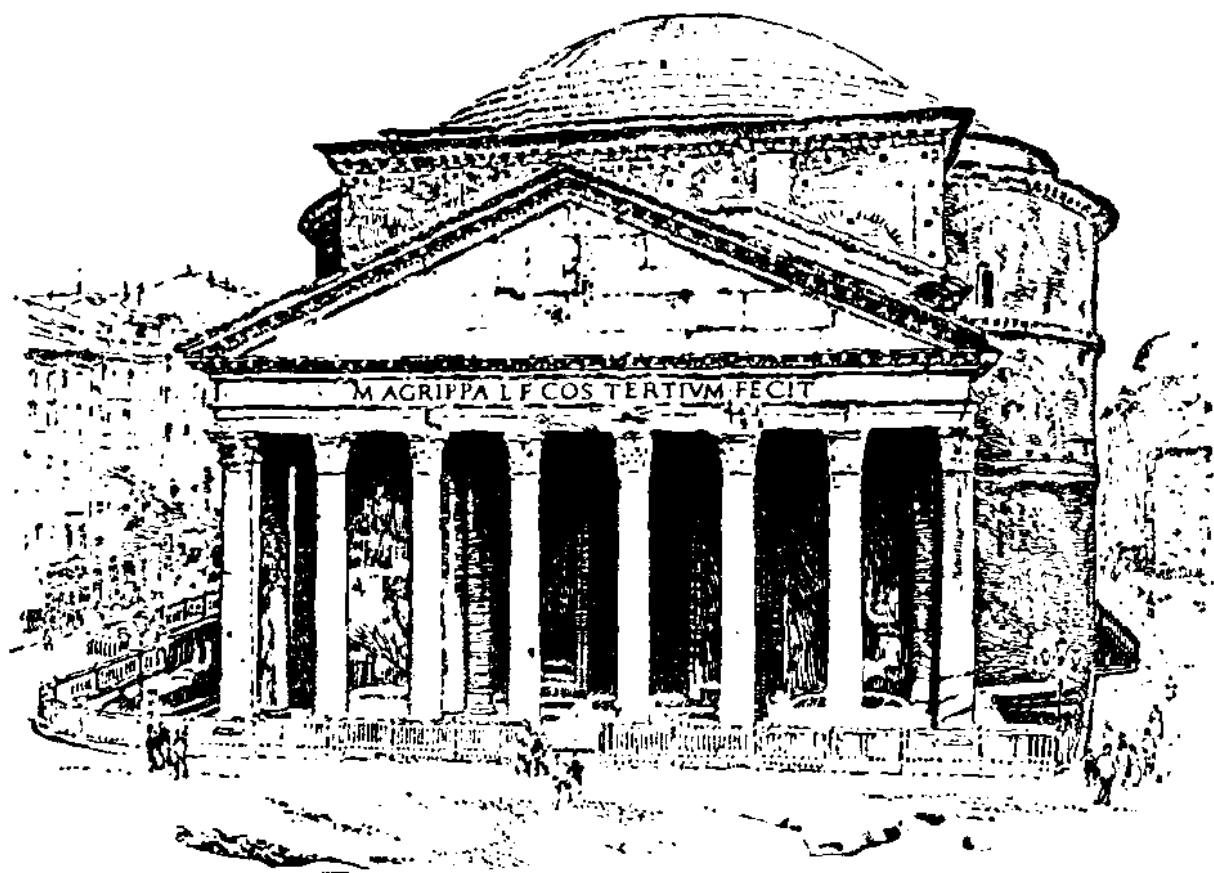
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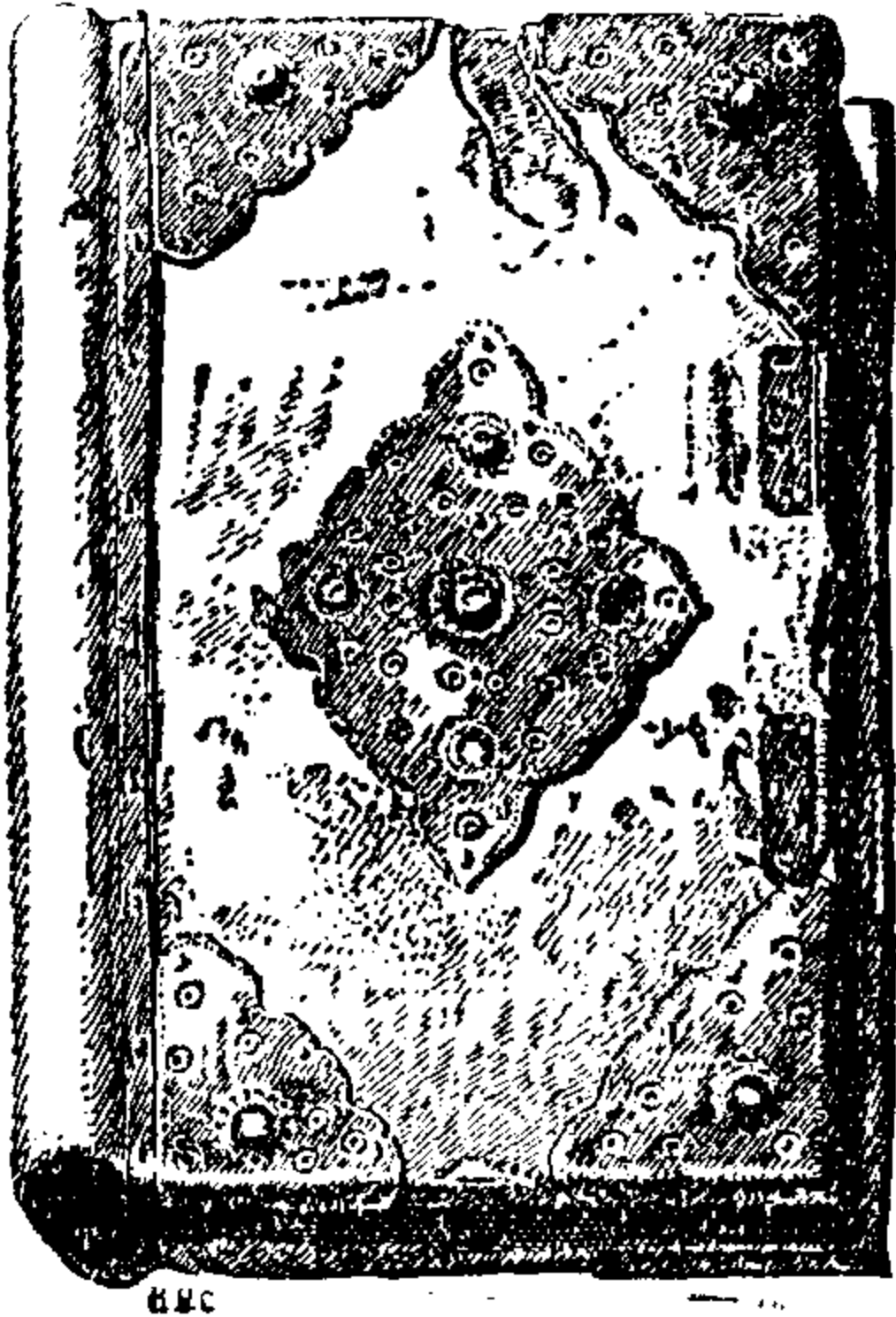
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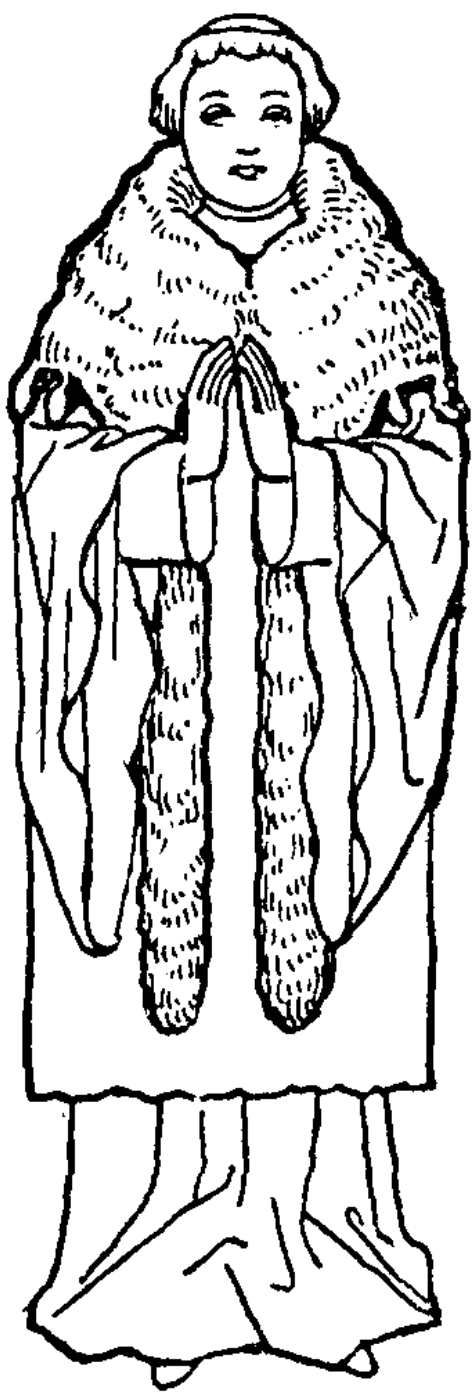
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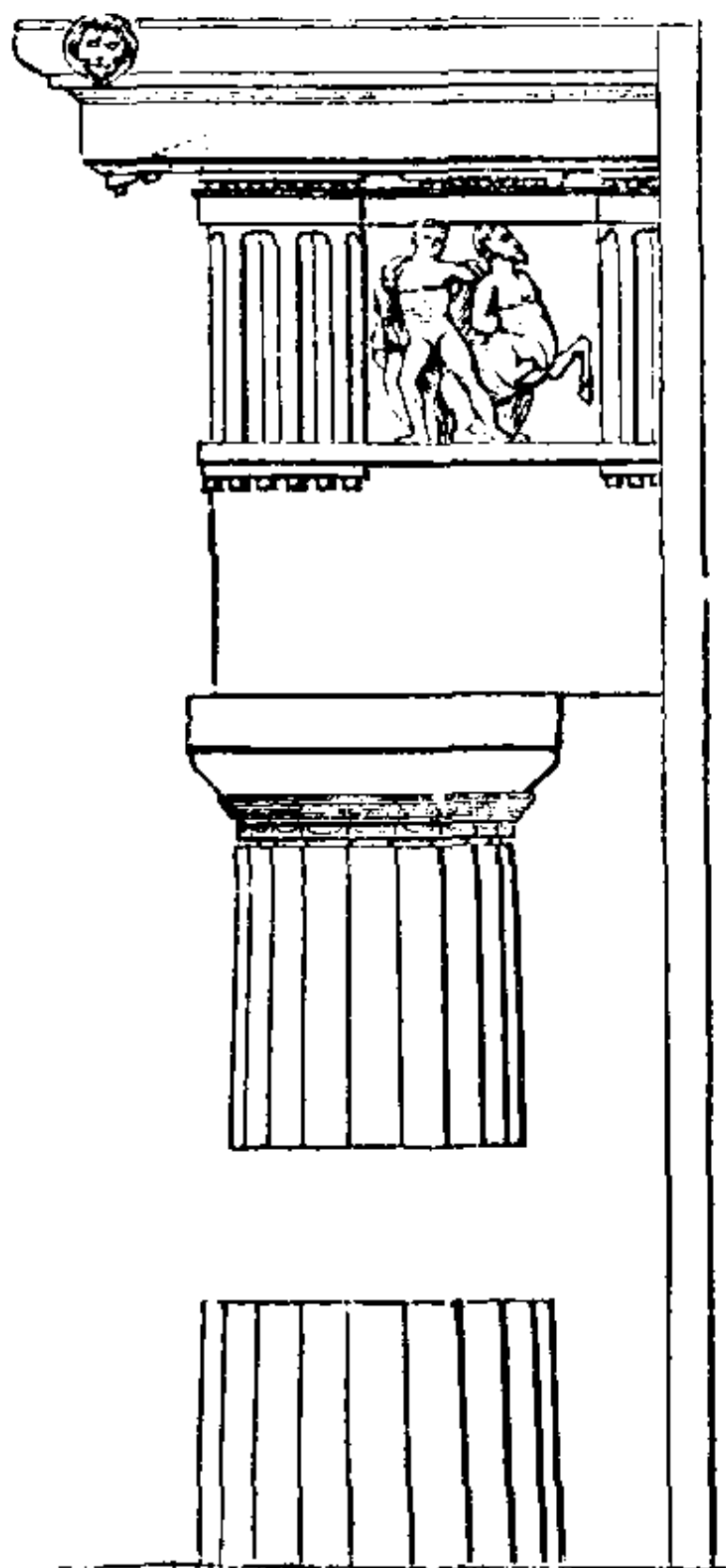
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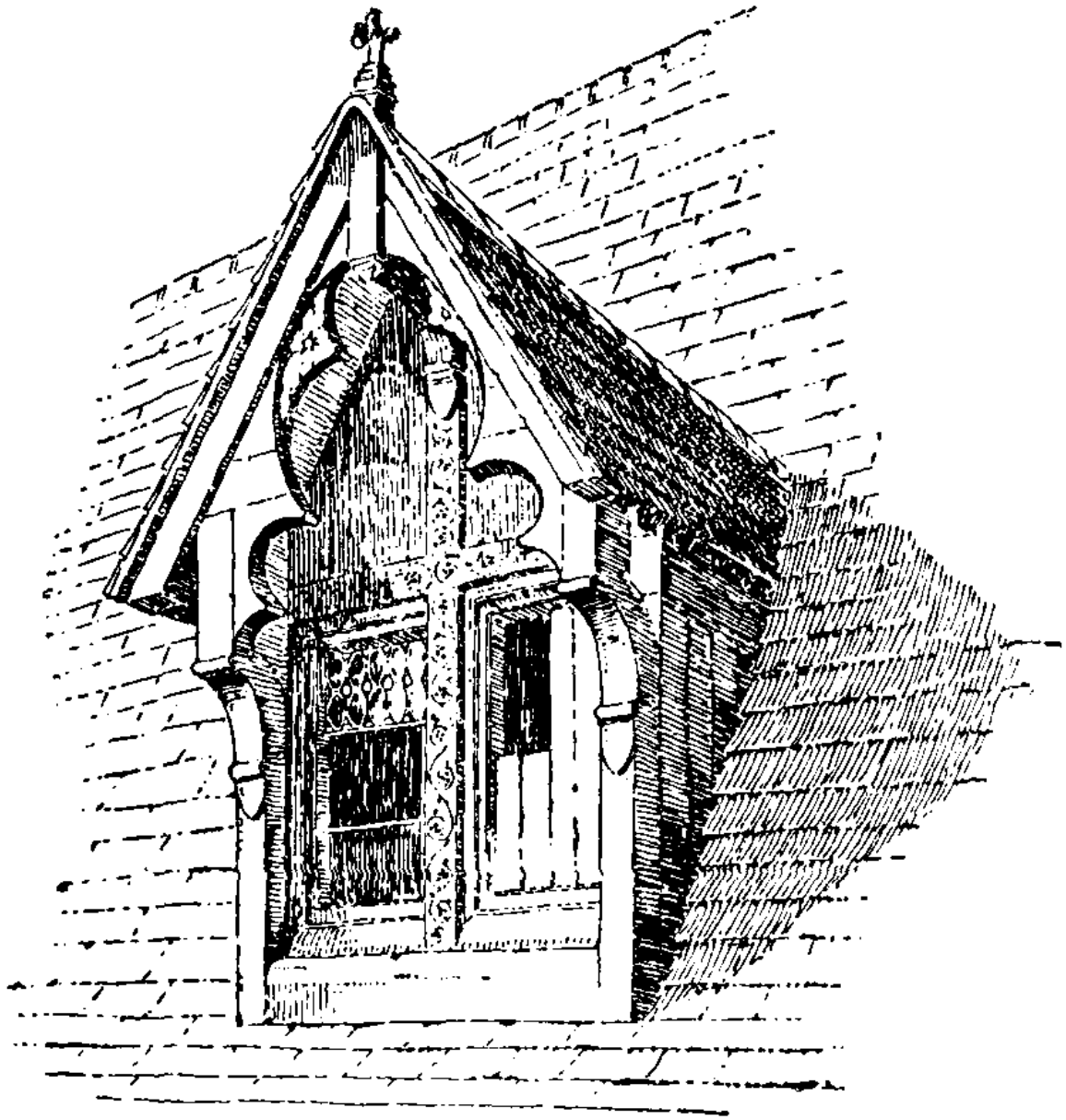
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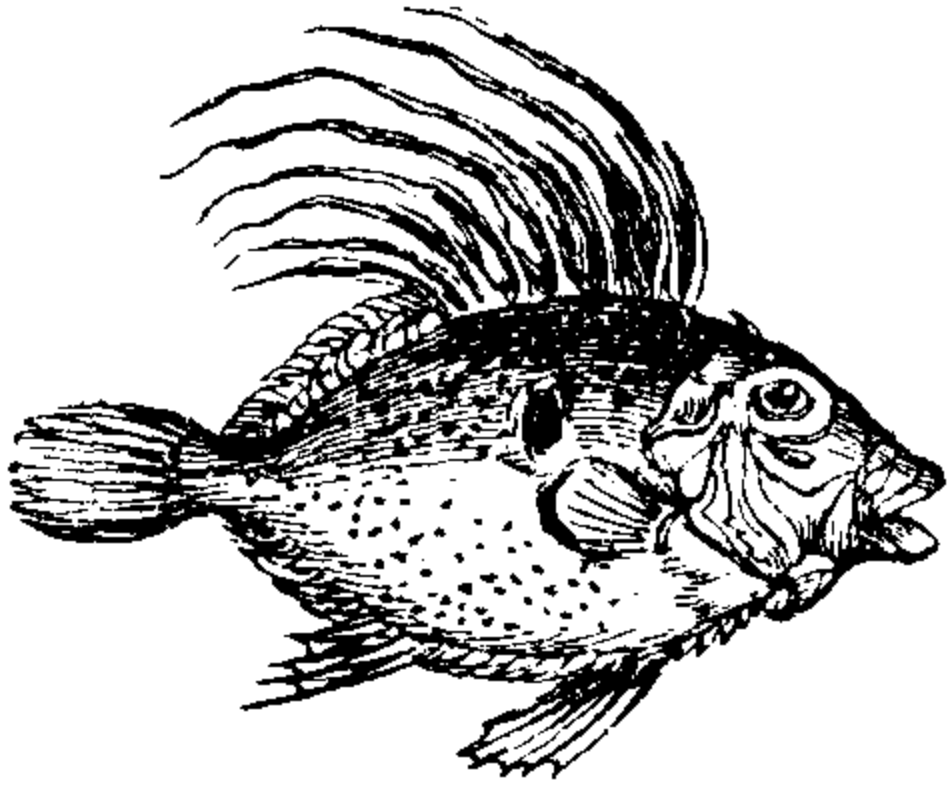
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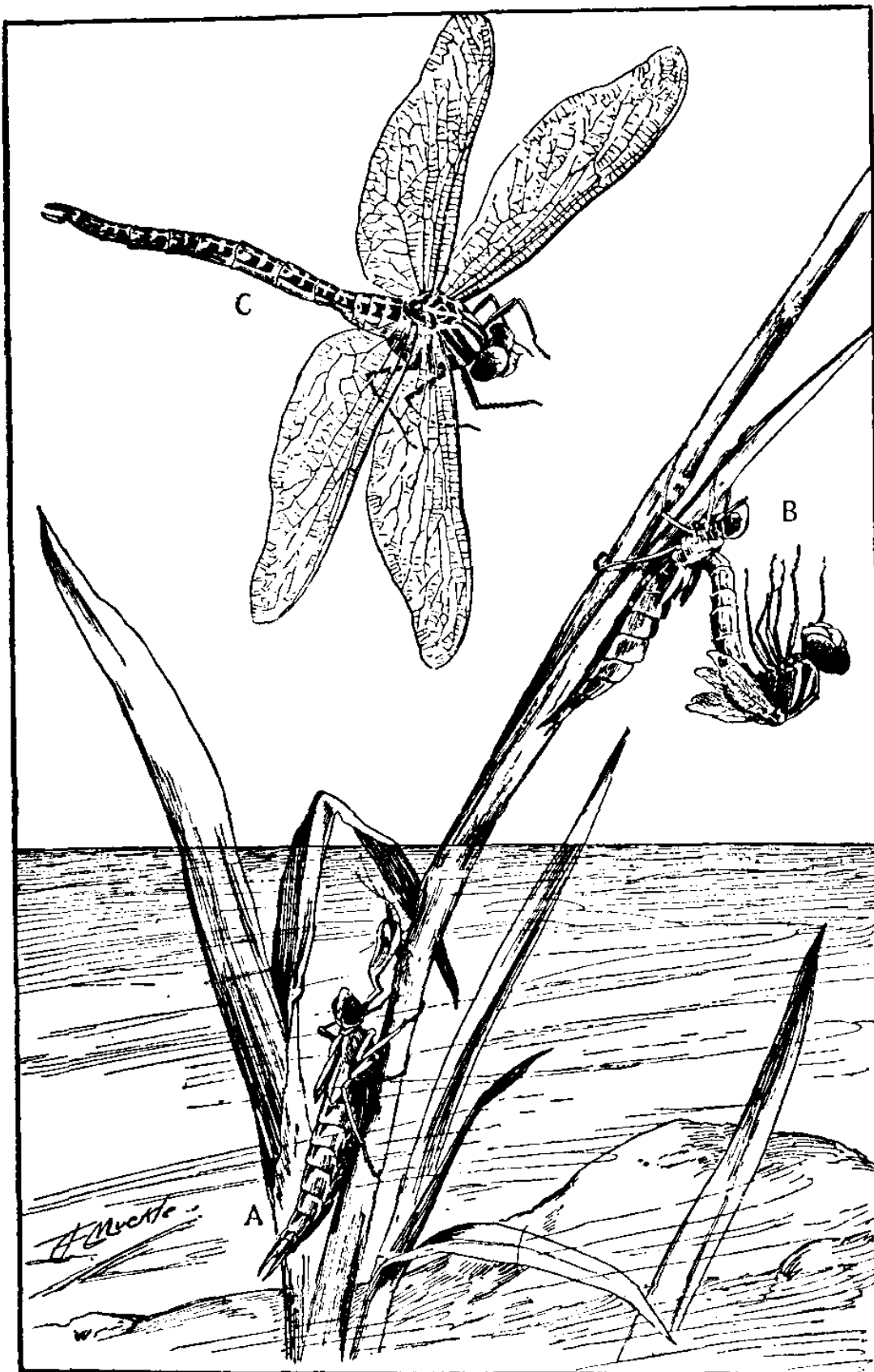
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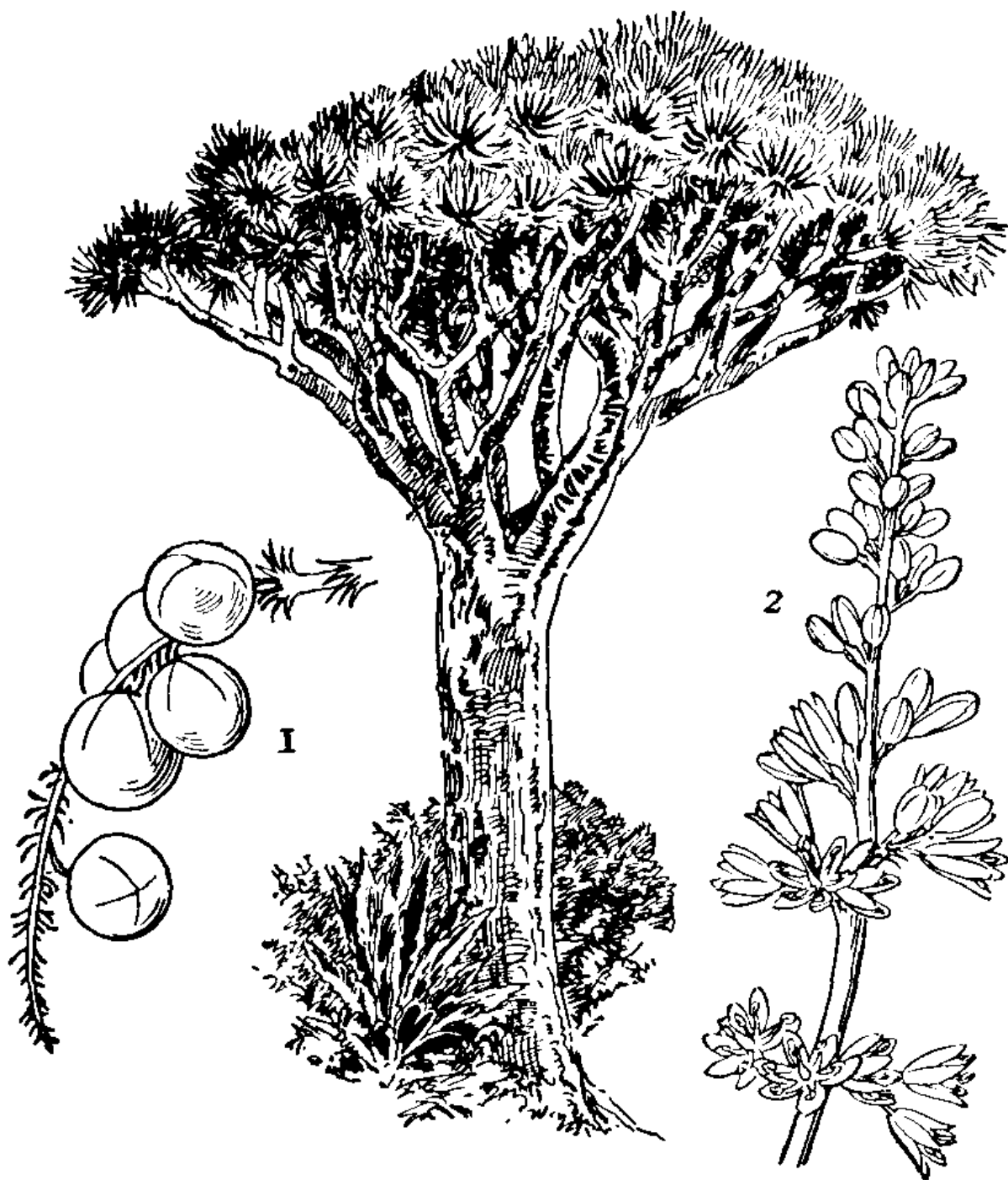
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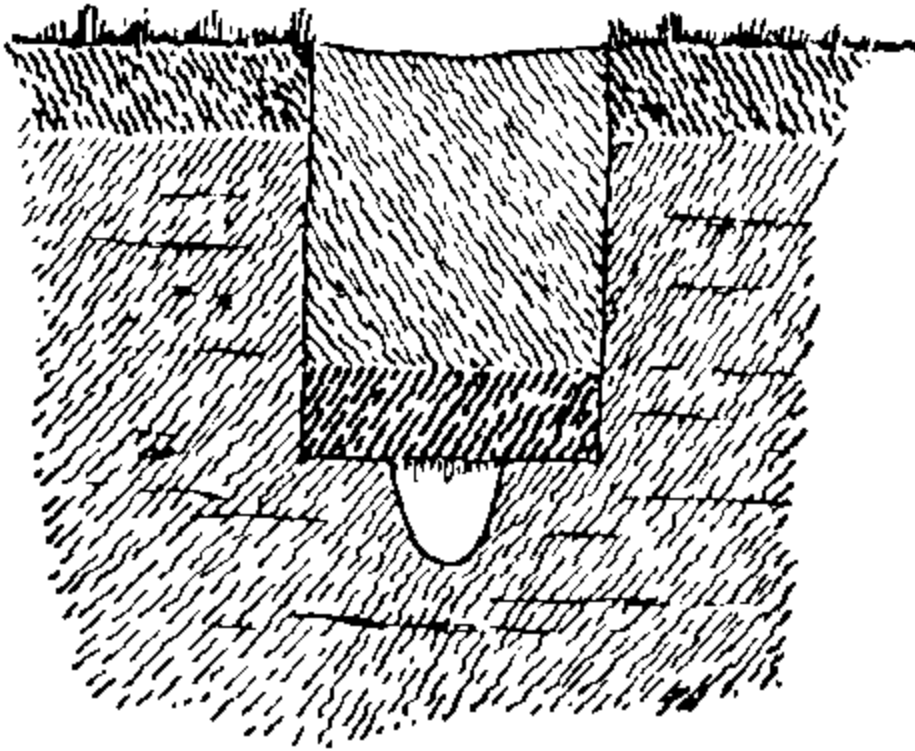
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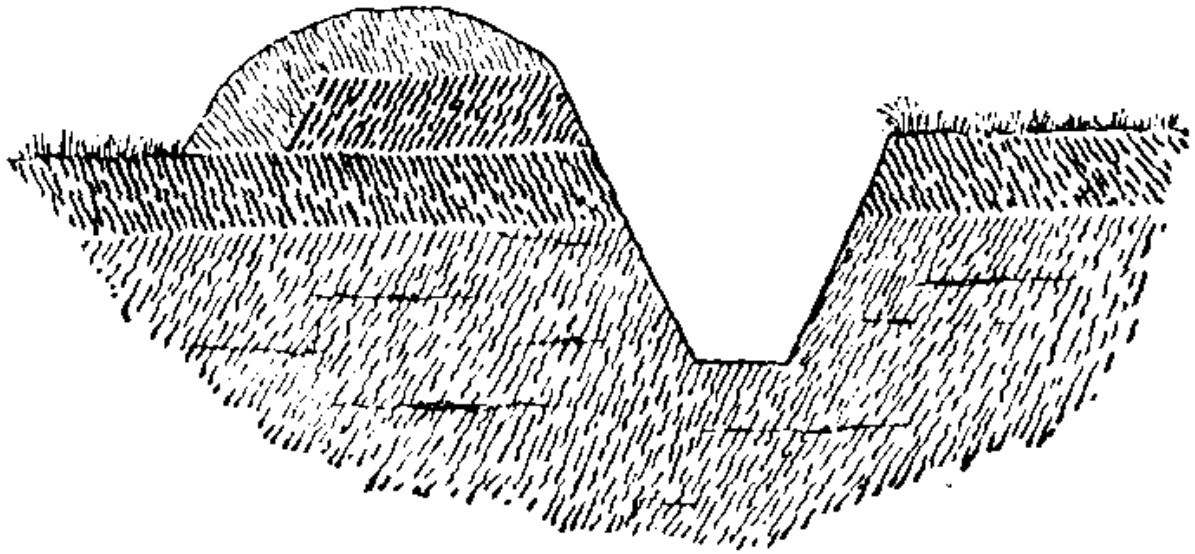
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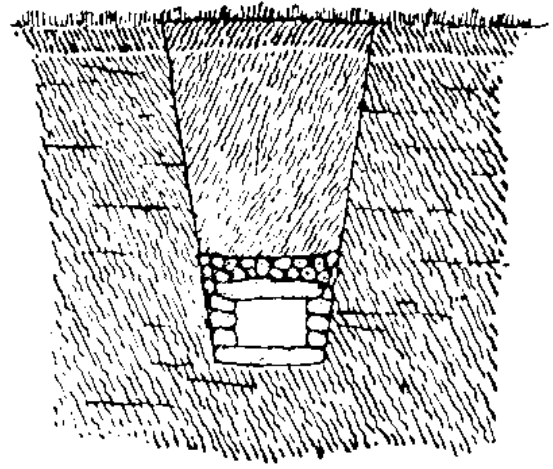
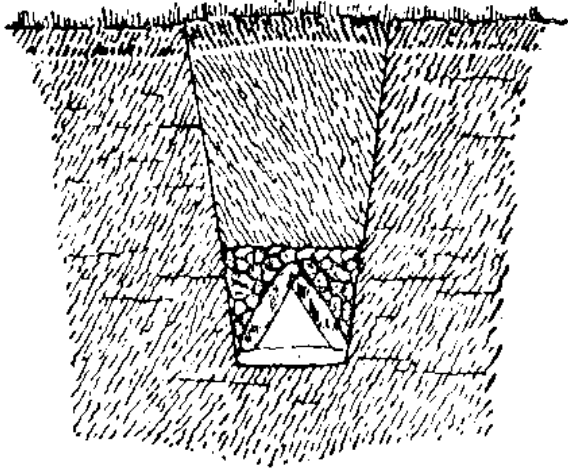
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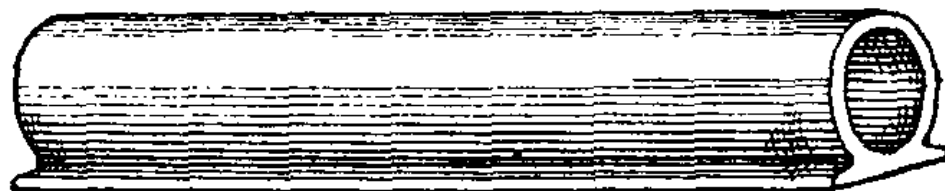
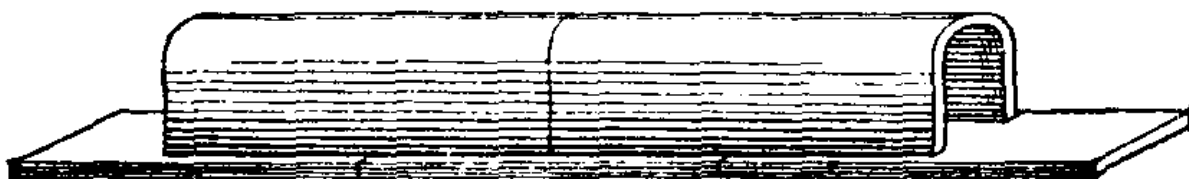
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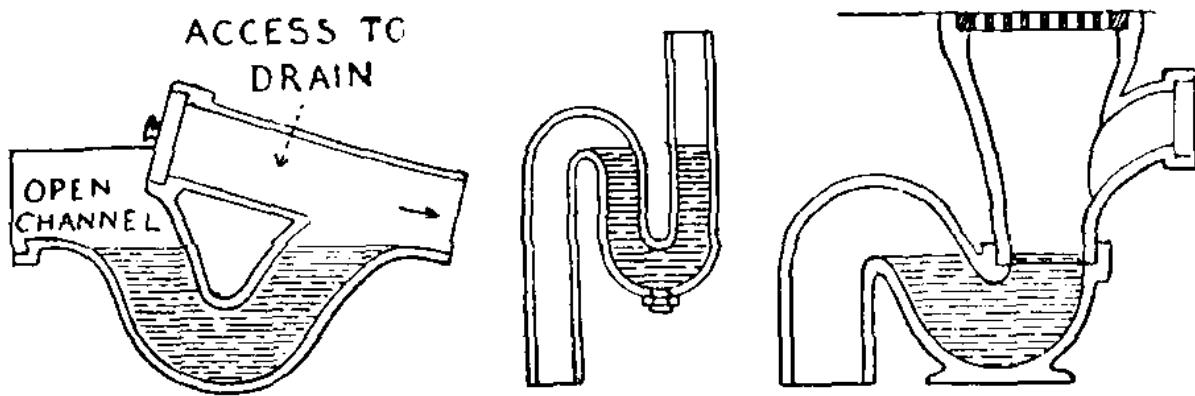
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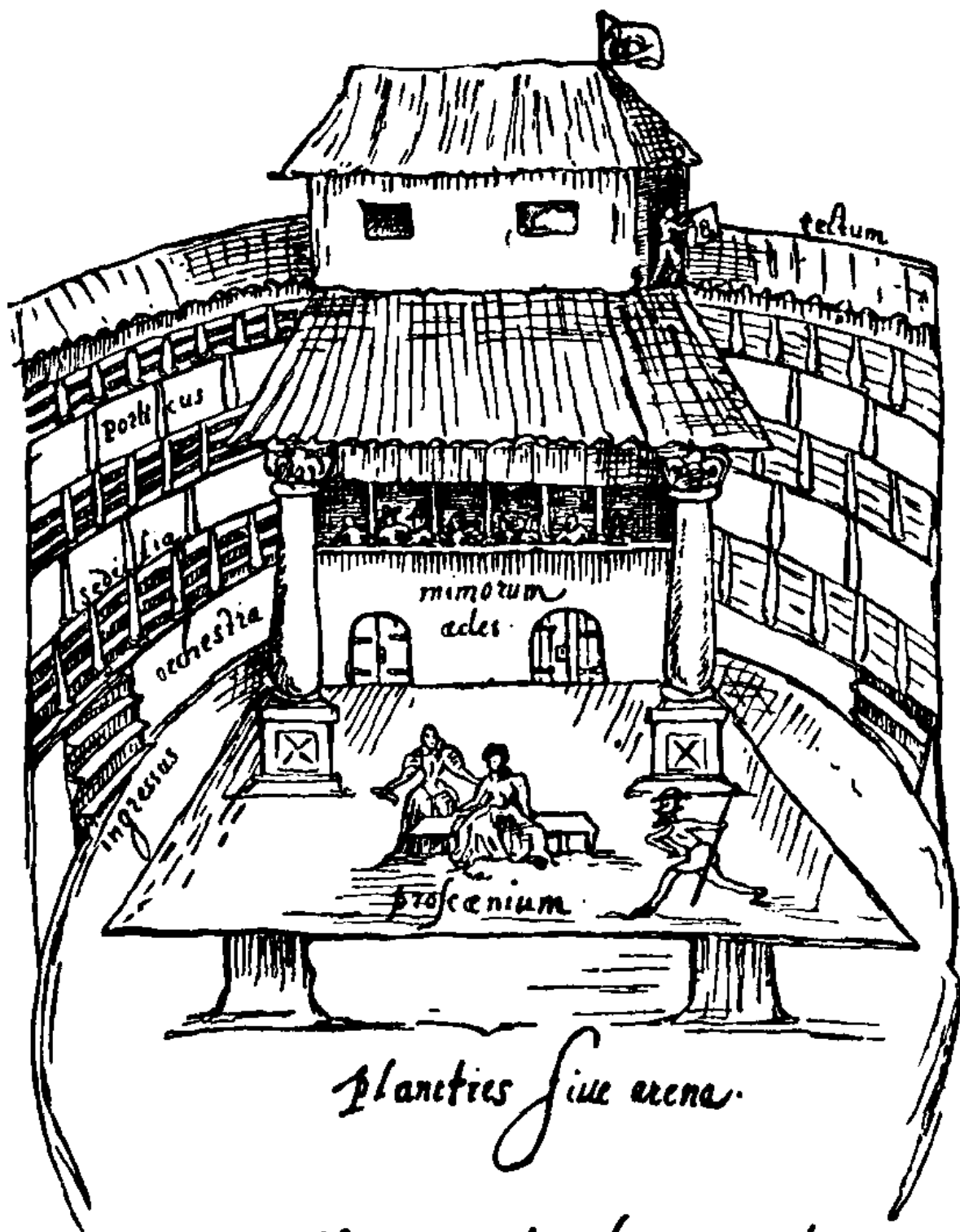
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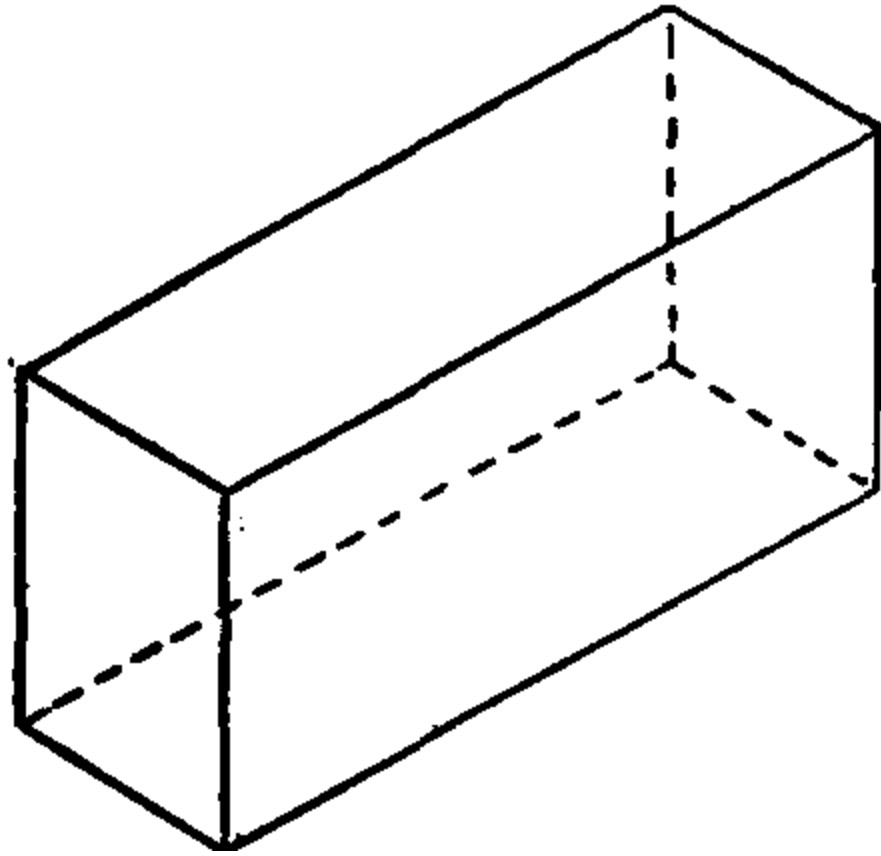


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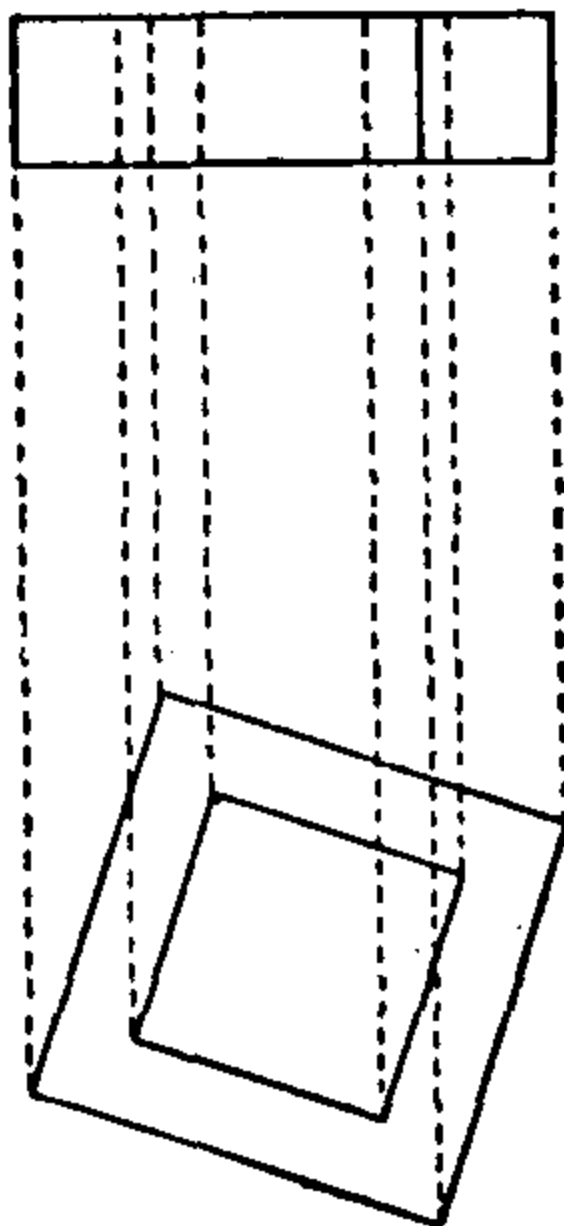


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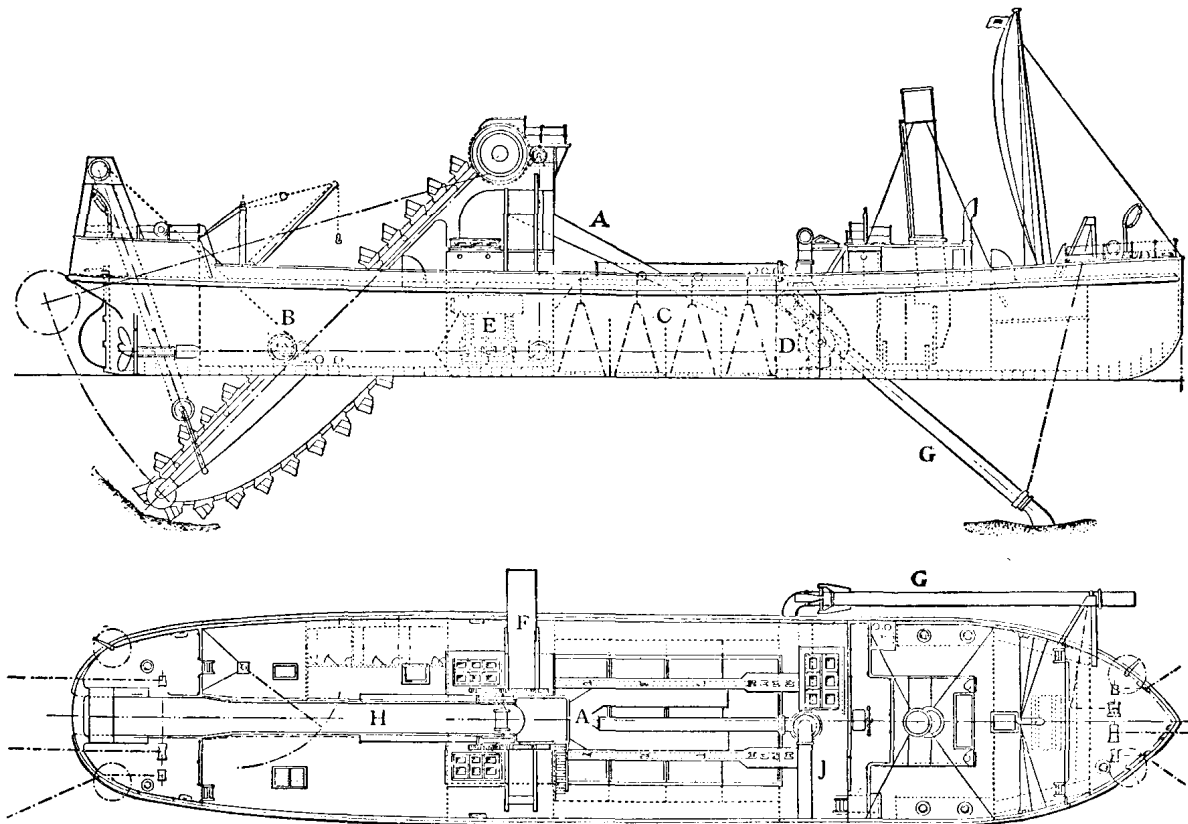
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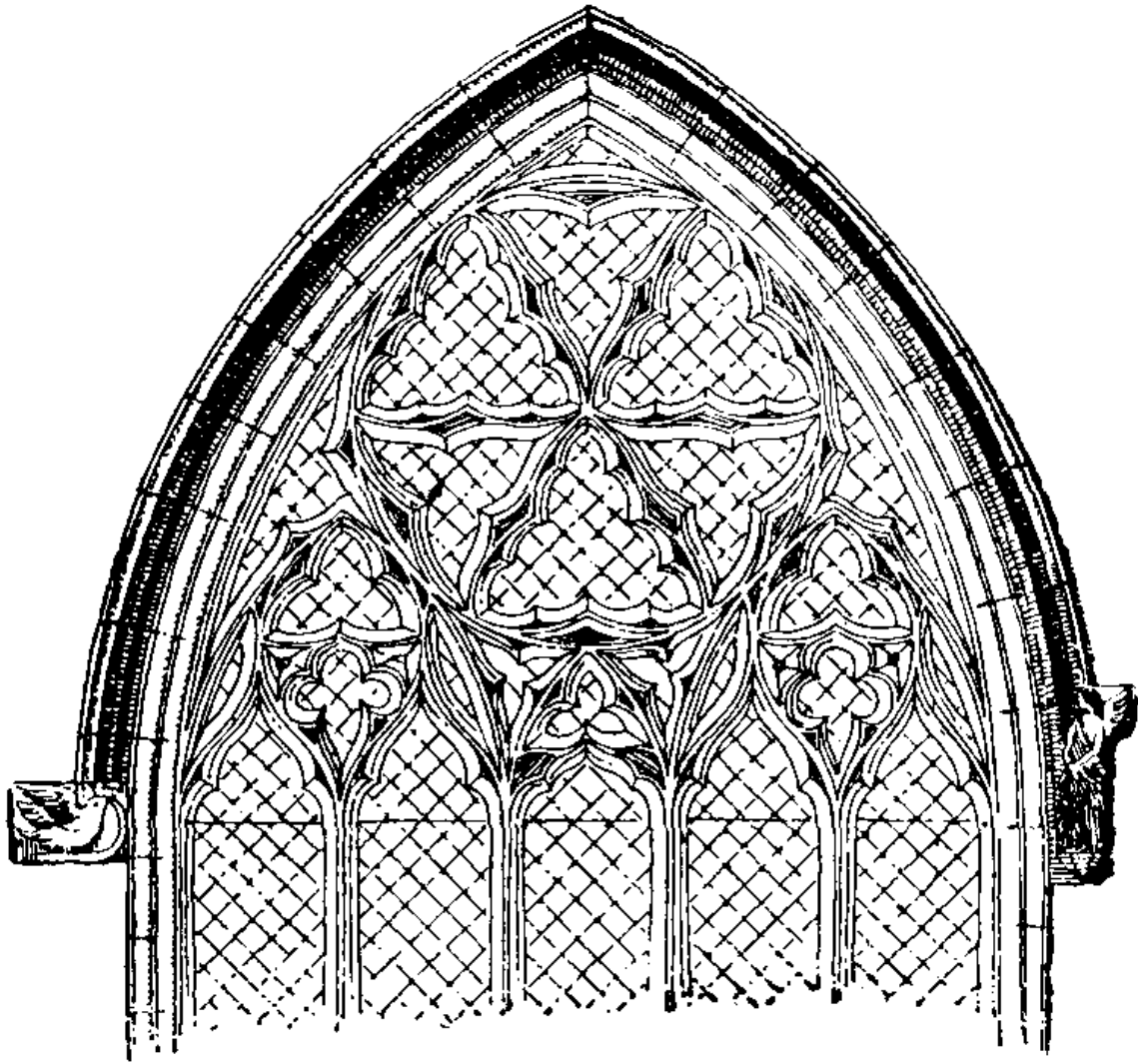
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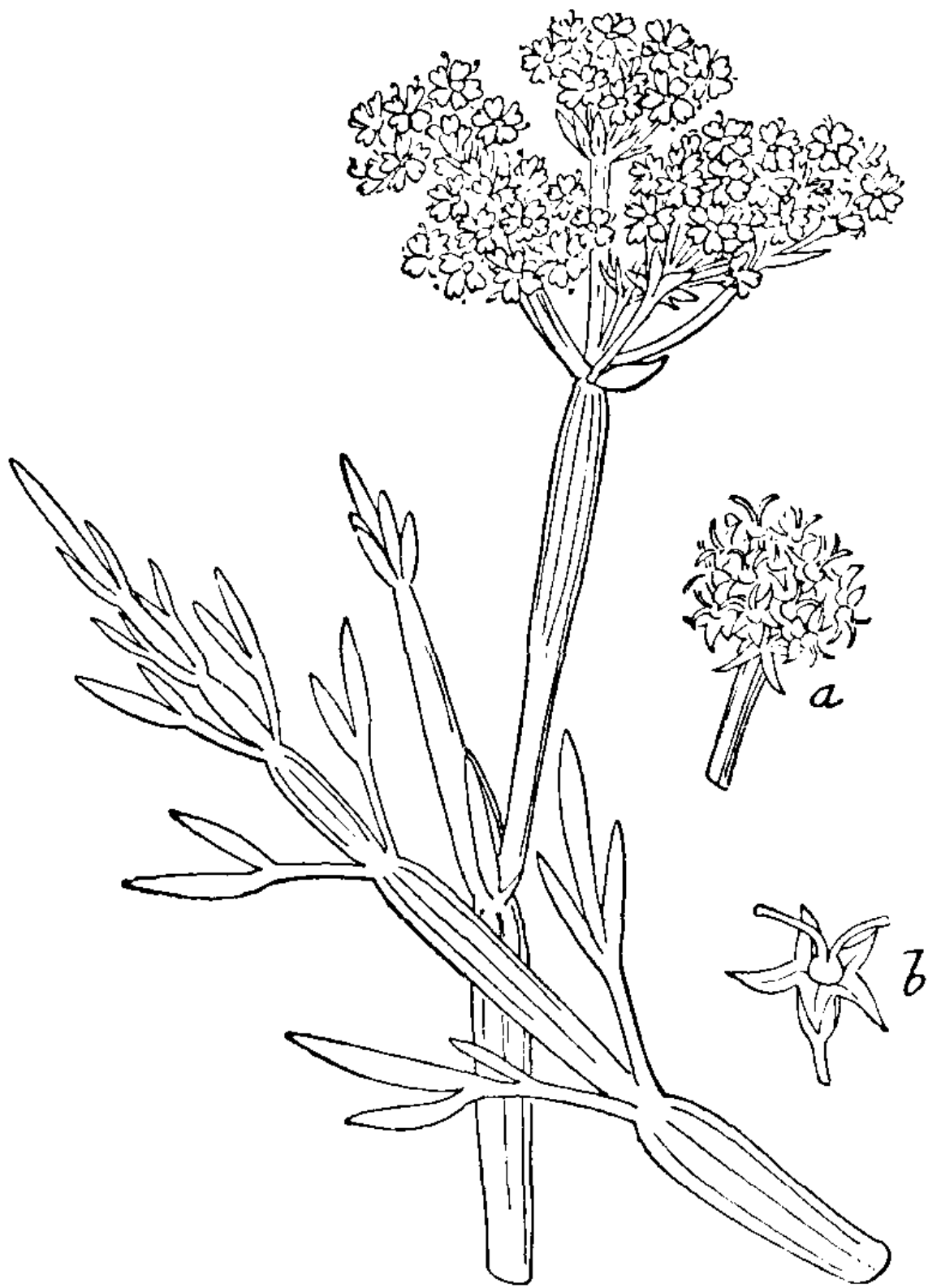
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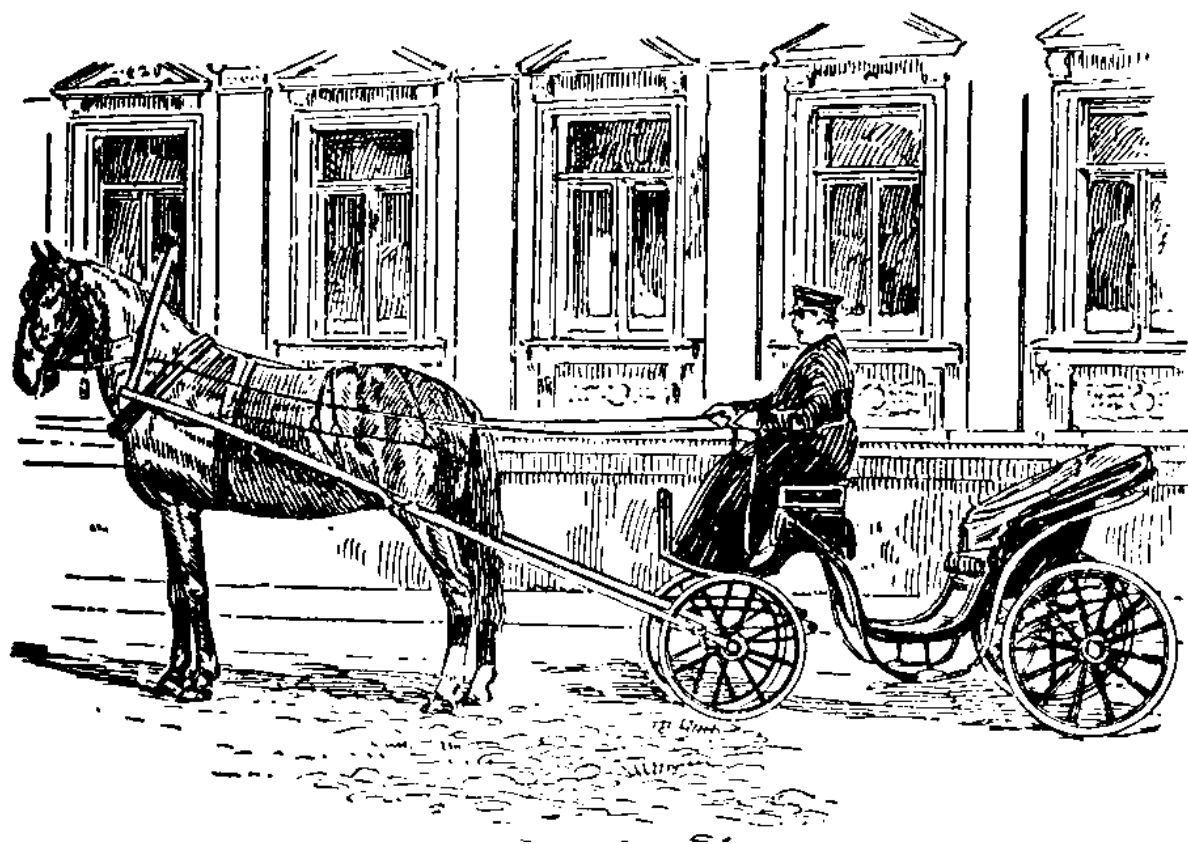
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Fig. 1

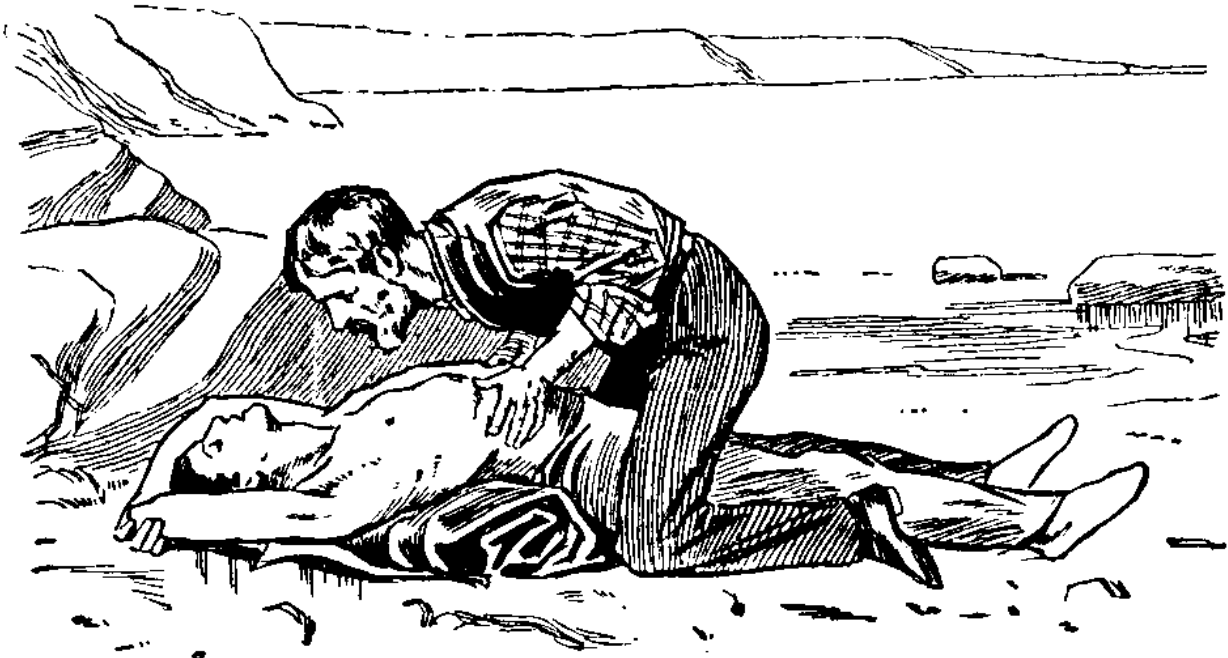
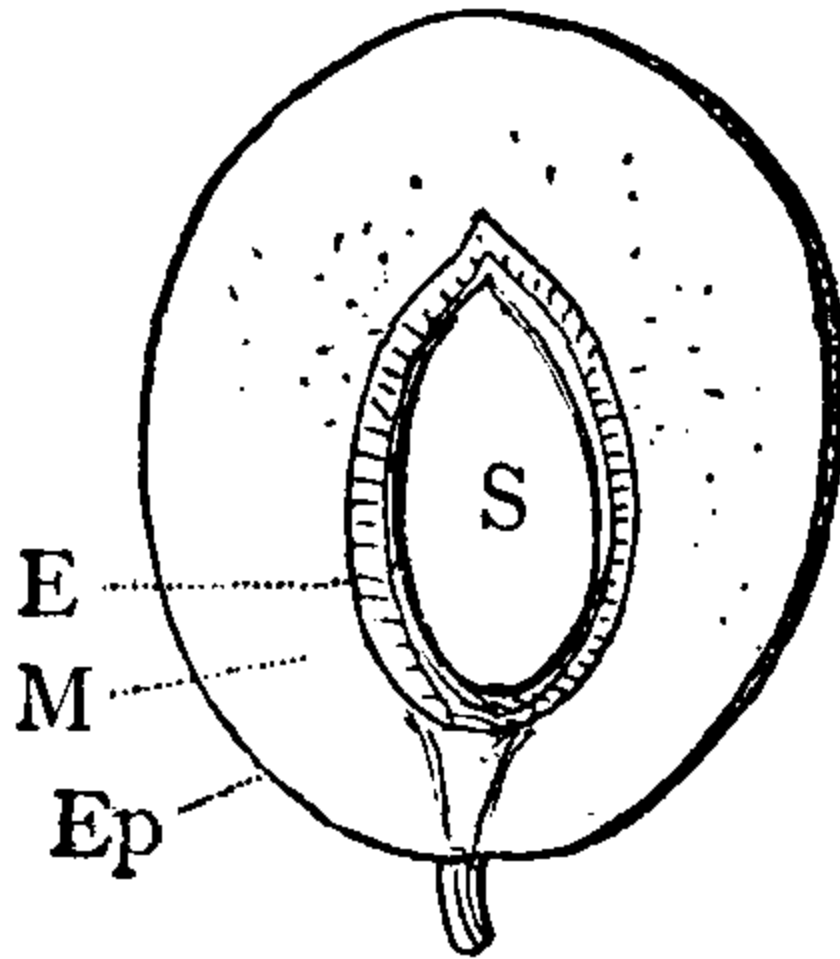
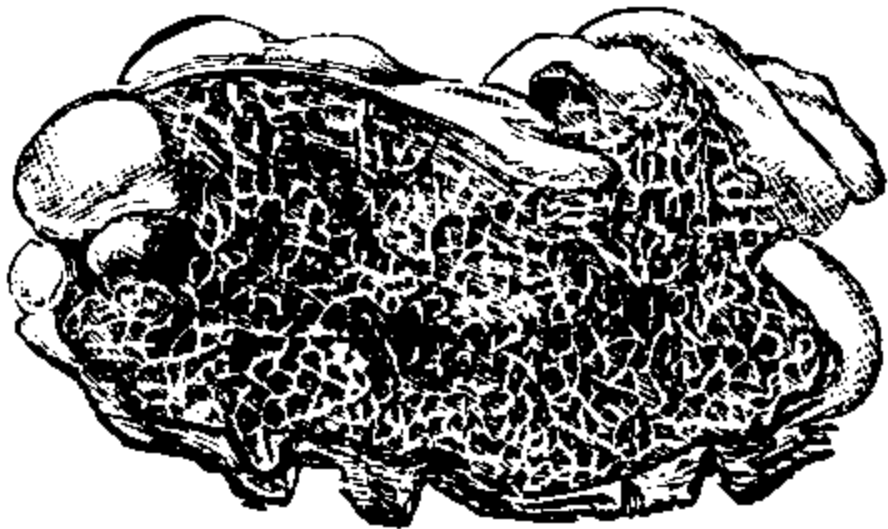


Fig. 2

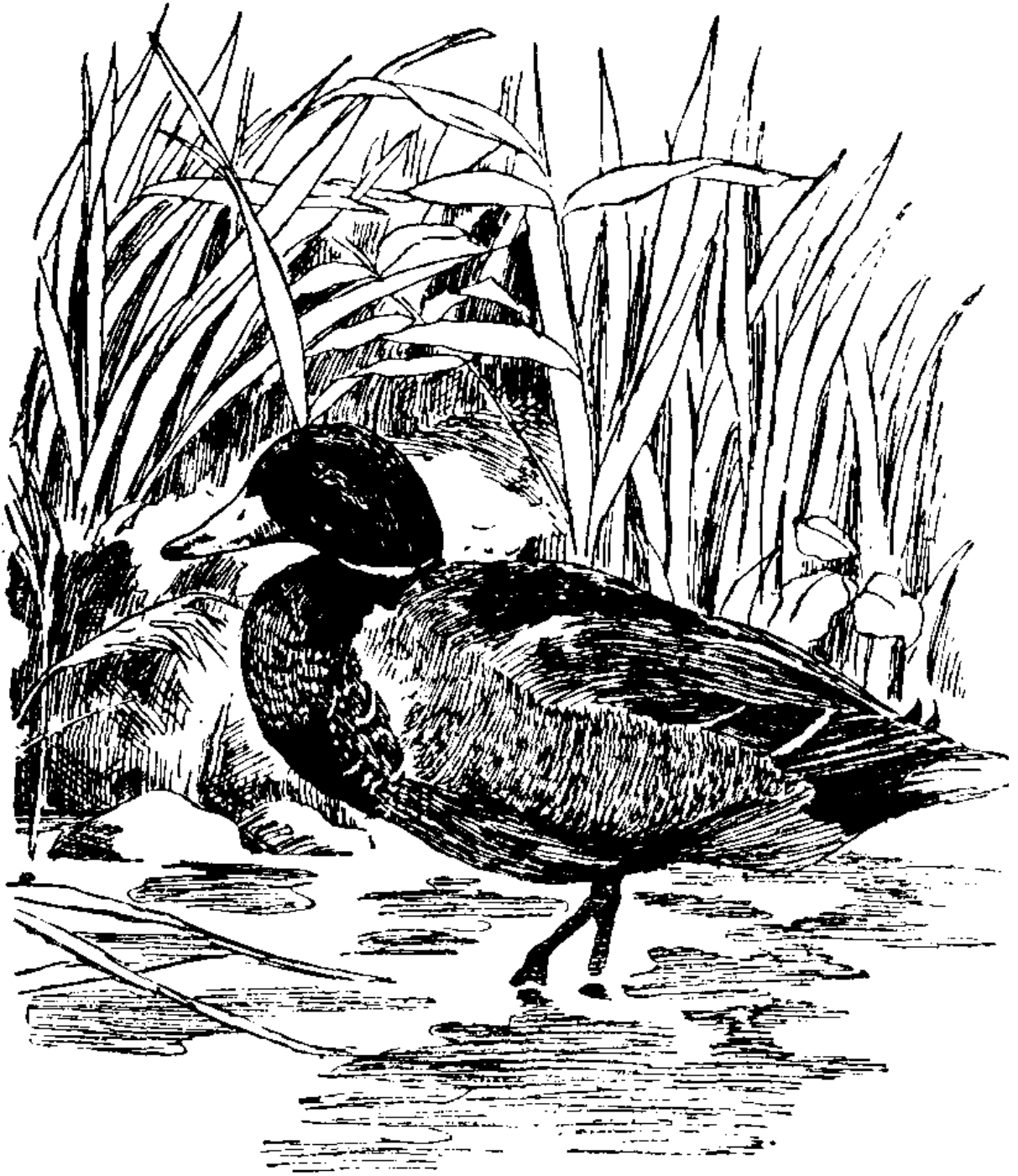
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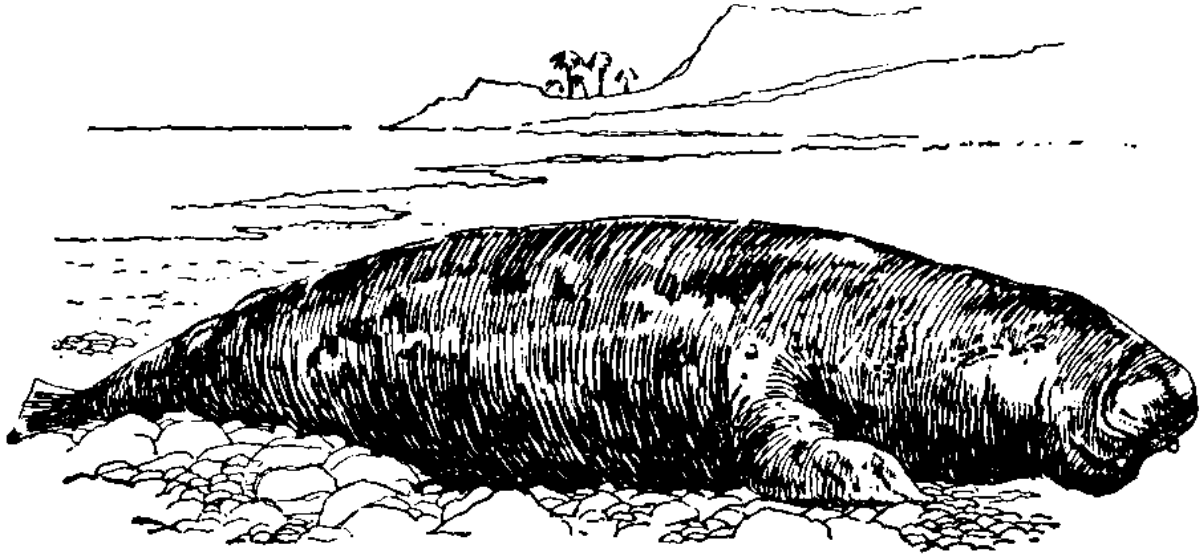
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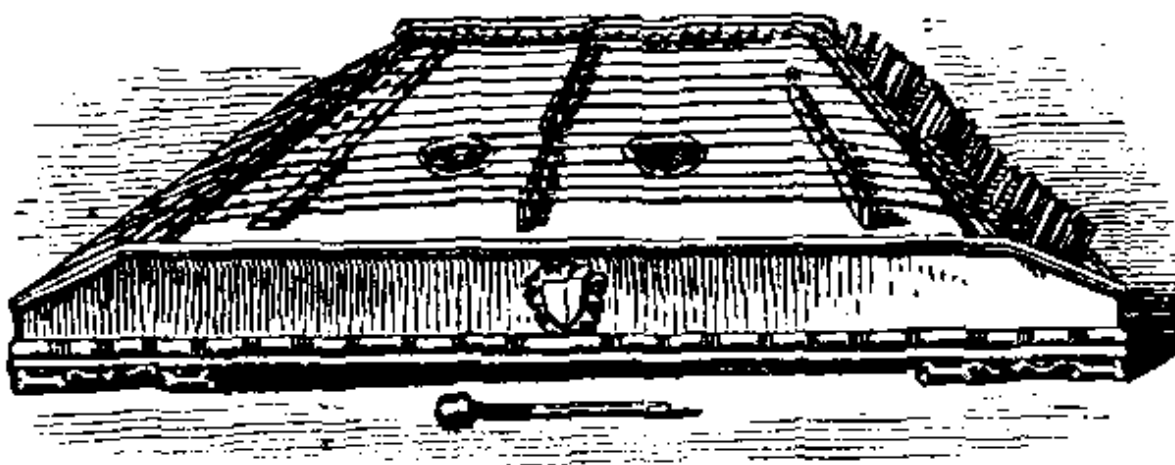
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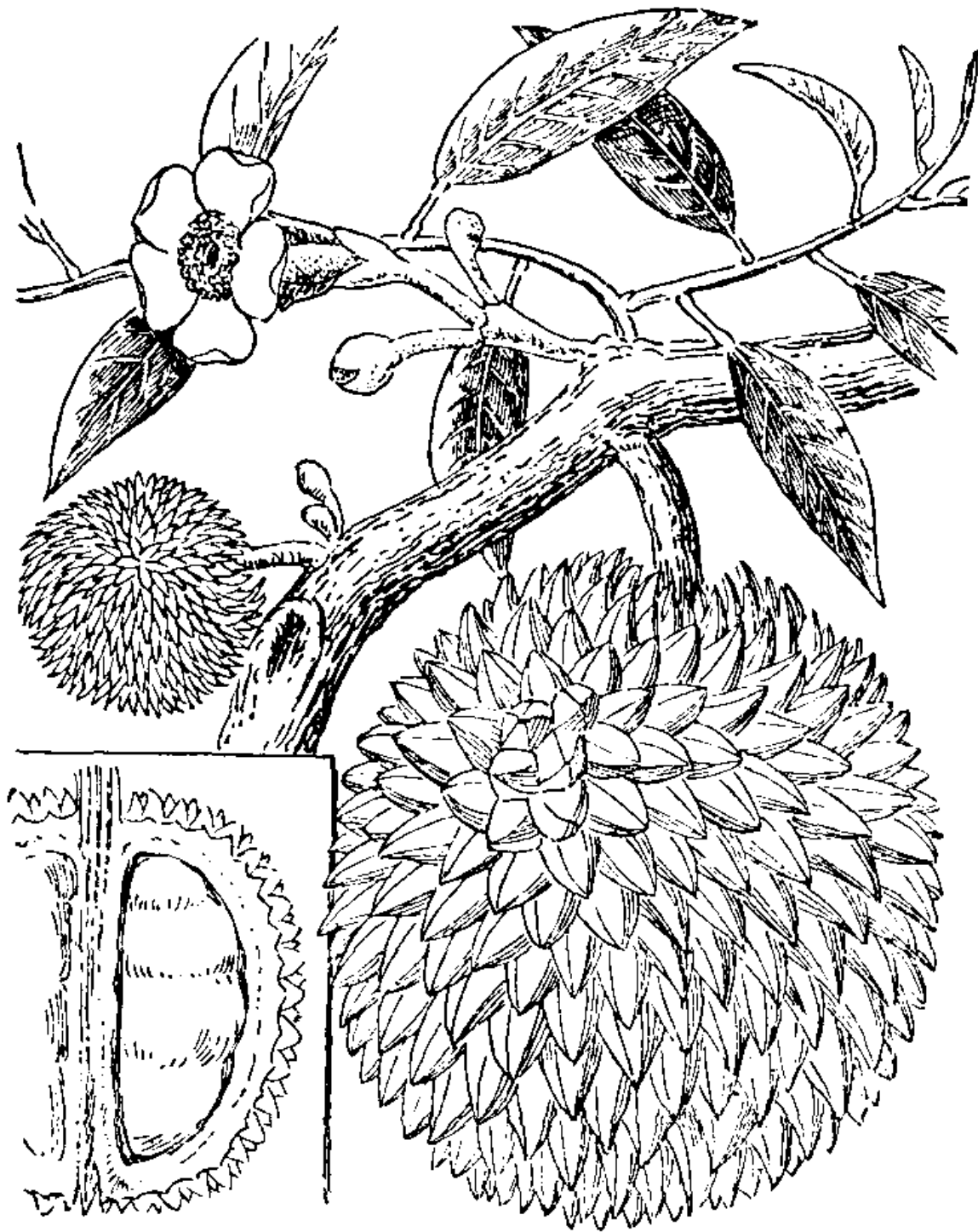
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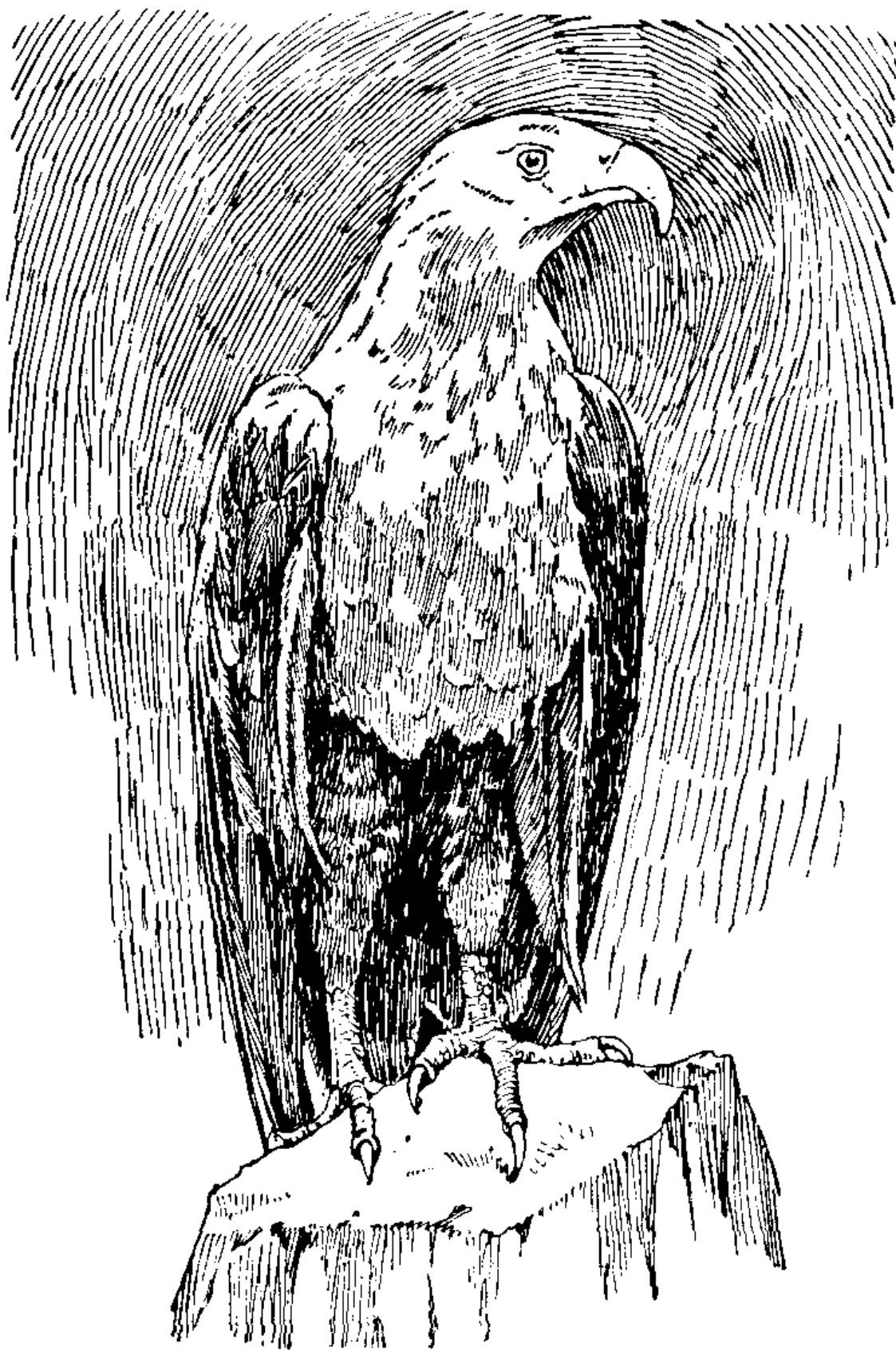
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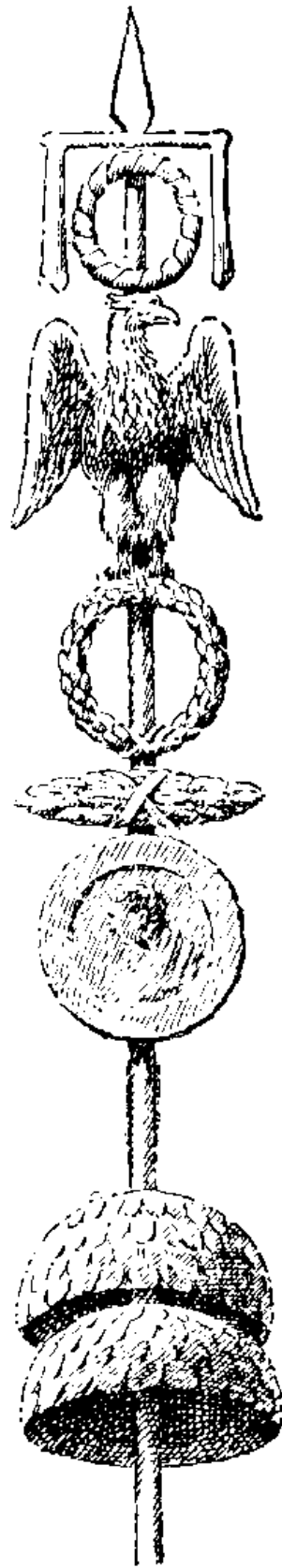
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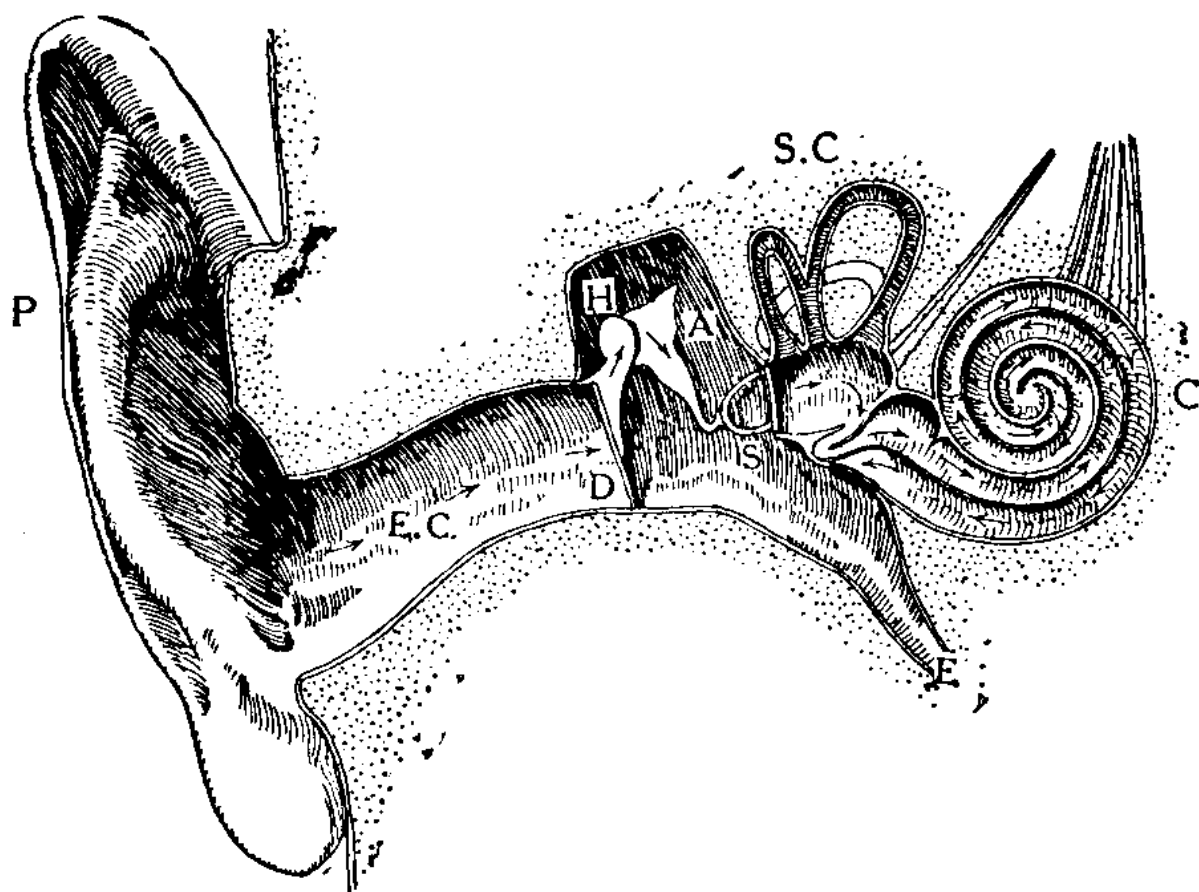
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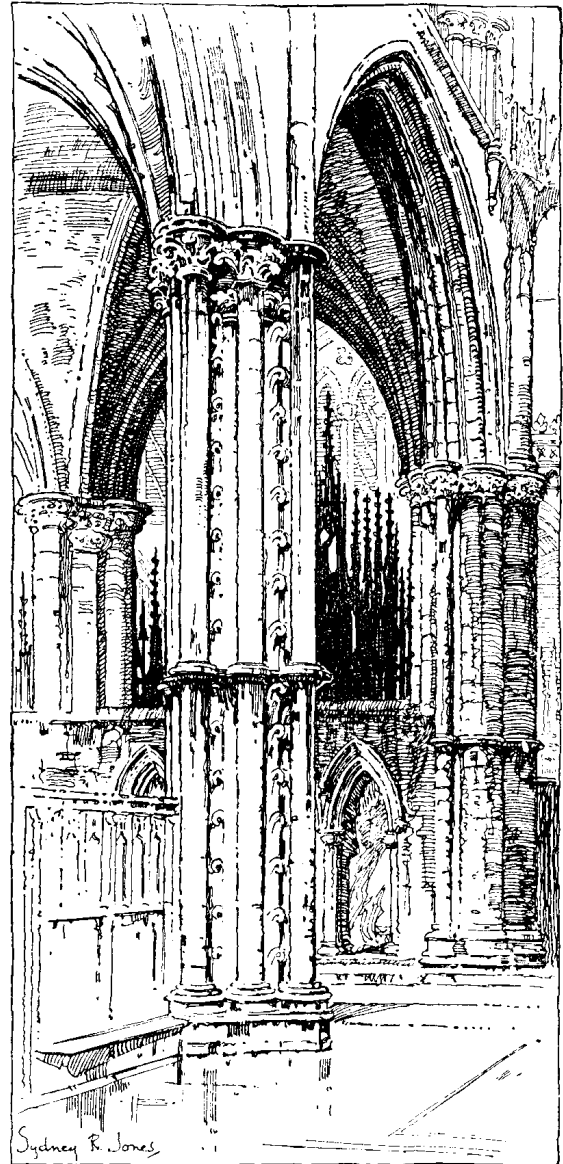
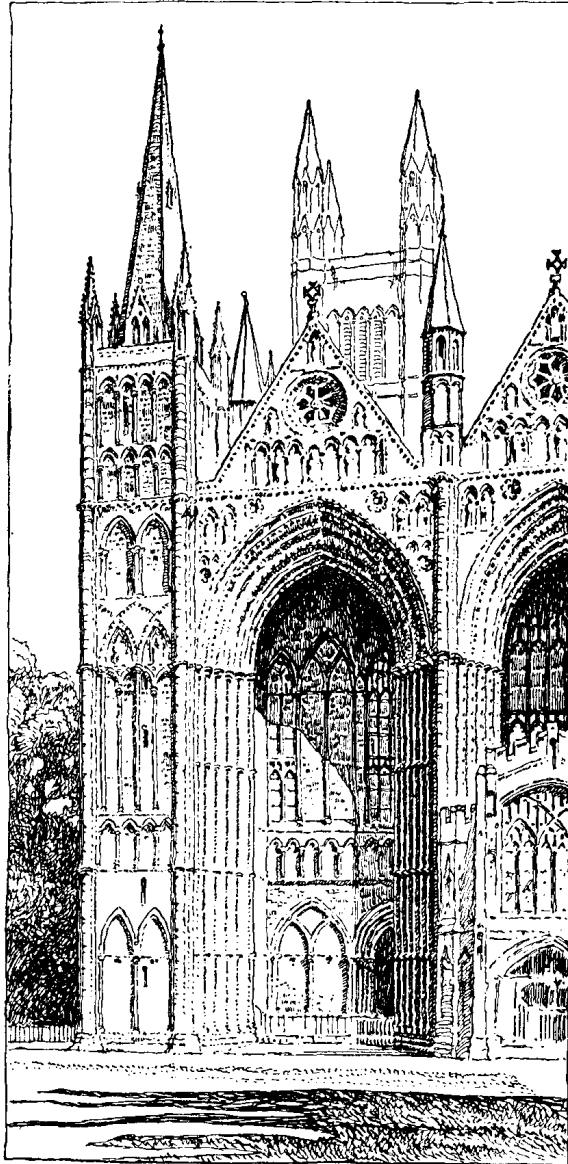
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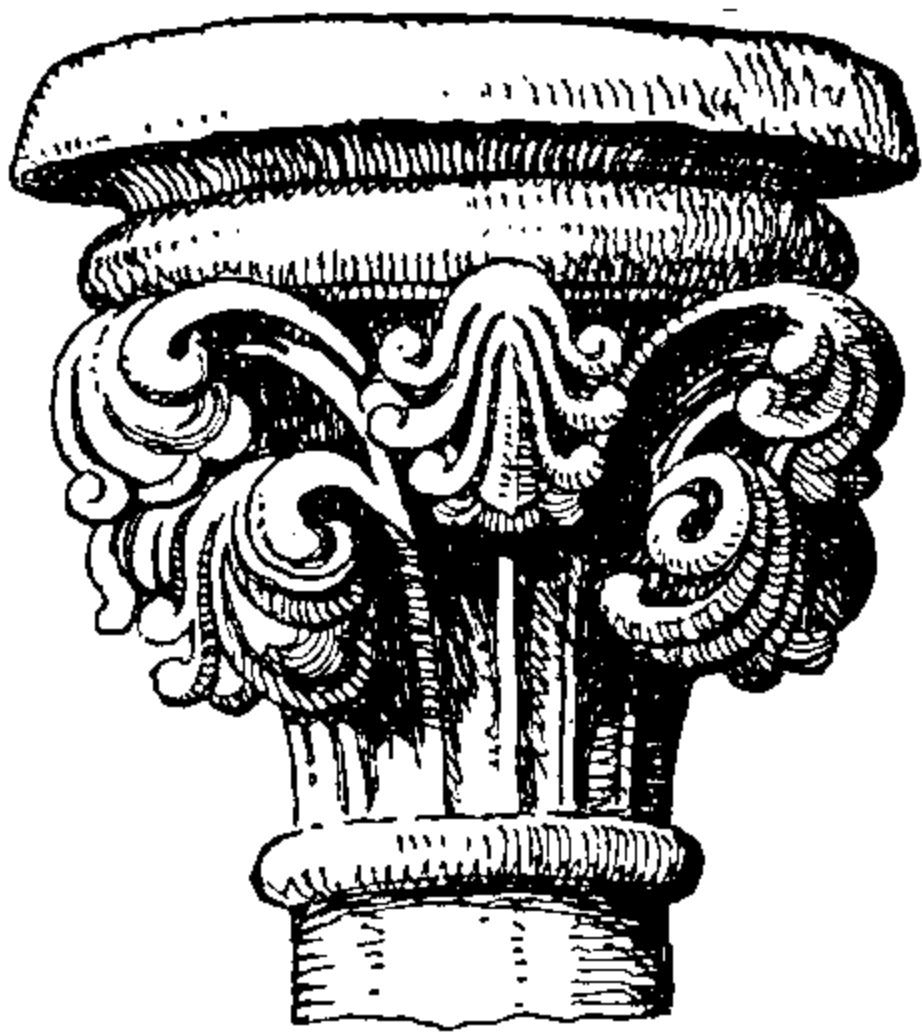
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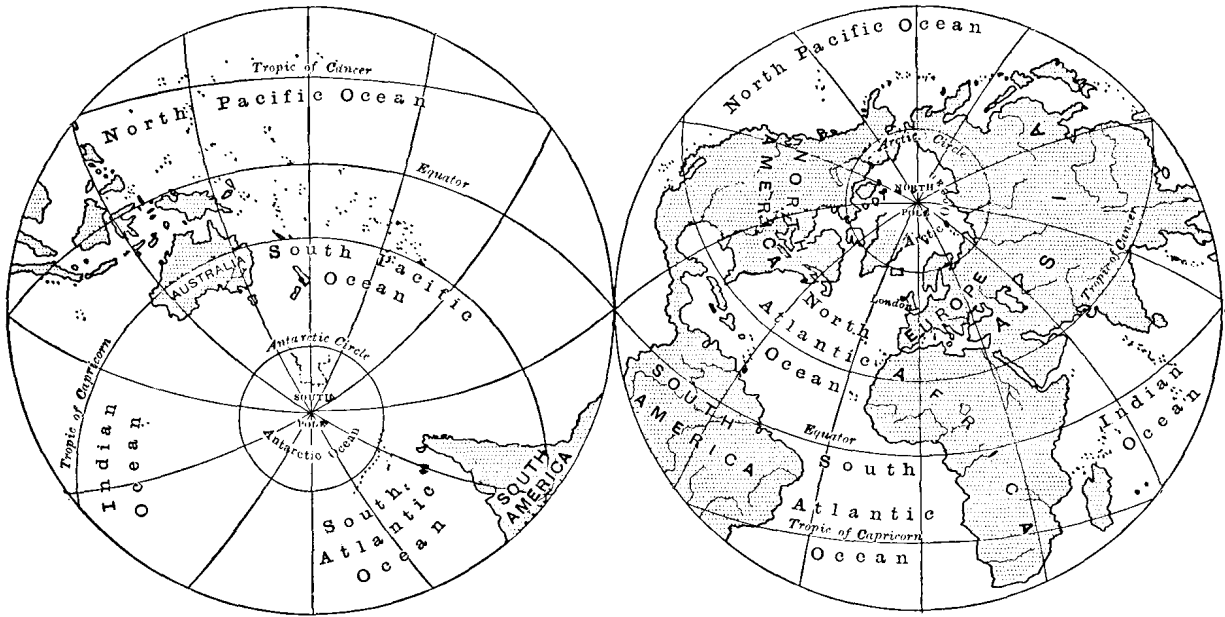
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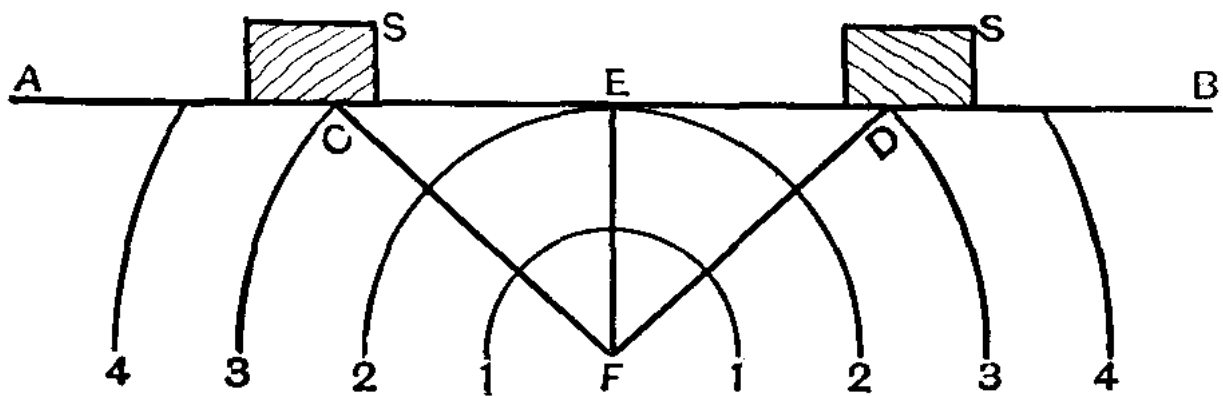
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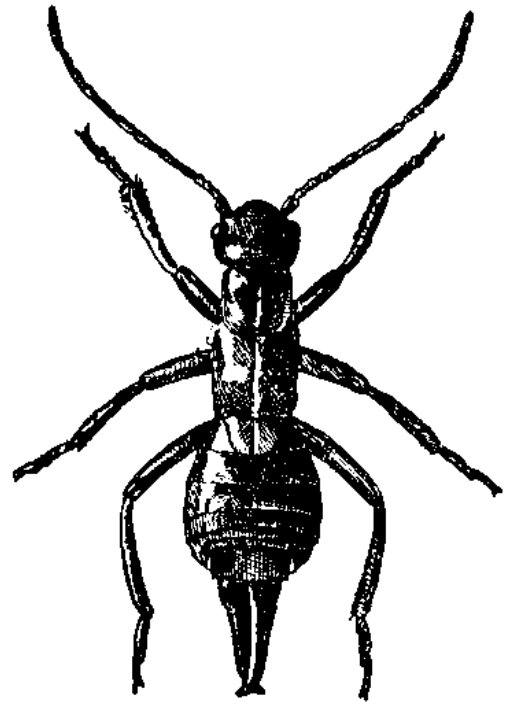
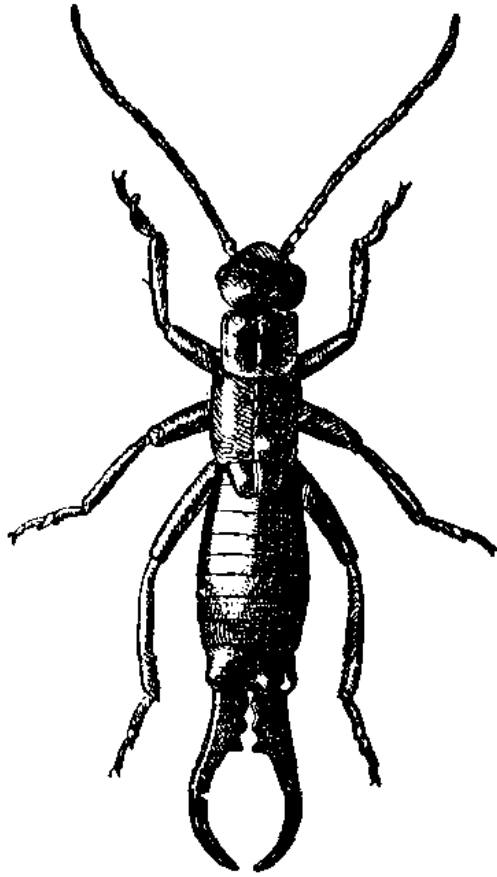
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